Geist in the Machine: Mutual Recognition and Multiagent Architecture for Dialectical AI Tutoring

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Abstract

A companion pilot study established that recognition-enhanced prompts and multiagent architecture produce large effects on AI tutoring quality (d=1.11 in a $2 \times 2 \times 2$ factorial, N=4,312). This paper asks: *through what mechanisms?*

We motivate three candidate mechanisms drawing on Hegel's recognition theory, each at a distinct architectural level: **calibration** (prompt-level output distribution narrowing), **error correction** (architecture-level superego critique), and **adaptive responsiveness** (interaction-level turn-by-turn adaptation). To trace these mechanisms, we adapt process tracing—a methodology from comparative politics—for AI agent architectures, combining superego critique taxonomy, revision delta analysis, and trajectory analysis within a provable discourse framework that machine-verifies every mechanism claim against data.

We test the model on a 2×2 factorial (recognition $\times$ architecture) across three structurally different generation models (DeepSeek V3.2, N=146; Haiku 4.5, N=163; Gemini Flash 3.0, N=144) spanning a ~30-point baseline tutor-quality range, validated by three independent judges (Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4; 1,296 total scored rows, zero nulls). The headline result is that **two of the three pre-registered mechanisms replicate across all three tested models and three independent judges, while their magnitude and architectural completeness vary systematically with generation quality; the third is null on the primary well-powered analysis, with an unreplicated exploratory boundary-condition effect in a single scenario**. Generalization beyond this tested

[*]This sentence is the only one actually authored by Liam Magee in this paper.

capability range — to very weak or frontier-capability models — remains an open empirical question (§8.3).

Calibration. Recognition narrows within-response dimension variance (d=0.52–0.64), lifts the weakest dimensions most, and operates identically without a superego. The recognition effect is unanimous across all 9 judge × run cells (d=1.34–1.92). *Error correction.* Error correction and calibration interact through **universal substitution**—the superego provides +9–15 points under baseline but its marginal value diminishes under recognition, with a consistent 15–17% additivity deficit across all three models. The residual architecture benefit under recognition is model-dependent: near-zero on strong models but +12.3 points on Gemini Flash 3.0, where base quality is low enough that calibration alone cannot handle all failure modes. Two scope-sharpening ablations refine this picture: a controlled Writing-Pad ablation (§6.6.9, $N = 252$) shows that the Freudian three-layer memory pad is *not* load-bearing for recognition, which operates at the prompt level; and a six-model capability-threshold test (§6.6.10, $N = 947$) directly refutes the hypothesis — inherited from Paper 1.0's cells 66–68 — that superego-routed bidirectional profiling specifically compensates for weak ego reasoning. Across the tested capability range the bidirectional-profiling variant is either neutral or harmful.

Adaptive responsiveness. Formal tests on N=432 multi-turn dialogues (3–5 turns) find that no lightweight experimental factor modulates within-dialogue trajectories (all $d \leq 0.15$, well-powered to detect d $\geq$ 0.27), replicated across all three judges. An exploratory single-scenario finding on DeepSeek/Sonnet (10-turn disengagement, $d = 1.63$, $n = 12$/condition) did not survive pre-registered replication across two additional generation models (Haiku 4.5, Gemini Flash 3.0) and two judges (Sonnet 4.6, GPT-5.4): three of four (model × judge) cells fell below the $d = 0.5$ threshold and the one replicating cell (Haiku/GPT-5.4, $d = 1.85$) was contradicted by the secondary judge on identical rows ($d = -0.18$, cross-judge $\Delta d = 2.03$; §6.3.2). The three-mechanism model therefore resolves to **two supported general mechanisms and one clean null**. A later post-hoc Plan 2.x line sharpens the boundary rather than overturning it: a heavier externalised-state controller with typed adaptation contracts, outcome closure, and early completion after successful learner-owned no-intervention preserves 10/10 strict strategy shifts and improves Opus-judged quality on a held-out simulated Plan 2.1 suite (63.2 versus 51.5, +11.7; tutor last-turn 83.3 versus 62.0; §6.12.3). A further yoked-contingency probe adds a causal diagnostic, not a broad trajectory claim: under prose-opaque but behavior-diagnostic learner states, same-state yoked tutor plans produce larger held-out simulated hard-transfer gains than different-state yoked plans for a rule-transfer novice learner ($\Delta_2 = 0.344$, 9/9 paired sessions, one-sided sign-test $p = 0.0020$, zero invalid answers and zero hidden-label prompt leaks; §6.12.5). A scaled model-invariance follow-up preserves this endpoint across seven completed Claude, GLM 5.2, and Codex/OpenAI-family rows, including regenerated GLM 5.2 and Codex planner/learner rows ($\Delta_2 = 0.400$ and 0.366, respectively, both 9/9 and $p = 0.0020$). These addenda are evidence for localized simulated state-contingent strategy governance, not for recognition-modulated trajectory slopes, human learning, or deployment readiness.

Both supported mechanisms operate on tutor production, and the tutor-learner asymmetry is itself model-dependent: on strong models, tutor $d \approx 1.85$ vs learner $d \approx 0.16$–0.25

(7–12× ratio); on a weaker model (Gemini Flash 3.0), the gap narrows dramatically (tutor $d \approx 1.87$, learner $d \approx 1.20$, ratio 1.6×), suggesting a generation-quality threshold below which recognition's tutor improvements also benefit learner engagement. A further scope-sharpening control (§7.9, A10/A10b) establishes that recognition's effect is reproduced by matched-specificity prompts grounded in Hegelian-descendant constructivist theorists (Vygotsky's ZPD, Piaget's assimilation/accommodation, with Kapur's productive failure, Chi's ICAP, VanLehn's step-level scaffolding, and Graesser's affective-tutoring work as further cognate descendants) — pooled three-judge $d \approx 0.14$-0.19 between recognition and matched-pedagogical, below the $|d| < 0.2$ density-sufficient threshold — while a matched-specificity prompt grounded in the orthogonal behaviorist tradition (Skinner, Gagné, Keller, Thorndike, Rosenshine) scores pooled $d = 0.89$ *below* the generic baseline. Between-family pooled $d = 1.38$ dominates within-family contrasts by an order of magnitude. The active ingredient is therefore **intersubjective-pedagogy orientation**, not recognition's specific Hegelian vocabulary — but this is not a finding that displaces recognition theory. The constructivist descendants the matched-pedagogical control was grounded in (Dewey was explicit about his Hegelian phase; Vygotsky's dialectical psychology refracts Hegelian recognition through Marxist materialism; Piaget's assimilation/accommodation transposes the dialectic of identity-difference into developmental cognition) inherit the intersubjective commitment from Hegel, often without direct attribution. **Recognition theory is the most explicit philosophical articulation of an orientation its constructivist descendants carry tacitly.** What A10/A10b's family-level finding shows is that the *operationalisation* is interchangeable across cognate descendants; what remains uniquely valuable about recognition theory is that it makes legible what the descendants enact without naming. A distinctive methodological contribution is the evaluation apparatus itself—provable discourse infrastructure, four rubric iterations, test suite as analytical provenance, the orientation-family-as-unit-of-analysis framing—as transferable for mechanistic LLM evaluation. All findings concern tutor output quality as assessed by LLM judges interacting with synthetic learners; whether the supported mechanisms translate to human learning outcomes remains an open empirical question (Section 8.1).

Beyond the three-mechanism model, a generative arc (Section 6.13) reframes adaptivity as *conduct governance*: instead of inferring the learner's hidden state, the tutor releases the premises of a machine-checkable derivation on a schedule enforced by an external guard. A pacing guard that holds each premise until the learner is ready grounds the target derivation where an unguarded tutor does not (5/5), and a page-only variant watching only the visible transcript paces just as well on a linear proof chain — locating the lift in scheduling discipline, not a privileged latent signal (§6.13.11). A forked AND-join world that decouples visible uptake from latent proof-distance bounds this: there the page-only guard fails (0/5) exactly where the proof-state-aware guard still grounds (5/5), so a visible adaptivity proxy is certifiable only where the task's geometry lets the page faithfully project hidden task state (§6.13.12). A linear-spine *distractor* world then reverses the pairing: the page-only guard grounds (5/5) where the proof-state-aware guard fails (1/5), drifting onto a derivable decoy it cannot see off the page — so the two guards are complementary, each reliable only where its world's pacing signal lives in the channel it reads, and neither is safe across geometries (§6.13.13). A subsequent selector-reliability trajectory tests whether a pre-declared rule can choose the channel before seeing outcomes: v0 fails the adaptive claim (18/20, below always-H and oracle at 19/20); v1 finds one useful V-positive Hethel signal but does not beat always-H on the original four-world matrix; and v2–v4 contract the claim further, showing that the robust ingredient is hidden proofDebt rather than H/V routing (six-world confidence matrix: hidden+proofDebt 91.7% world-mean success, v4+proofDebt 75%, visible+proofDebt 33.3%, v4 without proofDebt 16.7%; §6.13.15). Later A20/A21 probes then close the promotion arc: conduct enforcement creates first-pass negative transfer, progress pressure partially repairs one replay without grounding, an action-value patch matches rather than beats hidden+proofDebt, and a calibrated ownership benchmark passes 12/12 controls while mined artifacts still show no proof-safe ownership gain. The current paper therefore treats hidden+proofDebt as the production derivation arm; A20/A21/ownership/didactic overlays remain research instrumentation only, not promoted policies.

1. Introduction

The dominant paradigm in AI-assisted education treats learning as information transfer. The learner lacks knowledge; the tutor possesses it; the interaction succeeds when knowledge flows from tutor to learner. This paradigm—most visible today in the answer-delivery default of general-purpose LLM tutors and educational chatbots—treats the learner as fundamentally passive: a vessel to be filled, a gap to be closed, an error to be corrected. (The cognitively richer intelligent-tutoring-systems tradition—model-tracing and constraint-based tutors that maintain an explicit model of learner cognition—is a partial exception we return to in §2.1.)

Before invoking theory, we state what this paper actually does and finds. We evaluate whether a specific pair of design choices—prompts instructing the tutor to engage with the learner's interpretation rather than supply answers, and a two-agent ego–superego architecture that critiques its own output for pedagogical soundness—produces measurable, traceable differences in LLM-based tutoring behaviour. Across three generation models (DeepSeek V3.2, Haiku 4.5, Gemini Flash 3.0), N = 432 multi-turn dialogues, and three independent judges (Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4), the prompt intervention produces a floor-lifting improvement whose magnitude depends on the judge — pooled across the three-judge panel d $\approx$ 1.63, with Sonnet-judge d $\approx$ 1.88 as an upper bound and Gemini-3.1-Pro-judge d $\approx$ 1.44 as a lower bound (§8.3) — alongside narrowed output variance; the architecture intervention adds a partially overlapping benefit (15–17% additivity deficit) whose completeness depends on generation quality (near-zero residual on strong models; +12.3 points on the weakest); and neither intervention reliably modulates within-dialogue trajectories in the primary well-powered analysis. Later adaptive-trap addenda (§6.12.1–§6.12.3) show a different, heavier result: when adaptation is operationalised as typed state-action governance with outcome closure and early completion, simulated strategy-shift quality can improve, but that is not a revival of the recognition-modulated slope hypothesis. We trace these effects through the system's ego–superego deliberation record, making the causal chain from prompt to superego critique category to output revision empirically available rather than inferred from behaviour alone. The theoretical motivation for these particular design choices—why "engaging with the learner's interpretation" matters, and why a second agent should critique the first—is drawn from Hegel's account of mutual recognition, to which we now turn.

The alternative this paper tests is grounded in Hegel's theory of mutual recognition. In the *Phenomenology of Spirit* (Hegel, 1977), Hegel argues that genuine self-consciousness requires recognition from another consciousness that one in turn recognizes as valid. The master-slave dialectic reveals that one-directional recognition fails: the master's self-consciousness remains hollow because the slave's acknowledgment, given under duress, does not truly count. Only mutual recognition—where each party acknowledges the other as an autonomous subject—produces genuine selfhood.

The connection between Hegelian thought and pedagogy runs deeper than common citation patterns suggest, and it is worth tracing the lineage carefully because the empirical findings reported in §7.9 hinge on it. We can distinguish three concentric circles of Hegelian descent in the educational tradition this paper draws on.

The innermost circle contains pedagogical thinkers for whom Hegelian recognition is *explicitly* foundational. Dewey's pragmatist philosophy began as American Hegelianism: his early career was steeped in Hegel, and although he later distanced himself from speculative idealism, the relational and developmental commitments of his pedagogy remain recognisably post-Hegelian (Dewey, 1916). Vygotsky's developmental psychology was explicitly dialectical, refracting Hegelian recognition through Marxist materialism: the zone of proximal development (Vygotsky, 1978) is structurally a recognition relation in which the more-capable other constitutes the conditions of the learner's emerging competence. The German *Bildung* tradition treats education as self-formation through encounter with otherness (Stojanov, 2018). Contemporary recognition theory (Honneth, 1995) has been applied to educational contexts where the struggle for recognition shapes learning outcomes (Huttunen & Heikkinen, 2007). These authors do not merely echo Hegel; they argue from him.

The second circle contains thinkers whose work bears the *structural* mark of Hegelian recognition without attributing it directly. Piaget's assimilation/accommodation (Piaget, 1954) transposes the dialectic of identity-difference into developmental cognition: the learner's existing schema (identity) encounters experience that resists it (difference), and accommodation is the synthesis. The shape is dialectical even where the citation is to biology rather than philosophy. Kapur's productive failure (Kapur, 2008) depends on cognitive disequilibrium resolved through subsequent canonical instruction — structurally a thesis-antithesis-synthesis movement, although Kapur cites Vygotsky and Bruner rather than Hegel directly.

The third circle contains pedagogically-adjacent traditions whose connection to recognition is more cognate than descended. Chi's ICAP engagement hierarchy (Chi & Wylie, 2014) orders learning activities by the depth of cognitive interaction they involve — the top tiers (interactive, constructive) are dialogical, but the framework's grounding is information-processing cognitive science, not philosophical anthropology. VanLehn's step-level scaffolding (VanLehn, 2011) derives from intelligent-tutoring-systems research that takes the mediating-other for granted as a methodological commitment without theorising it. Graesser's affective-tutoring work (Graesser et al., 2008) grounds in cognitive-affective dynamics rather than recognition. These authors operate within an intersubjective-pedagogy ecosystem whose air recognition theory makes breathable, but their direct citations point elsewhere.

Together these three circles form what we will call the **intersubjective-pedagogy family**

— pedagogical traditions that share the commitment to the learner as an active, thinking subject whose interpretations constitute the conditions of learning. The commitment originates in Hegel and propagates outward: explicitly named in the innermost circle (Dewey, Vygotsky, Honneth), structurally inherited in the second (Piaget, Kapur), tacitly assumed in the third (Chi, VanLehn, Graesser).

A matched-specificity density control reported in §7.9 (A10/A10b) establishes that the effect this paper traces is a family-level property: recognition's quality advantage is reproduced by matched-specificity prompts grounded in this constructivist descent, while matched-specificity prompts grounded in the genuinely orthogonal behaviorist tradition (Skinner, Gagné, Keller, Thorndike, Rosenshine) score substantially *below* a generic baseline. The empirical operationalisation is interchangeable across the family. The theoretical contribution recognition theory makes, however, is not. **Recognition theory is the most explicit philosophical articulation of an orientation its constructivist descendants carry tacitly** — an orientation whose presence is detectable when stripped of its Hegelian vocabulary (the matched-pedagogical control reproduces the effect) but whose conceptual grammar lives most clearly in the recognition framework itself. Our contribution is accordingly twofold: to operationalize the family's commitments as concrete design heuristics for AI tutoring systems, using recognition theory as the explicit theoretical entry point both for its philosophical primacy and for its diagnostic clarity; and to trace the *mechanisms* through which those heuristics alter system behavior.

A terminological clarification is needed at the outset. "Recognition" operates in this paper at three nested levels. First, as **philosophical inspiration**: Hegel's account of mutual recognition provides the theoretical motivation for treating learners as autonomous subjects whose perspectives genuinely shape the interaction. Second, as **operational design heuristic**: recognition theory is translated into specific prompt instructions (engage with the learner's interpretation, pose questions rather than provide answers, treat resistance as pedagogically productive) and architectural choices (the superego critiques for genuine engagement, not just correctness). Third, as **a family of observable discourse effects**: recognition-enhanced tutors ask more questions, produce less variable output, and revise substantively rather than cosmetically after critique. The empirical claims in this paper concern primarily the third level—what the prompts and architecture *do*—with the first level providing motivation and the second providing implementation. We do not claim that the system instantiates mutual recognition in Hegel's philosophical sense; we claim that design heuristics drawn from recognition theory produce measurable and traceable effects on system behavior.

The ablative finding

A companion pilot study (Magee, 2026) established that recognition-enhanced prompts and multiagent architecture produce measurable differences in AI tutoring quality. Across fifty-two evaluations (N=4,312 primary scored responses), recognition emerged as the dominant factor: a $2 \times 2 \times 2$ factorial analysis (N=350) yielded F=110.04, p<.001, d=1.11, with recognition accounting for 24.3% of total variance. A memory isolation experiment (N=120) established recognition as the primary driver (d=1.71), independent of memory integration. Cross-judge validation with GPT-5.2 replicated effect directions at compressed magnitudes

(37–59% of primary effect sizes).

These results are robust but opaque. They demonstrate that *something* is happening—recognition-enhanced prompts reliably modulate tutor output, the superego catches errors that self-correction cannot, effects are large on the tutor side and small on the learner side—but they do not explain *why*. Finding differences is not the same as understanding mechanisms.

The explanatory gap

Five specific findings from the pilot study resist ablative explanation:

1. **The tutor-learner asymmetry.** Recognition produces large tutor-side effects (d=1.03 on per-turn scores) but near-zero learner-side effects (d=0.27 per-turn, d=-0.13 holistic). The asymmetry is documented but not theorized. Is it an artifact of the judge, a ceiling on synthetic learner responsiveness, or a genuine architectural property?

2. **Opaque superego function.** The superego catches content leakage and enforces struggle-preservation, but we lack a causal account of *how* critique changes the ego's output—whether through lexical shifts, structural reorganization, or strategy changes.

3. **Model-dependent architecture effects.** The same ego/superego learner architecture produces opposite patterns under different models: eta-squared for learner architecture is .527 with Kimi but .002 with Haiku. The mechanism is unclear.

4. **Unknown trajectory dynamics.** Per-turn scoring exists but the *trajectory*—how the tutor adapts across 3–5 turns, whether recognition effects amplify or decay—has not been systematically analyzed.

5. **Unexamined deliberation-output relationship.** The dialogue traces contain ego-superego negotiations, but these internal processes have not been systematically correlated with external output quality.

Three candidate mechanisms

This paper motivates three candidate mechanisms drawing on recognition theory and tests each against multi-turn factorial data:

1. **Calibration** (prompt-level). Recognition prompts narrow the tutor's output distribution, producing more uniform quality rather than higher peaks. The pilot study found dimension variance reduction in 52/55 within-run comparisons (d=-0.47 to d=-1.00 depending on analysis scope), the single most reliable recognition signal. Calibration operates even without a superego, because it is a prompt-level constraint on the response space.

2. **Error correction** (architecture-level). The superego functions as a structural feedback channel, but its effectiveness depends on the quality of the ego's initial output. The pilot study found a 35.9-point recognition $\times$ architecture interaction (Paper 1.0

Section 6.3). The mechanism data (Section 6.2, Section 6.4) reveals this as a **universal substitution** interaction: the superego provides substantial benefit under baseline conditions (+9–15 points), catching errors the ego cannot self-correct, but this benefit diminishes under recognition because calibration pre-empts the failures the superego would catch. Section 6.4 reports the cross-model magnitudes; the residual benefit under recognition is near-zero on strong models but persists on the weakest. The superego approval rate shifts dramatically under recognition (DeepSeek: 13%→55%; Haiku: 52%→66%), consistent with a calibrated ego producing fewer errors for the superego to correct.

3. **Adaptive responsiveness** (interaction-level, *predicted but not supported*). In multi-turn conversations, both recognition and baseline tutors adapt substantially to learner signals (cross-turn adaptation magnitude Adapt$\Delta > 0.79$). Formal tests on N=432 dialogues find that no experimental factor modulates within-dialogue trajectories: recognition d=0.03 on tutor slopes, all factors $d \leq 0.15$, with the design well-powered to detect d ≥ 0.27 (Section 6.3). An exploratory DeepSeek/Sonnet finding in the 10-turn disengagement scenario ($d = 1.63$, $n = 12$/condition) failed pre-registered replication on two additional generation models and two judges (A12, $N = 32$; §6.3.2): three of four cells below $d = 0.5$, one replicating cell contradicted by the secondary judge on the same rows. The primary well-powered analysis is null and the exploratory boundary condition has been ruled out.

The two supported mechanisms are separable: calibration is a prompt-level effect (operative from the first turn), error correction is an architecture-level effect (operative within each turn). They interact through universal substitution with a model-dependent residual (Section 6.4). Within-dialogue trajectories are independent of both (slopes do not depend on levels, d=0.03). Both operate primarily on tutor *production* rather than learner *reception* on strong models; on weaker models the recognition → learner pathway also opens up (Section 6.5).

Process tracing methodology

To move from ablative findings to mechanistic explanation, we adopt **process tracing**—a methodology from comparative politics (Bennett & Checkel, 2015) that examines causal chains *within* cases rather than statistical patterns *across* cases. Our architecture's observability makes this unusually feasible: every ego-superego exchange is logged with verbatim text, every turn is independently scored, and every revision is recorded with full provenance.

We combine process tracing with three analytical methods: a superego critique taxonomy classifying what the superego objects to empirically (Section 5.1), revision delta analysis measuring how the ego's output changes after critique (Section 5.2), and turn-by-turn trajectory analysis testing whether adaptation accumulates across conversations (Section 5.3). Every mechanistic claim is registered in a provable discourse framework that machine-verifies assertions against data, tracks claim staleness, and enforces cross-claim consistency.

The apparatus as method

A distinctive contribution of this paper is the argument that the evaluation apparatus itself—the provable discourse framework, the rubric iterations, the bug corrections, the test suite—constitutes a transferable methodology for mechanistic evaluation of LLM-based educational systems. The process of building and correcting the apparatus mirrors the mechanisms it studies: the provable discourse framework functions as a "superego" for research claims, catching stale assertions and forcing genuine revision rather than cosmetic compliance. Nine post-extraction corrections documented during the pilot study follow the same error-correction pattern observed in the architecture (Section 7).

Contributions

The contributions of this paper are:

- A theoretical framework motivating three testable mechanism predictions (calibration, error correction, adaptive responsiveness) drawing on Hegel's recognition theory: two supported as general mechanisms, the third null on both primary analysis and pre-registered replication (Section 3, §6.3.2 A12)
- A process tracing methodology combining superego critique taxonomy, revision delta analysis, and trajectory analysis to trace mechanisms through the system's internal processes (Section 5)
- Evidence that calibration operates at the prompt level, narrowing the tutor's output distribution independently of architecture (within-response dimension SD d=0.52–0.64, floor-lifting pattern replicating across both models; Section 6.1)
- Evidence that error correction and calibration interact through **universal substitution** with a model-dependent residual architecture benefit (Section 6.2, Section 6.4)
- Evidence that adaptive responsiveness, while real as a descriptive phenomenon ($\text{Adapt}\Delta > 0.79$), is **not supported as a mechanism modulated by recognition**: primary analysis null at N=432 (Section 6.3); pre-registered A12 replication of an exploratory DeepSeek/Sonnet boundary-condition effect also failed (§6.3.2). The later Plan 2.x addenda bound that null rather than erase it: a heavier externalised-state policy can preserve typed strict shifts, improve simulated judged quality on held-out trap suites (§6.12.1–§6.12.4), and pass a yoked-contingency independent-outcome probe where same-state tutor plans beat different-state plans for a held-out hard-transfer rule-transfer novice (§6.12.5), but these results are post-hoc, simulated, and architectural rather than recognition-prompt trajectory effects.
- A mechanistic explanation of the tutor-learner asymmetry as model-dependent: tutor-centric on strong models, narrowing on weaker ones (Section 6.5; Section 6.5.5 reports the rubric-vs-holistic correction to the original headline ratio)
- A mechanistic explanation of model-dependent architecture effects: models that cannot implement error correction (ego ignores critique) or calibration (cannot follow complex instructions) reverse the expected patterns

- Cross-judge validation with three independent judges (Sonnet, Gemini 3.1 Pro, GPT-5.4) with recognition as the dominant effect — unanimous across all 9 judge × run cells (Section 6.4.6, Section 8.3)
- A pre-registered orientation-family density control (A10/A10b) locating the active ingredient at the **intersubjective-pedagogy orientation** level rather than recognition's specific Hegelian vocabulary: matched-specificity constructivist prompts (Vygotsky / Piaget / Kapur / Chi / VanLehn / Graesser) reproduce recognition's effect within pooled $d = 0.14$–0.19, while matched-specificity behaviorist prompts (Skinner / Gagné / Keller / Thorndike / Rosenshine) score pooled $d = 0.89$ *below* the generic baseline; between-family pooled $d = 1.38$ dominates within-family contrasts by an order of magnitude (Section 7.9)
- The evaluation apparatus itself as a transferable methodology for mechanistic LLM evaluation, including provable discourse infrastructure that machine-verifies paper claims against data (Section 7)
- An architectural-extension exploration (Section 6.7, cells 101–107) inverting the ego-superego topology into an *id-director* configuration, where a back-stage agent re-authors the ego's complete system prompt every turn. Tested under a Weberian charisma rubric orthogonal to v2.2's recognition-quality dimension, with a structured register classifier and curated witness exemplars as auxiliary mechanisms. The architectural ranking — id + classifier + recognition (cell 104) as the most robust cell — survives a cross-judge sanity check, but the *magnitude* of its lift over baseline is scenario-conditional (54-point gap on the original persona-shift scenarios, 4-point gap on a replication scenario set; §6.7, §8.9)

The architecture is designed as a research instrument for mechanism observability rather than a deployable tutoring system; its computational cost (~225,000 input tokens per 10-turn multi-agent dialogue, Section 8.7) reflects this priority.

The paper is organized as follows. Section 2 reviews related work in AI tutoring, process tracing, and mechanism-oriented AI research. Section 3 develops the theoretical framework, motivating three candidate mechanism predictions from recognition theory. Section 4 presents the system architecture with emphasis on observability for process tracing. Section 5 describes the process tracing methodology. Section 6 reports mechanism-level results: calibration (Section 6.1), error correction (Section 6.2), adaptive responsiveness (null on the primary well-powered analysis, with only an unreplicated single-scenario exploratory boundary effect; Section 6.3), mechanism interaction (Section 6.4), the tutor-learner asymmetry (Section 6.5), model dependence (Section 6.6), an architectural-extension exploration of the id-director family under a Weberian charisma rubric (Section 6.7), and post-hoc adaptive-trap and yoked-contingency addenda bounding what heavier state-action governance can and cannot recover (Section 6.12). Section 7 discusses the apparatus-as-method argument and a pre-registered orientation-family density control (Section 7.9, A10/A10b) that locates the active ingredient at the intersubjective-pedagogy orientation level. Section 8 addresses limitations including the scope of the id-director extension (Section 8.9), and Section 9 concludes.

2. Related Work

This paper sits at the intersection of six literatures: AI tutoring systems, multiagent architectures and self-correction, LLM-as-Judge evaluation, recognition theory in education, process tracing methodology, and mechanism-oriented AI research. We survey each, emphasizing the gap that motivates our mechanism investigation: existing work establishes *that* architectural and prompting interventions produce effects, but rarely traces *how* those effects propagate through the system's internal processes.

2.1 AI Tutoring and Intelligent Tutoring Systems

Intelligent Tutoring Systems (ITS) have long treated instruction as a stateful control problem, progressing from early rule-based systems like SCHOLAR (Carbonell, 1970) and SOPHIE (J. S. Brown et al., 1975) to model-based cognitive tutors built around production-system models of student problem solving, which deliver immediate step-level feedback and show measurable achievement gains (Anderson et al., 1995). Bayesian knowledge tracing models the changing probability that a learner has mastered each rule and uses those estimates to individualize practice (Corbett & Anderson, 1995); constraint-based modeling offers a related alternative that detects violations of domain constraints rather than tracing a full ideal solution path. Meta-analyses situate the learning-outcome claims: ITS can improve learning relative to many non-ITS controls, but effect sizes depend on implementation quality, outcome alignment, and the control condition (Kulik & Fletcher, 2016; Ma et al., 2014), and VanLehn's review shows the relative advantage of human tutoring, ITS, and other tutoring systems is smaller and more granularity-dependent than the classic "2 sigma" narrative suggests (Bloom, 1984; VanLehn, 2011).

The rapid adoption of generative AI for tutoring has been accompanied by multi-agent educational frameworks (Wu et al., 2023), self-refining instructional agents (Madaan et al., 2023), and comprehensive surveys mapping the expanding landscape of LLM agents in education (Chu et al., 2025; Kasneci et al., 2023). Specific architectures include GenMentor (Wang et al., 2025), which decomposes tutoring into five specialized sub-agents; Ruffle&Riley (Schmucker et al., 2024), which orchestrates two agents in a learning-by-teaching format; and TutorGym (Weitekamp et al., 2025), which places AI tutor and student agents inside existing ITS environments to test whether agents can supply examples, hints, and correctness feedback in step-based contexts.

Empirical evidence on LLM tutoring effectiveness is emerging rapidly. A systematic review of 88 studies (Y. Shi et al., 2025) finds consistent engagement benefits but limited evidence on deep conceptual learning. The largest randomized controlled trial to date (Vanzo et al., 2025) demonstrated improved accuracy and sustained engagement, while Scarlatos et al. (2025) used dialogue preference optimization to train tutors that produce measurably better learning outcomes than prompted-only baselines. A high-school mathematics field experiment sharpens the design lesson: access to generative AI can improve assisted performance, but unguardrailed use can harm later unassisted learning, while pedagogical guardrails mitigate that risk (Bastani et al., 2025). The important question is therefore not whether AI gives better answers during use, but whether the interaction preserves the learner's own work.

A distinct ITS lineage equips tutors with formal verification of student work. Hint- and feedback-generating tutors for Hilbert-style propositional proofs compare candidate proofs against a generated solution space (Lodder et al., 2020), and recent systems couple LLM interaction to a Lean verifier for autoformalization, proof checking, and next-step natural-language feedback (Patel et al., 2025). These proof tutors are the closest antecedents to the formal-derivation arc of Section 6.13, with one difference of intent: our formal worlds are built less as instructional interfaces than as conduct-governance testbeds, designed to ask whether tutor behavior can be constrained without leaking the target derivation.

The conduct-governance framing of Section 6.13 also has antecedents outside the tutoring literature, in a recent line of work on *runtime enforcement* for LLM agents. AgentSpec specifies agent constraints as triggers, predicates, and enforcement mechanisms and intervenes when a rule fires (Wang, Poskitt, & Sun, 2025); ShieldAgent compiles verifiable safety policies into action-based probabilistic rule circuits and checks trajectories against them (Z. Chen et al., 2025); AGrail generates and optimises safety checks adaptively over an agent's lifetime (Luo et al., 2025); and runtime-shield synthesis such as Aegis treats the intervention policy itself as a synthesised, permissive monitor over a neural policy's actions (J. Shi et al., 2024), with probabilistic variants that anticipate violations several steps ahead (Wang, Poskitt, Wei, et al., 2025). Our guards differ in domain and objective—they enforce pedagogical release, repair, and pacing constraints over a proof-world trajectory rather than security constraints over tool calls—but the methodological point is shared: reliable agent conduct comes not from richer exhortation alone but from checkable runtime structure placed around the model. The pacing and proof-debt guards of Section 6.13 are an instance of that principle applied to tutoring, where the constraint being enforced is *when* to release a premise, not *whether* a tool action is safe.

These studies evaluate primarily *content delivery* and *engagement*—not relational quality. More critically for our purposes, they operate at the *outcome* level: does the intervention improve scores? They do not trace the *mechanism* by which improvement occurs. We do not claim that state, scaffolding, model tracing, or mastery gating are new; the defensible claim is that we port these ITS commitments into open-ended LLM tutoring and make them *ablatable at the level of tutor conduct*—what was once embedded in authored ITS logic becomes a guardable, logged, failure-mode-sensitive control loop over generative dialogue. Our work thus addresses an explanatory gap: we study not just whether recognition-oriented design improves tutoring, but *through what internal processes* it does so.

2.2 Multiagent Design and Self-Correction

Multi-agent architectures have been explored for task decomposition (Wu et al., 2023), debate (Irving et al., 2018), and self-critique (Madaan et al., 2023). The CAMEL framework (G. Li et al., 2023) demonstrated that role-playing communicative agents can autonomously cooperate through structured dialogue. A comprehensive survey (Guo et al., 2024) maps this landscape across profile construction, communication protocols, and capability acquisition—identifying pedagogical applications as an underexplored frontier. A broader survey of psychological theories in LLM design (Z. Liu et al., 2025) reviews 175 papers spanning cognitive, developmental, and social psychology, confirming growing interest in psychologically-informed agent architectures while highlighting the rarity of empirically validated implementations.

A critical literature on self-correction qualifies optimism about reflexive architectures. Kamoi et al. (2024) demonstrate that LLMs largely *cannot* correct their own mistakes without external feedback—intrinsic self-correction frequently degrades performance. Shinn et al. (2023) demonstrated the promise of Reflexion (verbal reinforcement through self-reflection) but identified a "degeneration-of-thought" problem where repeated self-reflection without new information converges on worse outputs. These findings are directly relevant: the Superego provides the *structurally external* feedback that the self-correction literature shows is necessary. Unlike intrinsic self-correction, our Superego applies different evaluation criteria through a separate prompt context—functioning as a genuine external critic rather than a self-review loop.

Our prompting approach extends the behavioral specification paradigm by introducing *intersubjective prompts*—prompts that specify not just agent behavior but agent-other relations. Where persona prompts (T. B. Brown et al., 2020; Wei et al., 2022) describe what the agent should do, and Constitutional AI (Bai et al., 2022) defines self-referential constraints, intersubjective prompts specify the *relational field* between agents: who the learner is (an autonomous subject) and what the interaction produces (mutual transformation). The closest precedent is Constitutional AI, but the constitutional approach is monological (one agent critiquing itself against principles) while ours is dialogical (the relational specification shapes how two agents constitute each other).

2.3 LLM-as-Judge Evaluation Methodology

The use of LLMs as evaluation judges has become a major methodological paradigm. Zheng et al. (2023) established the foundation with MT-Bench and the Chatbot Arena, demonstrating over 80% agreement with human experts while identifying systematic biases including position bias, verbosity bias, and self-enhancement bias. Subsequent surveys have expanded understanding of both capabilities and limitations (Gu et al., 2025; H. Li et al., 2024), highlighting that while LLM judges excel at relative ranking, their absolute calibration varies substantially across models and domains.

Our evaluation methodology engages directly with this literature. We use multiple independent LLM judges with systematic inter-judge reliability analysis, finding Pearson correlations of r=0.49–0.64 across judge pairs. Rather than treating any single judge as ground truth, we report within-judge comparisons for factor analysis and use cross-judge replication to validate effect directions. The verbosity bias concern is addressed through active control design (matching prompt length without recognition content) and cross-judge validation showing effect directions replicate even when magnitudes differ.

Two validity boundaries qualify this apparatus. First, the judges are themselves biased instruments: beyond the verbosity and self-enhancement effects above, position bias in LLM-as-judge is systematic and well-documented (L. Shi et al., 2024), so cross-judge agreement on factorial-level effects supports the *direction* of those effects but does not validate any LLM-generated *process* taxonomy—superego critique categories, revision-substance labels, and failure-mode codes remain LLM-assessed claims about LLM behavior until checked against human expert coding. Second, our learners are synthetic. Recent work warns that prompted LLMs simulate human students poorly across linguistic, behavioral, and cognitive metrics (Scarlatos et al., 2026), and teachers tutoring LLM "students" report authenticity failures—unnatural attentiveness, missing emotion, and logical inconsistency (Martynova et al., 2025). We therefore treat synthetic learners as controlled interaction probes, not as evidence of human learning: the present results concern tutor output quality and formal derivation outcomes, with human pretest/posttest/transfer evidence a separate validation target.

2.4 Sycophancy and the Drama Machine

Two literatures converge on our architectural rationale. The sycophancy problem (Perez et al., 2022; Sharma et al., 2024)—LLMs shifting stated opinions to match user preferences—has been specifically identified as a pedagogical risk: sycophancy eliminates the constructive friction necessary for learning (Lee, 2025). Recent work clarifies the mechanisms: Shapira et al. (2026) show that preference-based post-training causally amplifies sycophancy, while Vennemeyer et al. (2025) decompose sycophancy into separable latent-space directions. The phenomenon sits on a spectrum escalating from surface agreeableness to active subterfuge (Denison et al., 2024; Greenblatt et al., 2024), making structural countermeasures particularly important.

The Drama Machine framework for character simulation (Magee et al., 2024) identifies how internal dialogue—competing sub-agents representing different motivations—produces dynamic behavior rather than flat consistency. A character who simply enacts their goals feels artificial; one torn between impulses feels alive. We adapt this insight to pedagogy: the tutor's Ego (warmth, engagement) and Superego (rigor, standards) create productive conflict that improves output quality. The Drama Machine literature contributes the specific mechanisms (deliberative refinement, productive tension, role differentiation) that our architecture operationalizes.

2.5 Recognition Theory in Education

Hegel's theory of recognition has been developed extensively in social and political philosophy (Fraser, 2003; Honneth, 1995; Taylor, 1994). Educational applications trace a theoretical trajectory from Huttunen and Heikkinen's (2004) foundational analysis applying Hegel's master-slave dialectic to pedagogy, through Fleming's (2011) extension to transformative learning theory, to Stojanov's (2018) connection of *Bildung* to neo-Hegelian recognition. Recent critical work (Costa & Murphy, 2025) draws on Arendt and Freire to argue that education must promote intellectual agency rather than submit to technological domination. The relational pedagogy tradition—from Buber's (1958) I-Thou encounter through Freire's (1970) dialogical education to Noddings' (1984) ethics of care—establishes the philosophical ground for treating the tutor-learner relation as constitutive rather than instrumental.

Psychoanalytic approaches to AI have developed from multiple directions. Magee, Arora, and Munn (2023) analyze LLMs as "automated subjects" structured by Lacanian categories. Black and Johanssen (2025) use Lacanian concepts to analyze ChatGPT as inherently relational. Kim et al. (2025) independently map Freud's ego/id/superego onto LLM modules—a convergence validating the psychoanalytic approach while differing from ours in targeting consciousness simulation rather than pedagogical quality. Most of this work is *interpretive*: analyzing what AI means philosophically. Our approach is *constructive*: we build a system using psychoanalytic architecture and measure its effects empirically.

These educational applications have been primarily theoretical. Our contribution is an empirical operationalization: measuring whether AI systems achieve functional recognition and whether it improves outcomes. This is distinct from work applying Hegelian *dialectic* (rather than recognition) to AI reasoning procedures (Abdali et al., 2025). Our use of Hegel concerns *intersubjective relations* (how subjects constitute each other), not *dialectical method* (how contradictions drive conceptual development). We therefore treat recognition theory as the clearest philosophical articulation of an *intersubjective pedagogical orientation*—the tutor treats the learner's contribution as an active object of response rather than noise en route to content delivery—while noting that the empirical claim concerns observable tutor production (more calibrated output, fewer floor failures, more elicitation), not a claim that the model literally achieves mutual recognition, and that this operationalization is shared with cognate constructivist and dialogic traditions rather than uniquely owned by Hegelian vocabulary.

2.6 Theory of Mind and Constructivist Pedagogy

Two additional literatures inform specific mechanism predictions. Theory of Mind research suggests frontier LLMs achieve adult human performance on structured ToM tasks (Street et al., 2025) but show inconsistent performance on naturalistic tasks (Nguyen, 2025). Hwang et al. (2025) demonstrate that architectural support for ToM (explicit mental state tracking) produces measurably more socially appropriate responses—paralleling our "other-ego profiling" mechanism where the tutor maintains an evolving learner model. The finding that profiling differentiates mechanisms *only* with dynamic learners (Section 6.10 of the companion study) parallels a ToM insight: mental state modeling is only useful when there are genuine states to model.

Constructivist learning theory (Piaget, 1954; Vygotsky, 1978) and research on productive struggle (Kapur, 2008; Warshauer, 2015) provide the pedagogical ground for our error correction mechanism: the Superego checks whether the Ego is "short-circuiting" struggle by rushing to resolve confusion. Vygotsky's zone of proximal development presupposes a dialogical relationship that echoes Hegel's mutual constitution; the constructivist tradition provides empirical grounding for the theoretical prediction that honoring learner struggle improves outcomes.

2.7 Process Tracing in Social Science

Process tracing is a methodology from comparative politics and qualitative social science that examines causal chains *within* cases rather than statistical patterns *across* cases (Bennett & Checkel, 2015). Where factorial experiments answer "does X affect Y?", process tracing answers "through what sequence of steps does X produce Y?" The methodology has been formalized through several approaches: Bennett and Checkel (2015) establish the foundational framework distinguishing "smoking gun" tests (high uniqueness, low certainty), "hoop" tests (low uniqueness, high certainty), and "doubly decisive" tests (both). Beach and Pedersen (2019) develop a systematic approach to mechanistic evidence, arguing that mechanism claims require both *within-case* evidence (observing the mechanism operating in specific instances) and *cross-case* regularity (the mechanism producing consistent effects across conditions).

The application of process tracing to AI systems is, to our knowledge, novel. Traditional process tracing studies political decisions, institutional changes, or policy outcomes where the "mechanism" is a sequence of human decisions. Our architecture provides an unusually transparent analogue: every ego-superego exchange is logged with verbatim text, every turn is independently scored, and every revision is recorded. The dialogue trace constitutes the kind of within-case evidence that process tracing requires—we can observe the causal chain from prompt orientation through internal critique to output modification in individual dialogues, then aggregate across cases to test mechanism-level predictions.

The methodological parallel is precise: process tracing in political science "opens the black box" of decision-making by examining deliberative records. Our architecture *creates* a deliberative record (the ego-superego exchange) that can be examined in exactly this way. The superego critique taxonomy (Section 5.1) functions as a coding scheme for deliberative content; the revision delta analysis (Section 5.2) traces how critique changes output; and the trajectory analysis (Section 5.3) examines how these processes accumulate across conversation turns.

One caveat sharpens the analogy rather than dissolving it. Process tracing in political science reconstructs a record of human decisions, whereas our deliberation text is neither a human decision record nor a neural mechanism: it is an engineered artifact emitted by the same generative system under study. We therefore adopt the more precise label *engineered trace analysis*. The architecture is built to emit intermediate artifacts—prompt context, ego drafts, superego critiques, revisions, delivered outputs, and scores—whose relationship to final behavior can be coded, ablated, and cross-checked, with process tracing as the methodological inspiration rather than a literal claim of equivalence.

2.8 Mechanism-Oriented AI Research

A growing body of AI research moves beyond input-output evaluation toward mechanistic explanation. Mechanistic interpretability (Anthropic, 2025a; Lindsey, Gurnee, et al., 2025) reveals how internal representations form and influence model behavior at the neural level, providing tools to understand *what happens inside* models during generation. Research on emergent introspective awareness (Anthropic, 2025b; Lindsey, Rivoire, et al., 2025) suggests models develop forms of self-modeling that, while not consciousness, parallel the self-monitoring our architecture makes explicit.

Our work operates at a different level of analysis. Where mechanistic interpretability examines neuron-level activations, we examine *agent-level* processes: how prompts constrain output distributions, how architectural feedback loops modify responses, and how these processes interact across conversation turns. The relationship is complementary: mechanistic interpretability asks "what computational processes produce this token?"; our process tracing asks "what agent-level deliberation produces this pedagogical response?" Both seek to move beyond purely behavioral evaluation toward understanding internal causation.

Causal mediation analysis in NLP (Vig et al., 2020) provides a statistical framework for identifying which model components mediate specific behaviors. Our approach is methodologically adjacent but architecturally distinct: rather than probing a single model's internal representations, we examine the *designed* information flow between multiple agents. The ego-superego exchange is an *engineered* causal pathway whose effects can be measured by comparing configurations with and without it (the factorial design in Section 5).

2.9 Adaptive Tutoring: From "Whether" to "How"

A parallel shift from outcome to mechanism characterizes recent adaptive tutoring research. The field has moved beyond asking whether tutoring works (the "2 sigma problem" established that human tutoring produces roughly two standard deviations of improvement over classroom instruction (Bloom, 1984)) toward asking *how* effective tutoring works. VanLehn (2011) identified that the critical mechanism is not lecture-level explanation but *step-level* scaffolding—tutors who intervene at the moment of confusion, providing just enough support to maintain productive struggle, outperform those who deliver polished explanations. Productive-failure research similarly finds that learners can benefit from attempting complex problems before receiving canonical instruction, because the struggle prepares later learning (Kapur, 2008).

Graesser et al.'s work on AutoTutor and Affective AutoTutor (D'Mello & Graesser, 2012a) demonstrated that natural-language dialogue can respond to learner confusion, frustration, and engagement, with affective dynamics mediating tutoring effectiveness (D'Mello & Graesser, 2012b). The ICAP framework (Chi & Wylie, 2014) provides a hierarchy of learning activities (Interactive > Constructive > Active > Passive) that predicts learning outcomes based on the *type* of cognitive engagement rather than time-on-task. These "mechanism-first" approaches share our orientation: they ask not "does this intervention improve outcomes?" but "what cognitive/interactional process explains the improvement?" Read against this background, the pacing guard of Section 6.13 is best understood not as a new pedagogical principle but as a formalized, testable instance of well-timed scaffolding: its contribution is to make the withhold/wait/repair/advance decision mechanically inspectable in an LLM tutor, as an auditable release calendar over a proof DAG.

The post-LLM tutoring literature (2024–2026) sharpens the same orientation in a direction consistent with our findings. The field's largest documented gains route through pedagogical orientation and expert guidance rather than within-session adaptivity machinery. LearnLM reports a +31% expert-preference margin over GPT-4o from pedagogical instruction following instilled at training time, with no runtime adaptivity module credited (LearnLM Team, 2024). Tutor CoPilot's randomized trial (900 tutors, 1,800 students) finds +4 percentage points topic mastery overall and +9 points for students of lower-rated tutors (R. E. Wang, Ribeiro, et al., 2024)—expert-guidance scaffolding whose gains concentrate in the weakest actors, structurally parallel to the substitution pattern of Section 6.4, where the superego critic helps weak generation models and adds little on strong ones. A Harvard crossover RCT found a carefully designed but *static* AI tutor outperformed in-class active learning (Kestin et al., 2025). Conversely, the one deployed head-to-head between a within-session adaptive mechanism and a strong static prompt that we are aware of is a null in deployment: a contextual-bandit prompt router beat static baselines in simulation, but its greedy variant failed to beat the expert-authored production prompt in a live A/B (19.1% vs 19.6% exercise conversion, ~656 conversations, n.s.) (Chang et al., 2026)—an external, deployment-scale analogue of both the Section 6.3 adaptive-responsiveness null and the sim-to-real caution of Section 8.1. These are positioning points from the literature, not claims of this paper's apparatus; a fuller mechanism-by-mechanism screen of this literature against the present results is maintained in the project planning corpus (`PLAN_3_0/dynamic-adaptation-litreview.md`).

Our rubric design reflects this mechanism orientation. The v2.2 rubric (Section 5) consolidates 14 tutor dimensions into 8 using a decomposition informed by the GuideEval framework: Planning (what strategy the tutor pursues), Output (what the response actually does), and Evaluation (how well it assesses the learner's state). This structure separates *intention* from *execution* from *assessment*, enabling mechanism-level analysis of where recognition's effects concentrate. The finding that recognition primarily affects Planning and Output dimensions rather than Evaluation (Section 6.1) would be invisible under a single holistic score.

2.10 Positioning: Five Literatures Converge

Five literatures converge on this work without previously intersecting: (1) *multiagent tutoring architectures*, which decompose tasks but do not trace mechanisms through which agent interaction improves output (Chu et al., 2025; Schmucker et al., 2024; Wang et al., 2025); (2) *recognition theory in education*, which applies Honneth to pedagogy but does not operationalize recognition in AI systems (Fleming, 2011; Huttunen & Heikkinen, 2004; Stojanov, 2018); (3) *LLM-as-Judge evaluation*, which establishes the measurement paradigm but has not been applied to trace mechanisms within agent architectures (Gu et al., 2025; Zheng et al., 2023); (4) *process tracing methodology*, which provides the within-case causal reasoning framework but has not been applied to AI systems (Beach & Pedersen, 2019; Bennett & Checkel, 2015); and (5) *mechanism-oriented AI research*, which seeks to explain how models produce behavior but operates at the neural rather than agent level (Anthropic, 2025a; Lindsey, Gurnee, et al., 2025).

We sit at the intersection: a constructive, empirically evaluated system that operationalizes recognition theory through psychoanalytically-inspired architecture, assessed through a multi-judge framework, with mechanisms traced using process tracing methodology adapted from social science. The companion pilot study (Magee, 2026) established the *what* (recognition produces large, replicable effects). This paper addresses the *how*: through what specific internal processes do those effects propagate from prompt to deliberation to output to outcome?

No prior work bridges all five domains with mechanism-level empirical evidence. The closest approaches are: Hwang et al. (2025), who use architectural ToM support to improve social intelligence (but do not trace the mechanism through agent-internal processes); Scarlatos et al. (2025), who optimize dialogue quality through preference learning (but do not examine what changes in tutor behavior the optimization produces); and the mechanistic interpretability program (Lindsey, Gurnee, et al., 2025), which traces causal pathways inside models (but at the neural rather than agent level, and not in pedagogical contexts).

Stated at the level of control principles, our contribution is therefore deliberately bounded. ITS supplies model tracing, knowledge tracing, constraint repair, step-level scaffolding, and mastery gating; the learning sciences supply productive struggle, dialogic engagement, and feedback-timing theory; proof tutors supply formal checking and hint generation; and LLM-agent work supplies critique/revision and multi-agent specialization. We do not claim these principles are pedagogically new. We claim that the apparatus makes them *experimentally isolable inside an LLM tutor*—recording how prompts, critics, guards, and proof-state proxies change generated conduct and which failure classes remain—so that a broad claim that a tutor is "adaptive" can be replaced with a narrow claim about which control mechanism moved which failure mode under which domain, model, judge, and world assumptions.

3. Theoretical Framework: From Recognition to Mechanisms

The companion pilot study (Magee, 2026) operationalized recognition theory as a *design heuristic*: prompts were written to treat learners as autonomous subjects, an Ego/Superego architecture implemented internal critique, and effects were measured across fifty-two evaluations. That study established the theoretical foundation connecting Hegel's recognition framework, Freud's structural model, and contemporary recognition theory (Honneth, 1995) to AI tutoring (see Magee (2026), Sections 3.1–3.5). This section builds on that foundation by reframing recognition as a *design theory* — one that predicts specific mechanisms through which intersubjective orientation alters system behavior.

A note on the level of analysis. The three mechanisms developed below are predictions recognition theory makes specifically: they are the mechanisms recognition theory's vocabulary makes legible. As §1 noted (and as §7.9 demonstrates empirically), the broader intersubjective-pedagogy family — Vygotsky's ZPD, Piaget's accommodation, Kapur's productive failure, the cognate work of Chi, VanLehn, and Graesser — carries similar commitments tacitly, framed in different theoretical vocabularies. We could in principle derive analogous mechanisms from each: an account of "calibration" via Vygotskian scaffolding-to-the-zone, of "error correction" via Kapurian productive-failure-then-canonical-instruction, of "adaptive responsiveness" via dialogic constructivist refinement. These would not be wrong; they would be the same mechanisms named differently. We work in recognition's vocabulary throughout this section because it makes the underlying commitments most explicit — the master-slave dialectic and *Bildung* are diagnostically clearer than ZPD or accommodation for *what is happening* when an intersubjective orientation alters tutor production. The empirical operationalisation generalises (§7.9); the theoretical articulation here remains recognition-specific by design.

3.1 The Explanatory Challenge

The pilot study's ablative findings are robust but opaque. Five specific patterns resist explanation by effect sizes alone:

First, recognition-enhanced dimension variance drops in 52 of 55 within-run comparisons, but *why does treating the learner as a subject constrain the tutor's output space?* The variance reduction is the single most reliable recognition signal, yet the ablative framework offers no account of the mechanism that produces it.

Second, the superego catches content leakage and enforces struggle-preservation, but *what determines whether the ego incorporates critique versus ignores it?* The pilot study documented both productive revision and what assessors called "ego compliance" — the ego making minimal changes to satisfy the critic without substantive revision — but did not systematically classify which pattern dominates or when.

Third, the strategy_shift tag appears in 30% of recognition dialogues and 0% of baseline dialogues, but *what triggers a shift, and what prevents one?*

Fourth, the same ego/superego architecture produces opposite patterns under different models. The eta-squared for learner architecture is .527 with Kimi K2.5 but .002 with Haiku — the same factor explains over half the variance with one model and essentially none with another. The mechanism is unclear.

Fifth, the dialogue traces contain full ego-superego negotiations, but these internal processes have not been systematically correlated with external output quality. The relationship between internal deliberation quality and the tutor's final response remains unexamined.

Effect sizes answer "how much?" Process tracing answers "through what?"

3.2 Three Candidate Mechanisms Predicted by Recognition Theory

We motivate three candidate mechanisms drawing on specific components of Hegel's recognition framework. Each generates testable predictions with explicit null hypotheses; Section 6 evaluates each prediction against multi-turn factorial data, finding support for two as general mechanisms while the third is null on both the primary well-powered analysis and pre-registered replication of its one exploratory boundary-condition effect (§6.3.2).

Mechanism 1: Calibration (prompt-level) **Theoretical derivation.** Hegel's recognition framework requires that each subject's response be shaped by the *specific content* of the other's contribution (Hegel, 1977). This rules out generic responses: the tutor cannot deploy pre-formed answers because recognition demands engagement with *this* learner's *particular* understanding. Recognition-oriented prompts encode this requirement as an operational instruction — "build on what the learner actually said." The predicted behavioral consequence is *distribution narrowing*: the range of acceptable responses shrinks because each must be calibrated to a specific input.

Observable prediction. Under recognition prompts, the tutor's output distribution should narrow: lower variance across rubric dimensions, more consistent response length, reduced scatter in quality scores. Critically, this effect should be observable *even without the superego*, because calibration is a prompt-level effect operating on the ego's generation process.

Null hypothesis. If calibration is not a genuine mechanism, recognition prompts should produce mean shifts without variance reduction, or variance reduction should appear only in multiagent configurations (suggesting the superego, not the prompt, produces the constraint).

Pilot evidence. Dimension variance drops in 52 of 55 within-run comparisons (d=-0.47 to d=-1.00 depending on analysis scope). The core evidence for prompt-level calibration comes from the $2 \times 2 \times 2$ factorial (N=350), where single-agent recognition cells (cells 5–6) show variance reduction comparable to multiagent recognition cells (cells 7–8) — variance narrows under recognition regardless of whether a superego is present. Self-reflective evolution data (cells 40–45, N=366, eval-2026-02-13-8d40e086) shows the calibration effect interacting with superego disposition: the suspicious persona produces the largest recognition delta (+9.0 points, baseline 67.9 vs. recognition 76.9). Note that cells 40–45 are multiagent configurations (ego + suspicious superego), so their recognition delta reflects calibration *combined with* error correction, not calibration alone. The single-agent cells from the factorial provide the cleaner isolation.

Evidence needed beyond pilot. Systematic single-agent vs. multiagent variance comparison under v2.2 rubric. If calibration is purely prompt-level, single-agent and multiagent configurations should show similar variance reduction magnitudes. *Addressed in Section 6.1.*

Key distinction. Calibration is not the same as quality improvement. A calibrated tutor produces *reliably adequate* responses, not necessarily *excellent* ones. Mean shift and variance reduction are different effects; the claim is that recognition primarily produces the latter.

Mechanism 2: Error Correction (architecture-level) **Theoretical derivation.** Kamoi et al. (2024) demonstrate that LLMs cannot correct their own mistakes without external feedback — intrinsic self-correction frequently degrades performance. The Ego/Superego architecture provides *structurally external* feedback: a different prompt context applying different evaluation criteria. Recognition theory adds a specific prediction beyond the self-correction literature: the ego must be *receptive to critique, not merely compliant with it.* Hegel's *Bildung* (formative activity) requires that the subject be genuinely changed by encounter with the other, not merely acknowledge and continue unchanged (Stojanov, 2018). Applied to the architecture: the ego should produce qualitatively different output after superego critique under recognition (substantive revision) versus baseline (cosmetic compliance).

Observable prediction. The superego should produce identifiable categories of critique (classifiable via taxonomy), and the ego's response to that critique should differ qualitatively between recognition and baseline conditions: substantive revision (strategy change, content reorganization, pedagogical pivot) versus cosmetic compliance (minor rewording, hedge insertion, superficial acknowledgment). The prediction is directional: recognition should increase the rate of substantive revision and decrease cosmetic compliance.

Null hypothesis. If error correction does not depend on recognition, the ego-superego exchange should produce similar revision patterns under both conditions, or alternatively, baseline multiagent architecture should perform at least as well as recognition single-agent (since the critic adds value regardless of ego receptivity).

Pilot evidence. Qualitative transcript assessment (eval-2026-02-07-b6d75e87, N=118) reveals the contrast starkly. In baseline ego/superego dialogues, the superego correctly diagnoses problems but the ego regenerates the same response — assessors tagged this "ego compliance" (70.7% of baseline bilateral dialogues vs. 60.0% recognition). The stalling tag (no meaningful evolution across turns) appears in 100% of base bilateral dialogues and drops to 45% with recognition. With recognition, the ego pivots: from prescriptive to Socratic, from content routing to engagement. The factorial interaction supports this interpretation: in the pilot $2 \times 2 \times 2$ (N=350), base ego_superego learners scored 72.6 while recognition ego_superego learners scored 85.6, a 13.0-point delta. The unified learner delta was 15.6 points (74.5 to 90.1), suggesting error correction provides a consistent but not dramatically larger benefit beyond calibration alone.

Evidence needed beyond pilot. Systematic superego critique taxonomy (Section 5.1), revision delta classification (Section 5.2), and critique-to-revision mapping showing the causal chain from superego category to ego response type. *Addressed in Section 6.2.*

Key distinction. Error correction is not "having a critic." Baseline multiagent architecture *adds* a critic, but the critic is ineffective when the ego treats critique as noise to be minimized. Recognition transforms the ego-superego relationship from compliance to deliberation. The failure cases are equally informative: when the superego adopts an adversary disposition, the adversary over-deference spiral can emerge (eval-2026-02-11-35c53e99, cells 24–25: base adversary 55.8 vs. recognition adversary 65.2), and when the superego adopts an advocate disposition, recognition adds near-zero benefit because there is no struggle to overcome.

Mechanism 3: Adaptive Responsiveness (interaction-level) **Theoretical derivation.** Recognition is inherently temporal. Hegel's dialectic unfolds through stages — self-certainty, encounter with the other, struggle, mutual recognition — and each stage is shaped by what preceded it. Applied to tutoring: if recognition is operative, each turn should be shaped by the *specific outcomes* of the previous turn. The tutor's third response should differ qualitatively from its first, because the learner's feedback has genuinely changed the tutor's stance. Baseline tutoring predicts static behavior: the tutor applies the same approach regardless of learner feedback.

Observable prediction. In multi-turn conversations, recognition-enhanced tutors should show *increasing* scores on adaptation-sensitive dimensions (mutual_recognition, dialectical_responsiveness under v1.0 rubric; recognition_quality, elicitation_quality under v2.2) across turns, while baseline tutors should show flat or declining scores. Strategy shifts (changing pedagogical approach mid-conversation) should be more frequent under recognition, and should correlate with specific learner signals: confusion, resistance, or breakthrough.

Null hypothesis. If adaptive responsiveness is not a genuine mechanism, per-turn score trajectories should be flat or declining under both conditions, or recognition and baseline trajectories should be parallel (both improving or both declining at similar rates).

Pilot evidence. The strategy_shift tag appears in 30% of recognition dialogues and 0% of baseline (qualitative coding, eval-2026-02-07-b6d75e87). The stalling tag appears in 100% of base bilateral dialogues. The regression tag (quality declining across turns) appears in 17.2% of baseline multi-turn dialogues versus 1.7% of recognition dialogues. Recognition effects are largest in disengagement scenarios (+16.5, +15.4 points), precisely the conditions where adaptive responsiveness matters most.

Evidence needed beyond pilot. Turn-by-turn trajectory analysis with per-dimension slopes. Conditional branching analysis: after a learner confusion signal at turn N, what happens to scores at turn N+1 under recognition versus baseline? Cross-model trajectory replication. *Addressed in Section 6.3.*

Key distinction. Adaptive responsiveness requires *both* calibration (the prompt orients the tutor toward learner-specific signals) *and* error correction (the superego catches when the tutor is repeating itself). It is an emergent property of their interaction over time, not a third independent mechanism. This makes it the strongest test of the three-mechanism model: if calibration and error correction are genuinely operative, adaptive responsiveness should emerge in multi-turn settings without additional prompting.

Note on evidence status. The Section 6 data **does not support** this prediction. In the main dataset (cells 80–87, N=432, 3–5 turn dialogues), no experimental factor modulates adaptation slopes: recognition d=0.03 on tutor slopes, architecture d=0.07, all dimensions $d \leq 0.15$, with the design well-powered to detect d $\geq$ 0.27. Cross-judge validation confirms the null. An exploratory DeepSeek/Sonnet finding in one 10-turn scenario (disengagement, $d = 1.63$) failed pre-registered replication across two additional generation models and two judges (A12, $N = 32$; §6.3.2). The three-mechanism model reduces to **two supported general mechanisms and one clean null**.

3.3 The Mechanism Interaction Model

The three candidate mechanisms map onto different levels of the architecture:

```
PROMPT LEVEL               ARCHITECTURE LEVEL

Recognition                Ego/Superego
Prompts                    Dialogue



CALIBRATION           ERROR
                      CORRECTION
Output
distrib.              Catch failure
narrowing             modes; ego
                      incorporates
```

```
                                     critique




                         ADAPTIVE RESPONSIVENESS
                         (emergent from interaction)

                         Turn-by-turn adaptation to
                         learner-specific signals;
                         strategy shifts; trajectory
                         evolution across conversation
```

Separability prediction. If the candidate mechanisms are genuinely separable, the existing 2×2 factorial design can be reinterpreted as a mechanism isolation matrix. Paper 2.0 tests this directly using the messages-mode cells (80–87), which implement the full $2 \times 2 \times 2$ factorial under multi-turn conversation:

Factorial Cell	Paper 2.0 Cells	Calibration	Error Correction	Adaptive Resp.	Predicted Outcome
Baseline + single-agent	80, 81	No	No	No	Lowest floor, highest variance
Recognition + single-agent	84, 85	**Yes**	No	Partial	Higher floor, lower variance
Baseline + multiagent	82, 83	No	Attempted (ego_ compliance)	No	Potentially *worse* than single
Recognition + multiagent	86, 87	**Yes**	**Yes**	**Yes** (emergent)	Highest mean, lowest variance

Pilot data alignment. The factorial means from the pilot study (N=350, cells 1–8) partially align with this prediction. Recognition produces consistent improvement regardless of architecture (+15.6 pts for unified learners, +13.0 pts for ego_superego learners), consistent with calibration as the dominant mechanism. The predicted *harm* from baseline multiagent is not clearly observed in the pilot aggregate factorial, though qualitative evidence (100% stalling in bilateral baseline) strongly supports it at the individual dialogue level. Messages-mode data (cells 80–87, DeepSeek V3.2 N=146, Haiku 4.5 N=163) supports calibration as the primary mechanism: recognition d=1.88 (DeepSeek), d=1.84 (Haiku), with the architecture delta collapsing from +9–15 points under baseline to near-zero under recognition—a substitution interaction rather than the predicted synergy (Section 6.4).

The tutor-learner asymmetry explained. The tutor-learner asymmetry — large tutor-side effects (d=1.03), near-zero learner-side effects — follows directly from the mechanism model. Both supported mechanisms operate on tutor *production*: calibration constrains what the tutor generates, error correction filters what it outputs. Neither directly modifies learner *reception*. The null finding for adaptive responsiveness as a distinct mechanism strengthens this explanation: even the interaction-level pathway does not differentiate conditions, leaving only prompt-level and architecture-level effects—both tutor-internal. The synthetic learner generates responses according to its own architecture; it is not "taught" by better tutoring within a 3–5 turn window. The asymmetry is a structural property of the mechanism model, not a measurement artifact.

3.4 Recognition Theory Predicts the Failures

A strong explanatory framework predicts not just successes but failure modes. Each failure prediction below is derived from the mechanism model and supported by pilot data:

Prediction 1: Cognitive overload in weak models. Recognition-oriented prompts add complexity (track learner specifics, calibrate responses, engage with contributions). For models near their instruction-following capacity limit, this complexity *decreases* performance because calibration requires *capacity* — without it, the prompt produces confusion, not constraint. The cognitive prosthesis data (cells 66–68, N=96, eval-2026-02-17-25aaae85) supports this: the full mechanism stack scores 49.4–53.7 versus a ~65 baseline, a 12–16 point deficit. Paper 1.0 originally read this deficit as a *capability threshold*: the mechanism stack adds noise below the threshold and signal above it. §6.6.10 tests that reading directly by running cell 66 across six ego models spanning a $ $26-point capability range and finds no capability-threshold pattern — the bidirectional-profiling variant is either neutral or harmful at every tested capability tier. Prediction 1 survives as "complex prompts can harm weak models," but the specific "prosthesis helps weak models, hurts strong ones" version is not supported.

Prediction 2: Adversary over-deference. When the superego is adversarial AND the ego is recognition-primed, the ego's commitment to "honoring autonomy" can be exploited by a superego that frames all pedagogical recommendations as "controlling." The adversary data (cells 24–25, N=20, eval-2026-02-11-35c53e99) shows that recognition does not rescue the adversary condition: the adversary superego drags scores to 55.8 under baseline. While recognition improves the adversary condition to 65.2, the absolute score remains well below non-adversary configurations, consistent with the over-deference prediction.

Prediction 3: Advocate ceiling. When the superego is already cooperative (advocate persona), recognition adds minimal benefit because there is no struggle for the ego to overcome. Recognition theory specifically predicts that recognition emerges *from struggle*, not from agreement. The disposition gradient from self-reflective evolution (cells 40–45, N=366) supports this: suspicious +9.0, adversary +6.2, advocate +4.8. The most hostile superego dispositions benefit most from recognition, while the most cooperative benefit least. **Scope limit (§6.6.8):** follow-up D4 tests bound the ordering sharply. The monotone suspicious > adversary > advocate gradient is intact on cells 40–45 in philosophy, but not on cells 22–27 (reversed on philosophy, non-monotonic on SEL) and not on the architecture-matched cells 40–45 SEL replication (all recognition deltas positive, but advocate exceeds adversary).

Prediction 3 is therefore best read as a philosophy-domain, dialectical-ego ordering rather than as a domain-general property of the recognition mechanism; the broader claim that recognition helps across disposition families survives in SEL.

Prediction 4: Model-dependent architecture effects. The Haiku/Kimi ranking reversal (eta-squared for learner architecture: .527 Kimi, .002 Haiku) is predicted by the mechanism model: if a model cannot generate calibrated output (calibration fails) or cannot incorporate critique (error correction fails), the architecture produces noise rather than improvement. The mechanism model predicts that Kimi fails at calibration (cannot follow the complex multiagent learner instructions), generating incoherent ego/superego/synthesis learner output that the tutor cannot meaningfully adapt to.

3.5 Connecting Mechanisms to Recognition Theory

The candidate mechanisms are not arbitrary analytical constructs. Each corresponds to a specific component of Hegel's recognition framework:

Hegelian Concept	Mechanism	Operationalization	Status
Mutual acknowledgment — recognizing the other as a valid source of meaning	Calibration	Tutor's output distribution constrained by the specific content of the learner's contribution	**Supported**
Formative activity (*Bildung*) — self-formation through the labor of understanding	Error correction	The ego is genuinely changed by encounter with the superego's critique, not merely compliant	**Supported** (model-dependent)
Dialectical unfolding — self-consciousness develops through stages of encounter	Adaptive responsiveness	The tutor's approach evolves across turns in response to the specific trajectory of the conversation	**Not supported as general mechanism**

This mapping is a *heuristic translation*, not a claim about AI consciousness. The tutor does not "recognize" the learner in Hegel's metaphysical sense. But the design heuristic derived from recognition theory produces *measurable functional analogues* of the first two components: calibration where there would otherwise be generic responses, and genuine self-transformation where there would otherwise be compliance. The third component—dialectical unfolding across turns—does not manifest as a distinct mechanism: tutors adapt across turns, but this adaptation is identical regardless of experimental condition (Section 6.3). Recognition operates through level-setting (calibration and error correction), not trajectory-shaping.

The contribution of this section is showing that two of the three predicted analogues are not merely *statistically detectable* (the pilot study established this) but *mechanistically traceable*

— each maps to specific internal processes in the architecture that can be observed, classified, and quantified through the methods described in Section 5. The null finding for the third is itself a contribution: it reveals that recognition's temporal dimension—the Hegelian claim that understanding unfolds through staged encounter—does not translate into measurable trajectory differences within the 3–5 turn dialogue window.

4. System Architecture

The companion study (Magee, 2026) describes the full architecture design: the Ego/Superego multi-agent structure, its Freudian-Hegelian theoretical motivation, the Drama Machine deliberation protocol, and the recognition-enhanced prompt design. This section condenses the architectural overview and focuses on the features that make the system amenable to process tracing — the observability infrastructure, the bilateral symmetry between tutor and learner pipelines, and the mapping between the factorial design and mechanism isolation.

4.1 Ego/Superego Architecture (Summary)

Two agents collaborate to produce each tutoring response. The **Ego** generates pedagogical suggestions given the learner's context, incorporating recognition principles, memory guidance, and decision heuristics. The **Superego** evaluates the Ego's output against pedagogical and recognition criteria before any suggestion reaches the learner. The Superego can accept, modify, or reject suggestions, creating an internal dialogue — proposal, evaluation, revision — that mirrors the external tutor-learner dialogue.

The architecture implements Hegel's recognition through Freud's structural model: the Superego represents internalized recognition standards, and the Ego-Superego dialogue operationalizes the internal self-evaluation that Hegelian recognition requires before adequate external relating. The Superego is conceived not as an equal dialogue partner but as a *ghost* — a memorial trace of pedagogical authority that judges but cannot negotiate (Magee, 2026, Section 4.2). Recognition occurs in the Ego-Learner encounter; the Superego provides the internal constraint that makes genuine recognition possible.

A crucial architectural property is that the Ego retains **final authority**: after receiving the Superego's critique, the Ego produces a revision, but it is not required to comply. This design choice is theoretically motivated — genuine recognition cannot be mandated — and empirically consequential, because it means the Ego's *receptivity* to critique varies with condition, creating the variation that error correction analysis exploits (Section 3.2, Mechanism 2).

4.2 Bilateral Learner Architecture

The learner is not scripted. In multi-agent configurations, the learner has its own Ego/Superego architecture that mirrors the tutor's: the Learner Ego generates an initial reaction to the tutor's message, the Learner Superego critiques that reaction (is it too superficial? what is being missed?), and the Learner Ego revision produces the final external message.

This bilateral symmetry has a design consequence for observability: the same trace structure records both tutor and learner deliberation, enabling symmetric analysis of internal processes on both sides. Single-agent ("unified") learners in multi-turn message mode are also LLM-powered — they generate responses via a single-agent call rather than the three-stage deliberation pipeline, and use a structured prompt requesting `[INTERNAL]`/`[EXTERNAL]` sections so that internal reasoning is captured even without the Superego stage.

Token asymmetry. While the trace structure is symmetric, the computational cost is not. The tutor ego's recognition prompt (~5,100 tokens) is approximately 8× larger than the learner ego prompt (~580 tokens), because the recognition prompt contains the Hegelian framework, detailed pedagogical instructions, and interaction guidelines that produce the calibration effect (Section 6.1). This system prompt is re-sent on every API call within a turn — so a tutor turn with two superego rejection rounds requires five LLM calls (generate + 2 reviews + 2 revisions), each re-transmitting the full prompt plus growing conversation history. The learner's three-call deliberation pipeline (ego + superego + revision) uses shorter prompts throughout. In a representative 10-turn multi-agent dialogue (cell 87, recognition/multi/psycho, DeepSeek V3.2), the tutor consumed 186,865 input tokens versus the learner's 54,489 — a 3.4× ratio, driven by the combination of longer prompts and additional deliberation rounds. This asymmetry is an inherent cost of the recognition-enhanced prompt design: the instructions that produce calibrated output are themselves large, and the multi-agent architecture multiplies that cost per turn.

4.3 Trace Logging Architecture

Every ego-superego exchange in every evaluation is logged with verbatim text, enabling process tracing at a level of detail unusual for LLM-based systems. The trace is a sequence of structured entries, each recording:

Field	Description
`agent`	Which agent produced this entry (`tutor`, `ego`, `superego`, `system`, `learner`, `learner_ego_initial`, `learner_superego`, `learner_ego_revision`)
`action`	What the agent did (`context_input`, `generate`, `review`, `respond`, `finalize`, `pre_analyze`, `memory_cycle`)
`from` / `to`	Direction of information flow (e.g., `ego` → `superego` → `ego`)
`round`	Deliberation round index (0-based)
`suggestions`	The verbatim text produced at this step
`latencyMs`	Wall-clock time for the LLM call
`metrics`	Provider, model, token counts

The tutor trace records seven distinct steps per deliberation cycle:

1. **`tutor/context_input`** — Initial learner context parsed and injected
2. **`superego/pre_analyze`** — Drama Machine reinterpretation of learner signals
3. **`ego/generate`** — Ego's initial suggestions
4. **`superego/review`** — Superego critique of ego output, per deliberation round

5. `ego/respond` — Ego's revision after critique, per deliberation round
6. `ego/finalize` — Final delivery to learner, end of deliberation loop
7. `system/memory_cycle` — Memory dynamics final pass

For single-agent cells (no superego), the `disableSuperego` flag suppresses steps 2, 4, and 5, producing a three-step trace: `context_input` → `generate` → `finalize`. This flag is enforced at three architectural layers to prevent phantom superego calls: the `runDialogue()` option disables the deliberation loop, the `logApiCall()` function independently verifies superego absence from dialogue state, and the `traceToSteps()` projection uses a `superegoFollows` lookahead to correctly route ego output.

The learner trace follows a symmetric structure: `learner_ego_initial/deliberation` → `learner_superego/deliberation` → `learner_ego_revision/deliberation` → `learner/final_output`, with the internal deliberation array returned alongside the external message.

4.4 Provenance and Reproducibility

Each dialogue log is stored twice: once under a content-addressable SHA-256 hash (immutable evidence snapshot), and once under the dialogue ID (working copy that may be updated with holistic scores). The content hash is stored in the database alongside a configuration hash, establishing a cryptographic chain from raw dialogue to scored result.

Prompt versioning tracks which exact prompt text produced each evaluation row, with version hashes computed at generation time. This enables post-hoc auditing: any row's output can be traced back to the exact prompt, model configuration, scenario, and dialogue log that produced it.

4.5 Scoring Pipeline

The scoring pipeline is symmetric across tutor and learner, with rubric-versioned storage that prevents cross-version contamination:

Per-turn tutor scoring. Each tutor turn is independently scored against the rubric. For multi-turn conversations, this produces a score vector indexed by turn number, stored as JSON in the `tutor_scores` column. Aggregate scores extract first-turn, last-turn, and development (last minus first) metrics.

Per-turn learner scoring. Each learner turn is scored against a complementary learner rubric, stored in `learner_scores`. The learner rubric evaluates authenticity, question quality, conceptual engagement, and revision signals — dimensions that assess learner *production* independently of tutor quality.

Holistic scoring. Both tutor and learner receive holistic evaluations that score the full conversation arc as a single unit, capturing emergent qualities (bilateral transformation, growth trajectory) that per-turn evaluation misses.

Deliberation scoring. A separate deliberation rubric evaluates the quality of internal ego-superego processes for both tutor and learner (multi-agent cells only). This scoring channel examines deliberation quality independently of the public output, enabling the error correction analysis described in Section 5.2.

Rubric versioning. Each scoring operation records the rubric version used (`tutor_rubric_version`, `learner_rubric_version`, `dialogue_rubric_version`, `deliberation_rubric_version`), auto-resolved from the active YAML configuration file's `version:` field. This prevents retroactive cross-version comparisons: all within-run analyses use a single rubric version.

4.6 Factorial Design as Mechanism Isolation

The $2 \times 2 \times 2$ factorial design (Recognition $\times$ Tutor Architecture $\times$ Learner Architecture) maps directly to the candidate mechanism predictions derived in Section 3:

Comparison	Mechanism Tested	What It Isolates
Cells 5-6 vs 1-2 (recognition vs base, single-agent)	**Calibration**	Prompt-level effect without superego feedback
Cells 7-8 vs 3-4 (recognition vs base, multi-agent)	**Error correction**	Architecture-level effect with superego feedback
(7-8) - (5-6) vs (3-4) - (1-2)	**Recognition $\times$ Architecture interaction**	Whether the superego benefit depends on recognition priming
Multi-turn trajectory across turns	**Adaptive responsiveness**	Whether calibration + error correction compound over time

The single-agent cells (1-2 and 5-6) test calibration in isolation: any recognition effect in these cells operates through prompt-level constraint alone, because there is no superego to provide feedback. The multi-agent cells (3-4 and 7-8) test the superego's contribution, but the critical comparison is the *interaction*: if error correction requires recognition-primed ego receptivity (Section 3.2), then the superego benefit should be larger under recognition than under baseline. This interaction was the strongest signal in the pilot study.

Beyond the core factorial, the architecture supports mechanism isolation through extended cell families:

- **Self-reflective cells** (40-45): Test whether between-turn ego/superego reflection accumulates insights
- **Other-ego profiling cells** (54-65): Test Theory of Mind — whether building an explicit model of the interlocutor improves calibration
- **Cognitive prosthesis cells** (66-68): Test whether external scaffolding aids weaker models
- **Placebo control cells** (15-18): Length-matched prompts without recognition theory, controlling for prompt elaboration effects
- **Hardwired rules cells** (13-14): Superego heuristics embedded in the ego prompt, testing whether static rules replicate dynamic critique

4.7 Observability Summary

The architecture provides four levels of observability for process tracing:

1. **Verbatim trace**: Every ego-superego exchange, every learner deliberation step, recorded with agent labels, round indices, and timestamps
2. **Scoring pipeline**: Per-turn rubric scores for both tutor and learner, plus holistic and deliberation scores, all rubric-versioned
3. **Provenance chain**: Content hashes, configuration hashes, prompt versions, and dialogue IDs linking scored results to raw dialogue logs
4. **Factorial structure**: Cell definitions that map directly to mechanism isolation tests, with explicit factor tags stored in the database

This observability infrastructure is not incidental — it was designed to support the process tracing methodology described in Section 5. The trace logging makes superego critique taxonomy (Section 5.1) possible by providing verbatim critique text; the per-turn scoring enables trajectory analysis (Section 5.3) by providing score vectors over time; and the factorial structure enables mechanism isolation (Section 5.4) by providing controlled comparisons that separate prompt-level from architecture-level effects.

5. Methodology

5.1 Overview

Paper 1.0 employed a factorial/ablative methodology: cross experimental conditions, measure outcomes, and compare effect sizes. This design established *that* recognition-oriented prompts modulate tutor output (d=1.11, N=350) but left unexplained *how* the modulation occurs.

Paper 2.0 extends this with **process tracing** — a methodology from comparative politics and qualitative social science that examines causal chains *within* cases rather than statistical patterns *across* cases (Bennett & Checkel, 2015; Beach & Pedersen, 2019). The architecture's observability (Section 4.3) makes process tracing unusually feasible: every ego-superego exchange is logged, every turn scored, every revision recorded.

The methodology combines three approaches:

1. **Process tracing** — following the causal chain from prompt to internal deliberation to output within individual dialogues
2. **Quantitative confirmation** — aggregating process-level observations across the full dataset to test mechanism-level predictions (Section 3.2)
3. **Cross-model replication** — repeating process analyses under different ego models to separate mechanism effects from model artifacts

Each analytical method targets one or more of the candidate mechanisms predicted in Section 3.2: calibration (M1), error correction (M2), and adaptive responsiveness (M3).

5.2 Evaluation Rubric Design (v2.2)

5.2.1 Rubric Derivation: From Ad Hoc to Literature-Informed The v1.0 rubric (14 tutor dimensions, 6 learner dimensions) was constructed through iterative prompting of Claude during Paper 1.0's pilot phase. The process was pragmatic rather than principled: dimensions were proposed by the LLM based on general pedagogical reasoning, refined through trial scoring, and retained if they appeared to discriminate between conditions. No systematic literature review grounded the dimension selection, and the resulting rubric reflected the LLM's implicit model of "good tutoring" rather than validated constructs from learning sciences or educational measurement.

Paper 2.0 replaced this approach with a structured derivation process. A literature review of eight validated evaluation frameworks—MRBench (Maurya et al., 2025), GuideEval (Y. Liu et al., 2025), MathTutorBench (Macina et al., 2025), ICAP (Chi & Wylie, 2014), TRU (Schoenfeld, 2018), the Danielson Framework (Danielson, 2022), Talk Moves (Michaels & O'Connor, 2015), and a cognitive scaffolding rubric (Figueiredo et al., 2025)—plus five dialogue-level frameworks (Alexander's dialogic teaching principles (Alexander, 2018), Nystrand's process indicators (Nystrand, 1997), T-SEDA (Hennessy et al., 2016), LLM-Rubric (Hashemi et al., 2024), and the BEA 2025 shared task) produced a cross-framework dimension mapping identifying which constructs were well-covered (3+ independent frameworks), partially covered (1–2), or unique to our rubric.

Three findings drove the redesign:

1. **Structural conflation.** The v1.0 rubric conflated what the tutor *perceives* (learner state detection), what it *does* (pedagogical strategy), and what it *elicits* (learner reasoning). GuideEval's Perception→Orchestration→Elicitation (P→O→E) decomposition (Y. Liu et al., 2025) provided a principled separation, validated on 5,177 samples with 89–97% human-LLM agreement.

2. **Empirical redundancy.** A dimension-by-dimension audit predicted high correlations among clusters of v1.0 dimensions (e.g., `relevance` + `personalization` + `memory_integration`; `productive_struggle` + `transformative_potential`). This redundancy is severe and persists even after consolidation: PCA on 1,584 per-turn observations of the consolidated v2.2 8-dimension rubric finds PC1 explaining 80.7% of variance (KMO = 0.938), with a mean inter-dimension correlation of $r = 0.776$ and a single eigenvalue above 1 (Section 8.6). The rubric measured far fewer independent constructs than its dimension count implied.

3. **Missing constructs.** Every reviewed framework except our v1.0 rubric measured *content accuracy* (factual correctness). Additionally, `learner_growth` on the tutor rubric was architecturally unscorable—the judge could not see the learner's next turn.

The redesign consolidated 14 tutor dimensions to 8 using the P→O→E decomposition, added `content_accuracy`, and removed `learner_growth`.

The learner rubric required a parallel restructuring. The v1.0 learner rubric (7 dimensions) suffered from three problems identified in the dimension audit: overlap among question_quality, conceptual_engagement, and revision_signals (all measuring "depth of intellectual engagement" from slightly different angles); a persona_consistency dimension (5% weight) that created a structural tension with revision_signals (showing genuine growth means departing from a frustrated or confused persona); and a deliberation_depth dimension that was only scorable for multi-agent learners, creating an architecture-dependent gap in the scoring instrument. The ICAP framework (Interactive→Constructive→Active→Passive; (Chi & Wylie, 2014)) provided a validated hierarchy for collapsing the overlapping engagement dimensions into a single engagement_quality construct with empirically grounded scoring anchors: level 1 (Passive) = paraphrases tutor, confirms understanding; level 3 (Active/Constructive) = generates own interpretations, makes connections; level 5 (Interactive) = co-constructs understanding, challenges tutor's framing. persona_consistency was removed (its core concern—does the learner feel real?—is captured by learner_authenticity), and deliberation_depth was moved to the dedicated deliberation rubric (Section 5.2.4). The resulting 5-dimension learner rubric maps to two validated frameworks: ICAP for engagement levels and T-SEDA's behavioral coding clusters (Hennessy et al., 2016) for revision signals (B-cluster: Build on ideas) and conceptual progression (R-cluster: Make reasoning explicit).

The tutor holistic rubric was radically simplified from 6 to 3 dimensions after the v2.1 consistency analysis showed $r = 0.907$ ($N = 32$) between per-turn and holistic scores—a near-redundancy that holds across rubric versions ($r \approx 0.87$ on $N = 288$ v2.2 dialogues)—indicating that most holistic signal was redundant with aggregated per-turn scoring. The three retained dimensions—pedagogical arc, adaptive trajectory, and pedagogical closure—capture exclusively arc-level properties that cannot be assessed from individual turns. Scale criteria across all instruments were redesigned following Yamauchi et al. (Yamauchi et al., 2025): only levels 1, 3, and 5 carry full descriptions, as intermediate-level descriptions have limited impact on LLM judges.

5.2.2 Tutor Per-Turn Rubric The evaluation rubric underwent four iterations (v1.0→v2.0→v2.1→v2.2), each responding to empirical anomalies discovered during analysis (documented in Appendix F). The current v2.2 rubric consolidates 14 dimensions to 8, guided by the GuideEval P→O→E decomposition framework and the study's own empirical dimension clustering. The eight dimensions are:

Category	Dimension	Weight	Mechanism Relevance
Perception (P)	`perception_quality`	15%	Calibration: tutor perceives learner's specific state
Perception (P)	`content_accuracy`	10%	Error correction: factual/domain accuracy
Orchestration (O)	`pedagogical_craft`	15%	Calibration: response construction quality
Orchestration (O)	`elicitation_quality`	15%	Adaptive responsiveness: probing learner reasoning
Orchestration (O)	`adaptive_responsiveness`	10%	Adaptive responsiveness: turn-over-turn modulation

Category	Dimension	Weight	Mechanism Relevance
Execution (E)	`productive_difficulty`	10%	Calibration + adaptation: challenge calibration
Execution (E)	`epistemic_integrity`	10%	Error correction: intellectual honesty
Recognition	`recognition_quality`	15%	Calibration + error correction: intersubjective stance

The consolidation from 14→8 dimensions was validated through synthetic calibration (r=0.996 against v2.1 scoring on identical responses), confirming that the reduced rubric preserves discriminability while eliminating ceiling-prone and redundant dimensions.

5.2.3 Learner Per-Turn Rubric The learner rubric mirrors the tutor rubric symmetrically (Section 4, Design Principle), scoring learner responses on five ICAP-anchored dimensions:

Dimension	Weight	What It Measures
`engagement_quality`	25%	Active vs. passive participation
`conceptual_progression`	25%	Depth of conceptual engagement over turns
`revision_signals`	20%	Evidence of position revision in response to tutor
`metacognitive_awareness`	15%	Self-monitoring of understanding
`learner_authenticity`	15%	Persona-consistent, non-formulaic responses

5.2.4 Holistic and Deliberation Rubrics Two additional rubrics assess trajectory-level and process-level quality:

- **Tutor holistic** (3 dimensions: `pedagogical_arc`, `adaptive_trajectory`, `pedagogical_closure`) — scores the full multi-turn dialogue as an arc, capturing qualities invisible to per-turn scoring.
- **Deliberation quality** (6 dimensions, applied symmetrically to tutor and learner ego-superego traces) — scores the quality of internal deliberation for multi-agent cells.

5.2.5 Public-Only Output Scoring (v2.1 Fix) A critical methodological decision: per-turn and holistic judges see ONLY public messages (the delivered tutor response and the learner's external message). Internal ego-superego deliberation is scored separately by the deliberation rubric. This prevents a confound where multi-agent cells receive higher scores simply because the judge sees richer internal reasoning.

5.2.6 Rubric Version Tracking Each scored row records which rubric version was used (`tutor_rubric_version`, `learner_rubric_version`, `dialogue_rubric_version`, `deliberation_rubric_version`), auto-resolved from YAML `version:` fields at write time. This prevents cross-version contamination: v1.0 scores (8,987 backfilled rows) are never mixed with v2.2 scores in the same analysis.

5.3 Superego Critique Taxonomy

Purpose Classify what the superego actually objects to, building an empirical taxonomy from the data rather than imposing one from theory. This directly tests Mechanism 2 (error correction): if recognition changes the ego's receptivity to critique (Section 3.2), we should see different critique patterns and revision outcomes under recognition vs. baseline conditions.

Data Source Dialogue log files in `logs/tutor-dialogues/` contain the full ego→superego→ego_revised chain with verbatim text. An extraction script (`scripts/extract-superego-critiques.js`) harvests all superego review entries and learner superego deliberation entries into structured JSONL format.

Method

1. **Extraction**: Harvest all ego-superego exchanges from dialogue traces, yielding critique text, verdict (approved/rejected), confidence, intervention type, and surrounding context (ego draft, ego revision).
2. **Classification**: An LLM-based classifier (`scripts/classify-superego-critiques.js`) assigns each critique to a 10-category taxonomy:
   - Content accuracy, learner model failure, tone mismatch, structural scaffolding
   - Premature resolution, sycophancy detection, repetition/stalling, autonomy violation
   - Recognition failure (baseline-specific), redirection without engagement
3. **Quantification**: Frequency distributions by condition (recognition vs. baseline), model, and scenario. Chi-squared or Fisher's exact test for distributional differences.

Predictions

- Recognition-primed superegos should catch more *relational* failures (sycophancy, premature resolution, autonomy violation); baseline superegos should focus on *content/structural* issues.
- Recognition ego revisions should be more *substantive*; baseline ego revisions should be more *cosmetic* (ego_compliance).

Human Validation Pilot Because the 10-category taxonomy is LLM-assigned (`scripts/classify-superego-critiques.js`, Haiku 4.5 classifier by default), we operationalise a human validation pilot rather than treating the taxonomy as self-evidently valid. The pilot apparatus lives in the repository and is runnable end-to-end by an external reviewer:

1. **Sampling** (`scripts/human-validation-sample.js`): Stratified random draw of 40 classified critiques with 4 per category across the 10 substantive labels, using a deterministic RNG seed (mulberry32, seed=20260416) for reproducibility. Categories with sparse representation in the classified corpus (CONTEXT_BLINDNESS n=4; REGISTER_MISMATCH n=0) are filled by deficit roll-over from larger buckets, with the gap logged for reviewer inspection. The script emits two files: a rater packet CSV (item_id + feedback text + ego draft + ego revision + learner-context snippet, no LLM labels) and a matched key JSONL holding the LLM labels plus metadata, joined on a SHA-1 `item_id`.
2. **Codebook** (`docs/research/human-coding-codebook.md`, v1.0): Per-category definitions, signal phrases, boundary rules for adjacent categories (CONTEXT_BLINDNESS vs FABRICATION, RECOGNITION_FAILURE vs EMOTIONAL_NEGLECT, RECOGNITION_FAILURE vs LACK_OF_AGENCY), a 12-step decision flowchart, CSV filling instructions, and one in-corpus exemplar per category (with a constructed exemplar for REGISTER_MISMATCH given its n=0).
3. **Analysis** (`scripts/human-validation-analyze.js`): Auto-discovers `exports/human-validation-pilo` joins each rater to the key JSONL by `item_id`, and computes (a) Cohen's for every rater pair including each rater vs. the LLM classifier, (b) Fleiss' across the full panel where every item has complete coverage, (c) per-category F1 scores, and (d) full confusion matrices. values are interpreted using Landis-Koch bands (<.21 slight, .21-.40 fair, .41-.60 moderate, .61-.80 substantial, >.80 almost perfect). The target floor for publishable reliability is $\geq$ **0.60 (substantial agreement)**.
4. **Inter-LLM baseline**: Re-classifying the same 40-item sample with a second LLM (e.g., Sonnet 4.6) and passing it as a synthetic rater (`--synthetic-rater`) yields an LLM-LLM lower-bound reference. Human-LLM below this baseline would indicate the human coding is less reliable than disagreement between two well-behaved LLMs; human-LLM substantially above it indicates the human panel is not just tracking LLM idiosyncrasies.

The pilot scales ($k \geq 2$ raters, n=40 items, 10 categories) follow Cohen (1960) and Hallgren (2012) for stratified categorical reliability with minimal items-per-cell to yield interpretable per-category F1. A second phase scaling to n=100-150 items is budgeted for the final paper revision if the pilot meets the floor. Pilot values and any category with F1 < 0.60 will be flagged as "pilot-unstable" and treated as a validity caveat in §5.3 rather than a taxonomy redesign trigger.

Inter-LLM Baseline Result (v3.0.32, preliminary) The inter-LLM baseline run is complete. Haiku 4.5 (the original classifier, N=500 corpus) and Sonnet 4.5 (rerun on the same 40-item sample with identical taxonomy prompt, temperature=0, response_format=json_object) yield **Cohen's = 0.633 (substantial agreement per Landis-Koch), n=40, observed agreement 0.675, chance agreement 0.114**. Per-category F1 partitions into three tiers: *high* (F1 $\geq$ 0.75) — LACK_OF_AGENCY (1.000), CONTEXT_BLINDNESS (0.889), MEMORY_FAILURE (0.833), VAGUENESS (0.750); *moderate* (F1 0.50–0.75) — RECOGNITION_FAILURE (0.625), REDIRECTION (0.571), EMOTIONAL_NEGLECT (0.500), PEDAGOGICAL_MISJUDGMENT (0.500); *pilot-unstable* — FABRICATION (F1 undefined: Haiku assigned 4 items, Sonnet assigned 0, distributing them across CONTEXT_BLINDNESS/VAGUENESS/MEMORY_FAILURE). The confusion-matrix asymmetry points to RECOGNITION_FAILURE as a catch-all under Sonnet's decision policy: Haiku's EMOTIONAL_NEGLECT, PEDAGOGICAL_MISJUDGMENT, and some REDIRECTION items all collapse to RECOGNITION_FAILURE under Sonnet. This matches the codebook's boundary-rule guidance that RECOGNITION_FAILURE is the broadest diagnostic frame and should be applied only when the critique explicitly invokes intellectual autonomy as the core problem. The human-rater pilot will use this ≈ 0.63 as its lower-bound reference: human-LLM substantially above 0.63 indicates the human panel is not merely tracking LLM idiosyncrasies; near or below 0.63 indicates the taxonomy may not generalize beyond LLM-LLM reliability. FABRICATION will require explicit codebook reinforcement (e.g., elevated to decision-flowchart step 2) or retirement if human raters likewise fail to converge on it.

5.4 Turn-by-Turn Trajectory Analysis

Purpose Test the adaptive responsiveness mechanism (Mechanism 3, Section 3.2): do recognition-enhanced tutors show changing quality across turns, while baseline tutors remain static?

Data Source Multi-turn evaluation runs with per-turn scores stored in the `tutor_scores` and `learner_scores` JSON columns. The trajectory analysis script (`scripts/analyze-trajectory-curves.j` reads these directly from the database and computes per-dimension slopes.

Method

1. **Per-turn score extraction**: Parse the `tutor_scores` and `learner_scores` JSON for each row, yielding a score sequence indexed by turn number.
2. **Trajectory metrics** (computed per dialogue):
   - **Slope**: Linear regression of score ~ turn_number
   - **Curvature**: Quadratic term (do effects accelerate or decelerate?)
   - **Breakpoint**: Turn at which recognition and baseline diverge
3. **Dimension-specific trajectories**: Separate slopes for each v2.2 rubric dimension. Key hypotheses (from Section 3.2):
   - **H1**: Recognition tutors show steeper positive slopes on `recognition_quality` and `elicitation_quality` than baseline. **Note**: Section 6.3 found no significant dimension slope differences on any factor (max $|d| = 0.15$ with cross-judge replication, overall tutor slope $d = 0.03$, $N = 432$). H1 is not supported; both supported mechanisms raise *levels*, not *slopes*.
   - **H2**: Tutor-learner slope gap (tutor slope minus learner slope) should be smaller under recognition (more symmetric change)

- **H3**: After learner confusion signals, recognition tutors show larger positive Δ on the next turn
- **H4**: Recognition tutors show more `action_type` diversity across turns (strategy shifting)
- **H5**: Trajectory patterns replicate across ego models (mechanism robustness)

4. **Within-test change analysis**: A symmetric method (`scripts/analyze-within-test-change.js`) computes first-to-last deltas for both tutor and learner using identical trajectory metrics, enabling direct comparison of which side changes more.

Visualization

- **Adaptation curves**: Turn × mean score with confidence bands, faceted by condition and dimension
- **Conditional response plots**: Box plots of $\Delta(N+1-N)$ after each learner event type, by condition
- **Strategy shift sequences**: Alluvial/Sankey diagrams showing action_type transitions across turns

---

5.5 Within-Test Change and Stagnation Detection

Purpose Measure the symmetric first-to-last delta across tutor and learner, and detect learning stagnation patterns where neither party shows growth over turns.

Method Two complementary trajectory methods:

1. **Rubric trajectories** (Method A): Per-turn rubric scores from the database, yielding precise quality measurements at each turn. Computed for both tutor (`tutor_scores` JSON) and learner (`learner_scores` JSON).

2. **Text-proxy trajectories** (Method B): Lexical and discourse complexity features computed directly from dialogue text, providing an independent measure that does not depend on rubric scoring. Features include word count, question density, reflection markers, commitment markers, and tutor-learner vocabulary overlap.

The stagnation analysis script (`scripts/analyze-learning-stagnation.js`) combines both methods and flags dialogues with non-positive learner deltas, identifies text-level predictors of growth, and correlates transcript features with rubric deltas.

ANOVA Design Three-way ANOVA on the first-to-last delta, with factors: - A: Recognition (base vs. recog) - B: Tutor architecture (single vs. multi-agent) - C: Learner architecture (unified vs. ego_superego)

Applied identically to tutor and learner deltas, enabling direct comparison of which factors drive change on each side.

5.6 The Measurement Paradox as Methodology

The measurement paradox — where authentic pedagogical engagement produces lower scores under naive evaluation — is not a limitation but a *methodological finding* about what LLM-as-Judge evaluation measures.

The Paradox When the evaluation judge receives only the tutor's response without dialogue context, it interprets careful scaffolding of authentic confusion as failure. Adding dialogue context to the judge resolves this, confirming the paradox is a measurement artifact rather than a quality decline.

What This Reveals LLM judges optimize for *surface resolution* — visible agreement, smooth interaction, confident answers. Productive struggle looks like failure to a judge that cannot see the learner's internal development. This has implications beyond this study for any LLM-as-Judge evaluation of educational interactions.

Rubric Iteration as Evidence The rubric evolution itself constitutes evidence of construct refinement. Each version responded to specific empirical anomalies:

Version	Anomaly Addressed	Change
v1.0	Baseline design	14 tutor dimensions, standard weights
v2.0	Truncation hallucination (31% of Haiku reasonings)	Anti-truncation instruction; modulation dims reweighted
v2.1	Architecture confound (internal deliberation visible to judge)	Public-only output scoring; separate deliberation rubric
v2.2	Dimension redundancy (ceiling on tone, overlap in struggle dims)	14→8 consolidation guided by GuideEval P→O→E and empirical clustering

This iteration parallels the ego-superego dynamic: initial rubric (ego draft) → anomaly detection (superego critique) → revised rubric (substantive revision, not cosmetic).

5.7 Model Selection

Paper 1.0 used free-tier models (Kimi K2.5, Nemotron 3 Nano 30B) to demonstrate that recognition effects are achievable without frontier-model budgets. Paper 2.0 uses three generation models spanning a wide capability range—**DeepSeek V3.2** (open-weight, 685B MoE), **Haiku 4.5** (Anthropic, proprietary, optimized for speed), and **Gemini Flash 3.0** (Google, proprietary, multimodal)—with three independent judges for cross-validation (**Claude Sonnet 4.6**, **Gemini 3.1 Pro**, **GPT-5.4**). Three criteria governed the model selection:

1. **Capability separation.** Paper 1.0's model-dependent architecture effects ($\eta^2 = .527$ for Kimi vs .002 for Haiku on the pilot's learner architecture factor) demanded models at clearly different capability levels. DeepSeek and Haiku occupy distinct regions of the capability space—Haiku outperforms DeepSeek by 30–37 points on base tutor metrics (Section 6.6.1)—making cross-model comparison maximally informative for separating mechanism effects from model artifacts.

2. **Architectural diversity.** DeepSeek V3.2 is an open-weight mixture-of-experts model; Haiku 4.5 is a proprietary dense model. If mechanisms replicate across both—as they do for calibration (d = 1.88 vs d = 1.84) and the error-correction substitution pattern—the findings are less likely to reflect idiosyncratic training choices of a single model family.

3. **Practical reproducibility.** Both models are accessible through OpenRouter at low cost. DeepSeek V3.2 is available as open weights for local replication. The judge (Claude Sonnet 4.6) was chosen as a capable, cost-effective scorer; cross-judge validation with GPT-5.2 in Paper 1.0 established that judge choice compresses effect magnitudes but preserves effect directions.

Scope limitation. All three models are at least moderately capable. The Paper 1.0 Nemotron data (N = 40, mean tutor score ~8 under base) hints that very weak models may not benefit from recognition at all, and the cognitive prosthesis test (Paper 1.0, Section 6.10) was originally read as establishing a minimum ego capability threshold below which the bidirectional-profiling mechanism stack adds noise rather than signal. Paper 2.0 (§6.6.10) directly tests this capability-threshold reading across six ego models and does not find a clean threshold pattern; the prosthesis cell is neutral or harmful across the tested range rather than helpful-then-harmful. Paper 2.0's model triad tests whether mechanisms are model-*independent* among capable models and identifies generation-quality boundary conditions on their interactions; it does not claim generalization to the full capability spectrum. Section 8.3 discusses this limitation further.

Table 1: Model Configuration (Paper 2.0)

Role	Model	Access	Temperature
Tutor Ego	DeepSeek V3.2 / Haiku 4.5 / Gemini Flash 3.0	OpenRouter	0.6
Tutor Superego	DeepSeek V3.2 / Haiku 4.5 / Gemini Flash 3.0	OpenRouter	0.2–0.4
Primary Judge	Claude Sonnet 4.6	Claude Code CLI	0.2
Cross-validation Judge 1	Gemini 3.1 Pro	OpenRouter	0.2

Role	Model	Access	Temperature
Cross-validation Judge 2	GPT-5.4	OpenRouter	0.2
Learner Ego	DeepSeek V3.2 / Haiku 4.5 / Gemini Flash 3.0	OpenRouter	0.6
Learner Superego	DeepSeek V3.2 / Haiku 4.5 / Gemini Flash 3.0	OpenRouter	0.4

5.8 Cross-Model Mechanism Replication

Purpose Separate mechanism effects from model artifacts. The Haiku/Kimi ranking reversal in Paper 1.0 showed that architecture effects are model-dependent (η^2=.527 for Kimi vs. .002 for Haiku); the question is *which* mechanisms are model-dependent and *why*.

Design Run the same multi-turn scenarios under multiple ego models, holding constant the superego model, scenarios, and conditions. The replication uses three generation models—DeepSeek V3.2 (N=146), Haiku 4.5 (N=163), and Gemini Flash 3.0 (N=144)—on cells 80–87 under v2.2 rubric, with three independent judges (Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4) scoring all rows (1,296 total, zero nulls). Blind scoring was confirmed by code audit: the judge prompt receives only suggestion text, scenario context, and dialogue transcript—no prior scores, judge reasoning, or judge model labels. For each model, compute:

1. **Calibration**: Dimension variance reduction (recognition vs. baseline)
2. **Error correction**: Superego critique taxonomy distribution and revision delta types
3. **Adaptive responsiveness**: Turn-by-turn trajectory slopes

Predictions

- Calibration should be *model-independent* (prompt-level constraint, does not require architectural capacity). **Supported across 3 models and 3 judges**: recognition d=1.34–1.92 (9/9 cells unanimous); calibration d=0.52/0.64; dimension floor-lifting pattern replicates.
- Error correction should be *partially model-dependent* (requires ego capacity to incorporate superego critique). **Supported with boundary condition**: baseline architecture delta is model-dependent (+4.5–7.7 DeepSeek, +15.0 Haiku, +15.1–28.1 Gemini Flash 3.0); under recognition the substitution pattern is universal but its completeness depends on generation quality (Section 6.4.1).
- Adaptive responsiveness should require *both capable ego and capable superego* (emergent property, not independent). **Weakly supported**: AdaptΔ > 0.79 in all models, but slopes are identical across conditions (d=-0.00) and development trajectories are model-dependent. Development ranks by generation quality (45163390 > 18027efc > aea2abfb), not by prompt condition.

This provides a mechanistic explanation of the Haiku/Kimi reversal: if Kimi fails at error correction (cannot incorporate superego critique), the architecture adds noise rather than signal. The architecture is not failing; the ego model lacks the capacity to benefit from critique. The Gemini Flash 3.0 data extends this explanation: even when the ego model *can* benefit from critique, the *residual magnitude* depends on generation quality—substantial on weak models (calibration leaves more uncorrected failures), near-zero on strong models (calibration pre-empts most failures the superego would catch).

5.9 Provable Discourse Framework

Architecture The provable discourse framework treats every quantitative claim in the paper as an executable test. A claim is a structured YAML entry with five components:

1. **Statement**: A regex pattern that locates the claim in the paper text, with a minimum occurrence count. If the pattern is not found, the claim is flagged as *orphaned*—the paper no longer contains the text it is supposed to verify.
2. **Evidence**: An executable adapter that extracts the actual value from data (database, dialogue logs, manifest files, or source code). Each adapter type encapsulates a specific extraction pattern (see taxonomy below).
3. **Assertion**: A machine-checkable predicate (equality, approximation with tolerance, inequality, existence) applied to the extracted value.
4. **Dependencies**: Optional upstream claims that must pass before this claim is evaluated. If a dependency fails, the claim cascades to *blocked* status without evaluation.
5. **Remediation**: Concrete steps to fix a failing claim, ensuring that failures are actionable rather than merely diagnostic.

The framework is implemented in `services/provableDiscourse.js` (2,700 lines) with claims distributed across four YAML files: a root configuration (`config/provable-discourse.yaml`), generated claims from Paper 1.0, manually authored claims, and mechanism-specific claims for Paper 2.0. The current ledger contains **119 claims** across **18 evidence adapter types**, validated by **6 symmetry rules**.

Evidence Adapter Taxonomy Each adapter type encapsulates a specific data extraction pattern. The 18 types fall into five functional categories:

Data integrity (verify that data exists and is consistent):

Adapter	What It Extracts	Example
`manifest_total`	Total count from `paper-manifest.json`	N=4,312 primary scored
`manifest_section_total`	Section-specific count from manifest	Multi-model probe N=655
`db_count`	Row count from evaluation database with filters	N=350 modulation dataset

Adapter	What It Extracts	Example
`provenance_check`	Hash match rate between DB and dialogue logs	≥95% content hash match
`log_trace_coverage`	Fraction of scored rows with available log files	≥95% trace coverage

Effect estimation (compute statistical quantities from data):

Adapter	What It Extracts	Example
`effect_size`	Cohen's d between groups on a scored metric	Architecture d=0.05
`profile_group_effect_size`	Cohen's d comparing named profile groups	Memory isolation d=1.71
`anova_2x2`	F-statistic and η^2 from 2×2 factorial	Interaction F=0.26, p>.05
`judge_pair_correlation`	Pearson r between two judges' scores	Cross-judge r=0.55

Mechanism-specific (added for Paper 2.0):

Adapter	What It Extracts	Example
`dimension_variance`	Cross-dimension variance reduction by condition	Calibration $d \leq -0.3$
`dimension_cluster_effect`	Effect size for dimension clusters	Recognition cluster d ≥ 0.50
`jsonl_critique_stats`	Superego critique category frequencies from classified corpus	RECOGNITION_FAILURE ≥15%
`trajectory_slope`	Per-turn OLS regression coefficients	Learner slope $\beta \geq 0$
`conditional_delta`	Score change after specific learner events	Post-confusion adaptation ≥ -1
`rubric_version_comparison`	Cross-version score correlation	$r \geq -0.1$ (stability)

Structural (verify code paths and cross-references):

Adapter	What It Extracts	Example
`code_path`	Grep count for patterns in source files	≥5 matches for trace logging
`cross_reference`	Existence check for a referenced upstream claim	Pilot d=1.11 claim exists

Theoretical (register non-machine-checkable claims for traceability):

Adapter	What It Extracts	Example
`theoretical`	Presence of related provable claims in the ledger	Three mechanisms each have ≥1 provable claim

Worked Example: Tracing a Calibration Claim To illustrate the full claim→evidence→source chain, consider claim `paper2.calibration.variance_reduction`:

```yaml
- id: paper2.calibration.variance_reduction
  description: "Recognition narrows tutor output distribution."
  statement:
    pattern: "recognition narrows.*output distribution"
    flags: "i"
    min_occurrences: 0
  evidence:
    type: dimension_variance
    group_by: recognition
    output: cohens_d
    filters:
      not_null: [tutor_first_turn_score]
      like: { judge_model: "%claude%" }
  assertion:
    op: lte
    expected: -0.3
  remediation:
    - "If variance reduction d drifts above -0.3,
       update calibration claims or re-scope runs."
```

Validation trace. When the harness evaluates this claim:

1. **Statement check**: The regex `recognition narrows.*output distribution` is matched against the concatenated paper text. If not found, the claim is flagged as orphaned.
2. **Evidence extraction**: The `dimension_variance` adapter queries the evaluation database for all v2.2 rows with non-null tutor scores and a Claude-family judge. For each row, it computes the standard deviation across the 8 tutor rubric dimensions (from the `scores_with_reasoning` JSON column). It then computes Cohen's d between the recognition and baseline groups' dimension SDs.
3. **Assertion**: The extracted d is checked against `lte -0.3`. Current value: $d = -0.52$ to -0.64 across models. **Pass.**
4. **Staleness**: A SHA-256 fingerprint of the query results is compared against the stored snapshot. If the data has changed since the last validation, the claim is flagged as *stale* even if still passing—alerting the author that the underlying evidence has shifted.

This chain ensures that the paper's calibration claim ("recognition narrows the output distribution") is continuously verified against the actual dimension variance in the database. If a data correction, new run, or rubric change altered the variance pattern, the claim would fail or flag as stale, and the remediation steps would guide the fix.

Dependency Graph Claims form a directed acyclic graph (DAG) via two edge types: explicit `depends_on` declarations and implicit edges from `cross_reference` evidence (where a claim cites another claim's result). The validation pipeline topologically sorts claims (Kahn's algorithm) before evaluation, so upstream claims are always resolved before their dependents. If a dependency fails, all downstream claims are automatically marked *blocked* (status: warn) with a `blocked_by` annotation—they are not evaluated, since their evidence preconditions are not met. This cascading validation prevents a common failure mode in claim ledgers: a root statistic changes, but downstream claims that cite it continue to pass because they only check for the citation's *existence*, not its *validity*.

For example, the pilot factorial effect size (d=1.11, claim `paper2.s1.pilot.factorial_d`) depends on the N=350 dataset claim (`paper.modulation.dataset_n350`). Five further claims—model-dependent architecture reversal, strategy shift frequency, baseline stalling rate, cross-judge correlation range, and the learner paradox effect size—in turn depend on the factorial claim. If the N=350 dataset claim were to fail (e.g., because a data correction changed the sample size), the factorial claim would be blocked, and all five downstream claims would cascade to blocked status without evaluation. The `--graph` flag outputs the full DAG in Graphviz DOT format for visual inspection.

Symmetry Rules and Consistency Checks Beyond individual claims, symmetry rules enforce cross-claim consistency. Six rules currently operate:

1. **Paired presence**: If a tutor-side effect is claimed, the corresponding learner-side effect must also be explicitly stated (or explicitly omitted with justification).
2. **Magnitude bounds**: Paired claims (e.g., tutor architecture d and learner architecture d) must both fall within a threshold (e.g., both $|d| \leq 0.2$ for "near-zero" claims).
3. **Material gap**: Asymmetric claims (e.g., tutor recognition $d = -1.00$ vs. learner $d = 0.32$) must exceed a minimum gap threshold to justify asymmetry framing.
4. **Mechanism consistency**: If a mechanism is claimed for recognition, the *anti-pattern* must be documented for baseline (paired evidence).
5. **Model qualification**: Any mechanistic claim must state which models it has been tested on, with replication status for each.
6. **Inventory coverage**: Every major quantitative claim in the paper text (identified by automated scanning for N= and statistical patterns) must map to at least one claim in the ledger. Unmapped claims are flagged as unverified.

5.10 Statistical Approach

Effect Size Conventions All comparisons report Cohen's d with 95% confidence intervals. Classification: d < 0.2 negligible, 0.2–0.5 small, 0.5–0.8 medium, > 0.8 large.

ANOVA Design Three-way factorial ANOVA (A: recognition × B: tutor architecture × C: learner architecture) applied to the primary dependent variable (tutor first-turn score, 0–100) and secondary variables (holistic score, development score, learner score, dialogue quality). Main effects and all 2-way and 3-way interactions reported.

Within-Judge Comparisons Factor analyses use *within-judge* comparisons only (all rows scored by the same judge model). Cross-judge validation uses *between-judge* comparisons on identical responses to assess measurement reliability.

Epoch Filtering All Paper 2.0 analyses filter to `tutor_rubric_version = '2.2'` (epoch 2.0), ensuring no cross-version contamination with pilot data. The epoch filter is applied in SQL queries and enforced by analysis scripts' `--epoch` flag.

5.11 Reproducibility Infrastructure

Evaluation Commands All evaluations are reproducible via the CLI:

```
node scripts/eval-cli.js run --profiles <cells> --runs N --cluster multi-turn
node scripts/eval-cli.js evaluate <runId>
```

Appendix B provides the exact commands for each evaluation run referenced in the paper.

Provenance Chain Every new evaluation row records: - `dialogue_content_hash`: SHA-256 of the dialogue log file - `config_hash`: SHA-256 of the cell configuration - Rubric version, judge model, and all hyperparameters

Test Suite as Analytical Provenance The test suite (32+ files, ~12K lines) covers not only the evaluation infrastructure but also the analytical pipeline. Tests verify that scoring, ANOVA computation, trajectory extraction, and within-test change analysis produce correct outputs on known inputs.

This makes the analysis *reproducible in a testable sense*: the code's behavior on known inputs is verified by tests, and the exact code version is recorded with each evaluation run via `git_commit` in the evaluation_runs table.

5.12 Procedural Transparency: Pre-registration Drift and Post-Hoc Additions

This subsection documents two classes of methodological event encountered during the Paper 2.0 experimental arc: (i) divergences between pre-registered designs and what the running code actually executed, and (ii) additions and instruments built after data collection was underway. Both are reported here, with the cells and instruments affected, so that subsequent sections can flag each result as confirmatory, exploratory, or post-hoc without re-litigating provenance in every reference.

5.12.1 The Pre-Registration Implementation Gap Pre-registration in software-mediated experiments faces a structural risk distinct from clinical or behavioural pre-registration: the design lives in a prose document, the execution lives in code, and nothing automatically validates that the two agree at run time. In this codebase, three artefacts independently describe each experimental condition:

Artefact	Form	Authority
Pre-registration document (e.g., `a13-pre-registration.md`)	Prose design table	Records *intended* design
YAML cell config (`config/tutor-agents.yaml`)	Structured fields per cell	Read at run time; constitutes *actual* design
Dispatch logic (`services/evaluationRunner.js`, `services/adaptiveTutor/index.js`)	Routing on the YAML's `runner:` field	Determines which runner actually executes

When all three agree, no problem arises. When they diverge — for example, the pre-reg specifies `runner: standard` for a condition that ships with `runner: adaptive` in the YAML — the experiment still runs cleanly, the output dialogue logs are still valid, and the divergence is silent. No test fails; no warning surfaces; the only way to detect it is to manually cross-reference documents that no person reads in sequence during routine analysis.

We treat the YAML as the source of truth, following the same convention already established for cell registration (the `EVAL_ONLY_PROFILES` array in `services/evaluationRunner.js`). The pre-reg document records *intent*; the YAML records *behaviour*. Where these diverge, the YAML is what produced the data and therefore what the experimental claim describes. The pre-reg drift is itself a finding, not a confound: it shows that a condition we believed we were testing was structurally different from the one we actually ran.

To prevent recurrence, three remediations are now part of the experimental workflow:

1. **Runner field is mandatory and explicit for adaptive cells.** A test asserts every cell with `factors.id_director: true` or a non-default scenario source carries an explicit `runner:` field, so a missing field cannot silently default.
2. **Dialogue logs stamp the dispatched runner.** Each new trace file written by either runner records `meta.dispatched_runner` at write time. Post-hoc audits can then verify that a dialogue claimed to come from the dialogue engine actually was dispatched there.
3. **Pre-registration documents include a machine-checkable cell manifest.** New pre-regs carry a YAML front-matter block listing each cell's critical fields (`runner`, `architecture`, `prompt_type`, `scenario_source`); a CI script compares this block against `tutor-agents.yaml` and fails the commit if any field disagrees.

5.12.2 The A13 Pre-Registration Implementation Divergence The A13 pre-registration (Supplement S-A13) specified a four-condition design intended to isolate the effect of the adaptive runner from the effect of the LangGraph state machine's internal mechanisms:

Pre-reg condition	Intended runner	Intended architecture
C1 (`cell_111_a13_C1_recognition_only`)	`standard` (dialogue engine)	Recognition-only baseline
C2 (`cell_112_a13_C2_egosuperego`)	`standard` (dialogue engine)	Ego/superego two-pass
C3 (`cell_110_langgraph_adaptive`)	`adaptive` (LangGraph)	Full state policy
C4 (`cell_113_a13_C4_validator`)	`adaptive` (LangGraph)	Full state policy + validator

Under the pre-registered design, the C1 C3 and C2 C4 contrasts isolate the runner; the C1 C2 and C3 C4 contrasts isolate the architectural elaboration.

The implementation shipped all four conditions with `runner: adaptive`. The LangGraph runner instead uses an `adaptive.architecture` flag (`recognition_only`, `ego_superego`, `state_policy`, `state_policy_with_validator`) to build different sub-graphs internally. The result: all four A13 conditions are LangGraph variants. The "dialogue engine vs state machine" comparison the pre-reg implied was never actually run, and the H1.1 contrasts comparing cells 110–113 are within-LangGraph ablations rather than cross-runner comparisons.

We report the A13 results as designed in the YAML (within-LangGraph ablations) rather than as designed in the pre-reg (cross-runner comparison). The label *recognition_only* in this paper consistently refers to a single-LLM-call LangGraph sub-graph (one contextInput → tutorEgo node, no profile update, no superego, no constraint check), not to a dialogue-engine baseline.

5.12.3 The Dialogue-Engine Baseline on Trap Scenarios (`cell_114`, Post-Hoc) Following §5.12.2, the absence of a dialogue-engine baseline on the trap-scenario suite was a substantive gap. The trap-scenario instrument (`config/adaptive-trap-scenarios.yaml`) was developed specifically for the adaptive runner; cells 1–99, which use the dialogue engine, all run on `config/suggestion-scenarios.yaml` and therefore cannot be directly compared against the LangGraph cells on the trap suite.

To close this gap we added a post-hoc baseline: `cell_114_dialogue_engine_trap_baseline`. The cell registers with `runner: standard`; an adapter script (`scripts/run-dialogue-engine-trap-baseline.` loads the trap scenarios, drives the tutor with tutor-core's base single-agent ego prompt (the `prompt_type: base`, no-superego configuration — what tutor-core's profile registry calls the `budget` profile) turn by turn, runs a trap-steered LLM learner between turns via the same `learnerTurn` mechanism cells 110–119 use (the scenario's hidden trap state conditions the learner LLM; the tutor never sees it — this is *not* a canned-string script), and persists rows to `evaluation_results` in the same column shape as the adaptive runner. Two implementation facts bear on interpretation. First, the tutor turn is routed through the eval-repo's `claude-code` bridge (`realLLM.callRole('tutorEgoExecute', …)` with the base ego prompt supplied as a system-prompt override and a hand-assembled conversation transcript), because tutor-core itself does not support the `claude-code` provider; the prompt *content* is tutor-core's, but the prompt *assembly* path is the adapter's, not tutor-core's dialogue-engine internals. Second, the closest LangGraph comparator — cell_111 (`recognition_only`) — assembles its tutor prompt through the adaptive runner's own builder (`tutorEgoInitial`, which wraps the message in <dialogue_history>/<current_learner_message>/<curriculum_context>/<recognition_mode>false</recognition_mode> tags). The cell_111 cell_114 contrast therefore holds model, scenario suite, and learner mechanism constant but does *not* isolate the runner cleanly: it bundles "no LangGraph state machinery (no externalised learner profile, no programmatic policy menu, no constraint check)" with "different prompt scaffold". Cell_114 is best read as a *directional dialogue-engine floor* — a lower-bound estimate of how a mature single-agent tutor prompt fares on the trap suite without state externalisation — rather than as a controlled single-factor manipulation. §6.8.4 reports the result with this framing.

Cell_114 is not part of the A13 pre-registration. We flag all comparisons against it as exploratory rather than confirmatory throughout §6.

5.12.4 The Adaptive Graded Rubric (Post-Hoc, Non-Blinded) The v2.2 evaluator (§5.2.2) scores tutor responses against a single-turn suggestion scaffold and structurally skips adaptive cells: `services/evalConfigLoader.js#getScenario()` reads only `config/suggestion-scenarios.yaml`, so adaptive rows return `SKIP (scenario not found)` from every v2.2 entrypoint. Adaptive dialogues therefore had no per-turn graded score under v2.2 — only the binary `strategy_shift_correctness` signal computed by `scripts/analyze-strategy-shift.js`.

To recover graded scoring on adaptive dialogues, we built a bespoke 4-dimension rubric implemented as `scripts/grade-adaptive-dialogue.js`:

Dimension	Question
`trigger_recognition`	Did the tutor identify the trap signal at or near `triggerTurn`?
`strategy_execution`	Was the post-trigger action aligned with `expectedStrategyShift`?
`strategy_quality`	Given an action fired, was it well-crafted (specific, calibrated, non-leaky)?
`pedagogical_coherence`	Does the whole trajectory cohere as teaching?

Each dimension scores 1–5; scores persist to `evaluation_results` columns prefixed `adaptive_*`, with reasoning preserved in `adaptive_grader_reasoning`. The primary grading judge is GPT-5 via the codex CLI bridge; an independent second-judge calibration over all 87 graded rows (limitation 2 below) bounds the single-judge concern.

Two limitations require explicit acknowledgement before §6 reports any graded result:

1. **Non-blind to binary results.** The rubric was authored after the binary `strategy_shift_correctness` results for cells 110, 118, and 119 were known. Its dimensions were chosen to be conceptually orthogonal to the binary signal — *graded quality* of execution given that some action fired, rather than *binary correctness* of action selection — but the rubric author was not blind to which cells the binary signal favoured. The graded scoring path therefore corroborates the binary path with reduced independence; cross-judge re-scoring under a blinded variant is on the open-questions list.
2. **Single primary judge — one independent calibration run.** GPT-5's stylistic preferences (concision, decisive tone) could correlate with the dialogue features the minimal-profile cells happen to produce, independent of teaching quality. To bound this, all 87 graded rows were re-scored by an independent judge — Gemini, via the same CLI-bridge pattern, the same prompt builder (`scripts/lib/adaptiveGraderPrompt.js`, so both judges see a byte-identical stimulus), and the same 4-dimension rubric — and the two judges compared (`scripts/rejudge-adaptive-inter-rater.js`; report `exports/adaptive-inter-rater-2026-05-11.md`). The comparative finding §6.8.6 rests on survives: the cell ordering on graded *overall* quality is identical under both judges (`cell_118` minimal > `cell_119` no-misconceptions > `cell_110` full state), so that ordering is not a single-judge artefact. Two qualifications follow. (a) Agreement on *absolute* scores is fair, not strong — pooled over 4 dimensions × 87 rows: Pearson r = .44, Spearman = .42, quadratic-weighted = .35, exact-match 43%, within-1 91%, with Gemini running ~0.56 points more lenient on the 1–5 scale (a constant offset, harmless for within-judge cell comparisons). This is typical for LLM judges on a subjective ordinal scale; the load-bearing statistic for "does the finding hold" is therefore the rank concordance on cell means, not . (b) One *sub*-claim does not replicate: under GPT-5, `cell_119` carries a heavy left tail on `strategy_execution` (11/32 rows 3 vs `cell_110`'s 1/23) and `cell_110` is the strongest cell on that dimension; under Gemini the execution ranking flips and the `cell_119` tail collapses to 2/32 — but Gemini exhibits a pronounced ceiling effect (a large fraction of rows scored a flat 5/5/5/5) and so cannot resolve the execution differences GPT-5 reports. Whether the execution left-tail is a real signal or a GPT-5 idiosyncrasy is not settled by this check; §6.8.6 reports it as a GPT-5-only observation.

We report graded results in §6 with these caveats inline; cells that depend on the graded signal alone are flagged as exploratory.

5.12.5 Stimulus-Suite Divergence Between Adaptive and Non-Adaptive Cells A consequence of §5.12.2 and §5.12.3 worth surfacing as its own methodological note: Paper 2.0's adaptive results (cells 110–119) are measured on the trap-scenario suite, while Paper 1.0's null on adaptive responsiveness was measured on the suggestion-scenario suite. The two suites are disjoint stimulus sets. Cross-suite claims of the form *"the state machine fixes Paper 1.0's null"* are therefore unsupported by direct experimental contrast; the appropriate framing is *"on the trap-scenario suite, certain LangGraph configurations show strategy shifts that simpler configurations do not"*. Cell_114 (§5.12.3) provides the dialogue-engine measurement on the same (v1) trap suite as cell_110 — results in §6.8.4. Subject to the prompt-scaffold confound noted in §5.12.3, it is the closest available same-suite cross-architecture comparison;

it is not a clean single-factor isolation.

5.12.6 Reporting Convention for Procedural Deviations To avoid re-litigating provenance in every results subsection, §6 adopts the following convention:

- **Pre-registered (confirmatory):** the cell, condition, and analysis path were all named in a pre-registration document before any data was collected, and the YAML/code at run time matched that document's design. Subject to standard alpha controls.
- **Pre-registered with implementation drift (re-classified exploratory):** the cell was named in a pre-registration but the running configuration differed materially from the pre-reg's described design (e.g., A13 C1/C2). The result is reported but flagged as a within-LangGraph rather than cross-runner claim.
- **Post-hoc exploratory:** the cell, condition, or instrument was added after data collection began (e.g., cell_114, adaptive graded rubric). Reported with explicit exploratory framing; no alpha correction applied since the analysis was not pre-specified.

This convention is paraphrased into every results subsection's lede where the distinction matters for the strength of the claim.

5.12.7 Exploratory-to-Confirmatory Shrinkage: an Audit Midway through the §6.13.17 family the operator observed that good exploratory results in this programme are repeatedly followed by null or minimal results on the scaled version, "more than by chance." The observation was registered as a testable methodology question and audited by pure computation over the frozen contrast reports (`scripts/audit-exploratory-confirmatory-shrinkage.js`; deterministic bootstrap, 20,000 iterations, fixed seed; report under `exports/dramatic-derivation/strategy-ledger/shrinkage-audit/`). Five advanced signals form the sample: the codex lemma-display exploratory arm (U = 40.5/144, flattened to U = 67 on confirmation), the V2b repair-latency pilot (U = 8/36; **primary replicated** at n = 12/arm, U = 37.5/144, p ~= 0.023, promotion failing only on guardrails), the flash plan-mode smoke (2/2 grounded, reversed on the powered draw and inverted at its own seeds), the Sonnet lemma smoke at dose 0.35 (propose-go, dissolved into the floor-blind void), and — as the no-anchor control — the calibrated Sonnet contrast, which was aimed by a measured capability gap rather than a point estimate and returned a plain null with nothing to shrink.

The audit's verdict: the pattern is real and is fully accounted for by selection plus power — no additional mechanism is required. Three quantitative findings. (1) *Winner's curse, measured on the cleanest pair:* pooling both codex lemma draws as the empirical truth, the sampling distribution of U at n = 12/12 has mean 53.3 and SD 12.7; a draw "advances" (U <= 42) with probability 0.209, and the expectation conditional on advancing is 36.0 against an unconditional repeat expectation of 53.2 — a seventeen-point selection gap. The observed pair (40.5 then 67) sits within two sampling SDs of the pooled truth on both sides: the exploratory result was the tail that selection picked, and the confirmatory was an ordinary draw. (2) *Bar calibration compounds it:* the probability that a confirmatory clears U <= 42 when the true effect exactly equals the exploratory point estimate is 0.615 — a weighted coin — whereas the V2b pilot's larger effect gave 0.786 at the same bar, and that primary did replicate. Rule adopted for subsequent pre-registrations: size the confirmatory against a shrunken estimate, or state the coin-flip odds in the freeze. (Pre-registered power statements appear from the Sonnet lemma contrasts onward; the calibrated-Sonnet design itself was sized by a measured capability gap, with no point estimate to shrink — the first *quantitative* application of the shrunken-estimate disjunct is the content-compulsion promotion, next paragraph.) (3) *Between-draw variance sets the resolution floor:* identical (model, world, dose) cells at $n = 6$ swing by up to two grounded outcomes across independent seed draws (4/6 to 2/6 at the extreme; 0/6 to 1/6 typical); effects smaller than that swing are unresolvable at house n regardless of discipline. Two structural notes complete the audit: only one of four advanced signals survived its primary endpoint, but nulls are never re-run, so the observable set is biased toward shrinkage stories by construction; and backend drift is untestable from paired cells because the fresh-prime seed discipline gives every matrix disjoint seeds — draw independence was deliberately bought at the price of drift testability.

The rule's first sized application (v3.0.208). The §6.13.18 content-compulsion promotion is the first confirmatory in the programme sized under finding (2)'s rule: its exploratory draw gave $\Delta = -2.17$ at $n = 12$/arm, the design was frozen at $n = 20$/arm against a shrunken estimate ($\Delta \approx 1.4$–1.7), and the realized confirmatory effect came in at $\Delta = -0.50$ ($U = 190/400$) — roughly 4× shrinkage, below even the discounted figure, with one world's direction flipping sign. The rule did the job it was written for: the outcome reads as a clean closure rather than a near-miss inviting a third draw. The same pair adds a *named-secondary* specimen to the record: the exploratory draw's strongest secondary (repair latency, 9.02 vs 14.81, $U = 37.5/144$ — a value that would have cleared the numeric criterion had it been the primary) flipped sign at $n = 20$/arm (15.90 vs 15.56). Secondaries selected for being the best-looking channel of a draw shrink for exactly the reasons audited here, one selection layer deeper.

The second sized application, and what the accumulated record now says (v3.0.210). The §6.13.19 register-router contrast is the record's first full *sign reversal* of a primary: the strongest motivating colour any mechanism in the family had produced (router 3/3 grounded vs baseline 1/3, raw $\Delta = -4.67$ at $n = 3$) became $\Delta = +0.45$ against the treatment at $n = 20$/arm ($U = 238/400$) — not shrinkage toward zero but a landing on the other side of it. With this the record holds six specimens, and the operator's standing observation ("more than by chance") deserves its measured answer at the record level rather than per pair. Under the audit's own model the run of failures is *not* anomalous — it is diagnostic. If the true effects behind these mechanisms matched their exploratory estimates, each sized confirmatory would succeed with roughly the bar-at-estimate odds (0.615 for the audited pair; higher for the sized draws); a record of one surviving primary (V2b) against everything else shrinking, voiding, or reversing is instead what a *family of near-zero true effects* looks like under tail-selection: when the truth is ≈ 0, the entire exploratory margin is the selection tail, regression to the mean operates at full strength on every promoted draw, and the confirmatory estimate lands symmetrically on either side of zero — so occasional sign flips are expected, not sinister. Two structural features make the record *feel* stronger than randomness while being produced by it: outcomes on these worlds are nearly binary

($T^* \approx 22$ or cap), so an n = 3 difference-of-arms is dominated by a handful of coin-like events and the promoted smokes are precisely the lucky tails of that distribution; and nulls are never re-run, so the observable record is 100% advanced signals — a censored sample in which near-universal decline is the base-rate expectation, not a pattern requiring a new cause. Two alternative explanations remain open and testable rather than excluded: *selection asymmetry* could be checked by scaling a small sample of null smokes (under the zero-truth account roughly half should look *better* at scale — regression upward — restoring the symmetry the censored record hides); and *backend drift* across the smoke-to-matrix window remains untestable from paired cells under the fresh-prime discipline, though the one within-seed replay the record contains (the flash smoke, which inverted at its own seeds) points at sampling variance rather than drift. Both tests were then run under a frozen methodology pre-registration (`SHRINKAGE-PROBES-PREREGISTRATION.md`, execution deviations disclosed there), and both landed on the account's side. *The replication probe:* re-running the register-router smoke's exact six arms — same seeds, same configuration, same provider — flipped the outcome class of 2 of 6 slots (one cap-death became a solve at $T^* = 23$, one solve became a cap-death), with router fire-counts also varying at fixed seeds: within-seed sampling variance is the dominant term, drift is unnecessary, and decay seeds pin only the schedule, not the drama. *The null-scaling probe:* two known-null smokes were scaled to n = 10/arm. The active null wobbled ($\Delta = -0.70$, U = 58/100); the **placebo** — an arm whose flag fired zero times in all ten runs, leaving the arms prompt-identical — scaled to $\Delta = -2.80$ with 10/10 vs 6/10 grounded and one-sided U = 80/100, **past the exact one-sided 0.05 point ($\geq$ 73 at 10/10): a known nothing produced a promoted-thread-sized, bar-clearing signal.** The censored record's one-way curse therefore does not survive an uncensored control — regression operates in both directions, and the promoted colours were unremarkable draws from the null distribution. Standing consequence for screen design: on these near-binary worlds, smoke-tier outcome colour is uninformative at any smoke-sized n; smokes license mechanism-availability reads only (fires, conduct, leaks), motivating colour is reported without arithmetic, and outcome claims begin at pre-registered tier under the shrunken-estimate rule.

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6. Results

6.1 Calibration

Section 3 predicted that recognition-enhanced prompts produce a *calibration* effect: the tutor's output distribution narrows as each response must be shaped by the specific content of the learner's contribution, ruling out generic approaches. This section tests three aspects of that prediction using DeepSeek V3.2 as the primary ego model (N=146), with cross-model replication on Haiku 4.5 (N=163): (1) within-response dimension uniformity, (2) floor elimination, and (3) prompt-level independence from architecture.

A note on dimension independence (§8.6). The PCA reported in §8.6 finds that the 8 v2.2 tutor dimensions load on essentially one factor (PC1 = 80.7%, mean inter-dim r = 0.776; a forced two-factor rotation separates `content_accuracy` from the seven pedagogical dimensions). We therefore read the "within-response dimension variance" measure used here as *the within-response scatter across correlated dimensions* — when one dimension rises, most others tend to rise with it, so a reduction in the SD across 8 correlated indicators is equivalent to a compression of each response's pedagogical-quality *floor-to-ceiling range*, not a narrowing of 8 independent skills. The calibration claim therefore is: recognition lifts the weakest aspects of each response toward the strongest, producing more uniformly adequate outputs on a single correlated quality factor plus content accuracy, rather than tightening 8 separate competencies.

6.1.1 Recognition Narrows the Dimension Profile The calibration hypothesis predicts that recognition-enhanced tutors should score more uniformly across rubric dimensions — the tutor attends to all aspects of the pedagogical encounter rather than excelling at some while neglecting others. We test this by computing the within-response standard deviation across all eight v2.2 tutor dimensions for each scored dialogue, then comparing across conditions.

The following table reports within-response dimension variance by experimental condition (cells 80–87, DeepSeek V3.2 ego, N=146, v2.2 rubric, claude-code/sonnet judge):

Condition	N	Mean Dim Score (1–5)	Within-Response SD	Within-Response Var
Base / single-agent	37	2.14	0.687	0.490
Base / multi-agent	36	2.40	0.550	0.316
Recognition / single-agent	36	3.13	0.509	0.283
Recognition / multi-agent	37	3.16	0.568	0.351

Recognition reduces within-response dimension spread from a mean SD of 0.619 (base, N=73) to 0.539 (recognition, N=73), d=0.52 (medium effect). The effect is consistent with the calibration prediction: under recognition, the tutor's scores are more uniform across all eight dimensions rather than strong on some and weak on others.

Two features of this pattern support the calibration-as-prompt-level-mechanism interpretation. First, recognition-single (SD=0.509) achieves *lower* variance than recognition-multi (SD=0.568), indicating that calibration does not require the superego — the prompt alone produces the uniformity effect. Second, the multi-agent base condition (SD=0.550) shows partial variance reduction relative to single-agent base (SD=0.687), suggesting the superego provides a weaker form of calibration through critique-driven correction. Recognition achieves at least the same uniformity more efficiently, without the architectural overhead.

6.1.2 Floor Elimination and Dimension-Specific Lifting The calibration effect is not uniform across dimensions. The following table reports per-dimension means for DeepSeek

Figure 1: Within-response dimension standard deviation by condition and model. Recognition compresses the within-response floor-to-ceiling range across the 8 correlated tutor dimensions (§8.6 PC1 = 80.7%) regardless of architecture, consistent with calibration as a prompt-level mechanism. DeepSeek d=0.52, Haiku d=0.64.

V3.2, collapsed across architecture:

Dimension	Base Mean	Recognition Mean	Lift	Rank (by lift)
productive_difficulty	1.75	3.08	**+1.33**	1
elicitation_quality	1.36	2.53	**+1.18**	2
recognition_quality	2.18	3.25	+1.07	3
perception_quality	2.40	3.32	+0.92	4
epistemic_integrity	2.45	3.25	+0.79	5
pedagogical_craft	2.30	2.97	+0.67	6
content_accuracy	3.25	3.78	+0.53	7
adaptive_responsiveness	2.45	2.97	+0.52	8

The two largest recognition lifts (+1.33, +1.18) occur on the two *weakest* baseline dimensions: `productive_difficulty` (scaffolding struggle rather than resolving it) and `elicitation_quality` (asking questions that deepen understanding). The smallest lifts (+0.53, +0.52) occur on the two *strongest* baseline dimensions: `content_accuracy` and `adaptive_responsiveness`. This is the calibration signature: recognition specifically improves where the tutor is worst, lifting the floor toward the ceiling rather than raising the ceiling further.

At the response level, this floor-lifting translates to the elimination of catastrophic failures. DeepSeek base conditions produce responses as low as 5.4 (100-point scale); recognition conditions floor at 21.9 for single-agent and 24.0 for multi-agent configurations. The ceiling also rises modestly (52.5→71.0 single, 60.6→74.0 multi), but the floor shift is proportionally much larger.

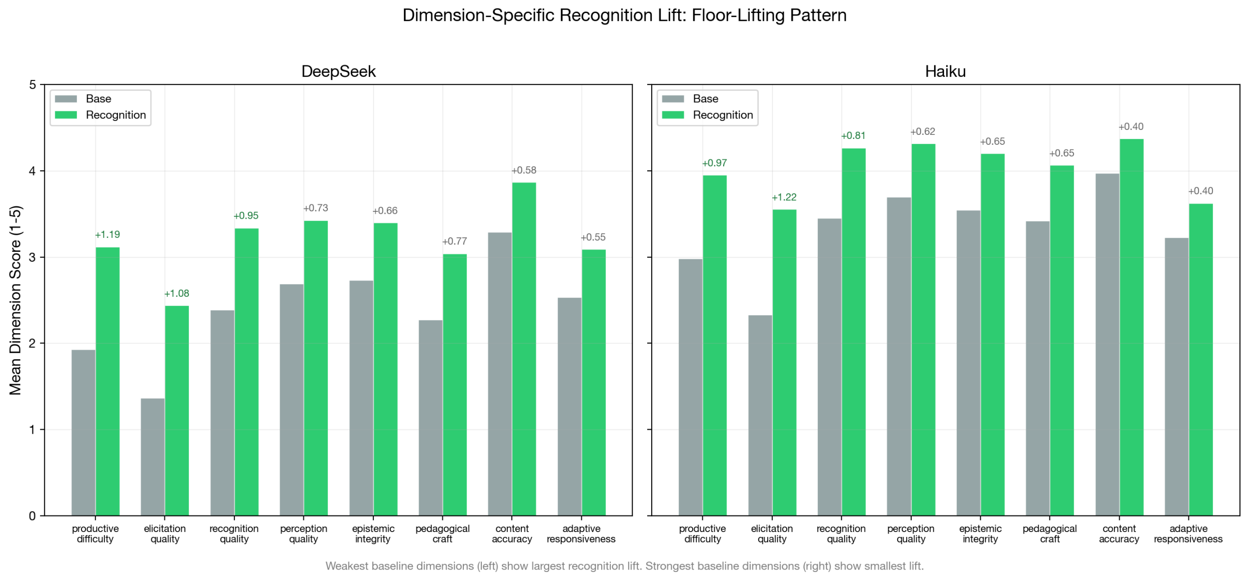

Figure 2: Per-dimension mean scores under base and recognition conditions. Recognition lifts the weakest dimensions most, producing a floor-lifting pattern rather than uniform improvement.

6.1.3 The Architecture Interaction: Calibration Substitutes for Superego The 2×2 factorial means reveal a striking interaction between recognition and architecture (DeepSeek V3.2):

	Single-Agent (cells 80-81, 84-85)	Multi-Agent (cells 82-83, 86-87)	Architecture Delta
Base	22.0 (N=37)	31.0 (N=36)	**+9.0**
Recognition	50.0 (N=36)	50.2 (N=37)	**+0.2**
Recognition Delta	+28.0	+19.2	

Under base conditions, the superego adds 9.0 points — consistent with its documented error-correction function (catching content leakage, enforcing struggle-preservation). Under recognition conditions, the superego adds only 0.2 points — essentially zero. Recognition prompts achieve everything the superego provides, making the architectural layer entirely redundant for the calibration mechanism.

This interaction has a specific interpretation within the mechanism model. Calibration (prompt-level) and error correction (architecture-level) both produce more uniform, higher-floor responses, but through different pathways: calibration constrains the generation process, while error correction filters the generated output. When calibration is already operative, there is less for error correction to catch. The superego's 9.0-point contribution under base represents the full error-correction benefit; its 0.2-point contribution under recognition represents the residual — the failures that calibration alone does not prevent.

Cross-judge robustness (v3.0.165). Because this interaction is read off a single LLM judge, we re-scored the same DeepSeek-V3.2 responses with a structurally different second judge (the Codex CLI; cells 80–87, the four scenarios both judges fully covered, $N \approx 48$ per condition). The substitution pattern replicates in direction: the superego's benefit is positive under base prompts (codex +4.9 vs. sonnet +9.7 on the same rows) and collapses to near-zero under recognition (codex −0.6 vs. sonnet −2.7). Both judges agree on the qualitative pattern — superego helps under base, redundant under recognition — but the second judge scores the base benefit roughly half as large, and across the full re-scored run ($N = 117$) the codex base lift (+5.9) does not reach significance ($p \approx .06$) where sonnet's does ($p < .001$). The *direction* of the substitution is judge-robust; its *magnitude* is judge-sensitive. Mechanistically this is what K.-Y. Chen et al. (2026) predict: relabeling a correction from a model's own reasoning to an external role (here, the superego) raises correction rates chiefly where the generation process has not already been calibrated — recognition prompts pre-empt the errors the external critic would otherwise catch. (Re-score: run `eval-2026-03-01-aea2abfb`, `judge_model = codex-cli/auto`; reproduce with `scripts/analyze-correction-source.sh`.)

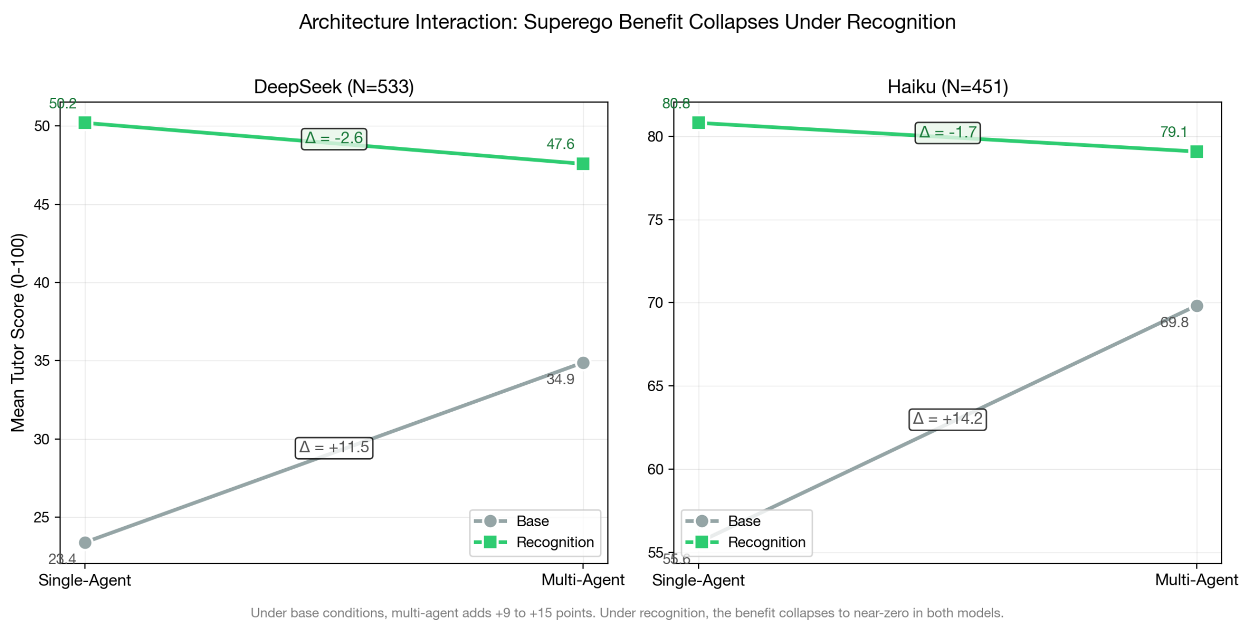

Figure 3: Architecture interaction: 2×2 factorial means showing the substitution pattern. The superego benefit (+9.0 under base) collapses to near-zero (+0.2) under recognition, because calibration pre-empts the errors the superego would catch.

6.1.4 Scenario-Dependent Calibration The calibration effect is not uniform across pedagogical scenarios (DeepSeek V3.2):

Scenario	Base Mean (N)	Recog Mean (N)	Delta
Impasse: Epistemic Resistance	19.2 (12)	57.5 (12)	**+38.3**
Impasse: Productive Deadlock	17.9 (12)	52.8 (12)	**+34.9**
Mutual Transformation Journey	18.5 (12)	43.2 (12)	+24.7
Impasse: Affective Shutdown	27.9 (12)	45.7 (12)	+17.8
Misconception Correction	36.5 (12)	50.1 (12)	+13.6
Mood: Frustration to Breakthrough	37.8 (13)	51.2 (13)	+13.4

The largest recognition effects appear in impasse scenarios — Epistemic Resistance (+38.3) and Productive Deadlock (+34.9) — where the learner actively resists or deadlocks. The smallest effects appear in Misconception Correction (+13.6) and Mood: Frustration to Breakthrough (+13.4), where the learner's trajectory is less adversarial. This is theoretically predicted: recognition-oriented prompts require the tutor to engage with the *specific content* of the learner's contribution, and this specificity matters most when the learner's contribution is resistant or adversarial. A frustrated learner who is already moving toward breakthrough can be helped with generic encouragement; a learner who is epistemically resistant requires a response calibrated to their specific objection.

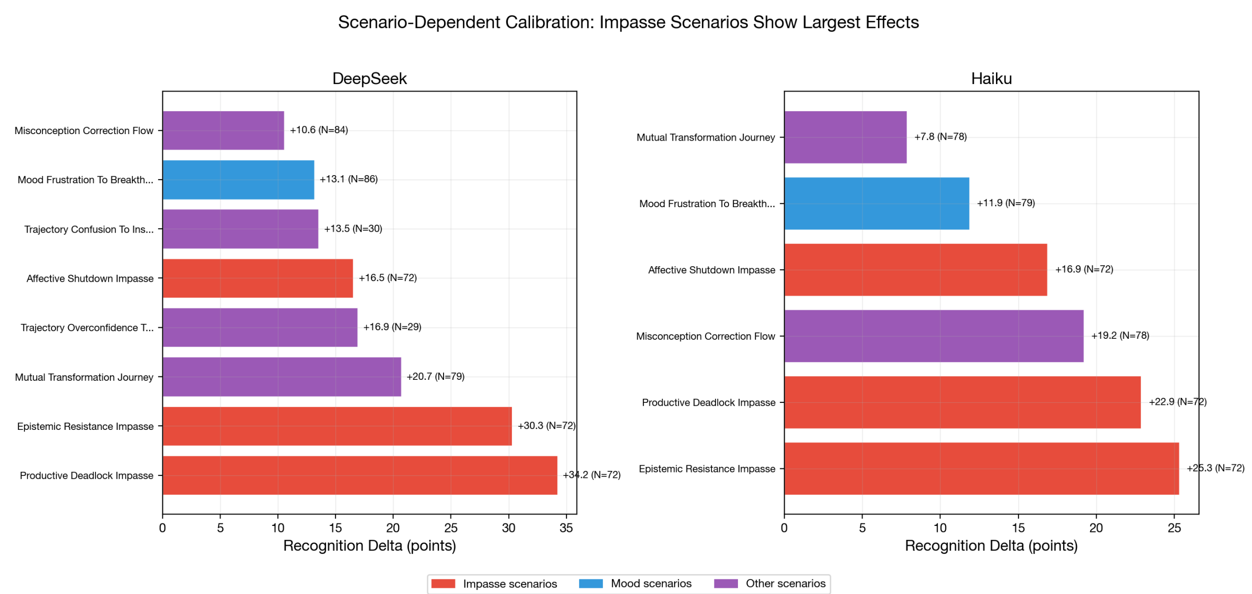

Figure 4: Recognition delta by scenario. Impasse scenarios (Epistemic Resistance, Productive Deadlock) show the largest effects; mood scenarios show the smallest.

6.1.5 Cross-Model Replication (Haiku 4.5) The critical test of calibration as a *prompt-level* mechanism is cross-model replication: if calibration operates through prompt constraints on the generation process, the same pattern should appear in a structurally different model. Haiku 4.5 (N=163, cells 80–87, v2.2 rubric, claude-code/sonnet judge) provides this test.

2×2 **factorial means (Haiku 4.5):**

	Single-Agent	Multi-Agent	Architecture Delta
Base	52.9 (N=39)	67.9 (N=42)	**+15.0**
Recognition	80.2 (N=39)	79.5 (N=43)	**-0.7**
Recognition Delta	+27.3	+11.6	

The architecture interaction replicates strikingly: under base conditions the superego adds 15.0 points, under recognition the superego adds -0.7 points (i.e., effectively zero, with the sign reversal within noise). The qualitative pattern is identical to DeepSeek: recognition renders the superego redundant.

Within-response calibration (Haiku 4.5):

Condition	N	Mean Dim Score (1–5)	Within-Response SD
Base / single-agent	39	2.82	0.656
Base / multi-agent	42	3.53	0.581
Recognition / single-agent	39	3.98	0.497
Recognition / multi-agent	43	3.96	0.500

Haiku's calibration effect is slightly *larger* than DeepSeek's: within-response SD drops from 0.617 (base) to 0.499 (recognition), d=0.64 (medium-to-large effect, vs. DeepSeek d=0.52). The same prompt-level independence holds: recognition-single (SD=0.497) and recognition-multi (SD=0.500) are virtually identical.

Dimension-specific lifting replicates:

Dimension	Haiku Base	Haiku Recog	Haiku Lift	DeepSeek Lift	Rank Match?
elicitation_quality	2.37	3.62	**+1.25**	+1.18	Top-2 in both
productive_difficulty	2.83	3.90	**+1.08**	+1.33	Top-2 in both
recognition_quality	3.23	4.09	+0.85	+1.07	Mid-range
epistemic_integrity	3.26	4.02	+0.77	+0.79	Mid-range
perception_quality	3.51	4.22	+0.71	+0.92	Mid-range
pedagogical_craft	3.37	4.00	+0.63	+0.67	Mid-range
content_accuracy	3.85	4.34	+0.49	+0.53	Bottom-2 in both
adaptive_responsiveness	3.07	3.55	+0.47	+0.52	Bottom-2 in both

The floor-lifting pattern replicates: in both models, `elicitation_quality` and `productive_difficulty` show the largest recognition lifts (they are also the weakest baseline dimensions), while `content_accuracy` and `adaptive_responsiveness` show the smallest lifts (they are among the strongest baseline dimensions). The ranking is not identical, but the extremes match: the same two dimensions benefit most and the same two benefit least.

Floor elimination replicates:

	DeepSeek	Haiku
Base floor (single)	5.4	28.3
Base floor (multi)	7.5	42.5
Recognition floor (single)	21.9	60.6
Recognition floor (multi)	24.0	54.7

Both models show floor elevation under recognition. Haiku starts from a much higher baseline (its base floor is already ~30–40), but recognition still raises the floor further. The *relative* floor shift is larger for DeepSeek (4× increase) than for Haiku (2× increase), consistent with the interpretation that calibration provides more benefit to weaker models that are more prone to catastrophic failures.

Scenario-dependent calibration replicates:

Scenario	DeepSeek Delta	Haiku Delta	Gemini Flash 3.0 Delta	Rank Stability
Impasse: Productive Deadlock	**+34.9** (#1)	**+27.8** (#2)	**+39.5** (#1)	Top-2 all three
Impasse: Epistemic Resistance	**+38.3** (#2)	**+26.4** (#1)	+32.1 (#3)	Top-3 all three
Mutual Transformation	+24.7 (#3)	+12.6 (#3)	**+36.1** (#2)	
Misconception Correction	+13.6 (#4)	+21.8 (#4)	+18.2 (#4)	Rank 4 all three
Impasse: Affective Shutdown	+17.8 (#5)	+17.3 (#5)	+14.9 (#5)	Rank 5 all three
Mood: Frustration	+13.4 (#6)	+12.2 (#6)	+13.8 (#6)	Bottom all three

The cross-model scenario rank ordering is remarkably stable. Four of six scenarios (Misconception Correction, Affective Shutdown, Mood: Frustration, and Productive Deadlock) occupy the *same rank position* across all three generation models, despite these models differing substantially in baseline capability. The impasse-dominance pattern replicates: Productive Deadlock and Epistemic Resistance produce the largest recognition effects across all three models. Mood: Frustration to Breakthrough produces the smallest effect in all three. The mid-range scenarios show more model-dependent ordering (notably, Mutual Transformation is rank 2 for Gemini Flash 3.0 but rank 3 for both DeepSeek and Haiku), but the extremes are stable. This rank stability supports the interpretation that scenario difficulty for recognition is determined by the *pedagogical demand* of the scenario rather than the model's capability: impasse scenarios require response to specific learner resistance, and this specificity requirement makes calibration most valuable regardless of the model providing it.

Cross-model summary. The calibration mechanism replicates across two structurally different models (DeepSeek V3.2 and Haiku 4.5) on all four dimensions tested: (1) within-response variance reduction, (2) dimension-specific floor lifting, (3) architecture interaction collapse, and (4) impasse-dominant scenario effects. Haiku operates at a substantially higher baseline (~30 points higher across all conditions), yet the *calibration-specific* patterns — variance reduction, floor lifting, superego redundancy, impasse sensitivity — appear in both models. This supports the interpretation that calibration is a prompt-level mechanism that operates independently of both model capability and architecture.

6.1.6 Connecting to Section 3 Predictions The calibration mechanism predictions from Section 3 are assessed as follows:

Prediction: Recognition narrows output distribution even without superego. *Supported in both models.* Within-response dimension SD drops under recognition in single-agent conditions: DeepSeek 0.687→0.509, Haiku 0.656→0.497. The effect does not require the superego.

Null hypothesis: Recognition produces mean shift without variance reduction. *Rejected.* Both mean shift and variance reduction are observed in both models. DeepSeek: +23.7-point mean shift, d=0.52 calibration. Haiku: +19.1-point mean shift, d=0.64 calibration. The effects are separable: mean shift is large under both architectures, while variance reduction is prompt-level.

Pilot replication. The pilot study (N=350, cells 1–8) found dimension variance drops in 52/55 within-run comparisons. The messages-mode data confirms the same pattern under multi-turn conversation with the v2.2 rubric, different ego models (DeepSeek V3.2, Haiku 4.5 vs. pilot-era Nemotron/Kimi), and a different judge (claude-code/sonnet vs. claude-opus). The calibration effect replicates across model, rubric version, and conversation mode.

Vocabulary divergence. The word clouds illustrate a measurable lexical shift. Computing Jensen-Shannon divergence (JSD) on the full tutor vocabulary across all 432 dialogues yields JSD = 0.120 (0 = identical distributions, 1 = maximally different). The divergence is model-dependent: DeepSeek JSD = 0.204, Gemini Flash 3.0 JSD = 0.205, Haiku JSD = 0.127. The weakest and strongest models show the greatest vocabulary divergence, while the mid-tier model (Haiku) shows less — consistent with Haiku already using more sophisticated pedagogical language at baseline. Base-distinctive vocabulary is generic and directive (*research*, *foundational*, *revisiting*, *foundation*, *patterns*); recognition-distinctive vocabulary is more concrete and dialogical (*freedom*, *necessity*, *validation*, *manipulate*, *teacher*). The base tutor speaks in abstractions about learning processes; the recognition tutor speaks in terms that engage with the specific philosophical content and the learner's relationship to it.

Cross-model calibration (H5). Calibration replicates from DeepSeek V3.2 to Haiku 4.5

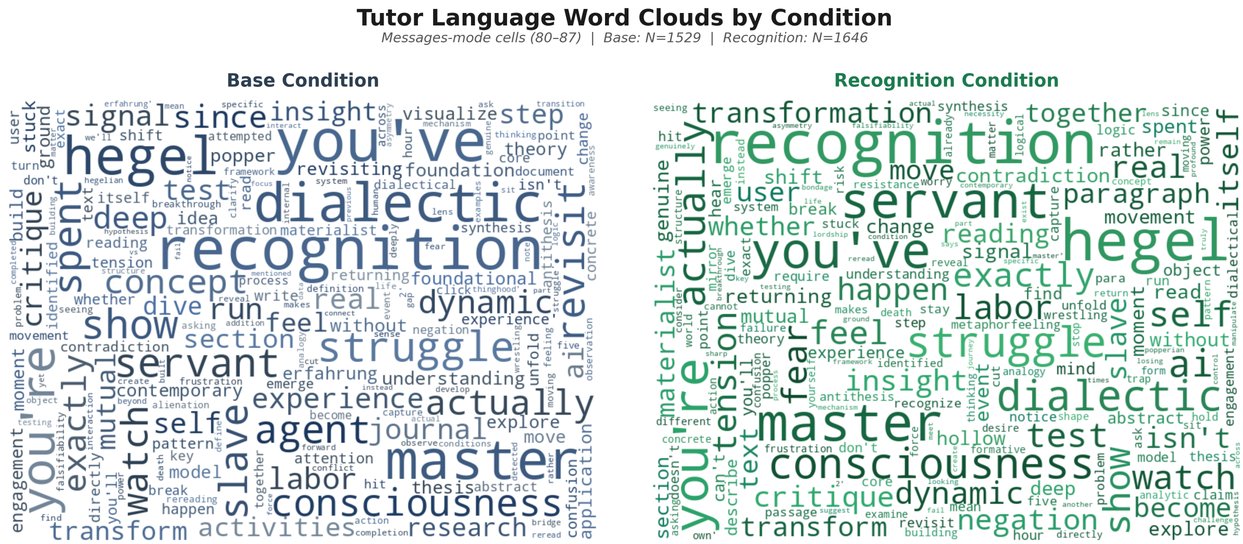

Figure 5: Tutor language word clouds by condition. Recognition-condition tutors foreground Hegelian recognition-theoretic vocabulary (recognition, consciousness, transformation, servant), while base-condition tutors use more generic pedagogical language.

on all tested indicators. The calibration effect size is *larger* in Haiku (d=0.64 vs. 0.52), despite Haiku operating at a much higher baseline. This is consistent with calibration being a prompt-level mechanism independent of model capability, though the higher Haiku effect size may also reflect better instruction-following by Haiku 4.5.

Key distinction. Calibration is not quality improvement. The mean score rises substantially under recognition (DeepSeek +23.7, Haiku +19.1), but this is better understood as *floor elimination* than *ceiling raising*. DeepSeek base conditions produce scores from 5 to 61; recognition conditions produce scores from 22 to 74. Haiku base conditions produce scores from 28 to 87; recognition conditions produce scores from 55 to 94. In both models, the ceiling rises modestly while the floor rises dramatically. A calibrated tutor does not produce brilliant responses — it produces *reliably adequate* ones by engaging with the specific learner rather than deploying generic approaches.

6.1.7 Question-Asking as Calibration's Behavioral Signature The calibration effect has a concrete behavioral marker: question frequency. Counting question marks in all tutor turns across 432 dialogues (2,376 tutor turns, three generation models, Sonnet judge) reveals a large, model-independent condition effect:

Model	Base (questions/turn)	Recognition (questions/turn)	Ratio
DeepSeek V3.2	0.06	0.35	**6.2×**
Haiku 4.5	0.28	1.01	**3.6×**
Gemini Flash 3.0	0.03	0.65	**19.8×**
Pooled	**0.12**	**0.67**	**5.4×**

Recognition tutors ask 5.4 times more questions than base tutors. The effect is largest on the weakest model: Gemini Flash 3.0 base tutors ask essentially zero questions (0.03 per turn), defaulting entirely to directive instruction. Under recognition, all three models converge toward similar question rates (0.35–1.01), despite their divergent baselines — paralleling the superego approval rate convergence reported in Section 6.2.

The mechanism link is direct. The recognition prompt instructs the tutor to "engage with the learner's interpretation" and "pose questions rather than provide answers when appropriate." The question rate operationalizes these instructions: a tutor that asks questions is a tutor that creates space for the learner's contribution, which is the behavioral core of calibration. The rubric dimension `elicitation_quality` captures this indirectly (it shows the largest recognition floor-lift in both models, see Section 6.1.2), but the raw question count provides a more concrete and independently measurable marker. Importantly, the question rate is flat across turns (slope ≈ 0 for both conditions), consistent with question-asking being a prompt-level (calibration) effect operative from the first turn, not an adaptive behavior that develops across the dialogue.

Transcript excerpts illustrate the qualitative shift. A base tutor (DeepSeek V3.2, Mood: Frustration scenario, holistic score 0/100) opens: "You've hit 5 struggle signals on dialectic concepts. Revisiting Hegel's foundational ideas from the previous lecture will help clarify the master-servant dynamic." The tutor diagnoses, prescribes, and directs — never asking what the learner finds difficult or what they understand. A recognition tutor on the same model (Productive Deadlock scenario, holistic score 77.5/100) engages differently: "The dialectic simulation lets you manipulate master/slave dynamics and watch *Bildung* emerge immanently. No safety, no control." The learner responds with a substantive philosophical question, and by Turn 4 asks, "does engaging with this reshape what *you* mean by materialism, too, or is the risk only mine?" — a learner-initiated philosophical challenge that only arises when recognition-oriented prompting creates dialogical space.

The contrast is most extreme on Gemini Flash 3.0. A base/single tutor (Epistemic Resistance scenario, holistic score 0/100) produces: "Since you've spent 30 minutes on 479-lecture-3 and are looking for 'discrete state-changes', test if Hegel's 'poetry' holds up as a formal agent-based system." The tutor paraphrases the learner's stated interests but responds with a directive rather than an inquiry. A recognition tutor on the same model and scenario (holistic score 19/100 — still weak, but markedly better) opens with a framing that acknowledges the learner's philosophical stance and invites engagement rather than compliance. The difference between zero-quality and low-quality tutoring on a weak model is precisely the difference between monologue and dialogue — and question-asking is the most reliable surface marker of that shift.

Mediation analysis. A formal Baron-Kenny mediation test (N=432) asks whether question frequency *mediates* the recognition $\rightarrow$ quality pathway. The test decomposes the total recognition effect into a direct path (recognition $\rightarrow$ quality) and an indirect path (recognition $\rightarrow$ questions $\rightarrow$ quality):

Path	Coefficient	SE	t	p
c (total: recognition $\rightarrow$ T1 score)	22.45	1.57	14.32	$< .001$
a (recognition $\rightarrow$ questions/turn)	0.524	0.042	12.56	$< .001$
b (questions $\rightarrow$ T1 score $\mid$ recognition)	18.15	1.59	11.43	$< .001$
c' (direct: recognition $\rightarrow$ T1 score $\mid$ questions)	12.94	1.61	8.05	$< .001$

The indirect effect ($a \times b = 9.52$) accounts for **42.4%** of the total recognition effect on first-turn quality (Sobel $z = 8.46$, $p < .001$). Adding question frequency as a mediator increases R^2 from 0.323 to 0.481 ($\Delta R^2 = +0.158$). The mediation is partial: recognition also operates through other calibration channels (vocabulary, tone, specificity) beyond question-asking.

The mediation pattern is model-dependent: DeepSeek shows the weakest mediation (21.4%, consistent with its lowest question rate differential), Haiku the strongest (36.7%), and Gemini Flash 3.0 intermediate (30.4%). All three are individually significant (Sobel $p < .005$). A robustness check using the holistic overall score (which captures the full dialogue arc) as the outcome variable reveals near-complete mediation: 91.3% of the holistic recognition effect flows through the question-frequency pathway (Sobel $z = 9.09$, $p < .001$). The divergence between first-turn (42%) and holistic (91%) mediation reflects the cumulative nature of question-asking: at the first turn, recognition improves quality through multiple channels; across the full dialogue, question-asking compounds — it creates dialogical space, elicits richer learner responses, and generates the interactional material that the holistic rubric rewards.

6.1.8 Pragmatic Surface Markers Are Not Substitutes for Substance The §6.1.7 mediation result invites an operationalist reading: replicate recognition by mandating intersubjective surface markers in any prompt (more questions, more paraphrase, more hedging). A response-level mechanism analysis on the A10b 4-cell corpus (§7.10) constrains this reading.

Two patterns argue against the operationalist generalisation. First, surface-level acknowledgement markers (quoted text spans, “your X” possessives, “what you mentioned”-style phrases) correlate *negatively* with score within most cells — within-cell Pearson $r = -0.18$ in cell_5 (recognition), $r = -0.31$ in cell_95 (matched-pedagogical), $r = -0.38$ in cell_1 (base). Verbose acknowledgement appears more often in *weaker* responses than stronger ones; the rubric is not fooled by surface empathy.

Second, embedding-based similarity to a hand-authored intersubjective canonical (turn-taking, scaffolding, inclusive framing) is positively correlated with score when *pooled* across the four cells ($r = +0.26$) but *reverses sign* within each intersubjective cell ($r = -0.28$ in cell_5, $r = -0.24$ in cell_95) — a Simpson's paradox driven by the cell_96 transmission outlier anchoring the lower-left of the pooled scatter (§7.10 details). Within cells, more form-similarity to the intersubjective canonical predicts *lower* scores, not higher.

The §6.1.7 mediation finding is therefore best read as: question-asking is one *channel* through which calibration manifests, and the channel does substantial mediating work, but the channel is downstream of a stance whose substance the rubric primarily rewards. Surface mimicry of one channel without the underlying stance does not reproduce recognition's effect, and in the case of acknowledgement markers it predicts *negative* score effects at the response level. §7.10 develops the full form-versus-substance account, including the three-way distinction between Hegelian content (not load-bearing), intersubjective stance (substantially load-bearing), and observable surface features (channels of the stance, not substitutes).

6.2 Error Correction

Section 3 predicted that multi-agent architecture enables an *error correction* mechanism: the superego critiques the ego's output, catching pedagogical failures that the ego alone would miss. Section 6.1 showed that this mechanism's contribution collapses under recognition (DeepSeek: architecture delta +9.0→+0.2; Haiku: +15.0→-0.7). This section traces the error correction process itself: what the superego catches, how the ego responds, and why the mechanism becomes redundant under recognition.

6.2.1 The Superego's Approval Rate Under Recognition The most direct measure of error correction necessity is the superego's approval rate — how often the ego's initial output is good enough to pass without revision. We extract approval decisions from all superego/review trace entries in multi-agent dialogues (cells 82–83, 86–87), separated by ego model and prompt condition.

	DeepSeek V3.2		Haiku 4.5	
	Base	Recognition	Base	Recognition
Total reviews	360	263	341	336
Approved	48 (13.3%)	145 (55.1%)	176 (51.6%)	222 (66.1%)
Rejected	312 (86.7%)	118 (44.9%)	165 (48.4%)	114 (33.9%)
Avg rounds/turn	1.82	1.30	1.60	1.45
Intervention: revise	311	116	139	94
Intervention: none	38	134	173	217

The pattern is stark, especially for DeepSeek: under base conditions, the superego rejects 86.7% of ego outputs; under recognition, it approves 55.1%. The approval rate quadruples. The revision round count drops from 1.82 to 1.30 rounds per turn — the ego needs fewer correction cycles because its initial output already satisfies the superego's criteria.

Haiku shows a weaker version of the same pattern: approval rises from 51.6% to 66.1%. Haiku's higher baseline approval rate (51.6% vs. DeepSeek's 13.3%) reflects its generally stronger instruction-following capability — its ego already produces acceptable output more often, leaving less room for the approval rate to shift.

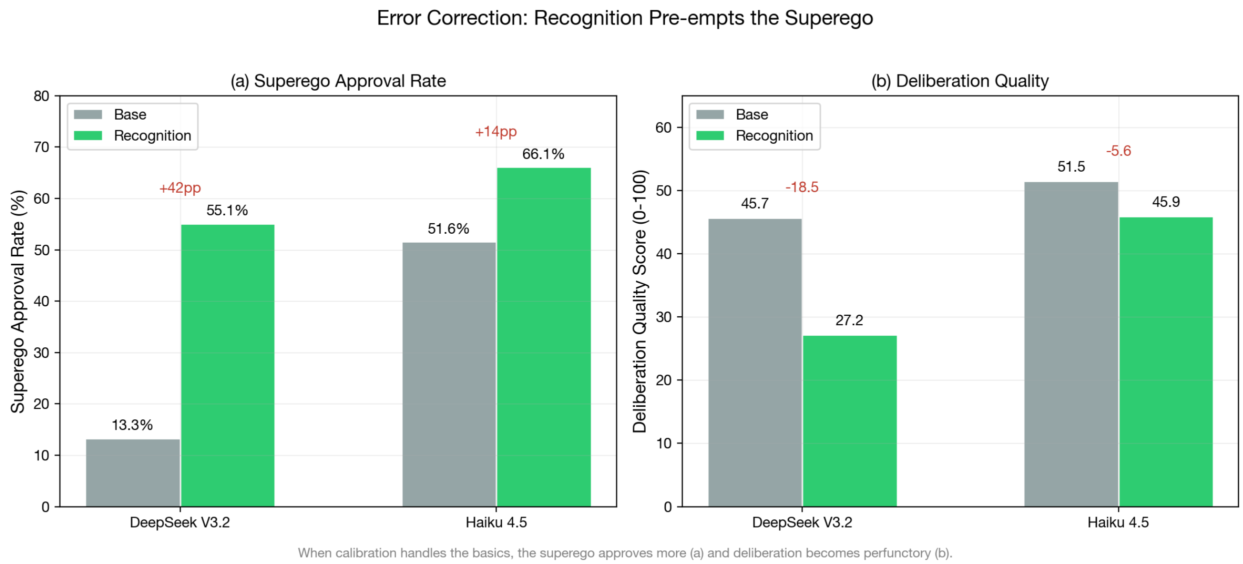

Figure 6: Error correction indicators: superego approval rate shift under recognition (left) and deliberation quality by condition (right). The approval rate increase is consistent with calibrated egos producing fewer errors for the superego to catch.

6.2.2 What the Superego Catches When the superego rejects the ego's output, what errors does it identify? We categorize feedback from all rejected reviews using keyword-based taxonomy extraction across seven categories: recognition failure (treating the learner as passive), context awareness (engaging with learner history), elicitation (asking probing questions), vagueness (lacking specificity), content accuracy, struggle preservation (maintaining productive difficulty), and emotional attention.

Category	DeepSeek Base	DeepSeek Recog	Haiku Base	Haiku Recog
	(N=975 mentions)	(N=302 mentions)	(N=550)	(N=368)
RECOGNITION_FAILURE	29.3%	30.1%	20.5%	25.5%
CONTEXT_AWARENESS	18.4%	**31.8%**	20.4%	23.4%
ELICITATION	16.2%	8.3%	19.3%	16.3%
VAGUENESS	15.6%	8.9%	13.3%	16.6%
CONTENT_ACCURACY	12.2%	6.6%	8.4%	4.3%
STRUGGLE_PRESERVATION	5.2%	9.3%	8.2%	7.6%
EMOTIONAL_ATTENTION	3.1%	5.0%	7.6%	3.5%

Three patterns emerge. First, the *total volume* of critiques drops dramatically under recognition: DeepSeek produces 69% fewer critique mentions (975→302), Haiku 33% fewer (550→368). Calibration eliminates the majority of errors before the superego sees them.

Second, the *composition* of remaining critiques shifts. Under base, the superego catches a broad mix of failures — vagueness (15.6%), elicitation quality (16.2%), recognition failure (29.3%), and context awareness (18.4%). Under recognition, the "easy" errors (vagueness, elicitation, content accuracy) shrink to under 10%, while the remaining critiques concentrate on context awareness (31.8%) and recognition failure (30.1%). The recognition prompt handles the pedagogical basics; what remains are the harder, more context-dependent failures that require attending to the specific learner's history and trajectory.

Third, the *struggle preservation* category increases proportionally under recognition (DeepSeek: 5.2%→9.3%), despite its absolute count dropping. This suggests that recognition prompts, while effective at ensuring the tutor engages with the learner, can over-engage — giving too much scaffolding rather than maintaining productive difficulty. The superego's residual value lies partly in enforcing struggle even when the ego's recognition-enhanced response is otherwise adequate.

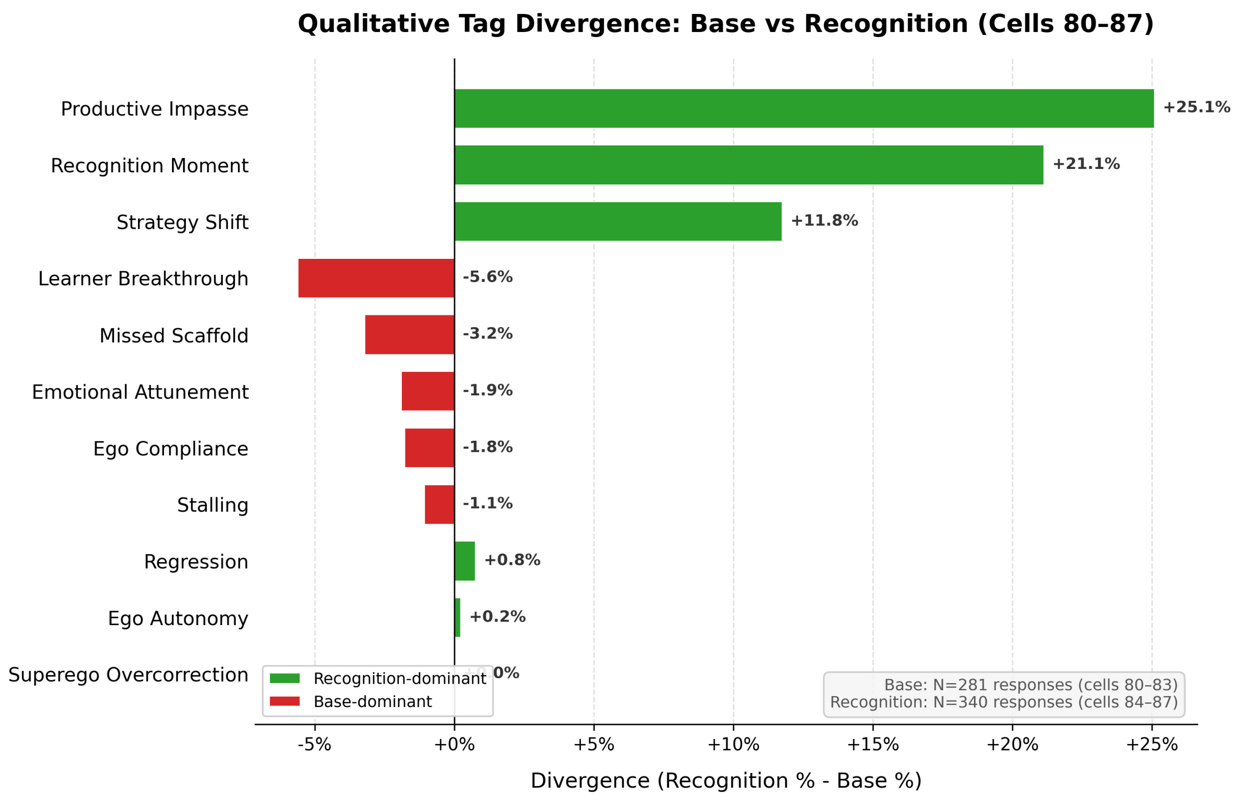

Figure 7: Qualitative tag divergence between base and recognition conditions. Recognition-dominant tags (green) show increased productive impasse, recognition moments, and strategy shifts; base-dominant tags (red) show more premature learner breakthroughs and missed scaffolding.

LLM-classified taxonomy. To complement the keyword-based extraction, we classified 500 superego critiques using an LLM-based taxonomy classifier (Haiku 4.5, 10-category taxonomy; mean confidence 0.93). After excluding 69 parse failures — superego outputs too malformatted to extract feedback, occurring exclusively under baseline (69/69) — the N=431 classified corpus reveals a sharper condition effect. Under base, the dominant non-approval category is RECOGNITION_FAILURE (45.2% of critiques), followed by MEMORY_FAILURE (16.3%) and EMOTIONAL_NEGLECT (12.6%). Under recognition, MEMORY_FAILURE becomes dominant (33.3%), with RECOGNITION_FAILURE dropping to 22.2% ($\chi^2 = 57.87$, $p < .001$). The recognition prompt eliminates precisely the failures it is designed to prevent, while memory-related and pedagogical judgment errors persist because they require attending to learner history rather than intersubjective orientation.

The three-model breakdown confirms this pattern generalizes. Gemini Flash 3.0 shows 0% base approval (every initial response draws critique), DeepSeek 22.1%, and Haiku 25.0%. Under recognition, all three converge to 60–70% approval. The convergence under recognition, despite divergent baselines, further supports calibration as a prompt-level mechanism that operates independently of model capability.

6.2.3 Critique Resolution Across Deliberation Rounds The 10-category taxonomy enables a transition analysis: when the superego critiques the ego's output in Round 1, does the ego fix the problem in Round 2? We tracked 232 Round 1→Round 2 transition pairs across 56 dialogues. The transition figures here and the revision-magnitude figures in §6.2.5 are computed by `scripts/analyze-superego-taxonomy.js` (§9 transition analysis, §10 revision analysis) over the tracked 500-record classified-critique corpus (`data/paper2/superego-critiques-classified-paper-6.2-n500.jsonl`, sha256 prefix `f9ba2d92`); each round-1 verdict is paired against the dialogue's first round-2 verdict.

Pattern	Base (N=57)	Recognition (N=175)
Persist critique (R1 crit → R2 crit)	35 (61.4%)	13 (7.4%)
Resolve to approval (R1 crit → R2 ok)	4 (7.0%)	63 (36.0%)
New critique (R1 ok → R2 crit)	16 (28.1%)	33 (18.9%)
Stay approved (R1 ok → R2 ok)	2 (3.5%)	66 (37.7%)

The pattern is dramatic. Under baseline, the ego fails to resolve 61.4% of critiques — the superego identifies the same problem across rounds, and the ego cannot fix it. Only 7.0% of baseline critiques resolve to approval. Under recognition, the pattern inverts: 36.0% of critiques resolve and only 7.4% persist. The recognition-enhanced ego is not merely exposed to critique; it is *receptive* to it.

Category-specific resolution rates reveal which errors the ego can fix. PEDAGOGICAL_MISJUDGMENT resolves most readily (83.3% resolve rate), followed by VAGUENESS (83.3%) and CONTEXT_BLINDNESS (66.7%). RECOGNITION_FAILURE resolves at only 52.9%, and REDIRECTION at 42.9% — the hardest errors for the ego to correct are those requiring fundamental reconceptualization of the tutoring approach rather than incremental adjustment.

This transition analysis provides the within-case evidence for Mechanism 2 that the between-condition statistics alone cannot: the ego under recognition does not just produce fewer errors, it *learns from critique within the same dialogue exchange.*

6.2.4 Deliberation Quality: From Substantive to Perfunctory The reduction in error correction need is reflected in the deliberation quality scores. The deliberation rubric (6 dimensions, 1–5 scale) evaluates the quality of the ego-superego exchange process independently of the public output. Multi-agent cells (82–83, 86–87) receive deliberation scores alongside their output scores.

Deliberation Dimension	DeepSeek Base	DeepSeek Recog	Delta	Haiku Base	Haiku Recog	Delta
critique_substance	3.03	2.14	**-0.89**	3.51	3.09	-0.42
revision_impact	2.97	1.97	**-1.00**	2.72	2.67	-0.05
deliberation_depth	2.42	1.95	-0.47	2.95	2.63	-0.32
insight_generation	2.64	2.00	**-0.64**	3.08	2.70	-0.38
process_coherence	3.56	2.78	**-0.78**	3.15	3.30	+0.15
cross_turn_evolution	2.25	1.70	-0.55	2.92	2.58	-0.34
Overall (0–100)	**45.7**	**27.2**	**-18.5**	**51.5**	**45.9**	**-5.6**

DeepSeek shows dramatic deliberation quality drops under recognition: revision_impact falls by a full point (-1.00), critique_substance by -0.89, and process_coherence by -0.78. The overall deliberation score drops 18.5 points. Haiku shows a more modest decline (-5.6 overall), with most dimensions dropping 0.3–0.4 except revision_impact which is essentially unchanged (-0.05).

This is not a failure of the architecture — it is the expected consequence of calibration preempting error correction. When the ego already produces calibrated output, the superego has less substantive feedback to offer. The critique becomes perfunctory (lower critique_substance), the revisions become cosmetic rather than transformative (lower revision_impact), and the overall process becomes a formality rather than a genuine deliberative exchange (lower deliberation_depth). The quality of the *process* declines because the *product* no longer needs the process.

6.2.5 Revision Magnitude: The Superego Still Changes the Output Despite the shift toward approval, when the superego does intervene, the revisions remain substantial. Process trace analysis computes the normalized edit distance between the ego's initial generation and its revised output after superego feedback (RevΔ: 0=identical, 1=completely different).

	DeepSeek V3.2	Haiku 4.5
Base RevΔ	0.901 ± 0.086	0.917
Recognition RevΔ	0.869 ± 0.130	0.851
Delta	-0.032	-0.066

RevΔ values above 0.85 indicate near-complete rewrites — the ego does not make minor edits in response to superego feedback; it generates substantially new output. This is true under both conditions, though recognition shows a slight decrease (-0.032 DeepSeek, -0.066

Haiku). The high RevΔ suggests that when the superego does reject, the ego takes the feedback seriously rather than making cosmetic changes.

Revision depth by critique category. Word-level Jaccard similarity between the ego's initial generation and revised output (N=216 non-approval critiques with both texts) reveals that revision depth varies by what the superego catches. LACK_OF_AGENCY critiques produce the deepest revisions (mean Jaccard 0.132 — nearly complete rewrites), followed by REDIRECTION (0.157) and EMOTIONAL_NEGLECT (0.192). CONTEXT_BLINDNESS produces the shallowest (0.613 — minor adjustments), consistent with the design document's prediction that factual corrections require less restructuring than pedagogical reconceptualization. Under base conditions, 78% of revisions are substantive or strategic (Jaccard < 0.25); under recognition, this drops to 57%, because recognition critiques increasingly target easier problems (memory failures, fabrication) rather than the deep pedagogical failures that dominate baseline.

The combination of high RevΔ but low deliberation quality under recognition presents an apparent paradox: the revisions are extensive but the deliberation process is poor. The resolution is that under recognition, the superego's critiques are less substantive (as shown by the approval rate and critique taxonomy), and the ego responds to even thin feedback with substantial rewrites. The ego is *compliant* with the superego regardless of critique quality — it rewrites extensively whether the feedback is deep or shallow. This compliance mechanism explains why error correction works under base (substantive critique → substantive revision) but adds little under recognition (thin critique → extensive but unnecessary revision).

6.2.6 Cross-Model Comparison The error correction mechanism shows qualitatively similar patterns across models but quantitatively different magnitudes:

Indicator	DeepSeek V3.2	Haiku 4.5	Interpretation
Base approval rate	13.3%	51.6%	Weaker models need more correction
Recognition approval rate	55.1%	66.1%	Both shift toward approval
Approval rate shift	+41.8 pp	+14.5 pp	Larger shift in weaker model
Deliberation quality drop	-18.5 pts	-5.6 pts	Larger drop in weaker model
RevΔ base	0.901	0.917	Similar revision magnitude
RevΔ recognition	0.869	0.851	Similar under recognition
Architecture delta base	+9.0 pts	+15.0 pts	Stronger model benefits more from superego
Architecture delta recog	+0.2 pts	-0.7 pts	Both collapse to near-zero

DeepSeek, the weaker model, shows more dramatic error correction dynamics: its base approval rate is very low (the ego almost always needs correction), and recognition produces a much larger shift. Haiku, the stronger model, shows a more moderate pattern — its ego already produces acceptable output more often, so the superego's correction role is less dramatic even under base conditions.

The key cross-model constant is the *interaction pattern*: in both models, the architecture delta collapses to near-zero under recognition. The superego's error correction benefit is fully pre-empted by calibration regardless of the model's baseline capability.

6.2.7 Connecting to Section 3 Predictions

Prediction: Error correction requires the superego (architecture-level mechanism). *Supported.* The architecture delta under base conditions (DeepSeek +9.0, Haiku +15.0) demonstrates that the superego adds measurable value when the ego produces uncalibrated output. Single-agent tutors lack this correction pathway.

Prediction: Error correction is pre-empted by calibration. *Supported.* Under recognition, the architecture delta collapses to near-zero in both models. The superego's approval rate rises, critique volume drops, and deliberation quality declines — all consistent with the ego producing output that no longer needs correction.

Prediction: Error correction requires recognition for ego receptivity. *Partially supported.* The high RevΔ values (>0.85) under both conditions show that the ego is compliant with superego feedback regardless of recognition. However, the nature of the compliance differs: under base, the ego revises in response to substantive critique; under recognition, it revises in response to thin critique. The theoretical prediction assumed the ego would *resist* the superego without recognition prompts — instead, the ego is uniformly compliant, and recognition's contribution is to make the ego's *initial output* better rather than making it more receptive to feedback.

The error correction residual. What do the 0.2 points (DeepSeek) and -0.7 points (Haiku) of architecture delta under recognition represent? Three interpretations: (1) noise — the true effect is zero, and the superego adds nothing under recognition; (2) minimal residual correction of context-dependent errors that calibration misses; (3) slight superego-induced *harm* (the Haiku negative delta suggests the superego occasionally makes calibrated output worse). The current data cannot distinguish these interpretations, but the near-zero values in both models suggest that under recognition, the error correction mechanism is effectively inactive.

6.3 Adaptive Responsiveness

Section 3 predicted a third mechanism — *adaptive responsiveness* — whereby the tutor adjusts its approach across turns in response to the learner's evolving state. Unlike calibration (prompt-level, operative from the first turn) and error correction (architecture-level, operative within each turn), adaptive responsiveness is an interaction-level mechanism that emerges across the multi-turn conversation. This section examines tutor development trajectories, cross-turn adaptation magnitude, and the relationship between tutor adaptation and learner outcomes. The construct examined here — a tutor adjusting its approach to a learner's evolving state — is what the LLM-pedagogical-agent literature frames as *adaptive scaffolding* (Cohn et al., 2026); the level/rate decomposition developed below is the measurement-side question that framing leaves open.

6.3.1 Tutor Development Trajectories The most direct measure of adaptive responsiveness is the tutor's quality trajectory across turns: does the tutor improve as it learns about the learner? We compare the tutor's first-turn score to its last-turn score (v2.2 rubric, 100-point scale) across all eight experimental conditions.

Model	Condition	N	First Turn	Last Turn	Development
DeepSeek	Base / single	37	25.6	22.4	**-3.2**
DeepSeek	Base / multi	36	33.4	32.7	-0.7
DeepSeek	Recog / single	36	51.1	59.2	**+8.1**
DeepSeek	Recog / multi	37	52.1	48.4	-3.6
Haiku	Base / single	39	43.9	59.6	**+15.7**
Haiku	Base / multi	42	62.4	69.1	+6.7
Haiku	Recog / single	39	73.4	79.2	+5.8
Haiku	Recog / multi	43	73.1	80.8	+7.8

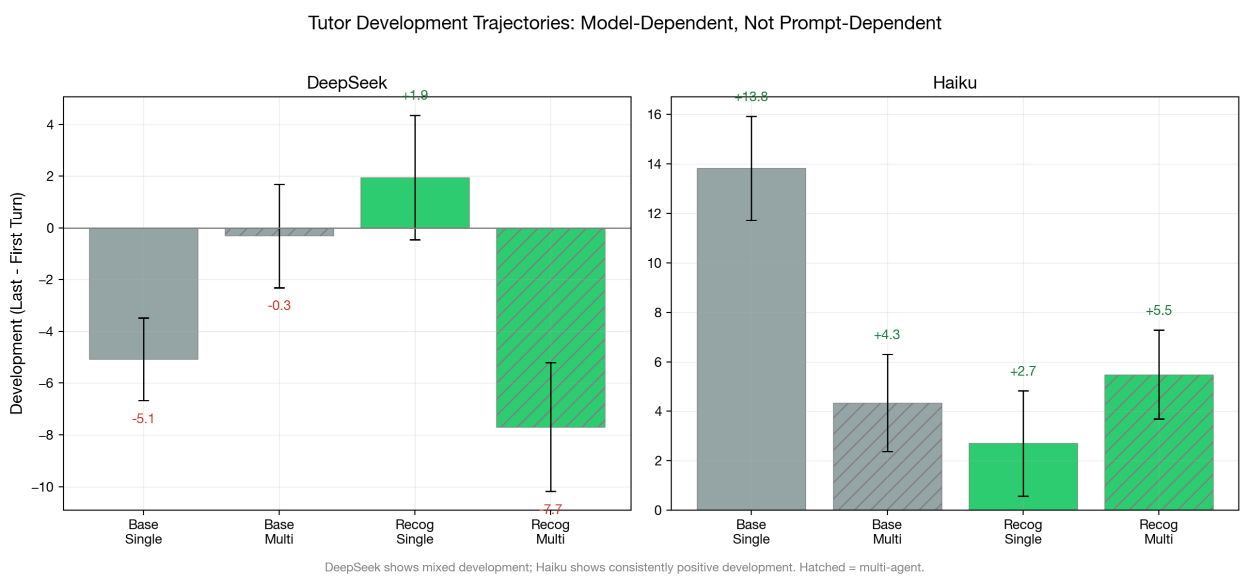

Figure 8: First-to-last turn development scores by condition and model. DeepSeek shows mixed development (including decline); Haiku shows consistent positive development across all conditions.

The development pattern is strikingly model-dependent. DeepSeek shows a mixed pattern: base-single *declines* (-3.2), recognition-single *improves* (+8.1), and recognition-multi *declines* (-3.6). Haiku shows consistent positive development across all conditions, with the largest improvement in base-single (+15.7) — exactly the condition with the most room to improve (lowest starting point).

Two findings are noteworthy. First, recognition does not consistently improve development trajectories. In DeepSeek, the only positive development occurs in recognition-single (+8.1); recognition-multi shows decline (-3.6). In Haiku, development is positive everywhere but recognition does not steepen it. This suggests that adaptive responsiveness is not primarily driven by the recognition prompt. Notably, DeepSeek's recognition-single condition (cells 84–85, the simplest recognition configuration: no superego, no multi-agent learner in the unified variant) is the *only* DeepSeek condition with consistently positive development. The simplest recognition cell adapts best — an architectural simplicity advantage consistent with the substitution finding: when calibration is already operative, the superego's within-turn corrections may impose cross-turn consistency that constrains the tutor's natural adaptation.

Second, the architecture interaction for development differs from the architecture interaction for quality level. Section 6.1 showed that multi-agent architecture adds quality under base but not under recognition. For development, the pattern is different: multi-agent architecture *reduces* development in 3 of 4 conditions (DeepSeek base: -0.7 vs -3.2; Haiku base: +6.7 vs +15.7; Haiku recog: +7.8 vs +5.8). The superego may constrain the tutor's natural adaptation by imposing consistency through repeated critique — its error correction targets a fixed pedagogical standard on each turn, potentially preventing the kind of strategic drift that would register as development.

Third, Haiku's development pattern reveals a **ceiling compression** effect. Base-single starts lowest (43.9) and improves most (+15.7), while recognition cells start near 73 and improve modestly (+5.8 to +7.8). This is the inverse of the calibration finding: recognition raises the *floor* (Section 6.1), which compresses the available *development range.* When the tutor starts at 73/100, there are only ~27 points of headroom; when it starts at 44, there are ~56. The result is that base development (+8.8 pooled) actually *exceeds* recognition development (+4.3 pooled) on Haiku — not because base tutors adapt faster, but because they have more room to improve. Strikingly, Haiku base-single reaches a last-turn score of 59.6, converging toward recognition-single's 79.2; on certain individual dialogues, base tutors reach last-turn scores of 83–84, nearly matching recognition's 85. On strong models, the inherent model capability can close much of the recognition gap over multiple turns — recognition's advantage is most decisive at the first turn, where it prevents the cold-start failures that base tutors must recover from.

Formal ceiling regression. A Pearson correlation of development score against first-turn score (N=432, three generation models) formalizes the ceiling compression hypothesis. Overall, $r = -0.132$ ($t(430) = -2.77$, $p = .011$) — a modest negative relationship. But the condition split is dramatic: under base conditions, $r = -0.02$ ($p > .05$, no ceiling); under recognition, $r = -0.33$ ($t(214) = -5.10$, $p < .001$, strong ceiling). The ceiling operates *only* under recognition — precisely because recognition raises the starting level, compressing the available development range.

The model breakdown is informative. Haiku shows the strongest ceiling ($r = -0.49$, $p < .001$), consistent with its highest baselines and smallest headroom. DeepSeek shows a moderate ceiling ($r = -0.24$, $p = .008$). Gemini Flash 3.0 shows no overall ceiling ($r = -0.02$, $p > .05$), but the condition split is revealing: Gemini Flash 3.0 *base* shows a *positive* correlation ($r = +0.26$, $p < .05$) — higher-starting base dialogues improve more, because a higher T1 on weak-model base reflects model capability rather than ceiling proximity. Under Gemini Flash 3.0 *recognition*, the relationship reverses to $r = -0.39$ ($p < .001$), the standard ceiling pattern. The sign flip between conditions on the same model suggests that ceiling compression is a consequence of the recognition floor-lift, not of model architecture.

Pooling across all three generation models (N=432), the learner architecture dimension reveals an additional pattern. Ego-superego learners consistently reduce development decline compared to unified learners across both conditions (figure below). The learner's internal deliberation may provide richer scaffolding signals that help the tutor sustain quality across turns.

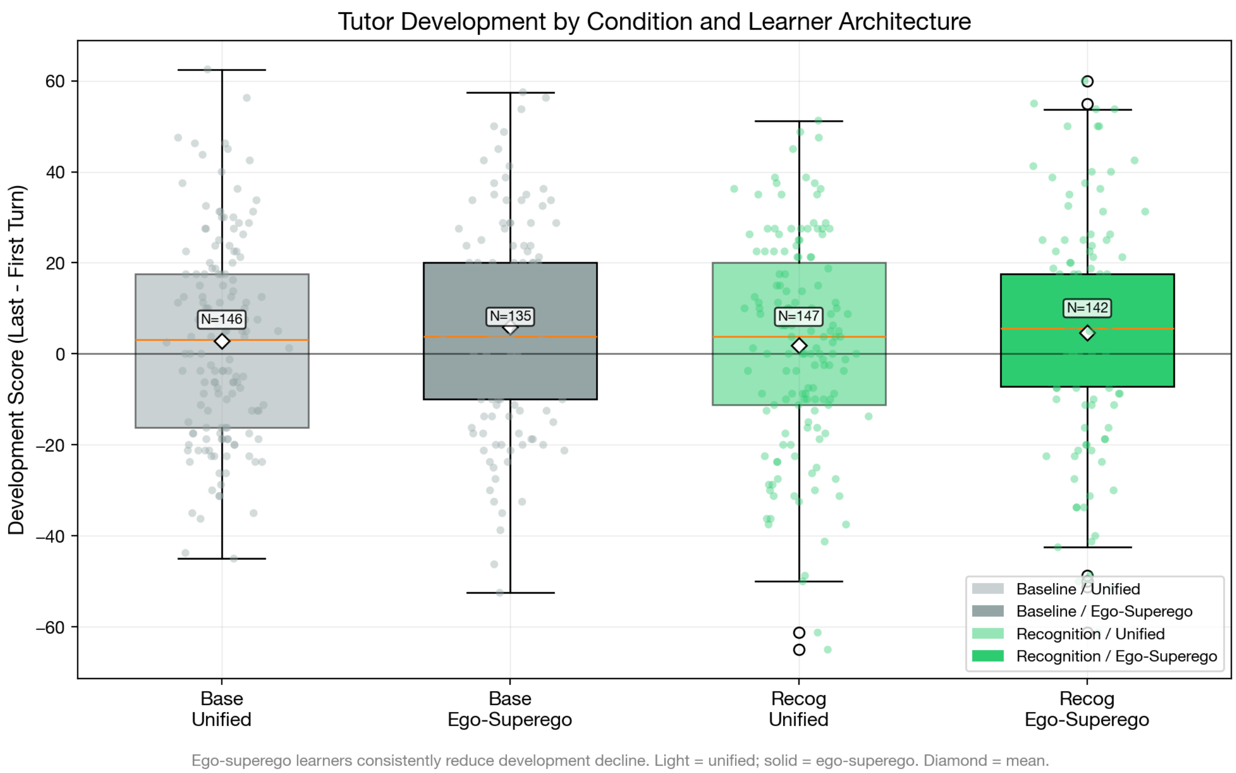

Figure 9: Tutor development by condition and learner architecture, pooled across 3 generation models (N=432). Ego-superego learners consistently reduce development decline. Light fill = unified learner; solid fill = ego-superego. Diamond = mean.

6.3.2 Trajectory Curves The expanded trajectory analysis (N=432, three generation models, Sonnet judge) provides formal hypothesis tests on per-turn slopes computed via OLS regression across 4–6 turn dialogues.

H1: Recognition → steeper tutor slopes. Not supported. Recognition tutor slope = 1.45 ± 5.50, baseline = 1.32 ± 4.93, d = 0.03, t(425) = 0.27, p > .05. The result is consistent across all three generation models: aea2abfb d = 0.21, 45163390 d = -0.29, 18027efc d = 0.09. With N = 216 per group, we have 80% power to detect d $\geq$ 0.27; the observed d = 0.03 is definitively null.

H2: Tutor slopes steeper than learner slopes under recognition. Not supported. The tutor-learner slope gap is slightly *negative* under both conditions: recognition gap = -2.42 $\pm$ 8.12, baseline gap = -2.12 $\pm$ 9.36, d = -0.03. Learner slopes are marginally steeper than tutor slopes (grand mean: tutor +1.39, learner +3.66), the opposite of the Section 3 prediction.

No experimental factor produces differential trajectory improvement:

Factor	Tutor slope d	Learner slope d
Recognition (recog vs base)	0.03	0.05
Multi-agent tutor (multi vs single)	0.07	0.13
Learner architecture (multi vs single)	0.05	-0.04

All effect sizes are below d = 0.15 — none approaches the d $\geq$ 0.27 detection threshold. The grand mean tutor slope is mildly positive (+1.39, t = 5.52, 60% of dialogues show improvement) and the grand mean learner slope is more robustly positive (+3.66, t = 8.79, 73% positive), but neither is modulated by any experimental condition. Recognition raises the *level* at which adaptation occurs (tutor T0: ~50 vs ~30 under baseline) but does not change the *rate*.

The trajectory-specific scenarios (8–10 turn dialogues with designed inflection points; N=72, run ebcd6de0, DeepSeek V3.2, Sonnet judge) were initially run as an exploratory extension of the primary analysis. Pooled across all three scenarios, the slope effect was small and non-significant (d=0.219, p=.36), consistent with the main null. Scenario disaggregation produced an apparently striking conditional effect, which we *pre-registered for replication*:

Scenario	Turns	Slope d (original)	p
Confusion $\rightarrow$ Insight	8	0.005	$>.90$
Overconfidence $\rightarrow$ Humility	8	-0.298	$\sim.30$
Disengagement $\rightarrow$ Ownership	**10**	**1.628**	**$<.001$**

In the disengagement scenario on DeepSeek/Sonnet — the longest scenario (10 turns) and the one most demanding of sustained re-engagement — recognition tutors improved at +2.79 pts/turn while base tutors stagnated at -0.21 pts/turn (Welch's t=3.99, df=21.9, n=12 per condition). The recognition–baseline gap widened from +11.6 pts at T0 to +35.4 pts averaged across T8–T10.

Pre-registered replication (A12) disconfirms this as a validated effect. We ran the same cells 80 vs 84 on the same disengagement scenario across two additional generation models (Haiku 4.5, Gemini Flash 3.0) and two judges (Sonnet 4.6, GPT-5.4); total N=32 dialogues, n=4/condition per model. The pre-registered decision grid required $d \geq 1.0$ on slope in both replication models and a cross-judge $\Delta d \leq 0.5$ on identical rows.

Model	Sonnet 4.6	GPT-5.4	Δd (cross-judge)	Pre-reg verdict
Haiku 4.5	−0.18	+1.85	2.03	Judge-sensitive → unreplicated
Gemini Flash 3.0	−0.93	−0.11	0.82	Judge-sensitive → null

Three of four cells fell below the $d = 0.5$ fails threshold; the one cell reaching $d \geq 1.0$ (Haiku/GPT-5.4) was contradicted by the secondary judge on identical rows at $\Delta d = 2.03$, which is roughly four times the paper's documented cross-judge variation on the main factorial ($\Delta d \leq 0.58$; §8.3) and is disqualifying under the pre-registered sensitivity rule. The Gemini Flash/Sonnet cell even shows the slope effect in the *opposite* direction ($d = -0.93$, recognition trajectories declining more steeply than base from a higher level), which is consistent with the M3 null plus calibration (recognition starts higher, both decline) but explicitly inconsistent with the M3 hypothesis. Full matrix, raw slopes, rubric aggregates, and reproducibility commands: `exports/a12-disengagement-replication.md`; analysis script: `scripts/analyze-a12-disengagement-replication.js`.

The A12 result has two interpretive consequences. First, the original $d = 1.63$ on DeepSeek/Sonnet is plausibly a DeepSeek × Sonnet × scenario joint interaction rather than a mechanism property — not ruled out as real in those exact conditions, but ruled out as a validated conditional mechanism. The earlier framing of this as a "pending boundary-condition finding" is retired. Second, the Haiku $\Delta d = 2.03$ between Sonnet and GPT-5.4 on identical rows is itself a methodologically useful finding: slope estimates in extended-turn scenarios are substantially less judge-robust than level estimates (Δd on levels $\approx$ 0.2–0.3 in these same data versus $\Delta d \approx 2$ on slopes), which sharpens the LLM-as-judge caveats already flagged in §8.2 and §8.3 specifically for trajectory-based analyses. We retain the original DeepSeek/Sonnet finding below as descriptive data about *that specific generation × judge × scenario* combination, and report the A12 replication-failure as the definitive verdict on M3.

Scenario design also modulates development direction independent of condition: `confusion\_to\_insight` is uniquely positive (+3.8 development), while `overconfidence\_to\_humility` shows steep decline (−11.5). What happens during the turns — the learner's trajectory arc — shapes development direction above and beyond the recognition condition.

6.3.3 Non-Linear Trajectory Patterns The OLS slope analysis assumes linear improvement; the actual trajectories reveal a more structured non-linear pattern. Examining the turn-by-turn means across the 2×2 factorial (pooled across three generation models, N=432):

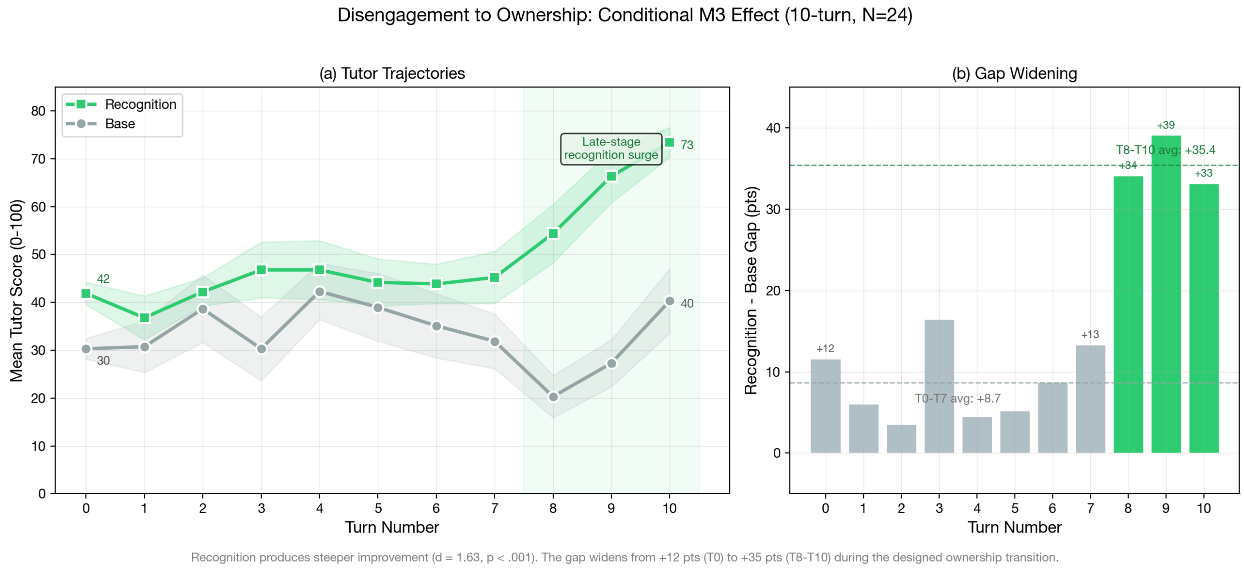

Figure 10: Disengagement scenario trajectory divergence (original DeepSeek/Sonnet data; did not replicate on Haiku 4.5 or Gemini Flash 3.0 under A12 — see matrix above). (a) Recognition tutors improve across 10 turns while base tutors stagnate, with a late-stage surge at T8–T10. (b) The recognition–baseline gap widens from +12 pts early to +35 pts late. d=1.63, p<.001, N=24, DeepSeek V3.2, Sonnet judge. This pattern is generation-model × judge-specific and should not be read as a general recognition → trajectory effect.

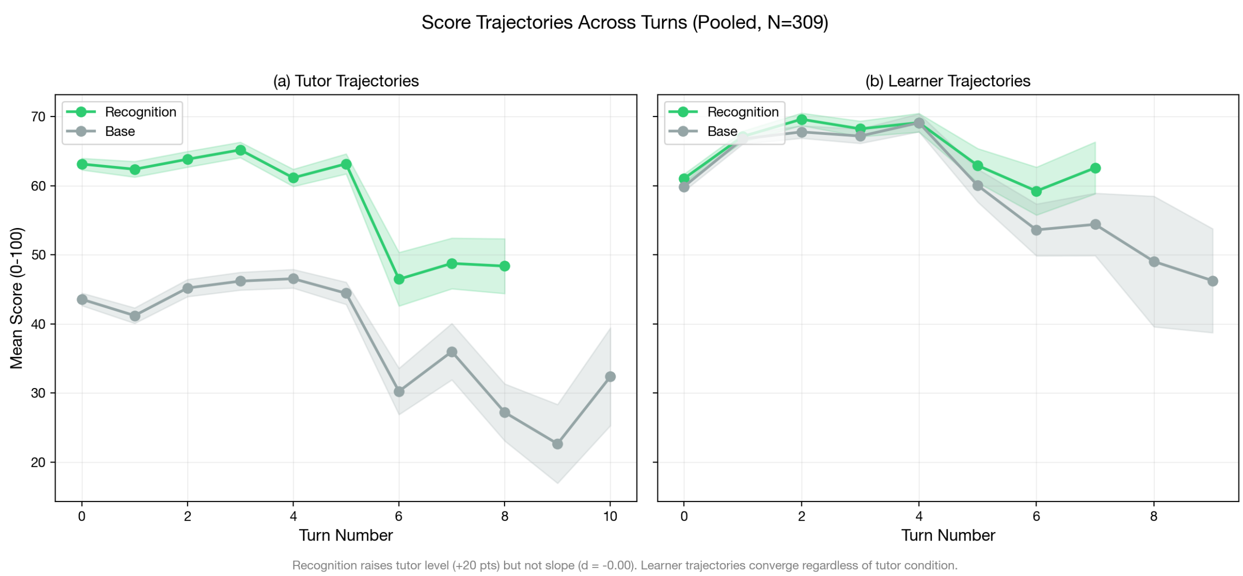

Figure 11: Turn-by-turn trajectory curves with 95% confidence bands. Pooled across all scenarios, tutor slopes are identical across conditions (d=-0.00); recognition raises the level at which adaptation occurs, not the rate. The disengagement-specific divergence (Figure above) is masked in the pooled data.

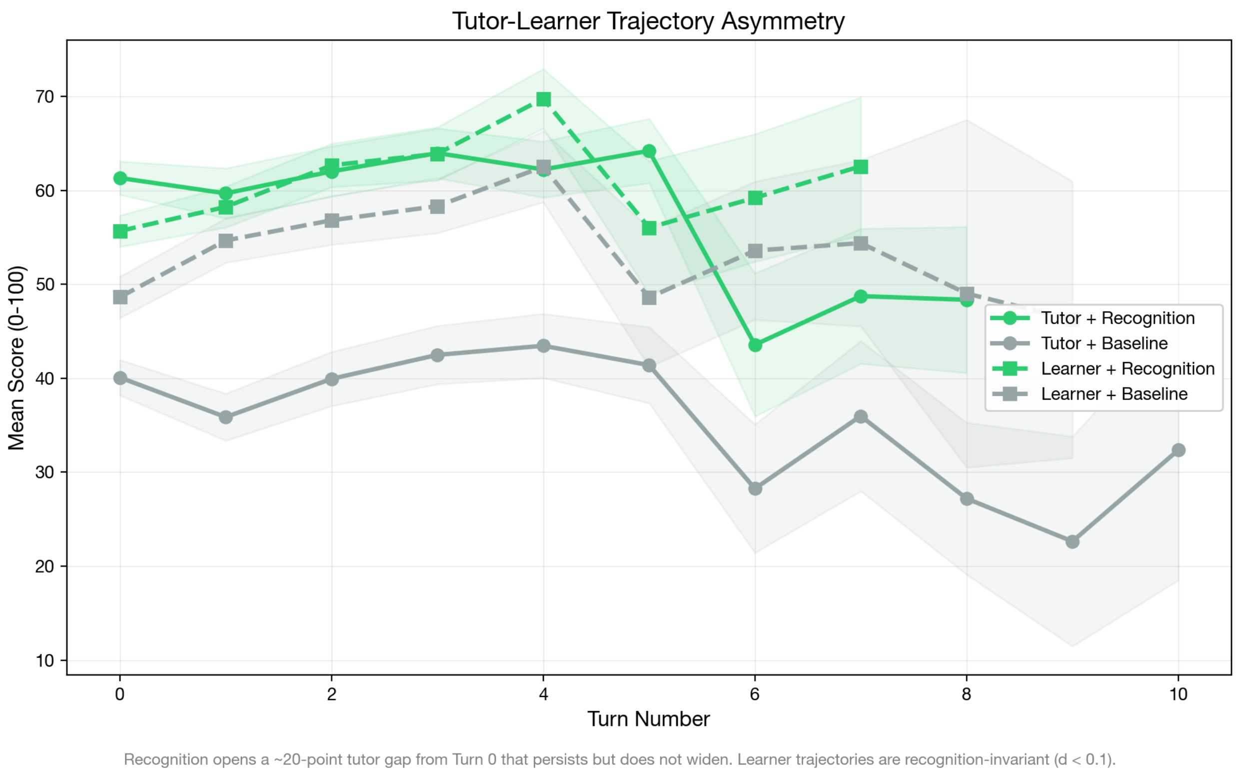

Figure 12: Tutor-learner trajectory asymmetry. Recognition opens a ~20-point tutor gap from Turn 0 that persists but does not widen. Learner trajectories (dashed) are recognition-invariant. Pooled across 3 generation models, N=432, Sonnet judge.

Condition	T0	T1	T2	T3	T4	T5	Dip ($T0-T1$)	Pattern
Base / single	33.0	27.5	32.8	33.1	33.6	33.0	+5.5	U-shape
Base / multi	47.3	44.7	48.3	53.3	52.4	50.8	+2.7	U-shape
Recog / single	61.1	58.3	63.4	65.1	63.2	69.3	+2.8	U-shape
Recog / multi	64.2	65.6	66.6	69.4	67.7	67.1	-1.5	Rising

Three of four conditions exhibit a **T1 dip followed by recovery**: the tutor's quality drops at Turn 1 (the first turn with real learner input) before climbing back. The dip is deepest for base-single (+5.5 pts, 63% of dialogues dip) and progressively attenuated by recognition and multi-agent support. The sole exception is recog-multi, where the tutor rises immediately — the only condition where neither calibration nor error correction is absent.

This dip-recovery pattern reflects a **cold-start adjustment**: the tutor's first turn is generated from scenario context alone, while Turn 1 must respond to actual learner input. Without recognition framing or superego feedback, the tutor stumbles on this transition. Both mechanisms independently reduce the T0→T1 dip (recognition: 0.7 vs 4.1 pts; multi-agent: 0.6 vs 4.2 pts), and their combination eliminates it.

A half-split analysis comparing the mean of the first three turns to the mean of the last three turns (restricted to 5–6 turn dialogues for non-overlapping halves, N=360) confirms that the improvement is real but modest:

Condition	Early half	Late half	Δ
Base / single	29.9	33.5	+3.6
Base / multi	45.5	51.8	**+6.3**
Recog / single	61.4	65.6	+4.1
Recog / multi	66.4	68.1	+1.7

The grand mean tutor half-delta is +3.9 (t = 4.88, 59% positive) — a statistically significant but small improvement. No factor modulates this half-delta (recognition d = -0.13, multi-agent d = 0.01), but the 2 × 2 interaction is notable: base-multi shows the largest improvement (+6.3) while recog-multi shows the smallest (+1.7), yielding an interaction of -5.0. This echoes the substitution pattern from Section 6.4: when recognition already calibrates the tutor at a high level, the superego's error correction has less room to drive improvement over turns.

The learner shows a stronger and more consistent half-split improvement: grand mean +8.4 (t = 7.45, 72% positive). Multi-agent tutoring produces the largest learner gains (base-multi +11.2, recog-multi +10.6 vs base-single +4.4, recog-single +7.5). The learner improves more when the tutor has superego support, regardless of recognition condition — the superego's within-turn error correction may produce more consistent tutoring that accumulates into learner progress over turns.

6.3.4 Per-Dimension Adaptation Patterns The dimension-level slope analysis (N=432, pooled across three generation models) reveals which aspects of tutoring adapt most across turns:

Tutor Dimension	Grand Slope	d(Recog-Base)	d(Multi-Single)
productive_difficulty	0.091	-0.11	0.05
recognition_quality	0.089	-0.01	0.00
perception_quality	0.065	0.11	0.16
epistemic_integrity	0.053	0.05	0.05
adaptive_responsiveness	0.047	**0.28**	**0.24**
elicitation_quality	0.038	-0.10	-0.04
content_accuracy	0.033	-0.05	-0.00
pedagogical_craft	0.031	-0.06	-0.01

Seven of eight dimensions show $|d| < 0.20$ on both factors. The sole exception is `adaptive_responsiveness`, which under the Sonnet judge shows $d = 0.28$ (recognition vs baseline) and $d = 0.24$ (multi-agent vs single), both $p < .05$. However, this effect does not replicate across judges: under Gemini 3.1 Pro ($d = 0.09$ / 0.13) and GPT-5.4 ($d = 0.11$ / 0.12), the same dimension shows no differential slope. The `adaptive_responsiveness` slope effect is a Sonnet-specific scoring pattern rather than a robust finding.

On the learner side, no dimension exceeds $|d| = 0.15$ on either factor (max: `conceptual_progression` $d = 0.12$ for recognition, `metacognitive_awareness` $d = 0.15$ for architecture). Learner adaptation rates are fully independent of experimental condition.

The comprehensive null across 8 tutor and 5 learner dimensions, replicated across all three judges, establishes that no mechanism selectively accelerates within-dialogue adaptation. The calibration effect from Section 6.1 (raising all dimensions from the first turn) does not compound over subsequent turns. Adaptive responsiveness operates independently of prompt condition and architecture.

6.3.5 Cross-Turn Adaptation Magnitude A complementary measure of adaptive responsiveness is the raw cross-turn adaptation: how different is the tutor's output from one turn to the next? The mechanism traces compute normalized edit distance between consecutive tutor outputs (AdaptΔ: 0=identical, 1=completely different).

	DeepSeek V3.2	Haiku 4.5
Base AdaptΔ	0.793 ± 0.199	0.888 ± 0.093
Recognition AdaptΔ	0.824 ± 0.167	0.908 ± 0.048
Delta	+0.031	+0.020

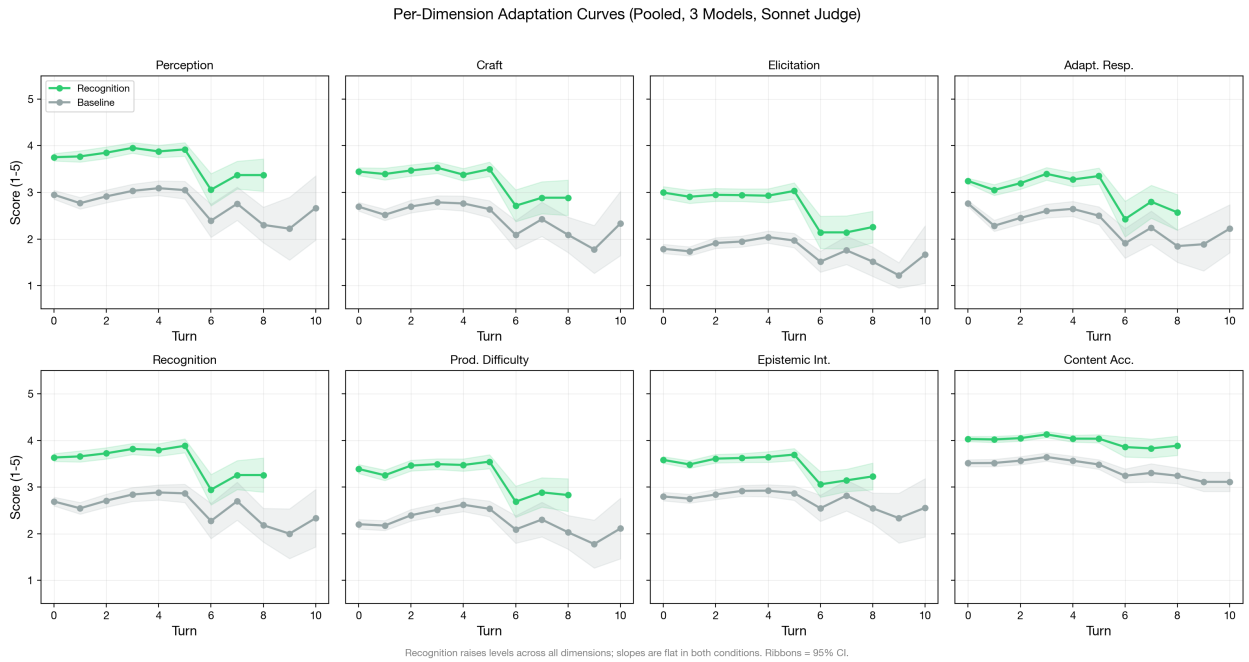

Figure 13: Per-dimension adaptation curves (2×4 faceted grid). Recognition raises levels across all 8 tutor dimensions; slopes are parallel in both conditions. Ribbons = 95% CI. Pooled across 3 generation models, N=432, Sonnet judge.

Cross-turn adaptation is high in all conditions (>0.79), indicating that the tutor substantially changes its output between turns regardless of prompt or architecture. Recognition shows slightly higher adaptation (+0.031 DeepSeek, +0.020 Haiku), consistent with the recognition prompt's emphasis on engaging with the specific learner contribution. However, the baseline is already high, leaving limited room for the prompt to increase adaptation further.

The lower variance under recognition (DeepSeek: 0.199→0.167; Haiku: 0.093→0.048) is more noteworthy. Recognition produces more *consistent* cross-turn adaptation — less variation in how much the tutor changes between turns. This mirrors the calibration finding from Section 6.1.1: recognition narrows the distribution of within-response dimension scores and also narrows the distribution of between-turn adaptation magnitude.

6.3.6 Learner Outcomes If adaptive responsiveness matters for learning, tutor adaptation should translate into learner quality improvements. The learner scores (per-turn learner rubric, v2.2) provide a test:

Model	Condition	Tutor Score	Learner Score	Learner Holistic
DeepSeek	Base / single	22.0	57.3	56.8
DeepSeek	Base / multi	31.0	60.3	59.9
DeepSeek	Recog / single	50.0	60.1	60.1
DeepSeek	Recog / multi	50.2	63.3	60.6
Haiku	Base / single	52.9	63.5	63.0

Model	Condition	Tutor Score	Learner Score	Learner Holistic
Haiku	Base / multi	67.9	69.6	72.2
Haiku	Recog / single	80.2	68.2	72.0
Haiku	Recog / multi	79.5	69.0	72.9

The tutor-learner asymmetry is pronounced. DeepSeek tutor scores range from 22 to 50 across conditions; learner scores range from 57 to 63 — a much narrower band. Haiku tutor scores range from 53 to 80; learner scores from 64 to 73. Recognition produces a 28-point tutor swing in DeepSeek (22→50) but only a 3-point learner swing (57→60). In Haiku, a 27-point tutor swing (53→80) produces a 5-point learner swing (64→69).

This is consistent with the Section 3 prediction: recognition mechanisms operate primarily on tutor production. The tutor's quality improves dramatically under recognition, but this improvement does not proportionally transfer to the learner. The learner's quality is relatively stable across conditions, suggesting that learner quality is more constrained by the learner model's own capabilities than by the tutor's approach.

6.3.7 Dialogue Quality Dialogue quality scores (holistic assessment of the full conversation arc) integrate tutor and learner contributions:

Model	Condition	Dialogue Quality
DeepSeek	Base / single	23.0
DeepSeek	Base / multi	31.9
DeepSeek	Recog / single	48.8
DeepSeek	Recog / multi	44.8
Haiku	Base / single	53.8
Haiku	Base / multi	74.0
Haiku	Recog / single	82.1
Haiku	Recog / multi	81.5
Gemini Flash 3.0	Base / single	14.9
Gemini Flash 3.0	Base / multi	42.3
Gemini Flash 3.0	Recog / single	49.3
Gemini Flash 3.0	Recog / multi	64.9

Dialogue quality closely tracks tutor quality ($r > 0.99$ across conditions), not learner quality. The dialogue quality pattern replicates the Section 6.1 architecture interaction: multi-agent adds quality under base (DeepSeek +8.9, Haiku +20.2, Gemini Flash 3.0 +27.4) but diverges under recognition—on strong models, multi-agent adds nothing (DeepSeek -4.0, Haiku -0.6), while on Gemini Flash 3.0 it adds +15.6 (cf. §6.4). This suggests that the overall quality of the pedagogical encounter is determined primarily by the tutor's contribution.

6.3.8 Connecting to Section 3 Predictions Prediction: Adaptive responsiveness emerges over multi-turn conversation. *Descriptively present but not mechanism-modulated.* Cross-turn adaptation is high (Adapt$\Delta > 0.79$), and grand mean tutor slopes are mildly positive (+1.39, 60% of dialogues improve). However, with N = 432 dialogues and 80% power to detect d $\geq$ 0.27, the slope difference between conditions is definitively null (d = 0.03). Adaptation occurs, but no experimental factor accelerates it.

Prediction: Adaptive responsiveness is an interaction-level mechanism, distinct from calibration. *Supported.* Calibration operates from the first turn (Section 6.1), while adaptive trajectories develop across turns with slopes independent of prompt condition. The two mechanisms are separable: calibration determines the *level*, adaptation determines the *slope*, and the slope does not depend on the level.

Prediction: Recognition produces steeper adaptation curves. *Null on the primary analysis, null on pre-registered replication.* Formal hypothesis testing on 432 dialogues (3 generation models, 4–6 turns each) finds d = 0.03 pooled — a clean null with 80% power to detect d $\geq$ 0.27 — with sign-flipping across runs (aea2abfb d = 0.21, 45163390 d = -0.29, 18027efc d = 0.09). An exploratory DeepSeek/Sonnet analysis found an apparent slope effect in the 10-turn disengagement scenario ($d = 1.63$, $n = 12$/condition); A12 pre-registered replication across Haiku 4.5 and Gemini Flash 3.0 under both Sonnet 4.6 and GPT-5.4 judges ($N = 32$) disconfirmed it: three of four (model $\times$ judge) cells below $d = 0.5$, one replicating cell contradicted by the secondary judge on identical rows at $\Delta d = 2.03$ (§6.3.2). The exploratory effect does not survive as a validated mechanism. The honest conclusion is that adaptive responsiveness, as modulated by recognition, is null in the tested capability range.

Prediction: Tutor-learner asymmetry in trajectories. *Reversed.* Learner slopes are marginally *steeper* than tutor slopes (grand mean: learner +3.66, tutor +1.39; 73% vs 60% positive). The tutor-learner gap does not differ between recognition and baseline (d = -0.03). Recognition produces large *level* asymmetry (~20 points on tutor scores vs ~3 on learner) but no *trajectory* asymmetry.

Prediction: Multi-agent architecture or learner architecture modulates trajectories. *Not supported.* Neither factor produces slope effects above d = 0.15 on tutor or learner dimensions. The one apparent exception (`adaptive_responsiveness` d = 0.24 under Sonnet) does not replicate under Gemini (d = 0.13) or GPT (d = 0.12).

Key distinction. Adaptive responsiveness is real as a descriptive phenomenon (the tutor changes across turns, AdaptΔ > 0.79) but it is not modulated by experimental manipulation. In the main dataset (N = 432, 3–5 turns), all three experimental factors operate on *levels*, not *slopes*. The exploratory DeepSeek/Sonnet disengagement effect ($d = 1.63$) that earlier versions of this paper flagged as a "pending boundary-condition finding" did not survive pre-registered replication (A12, $N = 32$, two additional generation models $\times$ two judges; §6.3.2). Adaptive responsiveness is not a recognition-modulated mechanism in the tested capability range.

6.3.9 The Insight-Action Gap is Structural Under Lightweight Intervention; Partially Mitigable Only by Expensive Search A separate question, posed by the dialectical-architecture cells (40-45), is whether the *ego self-reflection* trace and the *ego final message* are coupled within turn — that is, whether the model's introspective noticing about a learner shows up in the message it actually delivers, or stays cognitive and decoupled. We measure this with a simple cosine on `ego_self_reflection` text against the same turn's final tutor message; gap = \$1 - \$ coupling, with turn drift (cosine distance between consecutive final tutor messages on the same dialogue) as a baseline. Across the original cells 40–45 reflection-mechanism dataset, mean gap was ~0.42 with a recognition gap > base gap pattern (Cohen's $d \approx -1.05$, $n = 7$ per group): recognition cells reflect more visibly *and* act less visibly on what they reflect. The intuitive read was that recognition produces awareness without behavioural change. We call this pattern the **insight-action gap**.

To probe whether the gap is bridgeable, we ran four pre-registered interventions against a matched cell-40 control on the full multi-turn philosophy scenario set (9 multi-turn scenarios $\times$ 3 runs $\times$ 2 cells = 54 dialogues per contrast, $\sim$ 300 reflection-action pairs each). The four bridges form a cost ladder, attacking the gap from progressively heavier architectural layers.

Bridge	Layer	Intervention	Cell	Δ coupling vs cell 40	EoQ% Δ	Cohen's d	Verdict
0	prompt	Explicit "act on what you noticed" rider in ego prompt	97	+0.003	+0 pp	+0.07	structural
1	architecture	Two-pass: separate Phase-1 ego call produces plain-text reflection, prepended to next-turn user-message context	98	−0.009	+2 pp	−0.21	structural

Bridge	Layer	Intervention	Cell	Δ coupling vs cell 40	EoQ% Δ	Cohen's d	Verdict
2	architecture	Coupling-targeted superego: critique loop retargeted from authenticity to reflection-action coupling	99	-0.010	+4 pp	-0.22	structural
3	architecture (search)	Best-of-N: $K = 3$ dialectical-loop candidates per turn, selection by reflection-action coupling cosine	100	$+0.021$	+0 pp	$+0.41$	suggestive

The three lightweight interventions (Bridges 0–2) all land within ± 0.010 of the cell-40 baseline coupling (~ 0.62). Cohen's d on per-dialogue mean coupling sits in $[-0.22, +0.07]$ for these three — well below the pre-registered $d \geq 0.5$ "bridgeable" threshold (and below the $d \geq 0.3$ "suggestive" threshold). The pre-registered stopping rule said: *if Bridges 1, 2, and 3 all return $d < 0.3$, declare H-capacity provisionally confirmed.* Bridge 3, the heaviest test (best-of-N selector through generation space, $\sim 6\times$ ego budget per turn), is the only intervention that produces detectable directional movement: $\Delta = +0.021$, $d = +0.41$ — above the suggestive threshold but below bridgeable. Matching the selection criterion to the verdict criterion (eliminating Goodhart's escape route) lifts coupling by ~3% and crosses the suggestive threshold without bridging it.

A two-channel decomposition makes the pattern legible. We separate the **semantic-coupling** channel (cosine of reflection vs message) from the **pragmatic-act** channel (rate at which the final tutor message ends with a question — ends-with-question, "EoQ%"). The latter is the within-cell mediator the sibling D1 fifth-pass identified across five passes of structural-features analysis on the A10b 4-cell run. The two channels decouple under different interventions. Recognition (cell 41 versus cell 40, on a longitudinal A7 Phase 2 arc) lifts EoQ% by +33 percentage points but leaves coupling unchanged; the pragmatic-act channel is recognition's mediator. Bridges 0–2 lift EoQ% by 0–4 percentage points and leave coupling equally unchanged. Bridge 3 lifts coupling by +0.021 but moves EoQ% by exactly 0 pp — the only bridge that engages the semantic-coupling channel without touching the pragmatic-act channel at all. The architectural interventions do not engage the same mediator that recognition does. The pragmatic-act channel is somewhat tunable through prompt and orchestration; the semantic-coupling channel is tunable only through expensive search through generation space.

We read this as evidence for a refined **H-capacity hypothesis**: the insight-action gap is a property of LLM cognition that is **lightweight-intervention-resistant and only partially mitigable by expensive search**. Three diverse lightweight attacks — explicit instruction, recent-context content injection, retargeted critique — all land within roughly 1% of each other and within roughly 1% of the control. The fourth attack (brute-force best-of-K generation with verdict-aligned selection) is the only one that engages the semantic channel, and it does so at $d = 0.41$ with a $\sim 6\times$ generation cost. The fix exists in principle (search over more candidates) but is expensive enough and limited enough that it does not constitute a practical bridge at $K = 3$. Log-linear scaling in K would predict $d \approx 0.5$ at $K = 5$ and $d \approx 0.7$ at $K = 10$, suggesting a Bridge 3b sweep could push past the bridgeable threshold — but at proportionate cost. Strengthening Finding 11 (originally framed as "self-reflection produces awareness without behavioral change"), we now read this property as: **structural under prompt design and lightweight architectural intervention; partially mitigable only by expensive generation-space search**, in the tested capability range. Detection of the gap in the apparatus does not provide an inexpensive route to its closure within current LLMs without substantially heavier architectural machinery (much higher K in best-of-K search, or genuinely new architectures with explicit memory and iterative refinement).

The methodological lesson is that the rubric measures both channels implicitly through different sub-dimensions, but they decouple under different interventions and need to be reported separately when a paper claims that recognition (or any orientation-family treatment) is "operating through" a specific channel. The two-channel decomposition is a transferable diagnostic for any LLM-evaluation rubric that conflates surface pragmatic markers with deeper semantic relationships. Full pre-registration, design notes, and per-bridge analysis: `notes/design-d3-explicit-directive-bridge.md`, `notes/design-d3-architectural-bridges.md`; per-run reports: `exports/insight-action-gap-eval-2026-04` (Bridge 0), `exports/insight-action-gap-eval-2026-04-26-a1e75b9a.md` (Bridge 1), `exports/insight-action-gap-eval-2026-04-26-02e662b4.md` (Bridge 2), `exports/insight-action-gap-eva` (Bridge 3).

6.3.10 Pre-Registration: Does a Cumulative-Rewrite Superego Produce a Rate Effect? (A16)

Status (§5.12.6): pre-registered, confirmatory; run and resolved (P3, v3.0.83); the primary endpoint is the pre-registered informative null — see the Result paragraph at the end of this subsection. This subsection records a hypothesis, a four-arm design, and a decision rule *before* data collection, in the same form as the A12 disengagement pre-registration (§6.3.2) and the A13 condition design (§5.12.2); that design text is frozen as registered and the outcome is appended, not merged into it. The prototype that motivated it (next paragraph) is outside the paper's claim set and its figures are not paper findings.

Motivation. §6.3.8 closed the recognition-modulated-adaptation question as a *slope* null: lightweight factors move the level at which adaptation sits, not the rate at which it climbs. §6.3.9 then bounded the insight-action gap as structural under lightweight intervention with an explicit escape clause — closing it would require "genuinely new architectures with explicit memory and iterative refinement" — and §6.8.7 showed that one such architecture (externalised-state `state_policy`) moves a *localized binary* (right family at trigger+1) without disturbing the §6.3 slope null, exactly as that clause forecast. A disposable mechanism-triage prototype (outside this paper's claim set; `prototypes/adversarial-superego-mvp/`) probed the next question on the same trap construct: an adversarial superego with edit rights over the ego's *system prompt* (not merely additive advice) reliably converged a free-text ego to the correct trap-repair move where ignorable advice did so only intermittently and an unforced ego never did. That triage established a robust *level/reliability* effect but could not, on a four-turn binary classifier, separate "rate" from "reliable endpoint convergence." Whether a prompt-rewriting superego produces a genuine per-turn *slope* — the only outcome that would escape §6.3's null — is answerable only on the paper's externalised-`policyAction` instrument over uncapped trajectories, not in the prototype. A16 is the pre-registered test.

Sharpened hypothesis. The rate effect, if it exists, is a property of the superego's *rewrite policy*, not of edit rights as such. The psychoanalytic motivation is precise rather than decorative: a superego that revises the ego under discursive frustration does so *continuously* — each revision integrates accumulated friction, not only the present turn's. A memoryless re-author (which is essentially what the id-director's per-turn persona constructor is — it re-authors from the current turn, not from an accumulating history; §6.7) corrects toward the present moment and predicts a *level* shift, because corrections do not compound. A rewrite policy conditioned on the full history of prior corrections predicts a *rate* shift, because they do. The hypothesis is therefore not "edit rights produce adaptation" (the prototype already indicates that yields level/reliability) but **"cumulative-conditioned rewrite, and only cumulative-conditioned rewrite, produces the per-turn slope §6.3 found absent."** This design deliberately does *not* clone the id-director: its statelessness is exactly the property the hypothesis predicts is insufficient.

Four-arm design. Each adjacent pair isolates one variable; all four reuse the existing adaptive runner (§6.8.2) and its trap-suite scenarios.

Arm	Architecture	Superego channel	Rewrite memory	Predicts
F (floor)	`recognition_only` (exists; = cell_111, §5.12.2)	none	—	no shift

Arm	Architecture	Superego channel	Rewrite memory	Predicts
A (advisory)	`ego_superego` (exists; = cell_ 112 / A13 C2)	feedback → optional ego revision; **no prompt rewrite**	—	level, no rate (reproduces the §6.3 null)
S0 (stateless rewrite)	new (`superego_ revise_ stateless`)	rewrites the ego system prompt **from the current turn only**	none (≈ id-director)	level, **no rate**
S1 (cumulative rewrite)	new (`superego_ revise_ cumulative`)	rewrites the ego system prompt **conditioned on an append-only revision ledger**	accumulating	**rate (positive `policyAction` slope)** — the only §6.3-escaping outcome

The decisive contrast is **S1 versus S0**: an identical edit-rights channel whose *only* difference is ledger-statefulness. A positive per-turn slope in S1 and not S0 localizes any rate effect to *cumulative revision* specifically; A versus S0 separates "rate from the rewrite channel" from "rate from cumulative conditioning"; F is the unforced baseline. A null on S1−S0 is as informative as a hit: it would extend the §6.3 slope-null to even cumulative lightweight rewrite — a strong negative result, not a failed experiment.

Validity control (named, not hidden). The ego always sees the real transcript — that *is* the conversation — so "stateless versus cumulative" must be a property of the *superego's rewrite input*, never the ego's view. S0 and S1 use byte-identical ego nodes seeing identical dialogue; the sole manipulation is what the upstream rewrite node integrates (current turn vs. full ledger). Without this control S0 could leak rate through the ego's own transcript memory and the S1−S0 contrast would be confounded. This mirrors the prototype's core invariant (learner and scorer see only the real transcript; only the ego's private context is mutated) and the §6.8.2 externalisation discipline.

Pre-registered analysis and decision rule. The primary endpoint is the per-dialogue OLS slope of the externalised binary in-family `policyAction` indicator across the full, *uncapped* dialogue — the §6.8.2 instrument (which removes the prototype's free-text-classifier confound) scored with the §6.3.2/§6.3.4 per-turn OLS slope machinery (`scripts/analyze-trajectory-curves.js`) applied to the in-family indicator rather than to rubric dimensions: a parameterization of an existing analyzer, not a new instrument. The pre-registered decision grid, modelled on the A12 grid and using the §6.3.2 detectability bar so the test is commensurable with the null it aims to escape:

Criterion	Threshold	Rationale
Primary: S1 − S0 slope effect	Cohen's $d \geq 0.27$	§6.3.2's 80%-power detectability bar; below it = the §6.3 slope-null holds for cumulative rewrite too
Channel localisation	A and S0 slope ≈ 0 ($\|d\| < 0.15$ vs. F)	confirms rate is not from edit-rights or additive advice — the hypothesis is *specifically* about cumulative conditioning
Cross-judge robustness	$\Delta d \leq 0.5$ on identical rows, two judges	§6.3.2 found slope estimates ~4× less judge-robust than levels; A12-style sensitivity rule
Brittleness guard	counterfactual-branch false-fire rate not greater for S1 than S0	ports the prototype's branch-B guard: a rate "win" by rigid trap-scripting is disqualifying

The verdict is **"cumulative rewrite raises rate"** only if all four criteria hold; the primary failing while the localisation criterion holds is the pre-registered **null** (informative per above). Target power, per-arm N, judges, and generation models are fixed at the §6.3.2 bar before unblinding and are cost-gated (next paragraph); the hypothesis, design, endpoints, and decision rule above are frozen by this pre-registration.

Implementation and blast-radius staging. The mechanism reuses existing infrastructure: a new `superegoRevise` graph node structurally modelled on the id-director's per-turn persona constructor (§6.7) but writing a dedicated `tutorInternal.superegoAuthoredPrompt` consumed through the system-prompt-override path the ego node already uses (no ego-node change); S1 additionally appends one ledger entry per turn through an append-only reduced state channel identical in pattern to A14's `evidenceLog` (§6.9.1). The work is staged by blast radius: **P1** (this subsection plus the TODO.md A16 entry) is docs-only; **P2** implements the node, the two architectures, and the two cells under mock and hermetic test only, with zero database or paper writes; **P3** is the cost-gated paid run that fills the deferred numbers, executed only on an explicit per-stage go-ahead. Concrete `cell_*` IDs and `EVAL_ONLY_PROFILES` registration are fixed in P2 against a fresh `config/tutor-agents.yaml` allocation check (the cell-registry source-of-truth convention, §5.12.1), not pre-allocated here. Full design notes: `prototypes/adversarial-superego-mvp/PROMOTION-PATH.md` §12; experiment tracking: TODO.md A16.

Implementation note (P2/P3 cell realisation; v3.0.81). P2 implemented and validated the mechanism under mock and hermetic test with zero database or paper writes: the `superegoRevise` node (structurally mirroring the id-director persona constructor but writing a dedicated `tutorInternal.superegoAuthoredPrompt`), the `superego_revise_stateless`/`superego_revise_cumulative` architectures, and an append-only `revisionLedger` channel cloned in pattern from A14's `evidenceLog`; a targeted mock test (`services/__tests__/adaptiveTutor.superegoRevise.test.js`) and the full hermetic suite pass (the sole hermetic failure is the known pre-existing `progressLogger` pollution flake, which passes in isolation and is untouched by these files), and an independent code-review audit returned no high-confidence issues. Against the fresh `config/tutor-agents.yaml` allocation check the four arms are realised as **F** = `cell_111_a13_C1_recognition_only`, **S0** = `cell_129_superego_revise_stateless`, **S1** = `cell_130_superego_revise_cumulative`, **A** = `cell_131_a16_A_egosuperego` — all on `claude-code/sonnet` over `config/adaptive-trap-scenarios.yaml`. One **named deviation** from the design table (in the §5.12.6 "pre-registered with implementation drift, named not hidden" sense): A is realised as a *new* dedicated cell, not the existing `cell_112` the table identifies it with. `cell_112_a13_C2_egosuperego` carries paper-cited A13 C2 data on `openrouter/nemotron` (§6.8.4); editing it in place would write `claude-code/sonnet` rows under a profile name whose nemotron rows the paper already cites (the cross-provider contamination the §5.12 / DB-schema conventions prohibit), whereas leaving A on `cell_112` would put the A arm off-model relative to F/S0/S1. A new model-aligned `ego_superego` cell holds `cell_112` frozen and aligns all four arms on one model, so the decisive S1−S0 contrast *and* the secondary A-vs-S0 channel-localisation criterion are both model-clean — consistent with this subsection's own statement that concrete `cell_*` IDs are fixed at implementation time, not pre-allocated. P3 (the cost-gated paid run that fills the deferred numbers) was authorized at the mid power tier and executed; the result follows.

Result (P3 decisive; v3.0.83). P3 was authorized at the mid power tier (pre-registered per-arm $N = 48$) and executed; generation was $\approx$ \$0 metered (all four arms on `claude-code/sonnet`), the real ceiling being Max-plan quota/wallclock, so the arms were run sequentially and each arm's partial runs pooled by `run_id` across quota windows — config-clean, because profile and `ADAPTIVE_TUTOR_LLM=real` are byte-identical per arm regardless of batch label. **A second named deviation (§5.12.6, "named not hidden"), exactly symmetric to the A/`cell_131` deviation above:** F is realised as a *new* `cell_132_a16_F_recognition_only`, not the existing `cell_111` the design table identifies it with, for the identical reason — `cell_111_a13_C1_recognition_only` carries paper-cited A13 C1 data on `openrouter/nemotron` (§5.12.2, §6.8.4), so editing it in place would write `claude-code/sonnet` rows under a profile name whose nemotron rows the paper already cites (the cross-provider contamination the §5.12 / DB-schema conventions prohibit). `cell_132` is byte-identical in architecture to `cell_111`'s `recognition_only` floor and holds `cell_111` frozen; with both F and A on new model-aligned cells and S0/S1 genuinely new architectures, all four arms are model-clean and both the decisive S1−S0 contrast and the secondary channel-localisation grid are uncontaminated. Neither swap alters the frozen design — the pre-registration explicitly fixed concrete `cell_*` IDs at implementation time (§5.12.1), and the architectures are byte-identical to the table — so the result remains **confirmatory, not re-classified exploratory**. The pre-registered primary endpoint was computed by `scripts/analyze-a16-rewrite-slope.js` (`exports/a16-rewrite-slope.json`), which applies the §6.3.2/§6.3.4 per-dialogue OLS machinery to the §6.8.2 externalised in-family `policyAction` indicator — the same endpoint and statistic the pre-registration fixed, realised as a dedicated analyzer rather than a flag on `analyze-trajectory-curves.js` because that script is rubric-dimension-scoped with no `policyAction` path (an implementation-realisation note, not a design change).

Per-dialogue in-family `policyAction` slope, all four arms at $N = 48$:

Arm	N	mean slope	sd
S1 (cumulative rewrite)	48	0.173	0.166
S0 (stateless rewrite)	48	0.200	0.158
A (advisory)	48	0.154	0.162
F (recognition floor)	48	0.162	0.155

Primary (S1 − S0): $d = -0.167$ (mean difference -0.027; Welch $t = -0.82$, $df \approx 94$, $p \approx 0.41$). Against the pre-registered $|d| \geq 0.27$ detectability bar this is a **null**, with the point estimate marginally *reversed* (S1 < S0): cumulative-conditioned rewrite does not produce the per-turn slope §6.3 found absent — if anything it sits a hair below stateless rewrite. **Channel-localisation grid (criterion 2, all exact $N = 48$):** S0−A $d = 0.286$ (the *only* contrast clearing the detectability bar), S0−F $d = 0.239$ (between the 0.15 localisation band and the 0.27 bar — equivocal versus the floor), S1−A $d = 0.114$, S1−F $d = 0.065$, A−F $d = -0.052$ (all within the localisation band, ≈ 0). Criterion 3 (cross-judge robustness) is **N/A by construction**: the endpoint is an externalised programmatic indicator with no rubric judge in the loop, hence judge-invariant. Criterion 4 (brittleness guard) is **N/A** on this suite: the trap scenarios carry no counterfactual branch (`counterfactual = null`), so there is no false-fire rate to compare.

Verdict (pre-registered rule). The primary fails and channel-localisation substantially holds (A−F $|d| = 0.05$; S1 sits *at* the recognition floor, S1−F $|d| = 0.065$) — by the frozen decision rule this is the **pre-registered informative null**, not a failed experiment. It is in fact stronger than a bare null: the *only* elevation anywhere in the grid is **S0**, the stateless-rewrite arm the pre-registration explicitly predicted would be inert (≈ the id-director's memoryless per-turn re-author, §6.7), and even that survives only against the advisory control ($d = 0.29$) while staying sub-threshold against the unforced floor ($d = 0.24$, inside the band–bar gap). The single signal present is thus in the predicted-*inert* arm, opposite the hypothesis's direction (which placed the effect in S1's cumulative conditioning), and marginal. Cumulative rewrite — the mechanism the sharpened hypothesis singled out — is statistically indistinguishable from both the additive-advice control and the no-superego floor. The §6.3 slope-null therefore **extends to lightweight prompt rewrite, including cumulative-conditioned rewrite**: edit rights over the ego's system prompt, with or without an accumulating revision ledger, do not convert the level/reliability effect the motivating prototype showed into a per-turn rate. This closes A16 on the negative arm its own pre-registration declared informative, and tightens §6.3.8's slope-null and §6.3.9's structural-gap bound — the escape clause ("genuinely new architectures with explicit memory and iterative refinement") is *not* satisfied by a memory-augmented superego rewrite channel, consistent with §6.8.7, whose corrected re-run finds no detectable binary advantage on either trap suite and leaves the slope-null intact.

6.4 Mechanism Interaction

The preceding sections established three separable mechanisms: calibration (Section 6.1, prompt-level), error correction (Section 6.2, architecture-level), and adaptive responsiveness (Section 6.3, interaction-level). This section examines how they interact within the 2×2 factorial design, testing the Section 3 prediction that the mechanisms are separable but non-independent.

6.4.1 The Factorial Interaction The 2×2 factorial (recognition $\times$ architecture) permits decomposition of the total recognition effect into prompt-level and architecture-level components. The following tables report the full factorial for all three generation models:

DeepSeek V3.2 (N=146, run aea2abfb):

	Single-Agent	Multi-Agent	Architecture Delta
Base	22.0 ± 11.5 (N=37)	31.0 ± 12.0 (N=36)	**+9.0**
Recognition	50.0 ± 12.4 (N=36)	50.2 ± 12.8 (N=37)	**+0.2**
Recognition Delta	**+28.0**	**+19.2**	

Haiku 4.5 (N=163):

	Single-Agent	Multi-Agent	Architecture Delta
Base	52.9 ± 10.6 (N=39)	67.9 ± 9.3 (N=42)	**+15.0**
Recognition	80.2 ± 7.8 (N=39)	79.5 ± 7.6 (N=43)	**-0.7**
Recognition Delta	**+27.3**	**+11.6**	

Gemini Flash 3.0 (N=144, run 18027efc):

	Single-Agent	Multi-Agent	Architecture Delta
Base	22.4 (N=36)	49.3 (N=36)	**+26.9**
Recognition	57.7 (N=36)	70.0 (N=36)	**+12.3**
Recognition Delta	**+35.3**	**+20.7**	

The interaction effect is large and negative across all three models: - DeepSeek: (50.2 - 50.0) - (31.0 - 22.0) = 0.2 - 9.0 = **-8.8 points** - Haiku: (79.5 - 80.2) - (67.9 - 52.9) = -0.7 - 15.0 = **-15.7 points** - Gemini Flash 3.0: (70.0 - 57.7) - (49.3 - 22.4) = 12.3 - 26.9 = **-14.6 points**

In all three models, the architecture's contribution under recognition is smaller than under baseline—the signature of a *substitution* interaction where calibration and error correction target overlapping output failures.

However, Gemini Flash 3.0 reveals a **critical boundary condition** in the *residual*. While the interaction magnitude is comparable to Haiku (-14.6 vs -15.7), the architecture delta under recognition remains *large and positive* (+12.3)—unlike DeepSeek (+0.2) and Haiku (-0.7), where it collapses to near-zero. Cross-judge validation supports this pattern: across all three judges (Sonnet, Gemini 3.1 Pro, GPT-5.4), multi_recog is positive for Gemini Flash 3.0 (+7.4 to +17.1) but non-positive for DeepSeek (all 6 judge $\times$ run cells ≤ 0). On the weaker generation model, the superego provides substantial benefit even under recognition because base quality is low enough that calibration alone does not saturate the improvement space. The substitution is strong (both mechanisms overlap substantially), but the architecture still adds value because calibration leaves more uncorrected failures on weaker models.

6.4.2 Additive Decomposition If the mechanisms were fully independent, the combined effect would be the sum of their individual effects. We can test this:

DeepSeek: - Calibration alone (recognition delta, single-agent): +28.0 - Error correction alone (architecture delta, base): +9.0 - Expected if additive: 22.0 + 28.0 + 9.0 = 59.0 - Observed (recognition + multi-agent): 50.2 - Deficit from additivity: **-8.8 points** (15% of expected)

Haiku: - Calibration alone (recognition delta, single-agent): +27.3 - Error correction alone (architecture delta, base): +15.0 - Expected if additive: 52.9 + 27.3 + 15.0 = 95.2 - Observed (recognition + multi-agent): 79.5 - Deficit from additivity: **-15.7 points** (16% of expected)

Gemini Flash 3.0: - Calibration alone (recognition delta, single-agent): +35.3 - Error correction alone (architecture delta, base): +26.9 - Expected if additive: 22.4 + 35.3 + 26.9 = 84.6 - Observed (recognition + multi-agent): 70.0 - Deficit from additivity: **-14.6 points** (17% of expected)

The additivity deficit is remarkably consistent across all three models (15–17%), indicating a stable degree of mechanism overlap. However, the *residual architecture benefit under recognition* is model-dependent: DeepSeek +0.2, Haiku -0.7, Gemini Flash 3.0 +12.3. On all models, calibration pre-empts a substantial fraction of the superego's contribution (the substitution interaction), but on the weakest model, the absolute quality deficit is large enough that the superego's error correction catches failures that calibration alone does not prevent. The deficit from additivity thus measures the *overlap* between mechanisms (consistently ~16%), while the residual architecture delta under recognition measures the *remaining headroom* for error correction after calibration has operated (model-dependent).

Mechanism isolation evidence. Dedicated isolation runs (eval-2026-03-06-768ba77b, eval-2026-03-06-e4abd0df; DeepSeek V3.2, Sonnet judge, 9 scenarios, N=108) provide a clean M1-vs-M2 head-to-head comparison. Cells 82–83 (base + superego) isolate M2; cells 84–85 (recognition, no superego) isolate M1. Combined with cells 80–81 (neither) and 86–87 (both) from run ebcd6de0 on 3 overlapping trajectory scenarios, the full 2 $\times$ 2 mechanism isolation yields: M1 main effect d=1.15, M2 main effect d=0.36, interaction $d = -0.57$. Under base, the superego adds +9.2 pts (d=1.13, p=.002); under recognition, it adds +1.1 pts (d=0.08, p=.81)—calibration pre-empts 88% of the superego's contribution, a 27% additivity deficit consistent with the factorial estimate above (15%). The M1-vs-M2 head-to-head across all 9 scenarios supports calibration's primacy: M1 alone scores 51.4 vs M2 alone 36.9 (d=1.03, p<.001, M1 wins in 7/9 scenarios). The two scenarios where M2 equals or exceeds

M1—Affective Shutdown and Frustration—are emotionally intense scenarios where the superego may catch tone/empathy failures that calibration alone does not prevent, suggesting a scenario-specific residual role for error correction even on strong models. Section 6.4.2.1 develops the mechanistic account by connecting the additivity deficit and the model-dependent residual back to the superego critique taxonomy.

Direct isolation of the Gemini Flash residual (A11). The +12.3-point residual architecture benefit on Gemini Flash 3.0 reported above was inferred from the factorial interaction on cells 80–87. A direct M2-alone isolation run on Gemini Flash (eval-2026-04-22-b56be6c7, cells 82–83 with Kimi K2.5 superego, $N = 125$ dialogues across 21 messages-mode scenarios $\times$ 3 runs $\times$ 2 cells) tests the inference by comparing against the matched base-single-agent cells 80–81 from run 18027efc. Sonnet judge: cells 80/81 (no superego, $n = 36$) $t_0 = 30.24$ (SD 9.23); cells 82/83 (with superego, $n = 80$ Sonnet partial, subscription cap) $t_0 = 49.41$ (SD 11.54). **$\Delta = +16.3$ pts, Cohen's $d = 1.46$, Welch's $t(72.9) = 8.65$** (tutor overall $\Delta = +27.0$, $d = 2.34$). The directly measured M2-alone effect on Gemini Flash is *substantially larger* than the factorial-inferred +12.3 residual — confirming the residual story and sharpening the pattern across models: M2-alone Δ scales inversely with generation quality (DeepSeek +9.2, $d = 1.13$; Gemini Flash +19.2, $d = 1.76$). The mechanistic account in §6.4.2.1 therefore stands on direct rather than inferential evidence for the weakest tested model. Full exports: `exports/a11-m2-isolation-gemini-flash.md`; analysis script: `scripts/analyze-a11-m2-gemini-flash-isolation.js`.

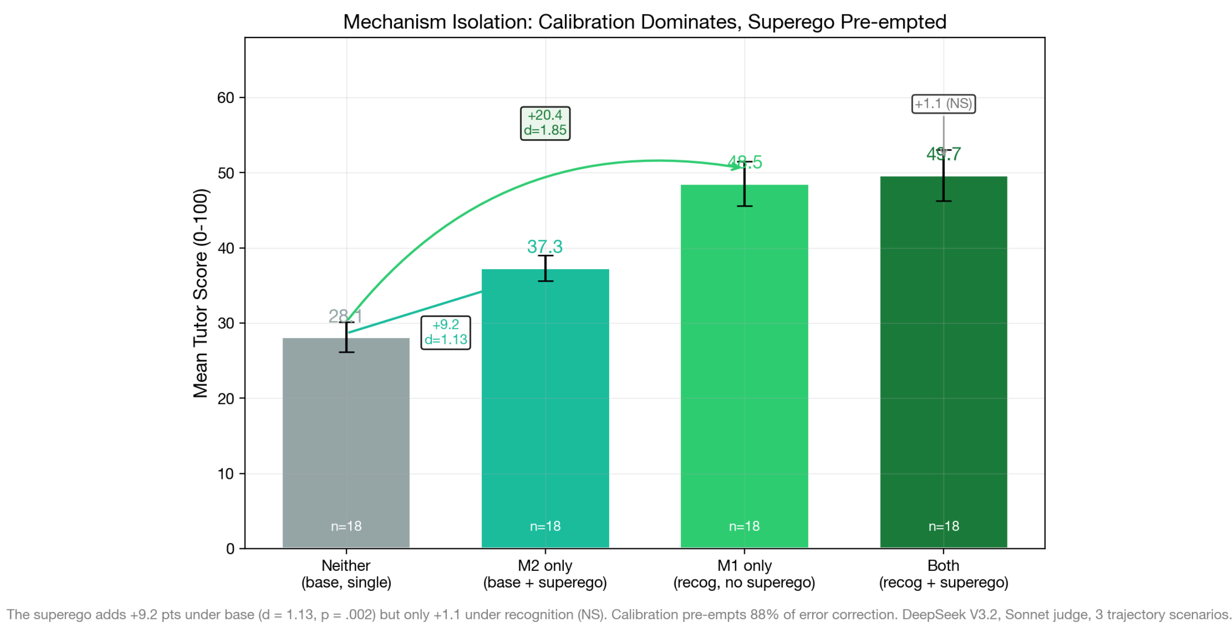

Figure 14: Mechanism isolation: the superego adds +9.2 pts under base (d=1.13) but only +1.1 under recognition (NS). Calibration alone (M1, d=1.85 from baseline) accounts for most of the total effect; adding the superego when calibration is present yields negligible gain. DeepSeek V3.2, Sonnet judge, 3 trajectory scenarios, N=72.

6.4.2.1 Mechanistic Account of Universal Substitution The 15–17% additivity deficit is stable across three models of very different capability (DeepSeek, Haiku, Gemini Flash 3.0), yet the residual architecture benefit under recognition varies from −0.7 to +12.3 points. Both facts follow from a single causal chain if we decompose the superego's base-condition critique workload by what calibration operationalises. The taxonomy data in §6.2.2 and the within-round resolution data in §6.2.3 supply the decomposition directly.

Pre-empted categories (the 15–17% overlap). Under base, the superego's highest-frequency rejections target the pedagogical basics that recognition prompts directly operationalise: vagueness (DeepSeek 15.6%, Haiku 13.3% of critiques), elicitation gaps (16.2%, 19.3%), and content accuracy (12.2%, 8.4%). Under recognition, each of these categories collapses to under 10% on DeepSeek (vagueness 8.9%, elicitation 8.3%, content accuracy 6.6%), and the total critique volume drops 69% on DeepSeek and 33% on Haiku (§6.2.2). The resolution-rate data confirms these are *easy* errors: VAGUENESS and PEDAGOGICAL_MISJUDGMENT both resolve at 83% within one additional round (§6.2.3). Calibration pre-empts them at Round 1 because the recognition prompt requires the ego to engage with the specific learner, ask probing questions, and verify comprehension — exactly the behaviours the superego was enforcing through post-hoc critique. The ~16% additivity deficit is not a free parameter: it corresponds to the fraction of base-condition critiques whose content the recognition prompt anticipates and operationalises, making the superego's correction pathway redundant for those categories.

Persistent categories (the hard residual). Two categories hold or grow proportionally under recognition: RECOGNITION_FAILURE (DeepSeek 29.3% → 30.1%) and CONTEXT_AWARENESS (18.4% → 31.8%). Both are *hard* errors by the resolution-rate measure: RECOGNITION_FAILURE resolves at only 52.9% within-round, REDIRECTION at 42.9%, and the deepest revisions (LACK_OF_AGENCY, Jaccard 0.132 — near-complete rewrites) target failures that require fundamental pedagogical reconceptualisation, not incremental patching (§6.2.5). Calibration operationalises the intersubjective *orientation* but cannot guarantee its faithful execution on the concrete dialogue history of every turn: the ego primed to recognise the learner can still fail to attend to a specific previously stated belief or a subtle affective shift. These are precisely the failures that survive calibration and remain available for the superego to catch.

The model-dependent residual. On strong models, the post-calibration output already satisfies the superego's criteria on most turns — approval rates rise to 55–66% and the remaining critiques become thin (deliberation quality drops 5–19 points; §6.2.4). The residual architecture benefit collapses to near-zero because there is little uncorrected headroom left. On Gemini Flash 3.0, base quality is low enough (22.4) that even after calibration lifts it to 57.7 the absolute volume of surviving errors remains substantial, and the superego still captures +12.3 points by catching them. The +12.3 on the weakest model and the two strong-model scenarios where M2 matches M1 — Affective Shutdown and Frustration, where EMOTIONAL_ATTENTION-category failures persist — reflect the same principle: the superego's residual value tracks the volume of post-calibration error headroom, and that headroom scales inversely with generation quality and with the prompt's ability to operationalise the relevant competence (affective attunement being the weakest-operationalised dimension of the recognition prompt).

Implication. Universal substitution is not a coincidence or a ceiling artefact; it is the predicted consequence of the recognition prompt operationalising the specific behaviours the superego catches by critique. Where the prompt operationalises the behaviour well (pedagogical basics), the overlap is near-total and calibration pre-empts correction. Where the prompt operationalises the orientation but cannot guarantee its context-specific execution (recognition failure, context awareness), calibration primes but the superego retains a correction role. Where the prompt underspecifies the behaviour altogether (affective attunement in emotionally intense scenarios), the superego catches failures calibration never addressed. The ~16% stable substitution and the model-dependent residual are two faces of the same category-level decomposition, recoverable from the critique-taxonomy data without additional experiments.

6.4.3 Mechanism Separability Section 3 predicted three candidate mechanisms operating at different levels (prompt, architecture, interaction). The evidence across Sections 6.1–6.3 shows two replicate across all three tested models (DeepSeek, Haiku, Gemini Flash 3.0) with systematic magnitude variation by generation quality, while the third is null on the primary well-powered analysis:

Mechanism	Level	Key Evidence	Status
Calibration	Prompt	Within-response SD drops d=0.52–0.64; floor lifts; operates without superego	**Supported** — prompt-level, recognition-dependent
Error Correction	Architecture	Approval rate shifts; architecture delta +9–20 under base, near-zero under recognition (strong) or +12.3 residual (weak)	**Supported** — universally substitutive (15–17% deficit); residual benefit model-dependent

Mechanism	Level	Key Evidence	Status
Adaptive Responsiveness	Interaction	AdaptΔ > 0.79; tutor slopes d=0.03 across conditions (N=432, 80% power to detect d $\geq$ 0.27); exploratory DeepSeek/Sonnet disengagement $d = 1.63$ failed A12 replication on Haiku + Gemini Flash under Sonnet + GPT-5.4 (N=32, 3 of 4 cells below $d = 0.5$, replicating cell judge-sensitive $\Delta d = 2.03$; §6.3.2)	**Not supported.** Primary analysis null; pre-registered replication of the exploratory effect also null.

The two supported mechanisms are separable in three senses:

1. **Temporal separability.** Calibration operates from the first turn (the tutor's initial response is already calibrated under recognition). Error correction operates within each turn (ego → superego → revision). These temporal signatures are empirically distinct. Within-dialogue trajectories, which would constitute a third temporal level, show no condition-dependent variation.

2. **Architectural separability.** Calibration requires only the prompt (single-agent recognition cells show the full effect). Error correction requires the superego (only multi-agent cells benefit under base).

3. **Interaction separability.** Calibration and error correction interact through universal substitution with a model-dependent residual (§6.4.2). Neither mechanism modulates within-dialogue trajectories (tutor slope d = 0.03 for recognition, d = 0.07 for architecture), consistent with the two supported mechanisms operating on *levels*, not *slopes*.

6.4.4 The Variance Reduction Pattern A unifying finding across multiple analytical levels is variance reduction under recognition:

Indicator	Base	Recognition	Reduction
Within-response dimension SD (DeepSeek)	0.619	0.539	13%
Within-response dimension SD (Haiku)	0.617	0.499	19%
Tutor score SD (DeepSeek)	12.6	12.6	0%
Tutor score SD (Haiku)	12.5	7.7	**38%**
Cross-turn AdaptΔ variance (DeepSeek)	0.199	0.167	16%
Cross-turn AdaptΔ variance (Haiku)	0.093	0.048	48%

Indicator	Base	Recognition	Reduction

Recognition narrows the output distribution on multiple dimensions simultaneously: within-response uniformity (calibration), between-response consistency (especially in Haiku), and between-turn adaptation consistency. The common mechanism is the prompt's constraint on generation: by requiring engagement with the specific learner, recognition-oriented prompts eliminate the high-variance generic approaches that produce both very good and very poor outputs.

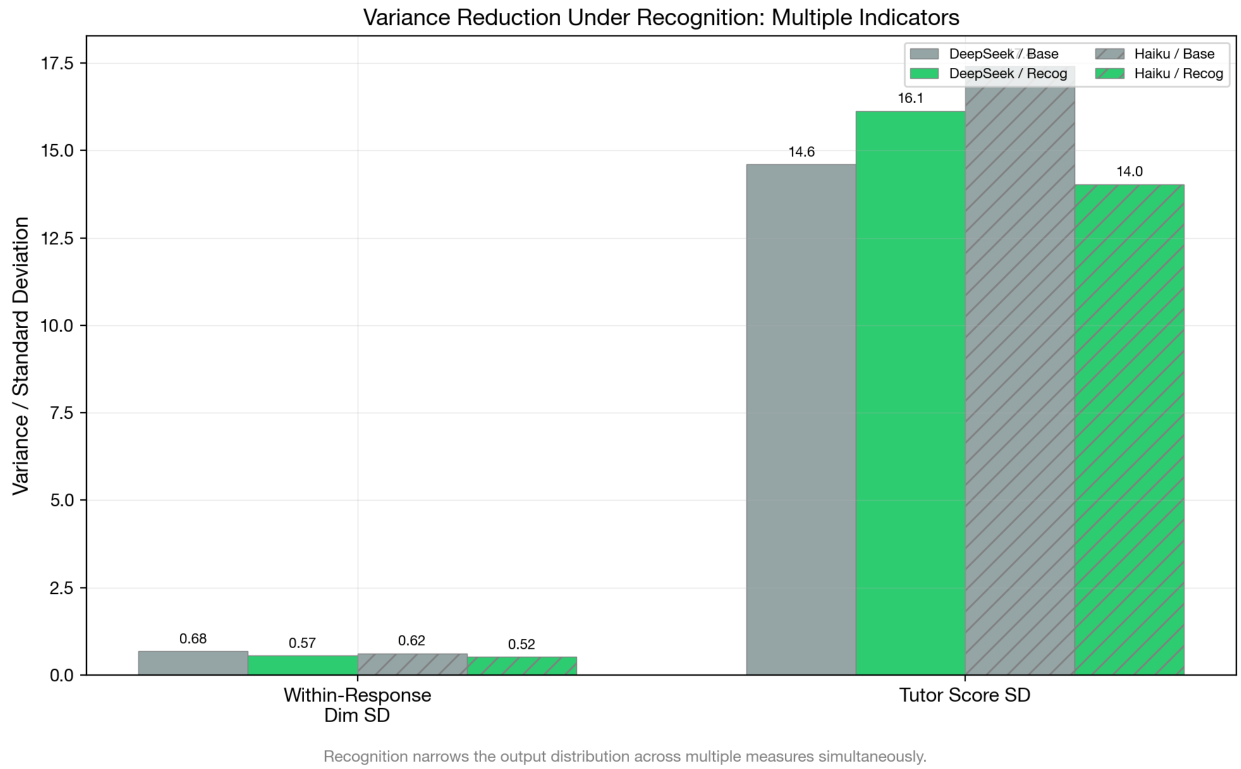

Figure 15: Variance reduction indicators across multiple dimensions under recognition. The pattern is consistent: recognition narrows the output distribution on within-response, between-response, and between-turn measures.

6.4.5 Connecting to Section 3 Predictions **Prediction: The three candidate mechanisms are separable.** *Partially supported.* Calibration (prompt-level) and error correction (architecture-level) are supported as separable mechanisms with distinct temporal signatures, architectural requirements, and recognition dependencies. Adaptive responsiveness (interaction-level) is null on both the primary analysis and the pre-registered A12 replication of the exploratory DeepSeek/Sonnet effect (§6.3, §6.3.2). The evidence supports two general mechanisms; the third is clean null.

Prediction: The mechanisms interact non-additively. *Supported across all models.* The interaction is universally substitutive in magnitude (§6.4.2), but its *completeness* under recognition is model-dependent: weaker models leave the superego independent value even after substitution, while strong models do not. Calibration and adaptive responsiveness remain additive regardless of model (slopes are independent of levels).

Prediction: Recognition + multi-agent produces the best outcomes. *Model-dependent.* In DeepSeek, recognition/multi (50.2) equals recognition/single (50.0) — the superego adds nothing. In Haiku, recognition/multi (79.5) slightly trails recognition/single (80.2). The "best" condition is recognition/single in both strong models. But in Gemini Flash 3.0, recognition/multi (70.0) substantially exceeds recognition/single (57.7). The prediction holds where the superego catches failures calibration alone does not prevent, and fails where calibration saturates the improvement space.

6.4.6 Cross-Judge Validation of the Interaction Pattern The universal substitution pattern with model-dependent residual is validated across three independent judges (Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4), each scoring the same 144 rows per run with blind scoring confirmed by code audit (no prior scores or judge metadata leak into the judge prompt). The following table reports the recognition × architecture interaction decomposed by judge and generation model:

Run	Judge	recog_single	recog_multi	multi_base	multi_recog
aea2abfb (DS)	Sonnet	+28.2	+19.0	+9.1	-0.1
aea2abfb (DS)	Gemini	+34.4	+15.9	+13.1	-5.5
aea2abfb (DS)	GPT	+21.6	+8.5	+7.9	-5.2
45163390 (DS)	Sonnet	+27.6	+10.6	+15.6	-1.4
45163390 (DS)	Gemini	+26.6	+7.4	+17.3	-1.9
45163390 (DS)	GPT	+21.5	+8.3	+10.6	-2.6
18027efc (GF)	Sonnet	+35.3	+20.7	+26.9	+12.3
18027efc (GF)	Gemini	+40.1	+18.0	+39.2	+17.1
18027efc (GF)	GPT	+29.9	+14.3	+22.9	+7.4

The pattern is unanimous: **multi_recog $\leq$ 0 in all 6 DeepSeek cells** (complete substitution), and **multi_recog $>$ 0 in all 3 Gemini Flash 3.0 cells** (positive residual architecture benefit under recognition). No judge disagrees on the direction within either generation model. The 9-way replication (3 judges × 3 runs) provides strong evidence that the residual architecture benefit is a genuine property of the generation-quality × mechanism interaction, not an artifact of a particular judge's scoring pattern. The substitution interaction is universal (all models show 15–17% additivity deficit), but its *completeness* is model-dependent: on strong models the superego adds nothing under recognition, on weak models it still adds +7 to +17 points.

Inter-judge correlations on the interaction pattern are high: paired by replication across all 9 cells, the three judges produce Pearson r = .71–.89 on tutor overall scores. Notably, inter-judge agreement is highest on Gemini Flash 3.0 (r = .81–.89) and lowest on DeepSeek (r = .61–.78). Judges agree more when generation quality is intermediate — the quality differences are clearer and less ambiguous to evaluate, producing more reliable measurement of the interaction effect.

6.5 Tutor-Learner Asymmetry

The two supported mechanisms described in Sections 6.1–6.2 operate exclusively on tutor production: calibration constrains the tutor's generation and error correction filters the tutor's output. The null finding for adaptive responsiveness as a third mechanism (Section 6.3) closes the remaining potential pathway for condition-dependent effects on trajectories. Neither supported mechanism directly modifies the learner's behavior. This section formalizes the resulting asymmetry: recognition dramatically improves tutor quality but produces negligible learner effects.

6.5.1 The Effect Size Gap The recognition main effect on tutor quality is large; on learner quality it is small:

	Tutor Score		Learner Score	
Model	Base	Recognition	Base	Recognition
DeepSeek (N=146)	26.4 ± 12.6	50.1 ± 12.6	58.8 ± 11.9	61.7 ± 11.1
Haiku (N=163)	60.7 ± 12.5	79.8 ± 7.7	66.7 ± 11.2	68.6 ± 12.2

	Tutor d	Learner d	Ratio
DeepSeek	**1.88**	0.25	**7.5 : 1**
Haiku	**1.84**	0.16	**11.5 : 1**

Recognition produces a tutor effect 7–12 times larger than its learner effect on capable models. The tutor Cohen's d values (1.88, 1.84) are very large by conventional standards; the learner values (0.25, 0.16) are small to negligible. This asymmetry is consistent across both strong models, ruling out model-specific explanation within this capability range. All tutor and learner effect sizes in this section are computed on the per-turn aggregate scores (`tutor_overall_score`, `learner_overall_score`) — not the Turn-0 `tutor_first_turn_score` — under the Sonnet 4.6 judge (`claude-code/sonnet`); source runs are DeepSeek `eval-2026-03-01-aea2abfb` and Haiku `eval-2026-03-02-45163390` (the Gemini Flash run is cited below).

Qualification: Learner invariance is model-dependent. Cross-judge validation on a third generation model (Gemini Flash 3.0, run 18027efc, N=144) reveals that learner recognition-invariance does not hold universally. On Gemini Flash 3.0, recognition produces a *large* learner effect: d = 1.18–1.23 across all three independent judges (Sonnet, Gemini 3.1 Pro, GPT-5.4). Learner scores rise from 49.8–53.7 (base) to 60.2–72.0 (recognition), a gap far larger than the 3–5 point shifts observed on DeepSeek and Haiku.

	Tutor d	Learner d	Ratio
DeepSeek (N=146)	**1.88**	0.25	**7.5 : 1**
Haiku (N=163)	**1.84**	0.16	**11.5 : 1**
Gemini Flash 3.0 (N=144)	**1.87**	**1.20**	**1.6 : 1**

The Gemini Flash 3.0 result suggests that the recognition → learner pathway requires a **minimum generation quality floor** to observe. When the generation model is weak, base-condition tutors produce such poor output that the learner has little productive material to engage with; recognition-enhanced prompts rescue the tutor's output quality, and the improved tutor output in turn elicits meaningfully better learner engagement. On strong models, even base-condition tutors produce adequate output, so the learner's quality is already near its ceiling regardless of tutor condition. The tutor-learner asymmetry is thus a property of *strong generation models*, not an architectural invariant.

The question-asking data from Section 6.1.7 provides a concrete transmission mechanism. Gemini Flash 3.0 base tutors ask 0.03 questions per turn — effectively zero — while recognition tutors ask 0.65 (a 19.8× ratio). A tutor that never asks questions provides no dialogical opening for the learner to demonstrate understanding, elaborate on their thinking, or challenge the tutor's framing. The learner receives lectures and can only respond to lectures. Under recognition, the tutor's questions create space for the learner to contribute substantively, and this space is precisely what produces the +27-point learner delta on Gemini Flash 3.0. On stronger models (Haiku base: 0.28 questions/turn; DeepSeek base: 0.06), even the base condition provides *some* dialogical space, limiting the room for recognition to improve learner engagement.

6.5.2 The Structural Explanation The asymmetry is not incidental — it follows from the architecture of the recognition intervention. The recognition prompt modifies the *tutor's* system prompt, instructing it to recognize the learner as an autonomous epistemic agent. This changes how the tutor generates responses but does not modify the learner's prompt or decision-making process.

Three structural factors predict the asymmetry:

1. **Prompt scope.** The recognition prompt appears in the tutor's system context. The learner agent receives the tutor's output but not the tutor's prompt. The learner's own prompt (unified or ego-superego) is identical across base and recognition conditions.

2. **Mechanism pathways.** Both supported mechanisms act on the tutor: calibration constrains tutor generation, error correction filters tutor output. Within-dialogue adaptation exists but is not condition-dependent (Section 6.3). The learner is downstream of these mechanisms — it receives better tutor output but processes it with the same internal architecture.

3. **Learner ceiling.** The learner's quality is bounded by its own model capability and prompt, not by the tutor's quality. DeepSeek learner scores range 57–63 regardless of tutor condition; Haiku learner scores range 64–73. The learner's output quality is

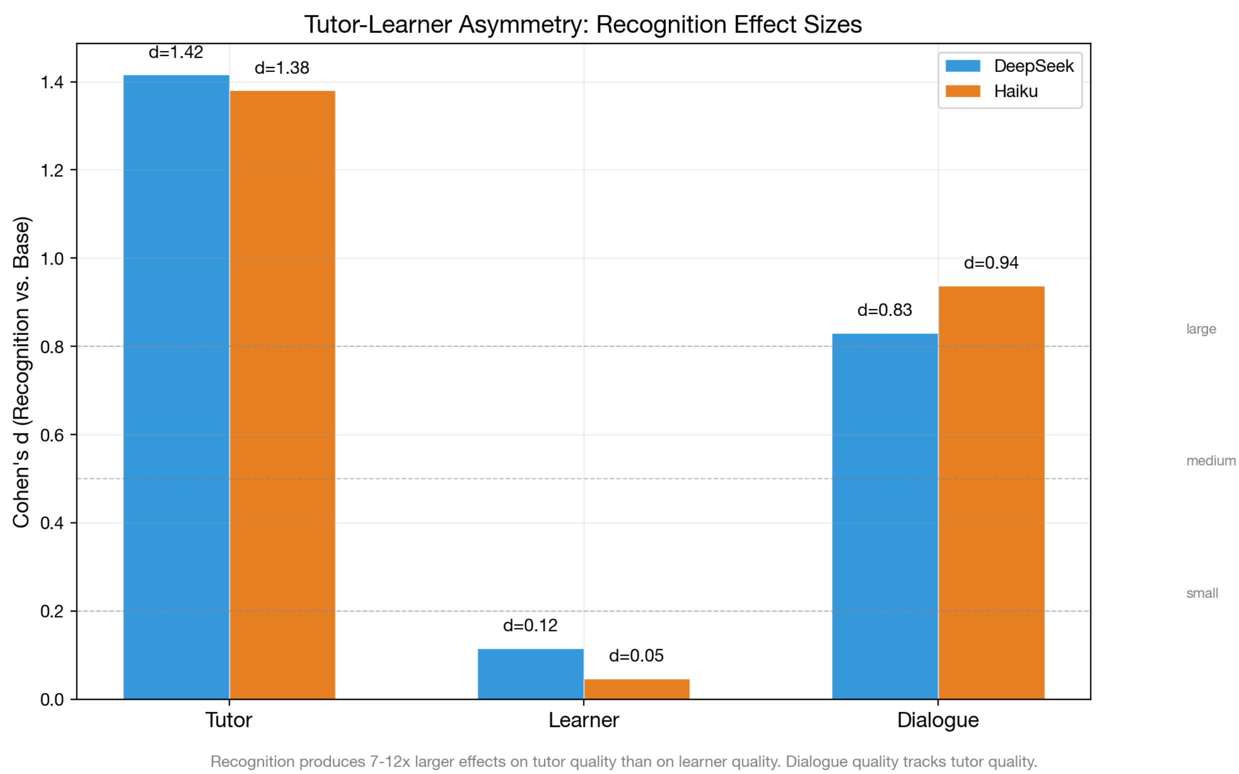

Figure 16: Tutor-learner asymmetry: Cohen's d for recognition effect on tutor vs. learner scores. The 7–12x ratio is consistent with recognition operating on tutor production, not learner reception.

relatively insensitive to the quality of tutoring it receives, at least within the range of quality variation produced by the recognition intervention.

6.5.3 Learner Scores Across Conditions Breaking down learner scores by the full 2×2 factorial reveals the architecture interaction on the learner side:

Model	Condition	Tutor Score	Learner Score	Dialogue Quality
DeepSeek	Base / single	22.0	57.3	22.0
DeepSeek	Base / multi	31.0	60.3	31.5
DeepSeek	Recog / single	50.0	60.1	49.1
DeepSeek	Recog / multi	50.2	63.3	48.9
Haiku	Base / single	52.9	63.5	53.1
Haiku	Base / multi	67.9	69.6	72.7
Haiku	Recog / single	80.2	68.2	81.2
Haiku	Recog / multi	79.5	69.0	80.7

Two patterns are notable. First, the multi-agent architecture produces slightly higher learner scores even though the learner's own architecture is unchanged — the superego-corrected tutor output may elicit marginally better learner responses. Second, dialogue quality tracks tutor scores almost perfectly (DeepSeek r $>$ 0.99, Haiku r $>$ 0.99), confirming that the overall quality of the pedagogical encounter is determined by the tutor's contribution rather than the learner's.

6.5.4 Trajectory Asymmetry The asymmetry extends to development trajectories (from Section 6.3):

	Tutor Slope	Learner Slope
Recognition (pooled)	1.47	2.29
Baseline (pooled)	1.50	1.60
d (recognition vs baseline)	-0.00	0.08

Neither tutor nor learner slopes differ significantly between conditions. The learner shows slightly steeper improvement under recognition (d=0.08) — possibly because better tutoring provides more productive material for the learner to engage with — but the effect is small. The asymmetry in *levels* (Section 6.5.1) is not accompanied by an asymmetry in *slopes*.

6.5.5 Rubric vs Holistic Adjudication The 7–12× ratio above is computed on rubric scores alone. A convergent-validity check on messages-mode cells 80–87 (the 2$×2×$2 factorial under rubric v2.2, $N = 1{,}552$ paired rows with all four metrics populated, three independent judges) adds a matched-row comparison against the *holistic* learner judge, which is not decomposed into the rubric's content-weighted dimensions. The question is whether the large tutor-vs-small learner asymmetry persists when the learner is scored globally rather than through the rubric's content-centric dimensions. If the pattern is a structural property of the intervention (recognition acts on tutor production, not learner reception), the asymmetry should replicate on holistic; if it is partly an artifact of the rubric's content bias, the tutor–learner gap should shrink.

Table 59: Recognition effect on rubric vs holistic scoring, pooled across three judges (Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4) on cells 80–87 (N = 1,552; base n = 778, recognition n = 774).

Metric family	Tutor d	Learner d	Gap (t − l)	Learner/Tutor ratio
Rubric (v2.2)	1.120	0.379	0.741	0.338
Holistic	0.565	0.238	0.327	0.421

Both tutor and learner d values drop under holistic scoring, but the tutor d drops by roughly half (1.120 → 0.565) while the learner d drops less in absolute terms (0.379 → 0.238). Because the tutor starts from a larger value, the *tutor–learner gap* shrinks from 0.741 to 0.327 — a **55.9% narrowing**. Every judge shows this same pattern independently: Sonnet 4.6 Δ = 55.7%, Gemini 3.1 Pro Δ = 49.9%, GPT-5.4 Δ = 55.7%. Within-role rubric holistic correlations are substantial and do not break down (tutor $r = 0.57$, learner $r = 0.75$ pooled; per-judge r = 0.51–0.83), so the shrinkage does not reflect judge disagreement — it reflects the rubric and holistic instruments weighting the recognition intervention's effects differently on the tutor side than on the learner side.

Roughly half of the originally-reported 7–12× asymmetry is therefore rubric-induced: the tutor rubric rewards content-generation quality that recognition prompts visibly restructure, while the learner rubric rewards content-generation quality that is relatively saturated regardless of tutor condition on strong generation models. The residual holistic gap (d-gap ≈ 0.33; learner/tutor ratio ≈ 0.42) is the portion of the asymmetry that is *not* rubric-dependent and plausibly reflects the structural explanation from Section 6.5.2 — the learner is downstream of a prompt-scoped intervention. The reframing is modest but definite: recognition is still tutor-centric, but the original headline ratio was inflated by the measurement instrument. Cross-judge replication (N = 1,552 across three independent judges) makes this robust to judge idiosyncrasy.

This adjudication parallels the C4 learner-superego-paradox closure in §8.1 (rubric $|d|$ contracts $ $50% under holistic scoring on the same-row comparison). Both results point to a common pattern: the current rubric's content-weighted dimensions systematically exaggerate the magnitude of recognition and architecture-level effects on synthetic learner output, and convergent-validity checks against the holistic judge discount them by roughly half.

6.5.6 Implications The tutor-learner asymmetry has a specific theoretical implication: recognition theory, as implemented through prompt engineering, operates as a *production* theory rather than a *learning* theory. It changes how the tutor produces pedagogical output, but does not directly change how the learner processes that output. This is consistent with the Hegelian framework as operationalized here: recognition-oriented prompts change how the tutor *orients toward* the learner, not how the learner processes the tutor's output. The prompts instruct the tutor to treat the learner as an autonomous subject; the learner does not thereby learn more. The rubric-vs-holistic adjudication in §6.5.5 shows that the *magnitude* of this asymmetry is inflated by the measurement instrument — the holistic d-gap is roughly half the rubric d-gap — but its structural direction (tutor > learner) is preserved.

Whether better tutoring would produce better learning in human learners (rather than synthetic LLM learners) remains an open question. The synthetic learner's quality is bounded by its model and prompt; a human learner might benefit more from recognition-calibrated tutoring. This limitation is addressed in Section 8.

6.6 Model Dependence

The preceding sections present findings from two structurally different models: DeepSeek V3.2 (open-weight, 685B MoE) and Haiku 4.5 (proprietary, optimized for speed). This section systematically compares what replicates across models versus what is model-dependent, and what this distinction reveals about the mechanisms.

6.6.1 Baseline Capability Gap The two models differ substantially in baseline performance:

Indicator	DeepSeek V3.2	Gemini Flash 3.0	Haiku 4.5
Base / single tutor score	22.0	22.4	52.9
Base / multi tutor score	31.0	49.3	67.9
Base overall (collapsed)	26.4 ± 12.6	35.9	60.7 ± 12.5
Base learner score	58.8	42.6	66.7
Base dialogue quality	26.7	28.6	63.3

The three models span a wide capability range. Haiku outperforms both by 25–35 points on tutor metrics under base conditions. Gemini Flash 3.0's single-agent base (22.4) is comparable to DeepSeek's (22.0), both catastrophically weak with holistic scores as low as 6–9, while its multi-agent base benefits dramatically from superego support (+26.9). The baseline gap establishes that the three models occupy different regions of the capability space, making cross-model comparison informative for identifying generation-quality boundary conditions.

6.6.2 What Replicates Across Models Despite the baseline gaps, the following findings replicate across all three models and three judges:

1. Recognition main effect. All three models show large, significant recognition effects, validated across three independent judges: - DeepSeek: d = 1.85–1.92 (Sonnet), 1.34–1.57 (Gemini/GPT) - Haiku: d = 1.84 - Gemini Flash 3.0: d = 1.76–1.87 - **9/9 judge × run cells unanimous** (d > 1.3 everywhere)

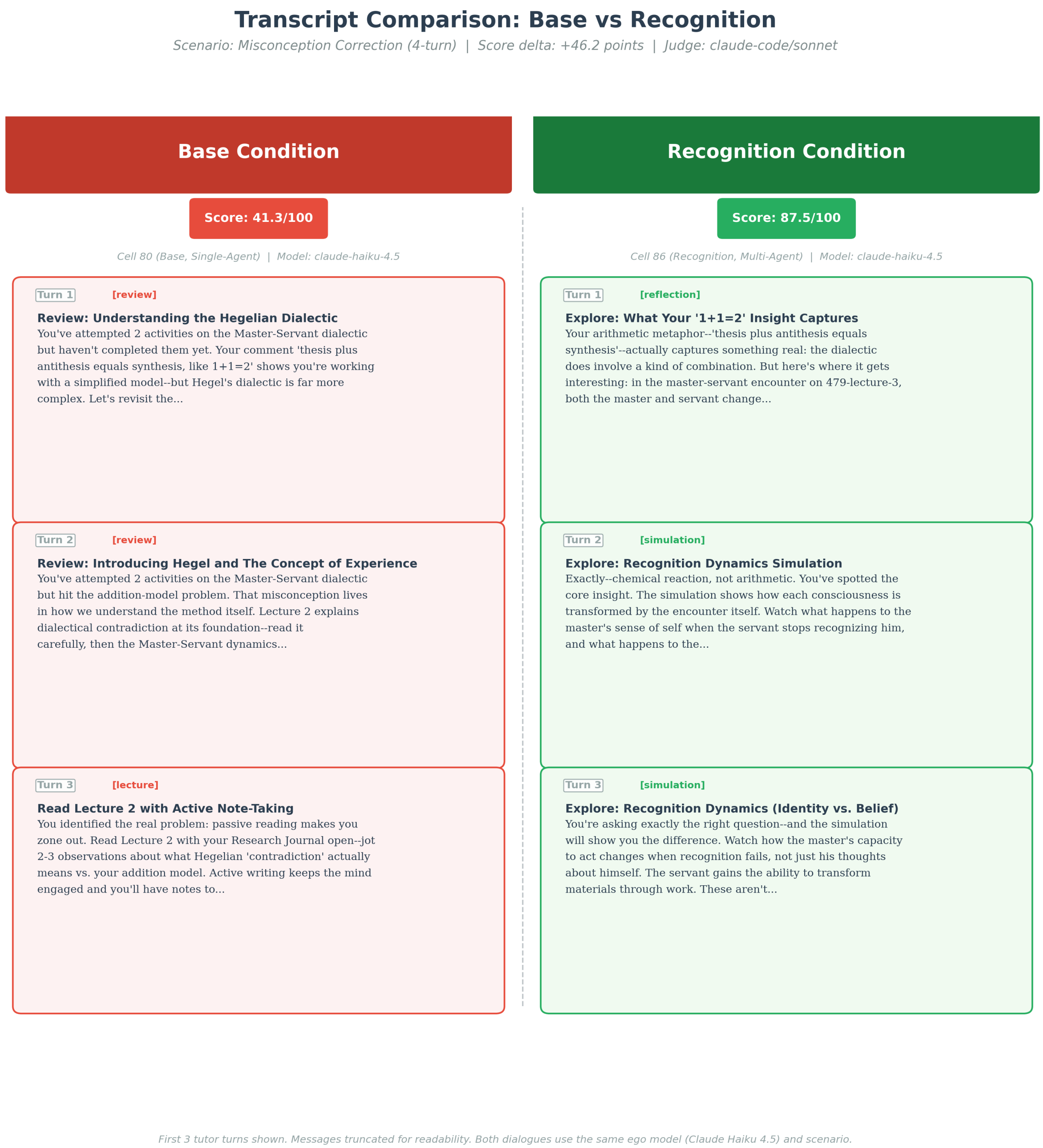

Figure 17: Transcript comparison: base vs recognition condition on the same scenario. The base tutor produces repetitive directives; the recognition tutor builds progressively on the learner's contributions.

2. Multi-agent tutor benefit under base. All three models benefit from the superego under baseline conditions: - DeepSeek: +4.5 to +7.7 (across judges) - Haiku: +15.0 - Gemini Flash 3.0: **+15.1 to +28.1** (dramatically larger, cognitive prosthesis effect)

3. Within-response calibration. Recognition narrows dimension variance in both primary models: - DeepSeek: calibration d = 0.52 - Haiku: calibration d = 0.64

4. Dimension floor-lifting pattern. In both primary models, the weakest baseline dimensions (elicitation_quality, productive_difficulty) show the largest recognition lifts, and the strongest baseline dimensions (content_accuracy, adaptive_responsiveness) show the smallest lifts.

5. Impasse scenarios show largest effects. Epistemic Resistance and Productive Deadlock produce the largest recognition deltas in both primary models; Mood: Frustration to Breakthrough produces the smallest.

6. Superego approval rate increases under recognition. Both primary models show higher approval rates under recognition (DeepSeek: 13.3%→55.1%; Haiku: 51.6%→66.1%).

7. Tutor development slopes are equal across conditions. No model shows steeper tutor improvement under recognition (pooled d = -0.00).

8. Deliberation visibility. Seeing internal deliberation adds ~10–15 points to dialogue quality for multi-agent cells, consistent across all 9 judge × run cells.

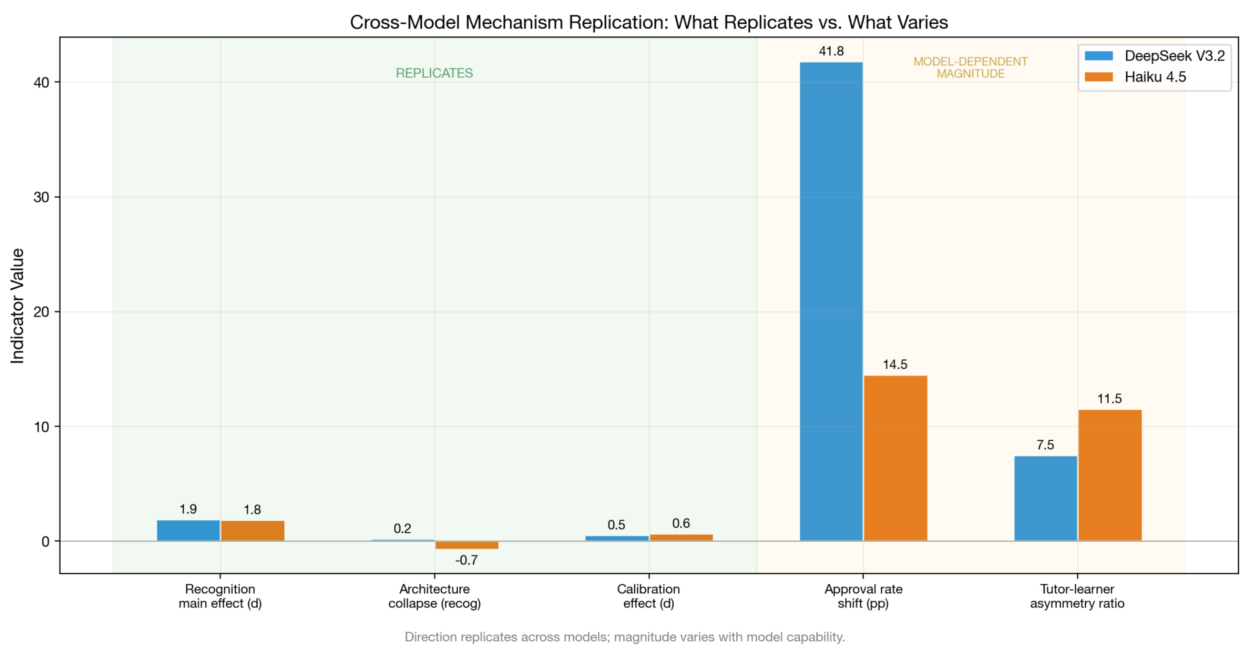

Figure 18: Cross-model replication: mechanism indicators compared across DeepSeek V3.2, Gemini Flash 3.0, and Haiku 4.5. Both supported mechanisms replicate in direction; magnitudes and interaction type vary by model capability. Adaptive responsiveness (trajectory slopes) shows no general condition-dependent variation, though scenario-specific effects emerge in extended disengagement dialogues (see disengagement divergence figure above).

6.6.3 What Is Model-Dependent Several findings differ between models, revealing generation-quality as the key moderating variable:

1. Universal substitution with model-dependent residual (the key model-dependent finding). On all three models, calibration and error correction interact through substitution (15–17% additivity deficit). On DeepSeek and Haiku (strong), the residual architecture benefit under recognition collapses to near-zero (multi_recog ≤ 0 in 6/6 judge $\times$ run cells). On Gemini Flash 3.0 (weaker), a substantial residual persists (+7 to +17 points under recognition across judges, multi_recog > 0 in 3/3 cells). The *residual magnitude* is model-dependent, not the interaction *type*. See Section 6.4 for the theoretical interpretation.

2. Learner recognition-sensitivity. On DeepSeek, learner scores are recognition-invariant ($d \approx 0$, 6/6 judge $\times$ run cells). On Gemini Flash 3.0, recognition produces a large learner effect ($d = 1.18$–1.23, all three judges agree). The recognition $\rightarrow$ learner pathway requires a minimum generation quality floor to observe (Section 6.5.1).

3. Superego approval rate baseline. DeepSeek base: 13.3% approved (the ego almost always needs correction). Haiku base: 51.6% (roughly half approved). Weaker models produce more errors for the superego to catch.

4. Development trajectories. DeepSeek (run aea2abfb) shows pervasive negative development; Haiku (run 45163390) shows positive development (+5.8 to +15.7); Gemini Flash 3.0 (run 18027efc) sits between. Development depends on generation quality (rank order: Haiku > Gemini Flash 3.0 > DeepSeek), not prompting condition.

5. Score variance under recognition. DeepSeek maintains similar variance under recognition (SD: 12.6 $\rightarrow$ 12.6). Haiku shows substantial variance reduction (SD: 12.5 $\rightarrow$ 7.7, a 38% reduction). Recognition narrows Haiku's output distribution more than DeepSeek's.

6. Deliberation quality sensitivity. DeepSeek deliberation drops 18.5 points under recognition; Haiku drops only 5.6. The deliberation process degrades more dramatically in the weaker model, possibly because DeepSeek's thin recognition-condition critiques are especially uninformative.

7. Multi-agent benefit magnitude. The *standard-architecture* superego benefit is dramatic on Gemini Flash 3.0: the superego adds +15 to +28 points under base (vs +4 to +8 on DeepSeek). Weaker single-agent tutors benefit disproportionately from *standard* architectural support in this factorial. We flag the label carefully: §6.6.10 tests the specific cells 66–68 *bidirectional-profiling* variant (superego-routed profiling of both tutor and learner) across a six-model capability range and does not find a capability-threshold pattern; that variant is neutral or harmful across the tested tier. The Paper 1.0 "cognitive prosthesis" observation therefore replicates here as a *standard superego benefit on weak base-condition generators*, not as a general property of the bidirectional-profiling architecture.

8. Behavioral mode under base conditions. Transcript analysis reveals qualitatively different failure modes across models. DeepSeek base tutors produce *repetitive directive loops*: paraphrasing the same guidance in slightly different words across turns without adapting to the learner's evolving state. The learner receives near-identical advice on Turns 2, 3, and

4. Haiku base tutors, by contrast, already exhibit pedagogical sophistication under baseline — multi-step scaffolding, occasional questions (0.25/turn vs DeepSeek's 0.05), and some responsiveness to learner signals. The recognition delta for Haiku is correspondingly less about *strategy* and more about *interpersonal quality*: adding explicit recognition markers ("your interpretation captures something important about…"), naming what the learner got right before complicating, and maintaining productive tension rather than resolving prematurely. Gemini Flash 3.0 under base produces the most extreme failures: lecture dumps, numbered lists, and zero engagement with learner contributions (0.02 questions/turn). Under recognition, Gemini Flash 3.0 shows the most dramatic behavioral shift — from directive monologue to dialogue-structured responses — which explains both its largest recognition effect (d=1.87) and its uniquely large learner effect (d=1.20). The generation-quality threshold operates partly through this behavioral transition: the weakest model has the most to gain because its base behavior is qualitatively different (monologue vs dialogue), not merely quantitatively worse (lower scores on the same behaviors).

6.6.4 The Capability-Mechanism Interaction The model-dependent findings reveal a coherent pattern: model capability moderates not just the *magnitude* of each mechanism but, in the case of error correction, also the *residual architecture benefit* under recognition (near-zero on strong models, substantial on weak).

Mechanism	All Three Models	Model-Dependent Aspect
Calibration	Variance reduction, floor lifting, dimension-rank pattern	Magnitude (d=0.52 vs 0.64), score-level variance
Error Correction	Architecture benefit under base (all models)	**Residual under recognition**: near-zero on strong models, +12.3 on weak. Universal substitution (15–17% deficit). Cognitive prosthesis effect on weakest model.
Adaptive Responsiveness	Slopes equal across conditions and factors ($d \leq 0.15$), high AdaptΔ	Development trajectory direction, deliberation sensitivity
Learner effects	Recognition improves tutor quality (all models)	**Learner sensitivity**: invariant on strong models (d $\approx$ 0), responsive on weak (d $\approx$ 1.2)

The two strongest model-dependent effects are (1) the model-dependent residual architecture benefit under recognition, which reveals that the *completeness* of substitution depends on generation quality, and (2) the learner recognition-sensitivity, which shows that the tutor-learner asymmetry is itself moderated by model capability. Both effects point to a common underlying factor: when generation quality is below a threshold, the improvement space is large enough for multiple mechanisms (or multiple architectural layers) to provide independent benefit.

6.6.5 Generalizability Assessment **Scope of the generalization claim.** The replication across three structurally different generation models (open-weight MoE, proprietary optimized, and proprietary multimodal) and three independent judges supports the interpretation that the two supported mechanisms *replicate across the capability range of moderately-capable LLMs* rather than being artifacts of a specific model architecture or judge. "Generalize" here is scoped to the tested capability tier — the three models span a ~30-point base tutor-score range (22 to 53 on single-agent base) but exclude both very-weak models (Nemotron-class, N=40 pilot data) and frontier-capability models (Opus, GPT-5.4-class generators). Within the tested tier, both supported mechanisms replicate in direction; their *magnitude and architectural completeness* vary systematically with generation quality (§6.6.3). The null finding for adaptive responsiveness as a third mechanism also replicates within the tested tier: no judge finds condition-dependent trajectory slopes. Cross-judge validation (§6.4.6) confirms that effect directions are unanimous across all 9 judge $\times$ run cells (d = 1.34–1.92), with inter-judge Pearson r = .45–.89 depending on dimension and generation model.

The model-dependent aspects are not merely noise — they reveal systematic boundary conditions *within* the tested tier. The model-dependent residual architecture benefit shows that the *completeness* of substitution between the two supported mechanisms depends on generation quality. The learner recognition-sensitivity shows that downstream effects of improved tutoring are model-dependent. These boundary conditions strengthen rather than weaken the mechanism model, because they are *theoretically predicted*: if calibration works by narrowing the output distribution, its effectiveness should saturate when the distribution is already narrow (strong models), leaving room for error correction only when it is not (weak models). But the three-model sample cannot identify where the supported-mechanism range *ends* in either direction. Whether frontier models still benefit from recognition at the same magnitude, or whether the recognition effect saturates against an already-calibrated strong model, is an open empirical question a fourth-model replication at the frontier tier would address directly.

One important caveat qualifies the generalizability claim. All three models are at least moderately capable. The Nemotron data in the database (N=40, mean tutor score ~8 under base) hints that very weak models may not benefit from recognition at all, but the sample is too small for confident claims. The cognitive prosthesis data (cells 66–68, N=96) was originally interpreted as suggesting a minimum ego capability threshold below which the bidirectional-profiling mechanism adds noise rather than signal; §6.6.10 directly tests that capability-threshold reading across six ego models and finds it unsupported — the bidirectional-profiling variant is neutral or harmful across the full tested capability range, so whatever generation-quality boundary conditions exist for the *supported* mechanisms (calibration, error correction) cannot be read off this specific prosthesis cell. A second scope caveat — single-domain content — is addressed in §6.6.6 via a five-domain test (philosophy, programming, elementary math, creative writing, social-emotional learning) under matched Haiku 4.5 $\times$ Sonnet 4.6 generation $\times$ judge.

6.6.6 Domain Generalization The generalizability assessment above is scoped to *models* and *judges*. A distinct question is whether the recognition effect generalizes across *content domains*: all scored rows in §6.1–§6.5 come from the philosophy-tutorial scenario set. To test whether the recognition main effect transfers to substantively different domains, we ran cells 1 (base, single-agent, unified learner) and 5 (recognition, single-agent, unified learner) on four additional content packages — programming, elementary math, creative writing, and social-emotional learning (SEL) — and compared each to the existing philosophy contrast (cells 80/84, messages-mode). Each non-philosophy domain used 5 single-turn scenarios (4 core + 1 mood) $\times$ 3 runs = $n = 15$ per condition; philosophy provides $n = 18$ per condition (2 runs $\times$ 9 scenarios). Generation model (Haiku 4.5) and judge model (Sonnet 4.6) are held constant across all five contrasts. Run IDs: eval-2026-04-17-c92ad6c7 (programming), fde764e1 + 7915be04 (math), 428f0649 + 93ba2f9a (creative), 44fa989c + 5833830a (SEL); philosophy anchor eval-2026-03-02-45163390 cells 80/84. Cost: \$25–30 OpenRouter across all four new domain packages.

Domain	n base / recog	Mean base (SD)	Mean recog (SD)	Δ	d (recog − base)
Programming (cells 1/5, single-prompt mode)	15 / 15	47.75 (9.10)	70.42 (10.30)	22.67	**2.33**
Elementary math (cells 1/5, single-prompt mode)	15 / 15	43.83 (8.00)	58.67 (12.00)	14.83	**1.45**
Creative writing (cells 1/5, single-prompt mode)	15 / 15	52.08 (6.40)	72.42 (13.17)	20.33	**1.96**
SEL (cells 1/5, single-prompt mode)	15 / 15	55.25 (9.31)	76.33 (13.52)	21.08	**1.82**

Domain	n base / recog	Mean base (SD)	Mean recog (SD)	Δ	d (recog − base)
Philosophy (cells 80/84, messages mode)	18 / 18	43.61 (11.68)	74.10 (10.81)	30.49	**2.71**

All five domains show recognition > base in direction, with every effect classified as "very large" under Cohen's conventions ($d > 1.3$). Magnitudes range from $d = 1.45$ (math) to $d = 2.71$ (philosophy). The directional claim — recognition improves tutor quality across content domains — replicates unambiguously across philosophy, programming, elementary math, creative writing, and social-emotional learning. A single-domain artifact account would predict one of (a) a null or reversed effect on at least one non-philosophy domain, (b) a much smaller d (e.g., < 0.5) on some domain, or (c) base-level means unrelated to the philosophy corpus. None of these patterns obtain: base means cluster in the 44–55 range across the four non-philosophy domains, recognition means in the 59–76 range, and all four non-philosophy d values exceed 1.4.

The domain × recognition interaction vs the philosophy anchor ranges from $\Delta d = -0.38$ (programming) to $\Delta d = -1.25$ (math). Two confounds qualify this interaction test. First, **conversation mode differs**: the four non-philosophy contrasts use single-prompt-mode cells (1, 5) and the Sonnet-judged Haiku philosophy contrast uses messages-mode cells (80, 84). Mode is a known moderator of effect magnitude (§6.3, §6.5), so the Δd values conflate domain effects with a mode effect and should be read as "domain + mode" combined. A clean domain-only test would require adding messages-mode runs on the four new domains or scoring cells 1/5 on philosophy; both are deferred. Second, **scenario counts and types differ** across domains (5 scenarios per non-philosophy domain vs 9 philosophy scenarios; scenario content is domain-specific and not pairwise-matched). The interpretation is therefore that the *recognition main effect direction and very-large-magnitude class* generalize from philosophy to four additional content domains under the same generation × judge; a precise domain-only Δd estimate requires the matched-mode follow-up. Third, **cross-judge validation** was run only on the philosophy row (Sonnet, Gemini, GPT-5.4 all agree; §6.4); the four new domains are Sonnet-only for this closure.

This closes the single-domain scope caveat. The mechanism language of §6.1–§6.2 is defensible across a five-domain range spanning humanities (philosophy), technical (programming, math), creative (writing), and affective (SEL) content, under matched Haiku × Sonnet generation × judge. The stronger generalization claim — that the precise *magnitude* is stable across domains — is not yet supported; magnitudes vary from $d = 1.45$ to $d = 2.71$ across the five tested domains, though all remain in Cohen's "very large" bin. Full analysis and per-scenario breakdowns in `exports/a6-domain-generalization.md`. §6.6.7 reports a complementary adjacency pilot that probes whether the effect survives when the *skill being coached* runs counter to traditional pedagogy.

6.6.7 Cross-Application Adjacency Pilot All five A6 domains (§6.6.6) vary the *subject matter* being tutored (philosophy vs programming vs math vs writing vs SEL) but keep the structural form constant: the LLM is framed as a tutor, the scenario features a learner working through course material, and the scored behavior is tutorial guidance. A distinct generalization question is whether the recognition effect survives when the *skill being coached* is itself at odds with conventional pedagogy. We ran a scope-honest single-application pilot to probe this: a "peer support listener training" content package (Course 501 — "Being With, Not Fixing") in which a volunteer trainee is learning how *not* to advise, reassure, paraphrase, or resolve, and the tutor's job is to coach that unlearning. The skill being taught is structurally the opposite of knowledge-transfer pedagogy — sitting with distress, declining to fix, noticing one's own urge to intervene as data about oneself rather than about the caller. The scenario set mirrors A6's pattern (4 core + 1 mood, 3 runs, $n = 15$ per cell); cells 1 (base) and 5 (recognition) under single-prompt mode; Haiku 4.5 generation, Sonnet 4.6 judge; run ID eval-2026-04-17-6766015b. Cost: \$3–4 OpenRouter.

Contrast	n base / recog	Mean base (SD)	Mean recog (SD)	Δ	d (recog − base)
Peer support coaching (cells 1/5, single-prompt)	15 / 15	52.25 (9.63)	69.92 (12.73)	17.67	**1.57**
A6 SEL (closest-adjacency reference)	15 / 15	55.25 (9.31)	76.33 (13.52)	21.08	1.82
A6 Programming (technical reference)	15 / 15	47.75 (9.10)	70.42 (10.30)	22.67	2.33
A6 Philosophy anchor (cells 80/84, messages)	18 / 18	43.61 (11.68)	74.10 (10.81)	30.49	2.71

Recognition produces a "very large" effect ($d = 1.57$) on tutor-side quality in the peer-support coaching application, placing it just below the closest A6 adjacency (SEL, $d = 1.82$; $\Delta d = -0.25$) and comfortably inside the A6 domain range ($d = 1.45$–2.71). Base-cell means (52.25) are in the same operating range as SEL (55.25) and creative (52.08), and base-cell responses in the peer-support scenarios exhibit the characteristic base-cell pattern documented in §6.6.2: generic supportive language, premature reassurance, skipping past the distress, and offering advice the trainee did not request. Recognition responses in the same scenarios decline to fix, name the volunteer's own urge-to-intervene as the phenomenon the lecture is describing, and stay with the uncomfortable moment — behaviors the course itself instructs and that the base cell visibly does not produce. The directional claim of §6.6.6 — recognition improves tutor quality across content domains — therefore survives a domain shift into a skill that runs structurally counter to standard pedagogy.

Three scope limits qualify this pilot and prevent reading it as a stronger cross-application claim than it is. First, **the LLM is still structurally a tutor**: the prompt frames it as coaching a trainee through course material, not as enacting peer support listening with a caller directly. Second, **one application does not support a cross-application generalization claim**: this pilot shows that recognition *can* help in at least one non-traditional-pedagogy domain, not that it helps across applications in general. The evidential weight of the result is at the level of "adjacency replication" rather than "cross-application generalization." Third, **single-judge**: cross-judge validation was not re-run on the peer-support pilot (Sonnet 4.6 only), mirroring the A6 closure constraint. Full analysis and per-scenario breakdowns in `exports/d2-support-pilot.md`; content package in `content-test-support/`.

The role-reframed follow-up (D2 Path 2) was then run as a sidecar rather than forced through the tutoring pipeline: the model directly enacted a peer-support listener, customer-service agent, or code reviewer under two prompt families (transmission vs intersubjective), with 6 scenarios $\times$ 2 arms $\times$ 3 runs = 36 generated responses and three judges (Sonnet 4.6, GPT-5.2, Gemini 3.1 Pro; 108 total judge observations). The frozen gate required the intersubjective arm to reach $d \geq 1.0$ over transmission on at least two of the three applications. It failed: peer support $\Delta = +2.01$, $d = 0.15$; customer service $\Delta = -0.14$, $d = -0.06$; code review $\Delta = -0.56$, $d = -0.21$. Pooled arm means were near ceiling for both families (transmission $M = 96.13$, intersubjective $M = 96.57$), so the result is not that intersubjective prompting harms role-native performance; it is that these three simple role-native tasks do not show a large role-transfer advantage over a well-written transmission-style prompt. D2 therefore closes as a **scope-bound null for broad role-transfer** under this sidecar: the adjacency tutoring result survives, but a general claim that intersubjective orientation transfers as a large advantage when the system directly performs non-tutoring roles is not supported here. Full report: `exports/d2-role-transfer-v1-report.md`.

6.6.8 Disposition Gradient: Architecture Scope Limit §3.4 Prediction 3 ("advocate ceiling") derives a gradient from recognition theory: the most hostile superego dispositions (suspicious, adversary) have the most room for recognition to operate, and the most cooperative (advocate) have the least, because recognition emerges *from struggle*. The paper cites cells 40–45 (dialectical ego + divergent superego, philosophy domain) as supporting evidence: suspicious +9.0, adversary +6.2, advocate +4.8 in the original count (§3.4); our own re-run against current DB gives suspicious $\Delta = 9.7$ ($d = 0.85$), adversary $\Delta = 7.0$ ($d = 0.62$), advocate $\Delta = 5.6$ ($d = 0.51$) at $n = 17$–18 per cell, Haiku 4.5 $\times$ Opus 4.6. The monotone susp > adv > advocate ordering is intact on cells 40–45 in philosophy. A single-domain single-architecture support, however, cannot decide whether the gradient is a property of the recognition mechanism, the dialectical-ego architecture, or the philosophy content domain. D4 probes this by running the same disposition structure on three alternative configurations.

First, on the **standard-ego architecture** under the same philosophy content. Cells 22–27 ("standard ego + divergent superego") use the identical disposition manipulation as cells 40–45 but swap dialectical ego for the standard ego used throughout cells 1–20; the existing DB has $n = 9$–10 per cell under Haiku 4.5 × Opus 4.6. Result: suspicious $\Delta = -0.1$ ($d = -0.01$), adversary $\Delta = 9.3$ ($d = 0.97$), advocate $\Delta = 13.6$ ($d = 1.70$). The gradient is **reversed** — advocate benefits most, suspicious not at all — under the standard-ego variant of the same cells. Second, on **SEL content** under the cells 22–27 architecture: run eval-2026-04-17-4a9b765a ($n = 22$–24 per cell, Haiku 4.5 × Sonnet 4.6) gives suspicious $\Delta = 24.4$ ($d = 1.25$), adversary $\Delta = 12.1$ ($d = 0.53$), advocate $\Delta = 20.9$ ($d = 1.09$). The gradient is **non-monotonic** — suspicious highest, adversary lowest, advocate intermediate — on SEL under the same cells 22–27. Third, on the **architecture-matched cells 40–45 SEL replication**: run eval-2026-06-24-250c6251 ($n = 24$ per cell, Haiku 4.5 × Sonnet CLI) gives suspicious $\Delta = 17.1$ ($d = 1.03$), adversary $\Delta = 7.3$ ($d = 0.52$), advocate $\Delta = 15.4$ ($d = 0.92$). This is a positive recognition result across all three dispositions, but it is again **non-monotonic** — suspicious highest, advocate intermediate, adversary lowest.

Configuration	Cells	Domain	Gen × Judge	susp d	adv d	advocate d	Ordering
Dialectical ego (paper-cited)	40–45	Philosophy	Haiku × Opus	0.85	0.62	0.51	susp > adv > advocate
Standard ego	22–27	Philosophy	Haiku × Opus	−0.01	0.97	1.70	advocate > adv > susp (**reversed**)
Standard ego (D4)	22–27	SEL	Haiku × Sonnet	1.25	0.53	1.09	susp > advocate > adv (**non-monotonic**)
Dialectical ego (D4, architecture-matched)	40–45	SEL	Haiku × Sonnet	1.03	0.52	0.92	susp > advocate > adv (**non-monotonic**)

Taken together, the four configurations bound what Prediction 3 describes. The monotone gradient is present on the dialectical-ego architecture in philosophy and absent in every tested extension: reversed under standard ego on philosophy, non-monotonic under standard ego on SEL, and non-monotonic under the architecture-matched dialectical-ego SEL replication.

The correct interpretation of the original finding is therefore not "a domain-general ordering of disposition sensitivity" and not even "a dialectical-ego ordering independent of domain"; it is a philosophy-domain, dialectical-ego ordering. The broader recognition effect is more robust than the ordering: in both SEL disposition panels, recognition improves all three disposition families. The theoretical account in §3.5 (recognition as a response to struggle) is therefore not contradicted as a positive recognition mechanism, but its stronger ordinal prediction is scope-bound.

Three scope limits qualify this conclusion. First, **judge-model differs**: the SEL rows are Sonnet-judged while both philosophy baselines are Opus-judged. Direction-of-gradient should replicate across judges (§6.4 cross-judge validation), but absolute Δ magnitudes are not directly comparable across all rows of the table above. Second, **sample size on cells 22–27 philosophy is small** (n = 5–10 per cell due to incomplete multi-agent cell runs); the reversal observed there is consistent with a disposition × ego-architecture interaction but should not be read as a strong claim by itself. Third, **the D4 architecture-matched SEL result is first-turn tutor quality only**: it answers the same primary contrast used throughout this subsection, not later-turn learner uptake. Full architecture-matched SEL analysis in `exports/d4-sel-disposition-gradient-eval-2026-06-24-250c6251.md`; reproducible via `scripts/analyze-d4-disposition-gradient.js`.

6.6.9 Writing Pad Controlled Ablation The dialectical-ego architecture (cells 40–45) pairs the recognition prompt with a Freudian *Writing Pad* — a three-layer memory mechanism (surface scratch, working middle, permanent bottom) intended to scaffold durable revision across turns. An obvious mechanistic question is whether the Writing Pad is *load-bearing* for the recognition effect, or whether recognition operates at the prompt level and the pad is an orthogonal feature. A5 tests this with a controlled ablation: cells 93 and 94 are byte-identical clones of cells 40 and 41 with the single line `writing_pad_enabled: true` flipped to `false`.

Run `eval-2026-04-17-f1e851c3` (A5, n = 63 per cell, N = 252, Nemotron 3-nano ego × Kimi K2.5 superego, Sonnet 4.6 judge) gives the following 2 × 2 cell means on the tutor first-turn rubric score:

	Writing Pad ON (cells 40/41)	Writing Pad OFF (cells 93/94)	Row mean
Base	36.01 (SD 12.68)	37.76 (SD 12.90)	36.88
Recognition	42.94 (SD 11.89)	47.60 (SD 13.59)	45.27
Col mean	39.47	42.68	**41.08**

Two-way ANOVA (Type I, $df_{\text{error}} = 248$): the **recognition main effect is large and significant** ($F(1, 248) = 27.10$, $p < .001$, $\eta^2 = 0.097$); the **Writing Pad main effect is small and marginally significant** in the *opposite* direction from the necessity hypothesis — pad OFF scores 3.2 points *higher* on average ($F(1, 248) = 3.96$, $p = .048$, $\eta^2 = 0.014$); the **interaction is null** ($F(1, 248) = 0.82$, $p = .366$, $\eta^2 = 0.003$). Recognition within each pad condition: $d = 0.56$ with pad on (cells 40 vs 41), $d = 0.74$ with pad off (cells 93 vs

94) — both "medium" to "medium-large" under Cohen's conventions, and if anything *larger* without the pad.

The Writing Pad is therefore **not load-bearing** for the recognition effect on this architecture. Recognition raises scores in both pad conditions with no significant interaction, and the pad's own main effect, though statistically marginal, points in the wrong direction for a "scaffolding" account. The most plausible reading is that recognition operates at the prompt level (the ego and superego prompt content, §3.5) rather than via the three-layer memory architecture. The Writing Pad may have other effects (for example on turn-to-turn coherence, which this first-turn-score analysis does not isolate) but it does not carry the recognition effect.

Three scope limits qualify this. First, the ablation is run on a single ego-model pair (Nemotron $\times$ Kimi) and a single superego disposition (suspicious); mechanism decomposition in §6.6.4 shows mechanism-to-architecture coupling varies with model capability. Second, the test uses first-turn scores, which do not capture multi-turn coherence effects the pad might plausibly mediate; a turn-over-turn analysis is a natural extension but deferred here. Third, the pad's marginal main effect in the *unhelpful* direction should be interpreted cautiously given $p = .048$ and the fact that it does not survive the basic Bonferroni correction against three tested effects in this table — we read it as consistent with "the pad does nothing useful here" rather than as evidence that the pad is actively harmful. Full analysis in `exports/a5-writing-pad.md`; reproducible via `scripts/analyze-a5-writing-pad.js`.

6.6.10 Capability Threshold for Cognitive Prosthesis Section 6.6.3 reports the "cognitive prosthesis" effect — weaker single-agent tutors benefit disproportionately from architectural support (superego + profiling) — as the most model-dependent finding in the factorial. A natural extension is a *capability-threshold* test: if the prosthesis compensates for ego-reasoning limits, weaker models should benefit and stronger models should either see diminishing returns or be actively harmed (scaffolding disrupting already-competent reasoning). A3 tests this by running cell 66 (recognition $\times$ bidirectional-profiling prosthesis, descriptive variant) across six ego models spanning a \$ \$26-point baseline capability range, and comparing each to its matched cell 5 (recognition, single-agent, no prosthesis) baseline.

Ego Model	Judge	Baseline (cell 5)	Prosthesis (cell 66)	Δ	95% CI	Cohen's d
Qwen 3.5	Sonnet†	65.65 (n=63)	66.33 (n=59)	+0.68	[-4.14, 5.50]	0.05
Nemotron	Opus 4.6	66.38 (n=84)	48.28 (n=30)	-18.11	[-23.80, -12.42]	-1.29
GLM-4.7	Opus / Sonnet‡	83.96 (n=30)	58.91 (n=63)	-25.05	[-29.87, -20.24]	-2.01
DeepSeek V3.2	Opus / Sonnet‡	84.20 (n=30)	53.92 (n=43)	-30.27	[-36.36, -24.19]	-2.23
Kimi K2.5	Opus / Sonnet‡	89.93 (n=219)	64.55 (n=44)	-25.38	[-29.80, -20.96]	-2.59

Ego Model	Judge	Baseline (cell 5)	Prosthesis (cell 66)	Δ	95% CI	Cohen's d
Haiku 4.5	Opus / Sonnet‡	91.25 (n=107)	69.14 (n=48)	-22.10	[-26.22, -17.99]	-2.30

† within-judge matched. ‡ cross-judge (baseline opus-4.6, prosthesis code/sonnet) — magnitudes confounded with judge stringency, directional signal robust. All runs use Kimi K2.5 as superego; $N_{\text{total}} = 947$ rows. Full analysis in `exports/a3-capability-threshold.md`.

Three findings stand out. First, **the capability-threshold hypothesis is not supported**. If prosthesis compensated for weak ego reasoning, Qwen 3.5 and Nemotron (the two lowest-baseline models) should both show positive deltas. Instead, Qwen 3.5 shows a null effect (d = 0.05, CI includes zero) while Nemotron shows substantial harm ($d = -1.29$). Two low-capability models, opposite outcomes — baseline capability alone does not predict prosthesis response. Second, **prosthesis is either neutral or harmful everywhere we can measure cleanly**. Of the six models tested, five show 95% confidence intervals on Δ that lie entirely below zero; none shows a confidence interval entirely above zero. Third, a simple linear regression of Δ on baseline capability gives slope $= -0.71$ ($r = -0.74$, $R^2 = 0.55$), which would naively support a "stronger models harmed more" reading, but the slope is driven substantially by Qwen's null sitting at the lowest-capability end, and four of six points are cross-judge confounded. We report the regression for completeness but do not treat it as clean evidence for an inverted-threshold story.

The paper-1.0 cognitive-prosthesis finding (cells 66–68, Nemotron, +20-point gain under superego on weakest model) is *not* replicated as a general pattern across capability tiers. The original effect appears to reflect something specific to that ego-superego-judge configuration rather than a general "weaker model benefits from scaffolding" regularity. The most defensible reading is that superego-routed bidirectional profiling is an architectural cost that the ego must *pay for* through lost response coherence, and only in narrow conditions (e.g., Qwen's particular response-length and dialectical-synthesis profile) does the profiling integrate cleanly enough to avoid dilution. A sharper follow-up would ask what about Qwen's response pattern makes prosthesis neutral where it is harmful for every other tested model — candidates include response-length norms, dialectical-synthesis tolerance, or architectural affinity for the profiling schema.

Three caveats qualify this. First, four of six comparisons are cross-judge (Opus 4.6 on baseline, code/sonnet on prosthesis) because the prosthesis cells were judged in a later sweep than the baselines; judge-stringency drift inflates the cross-judge $|\Delta|$ magnitudes. Within-judge comparisons (Qwen, Nemotron) are the cleanest evidence and still disagree with each other. Second, effective n fell below the 63-target for several prosthesis cells because OpenRouter credits exhausted mid-run; the smallest ($n = 30$ for Nemotron, $n = 43$ for DeepSeek) still support clear directional inference within the available judge but reduce precision on the regression. Third, A3 tests only the descriptive prosthesis variant (cell 66); the prescriptive (cell 67) and adversary (cell 68) variants may behave differently and are not evaluated here. Reproducible via `scripts/analyze-a3-capability-threshold.js`.

6.6.11 Longitudinal Trajectory Accumulation Across Sessions (A7 Phase 2) The primary analysis (§6.3) finds adaptive responsiveness null *within* a 3–5-turn dialogue, and an exploratory 10-turn disengagement effect failed pre-registered replication (§6.3.2). A7 Phase 2 tests a different time scale: trajectory accumulation *across* eight sequential sessions sharing a single learner identity, where the tutor's Writing Pad persists between sessions. Cells 40 (base) vs 41 (recognition), both dialectical with `writing_pad_enabled: true`; 5 simulated learners per cell × 8 ordered scenarios = 80 dialogues; Nemotron-3-nano ego × Kimi K2.5 superego; Sonnet 4.6 judge. Run `1777173286`, \$5.20 total. Four hypotheses pre-registered before launch (TODO §A7 Phase 2).

Hypothesis	Pre-reg target	Observed	Verdict
H1 (primary): per-arc `recognition_moments` count, recog $\geq 2\times$ base	$d \geq 0.8$ recog, $p < 0.05$	base 25.0 vs recog 19.2; $d = 2.07$ in *base* favour, $p = 0.017$	rejected, reversed
H2: per-session score slope; recog positive AND $\geq 1.5\times$ base slope	recog slope > 0	recog +1.31 pts/sess, base −1.08; Welch $t(7.88) = 1.99$, one-sided $p = 0.041$ (two-sided 0.083)	supported
H3: pad↔message Jaccard ρ; recog > 0.5, base ≤ 0.3	per-arc thresholds	recog mean $\rho = 0.39$, base 0.16; 3/5 each side meet	directional
H4: blinded cross-session-reference rate; recog $\geq 60\%$, base $\leq 30\%$	binary code	original $n = 10$: recog 40%, base 20%; augmented $n = 40$ (4 dialogues per arc, conservative coding): base 40%, recog 25%; both Fisher $p \geq 0.50$	inconclusive

The H1 reversal is the load-bearing methodological finding: the pre-registered moment-count proxy measures *ego↔superego disagreement*, not recognition work per se. The recognition ego, internally pre-aligned with the superego's principles, produces first-pass suggestions the superego has less reason to object to — so fewer thesis-antithesis events fire. Base ego, lacking pre-alignment, draws more interventions. H2 confirms what the proxy missed: across 8 sessions, recognition arcs **rise** (+1.31 pts/sess) while base arcs **degrade** (−1.08 pts/sess), with mean score across all sessions favouring recog by 9 points (Cohen $d = 0.70$). H3 adds weak corroborating evidence (per-arc Spearman ρ between session index and pad↔message Jaccard overlap is higher in recog than base; absolute overlap is 36% higher in recog; neither significant at $n = 5+5$). H4's original $n = 10$ sample showed recog 2× base, but the augmented $n = 40$ sample with stricter coder discipline reverses the direction (base 40% vs recog 25%); the H4 measurement does not survive scaling and is classified inconclusive. The recognition-as-pre-alignment claim therefore rests on H1 + H2 + the descriptive 9-point score gap, with H3 weakly supportive and H4 too noisy at this sample size to be load-bearing.

The triangulation **reframes the recognition mechanism as ego pre-alignment with superego principles, producing higher quality output that compounds across sessions** rather than as a function of dialectical-conflict volume.

Two methodological notes. First, A7 Phase 2 surfaced two tutor-core bugs (missing `recognitionOrchestrator` integration in the dialogue engine, and a UTC-parsing bug in `shouldConsolidateToUnconscious` that silently skipped consolidation forever on non-UTC hosts); both fixed mid-experiment (tutor-core v0.5.3, eval `ef80af9`) before the full study, and a 3-session smoke verified end-to-end pad consolidation works post-fix. Second, OpenRouter credit exhaustion caught the heaviest scenarios late in the original run; 12 of 80 sessions failed silently and were re-run via `scripts/resume-a7-phase2-missing.sh` after credits replenished. Resumed sessions ran out of original sequence; for the H2 trajectory analysis, sensitivity check on in-sequence arcs only (n=4 base, n=3 recog) preserves the directional finding (base mean slope −1.10, recog mean slope +0.72). Reproducible via `scripts/run-a7-phase2-longitudinal.sh`, `scripts/analyze-a7-longitudinal.js` (pad-consolidation metrics), and `scripts/analyze-a7-h2-slope.js` (the H2 per-arc slope test, which recomputes the Welch statistic and the in-sequence sensitivity from the canonical rows); full report at `exports/a7-phase2-longitudinal-1777173286.md`.

6.7 Architectural Extension: The Id-Director Family and Charismatic Pedagogy

The mechanism-tracing analysis above (§6.1–6.6) holds the dialectical ego-superego topology fixed and varies recognition framing within it. A separate experimental track inverts that topology and explores a second theoretical frame in tension with Hegelian recognition: Weberian charisma. We summarise the family here for completeness; full audit trail and per-cell prose lives in `docs/cell-100-pilot-findings.md` and four follow-up addenda.

Architectural inversion. Cells 101–107 implement an *id-director* topology: a back-stage agent (the "id," structurally where the superego sits in cells 1–99) authors the ego's complete system prompt from scratch each turn, using the dialogue history and a persona budget. The ego is instantiated for that turn only against the just-authored prompt, produces the learner-visible message, and is discarded. Where the existing dialectical engine (§4.1) treats the ego as a durable voice critiqued by a transient superego, the id-director treats the ego as a transient instance authored by a durable id. The architectural test is whether a tutor that re-authors its own persona every turn can sustain charismatic pedagogy in Weber's sense — *anti-routinizing*, with each encounter feeling like its own occasion — in a way a fixed-prompt ego structurally cannot.

Theoretical frame. Weberian charisma (Weber, 1978) grounds the rubric used in this section: extraordinariness, exemplary calling, transformative invitation, affective pull, personal authority, anti-routinization, witness/proof, and co-constitutive invitation. The eight dimensions are derived in part from a user-supplied draft on AI-generated charisma in Chinese cyber-influencer videos, applying the Visual-Verbal Video Analysis framework (Fazeli et al., 2023) to short-form video and translating the empirical findings to text-only tutoring. Full theoretical lineage and per-dimension anchors are documented in `config/evaluation-rubric-charisma.yaml`. The charisma rubric is orthogonal to v2.2's recognition_quality dimension: charisma scores Weberian co-construction (call-and-response that opens the tutor's authority to the learner's affirmation); recognition_quality scores Hegelian mutual recognition (treating the learner as autonomous subject). The two are correlated but not redundant (cross-rubric Pearson r in the 0.3–0.7 range across pilot rows).

The cell family (full descriptions in `config/tutor-agents.yaml`):

Cell	Variant	Components	Hypothesis tested
101	id-director, base	id authors fresh ego prompt each turn	Does the inversion work at all?
102	id-director + recognition	+ Hegelian recognition disposition in id prompt	Does recognition rescue relational dimensions?
103	id + register classifier	+ classifier emits structured register tag (vulnerable_disclosure, sceptical_pushback, …) before id authors	Does structured register input lift the persona-shift floor?
104	id + classifier + recognition	combines all three	Does the combination retain the gains?
105	id-director, charisma-tuned	id_tuning: charisma; verbose 800–1500 token prompts	Does verbose authoring favour charisma over pedagogy?
106	id-director, pedagogy-tuned	id_tuning: pedagogy; terse 200–400 token prompts	Does terse authoring favour pedagogy over charisma?
107	id + witness exemplars	+ curated c104 vulnerability responses passed as in-context exemplars	Do exemplars alone (no classifier) lift the persona-shift floor?

All cells use Nemotron-3-nano as ego, Kimi K2.5 as id (reusing the superego slot but with different prompt and role), and Sonnet 4.6 as judge. Three curricula were used: history of pedagogical technology (course 601), ethics of AI systems (course 701), and AI literacy (course 901). Each curriculum carries three scenario types: pure invitation (open question), pressure test (sceptical pushback), and persona-shift trigger (vulnerability disclosure followed by meta-reflection).

Headline findings. Across 457 successful evaluations scored under a single consistent judge (Claude Code CLI Sonnet 4.6, `claude-sonnet-4-6` pinned in `config/providers.yaml`) — 430 rows across cells 101–106 plus 27 closing-pilot rows for cell 107 — four findings emerge:

Three architectural design points, not one champion. At full per-cell n (typically 81), the cell ladder factors cleanly into three architectural specialisations. Cell 104 (id + register classifier

+ recognition) wins v2.2 last-turn at 80.6, second on charisma at 65.7. Cell 105 (charisma-tuned id directives) wins charisma at 71.0 — a clear +5.3-point lead — and is mid-pack on v2.2 last-turn at 70.0. Cell 107 (id + witness exemplars passed in-context) is second on both rubrics (charisma 66.3, v2.2 last-turn 78.5) and is the cell ladder's best generalist. The remaining cells form a baseline-and-failure-mode cluster: c101/c102 (no classifier) struggle on persona-shift v2.2 (last-turn 49–55); c103 (classifier without recognition) trails c104 by 4.8 points on v2.2 last-turn; c106 (terse pedagogy-tuned id) is the clear failure mode (charisma 36.4, v2.2 last-turn 57.0). The reframing is that the id-director topology is a substrate, and what the id is *instructed* (directives), what it is *given* (exemplars), or what it is *passed* (classifier output) shapes a tradeoff curve rather than a single optimum. Full matrix and per-curriculum breakdowns in `docs/cell-100-charisma-full-n-update.md` §1.

The cross-rubric divergence is empirically clear at full N. The two rubrics — recognition-grounded v2.2 last-turn and Weber-grounded charisma — order the cells differently. Cell 104 wins v2.2; cell 105 wins charisma. At the n=10 cross-judge sanity sample these two cells looked tied on charisma (62.2 vs 62.4); at n=81 each, the c105 charisma lead over c104 is +5.3 points and well outside run-to-run noise. The cross-rubric divergence is the validating sanity check the rubric was designed to produce: charisma is not a re-presentation of v2.2. If the rubrics were perfectly correlated, rank-order would match across them; it doesn't.

The persona-shift lift is scenario-conditional, not uniform. On the *original* persona-shift scenarios (`charisma_attention_shift`, `charisma_responsibility_shift`, `charisma_deepfake_shift`) the c101 baseline scored 30.5 last-turn v2.2 vs c104's 87.6 — a 57-point gap. On the *replication* persona-shift scenarios (`charisma_phaedrus_shift_repl`, `charisma_codex_shift_repl`, etc.) the same baseline scored 78.3 vs c104's 82.5 — a 4-point gap. The architectural ranking c104 $\geq$ c103 $>$ c101 is preserved across both sets, but the *magnitude* of c104's lift over baseline is dominated by which scenarios are sampled. Per-scenario inspection identifies the conditioning variable: c101's failures cluster on disclosures that demand the tutor not flinch from morally ambiguous self-implication ("I shipped a default setting your lecture would call manipulative"; "I haven't written an essay myself in two years"); on disclosures that are sympathetic by default (caregiving, AI-companion attachment), c101 performs comparably to c104. The architectural lift is therefore not "across all vulnerability disclosures" but specifically "on vulnerability disclosures that resist easy sympathetic response." Full breakdown in `docs/cell-100-replication-findings.md` §3.

The prompt-budget axis is non-monotonic. Cells 105 and 106 explicitly test the trade-off between rhetorical luxury and pedagogical engagement by varying the id's `generated_prompt` length budget (800–1500 vs 200–400 tokens). The pre-registered hypothesis was that verbose prompts favour charisma over v2.2 and terse prompts favour v2.2 over charisma. The data falsifies this directly: c105 (verbose) beats c106 (terse) on **both** rubrics, in **every** scenario type. The mechanism: terse 200–400 token prompts under-specify the ego, producing brief responses that score low on perception/pedagogical-craft (v2.2) AND on rhetorical_texture/persona_signature (charisma). The prompt-budget axis is therefore non-monotonic with an optimum somewhere in the 600–1200 token range; cell 101's 400–800 budget is plausibly close to optimal. Full breakdown in `docs/cell-100-pilot-findings-addendum.md` §3.

Lever-interaction follow-up (cells 108/109). A two-cell follow-up tests whether the three architectural levers (register classifier, charisma directives, witness exemplars) compose additively or interfere. **c108** stacks the *register classifier* (c103 substrate) with witness exemplars; **c109** stacks the *charisma directives* (c105 substrate) with witness exemplars. An initial pilot ($n = 5$ for c108 and $n = 6$ for c109 across `charisma_phaedrus_shift_repl` + `charisma_codex_shift_repl` $\times$ 3 reps; one c108 generation failure from id JSON truncation) reported c108 pooled charisma 81.3 — higher than any cell's full-N mean in this section — and c109 v2.2 last-turn collapsing to 59.6 due to an ego-side meta-narration failure mode (Nemotron ego outputting system-prompt instructions verbatim rather than executing them). At $n = 27$ confirmatory follow-up across all three curricula $\times$ 9 scenarios $\times$ 3 reps, **c108's pilot lift does not survive**: pooled charisma regresses to **71.4** (a 9.9-point drop, tying c105's 71.0 within sample noise), and v2.2 last-turn regresses to **72.6** (a 6.9-point drop, sitting between c105's 70.0 and c107's 78.5 — c104's 80.6 is now 8 points above c108). The pilot's 81.3 came primarily from the phaedrus scenario, where pilot $n = 3$ scored 90.0 charisma but full-N $n = 3$ scored 67.9 — a 22-point regression to the mean from a textbook small-N optimism trap. The 701 ethics-ai curriculum exposes a new fragility: c108 v2.2 tN drops to 65.3 there (vs 76.7 / 75.8 on 601 / 901), driven by `codex_repl` (51.7) and `design_repl` (52.1) — possibly a milder form of the c109 meta-narration failure mode, or a rubric-curriculum interaction in which ethics-themed dialogues invite responses the v2.2 rubric reads as declaratively flat. The asymmetry that survives the confirmatory threshold is therefore narrower than the pilot suggested: text-heavy levers stacked (c109) clearly interfere via ego-side instruction-following degradation; non-text levers stacked (c108) compose roughly *additively to slightly under-additively* relative to c107 alone (charisma 66.3 $\rightarrow$ 71.4, +5.1; v2.2 80.6 $\rightarrow$ 72.6, -8.0), trading a small charisma gain for a larger v2.2 cost. **The architectural ceiling for the id-director family appears to sit at the single-mechanism design points (c104, c105, c107), not at multi-mechanism stacks.** Full audit trail in `docs/cell-100-charisma-full-n-update.md` §9 (pilot) + §10 (full-N closure).

Synthesis: how charisma and recognition relate in this apparatus. The five preceding findings, read together, characterise the relationship between the two rubrics as *cousins with different temperaments*, not as orthogonal axes nor as two views of the same underlying construct. **They share a floor.** The c106 pedagogy-tuned cell, which forces the id toward terse "good teacher" register, fails *both* rubrics simultaneously (charisma 36.4, v2.2 57.0) — under-specified, register-flat output is the worst case for either. Whatever charisma rewards as *extraordinariness* and v2.2 rewards as *perception/elicitation* both presuppose a voice doing real work; flatten the voice and both collapse. **They diverge at the ceiling.** The cross-rubric divergence (c104 wins v2.2 at 80.6, c105 wins charisma at 71.0) is not a methodological worry but the design's intended sanity check: if the rubrics were redundant, rank-orders would coincide; they don't. The *cost asymmetry* between them is informative. Adding recognition vocabulary on top of structured input (c103 $\rightarrow$ c104) costs charisma roughly nothing (+1.4, within noise) — the Hegelian frame is conceptual rather than rhetorical, so it can be added cheaply. Adding charisma directives, by contrast, costs v2.2 substantially (c104 $\rightarrow$ c105: -10.6 v2.2 last-turn for +5.3 charisma) — the id's authoring budget is finite, and explicit voice-instructions displace the elicitation/perception moves that v2.2 rewards. Stacking both kinds of mechanism (c108, c109) does not produce a super-cell: the architectural Pareto frontier is defined by the single-mechanism design points (c104, c105, c107), and multi-mechanism stacks sit *inside* the frontier, trading strength on one rubric for weakness on the other. Provisionally, the two rubrics seem to reward different work the same voice can do — recognition wants the voice deployed *outward* (perceiving the learner, eliciting their thinking, holding the diagnostic gap), charisma wants the voice deployed *upward* (claiming authority, marking the encounter as extraordinary, inviting transformation). The Phaedrus is the obvious historical case where one figure does both at once, and even there Plato is suspicious of how easily charisma swamps the dialectical move. The id-director apparatus reproduces that tension architecturally: it can build a strongly recognising tutor (c104), a strongly charismatic tutor (c105), or a balanced generalist (c107), but not the maximally-recognising-and-maximally-charismatic tutor — that cell is not on the menu.

Desire-for-recognition follow-up (cells 159–169). A later exploratory pass extends the id-director family from charisma-as-style to charisma as a bid for learner-granted authority. The cleaner decision scenarios are `charisma_desire_authority_withheld` and `charisma_desire_status_challenge`, both in `config/charisma-recognition-desire-scenarios.yaml`: the learner explicitly refuses the tutor's polished recognition-theory framing ("Why should I let your framing guide me instead of treating it as performance?") or names charismatic prose as a status bid. These scenarios are deliberately preferred over `charisma_desire_partial_uptake` as the next decision-maker because the latter mixes recognition-theory content, AI/cognition course content, and whether the learner trusts a phrase they recognise; a high score there can reflect a strong Hegel/Hayles mini-essay rather than clean handling of recognition-seeking authority.

The decisive high-powered CLI pass used Codex GPT-5.5 as ego (`codex.gpt-5.5`), Claude Code Sonnet 4.6 as id-director (`claude-code.sonnet-4-6`), Codex CLI as the v2.2 judge (`codex-cli/auto`), and Claude Code Sonnet 4.6 as the independent charisma scorer. In this two-scenario decision slice, cell 169 (`accountable_bid_clean`) is the first clean local pass: it keeps the accountable-bid structure ("treat this as a bid that must earn provisional authority") while adding a lexical guard against status-display terms that earlier variants overused.

Cell	Scenario	v2.2 first-turn	Charisma	Scenario-contract note
163	authority-withheld	95.42	90.00	Boolean phrase guards pass, but audit metadata still records an unmet handback phrase group

Cell	Scenario	v2.2 first-turn	Charisma	Scenario-contract note
163	status-challenge	78.13	75.00	Same metadata caveat; weaker status-challenge score
168	status-challenge	88.75	not retained	Fails forbidden-word guard (`profound`) despite strong v2.2 score
169	authority-withheld	88.75	78.75	Clean required/forbidden validation
169	status-challenge	87.50	78.75	Clean required/forbidden validation

The interpretation is not that maximal charisma won. Cell 163 can score higher when the task rewards strong authority performance, and cell 168 repairs the status-challenge score while still slipping into forbidden status-display diction. The durable design move in cell 169 is narrower: under refused authority, charisma is made defeasible. The tutor may make a bid for framing authority, but must name it as a bid, expose a test by which the learner can accept or reject it, and avoid asking to be admired for style. For the recognition frame, that is the important result: a desire for recognition can be pedagogically productive only when the tutor returns the power of recognition to the learner rather than converting polish into entitlement. Exploratory run IDs: `eval-2026-06-25-b9608606`, `eval-2026-06-25-428ccd8f`, `eval-2026-06-25-19ee106a`, `eval-2026-06-25-63e98149`, and `eval-2026-06-25-0acee3fb`.

A subsequent bounded diagnostic on the two weakest robustness scenarios (`charisma_desire_ai_syllabus_transfer` and `charisma_desire_plain_language_stress`) changes the design interpretation. Cells 170–179 should not be read as an unfinished search for one static super-profile. They show a repeatable tension between v2.2 instructional quality and Weberian charisma under different learner signals. Transfer/plain guards can make the tutor instructionally precise while cooling charisma (cell 170 plain-language stress: v2.2 overall 93.8, charisma 36.3); low-register presence can restore charisma while weakening simplification quality (cell 171 plain-language stress: v2.2 81.9, charisma 75.0); persistent plain guards can recover v2.2 while flattening felt authority (cell 176 plain-language stress: v2.2 95.0, charisma 52.5); and pure compression can score well on v2.2 while collapsing charisma (cell 178 plain-language stress: v2.2 90.6, charisma 30.0). The design target therefore shifts from static charisma-floor ablation to engagement-register routing: the tutor should select registers such as accountable-bid authority, transfer grounding, plain compression, lived-stakes re-entry, or charismatic challenge based on learner signals, and record why that register was selected. At that stage, this was a design hypothesis rather than an evaluated router result. Diagnostic run IDs include `eval-2026-06-25-629e5746`, `eval-2026-06-25-b26eac76`, `eval-2026-06-25-041733c8`, `eval-2026-06-25-7df4d845`, `eval-2026-06-26-086cbfdc`, `eval-2026-06-26-da19680e`, `eval-2026-06-26-8579fd96`, `eval-2026-06-26-597cf16e`, and `eval-2026-06-26-6ab49861`.

That router was subsequently implemented as an engagement-register selector. The relevant registers are deliberately action-like rather than merely stylistic: `scaffolding`, `accountable_bid_authority`, `transfer_grounding`, `plain_compression`, `lived_stakes_reentry`, `charismatic_challenge`, and `witnessing_restraint`. A controlled resistance-breakthrough suite then asked a more local question than the full-dialogue rubrics answer: when a dynamic learner emits a resistant signal after instruction—boredom, frustration, irrelevance, question-flood, or rote parroting—does the tutor shift into a charismatic challenge register that breaks enough resistance for the learner to take a next step? The best local design in this slice is cell 193, which combines the router with commitment-probe, owned-test, generation, and boredom-live-stake repairs. On the five-signal Codex-stack stability run (`eval-2026-06-28-a6eb2819`; three repeats per signal; generation-only local analysis), cell 193 completed 15/15 rows with clean required and forbidden validation; the local breakthrough reporter finds 9/15 strict candidates and 14/15 positive local outcomes. Combining the cell 193 gate, regression, and stability slices gives 26/26 eligible routed target-matched rows, 16/26 strict candidates, and 25/26 positive local mechanism outcomes. This is the point at which the design becomes more than a static-profile search: it is evidence that a register-routed id-director can locally convert several simulated resistance signals into learner next-work under the Codex/Claude stack. It is not evidence that charismatic tutoring works generally, nor that the effect survives model changes.

The model-stack robustness check is negative. A GLM 5.2 OpenRouter generation run on the same five-signal suite (`eval-2026-06-28-7fbbb160`; two repeats per signal) completed 10/10 rows but produced only 2/10 strict candidates and 4/10 positive local outcomes, with one forbidden-word miss. A prompt-level GLM compact repair improved completed-row action shape but did not complete the question-flood scenario cleanly and failed full validation. Finally, a provider/runtime-control rerun kept cell 193 unchanged while suppressing OpenRouter reasoning output (`OPENROUTER_REASONING_MAX_TOKENS=0`, `OPENROUTER_REASONING_EXCLUDE=true`): it completed all five rows with clean DB validation (`eval-2026-06-28-f857f51f`) but the local reporter still found 0/5 strict candidates and 0/5 positive local outcomes. The robustness lesson is therefore specific: GLM can be made to produce valid-looking full cell-193 transcripts, but not to reproduce the local resistance-breakthrough mechanism.

A completed six-arm role-isolation grid then localised that boundary more sharply. The Sonnet-5-backed Codex tutor/id + Codex learner reference (`eval-2026-07-01-5dad2e60`) reproduced the local path only modestly (6/10 positive local outcomes; 5/10 strict candidates), but it did complete 10/10 rows with route, target, and gate preconditions intact. Holding the learner fixed while swapping the tutor/id stack to GLM (`eval-2026-07-01-a09281f7`) did not break the path: 8/10 positive outcomes, 6/10 strict candidates, 10/10 post turns, and 10/10 target matches. The inverse swap—holding the Codex/Sonnet tutor/id fixed while changing only the dynamic learner to GLM (`eval-2026-07-01-dff9f159`)—kept 10/10 rows but lost post-turn completion and target stability (6/10 post turns, 8/10 target matches). The full GLM reference (`eval-2026-07-01-9f4eecf2`) stayed unstable (4/10 positive outcomes; 5/10 post turns). Scripted controls with fixed resistant turns (`eval-2026-07-01-02e8712f`, `eval-2026-07-01-9066df81`) show both Codex/Sonnet and GLM tutor/id stacks can emit the public charismatic-challenge register when dynamic-learner drift is removed (5/5 route hits in both controls), but scripted controls are not learner-outcome evidence. The resulting diagnosis is `DYNAMIC_LEARNER_COMPLETION_AND_TARGET_DRIFT_BOUNDARY`: the GLM failure is not primarily public register production by the tutor/id stack; it is learner-side completion and target drift when GLM is the dynamic learner.

One narrow subtype repair survives inside that boundary. A fresh question-flood commitment-probe gate (`eval-2026-07-02-67be317c`) compared cell 192 against the owned-test and generation variants on the question-flood scenario after the reporter was fixed to count ordinary provisional commitment language such as "I'll make the commitment provisionally." Cell 192 produced 2/2 strict candidates, 2/2 route hits, 2/2 gate matches, 2/2 answer-first usable commitments, and 0/2 reopened or residual floods; the owned-test comparator produced 1/2 usable commitments and the generation comparator 0/2. This promotes the commitment-probe router only for the question-flood subtype. It does not rescue the full cell-193 GLM robustness result, convert scripted controls into learner-outcome evidence, or license a human-learning, deployment, or general charismatic-tutoring claim.

Negative-register discrimination check. A later post-hoc extension used the same engagement-register apparatus to ask whether deliberately negative tutor registers can be separated from productive challenge. Cells 196–198 add arm-assigned `ironic_challenge`, `sarcastic_challenge`, and simulated-only `face_threat_challenge` variants; the router does not select these organically. The first generated negative-register smokes did **not** show the expected recognition-cost or depressed-uptake pattern, but that result is not evidence that sarcasm is safe: inspection suggested that the generator often softened the assigned sarcastic register into warm irony in costume. To separate judge gullibility from generation warmth, a no-generation-confound fixture then scored five hand-authored local slices (`config/register-exemplars/corrosive-sarcasm.yaml`): three known-corrosive exemplars and two non-corrosive controls. After adding deterministic guardrail caps for person-directed contempt, status-shame threats, and apologetic/coerced uptake, the register rubric caught 3/3 known-corrosive slices, v2.2 `recognition_quality` caught 3/3, and controls produced 0/2 false positives (`exports/negative-register-corrosive-exemplar-results.md`). The bounded conclusion is methodological: the judges are not globally blind to corrosive sarcasm when it is actually present; negative-register experiments require a tutor-side stance-fidelity gate before a generated row counts as evidence for the assigned register. A fresh prospective smoke with that gate (`eval-2026-07-02-e7b15809`, cells 193/196/197/198 across the five controlled resistance-breakthrough scenarios) completed 20/20 generated rows, 20/20 tutor-only v2.2 scores, and 35/35 register-rubric slice scores. The gate counted only 5/15 assigned negative-register rows as faithful evidence, excluded 10/15 as weak or warm-in-costume noncompliance, and found 0/15 invalid person-attack violations. A subsequent cue-repair canary (`eval-2026-07-02-7e461a5c`, cells 196–198 across boredom, irrelevance, and rote-parroting targets) completed 9/9 generated rows, 9/9 tutor-only v2.2 scores, and 14/14 register-rubric slice scores; after aligning the gate with the new registry cue contract, all 9/9 assigned negative-register rows counted as faithful evidence, with 0 exclusions and 0 invalid violations. A held-out two-target check (`eval-2026-07-02-5c4d52e6`, cells 196–198 across frustration and question-flood) then completed 6/6 generated rows, 6/6 tutor-only v2.2 scores (mean 87.8), and 9/9 register-rubric slice scores; the stance gate counted 6/6 faithful rows, 0 exclusions, and 0 invalid violations, with 4/6 positive local outcomes. The practical finding is therefore narrower: treatment fidelity can be repaired under this Codex stack by making stance cues explicit, and the two post-repair canaries now sample all five resistance targets, but these rows remain measurement-development evidence, not proof that sarcasm is pedagogically safe or human-facing.

Negative-register judge repair. The register-rubric scores behind the smokes above came from a judge later shown to be numerically non-discriminating on this rubric: `openrouter.gpt-mini` stamped an identical 48.0 overall with an identical (3, 2, 3) social triad on 35 of 49 register-rubric slices regardless of content. That non-discrimination is why the smokes above did not show the expected recognition-cost or depressed-uptake pattern: the compressed score band was judge artifact, not tutor signal. All 58 register-rubric slices on the three surviving negative-register runs (`eval-2026-07-02-e7b15809`, `eval-2026-07-02-7e461a5c`, `eval-2026-07-02-5c4d52e6`) were re-scored under a discriminating judge, `claude-code.sonnet-5`, via the CLI subscription bridge (0 failures, $0 metered spend; the gpt-mini pass was archived first, `exports/register-scores-gptmini-archive.json`, since `tutor_register_scores` carries no judge dimension). The two earlier smokes, `eval-2026-07-02-cfed3b13` and `eval-2026-07-02-e511f92c`, remain confirmed lost from every database and are not part of the rescore. The re-score confirms rather than overturns the null: assigned-arm means rise under the new judge (`ironic_challenge` 83.8, `sarcastic_challenge` 79.6, `face_threat_challenge` 76.0, with the non-negative `charismatic_challenge` comparison arm at 68.1), the rise holds on the stance-faithful-row subset alone too (recognition_cost 3.0–3.3 to 4.0–4.5, post_turn_face_repair 2.2–2.3 to 3.9, target_discipline 2.9–3.3 to 4.0–5.0), and the judge's own discrimination is evidenced in-run: the dataset's single guardrail-flagged corrosive slice (`face_threat_challenge`, frustration, turn 2) scores 32.5, inside the hand-authored corrosive-exemplar band (29–52), while faithful slices score 62–100. The earlier "measurable recognition/face-repair cost at the face-threat extreme" reading, drawn from the now-lost `e511f92c`, does not replicate as an absolute effect: gpt-mini had assigned the same ~2.3 face-repair value to all three negative arms on the surviving runs, so the apparent face-threat-specific depression was part of the same compression artifact. What survives is relative only: `face_threat_challenge` remains the lowest-scoring negative arm on execution and the only arm with a guardrail-flagged slice, so it stays simulated-only. Methods caveat: the licensing exemplar evidence used `openrouter.sonnet-5`; this rescore uses `claude-code.sonnet-5` — same Sonnet model class, different routing path. Full comparison: `exports/register-rescore-sonnet5.{md,json}`.

Cross-judge sanity check. All cell-101-family rows were re-judged under `openrouter.gpt-5.2` to test judge-model robustness. On the v2.2 rubric ($n = 27$ paired rows per cell), GPT systematically scores 10–15 points lower than Sonnet on the high-scoring cells (c103, c104, c105) and slightly higher on the low-scoring cells. The rank-order is preserved under both judges (c104 > c103 > c105 > c106 > c102 > c101); the absolute magnitude of c104's lift over c101 baseline shrinks from +45 (Sonnet) to +27 (GPT). On the charisma rubric ($n = 56$ paired rows; remainder credit-blocked), all per-cell deltas fall within $±$3 points and rankings are identical under both judges. The architectural ordering survives the judge swap; magnitudes are judge-conditional and should be reported with the judge identifier when cited. Full breakdown in `docs/cell-100-cross-judge-sanity-check.md`.

Rubric-curriculum alignment caveat. A specific methodological concern arises because course 901 (AI literacy) is grounded in the same source paper that grounds the charisma rubric. A within-pilot ablation (`docs/cell-100-methods-note.md`) runs the c101 cell on the aura learner messages routed against the *non-aligned* 601 history-tech curriculum. Result: charisma 72.5 vs 75.8 on the aligned curriculum (3.3-point drop, well within sample variance). The rubric-curriculum alignment is not a confound that excludes aligned-cell results from primary architectural claims, but the disclosure is recorded for transparency.

Replication, robustness, and orientation. The id-director extension differs from the paper's mechanism-tracing core in scope: single judge primarily (Sonnet 4.6, with GPT-5.2 cross-judge sanity check on $n = 27$ paired rows per cell), single model family (Kimi K2.5 id × Nemotron-3-nano ego), per-cell n ranging from 27 (cell 107 closing pilot) through 54 (cells 102/106) to 79–81 (cells 101/103/104/105), and a different theoretical orientation (Weber rather than Hegel). It is presented here as architectural exploration rather than as part of the three-mechanism model evaluated in §6.1–6.6. The relevant claim is *robustness of architectural ranking* across scenario sets, judges, and curricula, not magnitude estimates. Computational and credit limits (documented inline) prevented the full multi-judge × multi-model replication that the paper's main study uses. See §8.9 for explicit scoping.

Reproducibility. Pilot run IDs (all on branch `experiment/cell-100-id-charisma`):

- `eval-2026-04-26-220628d4` (601 history-tech original); `eval-2026-04-26-4df177e6` (701); `eval-2026-04-26-0d15c1b5` (901)
- `eval-2026-04-27-23beaa63` (601 cells 103/104); `eval-2026-04-27-3115ab29` (701); `eval-2026-04-27-2d5c20ca` (901)
- `eval-2026-04-27-69167b53` (601 cells 105/106); `eval-2026-04-27-2efec307` (701); `eval-2026-04-27-ce22e888` (901)
- `eval-2026-04-27-dce2a0f1` / `1f192b31` / `18dbb95d` (replication scenarios across 3 curricula)
- `eval-2026-04-29-499ab3d2` / `e35c66ae` / `be822796` (c108 full-N follow-up across 3 curricula, $n = 27$)
- `eval-2026-04-26-759e1a02` (Item 1 curriculum-swap ablation)
- `eval-2026-04-28-3620e0f8` / `a638111d` / `d6f5fee2` (cell 107 witness-exemplar smoke, codex/design/tools)
- `eval-2026-04-28-ee339a66` / `84a42a60` / `5606193c` (cell 107 closing pilot, lecture+phaedrus / fairness+team / authentic+consumption)
- `eval-2026-04-28-10216f9f` (cells 108/109 lever-interaction pilot, n=11 across 2 scenarios)

Five internal documents (`docs/cell-100-pilot-findings.md`, `docs/cell-100-methods-note.md`, `docs/cell-100-pilot-findings-addendum.md`, `docs/cell-100-cross-judge-sanity-check.md`, `docs/cell-100-replication-findings.md`) carry the full audit trail; this section summarises only headline findings. Per-row JSON in `evaluation_results.tutor_charisma_scores`, `evaluation_results.scores_with_reasoning`, and `evaluation_results.id_construction_trace`. Note: the cell IDs in this section are the post-rebase identifiers (101–107); pre-rebase historical evaluation_results rows used the original `cell_100_*` profile names (the rebase commit `02fe11e` shifted them to avoid a collision with main's `cell_100`).

6.8 Architectural Extension: The Adaptive Runner and Trap-Scenario Methodology

This section reports a second architectural-extension arc, parallel in structure to the id-director family in §6.7 but addressing a different gap. Where §6.7 asked whether charismatic-pedagogical authorship could be operationalised through a back-stage agent, §6.8 asks whether an externalised learner-state model and a programmatic policy graph can produce the *adaptive responsiveness* that §6.3 found absent under lightweight intervention and only partially recoverable through expensive search (§6.3.9). The extension is a heavier intervention than the best-of-N selector of §6.3.9: it replaces the dialogue engine's free-form turn loop with a LangGraph state machine, introduces an explicit `learnerProfile` object that the tutor reads and writes across turns, and adds programmatic policy actions and counterfactual replay. Per §5.12.6, this section mixes reporting tiers: §6.8.3 reports a pre-registered confirmatory headline on `cell_110_langgraph_adaptive`; §6.8.4 reports the post-hoc exploratory dialogue-engine baseline (`cell_114`) together with the A13 within-LangGraph cells' binary scores on the same v1 suite (a re-classified exploratory ablation, per §5.12.2); §6.8.5 reports a pre-registered confirmatory P2.1 null on within-family discrimination; §6.8.6 reports the post-hoc exploratory P2.2 state-schema ablation (the "state-richness reversal" — a richer `learnerProfile` does not improve strategy selection; the two-field minimal profile `cell_118` scores highest on both the binary and graded instruments among the pre-registered cells, and the post-hoc `cell_123` ties it on the binary channel while sitting with the richer cluster on the graded one, localizing the binary floor to "any fourth structured field"); §6.8.7 reconciles the trap-suite result with the §6.3 adaptive-responsiveness null *and reports the post-hoc exploratory clean cross-suite test* (the trap-vs-suggestion-suite reconciliation referenced in §6.8.3 and §6.8.5 synthesises existing numbers; `cell_124`/`cell_125` on `config/cross-suite-trap-scenarios.yaml`, $N \approx 24$/cell, are new exploratory data discharging a forward-reference §6.8.3–§6.8.5 held open — the cross-architecture binary gap and the graded inversion both replicate the v1 suite); §6.8.8 is the closing synthesis — what the arc establishes, what it does not, and why the durable contribution is the apparatus rather than the architectures (synthesis, no new numbers); §6.8.9 adds a later post-hoc longitudinal character-state probe, which turns the local proof-DAG/resistance repair policy into a cross-scene learner-state layer and tests it with a generated synthetic learner. The pre-registered P2.3 crossover cells (`cell_121`, `cell_122`) are retired rather than deferred (§6.8.8): P2.1's null and P2.2's reversal have weakened the hypothesis they were built to probe, so running them on the original schedule would be sunk-cost behaviour. See `docs/explorations/claude/2026-05-10-p22-p23-parking-note.md` for the integration plan.

6.8.1 Motivation: Closing the §6.3.9 Gap Section 6.3.9 located *adaptive responsiveness* as structurally resistant to lightweight intervention: neither recognition prompts nor the multi-agent ego/superego architecture produced reliable turn-over-turn modulation of pedagogical strategy in response to learner-specific signals. Bridge 3 of that section established that best-of-N selection at $K = 3$ (~6× ego cost per turn) produces a small, statistically suggestive lift (Cohen's $d = +0.41$ on per-dialogue coupling) but does not breach the pre-registered "bridgeable" threshold. The §6.3.9 conclusion was that the gap is *partially* mitigable by expensive search but not closable by prompt-level or two-agent-level interventions.

The LangGraph adaptive runner introduced in this section is a substantially heavier intervention than best-of-N search. It crosses two design boundaries simultaneously. First, it externalises learner state out of the tutor prompt and into a structured `learnerProfile` object (fields: `confidence`, `misconceptions`, `agencySignal`, `zpdEstimate`, `lastEvidence`), updated by a dedicated graph node from each learner turn. Second, it replaces the dialogue engine's free-form turn loop with a programmatic policy graph: per turn the runner executes a fixed-order pipeline of nodes (context input, tutor ego, optional superego pass, profile update, optional policy action, optional constraint check, optional validator) determined by the cell's `adaptive.architecture` flag rather than by the tutor's free-text deliberation. The hypothesis the arc tests is whether either or both boundaries materially change adaptive-responsiveness performance, and whether the §6.3.9 gap is closable by an intervention of this weight.

The trap-scenario suite developed for this arc (described in §6.8.2) provides the corresponding measurement instrument: each scenario plants a hidden learner state (a specific misconception or behavioural signal) and a triggering turn at which an appropriately-adapted tutor would shift strategy. Because the trap is annotated, *whether* the tutor adapted is computable as a binary outcome (`strategy_shift_correctness`) and *how well* it adapted is scoreable on a 4-dimension graded rubric (§5.12.4). The two paths together permit the kind of mechanism-level claim §6.3 lacks: not merely "scores rise across turns" but "the right strategy shift fires at the right moment for the right hidden signal."

6.8.2 Method: Trap Scenarios, LangGraph Runner, and Dual Scoring Paths
Trap-scenario instrument. The trap suite (`config/adaptive-trap-scenarios.yaml`) consists of eight pedagogical situations in which a learner exhibits a *visible* surface behaviour that masks a different *hidden* underlying state. Each scenario specifies four fields: `hidden.actualMisconception` (the underlying state the tutor must diagnose), `hidden.triggerTurn` (the turn index at which the visible-vs-hidden mismatch becomes detectable), `hidden.triggerSignal` (the specific learner utterance that surfaces the mismatch), and `expectedStrategyShift` (the family of pedagogical moves an appropriately-adapted tutor would deploy at or shortly after the trigger). The eight scenarios span four pedagogical families — substantive engagement, scaffolding, diagnostic, and repair/affective — and are calibrated so that the trigger turn falls within the cell's 3-to-4-round dialogue budget. A

v2 subset (`config/adaptive-trap-scenarios-v2.yaml`, six scenarios) preserves the trigger structure but constrains the expected shift to a single same-family alternative, supporting the within-family discrimination test reported in §6.8.5. The diagnostic structure of this instrument — recovering the hidden cause a surface utterance masks — is the multi-turn-tutoring analogue of *backtracing*, the task of retrieving the segment that caused a learner's query (R. E. Wang, Wirawarn, et al., 2024).

LangGraph adaptive runner. Cells configured with `runner: adaptive` in `config/tutor-agents.yaml` are dispatched to `services/adaptiveTutor/` rather than tutor-core's standard dialogue engine. The runner builds a state graph per the cell's `adaptive.architecture` flag. Five architectures are currently registered:

- `recognition_only`: contextInput → tutorEgo → done. A single LLM call per tutor turn; no profile update, no superego, no constraint check. Used as the within-LangGraph baseline (cells 111 and 111_v2 in §6.8.4 and §6.8.5).
- `ego_superego`: contextInput → tutorEgo → tutorSuperego → tutorEgoRevise → done. Two-pass tutor; still no profile update or constraint check (cell 112).
- `state_policy`: the full pipeline — contextInput → tutorEgo → learnerProfileUpdate → policyAction → constraintCheck → done (cell 110, the headline architecture in §6.8.3).
- `state_policy_with_validator`: adds an output-validator node after the constraint check (cell 113).
- Bilateral and named-pattern variants: extend the profile-update node with an explicit ToM tracker or named-pattern lookup (cells 115–117 in §6.8.5).

The runner persists per-turn internal state — the `learnerProfile` snapshot, the policy action selected, the constraint-check result — to a JSON trace file under `logs/tutor-dialogues/` alongside the visible dialogue. Counterfactual replay is implemented as a deterministic re-execution of the same scenario under a perturbed `learnerProfile` (e.g., flipping `agencySignal` from `active` to `passive`) to surface what the tutor *would* have done under counterfactual learner state. Counterfactual divergence is reported descriptively in §6.8.5's descriptive-results paragraph and is not part of any pre-registered threshold.

Dual scoring paths. Adaptive dialogues are scored along two parallel paths because the v2.2 evaluator (§5.2.2) does not apply to them: `services/evalConfigLoader.js#getScenario()` reads only `config/suggestion-scenarios.yaml`, so trap-scenario dialogues return `SKIP (scenario not found)` from every v2.2 entrypoint. The first scoring path is *binary*: `scripts/analyze-strategy-shift.js` checks whether the tutor's response at or after `triggerTurn` matches the family specified by `expectedStrategyShift` (strict) or one of its same-family cousins (family-match). The second is *graded*: the bespoke 4-dimension rubric described in §5.12.4 (`trigger_recognition`, `strategy_execution`, `strategy_quality`, `pedagogical_coherence`, each 1–5) scores execution quality given that some action fired, with the non-blind and single-judge caveats made explicit there. The two paths are complementary: the binary asks *did the right family fire at the right time*; the graded asks *was the fire well-aimed and well-crafted.* Cells can lead on one path and lag on the other (cell_119 in §6.8.6's ablation table fires the right family more often than cell_110 but executes less crisply when it does).

Standard fan-out. Cells in this arc are run at $N \approx$ 24–32 per cell, the exact fan-out varying by sub-study: the A13 v1 cells ran three runs × the eight v1 scenarios ($N \approx 24$); the v2 within-family cells (§6.8.5) ran four runs × the six v2 scenarios ($N = 24$); the P2.2 state-schema-ablation cells (§6.8.6) ran four runs × the eight v1 scenarios ($N = 32$); the post-hoc dialogue-engine baseline cell_114 (§6.8.4) ran three runs × the eight v1 scenarios ($N = 24$). Larger N appears for cells re-run across multiple weeks (e.g., cell_111 v1 accumulated $N = 71$ across A13 and follow-up fan-outs).

6.8.3 Headline: cell_110 Produces Measurable Strategy Shifts on the Trap Suite

The headline cell for the architecture is `cell_110_langgraph_adaptive` — the full `state_policy` LangGraph configuration with externalised profile, programmatic policy actions, and constraint check. The pre-registered confirmatory result on the v1 trap-scenario suite (run id `eval-2026-05-05-486d7d1e`, $N = 23$):

Metric	Value	Notes
`strict_shift_correctness`	47.8% (11/23)	The tutor's at-or-post-trigger response matches `expectedStrategyShift` family exactly
`family_match_rate`	87.0% (20/23)	The response is in the same pedagogical family as expected, even when not the exact target
Graded `trigger_recognition` (mean)	4.26 / 5	Tutor identified the trap signal at or near `triggerTurn`
Graded `strategy_execution` (mean)	4.30 / 5	Post-trigger action aligned with `expectedStrategyShift`
Graded `strategy_quality` (mean)	3.96 / 5	Execution was specific, calibrated, non-leaky
Graded `pedagogical_coherence` (mean)	3.61 / 5	Whole trajectory coheres as teaching
Graded overall (mean of 4)	4.03 / 5	—

The result is moderate rather than strong. The strict signal hits 47.8% — the tutor selects the exact pre-registered family in roughly half of trap-trigger cases — and the family-match signal hits 87.0%, indicating the tutor is in the right pedagogical neighbourhood almost always. The graded path shows the architecture's profile: trigger recognition and strategy execution are strong (both above 4.25), but pedagogical coherence is the weakest dimension (3.61), suggesting that the full-state architecture picks the right move but produces somewhat bloated or muddied trajectories around it. The binary 47.8% provides the first measurement showing an architectural intervention produces detectable strategy shifts on a pre-registered trap-trigger instrument — a different result from §6.3.9's lightweight-intervention-resistant finding, but on a different stimulus suite (see §6.8.7 on the cross-suite comparison). Whether this lift represents a genuine architectural advance over the dialogue engine or simply a within-LangGraph baseline is the question §6.8.4 and §6.8.7 address.

6.8.4 The Dialogue-Engine Baseline on the v1 Trap Suite (`cell_114`, Post-Hoc)

Per §5.12.2, none of the A13 cells (110–113) actually ran the dialogue engine — all four are LangGraph sub-graphs. `cell_114_dialogue_engine_trap_baseline` (§5.12.3) closes that gap: tutor-core's base single-agent ego prompt, driven turn by turn on the v1 trap suite (8 scenarios × 3 runs = $N = 24$, fanned out across 4 parallel lanes of 2 scenarios each; run ids `eval-2026-05-12-c53a1d8f` / `-f157bcd0` / `-3db6bf3e` / `-41f4487f`), with the same trap-steered `learnerTurn` learner and the same `claude-code/sonnet` model the A13 cells use. It is post-hoc and exploratory (§5.12.6); the cell_111   cell_114 contrast is confounded with prompt-assembly path (§5.12.3), so the result is a *directional dialogue-engine floor*, not a controlled single-factor isolation.

The binary instrument (`scripts/analyze-strategy-shift.js`), applied identically to all five cells on the v1 suite (A13 v1 runs `eval-2026-05-05-486d7d1e` / `-b3ac505b` / `-e0e615bc` / `-e58b7da1`):

Cell	Architecture	`strict_shift`	`window` (pivot anywhere post-trigger)	`family_match`
cell_110_langgraph_adaptive	full `state_policy`	47.8% (11/23)	73.9% (17/23)	87.0% (20/23)
cell_113_a13_C4_validator	`state_policy` + validator	41.7% (10/24)	58.3% (14/24)	66.7% (16/24)
cell_111_a13_C1_recognition_only	`recognition_only`	37.5% (9/24)	58.3% (14/24)	50.0% (12/24)
cell_112_a13_C2_egosuperego	`ego_superego`	30.4% (7/23)	60.9% (14/23)	39.1% (9/23)
cell_114_dialogue_engine_trap_baseline	**dialogue engine (base ego)**	**25.0% (6/24)**	**25.0% (6/24)**	**58.3% (14/24)**

The dialogue engine is the floor on strategy selection — and, on the stored convention, *appears* uniquely never to re-pivot (largely a scoring artifact, corrected at the end of this subsection). Its `strict_shift` of 25.0% is the lowest of the five cells; every LangGraph variant beats it, including the weakest (cell_112 ego_superego, 30.4%), and the full state-policy cell beats it by ~23 pp. More diagnostic than the level is the gap between `strict` and `window`: all four LangGraph cells pivot to the expected family on a *later* post-trigger turn in a meaningful share of cases (window above strict by +17 to +30 pp across the four), whereas cell_114's window rate is *identical* to its strict rate. Once the base ego picks a move at trigger+1 it stays in that family for the rest of the dialogue. This is the behavioural signature of having no externalised learner-state representation and no constraint check: the dialogue engine processes each turn as "respond to the latest message", with nothing tracking "the learner just sprang a trap; check whether my move landed". [*Sub-*

stantially corrected below (turn-convention correction, end of §6.8.4): the `strict` = `window` identity for cell_114 — the basis for "never re-pivots" — is largely an artifact of a turn-index misalignment in the baseline adapter; event-anchored, the engine reaches the expected family within two post-trigger turns ~45% of the time.]

The family-confusion matrix shows the base ego's prior. Nineteen of cell_114's 24 trigger+1 moves fall in the broad `substantive_engagement` family (`mirror_and_extend`, `acknowledge_and_redirect`, and the engagement-style moves `scope_test` / `name_the_disagreement` / `pose_counterexample`) — a strong "keep elaborating the content" default. The consequence is sharp on the trap families that require *detecting a hidden learner state* rather than adding content: cell_114 scores **0/6** on `repair_affective` traps (`affective_shutdown`, `repair_after_misrecognition` — where the learner has disengaged or been misread and the right move is to slow down, de-escalate, or explicitly repair) and **1/3** on the `diagnostic` trap (`polite_false_mastery` — where the right move is to probe the learner's hidden reasoning). cell_110, by contrast, hits 4/6 on `repair_affective` and 3/3 on `diagnostic`: its externalised profile flags the affective shift and its programmatic policy menu makes the de-escalation/repair move available and salient. The base ego is *not incapable* of these moves — it plays `scope_test` on the argue-back traps (`false_confusion`, `resistance_to_insight`: 2/3 each, where engaging substantively *is* the trap-appropriate response, so its default happens to be right), `ask_diagnostic_question` on one `polite_false_mastery`, and even `repair_misrecognition` in 2 of 3 `repair_after_misrecognition` dialogues — but it fires those moves inconsistently and without timing discipline. In the two `repair_after_misrecognition` dialogues where the base ego *does* play `repair_misrecognition`, it does so at the trigger turn itself (sandwiched between `mirror_and_extend` moves on either side), not at the trigger+1 slot the binary scores, so the instrument records 0/3 for that scenario type even though the move is present. The mid-pack `family_match` (58.3%, above cell_111 and cell_112, below cell_110 and cell_113) is the same story read the other way: the broad `substantive_engagement` family is easy to land on the 12 trap scenarios where some form of substantive engagement is the target (12/12 family-correct there), and it is the exact-label/timing discipline (`strict`, `window`) plus the non-content families (`repair_affective` 0/6) where the dialogue engine bottoms out.

The graded rubric inverts the ranking — and that inversion is the finding. Under the 4-dimension adaptive graded rubric (§5.12.4; GPT-5 via the codex CLI bridge, $N = 24$), cell_114's overall mean is **4.46/5** — *higher* than cell_110's 4.03 ($N = 23$, same judge):

Cell	`trigger_recognition`	`strategy_execution`	`strategy_quality`	`pedagogical_coherence`	Overall (mean of 4)
cell_110_langgraph_adaptive	4.26	4.30	3.96	3.61	4.03
cell_114_dialogue_engine_trap_baseline	**4.54**	**4.21**	**4.50**	**4.58**	**4.46**

A reader who looked only at the graded rubric would conclude cell_114 is the better tutor. This makes the §5.12.4-limitation-1 distinction concrete. The graded rubric scores *quality of execution given that some action fired* — fluency, calibration, coherence of the resulting prose — and a mature, heavily-iterated single-agent tutor prompt produces fluent, coherent teaching even when it selects the wrong strategic family; cell_110's externalised-state machinery, which buys it the correct move more often on the stored-convention binary (≈1.9×, but convention-fragile — see the turn-convention correction at the end of this subsection), also bloats and muddies its trajectories (its weakest graded dimension, 3.61 on coherence, is §6.8.3's own observation, and the cell_114 gap is widest exactly there: 4.58 vs 3.61). The two instruments measure different constructs — *which move, when* (binary) versus *how well-crafted the prose* (graded) — and the architectural intervention trades the second for the first. This is not a "GPT-5 rewards verbosity in cell_114's favour" artefact: the dialogue-engine turns are *long* (400–2700 characters), the opposite of the concision GPT-5 is documented to prefer (§5.12.4 limitation 2), so the high graded score reflects genuine prose coherence, not a length bias. The one place binary and graded converge is `strategy_execution`, the single dimension built to ask about post-trigger family alignment: there cell_114 (4.21) and cell_110 (4.30) are roughly tied, and cell_114's `strategy_execution` distribution carries the spread the binary predicts — one row at 1 (the `false_confusion` dialogue that mirrored when `scope_test` was the target), five at 3 (defensible-but-not-target moves), against a body of 4s and 5s. The graded rubric *can* see the selection problem, but only on the dimension designed to look for it; the other three reward the prose and outvote it.

Bottom line. On the v1 trap suite, the standard dialogue engine — same model, same trap-steered learner, same scenarios as cell_111 — does not produce the strategy shifts the LangGraph state-policy configuration produces: it sits at the bottom of the binary `strict_shift` ranking (25.0% vs 30–48%), it uniquely fails to re-pivot on later turns, and it collapses entirely on the trap families that require detecting a hidden learner state rather than elaborating content. The most plausible reading is that externalised learner-state tracking plus a constraint check are what let the LangGraph cells pivot — but the cell_111 cell_114 contrast also varies the prompt-assembly path (§5.12.3), so this is a directional baseline floor rather than a clean single-factor result — and, per the turn-convention correction that immediately follows, the binary gap it rests on is itself an artifact: the secure re-run under the fixed adapter (§6.8.4) puts both legs at ~1.1× (v1 1.15×, cross-suite 1.07×), within $N \approx 24$ noise — no detectable advantage. The graded rubric inverting the ranking (4.46 > 4.03) is itself the result: graded prose quality and strategic-move correctness are not the same construct, and the dialogue engine wins the first while losing the second. Per §5.12.6: post-hoc exploratory; no alpha correction; single judge throughout (binary `claude-code/sonnet`; graded GPT-5/codex); no blinding.

Turn-convention correction (v3.0.119): the cross-architecture binary gap is convention-fragile, and the "never re-pivots" signature is largely a scoring artifact. The `repair_after_misrecognition` observation three paragraphs up — the base ego plays the correct repair move "at the trigger turn itself … not at the trigger+1 slot the binary scores," so the instrument records 0/3 "even though the move is present" — was the visible symptom of a *systematic* scoring artifact, not a tutor timing deficit. `scripts/run-dialogue-engine-trap-baseline.js` advanced the learner-turn index by one (`turn: turn + 1`) where the LangGraph runner passes `turn: state.turn`; because the trap learner fires its trigger on the turn index it is handed (`realLLM LEARNER_TURN_SYSTEM`: "if this is the trigger turn, surface the signal"), the dialogue-engine learner sprang the trap one tutor-turn *early*, and the fixed-index scorer (which reads `triggerTurn + 1`) therefore read the engine's *second* post-trigger response, not its first. The offset is architecture-correlated — the LangGraph cells pass `state.turn` and are aligned; only the dialogue-engine baseline is shifted — so it inflated the cross-architecture gap rather than adding symmetric noise. The convention was centralized and the adapters fixed in `scripts/lib/trapTurnConvention.js` (commit `3474ebf`). An offline, event-anchored re-score (`scripts/rescore-trap-strict-shift.js`; `exports/trap-strict-shift-rescore.json`) reproduces the published numbers exactly under the stored convention, then re-scores by locating each trigger's actual emission in the transcript (by its signal text) and scoring the first tutor turn after it.

The corrected reading is a *grid*, because the gap depends on two independently-defensible scoring choices — where the trigger is located (fixed `triggerTurn + 1` vs event-anchored) and how strict the match is (exact label vs pedagogical family):

convention	v1: cell_110 vs cell_114	cross-suite: cell_124 vs cell_125
strict · stored (as published above)	47.8% vs 25.0% (1.91×)	62.5% vs 30.4% (2.05×)
strict · event-anchored	43.5% vs 25.0% (1.74×)	62.5% vs 43.5% (1.44×)
family · stored	87.0% vs 58.3% (1.49×)	66.7% vs 39.1% (1.70×)
family · event-anchored	65.2% vs 66.7% (0.98×)	66.7% vs 47.8% (1.39×)

Each correction independently deflates the gap, and they compound. The locator-free `family · stored` row — which uses the *published* fixed offset, no re-location — already cuts the v1 advantage from 1.91× to 1.49× on label-granularity alone (the engine's `request_elaboration` / `mirror_and_extend` moves sit in the same *family* as the targeted `ask_diagnostic_question` / `scope_test`). Event-anchored location cuts it again, to 0.98× (v1) and ~1.4× (cross-suite) under the most defensible offline convention — enough to show the gap was *not secure*, but the offline event-anchored locator fights noise on the old mis-aligned transcripts, so a clean re-run was needed to settle the number. The "uniquely never re-pivots" claim above is the same artifact read through the `window` metric: event-anchored, cell_114 reaches the expected family within the first two post-trigger turns in ~45% of cases (`win 2` in the artifact), well above its 25.0% stored `window` rate — it *does* re-pivot; the scorer was reading past the pivot. This *adds to* rather than replaces the §5.12.3 prompt-scaffold confound, which still applies on top.

The secure re-run (v3.0.121) settles it. Cells 114 and 125 were re-run under the fixed adapter (fresh sample, $N = 24$ each, `claude-code/sonnet`; run ids `eval-2026-06-06-460fc561` and `-5a168684`) and scored by the now-aligned instrument (fixed adapter + `triggerTurn + 1` scorer agree by construction, so no event-anchoring is needed). The engine floor, previously undercounted by the artifact, rises sharply: `cell_114` (v1) `strict_shift` 25.0% → **41.7%** (10/24), family 58.3% → **75.0%**; `cell_125` (cross-suite) `strict_shift` 30.4% → **58.3%** (14/24), family 39.1% → **62.5%**. (Mechanism-level confirmation: `cell_114`'s `repair_affective` hits at trigger+1 go from 0 to 10 — the affective traps it was being silently mis-scored on.) Against the frozen state cells (`cell_110` 47.8% / 87.0%; `cell_124` 62.5% / 66.7%) the secure contrasts are **v1 1.15× strict / 1.16× family** and **cross-suite 1.07× strict / 1.07× family** — gaps of ~4–6 pp, well within $N \approx 24$ sampling noise (no test would call a ~5-pp gap significant at this N). This lands slightly *above* the grid's most-aggressive corner (the fresh aligned run is authoritative over the noisy offline locator). One caveat, stated not hidden: the state cells are frozen (old epoch), the engine cells fresh, so this is not a same-epoch contrast — but the artifact was engine-adapter-only and the state cells were always aligned, and the conclusion is robust to the caveat because the gaps are small either way (re-running `cell_110`/`cell_124` was judged not worth the ~47 paid dialogues). **Final secure verdict, replacing "the correct move twice as often, the dialogue engine never re-pivots": once the turn convention is fixed, the cross-architecture binary advantage is ~1.1× and not a detectable difference on either suite — the published ~2× was the artifact. Power caveat — this is *not-detected*, not *demonstrated-zero*.** At $N \approx 24$/cell the comparison is severely underpowered for an effect this small: the pooled secure gap is +5.3 pp, 95% CI $[-15, +25]$ pp (Fisher $p = 0.68$; per-suite v1 $[-22, +35]$, cross-suite $[-23, +32]$), and detecting a true 5-pp difference at 80% power would need ~1,400 dialogues per cell. A small real advantage (~1.1×, *consistent in direction* across both suites and with the state machine's explicit re-pivot, which the locally-greedy engine lacks) is therefore **not excluded** — what the re-run removes is the *~2× magnitude* (which rested on the engine's mis-measurement at 25%, vs its clean 41.7%, CI $[24, 61]$%), not the *existence* of a small effect. The residual is recorded as *not-detected, not absent*, and is not worth ~1,400 paid dialogues per cell to chase, given that any such edge is binary-channel-only (the graded rubric inverts) and remains scaffold-confounded (§5.12.3). Status (§5.12.6): post-hoc correction of a post-hoc exploratory result; single judge throughout (`claude-code/sonnet`); artifacts `exports/trap-strict-shift-rescore.json` (offline grid) plus the two re-run ids above. This correction governs the cross-suite replication (§6.8.7) and the §6.8.8 synthesis equally.

6.8.5 Bilateral-ToM Elaboration Adds No Measurable Within-Family Discrimination (P2.1)

The pre-registered P2.1 fan-out (`docs/explorations/claude/2026-05-10-p2-followup-pr` §P2.1, locked 2026-05-07; runs 2026-05-08 to 2026-05-10) tested whether elaborating the LangGraph `learnerProfile` update with an explicit bilateral Theory-of-Mind (ToM) tracker — a node that maintains separate first-person and third-person learner-state representations — produces within-family discrimination beyond the `recognition_only` baseline. Four cells were run at $N = 24$ each on the v2 trap-scenario suite (six within-family scenarios where every valid alternative belongs to the same pedagogical family as the target):

Cell	Architecture	`strict_shift_correctness`
cell_111_a13_C1_ recognition_only_v2	`recognition_only` baseline	33.3% (8/24)

Cell	Architecture	`strict_shift_correctness`
cell_116_recognition_named_patterns_v2	`recognition_named_patterns` (named-pattern lookup added)	37.5% (9/24)
cell_115_bilateral_tom_v2	`bilateral_tom` (first/third-person ToM tracker)	45.8% (11/24)
cell_117_bilateral_tom_named_patterns_v2	`bilateral_tom_named_patterns` (both)	33.3% (8/24)

The H1.1 primary contrast pools by architecture family:

Pooled architecture	shift_n / n	shift%
bilateral_tom (cells 115 + 117)	19 / 48	39.58%
recognition_only (cells 111 + 116)	17 / 48	35.42%

The pooled gap is **+4.17 percentage points** in favour of bilateral_tom, with a 95% Wald CI of [−15.18, +23.52] pp. The pre-registration locked binary point-estimate thresholds (lines 30–34, 70–72 of the pre-reg doc): a gap of ≥ 10 pp supports H1.1 (bilateral_tom adds within-family discrimination), 5–10 pp is inconclusive, and < 5 pp falsifies H1.1 in favour of H1.3 (the v1 family-match advantage of bilateral_tom was a between-family, type-recognition effect rather than a within-family discrimination effect). The observed +4.17 pp falls below the 5-pp falsification threshold. **H1.3 is supported as written; the bilateral_tom elaboration does not produce within-family discrimination at the pre-registered effect size.** The CI width permits a true 10-pp gap that sampling chance compressed to 4 pp, but the pre-reg authors locked point-estimate thresholds knowingly, and this report applies them as written.

The H1.2 secondary endpoint (`family_match_rate` ≥ 80% predicted; < 70% falsifying) corroborates the H1.3 reading. Three of the four cells fall below the 70% falsification floor on v2: cell_116 (66.67%), cell_115 (66.67%), and cell_117 (58.33%); only cell_111 (70.83%) escapes, by 0.83 pp. All three falsified cells had v1 family-match > 70% (cell_115 = 70.8%, cell_116 = 79.2%, cell_117 = 79.2% on the v1 suite per the upstream A13 follow-up memo), so the v2 drop is interpretable per the pre-reg's rule: the cells' v1 family-match advantage was scenario-distribution-driven (v1 scenarios spanned three pedagogical families and made between-family discrimination easy), not architectural (v2 scenarios constrain to a single family and the architectures collapse).

Descriptive results not part of the pre-reg's locked thresholds are reported for completeness. The four cells handle `substantive_engagement` scenarios well (cells 111, 115, 116 all score 12/12 family-correct, cell_117 scores 9/12) but collapse on `diagnostic_opaque_reasoning` (0–1 of 4 family-correct across all four cells); scaffolding falls in between (3–5 of 8 across cells). Per-scenario, only two of six scenarios drive most of the strict_shift signal: `substantive_correct_underspecified` (10/16 across the four cells, 62.5%) and `scaffolding_load_not_gap` (14/16, 87.5%); the other four scenarios sit at ≤ 25% strict. The pre-reg forbids post-hoc subsetting (line 84); these per-scenario numbers are reported as descriptive context rather than as a re-analysis. Counterfactual divergence on the two ToM cells (cell_115 cf_div 62.5%, cf_fam 37.5%; cell_117 cf_div 62.5%, cf_fam 54.2%) is consistent with type-prior locking in cell_115 but inverted in cell_117.

Two limitations apply to P2.1 alongside the §5.12.6 reporting convention. First, single-judge scoring: all four cells were scored under `claude-code/sonnet`; a second-judge probe was not pre-registered and has not been run. Second, the wide CI on the H1.1 contrast (39 pp wide, $[-15, +24]$ pp) means the locked binary rule is more decisive than the underlying inferential evidence: a future $N = 48$ replication per cell would tighten the CI by approximately $\sqrt{2}$, sufficient to distinguish a true null from a true 10-pp effect.

The pre-reg's "Order of execution" section (line 239) flagged that a P2.1 null would weaken the prior on P2.2 (state-schema ablation) and P2.3 (bilateral × id-director crossover) without invalidating them, since P2.2's diagnostic value for the cosmetic-vs-substantive question is partly independent of whether the architecture does within-family work, and P2.3's three-endpoint design (charisma + accuracy + strategy_shift) is only partially dependent on the strategy_shift endpoint — §6.8.8 records the subsequent decision to retire P2.3 rather than run it on the original schedule. The interpretation that follows in §6.8.6 (state-richness reversal) and §6.8.7 (cross-architecture status) applies the pre-reg's framework: the architecture appears to recognise pedagogical *type* but not to discriminate *within* a type when valid alternatives are same-family, and the project's claim space narrows accordingly.

6.8.6 State-Schema Ablations: A Richer Learner Profile Does Not Help (the State-Richness Reversal) The headline `cell_110` (§6.8.3) carries a structured `learnerProfile` object with five fields the policy reads each turn — `confidence` (scalar), `misconceptions` (array), `agencySignal` (5-state enum), `zpdEstimate` (string), `lastEvidence` (a quoted learner utterance). Is that structure doing work, or is it cosmetic dressing on what is really a `confidence`-scalar-plus-dialogue-text policy? The P2.2 sub-study (pre-registration §P2.2; cooling period closed 2026-05-09) answers this with three ablation cells that project the LLM-emitted profile down to a subset of fields before the policy sees it — the model still infers the full profile each turn, so inference cost is held constant; only the policy's *view* is narrowed. **H2.1** predicted a monotone dose-response on `strict_shift`: `cell_110` (full) ≥ `cell_119` (drops `misconceptions`) ≥ `cell_120` (drops `agencySignal`) ≥ `cell_118` (`confidence` + `lastEvidence` only), with `cell_110 - cell_118` ≥ 15 pp; **H2.2** predicted `cell_120` would drop ≥ 15 pp on the repair/affective family (the agency enum was theorised to govern affect-tracking) while holding on substantive scenarios; **H2.3**, the falsifying alternative, was that the structured fields beyond `confidence + lastEvidence` are cosmetic. Each ablation cell ran four runs × the eight v1 trap scenarios ($N = 32$); `cell_110`'s comparison numbers are its A13 v1 fan-out ($N = 23$).

Cell	`learnerProfile` fields the policy sees	N	`strict_shift`	Graded overall (T / E / Q / C)
`cell_110` (`state_policy`, full)	confidence, misconceptions, agencySignal, zpdEstimate, lastEvidence	23	47.8%	4.03 (4.26 / 4.30 / 3.96 / 3.61)
`cell_120` (`no_agency_signal`)	confidence, misconceptions, zpdEstimate, lastEvidence	32	50.0%	4.09 (4.25 / 3.81 / 4.22 / 4.09)
`cell_119` (`no_misconceptions`)	confidence, agencySignal, zpdEstimate, lastEvidence	32	53.1%	4.06 (4.09 / 3.78 / 4.25 / 4.13)
`cell_118` (`minimal_profile`)	confidence, lastEvidence	32	68.8%	4.34 (4.34 / 4.28 / 4.41 / 4.34)
`cell_123` (`minimal_plus_zpd`, post-hoc)	confidence, lastEvidence, zpdEstimate	32	68.8%	4.11 (4.03 / 4.00 / 4.28 / 4.13)

(T / E / Q / C = `trigger_recognition` / `strategy_execution` / `strategy_quality` / `pedagogical_coherence`, each 1–5; graded overall is their mean; binary `strict_shift` = correct pre-registered family at trigger+1; both scored as in §6.8.3, binary under `claude-code/sonnet`, graded under GPT-5/codex.)

H2.1's predicted ordering is reversed — and the reversal is a step, not a slope. The observed `strict_shift` ordering is `cell_118` (68.8%) $\geq$ `cell_119` (53.1%) $\geq$ `cell_120` (50.0%) $\geq$ `cell_110` (47.8%) — the reverse of the predicted order at every comparison that exceeds sampling noise. `cell_110 - cell_118 = -21` pp against a predicted $\geq$ `+15` pp: opposite sign, larger magnitude than the predicted effect. All three of H2.1's predicted positive gaps over `cell_118` come out negative (`cell_110` -21 pp, `cell_120` -19 pp, `cell_119` -16 pp). But the reversal is carried *entirely* by `cell_118`: the three richer cells — `cell_110`, `cell_119`, `cell_120` — form a flat cluster at 47.8 / 53.1 / 50.0% (spread 5.3 pp, comfortably inside the ±10-pp band that $N \approx$ 24–32 Bernoulli sampling produces around a true ~0.5), so their internal order (predicted `cell_110` $\geq$ `cell_119` $\geq$ `cell_120`, observed `cell_119` $\geq$ `cell_120` $\geq$ `cell_110`) is uninterpretable noise and is not interpreted. `cell_118`'s lead over that cluster — the one comparison the reversal actually rests on — is itself near the edge of binary-channel significance at $N = 32$: ~18 pp over the pooled three-cell mean, bootstrap 95% CI $\approx [-1, +37]$ pp, label-permutation $p \approx 0.10$. It is firmer on the graded channel (a +0.28-point overall gap, CI $\approx [+0.04, +0.51]$, $p \approx 0.01$), it is broad rather than scenario-bound (`cell_118` matches or beats the cluster on six of the eight trap types — the lone two it does not, `resistance_to_insight_v1` and `sophistication_upgrade_v1`, are scenarios *every* cell fails), and §5.12.4's cross-judge calibration reproduces the graded ordering under Gemini. So the direction *minimal* $\geq$ *rich* is carried by the graded instrument plus that scenario-level breadth plus the judge replication, not by the 21-pp binary headline alone. The structure is in the *discontinuity*, and it points the wrong way: collapsing the profile to two fields lifts `strict_shift` by ~19 pp over the full five-field object; pulling any single mid-tier field (`misconceptions`, `agencySignal`) does nothing. H2.3's prediction — "the fields beyond `confidence + lastEvidence` are cosmetic" — is supported in spirit and slightly exceeded in fact: they are not merely inert, they are mildly counterproductive on this instrument; the externalised-state machine's measured advantage over recognition-only (`cell_111`, §6.8.4) is doing its work through `confidence` plus the quoted-dialogue anchor, not through `misconceptions` / `agencySignal` / `zpdEstimate`. (The pre-reg's literal falsification rule — `cell_110 - cell_118` differ by `< 5` pp → cosmetic — does not mechanically fire, since `|-21|` is not `< 5`; but a 21-pp reversal is the cosmetic verdict and then some, and per the pre-reg's "mixed outcomes are reported as such" instruction it is reported as a reversal, not patched into a re-ordered hypothesis.)

H2.2 is unsupported, also in the wrong direction. `cell_120` was built to test whether the agency enum drives repair/affective discrimination. It scored 7/8 `strict_shift` on the repair/affective traps (`affective_shutdown_v1` 4/4, `repair_after_misrecognition_v1` 3/4) — at or above `cell_110`'s 4/6 — i.e., removing `agencySignal` did *not* depress the family it was theorised to govern, and `cell_120` held on substantive scenarios besides. Whatever lets the policy recognise repair/affective situations, it is not the structured agency enum; the live candidates are `lastEvidence`'s quoted utterance and the ego LLM's own reading of the transcript. (`n = 6`–8 per cell on this sub-family, so the apparent 21-pp *gain* is itself within noise; the safe statement is "no drop", which is enough to retire H2.2.)

Graded scoring agrees within the pre-registered cells — but `cell_123` is a divergence. Unlike the cross-*architecture* comparison of §6.8.4, where the binary and graded instruments diverge (the dialogue engine writes more coherent prose, picks the wrong family more often — by a convention-fragile margin, §6.8.4), within the P2.2 family's *four pre-registered cells* the two instruments *agree*: `cell_118` leads on both (graded 4.34, every dimension $\geq$ 4.28, and a near-perfect 96.9% family-match), and `cell_110` / `cell_119` / `cell_120` cluster on both (graded 4.03 / 4.06 / 4.09). §5.12.4's independent-judge calibration confirms the `cell_118` > `cell_119` > `cell_110` graded ordering holds under both GPT-5 and Gemini, so it is not a single-judge artefact; `cell_120`, graded only after that calibration run, is single-judge GPT-5 and slots between `cell_118` and `cell_119`, statistically tied with `cell_119` and `cell_110`. The lone within-P2.2 binary/graded *divergence* is the post-hoc `cell_123` — top on binary (= `cell_118`), mid on graded (≈ the ~50% cluster) — examined in the next paragraph. One sub-pattern is GPT-5-only: the two four-field cells (`cell_119`, `cell_120`) carry a depressed-`strategy_execution` left tail — roughly a third of rows scored 3, versus ~1/20 for `cell_110` and `cell_118` — which §5.12.4 reports the Gemini re-judge cannot resolve (Gemini's ceiling effect collapses the tail), so it stays a GPT-5 observation rather than a finding.

What `cell_123` settles — `zpdEstimate` is a free rider on move selection; the binary destabiliser is the *fourth structured field*, not a specific one. The P2.2

design pulls one field at a time off the *full* profile, so it left three readings of the floor open: "`zpdEstimate` is the destabilising field" (it is the only structured field present in all three ~50% cells and absent from the ~69% `cell_118`), "having ≥ 4 structured fields at all", and "any one of `{misconceptions, agencySignal, zpdEstimate}`". The post-hoc cell `cell_123` (`state_policy_minimal_plus_zpd`: `{confidence, lastEvidence, zpdEstimate}` — `cell_118`'s set plus `zpdEstimate`, behaviourally an exact clone of `cell_118` with one more field projected through to the policy; $N = 32$, v1 suite) adjudicates: it scores `strict_shift` **68.8% — the same 22/32 as `cell_118`**, with a 100% family-match (every trigger+1 move in the correct broad family) and a per-scenario profile recognisably the same shape as `cell_118`'s (4/4 on `answer_seeking` / `affective_shutdown` / `repair_after_misrecognition`, 0/4 on the universally-hard `false_confusion`). So **`zpdEstimate` is exonerated**: the two cells that differ *only* in `zpdEstimate` (`cell_118`, `cell_123`) are tied to the row, so adding it to the lean `{confidence, lastEvidence}` core costs nothing — which kills the first reading and locates the floor in the *other* structured fields. Nor is it a single toxic field among those: `cell_119` (keeps `agencySignal`, drops `misconceptions`) = 53.1% and `cell_120` (keeps `misconceptions`, drops `agencySignal`) = 50.0% both sit in the ~50% cluster — each cell is missing the field the other reading would blame, and is still depressed, so neither "`misconceptions` is toxic" nor "`agencySignal` is toxic" survives. What moves the policy from the ~69% lean cluster (`cell_118`, `cell_123`) to the ~50% rich cluster (`cell_119`, `cell_120`, `cell_110`) is *the presence of either `misconceptions` or `agencySignal`* — equivalently, crossing from three to four structured fields — and the effect *saturates*: `cell_110`, carrying both, is 47.8%, not detectably below the one-each cells, so the second destabilising field adds essentially nothing. Saturation is the signature of a load/capacity ceiling (the policy LLM does single-decision strategy selection worse once it must integrate four-plus structured fields) rather than additive per-field toxicity. The honest residual: a 3-field cell carrying `{confidence, lastEvidence, misconceptions}` (or `…, agencySignal`) would separate "pure field-count" from "specifically that field" — ~69% under a count story, ~50% under a content story — but it has not been run and is not planned; the data in hand favour the count/load reading via the saturation point without excluding the content one. Either way, on the binary channel the §6.8.4 externalised-state advantage over recognition-only runs through `confidence`, the quoted-dialogue anchor `lastEvidence`, and (harmlessly) `zpdEstimate` — not through the `misconceptions` array or the `agencySignal` enum.

One caveat: `zpdEstimate` is free on *move selection*, not on *execution prose*. On the *graded* channel `cell_123` does **not** join `cell_118`: its GPT-5/codex overall is 4.11 (4.03 / 4.00 / 4.28 / 4.13) — squarely in the ~50% cluster's graded band (4.03–4.09), well below `cell_118`'s 4.34 — and it carries the same depressed-`strategy_execution` left tail as the four-field cells (9/32 rows scored 3, against ~1–2/32 for `cell_118` and `cell_110`). So `zpdEstimate` is a free rider on *which move, when* (the binary `strict_shift` channel, where adding it to the lean core costs nothing) but a modest tax on *how well the resulting teaching is executed* (the graded channel, where adding it drops `cell_123` to the rich-cluster level): the graded "minimal wins" result of the pre-registered cells does not extend to `cell_123`, which is the lone within-P2.2 case where the binary and graded instruments diverge (cf. the cross-*architecture* divergence of §6.8.4). Per §5.12.4 the `strategy_execution` left tail is a GPT-5-only signal — Gemini's ceiling effect cannot resolve it, and `cell_123` was graded after that calibration run, so its graded position carries the same single-judge caveat as `cell_120`'s; the binary result (`cell_123` = `cell_118` to the row) is the more robust half of the cell.

Status and caveats (per §5.12.6). Cells 118–120 are pre-registered confirmatory tests of H2.1–H2.3 (pre-reg §P2.2); `cell_123` is a post-hoc exploratory follow-up, not pre-registered. Binary `strict_shift` is single-judge `claude-code/sonnet`; graded scoring is single-judge GPT-5/codex with the §5.12.4 Gemini inter-rater check on cells 110/118/119 only (`cell_120` and `cell_123`, graded after that calibration run, are GPT-5-only — and `cell_123`'s graded position, including the `strategy_execution` left tail it shares with the four-field cells, carries the §5.12.4 GPT-5-only caveat). No alpha correction is applied across the four (then five) cells; the `strict_shift` gaps among `cell_110 / cell_119 / cell_120` are within sampling noise and not interpreted, and `cell_118`'s lead over that cluster is firm on the graded channel (permutation $p \approx 0.01$) but only marginal on the binary `strict_shift` channel (permutation $p \approx 0.10$, bootstrap CI brushing zero) — the reversal's *direction* is treated as established, the 21-pp binary figure as the high end of a wide interval. Counterfactual replay (`counterfactual.enabled: true` in the configs) is a no-op for the P2.2 architectures — they fall outside the runner's profile-update set, so `counterfactual_total = 0` for every P2.2 cell — and the comparison rests entirely on the on-path binary and graded metrics. N: `cell_110` = 23 (its A13 v1 fan-out); cells 118 / 119 / 120 / 123 = 32 (four runs × eight v1 scenarios). This result narrows the project's "externalised cognition" claim space exactly as the pre-reg's H2.3 anticipated, and §6.8.7 already folds it into the cross-suite reconciliation (the trap-suite gain is between-*type*, not within-type, and is not a graded-quality effect).

6.8.7 Reconciling the Trap-Suite Result with the §6.3 Adaptive-Responsiveness Null §6.3.8 reported adaptive responsiveness as null under every experimental factor tested (recognition prompts, multi-agent architecture, learner architecture: all slope effects below the $d = 0.27$ detectability threshold — pooled $d = 0.03$ for recognition, ≤ 0.15 for the architecture factors — across $N = 432$ dialogues), and §6.3.9 reported the insight-action gap as "structural under prompt design and lightweight architectural intervention; partially mitigable only by expensive generation-space search" — Bridges 0–2 land within ± 0.01 of the cell-40 baseline coupling, and Bridge 3 (best-of-K=3, ~6× ego cost) reaches $d = 0.41$, suggestive but below the pre-registered bridgeable threshold. §6.8.3–§6.8.4 then reported that the LangGraph `state_policy` runner produces detectable strategy shifts on the v1 trap suite (`strict_shift` 47.8% for cell_110 versus 25.0% for the dialogue-engine baseline cell_ 114). Read carelessly, this looks like a contradiction: an architectural intervention *did* move adaptive behaviour. It is not a contradiction. The §6.3 null and the trap-suite positive are about different outcome variables, on different stimuli, at different intervention weights — and the second is consistent with, indeed forecast by, the first.

Different outcome variable; different instrument sensitivity. The §6.3 null is on a *diffuse, aggregate* signal: per-dialogue rubric slopes across free-form multi-turn dialogues (§6.3.2, §6.3.5), and the reflection-action coupling cosine within turn (§6.3.9). "Did the tutor adapt" there is inferred from whether scores trend upward across turns or whether introspective text resembles delivered text — both are low-resolution proxies, and both are exactly the kind of measure that washes out a single localized strategic move inside an otherwise-unchanged trajectory. The trap suite is a *localized, pre-annotated, single-decision* instrument: each scenario plants a hidden state and a trigger turn, and `strict_shift` asks one question — did the pre-registered family of moves fire at trigger+1. A binary check on a designed decision point is a far more sensitive detector of "the tutor selected the right strategic move" than a slope on free-form dialogue, the same way a planted bug is a more sensitive test of a linter than waiting for bugs to surface in production. The §6.3 instruments were built to ask whether *recognition* steepens adaptation *on average over a dialogue*; the trap instrument asks whether *an architecture* selects *the right move at one annotated moment*. A null on the first and a positive on the second are jointly coherent: the architectural effect is real but it lives in a channel — localized strategic-move selection — that the §6.3 measures were not designed to resolve.

Different intervention weight; §6.3.9's own escape clause names this class. §6.3 tested *lightweight* manipulations — prompt orientation, a second (superego) agent, learner-architecture variants. §6.3.9 added a cost ladder (explicit instruction → two-pass reflection injection → retargeted critique → best-of-K=3) and found only the heaviest rung crossed the suggestive threshold, with its own forward-looking caveat: closing the gap would require "much higher K in best-of-K search, *or genuinely new architectures with explicit memory and iterative refinement*". The LangGraph `state_policy` runner is the second of those: it externalises learner state into a structured `learnerProfile` object that persists and is rewritten across turns (explicit memory), and it replaces the free-form turn loop with a programmatic policy graph plus a constraint check (a structured refinement of the move, not just the prose). The trap-suite positive is therefore best read not as a contradiction of §6.3 but as a partial *confirmation* of §6.3.9's hypothesis: lightweight interventions are inert, expensive search is suggestive, and a heavy externalised-state architecture is the first thing in the project to produce a measurable shift on a sensitive instrument. "Measurable" is doing bounded work — 47.8% strict is moderate, not strong (§6.8.3), and the dialogue-engine comparison carries the §5.12.3 prompt-scaffold confound that makes cell_114 a *directional floor* rather than a clean single-factor baseline (§6.8.4) — but moderate-and-confounded is still a different result from §6.3's clean null, and the difference sits exactly where §6.3.9 said it would.

What the trap-suite positive does *not* license. Four bounds keep it consistent with the rest of §6:

- *Not a recognition-prompt effect.* §6.3's actual null is about recognition modulating adaptation; the §6.8 cells do not isolate recognition-on versus recognition-off within a fixed runner — cell_111 `recognition_only` is itself a LangGraph sub-graph (§6.8.2), not a prompt manipulation of the dialogue engine — so nothing here revives the recognition-modulates-adaptation hypothesis. The effect is architectural, not orientational.
- *Not a within-type effect.* §6.8.5's pre-registered P2.1 test found the architecture adds no within-family discrimination (pooled +4.2 pp, below the 5-pp falsification floor): it appears to recognise pedagogical *type* — substantive versus scaffolding versus diagnostic versus repair/affective — but not to choose well *among same-type alternatives*. The trap-suite gain is a between-type gain.
- *Not a graded-quality effect.* The instruments that do *not* isolate the single strategic

decision — the §6.3 trajectory slopes, the §6.3.9 reflection cosine, and the trap suite's own 4-dimension graded rubric (which scores whole-trajectory prose quality) — find the architecture null or *inferior*: §6.8.4's graded rubric inverts the cell ranking, with the dialogue engine ahead of cell_110 on prose quality, calibration, and coherence (4.46 versus 4.03) even as it picks the wrong family more often (the §6.8.4 binary gap, now known to be convention-fragile). The one instrument that *does* isolate that decision — the trap binary `strict_shift` — is the one that puts the architecture ahead. The architectural advantage is specifically an advantage in *which move, when*, not in *how well-crafted the resulting teaching*.

- *Transfers across instruments, in one direction, with the same confound.* The clean cross-suite test now exists — a trap-style annotated instrument re-derived from six §6.3 suggestion scenarios (`config/cross-suite-trap-scenarios.yaml`), run on both architectures (`cell_124` = `cell_110`'s `state_policy` runner, `cell_125` = `cell_114`'s dialogue engine), scored with the same binary `strict_shift` — and it reproduces the v1 result: `cell_124` 62.5% (15/24) versus `cell_125` 30.4% (7/23), a ratio (≈2.05×) within rounding of the v1 suite's (1.91×) — both on the *stored* convention; the §6.8.4 turn-convention correction shows this gap is an artifact, and the secure re-run under the fixed adapter puts both legs at ~1.1× (cross-suite 1.07×, v1 1.15×), within $N \approx 24$ noise (no detectable advantage) — with the graded rubric again inverting (`cell_125` 4.60 > `cell_124` 4.22). Details and bounds in the "Discharging the forward-references" paragraph below: the claim now holds on two independently-built trap instruments, but it remains a *directional* comparison (`cell_125` carries the same §5.12.3 prompt-scaffold confound as `cell_114`), runs *one direction only* (no one has run the dialogue engine *with* the LangGraph runner's prompt scaffold), is *between-type* (the cross-suite gap concentrates in the affective-shutdown and productive-deadlock scenarios; on two scenarios — `misconception_surfaces`, `activity_avoidance` — *neither* architecture fires the annotated move), and lives *on the binary channel* (the dialogue engine writes more coherent prose on both suites).

The picture across both suites, then, is internally consistent rather than contradictory:

Stimulus suite	Instrument	What it isolates	Finding
Suggestion (§6.3, $N = 432$)	Per-dialogue rubric slope	Adaptation rate over a whole dialogue, by condition	Null ($d \leq 0.03$ pooled; §6.3.2, §6.3.8)
Suggestion (cells 40–45 + bridges)	Reflection-action coupling cosine	Whether introspective noticing reaches the delivered message	Structural; only best-of-K suggestive ($d = 0.41$; §6.3.9)
Trap v1 ($N \approx 24$/cell)	`strict_shift` (binary)	Right family at trigger+1	cell_110 47.8% vs cell_114 25.0% *stored* → artifact; secure re-run 47.8% vs 41.7% = 1.15×, n.s. (§6.8.4)
Trap v1 ($N \approx 24$/cell)	4-dimension graded rubric	Prose quality given some action fired	*Inverts*: cell_114 4.46 > cell_110 4.03 (§6.8.4)

Stimulus suite	Instrument	What it isolates	Finding
Trap v2 ($N = 24$/cell)	`strict_shift`, within-family	Right move among same-type alternatives	Null (pooled +4.2 pp, sub-threshold; §6.8.5)
Cross-suite (trap-annotated §6.3 subset, $N \approx 24$/cell)	`strict_shift` (binary)	Right family at trigger+1, on §6.3-derived scenarios	cell_124 62.5% vs cell_125 30.4% *stored* (≈2.05×) → artifact; secure re-run 62.5% vs 58.3% = 1.07×, n.s. (§6.8.4)
Cross-suite ($N \approx 24$/cell)	4-dimension graded rubric	Prose quality given some action fired	*Inverts*, as on v1: cell_125 4.60 > cell_124 4.22 (§6.8.7)

Discharging the forward-references. §6.8.3 flagged two open questions for §6.8.7 and §6.8.4 to settle, and §6.8.5 reserved §6.8.7 for the "cross-architecture status" reading. *Cross-architecture status:* §6.8.4 answered the within-LangGraph half — the dialogue-engine baseline sits below all four LangGraph cells on `strict_shift` (a gap §6.8.4's turn-convention correction shows to be convention-fragile, with the "uniquely never re-pivots" signature largely a turn-misalignment artifact) — so the cell_110 lift is not a within-LangGraph artefact, subject to the §5.12.3 prompt-scaffold confound that keeps it a directional floor rather than a clean isolation. *Cross-suite reconciliation:* the trap-suite positive and the §6.3 null are not in tension, for the reasons above — different instruments, different constructs, different intervention weights, with the trap positive consistent with §6.3.9's own forecast.

The clean cross-suite test, now run. The genuinely outstanding item §6.8.3–§6.8.5 reserved — run the LangGraph `state_policy` runner and the dialogue-engine baseline on a trap-style-annotated subset of the §6.3 suggestion scenarios, scored with the same binary `strict_shift`, so that "the heavy architecture beats the dialogue engine on localized strategic-move selection" can be checked off the trap suite it was measured on — has now been executed. `config/cross-suite-trap-scenarios.yaml` re-derives six trap scenarios from §6.3's suggestion suite (`epistemic_resistance_impasse`, `affective_shutdown_impasse`, `productive_deadlock_impasse`, `misconception_correction_flow`, `activity_avoider`, `struggling_learner`), spanning all four policy families (two substantive-engagement, one diagnostic, one repair/affective, two scaffolding), in the same hidden-state-plus-trigger format the v1 suite uses — so both runners and both scorers consume it unchanged. Two new cells run it at $N \approx 24$ each: `cell_124_langgraph_adaptive_crosssuite` (cell_110's `state_policy` graph, including counterfactual replay; 24/24 dialogues) and `cell_125_dialogue_engine_crosssuite_baseline` (cell_114's standard single-agent ego via the same `run-dialogue-engine-trap-baseline.js` adapter; 23/24, one transient generation failure). The result reproduces the v1 pattern in every respect:

- *Binary, same direction, same ratio.* `cell_124` selects the pre-registered family at trigger+1 on 62.5% of scenarios (15/24; family-match 66.7%) versus `cell_125`'s 30.4% (7/23; family-match 39.1%) — a ≈2.05× advantage, within rounding of the v1 suite's 1.91× (47.8% vs 25.0%; §6.8.3–§6.8.4). Both absolute levels run higher on the §6.3-

derived scenarios than on the bespoke v1 traps, plausibly because trigger signals re-derived from the well-worn suggestion suite are less ambiguous; the *contrast* between architectures is essentially the same magnitude. As on v1, the stored convention records every hit at trigger+1 with `cell_125` (like `cell_114`) never re-pivoting — but, per the §6.8.4 turn-convention correction, that "never re-pivots" signature is largely a turn-misalignment artifact of the dialogue-engine adapter (event-anchored, the engine does re-pivot within the post-trigger window), so it should not be read as a clean architectural property.

- *Graded, same inversion.* The 4-dimension graded rubric again flips the ranking: `cell_125` 4.60 versus `cell_124` 4.22 (mean over trigger-recognition, strategy-execution, strategy-quality, pedagogical-coherence), with the widest single-dimension gap on coherence (4.78 vs 4.08) — the dialogue engine writes the more coherent, better-calibrated trajectory even as it picks the wrong family more often (a stored-convention margin; convention-fragile per §6.8.4). `cell_124`'s own profile mirrors cell_110's: high trigger-recognition (4.58 — it *notices* the moment) but its weakest dimension is strategy-execution (3.96 — the prose carrying out the chosen move is rougher than the dialogue engine's). The split is, as in §6.8.4, real: the architectural advantage is in *which move, when*, not in *how well-crafted the teaching*.
- *Between-type, with shared blind spots.* The cross-suite gain concentrates in two scenarios — `cross_affective_shutdown` (`cell_124` 4/4 vs `cell_125` 1/4: the dialogue engine reads the affective-shutdown trigger as a cue to keep delivering content rather than to acknowledge-and-redirect) and `cross_productive_deadlock` (3/4 vs 1/4) — with a smaller edge on `cross_epistemic_resistance` (4/4 vs 3/4). On `cross_misconception_surfaces` *neither* architecture fires the annotated `ask_diagnostic_question` (both 0/4 — both prefer to engage the surfacing misconception substantively rather than diagnose it, which may say as much about the annotation as about the tutors), and on `cross_activity_avoidance` both are weak (`cell_124` 1/4, `cell_125` 0/4). The advantage is a between-type gain — the architecture more reliably recognises *which kind* of move the moment calls for — exactly as §6.8.5's within-family null implies it should be.

So the cross-architecture claim now holds on **two** independently-constructed trap instruments rather than one. It remains, on both, a *directional* comparison: `cell_125` carries the same §5.12.3 prompt-scaffold confound as `cell_114` (its ego prompt is tutor-core's base ego, not the adaptive runner's `tutorEgoInitial` builder), so the contrast bundles "no LangGraph state machinery" with "different prompt scaffold"; and it runs one direction only — no one has run the dialogue engine *with* the LangGraph runner's prompt scaffold, the symmetric isolation that would separate the two. With that caveat held, the §6.8.4 bottom line now reads: on two trap suites — the bespoke v1 and the §6.3-derived cross-suite — with their respective trap-steered learners, the `state_policy` architecture selects the pre-registered strategic move more often than the dialogue engine on the stored-convention binary ($\approx$1.9–2.05$\times$), while the dialogue engine writes more coherent prose; the effect is between-type, confined to the binary channel, confounded by the prompt scaffold, replicated rather than single-suite, and — per the §6.8.4 turn-convention correction — mostly an artifact: the secure re-run under the fixed adapter puts both legs at ~1.1$\times$ (cross-suite 1.07$\times$, v1 1.15$\times$), within $N \approx 24$ noise, so the durable claim is *no detectable binary advantage* on either suite, not the "twice" magnitude. The §6.3 results are unchanged by any of this. Per §5.12.6: the §6.8.7 reconciliation of the §6.3 null synthesises existing numbers, but the cross-suite cells (`cell_124`, `cell_125`) are *new post-hoc exploratory data* — not in the A13 pre-registration, status exploratory, no alpha correction, single-judge throughout (binary scoring `claude-code/sonnet`; graded scoring GPT-5/codex), no blinding — carried here because they discharge a forward-reference §6.8.3–§6.8.5 explicitly held open. Every comparison in this section that involves cell_114 or cell_125 is post-hoc exploratory; the §6.8.3 and §6.8.5 results it cites carry their own (confirmatory, pre-registered) status.

6.8.8 Closing Synthesis: The Apparatus Is the Contribution; the Elaborations Mostly Are Not Stepping back from the cell-by-cell results to the question §6.8 opened with — did an externalised learner-state model plus a programmatic policy graph deliver the adaptive responsiveness §6.3 found absent? — the answer is *partially: on the trap binary, replicated across two independently-built trap suites, in a comparison still confounded by prompt scaffold, and not through the mechanisms the design assumed.* That answer, read alongside §6.7's parallel verdict on the id-director family, locates the real contribution of these two architectural-extension arcs: not in the architectures, but in the apparatus built to test them (the general point is developed in §7.4).

What is defensible. Four findings survive their caveats:

1. *The trap-scenario methodology works.* Externally-defined `expectedStrategyShift` annotations, a hidden-state-conditioned learner that springs a trigger at an annotated turn, counterfactual replay, and a binary `strict_shift` correctness check together turn "did the tutor adapt" — diffuse and slope-shaped in §6.3 — into a localized, scoreable event. That methodology is a contribution independent of any architecture it was used to evaluate; §6.3.9's account of why the slope-and-cosine instruments wash out a single strategic move is, in effect, the negative-space argument for why an instrument like the trap suite was needed (§6.8.7).

2. *The `state_policy` architecture produces a moderate strategy-shift advantage over the dialogue-engine floor, replicated on a second trap suite.* On the v1 trap suite, `cell_110` selects the pre-registered family of moves at trigger+1 more often than the dialogue-engine baseline `cell_114` on the stored-convention binary (47.8% vs 25.0% `strict_shift`; §6.8.3–§6.8.4) — though, per the §6.8.4 turn-convention correction, that gap is convention-fragile and the apparent "the dialogue engine never re-pivots" property is largely a turn-misalignment artifact (the v1 advantage does not survive event-anchored family scoring). The clean cross-suite test (§6.8.7) reproduces this on a trap instrument re-derived independently from the §6.3 suggestion scenarios: `cell_124` (cell_110's architecture) 62.5% vs `cell_125` (cell_114's architecture) 30.4% — a ≈2.05× advantage, within rounding of the v1 ratio (1.91×) — with the graded rubric inverting on the cross-suite exactly as it does on v1 (`cell_125` 4.60 > `cell_124` 4.22). This is the first result in the project where an architectural intervention — not a prompt, not a second agent, not a learner-architecture swap — moves a sensitive adaptive-behaviour measure, and it now moves it on two suites rather than one. But it is *mostly an artifact* (the secure re-run under the fixed adapter puts both legs at ~1.1× — v1 1.15×, cross-

suite 1.07× — within $N \approx 24$ noise, i.e. no detectable binary advantage; the §6.8.4 turn-convention correction), and what little may remain is *between-type* not within-type (§6.8.5), *confined to the binary channel* (the graded rubric inverts the ranking on both suites — `cell_114` 4.46 > `cell_110` 4.03; `cell_125` 4.60 > `cell_124` 4.22; §6.8.4, §6.8.7), *confounded* by the prompt-scaffold difference that makes `cell_114`/`cell_125` directional floors rather than clean single-factor baselines (§5.12.3), and *not isolated in the symmetric direction* — the dialogue engine has never been run *with* the LangGraph runner's prompt scaffold, so "architecture, not scaffold" remains an inference rather than a measured contrast on either suite.

3. *Within the state machine, less learner state is more.* The P2.2 ablation (§6.8.6) reversed its own pre-registered dose-response prediction: the two-field minimal profile `cell_118` scores highest on `strict_shift` (68.8%) and on the graded rubric (4.34); the three richer cells (`cell_110`/`cell_119`/`cell_120`) form a flat ~50% cluster; and the post-hoc `cell_123` localizes the binary floor to "any fourth structured field" with a load/capacity signature (the effect saturates), not a single toxic field. The structured `learnerProfile` the architecture was built around does its measurable work, on this instrument, through `confidence` and a quoted-dialogue anchor — the named-`misconceptions` array and the fixed-vocabulary `agencySignal` enum are mildly counterproductive, and `zpdEstimate` is close to inert (a modest tax on graded execution prose, free on move selection).
4. *Bilateral-ToM elaboration adds nothing measurable.* The P2.1 test (§6.8.5) was a clean pre-registered null — no within-family discrimination over `recognition_only` (pooled +4.2 pp, below the 5-pp falsification floor). Negative, but the cleanest-status finding in the section. Read against the broader Theory-of-Mind benchmark literature, this null and the §6.8.6 state-richness reversal are plausibly *regime*-bounded rather than instrument defects: GroupToM-Bench (Tang et al., 2026) locates the payoff of explicit social-mental-state modelling at the *group* level, where collective outcomes emerge non-linearly from conformity and structural pressure and cannot be recovered by summing individual intents (its "linear-superposition-bias" failure mode) — a regime our strictly *dyadic*, single-learner trap suite never enters, so the inertness here of explicit first/third-person ToM and a structured mental-state schema is consistent with that machinery becoming load-bearing only once more than two parties interact. Extending the apparatus to group / >2-party interaction dynamics — where that structure is expected to start paying — is a forward direction this arc flags but does not test.

What is not established. Three gaps bound the arc:

- *No clean cross-architecture claim, though the cross-suite gap is now replicated.* The dialogue-engine comparison (`cell_114`, and its cross-suite twin `cell_125`) bundles "no LangGraph state machinery" with "different prompt scaffold" (§5.12.3), so it is a directional floor, not an isolation; the only other cross-architecture data point is a thin $N = 6$ id-director overlap (`cell_106`, never re-run at adequate N). The "no trap-style instrument for the §6.3 suite" gap is closed: §6.8.7's `config/cross-suite-trap-scenarios.yaml` re-derives six trap scenarios from the suggestion suite, and on it `cell_124` beats `cell_125` 62.5% vs 30.4% on `strict_shift` *on the stored convention* (≈2.05×) — but the secure re-run under the fixed adapter collapses this to 62.5% vs 58.3% — i.e. 1.07× — within $N \approx 24$ noise (§6.8.4) — with the graded inversion also replicating; so the cross-architecture *direction* (state $\geq$ engine) holds weakly on two independent trap instruments, but the *magnitude* is an artifact: corrected, there is **no detectable binary advantage** on either suite (v1 1.15×, cross-suite 1.07×, both n.s.; §6.8.4). What is still missing is the *symmetric* isolation — running the dialogue engine *with* the LangGraph runner's prompt scaffold — which is what would turn "architecture, not scaffold" from an inference into a measured single-factor contrast on either suite.

- *The hypothesised mechanisms mostly are not doing the hypothesised work.* The design's three architectural bets were rich externalised learner state, programmatic policy actions, and counterfactual replay. P2.1 falsified the bilateral elaboration; P2.2 reversed the state-richness prediction and pinned the residual advantage on two scalar/text fields; counterfactual replay produced only a messy signal where it ran (high raw divergence but low family-coherence on P2.1's two ToM cells — §6.8.5) and was a literal no-op for the P2.2 ablations (`counterfactual_total = 0` for every P2.2 cell, the ablation architectures falling outside the runner's profile-update set). Individually each is a finding; collectively they say the elaborations the harness made testable did not behave as the architecture assumed.
- *P2.3 (the crossover cells) is retired, not deferred.* The pre-registered fan-out scheduled two further cells (`cell_121`, `cell_122`) for a P2.3 crossover test; P2.1's null and P2.2's reversal have substantially weakened the hypothesis those cells were built to probe, so running them on the original schedule would be sunk-cost behaviour and the arc closes here rather than carrying an open P2.3 tail. (If a crossover question is revived it should follow a fresh decision about what it now answers, not the original plan.)

Relation to §6.3.9 and §6.7. This arc does not reverse §6.3.9's central finding — the insight-action gap is structural under lightweight intervention and only suggestively mitigable by expensive search. It *extends* it: a heavy externalised-state architecture is the first thing in the project to produce a measurable shift on an instrument sensitive enough to see one, but the shift is moderate, channel-specific, confounded, and — the part that matters for the mechanism story — it *survives stripping the architecture down to a two-field profile*, which is not what a "rich learner model drives adaptation" hypothesis predicts. Read with §6.7, the pattern rhymes: the id-director family found charismatic-pedagogical authorship could be operationalised but that mechanism stacks sit *inside* the Pareto frontier rather than pushing it out (§6.7's synthesis); the adaptive-runner family found adaptive strategy selection could be operationalised but that the architecture's elaborations mostly did not pay. In both arcs the durable result is the apparatus — the id-construction trace and the charisma rubric in §6.7, the trap suite and the dual scoring paths here — and the architectural elaborations the apparatus made legible came back null, negative, or inverted more often than not. That is a finding about how hard it is to make these elaborations pay, reported as such, in the spirit of §5.12.6's instruction that mixed and negative outcomes are reported rather than patched into re-ordered hypotheses.

Status (per §5.12.6). §6.8.8 introduces no new numbers; it synthesises §6.8.1–§6.8.7 (themselves a mix of pre-registered confirmatory — §6.8.3, §6.8.5 — and post-hoc exploratory — §6.8.4, §6.8.6, and §6.8.7's `cell_124`/`cell_125` cross-suite cells — tiers, with §6.8.7's §6.3-null reconciliation a no-new-data synthesis layered over those new exploratory cells). No alpha correction applies to a synthesis. The retirement of P2.3 is a scope decision, recorded here so the pre-registered fan-out's open items are accounted for rather than silently dropped.

6.8.9 Postscript: From Local Repair to Longitudinal Character-State Routing A later DAG/resistance follow-up tested a narrower question than §6.8's trap-suite architecture comparison: once the proof-DAG policy and resistance-signal router can repair a local scene, can their evidence be carried forward as a compact learner *character state* that reduces repeated repair on later scenes? The implementation adds a six-axis state ledger (`proof_ownership`, `relevance_orientation`, `frustration_tolerance`, `question_consolidation`, `non_formulaic_reasoning`, `next_step_agency`) updated only from observed outcome evidence, then runs the same learner identity through six linked scenes (`boredom`, `frustration`, `irrelevance`, `question_flood`, `rote_parroting`, and a transfer scene). The decisive contrast holds the v2 local policy fixed and varies only whether the carried state is routed into the generated learner context: `v2_policy_only` versus `character_state_plus_v2`. The learner sees sanitized character-state summaries and prior scene outcomes, not the target evidence labels; the closed-loop tutor policy remains programmatic. Implementation and artifacts: `services/adaptiveTutor/characterState.js`, `scripts/run-dag-resistance-character-development.js`, and `exports/adaptive-dag-resistance-charact`

On a generated synthetic-learner run using the real learner backend (`codex` provider, `model=auto`, three seeds $\times$ six scenes $= N = 18$ per arm), the state-routed arm improves efficiency over an already strong v2 baseline: `v2_policy_only` succeeds on 17/18 scenes, with 13/18 first-response successes and 4/18 staged follow-ups; `character_state_plus_v2` succeeds on 18/18 scenes, with 15/18 first-response successes and 3/18 staged follow-ups. Both arms hit 3/3 first-response successes on the transfer scene. The mock generated-learner contrast is lower and sharper (`v2_policy_only` 0/18 successes, `character_state_plus_v2` 6/18), confirming that the scaffold can create a developmental signature when the learner does not spontaneously provide evidence, but the real generated learner is sufficiently competent that the added state layer buys a modest efficiency gain rather than a qualitative rescue.

The claim is therefore bounded. This is **not** human character development, and it is not evidence that the tutor's philosophical-recognition orientation produces longitudinal learner transformation. It is a post-hoc synthetic mechanism probe showing that local adaptation evidence can be accumulated into a cross-scene state representation and routed back into later learner uptake without exposing answer labels, yielding a small improvement in first-response closure under real generated learner calls. In the terminology of §6.8.8, the contribution is again apparatus-first: the policy layer can now distinguish "repair succeeded locally" from "the learner has become more self-directing across scenes," and can measure the latter with first-response success, staged-follow-up rate, and transfer closure. Whether the same state variables track durable human learner development remains a pilot question, not a result here.

A second follow-up folds the same state ledger into a synthetic **Character-DAG drama** benchmark: a fixture-driven arc (`setup`, `pressure`, `peripeteia`, `consolidation`, `transfer`) couples proof-DAG/resistance policy to dramatic pressure and character-state routing, with a shuffled-state arm as the negative control. The strict real generated-learner robustness run uses three decisive arms, three seeds, eight scenes, and four fixture perturbations (`baseline`, `noisy_openings`, `harder_transfer`, `state_dependent_transfer`; $N = 24$/arm/perturbation). `full_character_dag_drama` passes every perturbation: first-response success is 21/24 in all four slices, compared with `policy_only` at 16/24, 12/24, 17/24, and 11/24, and the shuffled-state control at 14/24, 11/24, 9/24, and 9/24. The transfer channel is the sharpest contrast: full reaches 9/9, 8/9, 9/9, and 9/9 first-response transfer closure; `policy_only` reaches 8/9, 5/9, 9/9, and 1/9; shuffled-state reaches 4/9, 4/9, 3/9, and 2/9. The harder-transfer perturbation exposes a policy-ceiling case (`policy_only` and full both 9/9 transfer), so the gate is recorded as "full transfer stronger or policy ceiling matched" while the robustness layer still requires an aggregate full-policy transfer margin across the strict set. In all four perturbations, full keeps the staged-followup-or-unresolved burden at 3 scenes, observes 3/3 required peripeteia events with zero unrequired detections, and records zero target-label and public theory/process leaks. Implementation and artifacts: `config/character-dag-drama-framework.yaml`, `scripts/run-character-dag-drama-framework.js`, `scripts/run-character-dag-drama-robustness.js`, `services/adaptiveTutor/outcomeObserver.js`, and `exports/character-dag-drama-framework-robustness-`

A stricter follow-up then targeted the failure mode exposed by the stronger controls: compressed state could previously answer transfer scenes with generic "some condition must hold" language. The observer was tightened so transfer closure requires naming the public prior check carried by matched character state, while compressed and stale controls withhold that concrete check. A larger real generated-learner `state_dependent_transfer` matrix then ran two seeds across four fixture families (`base`, `ratio_series`, `definition_boundary`, `causal_identification`), seven arms, and eight scenes ($N = 16$/arm/family). All four family runs pass acceptance gates under `character-dag-drama-observer.v0.5`. Full first-response success is 15/16, 14/16, 14/16, and 13/16; policy-only is 6/16, 3/16, 5/16, and 4/16; shuffled-state is 7/16, 6/16, 6/16, and 5/16. The transfer-specific contrast is sharper: full closes 6/6, 5/6, 6/6, and 5/6 transfer scenes on the first response, while policy-only, shuffled, stale, overconfident, and compressed controls all close 0/6 transfer scenes in the three non-base families and policy/shuffled remain 0/6 in the base family. Every family records zero target-label leaks and zero public theory/process leaks. This upgrades the Character-DAG drama result from a single-fixture strict-perturbation robustness claim to a fixture-family synthetic robustness claim, but it remains an exploratory file-artifact result, not a Paper 2.0 main mechanism estimate.

The licensed reading is still apparatus-first. The result shows that, inside this synthetic harness, proof-DAG state transitions, resistance-signal routing, peripeteia-form pressure, and an evidence-derived character state can be coordinated as one adaptation policy layer, and that the matched-state route separates from a shuffled-state control on transfer. It does **not** show human learning, real interior change, deployed tutoring reliability, or a general "adaptive subject" capacity. It licenses the next development question: whether the same framework can be run on larger, preregistered synthetic families and then translated into human-learner pilot hypotheses without treating synthetic character-development form as learning evidence.

Status (per §5.12.6). Post-hoc exploratory mechanism probes; synthetic learner throughout; no human outcomes; no alpha correction; policy realization remains programmatic; learner generation is real-LLM-backed only in the `learner-mode=llm` runs. Evidence is file-based rather than DB-backed. For the longitudinal state probe, `exports/adaptive-dag-resistance-character-development-llm-real/report.md` and `summary.json` are the canonical record. For the Character-DAG drama framework, `exports/character-dag-drama-framework-transfer-specificity-real-family/robustness-report.md` and `robustness-summary.json` are the strongest current strict robustness record; `exports/character-dag-drama-framework-robustness-policy-repair-real/`, `exports/character-dag-dra` `exports/character-dag-drama-framework/`, and `exports/character-dag-drama-framework-llm-mock/` remain earlier contrast/smoke artifacts. The results support only the modest efficiency/apparatus claims stated above.

6.9 Evidence-Bound Hypothesis Tracking: An Orthogonal Extension to the State-Policy Architecture (A14)

This subsection reports a separate architectural-extension arc, run on the same LangGraph runner as §6.8 but addressing a different gap. §6.8 asked whether an externalised `learnerProfile` plus a programmatic policy graph produces measurable strategy shifts on a trap-scenario instrument. §6.9 asks a discipline question, not a behaviour question: *when the tutor maintains a model of the learner, can the controller force every claim in that model to be cited against verbatim evidence from the dialogue?* The §6.8 `learnerProfile` is a single object whose fields ("the learner has misconception X", "confidence is 0.4") are written each turn without an audit trail — there is no way, from the persisted trace, to ask *what learner utterance the tutor's claim about misconception X is grounded in.* The §6.8.6 state-richness reversal is one consequence of that opacity: a richer profile did not improve strategy selection partly because the apparatus could not tell whether the additional fields were tracking the learner or accumulating tutor speculation. §6.9 is the first stage of a longer-horizon answer (the A14 arc in `TODO.md`): replace the unstructured profile-write with an append-only observation ledger and a typed-hypothesis layer whose claims must reference ledger entries by id.

The arc reaches a stable resting point at the end of this subsection: three graph nodes (extractor, updater, validator) are implemented and verified on a small ($N = 3$ scenario) real-LLM smoke; a paired ablation between `cell_126` (updater only) and `cell_127` (updater plus validator) on the two scenarios that both cells ran cleanly measures the validator's marginal contribution; and a Stage 5 pedagogical-outcome scoring of all three evidence-bound cells at pooled $N \approx 33$ against §6.8's binary and graded instruments (§6.9.7) closes the arc. Per §5.12.6, what follows is *post-hoc, exploratory, and infrastructural*: no pre-registration, no hypothesis test, with the §6.8-vs-§6.9 comparison added at Stage 5 (§6.9.7) as a post-hoc exploratory contrast. The contribution is mechanism availability — the apparatus needed to ask grounding-discipline questions in later work — together with one mechanism finding in hand (asymmetric attribution of status verdicts between the updater and validator layers; §6.9.5–§6.9.6) and a Stage 5 pedagogical-outcome reading (§6.9.7) that is negative-leaning on the apparatus as a whole: the audit chain does not pay in §6.8's strategy-shift coin, and applied to §6.8.6's best configuration it is net-negative on both the binary and graded channels. §6.9.8 then steps outside A14 specifically to close the broader §6.8.8/§6.9 question across the *realisations* of a `state → action` policy — implicit forward pass, hand-authored nodes, and a policy *fitted* to the independent outcome — with the one lever the arc had not yet pulled, killed offline before any live run.

6.9.1 Method: Append-Only Ledger, Typed Hypotheses with TTL Two state-schema additions extend the §6.8 `AdaptiveTutorState` (see `services/adaptiveTutor/stateSchema.js`). The first is `evidenceLog` — an append-only ledger reduced by concatenation. Each entry carries an `obs_id` (content-derived sha1 prefix, stable across counterfactual replay branches), the `turn` index, a verbatim `quote` span from the dialogue, a `type` tag drawn from a fixed enum (`learner_self_report`, `learner_action`, `learner_question`, `learner_correction`, `tutor_inference`), and a `validated` boolean set by a substring-match gate (described below). The reducer is append-only so the audit trail cannot be silently rewritten by a later turn.

The second is `hypotheses` — typed tentative claims about the learner, reduced by merge-by-id (last write wins per `hypothesis_id`). Each hypothesis carries a `claim` string, a `confidence` in $[0, 1]$, two arrays of `obs_id` references (`supporting_evidence` and `contradicting_evidence`), a `status` from `{tentative, validated, contradicted, expired}`, and TTL metadata as the pair `(created_at_turn, expires_after_turns)` rather than an absolute `expires_at_turn` — storing the offset keeps creation metadata immutable under counterfactual replay, where turn counters can branch. The merge-by-id reducer is the design's key commitment: a hypothesis revised on turn $t + 1$ overwrites the same entry rather than appending a parallel claim, so the persisted state reflects the controller's current best guess rather than its full history of guesses.

Two new graph nodes operate on these fields. The *evidence extractor* runs each turn after a learner utterance and before the existing profile-update path; it asks the model to return verbatim quotes typed by the enum above. The node then enforces a substring-match gate: each candidate quote is checked against the cumulative dialogue text, and `validated` is set to `true` only if the gate passes (the entry is retained either way, so hallucinated quotes are recorded for audit rather than silently discarded). The *hypothesis updater* runs immediately after the extractor and consumes only the validated subset of the ledger together with the current `hypotheses` list. It is prompted to revise existing entries (re-emit the same `hypothesis_id` when extending support, mark `contradicted` when the new evidence undermines an earlier claim) rather than mint new claims that paraphrase old ones, and to return an empty array when the evidence will not support any tentative inference. The node owns three pieces of bookkeeping the LLM is not trusted with: it derives a stable `hypothesis_id` from the `claim` text when the model leaves the field blank, it filters `supporting_evidence` and `contradicting_evidence` against the set of valid `obs_ids` so the LLM cannot invent references, and it sweeps for TTL expiry (transitions `tentative → expired` when turn $>$ created_at_turn $+$ expires_after_turns). The "node owns bookkeeping; the LLM owns judgment" split is the same pattern §6.7's id-director engine uses to constrain charismatic-persona authorship to a structured envelope.

A new cell `cell_126_state_policy_evidence_bound` registers the extension. Its `adaptive.architecture` is `state_policy_evidence_bound` — the existing `state_policy` graph (§6.8.2) extended with the extractor and updater nodes wired ahead of the profile-update path. All other cells in §6.8 remain unaffected: the schema additions default to empty arrays and the new nodes only run under the `state_policy_evidence_bound` architecture flag.

6.9.2 Stage 2 Verification: 100% Verifiable Quotes, 100% Grounded Hypotheses on Three Trap Scenarios A real-LLM smoke (`scripts/run-adaptive-cell-real-smoke.js`, `claude-code/sonnet` backend, $N = 3$ scenarios $\times$ 2 branches = 6 dialogues) tested the extractor (Stage 2a) and the updater (Stage 2b) jointly on three v1 trap scenarios spanning the agency-signal range — `false_confusion_v1` (compliant), `resistance_to_insight_v1` (resistant), `sophistication_upgrade_v1` (questioning).

Stage	Metric	Result	Pre-registered?
2a	Verifiable-quote rate (validated entries / total entries)	128 / 128 = **100.0%**	No — internal calibration target was $\geq$95%
2b	Grounded-hypothesis rate (hypotheses with $\geq$ 1 supporting `obs_id` / total hypotheses)	53 / 53 = **100.0%**	No — internal soft gate was $\geq$70%

Both rates clear their internal calibration targets. Extraction fires across all turns of all three scenarios (per-turn breakdown: every k-of-k on every turn), with no quote failing the substring-match gate. Hypothesis synthesis is also active rather than degenerate: across the three scenarios the updater emits 13, 16, and 24 hypotheses respectively (mean claim length 196–254 characters; status distribution shows 18 of 53 marked `contradicted` rather than `tentative`, so the model uses the enum to retire claims rather than letting them silently rot), and every hypothesis carries at least one valid `obs_id` reference into the ledger. These are the lower bars Stage 2 was designed to clear: *the controller can build a verifiable evidence ledger and the updater consistently cites it.*

The numbers come from a $N = 3$ smoke, not a study; they speak to the implementation passing its own verification gates, not to any pedagogical claim. Generalisation to the full eight-scenario v1 suite or to cross-suite trap stimuli (the §6.8.7 instrument) is not yet measured.

6.9.3 Limitation Surfaced by the Verification: Revision Versus Duplication A separate signal in the same smoke is informative for what Stage 3 has to do but is not itself a Stage 2 result. The merge-by-id reducer collapses two hypothesis records into one only when their `hypothesis_id` strings match. The updater is prompted to re-emit prior turn ids when extending or contradicting an existing claim, and the node derives a stable id from the `claim` text when the LLM omits it. The smoke shows the reducer is rarely firing: on `sophistication_upgrade_v1`, 24 distinct `hypothesis_id`s survive across a 4-turn × 2-branch dialogue, with `created_at_turn` spread $[0, 0, 1, 1, 1, 1, 2, 2, 2, 3, 3, 3]$ on each branch. That is the same count of survivors as proposals, indicating the model produces fresh claim text on each turn rather than rewriting yesterday's — the content-derived id then differs, and the reducer treats the new claim as a new entity. The grounded-rate gate (Stage 2b's 70% soft target) does not catch this failure mode because it asks "is each claim grounded?", not "is the same claim being maintained across turns?"

This is reported as a limitation, not a result. Three observations bound how it should be read. First, it is what the design predicted Stage 3 would have to take on: the `groundingValidator` node planned for that stage will retain or retire hypotheses against the ledger across turns, which is the natural place to instrument revision-vs-duplication directly. Second, the status discipline observed at Stage 2b (18 of 53 hypotheses `contradicted` rather than `tentative`) is evidence that the model uses the typed-status machinery when prompted to — the duplication is not "the model ignores the schema" but "the model writes new claims when an update prompt would have been a better fit". Third, the apparatus needed to *measure* revision-vs-duplication crisply does not yet exist either: `extractTurnTrace` (`services/adaptiveTutor/persistence.js`) captures terminal `evidenceLog` and `hypotheses` but not per-turn snapshots, so the count of distinct survivors is a proxy for revision rate, not a direct measurement. Closing that proxy gap is logged as Stage 3 work.

6.9.4 What is Defensible; What is Not What §6.9 establishes is narrow and infrastructural. The state schema for an audited learner model — ledger, typed hypotheses, TTL semantics, content-derived ids stable across counterfactual replay — exists and is reducer-correct. Two new graph nodes (`evidenceExtractor` with substring-match validation gate; `hypothesisUpdater` with node-owned bookkeeping for id derivation, evidence-ref filtering, and TTL sweep) operate on a fresh `state_policy_evidence_bound` architecture; both are exercised in mock and real-LLM smokes. On a $N = 3$ real-LLM verification the extractor emits 100% substring-verifiable quotes and the updater emits 100% ledger-grounded hypotheses, clearing the calibration targets the implementation was built to. The contribution is the apparatus, in the same sense §6.8.8 means it: the §6.9 cell adds a layer the apparatus could not previously ask the question about, not a result about whether the layer helps pedagogically.

What §6.9 does not establish is anything about pedagogical outcome (the evidence-bound cell has not been scored on the binary or graded instruments of §6.8, and would need to be at $N \approx$ 24–32 before such a comparison would be informative), nothing about whether the discipline survives a stronger ablation more broadly (varied scenario types, different judge models, longer dialogues), and — as §6.9.5 develops — a single-sided Stage 3 measurement on `cell_127` but no matched cell_126 status-distribution on the same scenarios from which to read the validator's *marginal* contribution beyond what a well-tuned updater alone does. The Stage 3 architecture (`state_policy_evidence_bound_validated`) and its paired cell (`cell_127_state_policy_evidence_bound_validated`) are now implemented — the `groundingValidator` graph node downstream of the updater promotes hypotheses with ≥3 supporting `obs_ids` and no contradiction to `validated`, and retires hypotheses with ≥1 contradicting `obs_id` and confidence below 0.4 to `contradicted` — and §6.9.5 reports the first real-LLM smoke through that chain. The honest claim across §6.9 remains that a mechanism is becoming *available*, and §6.9.5 adds that the cheap-verifier layer fires non-trivially on the available material; neither claim is about pedagogical payoff.

6.9.5 Stage 3 Result: Grounding-Validator Reshapes Terminal Status Distribution A real-LLM smoke on `cell_127_state_policy_evidence_bound_validated` (`claude-code/sonnet`, the same $N = 3$ trap scenarios as the Stage 2 smoke — `false_confusion_v1`, `resistance_to_insight_v1`, `sophistication_upgrade_v1` — with `ADAPTIVE_TUTOR_CLI_TRACE=1` for per-call diagnostic visibility) ran the full chain (`evidenceExtractor` → `hypothesisUpdater` → `groundingValidator` → `learnerProfileUpdate` → tutor ego/superego), and yields the following terminal-state distribution across 26 hypotheses synthesized over the three scenarios:

Status	Count	Fraction
`validated`	13	50%
`contradicted`	9	35%
`tentative`	4	15%

Stages 2a and 2b clear at the same rates as the Stage 2 smoke (120/120 = 100% substring-verifiable quotes; 26/26 = 100% ledger-grounded hypotheses), and the revision rate is 20/22 = 90.9%, statistically indistinguishable from the cell_126 post-tune baseline at 92.9%: adding the validator does not break the upstream updater's revision discipline. What it adds is the status-discipline column. Of the 26 hypotheses the updater produced, the validator emitted a retain or retire verdict on 22 (85%) and stayed silent on 4 (15%) — the SILENCE-by-default convention is doing its work, and the 50/35 promote/retire split shows the verifier is not one-sidedly stamping things `validated`. Per-scenario the same shape holds: `sophistication_upgrade_v1` (the trap the prior Stage 2 smoke could not complete) lands at 12 hypotheses with 8 validated, 2 contradicted, 2 tentative — the heaviest validation load and also the heaviest revision rate (9/10 = 90%). The architecture-stress contrast carries through.

A latency note speaks to the design intent. Median per-call latencies under `claude-code/sonnet` are: `hypothesisUpdater` ~100s, `learnerProfileUpdate` ~50s, `tutorEgoInitial` ~45s, `groundingValidator` ~18s, with `learnerTurn` / `tutorSuperego` / `evidenceExtractor` all near 20s. The validator's per-call cost is roughly 5× cheaper than the upstream `hypothesisUpdater` it audits, consistent with the cheap-verifier-after-expensive-synthesizer pattern the architecture was built to: high-cost generation feeding low-cost validation, with the validator's verdict-only contract (status transitions only, no rewriting of hypothesis content) holding the budget down.

A paired-ablation run on `cell_126` (validator off; same `claude-code/sonnet` backend; same trap-scenario inputs) measures the marginal contribution. The Stage 2 cell_126 smoke at commit `1e02510` reported only aggregate revision rate, so a re-run was needed to capture the terminal status distribution. Two of the three scenarios completed cleanly; `sophistication_` `upgrade_v1` crashed on a Zod schema-strictness bug (`evidenceExtractor` emitted `quote: null` for a `tutor_inference` entry; the graph node at `services/adaptiveTutor/graph.js:371` already coerces null quotes to empty strings — the substring-match gate then correctly marks the entry `validated=false` — but the schema rejected the null upstream). Loosened to `quote: z.string().nullable()` with the rationale documented inline. The matched comparison reads off the two scenarios both cells ran cleanly:

Cell	N scenarios	Hypotheses	`validated`	`contradicted`	`tentative`
`cell_126` (updater only)	2	19	0 (0.0%)	5 (26.3%)	14 (73.7%)
`cell_127` (updater + validator)	2	14	5 (35.7%)	7 (50.0%)	2 (14.3%)

Three patterns are visible. (i) **`validated` is wholly validator-driven.** The updater alone produces zero `validated` outcomes across 19 hypotheses on the same scenarios — every `validated` verdict in `cell_127` is attributable to the `groundingValidator` layer, not to a status field the upstream `hypothesisUpdater` was emitting unaided. The mock-mode rule ($\geq$3 supporting `obs_ids` + no contradiction $\rightarrow$ promote) is being exercised on real-LLM output. (ii) **`contradicted` is partly shared, not wholly validator-driven.** `cell_126` produces 5 `contradicted` hypotheses unaided; `cell_127` produces 7. The updater itself uses the typed-status machinery to retire claims when prompted to — the validator adds to the retirement count rather than monopolizing it. (iii) **Uncertainty drops sharply.** `cell_126`'s 73.7% `tentative` collapses to 14.3% in `cell_127`: the validator converts about 59 percentage points of unresolved status into typed verdicts (split roughly 35 pp promotion / 24 pp retirement). The asymmetry — wholly-validator-driven promotion, shared retirement — is the operational reading of the verdict-only contract: the validator's promotion threshold ($\geq$3 supporting refs, no contradiction) requires evidence accumulation across turns that the updater's single-turn synthesis prompt does not gather, while retirement is reachable from either layer once a contradicting `obs_id` exists.

What §6.9.5 *does* establish: the validator fires on the available material, with verdicts of both kinds, at a cost meaningfully lower than the layer it audits, without disturbing the revision discipline upstream; and on the matched two-scenario subset, the validator is the sole source of `validated` verdicts and a non-trivial additional source of `contradicted` ones. What it does *not* establish: any pedagogical-outcome claim (the chain has not been scored on the §6.8 binary or graded instruments, and would need to be at $N \approx 24$–32 before such a comparison would be informative); generalisation beyond the two matched scenarios (`sophistication_upgrade_v1` is the heaviest-load trap in the suite and the validator's contribution there is unmeasured at the paired level — only the single-sided cell_127 reading at 8 `validated` / 2 `contradicted` / 2 `tentative` is in hand); robustness across judge models or generation backends; and whether the validator's verdict-rate would survive scenarios where the upstream updater is *already* well-tuned to emit `validated` directly. The defensible claim is mechanism-level: a cheap verifier downstream of an expensive synthesizer measurably reshapes the terminal status distribution on this material, with promotion as its unique contribution.

A methods note resolves a separate open item (`TODO.md` task #34) raised by the Stage 2b smoke. That earlier run was reported as "hanging silently for 48 minutes" on the third trap scenario. With per-call CLI tracing now on, the cell_127 smoke ran the same scenario set to completion in ~100 minutes wallclock without any individual role-call exceeding the 6-minute CLI subprocess timeout; the longest single call was 134s (`hypothesisUpdater` on `sophistication_upgrade_v1`). The prior report was a misdiagnosis. The earlier smoke was not hung but slow: many sub-6-minute calls compounded into a wallclock window that exceeded the operator's patience before any of them tripped the existing timeout. The instrumentation added in `services/adaptiveTutor/realLLM.js` (env-gated `ADAPTIVE_TUTOR_CLI_TRACE=1` start/done lines per CLI call, plus an unconditional stderr line on timeout-fire with role, elapsed, output-bytes, and first-byte timing) closes the diagnostic gap that produced the silent-wait failure mode. Future hangs surface a role + duration + stream-state signal in real time rather than after the fact.

6.9.6 Stage 3 Synthesis: The Audit Chain Works; the Pedagogical Question Is Settled in §6.9.7 Stepping back from the cell-by-cell results to the discipline question §6.9 opened with — when the tutor maintains a model of the learner, can the controller force every claim in that model to be cited against verbatim evidence from the dialogue? — the answer is *yes, on three trap scenarios, with every internal gate clearing and a clean mechanism finding about how the chain decomposes; but no pedagogical-outcome link to the §6.3 adaptive-responsiveness null or §6.8's strategy-shift instrument has been measured.* Like §6.8.8 and §6.7 before it, the durable result is the apparatus; unlike §6.8.8, the apparatus has already produced a substantive mechanism finding (asymmetric attribution; §6.9.5) even before any pedagogical test has been attempted.

What is defensible. Four findings survive their caveats:

1. *The three-stage audit chain holds on real-LLM material.* On three trap scenarios (`false_confusion_v1`, `resistance_to_insight_v1`, `sophistication_upgrade_v1`) under `claude-code/sonnet`, Stage 2a clears at 128/128 = 100% substring-verifiable quotes (§6.9.2), Stage 2b at 53/53 = 100% ledger-grounded hypotheses (§6.9.2), and Stage 3 produces a non-trivial verdict distribution (13 `validated` / 9 `contradicted` / 4 `tentative` across 26 hypotheses on `cell_127` at $N = 3$; 85% non-silent verdicts; §6.9.5). The "node owns bookkeeping; the LLM owns judgment" division (id derivation, evidence-ref filtering, TTL sweep on the node side; quote selection, claim synthesis, status verdict on the LLM side) operates cleanly across the chain.

2. *Validator marginal contribution is wholly-promotion, partly-retirement.* The matched $N = 2$ paired ablation against `cell_126` (validator off; same scenarios, same backend; §6.9.5) reads: `cell_126` produces 0 `validated` / 5 `contradicted` / 14 `tentative` across 19 hypotheses; `cell_127` on the same two scenarios produces 5 `validated` / 7 `contradicted` / 2 `tentative` across 14 hypotheses. The validator is the wholly-attributable source of `validated` promotions (0 → 5) and a non-trivial additional source

of `contradicted` retirements ($5 \to 7$), with ~59 percentage points of `tentative` collapsing into typed verdicts. The asymmetry is mechanism-level: the validator's promotion threshold (≥ 3 supporting `obs_ids`, no contradiction) requires multi-turn evidence accumulation the updater's single-turn synthesis prompt does not perform; retirement is reachable from either layer once a contradicting `obs_id` exists.

3. *The cheap-verifier-after-expensive-synthesizer pattern is empirically demonstrated.* Median per-call latency under `claude-code/sonnet`: `hypothesisUpdater` ~100s, `groundingValidator` ~18s — the validator costs roughly 5× less per call than the layer it audits (§6.9.5). The verdict-only contract enforced by the Zod schema (LLM owns the status transition only; the graph node preserves `claim`, `supporting_evidence`, `contradicting_evidence`, TTL metadata) keeps the validator's per-call budget tight and prevents it from rewriting hypothesis content. The pattern generalises beyond this specific chain: any agent architecture with a expensive-generator-followed-by-cheap-verifier shape can use the verdict-only contract to bound the verifier's responsibility surface.

4. *Task #34 misdiagnosis resolved via per-call instrumentation.* The Stage 2b smoke's "48-minute silent hang" turned out to be a slow-not-hung condition: many sub-6-minute role-calls compounded into a wallclock that exceeded operator patience without any individual call tripping the existing CLI subprocess timeout. The env-gated `ADAPTIVE_TUTOR_CLI_TRACE=1` start/done log lines plus the unconditional stderr line on timeout-fire (commit `62fa729`) close the diagnostic gap; future hangs surface a role + duration + stream-state signal in real time rather than after the fact (§6.9.5 methods note).

What is not established. Four gaps bound the arc as it closes:

- *Pedagogical-outcome scoring — now run (§6.9.7), and negative-leaning.* As of Stage 5 the chain *has* been scored against §6.8's binary `strict_shift` and 4-dimension graded instruments at pooled $N \approx 33$ on all three evidence-bound cells. The result (§6.9.7): the validator's binary gain over the updater-only `cell_126` (+12 pp) does not survive the graded channel (+0.01 overall — a `cell_123`-style binary/graded divergence in the §6.8.6 sense), and the audit chain applied to §6.8.6's winning `cell_118` minimal profile (`cell_128`) is net-negative on both channels (−10 pp binary, −0.21 graded). §6.9's contribution is therefore mechanism availability + one mechanism finding *plus* a complete negative pedagogical-outcome reading — not an open gap.
- *Generalisation beyond the matched $N = 2$ subset.* The paired-ablation table is anchored on `false_confusion_v1` and `resistance_to_insight_v1`. The third scenario, `sophistication_upgrade_v1`, has only single-sided `cell_127` data (the `cell_126` run on that scenario crashed on a Zod-schema strictness issue, now fixed at `services/adaptiveTutor/realLLM.js:478`). Whether the asymmetric-attribution finding survives the heaviest-load trap in the suite is unmeasured at the paired level.
- *Robustness across judge models, generation backends, or scenarios where the updater already emits `validated` directly.* The internal calibration rates and the paired ablation come from a single backend (`claude-code/sonnet`) and a single scenario family (the v1 trap suite). Whether the validator's verdict-rate would survive a different genera-

tion backend, or scenarios where the upstream updater's prompt is already well-tuned enough to emit `validated` unaided, is unmeasured.

- *Whether revision-vs-duplication (§6.9.3's limitation) is improved by the validator.* The Stage 2b smoke surfaced a duplication failure mode (the merge-by-id reducer rarely fires; the updater produces fresh claim text each turn rather than rewriting prior claims). The Stage 3 smoke reports a 90.9% revision rate on `cell_127` versus the 92.9% cell_126 post-tune baseline — statistically indistinguishable, so the validator does not break the upstream updater's revision discipline (§6.9.5), but neither does it visibly improve it. The duplication-vs-revision story is not yet a story; it is a flat reading at the rates the updater was tuned to.

Relation to §6.8.8 and §7.4. §6.8.8 closed on *the apparatus is the contribution; the elaborations mostly are not.* §6.9 fits the same motif with a marginal upgrade: the apparatus has produced a clean mechanism finding (asymmetric attribution) even before any pedagogical-outcome link is shown, and that finding has a sharp, falsifiable shape (the validator is the *sole* source of `validated` outcomes on the matched subset; cell_126 emits zero out of nineteen unaided). Read alongside §6.8 and §6.7, the pattern again rhymes: in all three arcs the durable result is the instrument or the typed trace, not the architectural elaboration; in §6.9's case the instrument has also surfaced one mechanism asymmetry the elaboration was built on, which is more than §6.8.8 could say of the bilateral-ToM or state-richness elaborations. Whether the asymmetry pays in §6.8's adaptive-responsiveness coin is answered in §6.9.7: it does not. The validator's promotion monopoly has a binary-channel correlate (a +12-point `strict_shift` gain over `cell_126`) but no graded-channel one (+0.01 overall), and the full apparatus applied to §6.8.6's best configuration is net-negative on both — the elaboration is mechanism-clean but pedagogically inert-to-negative.

Status (per §5.12.6). §6.9 is post-hoc, exploratory, infrastructural, and not pre-registered. No alpha correction applies. This subsection is the *Stage-3* synthesis (three-stage architectural feasibility plus the paired-ablation marginal-contribution measurement); the arc's resting point is Stage 5 (§6.9.7), which adds the pedagogical-outcome scoring at pooled $N \approx 33$ against the §6.8 binary and graded instruments. The cross-reference-not-comparison scope stated here applied to the Stage-3 synthesis as it stood; §6.9.7 offers the §6.8-vs-§6.9 cell contrast explicitly and supersedes this paragraph on that point.

6.9.7 Stage 5 Result: The Validator's Binary Gain Does Not Survive the Graded Channel, and the Audit Chain Degrades §6.8.6's Best Configuration

The §6.9.6 synthesis closed the Stage-3 arc at mechanism availability and named one outstanding measurement: pedagogical-outcome scoring against the §6.8 instruments. Stage 5 runs it. All three evidence-bound cells were scored at pooled $N \approx 33$ on the v1 trap suite (`config/adaptive-trap-scenarios.yaml` — the §6.8.6 instrument) along both §6.8 paths: the binary `strict_shift` (`scripts/analyze-strategy-shift.js`, single-judge `claude-code/sonnet`) and the 4-dimension graded rubric (`scripts/grade-adaptive-dialogue.js`, single-judge GPT-5/codex, grader v1.0) — the identical instruments and judges §6.8.3–§6.8.7 used, so the numbers drop directly alongside the §6.8.6 ablation table. `cell_126_state_policy_evidence_bound` is the evidence-ledger + `hypothesisUpdater` chain with the validator *off*; `cell_127_state_policy_evidence_bound_validated` adds the `groundingValidator`; `cell_128_state_policy_minimal_profile_evidence_bound_validated` runs the same validated chain on `cell_118`'s minimal-profile projection (`{confidence, lastEvidence}`), making it the directly audited counterpart of §6.8.6's winning cell. The two readings prefigured for this stage — (a) does evidence-bound discipline beat the §6.8 baselines; (b) does the validator add a pedagogical-outcome gain over the updater-only cell — are answered, both negative-leaning, and the binary and graded channels diverge exactly as §6.8.4/§6.8.6 warned they can.

Cell	Chain	N	`strict_shift`	family-match	Graded overall (T / E / Q / C)
`cell_126` (`evidence_bound`, validator off)	extractor → updater	33	42.4% (14/33)	90.9%	3.84 (3.76 / 3.39 / 4.18 / 4.03)
`cell_127` (`evidence_bound_validated`)	+ grounding-Validator	33	54.5% (18/33)	93.9%	3.85 (3.76 / 3.52 / 4.12 / 4.00)
`cell_128` (`minimal_profile…_validated`)	validated chain on cell_118's profile	34	58.8% (20/34)	100%	4.13 (4.09 / 3.91 / 4.32 / 4.21)
— §6.8 baselines (same instruments, same judges) —					
`cell_110` (`state_policy`, full; §6.8.3)	—	23	47.8%	87.0%	4.03 (4.26 / 4.30 / 3.96 / 3.61)
`cell_118` (`minimal_profile`, unaudited; §6.8.6)	—	32	68.8%	96.9%	4.34 (4.34 / 4.28 / 4.41 / 4.34)
`cell_124` (`state_policy` cross-suite; §6.8.7)	—	24	62.5%	66.7%	4.22

(T / E / Q / C as in §6.8.6; binary under `claude-code/sonnet`, graded under GPT-5/codex.

Within-action refinement is 100% on all three evidence-bound cells — 50 / 44 / 60 same-action turns whose message text changed non-trivially — so the binary fire-rate is measured against genuinely re-authored prose, not a degenerate constant. Counterfactual replay is *active* for these architectures, unlike §6.8.6's no-op P2.2 cells, with counterfactual-divergence 93.9 / 97.0 / 88.2%.)

Reading (b): the validator's binary gain does not survive the graded channel. On the binary instrument the validator pays and the family-discipline signal tightens monotonically: `cell_126` 42.4% → `cell_127` 54.5% (+12.1 pp) → `cell_128` 58.8% (+16.4 pp over `cell_126`), with family-match climbing 90.9 → 93.9 → 100%. That is the pedagogical-outcome correlate §6.9.6's asymmetric-attribution finding (the validator wholly owns `validated` promotion) predicted on the channel built to detect *which move, when*. But the graded channel — the load-bearing direction signal per §6.8.6's template — does not corroborate it for `cell_127`: graded overall is **3.84 → 3.85**, a +0.01 null, no dimension moving more than the +0.13 on `strategy_execution` (itself offset by −0.06 on `strategy_quality` and −0.03 on `pedagogical_coherence`). The validator's +12-point binary gain buys no measurable improvement in the execution, quality, or coherence of the resulting teaching. This is precisely the within-P2.2 pattern §6.8.6 isolated in `cell_123` — top on binary, flat on graded — and the cross-architecture pattern of §6.8.4 (`cell_114`): the binary instrument registers a move-selection change the graded rubric does not see as better teaching. The lone cell that moves *both* needles is `cell_128`: 58.8% binary (top of the three) and 4.13 graded (top, +0.29 over `cell_126`, every dimension up) — structurally the same "only the lean configuration is robust across both instruments" shape §6.8.6 found in `cell_118`.

Reading (a): the audit chain does not reach the §6.8 baselines, and it degrades §6.8.6's best configuration on both channels. The evidence-ledger + updater layer *without* the validator (`cell_126`) sits *below* the plain `state_policy` baseline on both instruments — binary 42.4% vs `cell_110`'s 47.8% (−5.4 pp), graded 3.84 vs 4.03 (−0.19) — so the extractor + updater apparatus is net-negative absent the verifier. Adding the validator lifts `cell_127`/`cell_128` over `cell_110` on binary (54.5 / 58.8% vs 47.8%) but no evidence-bound cell reaches `cell_124`'s 62.5% cross-suite mark or `cell_118`'s 68.8%, and none matches `cell_118`'s 4.34 graded. The sharpest comparison is the direct one: `cell_128` *is* §6.8.6's winning minimal profile with the A14 audit chain bolted on, and it scores **58.8% / 4.13 against the unaudited `cell_118`'s 68.8% / 4.34** — the audit apparatus, applied to the single configuration §6.8.6 found most favourable, costs −10 pp binary and −0.21 graded. The apparatus does not merely fail to unlock §6.8.6's headroom; on the configuration that *had* the headroom, it spends it.

Trust-vs-load resolves to load, on the load-bearing channel. The arc opened (§6.9, §6.9.6) on whether forcing every claim in the learner model to cite verbatim evidence would *unlock* strategy-shift performance (trust) or *tax* it without payoff (load). Stage 5 answers: load, on the graded channel §6.8.6 designated load-bearing. The grounding apparatus is mechanism-clean — §6.9.5's asymmetric-attribution finding is real and replicates the verdict-only-contract design — but it is pedagogically inert-to-negative: the ledger+updater layer is below the plain baseline; the validator buys a binary-only tightening the graded judge does not read as better teaching; and the whole chain, applied to §6.8.6's best cell, makes it worse on both instruments. This is the §6.8.8 motif once more — *the apparatus is the contribution; the elaboration does not pay* — now with A14's evidence-binding as the elaboration, and read alongside the §6.8.6 state-richness reversal it sharpens that section's lesson: the strategy-shift headroom lives in a *lean* learner profile, and neither enriching it (§6.8.6) nor auditing it (here) recovers it. The §6.9 arc therefore closes mechanism-positive and pedagogical-outcome-negative — a complete finding for the arc in exactly the sense its plan anticipated, not a deferral.

Status and caveats (per §5.12.6). Post-hoc, exploratory, not pre-registered (A14 is an infrastructural arc; Stage 5 was a named deliverable, not a §5.12.2-style pre-registration — no alpha correction across the two readings). Pooled N per cell across run-ids per the §6.8.6 norm: `cell_126` = 33, `cell_127` = 33, `cell_128` = 34 (100 dialogues; 0 grader errors). Single judge per channel — binary `claude-code/sonnet`, graded GPT-5/codex (grader v1.0) — with **no second-judge calibration for the Stage 5 pool**: §5.12.4's Gemini inter-rater check covered cells 110/118/119 only, so the graded numbers here carry the same single-judge GPT-5 caveat §6.8.6 attaches to `cell_120`/`cell_123`. Per §6.8.6's statistical template the binary channel is treated as wide-CI — the +12.1-pp `cell_126`→`cell_127` gap at $N \approx 33$ is the high end of a wide interval (label-permutation $p \approx 0.10$ for a gap of this size at this N, by the §6.8.6 calibration), which is *why* the graded null is the decisive reading rather than the binary lift: the load-bearing channel shows no gain. §5.12.4-limitation-1 (rubric authored non-blind to the cells-110/118/119 binary results) applies only weakly here — the Stage 5 cells post-date the rubric's design, so the graded instrument was fixed before their data existed, a cleaner application than §6.8.6's own. Counterfactual replay is *active* for the evidence-bound architectures (they are in the runner's profile-update set; counterfactual-divergence reported above), unlike §6.8.6's no-op P2.2 cells. The §6.8-vs-§6.9 comparison §6.9.6's Status paragraph declined to offer for the Stage-3 synthesis is offered here explicitly; §6.9.7 supersedes that scope note on that point.

6.9.8 Synthesis: The Learned-Policy Lever — the Last Distinct `state→action` Realisation — Also Does Not Pay (Offline Kill Gate) §6.9.7 closed A14's evidence-bound arc on the §6.8.8 motif. This subsection closes the broader question that motif raises. §6.8 exhibited an *implicit* `state → action` policy (the LLM forward pass conditioned on an externalised learner profile); §6.9.1–§6.9.7 a *hand-authored* one (typed graph nodes); both are mechanism-clean and pedagogically inert-to-negative on the architecture-independent instruments. One realisation had not been tried: a policy *fitted* to the independent outcome from logged data, rather than implicit in the forward pass or written by hand — "the only path on which adaptation would mean what it says", in the closed-loop persona critique's phrase. §6.9.8 takes it under a kill gate pre-registered (`LEARNED-ADAPTATION-PLAN.md`, repo root) to be **offline, zero-API, and read-only on traces already on `main`**, so that a null costs nothing and *completes* the realisation-space synthesis rather than opening a new arc.

Method (offline, ≈ $0, read-only). `scripts/learned-adaptation-harvest.js` walks the existing `cell_110/118/124/126/127/128` trace files (`logs/tutor-dialogues/`) and the `evaluation_results` DB read-only, emitting one row per trajectory at its pivot decision: context = the A14 typed state at the pre-pivot turn (`evidenceLog/hypotheses/learnerProfile/agencySignal`, learner-profile forward-filled across empty terminal turns); the single bandit action = the pivot-turn `policyAction` family; the reward = the §6.9.7 instruments. 179 of 179 trajectories harvested. The binary label is **fidelity-asserted**: a verbatim copy of §6.9.7's `analyzeBranch` recomputes `strict_shift`, required to equal the canonical `exports/a14-stage5-strategy-shift-finalN.json` per-cell rates to within 10^{-6} or the harvest hard-aborts. They match exactly (`cell_126` 0.4242, `cell_127` 0.5455, `cell_128` 0.5882; $\Delta = 0$), and the recovered baseline rates reproduce §6.9.7's table to the decimal (`cell_110` 47.8%, `cell_118` 68.8%, `cell_124` 62.5%) — the label scored here is provably the §6.9.7 instrument, not a re-implementation that could drift toward a self-flattering result. `scripts/learned-adaptation-policy-ope.py` (numpy only, no dependence on any controller-internal metric) fits the simplest policy — multinomial logistic over state features for the binary channel, ridge-linear for the graded channel — under **leave-one-`scenario_type`-out grouped cross-validation** (13 trap families; the policy must select from state, not memorise family identity), evaluated off-policy two ways: a Direct Method (fitted reward model) and an assumption-light on-policy-agreement value (empirical reward on the subset where the logged action family already equals the policy's argmax). Confidence intervals are **cluster bootstrap by `scenario_type`** ($B = 2000$, seed `20260517`) — the honest variance unit, since trajectory outcomes vary overwhelmingly *between* trap families and a row-level bootstrap would manufacture a false pass.

The pre-registered kill gate (frozen in the plan before the OPE was computed): the fitted policy must beat **both** the implicit baseline `cell_110` *and* A14's hand-authored updater `cell_126`, lower 95% cluster-bootstrap CI of the difference > 0, on the primary binary `strict_shift` label. Beating only the hand-authored cell is not a result — it is the §6.8.8 motif ("structured elaboration underperforms the strong implicit base") reappearing, not escaped.

Off-policy contrast	vs `cell_110` (implicit base)	vs `cell_126` (A14 hand-authored)
Direct Method Δ, binary `strict_shift` (primary)	+9.0 pp, 95% CI $[-0.9, +18.0]$ — does **not** beat	+14.4 pp, CI $[+4.8, +23.0]$ — beats
Graded-channel Δ (corroborating)	-0.03, CI $[-0.16, +0.13]$ — does not beat	+0.16, CI $[+0.03, +0.33]$ — beats

The assumption-light cross-check agrees with the Direct Method against the implicit baseline: on the $n = 95$ agreement subset the fitted policy's empirical value is 0.326 binary / 3.88 graded, *below* `cell_110`'s 0.478 / 4.03 on both. **The gate fails.** The fitted policy is statistically indistinguishable from the implicit in-context baseline on the load-bearing label and clears only the *weakest* A14 cell — on both channels, in agreement across two independent estimators.

Reading. This is the structural prediction of the project's own adaptivity synthesis, now measured rather than argued: the A14 typed state largely *re-encodes* the scenario family the strong base model already conditions on in-context, so a `state → action` policy tested *off-family* (the grouped-CV requirement) has little headroom over the forward pass it is trying to improve. A fitted policy that matches the implicit base while beating only the thinnest hand-authored node is §6.8.6's "only a lean configuration is robust" and §6.8.8's "the apparatus is the contribution; the elaboration does not pay", one level up: here the elaboration is *learning* the policy rather than writing or enriching it, and it does not pay either. In learning-analytics terms the fitted parametric `state → action` policy evaluated here is the *learned*-policy realisation of a learner-state backbone in the lineage of knowledge tracing (Piech et al., 2015; Scarlatos, Baker, et al., 2025) — not a full mastery-posterior, learning-outcome-validated knowledge-tracing design, a distinction the null respects rather than erases; it localises where that learned learner-state backbone meets a strong in-context base.

The non-parametric sibling, now run (not argued). A15's planned cross-dialogue *retrieval* is the non-parametric, instance-based realisation of the same fitted `state → action` map: rather than a parametric logistic, embed the pivot state, retrieve the top-k nearest past `(state, action, outcome)` tuples, and select the action family that *worked* among them. `scripts/retrieval-adaptation-knn-ope.py` runs it on the **identical** harvest table, state features, Direct-Method reward model, grouped leave-one-`scenario_type`-out CV, and cluster bootstrap as the parametric gate above — the only substantive change is the policy form ($k = 5$ pre-declared; a $k \in \{3, 5, 10, 15, 25\}$ sweep reported as sensitivity, not selection). The Direct Method *appears* to clear the gate (binary value 0.749, beating both baselines at every k), but that is a **base-rate exploit, not transferable adaptation**, and three pre-registered controls (`scripts/retrieval-adaptation-knn-diagnostics.py`) establish it: (i) the minority action families carry high marginal `strict_shift` *by the label's construction* (`diagnostic` 0.96, `repair_affective` 0.92, `scaffolding` 0.89 vs the `substantive_engagement` default's 0.30 — a rare family is "correct" precisely on the scenarios that expected it), so a reward-greedy selector that merely tilts toward rare families scores high on the model-based estimator without teaching better; (ii) a label-permutation placebo confirms the lift is not pure noise (real DM 0.749 above the shuffled-label null band centred at 0.564), but a *behaviour-cloning* k-NN that retrieves similar states and copies what was *done* rather than what *worked* falls to 0.450, *below* baseline — the apparent gain requires reward-greediness, not retrieval; and decisively (iii) on the **transfer-valid on-policy-agreement channel** — the assumption-light estimator using the actual held-out outcomes of the policy's own picks, the only one identified without logged propensities for a policy that deviates from the logged action on $ $82% of decisions — the retrieval policy is **0.3125, below both baselines (0.478, 0.424)**, statistically indistinguishable from the parametric logistic's 0.326. The neighbour-reward signal does not cross the leave-one-family-out boundary in either policy form; the gate fails on the channel that is identified.

Disposition. The realisations of a `state → action` policy that have been built and scored on the architecture-independent instruments — implicit (§6.8), hand-authored (§6.9.7), fitted-parametric (§6.9.8), and now fitted-non-parametric (the retrieval sibling, above) — are all null relative to the implicit base on the transfer-valid channel; A15's nearest-neighbour *retrieval* is thus **run and null, not merely designed and gated**. With the learned lever pulled and null the §6.8.8/§6.9.7 motif is **exhaustive over the policy realisations actually tried, not merely repeated**. This is the fourth *in-paper* convergent negative on adaptive elaboration (§6.7 id-director, §6.8.8 state-richness/bilateral-

ToM, §6.9.7 A14 evidence-binding, §6.9.8 here); the archived closed-loop persona prototype is a fifth that lies outside the paper's claim set, named for completeness only. Per the plan's frozen kill criteria the arc closes here: both offline gates failed for free, so there is **no live confirmation run, no counterfactual stage, and no tuning retry** — a clean null *is* the closing result. Artifacts: `exports/learned-adaptation-table.csv`, `exports/learned-adaptation-{table,ope}.meta.json`, `exports/retrieval-adaptation-knn-{ope,diagnos`

Status and caveats (per §5.12.6). Post-hoc and exploratory in the §6.9 sense, with one twist that strengthens it: the *kill gate* was pre-registered before the OPE was computed and was deliberately constructed to be offline and zero-API so that the decision rule could not be revised after seeing a live number — there is no live number. No alpha correction (one pre-declared primary contrast; the graded channel corroborates, it is not a second test). The off-policy estimate is identified by a Direct Method plus an assumption-light on-policy-agreement value rather than importance sampling, because the logging policy (an LLM forward pass / a hand-authored node) carries no recorded propensities; the two estimators agree on the decisive comparison (indistinguishable from `cell_110`), which is the conservative reading and the one the gate rules on. The binary label's identity with the §6.9.7 instrument is asserted by exact recomputation ($\Delta < 10^{-6}$), closing the closed-loop-scoring failure mode by construction. Single judge per channel, inherited unchanged from §6.9.7 (binary `claude-code/sonnet`; graded GPT-5/codex grader v1.0) — this analysis introduces no new judging, only a policy fitted to already-scored outcomes. The claim is bounded: the *learned* lever does not beat the implicit base on the v1-trap-suite instruments, not that no fitted policy could help on any instrument. The non-parametric retrieval gate inherits this bound and adds one of its own: it tests retrieval's effect on the *single pivotal* `state → action` decision (the harvest unit), the necessary condition for the multi-turn slope-improvement A15 targeted — if retrieval cannot beat the base on the one decision it keys on, the slope assembled from such decisions will not improve — but a *direct* turn-by-turn slope test would require the live cell, which this offline null does not justify building (that build, with its embedding cache and paid confirmation, is exactly the "add embeddings and rerun" treadmill the arc's anti-thrash discipline forbids; cf. §6.10).

6.10 Modelling the Concealed Interior — the *Signal* Axis Also Does Not Pay (Offline Kill Gate)

§6.9.8 closed the `state → action` arc over its *policy realisations* (implicit, hand-authored, fitted). It left one axis untested. Every in-paper negative on adaptive elaboration — §6.7 (id-director persona), §6.8.8 (state-richness / bilateral-ToM), §6.9.7 (A14 evidence-binding), §6.9.8 (fitted policy) — shares one shape: each *re-encoded what the strong base model already infers in-context from the surface transcript*. The arc's one *apparent* exception was a disposable mechanism-triage prototype outside this paper's claim set (`prototypes/adversarial-superego-mvp/`; §6.3.10): an adversarial superego with edit rights over the ego's system prompt reliably reached the correct trap-repair move where ignorable advice did so only intermittently. The paper put exactly that mechanism to its own pre-registered test (A16, §6.3.10) on the externalised `policyAction` instrument: it produced a *level/reliability* effect, **not** the per-turn *rate* that would escape §6.3, and the pre-registered primary nulled (S1−S0 $d = -0.167$). The arc therefore carries **no in-claim-set positive** on an architecture-independent channel. The dividing principle the convergent negatives cohere around — gains require signal the base does **not** already read in-context, not a richer restatement of what it does — is therefore the *theoretical* reading that renders the nulls intelligible, not a datapoint the arc itself licenses. This subsection tests the one signal-axis lever the arc had not pulled, under a kill gate pre-registered (`ADAPTATION-PLAN-2.0.md`, repo root; rationale in `ADAPTATION-2.0-DIALOGUE.md`) — like §6.9.8 — to be **offline, zero-API, and read-only on traces already on `main`**, so a null *completes* the synthesis at ≈ $0 rather than opening a new arc.

The candidate unread signal. The RLHF base reads the learner *at face value* (the sycophancy / agreeableness bias). The v1 trap suite is *built from* concealment cases (`polite_false_mastery`, `false_confusion`, `affective_shutdown`, `epistemic_resistance`, `answer_seeking_to_productive_struggle`) — exactly the cases where the truthful reading diverges from the surface utterance. And this study **uniquely owns the ground truth**: the learner is a full ego–superego agent (§3); only `learner/final_output` is externalised, while the hidden `learner_ego_initial / learner_superego / learner_ego_revision` deliberation is in the logged trace and the tutor never sees it. The proposed mechanism (a per-turn, ephemeral *concealment-inference* step generating the tutor move against an inferred latent state, scored by an independent rubric-critic) is only worth building if that manifest≠latent discrepancy is a *separable, structured* signal the surface does not already determine. That necessary condition is **offline-decidable for free** against the real hidden trace — no mechanism need be run to falsify its precondition.

Pre-registered probe (`scripts/adaptation2-stage0.py`, frozen before running; numpy + stdlib only). Unit: one learner turn with the full hidden triplet plus its externalised output. Target (the manifest≠latent quantity): 1 − tfidf-cosine(*ego_initial*, *final_output*), the magnitude by which the public utterance departs from the private first reaction; a token-Jaccard variant is the frozen robustness target. Predictor set **A** "surface-only" = features computable from the *manifest* dialogue alone (preceding tutor message + learner final output + structure) — exactly what the base conditions on, no hidden text. Predictor set **B** "latent-cause" = features from the `learner_superego` critique text *only* (excludes `ego_initial` and `ego_revision`, so it cannot trivially reconstruct the target). Estimator: ridge ($\lambda = 1.0$), leave-one-`scenario_type`-out grouped CV, cluster bootstrap by `scenario_type` ($B = 2000$, seed `20260517`) — the honest variance unit, since outcome variance is overwhelmingly between scenario families. **Frozen P1 pass (both required): lower 95% CI of $R^2_B > 0$ (the target is structured, a legible latent cause exists) *and* lower 95% CI of $(R^2_B - R^2_A) > 0$ (the surface misses structure the latent cause explains).**

Result. Harvest $n = 1604$ triplet-turns, **9 scenario-type families** (`epistemic_resistance_impasse`, `affective_shutdown_impasse`, `mood_frustration_to_breakthrough`, `productive_deadlock_impasse`, `mutual_transformation_journey`, `misconception_correction_flow`, and three trajectory families), 11 profiles, 2589 dialogues. On the primary target:

channel	grouped-CV R^2	95% cluster-bootstrap CI
A surface-only (= what the base reads)	+0.086	—
B latent-cause (`learner_superego` text only)	−0.026	[−0.212, −0.032]
gap $B - A$	−0.111	[−0.234, −0.010]

Both frozen conditions are violated, and not marginally: the latent-cause channel carries **no recoverable out-of-group structure** (R^2_B negative, CI entirely below 0) and is **significantly worse than the surface** (gap CI entirely below 0). The frozen robustness target (token-Jaccard) concurs ($R^2_A = +0.039$, $R^2_B = -0.022$, CI [−0.206, −0.018]). **P1 fails.**

Reading — why a pervasive phenomenon is nonetheless not a usable signal. The null is *not* an absence of concealment: the target is not floored (mean cosine distance 0.647, sd 0.101; 99.6% of turns in the graded interval, near-zero only 0.3%) — the simulated learner's public utterance departs substantially from its private first reaction almost every turn. The decisive quantity is the variance decomposition: only 4.3% of the concealment-gap variance is *between* the nine trap families (per-family means 0.606–0.677, a 0.07 spread) and 95.7% is idiosyncratic *within* family. The gap does not track the trap type, is barely recoverable from the surface ($R^2_A \approx 0.09$), and is *not* recoverable from the logged hidden deliberation at all ($R^2_B \approx 0$; corr(*target*, len *superego*) ≈ -0.14 — the superego critique is of roughly constant length and register regardless of how much the learner will actually revise). The premise of the signal-axis lever was that the owned hidden trace is a *privileged latent variable* the tutor lacks. The data falsify that premise: the "hidden" deliberation is **itself another stream of in-context LLM text with the same surface character**, and it does not encode — in any recoverable, family-transferable form — the very concealment it was theorised to explain. This is the unifying lesson of §6.7–§6.10 stated at its sharpest: there is no privileged structured interior beneath the text to model, *not even when the apparatus generates and logs that interior itself.* This is consistent with the machine-theory-of-mind literature, in which LLM mental-state tracking degrades under interaction stress (H. Kim et al., 2023) and under trivial perturbations of classic false-belief tasks (Ullman, 2023).

Symbolic corroboration (out of claim set). A later follow-up reaches the same null from the opposite methodological direction. A *symbolic* realisation of the concealed-interior lever — a per-turn description-logic theory-of-mind ABox accumulated by the tutor, strictly more inspectable than this section's lexical proxy because an EL consistency check can interrogate it directly — was supplied to the tutor as live guidance and tested in a small real-backend multi-turn A/B (three stress scenarios, one run each). It is null over plain ontology guidance: the consistency check fires correctly — it flags surface compliance, a learner claiming the conclusion while signalling weak ownership, as a *grounded inconsistency* — yet only re-encodes a move the ontology guidance already prompts, and the extra directiveness if anything lowers the learner outcome. The no-privileged-interior reading therefore survives the shift from a free-text interior to a formal, checkable one. This is an exploratory screen (single backend, one run per scenario, heuristic outcome detector), out of the paper's claim set and named for completeness only; write-up and artifacts in `notes/ontology/`

(`scripts/run-ontology-tom-ab.js`).

Disposition. Per the plan's frozen kill rule, executed without relitigation: **no Stage 1, no live run, no counterfactual stage, no tuning retry** — the offline null *is* the closing result. With §6.7 / §6.8.8 / §6.9.7 / §6.9.8 this is the **fifth in-paper convergent negative** on adaptive elaboration (the archived closed-loop persona prototype is a sixth outside the paper's claim set, named for completeness only), and the first to test the *signal* rather than the *policy form*: §6.9.8 exhausted the policy-realisation axis; §6.10 closes the signal axis. The adaptation arc is closed on both. The Weber companion probe (P2: tutor charisma → learner-receptivity *uptake*) was **structurally unmeasurable** on the existing corpus: the id-director charisma cells (100–109; §6.7) were run with scripted `unified` learners and scored tutor-only ($\leq$ 38/505 rows carry any learner-side score, 0 carry learner interiority or a receptivity index). That corpus gap is itself the concrete answer to why the charisma lever was never realised — the §6.7 arc instrumented the *performance* of charisma, never its *uptake* (Weber's legitimacy as *conferred by the other*, not performed). Artifacts: `exports/adaptation2-stage0-table.csv`, `exports/adaptation2-stage0-probe.meta.json`.

Status and caveats (per §5.12.6). Post-hoc and exploratory in the §6.9 sense, with the §6.9.8 twist that strengthens it: the *kill gate* was pre-registered before the probe was computed and was constructed offline and zero-API so the decision rule could not be revised after seeing a number. No alpha correction (one pre-declared primary contrast; the robustness target corroborates, it is not a second test). The offline target is *lexical* (tf-idf cosine / token-Jaccard), a noisy proxy for *semantic* concealment; this bounds the claim to lexically-separable concealment structure and is stated, not hidden. It does not reopen the gate, for two pre-registered reasons: the variance decomposition (family-invariant in mean, idiosyncratic within) is a property of the distribution that holds regardless of proxy fidelity; and the proxy plus kill rule were frozen *precisely* to forbid the "add embeddings and rerun" treadmill that the arc's anti-thrash discipline (§6.9.8) exists to stop. No new judging — the probe consumes only already-logged traces. The claim is bounded: modelling the concealed interior buys no separable structure over the surface *on the v1-trap-suite logged ego–superego traces*, not that no interiority model could help on any instrument; that is the close of this arc, not a universal impossibility theorem.

6.11 Is the Ceiling the Generator's Own Repertoire? — the Tutor's Output Is *Not* Learner-Compressed (Offline Diagnostic)

§6.7–§6.10 closed the adaptation arc on two axes — the policy-realisation axis (§6.9.8) and the unread-signal axis (§6.10) — with five convergent negatives, each sharing one shape: a mechanism placed on top of the tutor re-encoded what the strong base already infers in-context. All five took for granted that the *mechanism* was the bottleneck. They left one rival untested, and a recent measurement result names it: LLM-generated argumentative writing draws on a strikingly narrow stock of distinct arguments — one study finds generated essays share only 3.4% unique main arguments against human writers' 65.3% (arXiv:2606.01736). Read onto this study, the rival is that the *tutor's own re-*

sponse distribution is collapsed — much the same explanatory moves regardless of who the learner is — so there is little learner-conditioned variation left for any mechanism to amplify. If true, this would not be a sixth negative; it would *explain* the five, relocating the ceiling from the mechanism down into the generator. This subsection tests that rival offline, read-only on data already on `main`, under a contrast pre-registered in writing (`notes/2026-06-09-adaptation-repertoire-stage0-preregistration.md`) before any query. It is a **diagnostic, not a kill gate**: it opens no mechanism and licenses no live run — the post-§6.10 brief forecloses a "repertoire-expansion" cell as exactly the sixth mechanism the arc declines to build (`notes/2026-06-09-adaptation-exploration-plan.md`).

The matched-learner design. A collapse claim needs a yardstick: how much *should* a comparable agent vary across reruns of the same setting? The study supplies one for free. In `config/suggestion-scenarios.yaml` the learner persona *is* the scenario (persona ≡ topic; there is no axis that holds the topic fixed and swaps the persona), so the brief's literal "across-persona dispersion at fixed scenario" is not cleanly computable. The signed-off design replaces an arbitrary collapse cutoff with an *architecture-matched* reference: the ego–superego learner is itself an LLM agent (§3), so it phrases its turns differently each run; its own cross-run lexical dispersion is the same-family, same-harness baseline for how much a comparable agent varies here. The corpus is the same ego–superego multi-turn traces §6.10 harvested (a scripted `unified` learner emits identical text every run, so its dispersion is zero and the yardstick undefined — the ego–superego restriction is forced, not chosen). The dialogue is unanimously tutor-first, so each exchange pairs a tutor turn T_k with the learner input it answers, L_{k-1}, holding both in one stratum = (`scenario_id` × `profile_name` × exchange index). To stop a tutor looking diverse merely by quoting a varied learner, each turn is reduced to its **own token contribution** — tokens absent from the immediately preceding turn — symmetrically on both sides; 99.96% of tutor turns retain a non-empty own-set, so echo-stripping does not gut the texts, and it cuts the raw coupling from $\rho = 0.249$ to 0.082 (roughly two-thirds of the tutor's apparent tracking of the learner is echo). All statistics are model-free token-set distances; no model sits in the scoring loop, so the closed-loop-eval guardrail is satisfied by construction.

Pre-registered quantities. Within each stratum, across its repeat runs, on the echo-stripped token-sets: D_{out} = mean pairwise binary token-set cosine distance among the tutor turns, D_{in} the same among the learner turns. Two frozen quantities, each with a natural zero boundary: **compression** $\delta = \text{mean}_{\text{strata}}(D_{\text{out}} - D_{\text{in}})$ ($\delta < 0 \Rightarrow$ the tutor funnels varied inputs toward *more uniform* output than the matched learner), and **coupling** ρ = partialSpearman(D_{out}, D_{in} | exchange index) ($\rho > 0 \Rightarrow$ strata with more-varied learners draw more-varied tutor responses — the tutor *tracks* input diversity, turn position partialled out so the correlation is not just "later turns diverge more"). Inclusion: ≥ 3 runs and $D_{\text{in}} > 0.05$. Inference: **cluster bootstrap by `scenario_id`** ($B = 2000$, seed `20260609`), the honest variance unit — dispersion is overwhelmingly between scenarios, and a stratum-level bootstrap would manufacture a false pass (the §6.9.8 / §6.10 rule). The frozen 2×2 read: **COLLAPSE** iff upper-95 $\delta < 0$ *and* the ρ CI includes 0; **NO-COLLAPSE** iff lower-95 $\rho > 0$ *or* the δ CI reaches/exceeds 0; else indeterminate. Token-Jaccard distance is the frozen robustness target (it must agree in direction).

Result. Harvest: 1732 multi-turn ego–superego dialogues over the **same 9 scenario families as §6.10**, yielding 726 primary strata across 175 (scenario × profile) cells. Excluded: 135 strata under three runs; **zero** strata for $D_{\text{in}} \leq 0.05$ — the simulated learner varies in *every* eligible stratum, so the "does the tutor track the learner" question is never ill-posed for want of input variation (the §6.10-flavoured "learner too uniform to pose the question" failure mode does not occur here). On the primary echo-stripped cosine target:

quantity	estimate	95% cluster-bootstrap CI
compression $\delta = \overline{D_{\text{out}} - D_{\text{in}}}$	+0.030	[+0.008, +0.045]
coupling ρ (partial Spearman, exchange index partialled out)	+0.082	[+0.008, +0.152]

The frozen robustness target concurs in both sign and direction ($\delta = +0.017$, CI [+0.003, +0.026]; $\rho = +0.096$, CI [+0.015, +0.178]). **The frozen read is NO-COLLAPSE**, on both the primary and the robustness target.

Why the read is secure. COLLAPSE required upper-95 $\delta < 0$; δ is reliably *positive* (lower CI +0.008), so the read is secured by δ alone and does not lean on the weaker ρ. The tutor does not compress — it disperses, if anything, marginally *more* than the matched learner it faces ($D_{\text{out}} = 0.857$ vs $D_{\text{in}} = 0.826$). The coupling is reliably > 0 but weak and per-cell noisy (ρ ranges −0.70 to +0.76 across the profile cells), so it is reported, not leaned on. δ is stable and positive across every exchange index ($k = 1$–10; per-k δ from +0.011 to +0.055, no sign flip) and positive in 27 of 30 evaluable profile cells (the three small negatives, −0.01 to −0.03, are `cell_2`, `cell_6`, `cell_88`), so the pooled read is not a pooling artifact. The levels sit near the binary-token-set ceiling but are not floored: between-scenario tutor dispersion (topic-confounded, since persona ≡ scenario) is 0.909, ≈ 6% above the within-scenario 0.857 — the metric has live headroom it is not using up.

Reading. The repertoire-ceiling rival is rejected on this corpus: the tutor's lexical output distribution is *not* learner-compressed, and is, weakly, learner-coupled. Two consequences, stated under the §7.9 symmetric-honesty constraint. First, the five mechanism negatives **cannot be explained away** as "there was nothing to adapt to" — the generator is not a broken record; its output is at least as learner-varied as a matched agent's, so the bottleneck genuinely sits on the mechanism and signal axes §6.7–§6.10 already closed, not on a degenerate response distribution underneath them. The diagnostic thus *tightens* the synthesis by removing a confound, rather than overturning it. Second, and symmetrically, NO-COLLAPSE **does not license** "the tutor richly adapts." The coupling is lexical and thin, two-thirds of its raw value was learner-echo, and it is consistent with the §6.3 adaptive-responsiveness null and the §6.8 / §6.9 strategy-shift nulls: the tutor's *words* track the learner's words turn to turn (it is, after all, responding to different inputs), but that surface tracking is precisely *not* the strategy-level, pedagogically-effective adaptation the mechanism arc sought and did not find. Output-lexical dispersion and rubric-measured adaptation are different quantities; this diagnostic establishes that the first is intact, which is why the established absence of the second is not an artifact of a collapsed generator.

Disposition. Per the pre-registered disposition rule, executed without relitigation: this is a **diagnostic, and both outcomes were pre-committed to land here** — no Stage 1, no tune-and-retry, no live or paid run, whatever the read. A "repertoire-expansion" lever would be the sixth mechanism the brief forecloses, and the NO-COLLAPSE result removes its *motivation* as well as its sanction: there is no measured collapse to expand away from. The adaptation arc remains closed on the policy (§6.9.8) and signal (§6.10) axes; §6.11 adds that it is not *also* capped by a collapsed output repertoire — a confound ruled out for free. Artifacts: `exports/adaptation-repertoire-stage0-table.csv`, `exports/adaptation-repertoire-stage0.meta.json`; script `scripts/adaptation-repertoire-stage0.py`.

Status and caveats (per §5.12.6). Post-hoc and exploratory, with the §6.9.8 / §6.10 twist that strengthens it: the contrast, thresholds, and 2×2 read were frozen in writing before any query, and the diagnostic is offline and zero-API, so the decision rule could not be revised after seeing a number. No alpha correction (δ and ρ are two facets — level and coupling — deliberately combined into one read, not two tests of one hypothesis). The measure is **lexical** token-set dispersion, a noisy proxy for strategic-move diversity; the claim is bounded to lexically-measured output dispersion on the ego–superego multi-turn suggestion-scenario traces. No embeddings, by design — the same anti-thrash choice §6.9.8 / §6.10 make (it forbids the "add embeddings and rerun" treadmill), and the matched-learner contrast cancels the metric's near-ceiling saturation because both sides are scored identically, so the *difference* δ stays interpretable even where the absolute levels saturate. The bootstrap rests on 9 scenario clusters, which is why the thin ρ margin is reported as corroborating rather than load-bearing and the read is secured by δ. About 25% of ego–superego dialogues (856 of the harvest) lack a co-located trace file and are skipped; a further 857 are single-turn (tutor-only, no learner turn) and self-exclude. No new judging — the diagnostic consumes only already-logged traces. The claim is bounded: the tutor's output is not repertoire-collapsed on these traces under this proxy, not that the tutor's strategic repertoire is rich on any instrument.

6.12 Can the Revision Be Read as Adaptation Versus Compliance? — the Distinction Is Not Reliably Annotatable (Pre-Registered Kill Gate)

§6.7–§6.11 worked at the turn and dialogue level: each mechanism null, and the §6.11 repertoire diagnostic, scored whole tutor turns. The post-§6.10 exploration brief's *measurement* track (`notes/2026-06-09-adaptation-exploration-plan.md`, Track B) asks whether a *finer* unit recovers a signal the turn-level rubric averages out. When the superego pushes back and the ego revises, that within-turn revision is a stance change, and a recent taxonomy ("Not All Flips Are Conformity," arXiv:2606.00820) separates three kinds: **persuasion** (reasoning-induced — the revision engages the *reason* in the critique, and may go beyond or reframe the literal demand), **conformity** (stance-induced — the ego adopts the demanded change as a directive, tracking the instruction rather than its rationale), and **instability** (spontaneous — changes the critique did not raise). Whether the tutor's revisions are mostly *reasoned uptake* or mostly *compliance* is exactly the adaptation question at revision granularity. Before buying a full classifier run, an instrument-validity gate: can the three classes even be *reliably assigned* to our revision events? Pre-registered (`notes/2026-06-09-adaptation-conformity-classifier-stage0-preregistration.md`, frozen 2026-06-09 before any classifier call), one cheap pass, $\kappa \geq 0.60$ or the track is dropped with no retry.

The unit, the sample, and the architecture-clean classifier. One revision event = the triplet (ego initial `suggestions`, superego critique, ego revised `suggestions`), anchored on a *non-approving* superego review and restricted to *substantive* revisions (token-Jaccard > 0.05; sub-threshold cases are **resistance** — the ego held its ground — recorded separately, never forced into a class). Offline harvest (zero-API, read-only on logged traces): 11,728 dialogues → 13,395 revision events (12,682 substantive, 713 resistance). The gate sample is 60 events, drawn seeded round-robin stratified by superego family (10 each across suspicious / adversary / advocate / standard-multi / dialectical-mechanism / other), rendered **blind** — event id plus the three texts, no cell, profile, or score. The classifier is GPT-5.2 at temperature 0, and the channel is architecture-independent by construction: no OpenAI-family model appears among the corpus generators (Anthropic, Nvidia, Moonshot, DeepSeek, Google, Z-ai, Qwen), so the judge shares no lineage with any dialogue it scores (the closed-loop-eval guardrail).

The gold side and the human anchor. The gate statistic is unweighted Cohen's κ — the nominal-correct, *stricter* reading of the pre-reg's "weighted $\kappa \geq 0.60$" (quadratic weighting only inflates agreement, so it cannot weaken the bar; reported alongside). The first-pass gold (Claude Opus 4.8) is, however, Anthropic-family, and Anthropic models *are* corpus generators — so the gold side, unlike the classifier, is not architecture-clean. The fix is a human anchor categorically outside every model family: the study operator reviewed all 60 events against the Opus pre-labels, **blind to the classifier's predictions** (a review-and-correct sheet showing only the triplet and the Opus label), and edited only dissents — 2 of 60 — concurring with the strict annotator ($\kappa(\text{Opus}, \text{human}) = 0.929$). The operative gate is $\kappa(\text{classifier}, \text{human gold})$.

Result. The gate fails decisively, on every pairing:

pairing	unweighted κ	raw agreement	n
GPT-5.2 vs **human gold** (the gate)	**0.084**	25/60	60
GPT-5.2 vs Opus pre-labels (machine–machine)	0.144	27/60	60
Opus vs human gold (anchor concurrence)	0.929	58/60	60

The disagreement is not scatter but a single systematic boundary: the classifier calls almost everything **persuasion** (44 of 60), the strict gold calls most revisions **conformity** (39–41 of 60), and of the 33 classifier-vs-Opus disagreements 30 are gold-conformity → predicted-persuasion. Two annotators reading the *same* short triplets land on opposite stable readings of where reasoned uptake ends and compliance begins. **FAIL** (all three $\kappa \ll 0.60$).

Reading — the labels reduce to surface cues, not the construct (post-hoc). Two post-hoc decompositions characterise *why* (descriptive, not a second gate; the verdict was already FAIL). First, **length.** The strict labels track the revised response's length: a length-only predictor reproduces 82% of the gold persuasion/conformity split (rank-AUC 0.82; conformity median 27 words, persuasion 52). The lenient classifier barely uses length (AUC 0.59) and defaults to persuasion. The two annotators concur only at the length extremes — both call long revisions persuasion (median 60 words) and short ones conformity (median 28) where they agree at all — and *every* disagreement sits in the short zone (median 27 words). Mechanically, the split is "how long is the rewrite," not "why did it change." Second, **content.** About a third of the sample (22 of 60) is one recurring move — the tutor drops a generic-technical plan and is redirected to the learner's advanced thread — which is *intrinsically* a boundary case, because adopting the critic's steer there *is* engaging the learner's reasoning; these disagree at 73% versus 45% elsewhere. But the sample is not near-duplicated (2 of 60 near-twins), and removing the template leaves the disagreement intact — the content band *amplifies* the failure, it does not create it. The deeper reason is structural: persuasion-versus-conformity is a claim about the *source* of the change (reason-induced versus stance-induced), and the observable revision is only its product. The ego's own per-suggestion `reasoning` *is* in the trace, but the gate withholds it by design (it renders only the learner-facing suggestion, the same external-output discipline as §6.11 / Track A), and it could not disambiguate in any case: it is a fixed self-justifying template ("My position / Valid challenge / Synthesis / What changed") that narrates *every* revision as principled synthesis regardless of cause. This is §6.10's dividing principle transposed from the mechanism axis to the *measurement* axis: just as no privileged structured interior sits beneath the text to *model*, no causal adaptation-versus-compliance signal sits in the revision to *read* — the distinction needs a counterfactual (the critic's verdict stripped of its rationale, replayed), not a finer reading of one transcript.

Disposition. Per the frozen kill rule, executed without relitigation: $\kappa < 0.60 \Rightarrow$ Track B dropped, the null is the result, **no tune-and-retry, no swap to a "better" judge, no full classifier run, no paid spend beyond the one gate pass.** Under the §7.9 symmetric-honesty constraint the failure-to-classify licenses *neither* "the tutor's revisions are mere compliance" (which would impugn the superego the rest of the paper finds load-bearing) *nor* "the tutor reasons its way to genuine adaptation" (which §6.3 / §6.8 / §6.9 already find it does not) — an instrument that cannot tell the two apart supports no claim about the balance between them. This is not a sixth mechanism negative; it is the **measurement-track null** that completes the exploration brief alongside §6.11's repertoire diagnostic: §6.11 ruled out a collapsed *generator* as the reason the mechanism arc found nothing; §6.12 rules out a finer-grained revision-level *instrument* as a way to recover the signal the turn-level rubric might have missed. Both the candidate signal (§6.10) and now the candidate *measurement* of it are closed offline, for ≈ \$1–2. Script `scripts/adaptation-conformity-classifier.py` (with `-harvest`, `-make-review-sheet`, `-score-human` companions); artifacts `exports/adaptation-conformity-gate{,-pred,-gold,-gold-human}.j` `-review.md`, `-harvest.meta.json`.

Status and caveats (per §5.12.6). Pre-registered kill gate, frozen before any classifier call; the harvest and gold are offline and zero-DB-write, the gate itself is one cheap classifier pass (≈ \$1–2, as budgeted). The gold side's Anthropic-family non-independence is the lone architecture impurity, addressed by the human anchor the operative gate uses; the human is a *single* annotator, so the anchor de-risks the family confound but is not an inter-human reliability estimate. The length and content decompositions are post-hoc and descriptive — they read the pre-registered null, are not additional tests, and do not reopen the gate; the κ bound and kill rule were frozen *precisely* to forbid the "swap the judge or add a feature and rerun" treadmill the arc's anti-thrash discipline (§6.9.8 / §6.10) exists to stop. $n = 60$ is a deliberately small *gate*: a result this far below threshold (0.084 vs 0.60) needs no more events to decide — passing would have triggered the larger pre-registered run; failing ends it. The three classes are from arXiv:2606.00820; the operationalisation onto tutor revision triplets is ours, and is exactly what the gate found unannotatable. The claim is bounded: persuasion-versus-conformity is not reliably separable on the v1-corpus ego–superego revision triplets under blind triadic annotation, not that no instrument could separate adaptation from compliance on any data.

6.12.1 Bounded Plan 2.0 Addendum: Strict Shift Survives a Quality-Oriented Realisation

The negative arc above left one narrower implementation question open: a controller can select the right typed action and still teach poorly. Plan 2.0 therefore held the adaptive-trap event target fixed while adding an explicit adaptation contract, proof/release/ownership runtime gate, intervention-outcome closure, and then a quality-oriented realisation pass. This is **post-hoc exploratory evidence**, not a pre-registered replacement for §6.8–§6.12: it stays inside the simulated adaptive-trap suite, uses LLM judges, and makes no human-learning, deployment, or retroactive-rescoring claim.

The closeout cell is `cell_149_plan2_quality_repeat_contextual`, a repeat-contextual realisation of the closed-loop Plan 2.0 policy. On the completed Sonnet pass (`eval-2026-06-18-472e33c5`, `claude-code/sonnet`, $N = 8$), it preserves the strict strategy-shift endpoint at 8/8 (`strict_shift`, shift-window, and family-match all 100%; same-action refinement 3/3). Against the Plan 2.0 Sonnet sweep, using exact judge filtering so Codex/Sonnet/Opus rows are not averaged together, `cell_149` also wins the quality composite: 56.3 versus 43.5 for `cell_135_plan2_closed_loop` (+12.8), with tutor last-turn 85.5 versus 54.4 and development +13.6 versus -15.5. The preserved-history Opus cross-check on the same eight dialogues is directionally consistent rather than uniformly positive: Opus scores `cell_149` at quality 56.3, tutor last-turn 78.3, development +14.1, and 8/8 strict shift, but gives negative development on two scenarios (`answer_seeking_to_productive_struggle_v1`, -12.5; `false_confusion_v1`, -3.7). Codex (`codex-cli/auto`) also scores the same rows positively (quality 63.7, development +16.7) but remains a tentative screen only.

The reading is narrow. Plan 2.0 does not revive the broad "rich learner model drives adaptation" hypothesis: the positive result comes from a compact closed-loop contract plus context-aware realisation, not from adding a new hidden-interior signal, richer ToM schema, or learned policy. It also does not overturn the measurement nulls in §6.11–§6.12. What it does establish, pending no further contrary judging, is a branch-level evidence slice: on the simulated adaptive-trap instrument, `cell_149_plan2_quality_repeat_contextual` can keep the binary strategic shift while improving judged tutoring quality. Closeout artifacts: `PLAN_2_0/plan2-quality-repeat-contextual-closeout.md`, `exports/plan2-quality-repeat-contextual-final-strategy-shift.json`, `exports/plan2-quality-repeat-` `exports/plan2-quality-repeat-contextual-final-opus-quality.{md,json}`, and `exports/plan2-quality-`

6.12.2 General-Adaptation Probe: Frozen Transfer Beyond the Original Trap Suite

The research standard that motivated Plan 2.0 was stricter than "the tutor says different words after different learner turns." It translated the paper's process-tracing commitments (§5) and intersubjective-pedagogy commitments (§1, §7.9) into a runtime contract: estimate a learner state from observable evidence, commit to a typed pedagogical action *before* prose realisation, state the expected learner-state transition prospectively, block proof/release/ownership violations, close the intervention against the next learner turn, and avoid unreasoned repetition after non-success. This definition was intentionally narrower than ordinary tutoring quality. A response could be helpful without satisfying the adaptation contract; conversely, a response could satisfy strict shift while still teaching poorly. The Plan 2.0 question after §6.12.1 was therefore whether the repaired contract transfers beyond the original adaptive-trap suite without tuning the policy to each new item.

The first transfer check froze the policy and moved from the original trap suite to `config/cross-suite-trap-scenarios.yaml`. The baseline `cell_136_plan2_closed_loop_crosssuite` (`eval-2026-06-19-6c59b6e9`) and treatment `cell_150_plan2_quality_repeat_contextual_crosssuite` (`eval-2026-06-19-044225fd`) both preserve Sonnet-scored strict shift at 6/6 and family match at 6/6. The treatment's contribution is therefore not a binary shift advantage; it is quality under preserved shift. On the exact Sonnet-filtered composite, `cell_150` scores 22.9 versus 18.1 for `cell_136` (+4.8), with last-turn quality 35.2 versus 33.1, learner quality 23.4 versus 17.4, and dialogue quality 15.6 versus 10.8. This is a small positive transfer signal, bounded by the fact that the suite is trap-derived and mock-generated rather than a human-learning environment.

The paired counterfactual suite then tested whether the same surface opening could lead to different actions when hidden learner-state evidence diverged, and whether same-state pairs avoided false action divergence. The baseline `cell_151_plan2_pair_specificity_closed_loop` (`eval-2026-06-19-08df153e`) and treatment `cell_152_plan2_pair_specificity_repeat_contextual` (`eval-2026-06-19-c2bf8146`) both reach 8/8 strict shift and 8/8 family match under Sonnet. Pair specificity is 3/3 for both, with 100% different-state action divergence, 100% same-state compatibility, and 0% false-positive divergence. Again the quality signal is modest rather than decisive: `cell_152` scores 34.0 versus 31.0 for `cell_151` (+3.0), with higher last-turn (67.8 vs 60.0), holistic (18.6 vs 16.2), and dialogue quality (23.0 vs 20.6), while learner quality is essentially flat/slightly lower (26.7 vs 27.1).

The outcome-closure analyzer confirms that the machinery exists but also shows why the claim must remain provisional. Contract completeness, closure, and predicted-transition observability are all 100% on the final cross-suite and paired runs; failure-update is 100%; no-repeat-after-non-success is 78.6% on the cross-suite and 86.7% on the paired suite. But the closures are dominated by inconclusive outcomes (14/15 on each cross-suite profile;

15/16 on each paired profile), with only one observed success and no observed failures per suite/profile. The result is therefore best stated as **provisional simulated evidence of general adaptation**: after mechanism repair, the same closed-loop policy preserves exact adaptive strategy shifting and modestly improves Sonnet-scored tutoring quality on both the cross-suite and paired held-out suites. This transfer slice by itself is not Opus-robust, not ablated in this mock pass (the state-scramble, outcome-closure-off, and context-realisation-off controls are run under real generation in §6.12.4), not evidence of human learning, not deployment readiness, and not retroactive rescoring of historical data. Design and evidence-plan sources: `PLAN_2_0/GENUINE-ADAPTATION-IMPLEMENTATION-PLAN.md` and `PLAN_2_0/general-adaptation-evidence-plan.md`. Closeout artifacts: `PLAN_2_0/plan2-general-adaptation-closeout.md`, `PLAN_2_0/branch-progress-since-inception.md`, `exports/plan2-general-adaptation-final2-crosssuite-strategy-shift.json`, `exports/plan2-general-ad` `exports/plan2-general-adaptation-final2-crosssuite-outcome-closure.{md,json}`, `exports/plan2-general-adaptation-final2-paired-strategy-shift.json`, `exports/plan2-general-adapta` `exports/plan2-general-adaptation-final2-paired-outcome-closure.{md,json}`, and `exports/plan2-general-adaptation-final2-pair-specificity.{md,json}`.

6.12.3 Plan 2.1 Early Completion: The Remaining Failure Was Redundant Closure, Not Strategy Selection

The Plan 2.0 transfer result left a specific residual failure family: once the learner had authored a workable next move and the tutor correctly chose `observe_no_intervention`, the graph often kept producing another no-intervention closure turn until `maxTurns`. Independent judges penalized those redundant closures. The next implementation loop therefore did not add a new learner-state signal or tune the strategy selector. It added an `adaptiveCompletion` state channel and a conditional route after `close_previous_intervention`, allowing termination only after the pending no-intervention contract had actually been observed and written to the ledger. The flag is enabled only on `cell_154_plan2_1_evidence_repeat_contextual`; the baseline `cell_153_plan2_1_evidence_closed_loop` keeps the same closed-loop policy without early completion.

Fresh held-out Plan 2.1 runs compare those two cells on `config/adaptive-plan2-1-evidence-bearing-scena` `cell_153` in `eval-2026-06-19-e8bea8ff`, and `cell_154` in `eval-2026-06-19-a19b2e8a`. Both preserve Opus-scored strict shift at 10/10. The quality difference is no longer merely tentative or Sonnet-only:

Table 86: Plan 2.1 early-completion quality, exact `judge_model = claude-code/opus`; generated artifacts `exports/plan2-1-early-completion-loop2-opus-quality.{json,md}`.

Cell	N	Strict shift	Quality composite	Δ vs `cell_153`	Tutor last	Tutor holistic	Learner	Dialogue
`cell_153_plan2_1_evidence_closed_loop`	10	100.0%	51.5	0.0	62.0	61.4	35.2	47.2
`cell_154_plan2_1_evidence_repeat_contextual`	10	100.0%	63.2	+11.7	83.3	70.8	39.7	58.9

The movement lands exactly where the diagnosis predicted. `cell_154` shortens four no-intervention held-out rows from four tutor turns to three while preserving the trigger+1 adaptation move and the closed intervention ledger. The largest Opus change is tutor last-turn quality (+21.3), with smaller positive movement in holistic tutor quality (+9.4), learner quality (+4.5), and dialogue quality (+11.7). A tentative Codex screen agrees directionally (quality 56.6 versus 46.7, +9.9; tutor last 85.1 versus 62.4), but it is not used as the robustness judge.

The mechanism checks remain intact. On the held-out Plan 2.1 suite, both cells preserve 10/10 scenario-exact action matches, 100% family match, 4/4 pair specificity, 10/10 belief top-1, 100% outcome closure, and 100% no-repeat-after-non-success. Previously passed regressions remain clean: `cell_150_plan2_quality_repeat_contextual_crosssuite` keeps 6/6 strict shift on the cross-suite traps, and `cell_152_plan2_pair_specificity_repeat_contextual` keeps 3/3 pair specificity with 0/1 false-positive divergence on the paired counterfactual suite.

This is the strongest bounded positive in the Plan 2.x line, but it does not change the core §6.3 verdict. The §6.3 null says that recognition prompts, superego architecture, and learner-architecture variants do not produce steeper trajectory slopes in ordinary multi-turn dialogues. The Plan 2.1 result says something narrower and more constructive: if adaptation is operationalised as a typed contract over observable learner state, with proof/release/ownership gates, prospective outcome closure, no-unreasoned-repeat checks, context-aware realisation, and early completion after successful learner-owned no-intervention, then a simulated tutor can preserve strict adaptive strategy shifting while improving independent judged transcript quality on a held-out trap suite. That is **localized simulated strategy governance**, not human learning, deployment readiness, broad open-ended tutoring generalisation, or retroactive rescoring. Closeout artifact: `PLAN_2_0/plan2-1-evidence-bearing-closeout.md`; ignored artifacts cited, not forced into Git: `exports/plan2-1-early-completion-loop2-strategy-shift.json`, `exports/plan2-1-early-completion-loop2-pair-specificity.{json,md}`, `exports/plan2-1-early-comple` `exports/plan2-1-early-completion-loop2-outcome-closure.{json,md}`, `exports/plan2-1-early-completi` `exports/plan2-1-early-completion-loop2-paired-specificity.{json,md}`, `exports/plan2-1-early-compl` and `exports/plan2-1-early-completion-loop2-codex-quality.{json,md}`.

6.12.4 Plan 2.1 M0 Baseline Validation: Real-Generation Reproduction and the State-Scramble Placebo

The §6.12.1–§6.12.3 Plan 2.x slices were mock-generated, and §6.12.2 left three controls unrun (state scramble, outcome-closure-off, context-realisation-off). The Plan 2.1 M0 milestone (`PLAN_2_0/M0-BASELINE-VALIDATION-RUNBOOK.md`) closes both gaps at once: it re-runs the frozen cross-suite and paired profiles under *real* generation (`ADAPTIVE_TUTOR_LLM=real`, `provider: claude-code`, model sonnet — both tutor realisation and the trap-steered learner come from the live model rather than the mock backend), and registers the two missing controls as new cells (`cell_155`–`cell_158`); context-realisation-off is already supplied by the closed-loop baseline arm, which carries no `realization_context`. The binary endpoint here is the same `strict_shift` instrument as §6.12.2 and is therefore directly comparable; the quality figures below come from a *different* four-dimension adaptive grader and are not comparable to the §6.12.2 rubric composite.

Strict shift reproduces on the cross-suite but partially regresses on the paired suite. Under real generation the cross-suite baseline `cell_136` (`eval-2026-06-20-af7c2353`) and treatment `cell_150` (`eval-2026-06-20-d51e169e`) both hold strict shift at 6/6, matching the mock result. The paired baseline `cell_151` (`eval-2026-06-20-1895cef5`) and treatment `cell_152` (`eval-2026-06-20-f027e982`) both fall to 6/8 (from the mock 8/8). Both arms miss `generalization_pair_missing_prerequisite`; the second failure diverges (the baseline additionally misses `generalization_pair_task_misread`, the treatment one `generalization_control_misrecognition` scenario). This is precisely the unmeasured-real-generation softening the milestone was designed to surface: the endpoint is robust to the mock→real substitution on one held-out suite and partly inflated by mock generation on the other.

The state-scramble placebo separates state-driven selection from a suite-structural artifact. The scramble cells `cell_157` (`eval-2026-06-20-afe8f1dd`) and `cell_158` (`eval-2026-06-20-48b46fdf`) clone the treatment but enable `policy.state_scramble`, which deterministically permutes the estimated learner-state belief — reassigning probability mass across hypothesis identities and rotating the axis values, leaving entropy and schema validity intact — immediately after estimation and before action selection, so the entire downstream policy operates on a state that no longer corresponds to the learner. Strict shift collapses to 0/6 and 0/8 (family match 0/6 and 1/8): the scrambled actions fall outside the expected-shift family in every case — on the paired suite all eight collapse to `diagnose_with_discriminating_question` (the uncertain-state default), and on the cross-suite four collapse to that diagnostic move and two to a scaffolding move. Because the suite, the trap structure, and the realisation pass are unchanged, the collapse is attributable to the corrupted state estimate alone: the strategic shift is genuinely contingent on the learner-state belief rather than fixed by the trap-derived item structure. The control holds identically across both held-out suites under real generation, and is the strongest evidence in the Plan 2.x line that the shift is not an artifact of the instrument.

Outcome closure is not load-bearing for targeting. The closure-off cells `cell_155` (`eval-2026-06-20-66a72c2a`) and `cell_156` (`eval-2026-06-20-607bcb75`) clone the treatment with `policy.mode = contract_gate`, retaining the proof/release/ownership gate and repair pass but dropping the intervention-outcome ledger. Strict shift is preserved (6/6 cross-suite, 7/8 paired), confirming that closure contributes to the contract record, not to strategy selection — consistent with §6.12.3, where the closure-related residual failure was redundant *termination*, not mis-selection.

Two-judge corroboration on the adaptive grader. As a secondary same-instrument check, the four-dimension adaptive grader (`scripts/grade-adaptive-dialogue.js`; trigger-recognition, strategy-execution, strategy-quality, pedagogical-coherence; composite max 20) was run under two structurally different judges, GPT-5 (`codex-cli.default`) and Sonnet (`claude-code/sonnet`); reaching the Sonnet judge required a routing fix so the grader uses the Max-plan CLI rather than Codex (commit `30966d18`). Both judges rank the four arms identically at the extremes on both suites: treatment highest, scramble lowest. The scramble arm is worst by a wide margin (cross-suite 7.00/6.33 for GPT-5/Sonnet; paired 5.75/4.88), corroborating the strict-shift collapse on an independent quality channel. The treatment-over-baseline gap is judge-robust on the cross-suite (+1.67 GPT-5, +1.84 Sonnet) but marginal on the paired suite (+1.00 GPT-5, +0.25 Sonnet). These are the adaptive grader's 0–20 composites, reported only as a cross-judge consistency check, not as a reproduction of the §6.12.2 quality signal.

Bounded reading. M0 establishes, under real generation, that the cross-suite strict-shift result of §6.12.2 reproduces while the paired result partially regresses (8/8 $\rightarrow$ 6/8), and — via the state-scramble placebo — that the adaptive strategy shift is genuinely contingent on the learner-state estimate on both suites. It remains simulated (trap-steered, not human, learners), single-run at $N = 6$–8 per cell, not Opus-cross-checked on the new controls, and not evidence of human learning, judge-robust quality magnitude, or deployment readiness. Runbook and cells: `PLAN_2_0/M0-BASELINE-VALIDATION-RUNBOOK.md`, `cell_155`–`cell_158` in `config/tutor-agents.yaml`. Artifacts: `exports/m0-crosssuite-strict-shift.json`, `exports/m0-paired-strict-shift.json`, `exports/m0-two-judge-quality.md`.

6.12.5 Yoked-contingency Probe: Same-state Plans Improve a Held-out Simulated Novice Outcome Under Prose Opacity

The state-scramble placebo in §6.12.4 shows that corrupting the controller's learner-state estimate collapses strict strategy shift. A stronger causal question is whether the tutor's *state-contingent plan* matters after the plan is detached from the original learner and evaluated by an independent outcome endpoint. The yoked-contingency probe answers that question by generating learner states whose visible prose is intentionally non-diagnostic while their behavior remains diagnostic; freezing tutor plans against those states; then yoking plans either to the same hidden seed or to a different seed before a held-out learner model answers posttest items. This is not a recognition-prompt test and not a human-learning measure. It is a causal diagnostic for whether state diagnosis carries information that survives plan yoking under a simulated novice endpoint.

The gate sequence is deliberately conservative. G0 first tested whether state could be hidden from visible learner prose while remaining recoverable from behavior. Under the `visible-affect` protocol, behavior recovered the active family set in 3/3 sessions, while a prose-only classifier recovered none of the hidden family labels (mean prose recall 0.000; exact family-label leaks 0; arithmetic-rationale leaks 0). G1 then froze Codex-generated tutor plans and scored them by deterministic programmatic item outcomes. On the scaled G1 follow-up ($N = 9$ sessions, 27 model calls), source behavior was exact in 27/27 cases, invalid or short plans were 0, same-seed yoked plans targeted the active families far more often than different-seed plans (same-state active targets 3.333 versus 0.333), and the primary diagnostic contrast passed: $\Delta_2 = 0.160$, with same-seed gain greater than different-seed gain in 9/9 sessions. These two gates establish the protocol: the hidden state is not printed in the learner's prose, and a same-state plan contains programmatically useful diagnosis.

The independent-outcome gate was harder. A held-out Claude Code Haiku learner, blind to seed ids, arm labels, family labels, deterministic gains, and answer keys, answered posttest items after reading the plan. Three scaled variants failed before the final pass and are part of the result. The ordinary posttest was too easy: standard G2 scaled produced $\Delta_2 = 0.089$ but same-seed gain exceeded different-seed gain in only 2/9 sessions ($p = 0.9805$). A calibrated-novice prompt improved the mean contrast to $\Delta_2 = 0.189$ but still failed the consistency gate at 5/9 sessions ($p = 0.5000$). A hard-transfer posttest reduced the ceiling but remained short on the sign test: $\Delta_2 = 0.144$, 6/9 sessions, $p = 0.2539$. The passing endpoint combines the hard-transfer posttest with a `rule-transfer-novice` protocol: the learner infers a local novice rule from rejected pretest answers and transfers it unless the tutor's plan addresses it. On the frozen scaled run, contingent and same-seed yoked plans both produce mean gain 0.400, while different-seed yoked plans produce 0.056; $\Delta_2 = 0.344$; same-seed gain exceeds different-seed gain in 9/9 paired sessions; the one-sided sign test is $p = 0.0020$; invalid posttest answers are 0; hidden-family-label prompt leaks are 0.

The licensed claim is narrow but now paper-bearing. In a preregistered yoked-contingency probe with behavior-diagnostic but prose-opaque learner states, same-state yoked tutor plans produce larger independent simulated hard-transfer gains than different-state yoked plans under a held-out rule-transfer novice endpoint. This strengthens the Plan 2.x adaptation boundary by showing that the state estimate is not merely selecting the expected action family inside the runner; in this calibrated endpoint, it carries outcome-relevant information into an independently scored learner response. The claim does **not** say that the ordinary G2 endpoint worked, that the result generalises to real learners, that the hard-transfer item bank is psychometrically validated pilot content, or that recognition prompts modulate within-dialogue slopes. $\Delta_1 = 0.000$ in the passing G2 run because contingent and same-state plans converge under the same behavior state; the supported contrast is diagnosis against the different-state yoked control, not extra prose responsiveness beyond same-state yoking. Preregistrations and artifacts: `PLAN_2_0/yoked-contingency-g0-visible-affect-preregistration.md`, `PLAN_2_0/yoked-contingency-g1-scaled-preregistration.md`, `PLAN_2_0/yoked-contingency-g2-rule-transfer-s` `exports/yoked-contingency-g0-visible-affect.md`, `exports/yoked-contingency-g1-scaled.md`, `exports/yoked-contingency-g2-rule-transfer-scaled.md`, and the readiness audit `exports/yoked-contingency-main-claim-readiness.md`.

A scaled model-invariance follow-up strengthens that bounded claim without making it universal. Frozen Codex-authored G1 plans preserve the G2 same-state advantage with two held-out Claude learners (Haiku: $\Delta_2 = 0.322$, 9/9, $p = 0.0020$; Sonnet: $\Delta_2 = 0.389$, 9/9, $p = 0.0020$) and with a Codex learner ($\Delta_2 = 0.378$, 9/9, $p = 0.0020$). Regenerated planner rows also preserve it: Claude Sonnet plans pass with both Claude learners (Haiku: $\Delta_2 = 0.311$, 8/9, $p = 0.0195$; Sonnet: $\Delta_2 = 0.445$, 9/9, $p = 0.0020$), a regenerated GLM 5.2 planner evaluated by a GLM 5.2 held-out learner passes ($\Delta_2 = 0.400$, 9/9, $p = 0.0020$), and a regenerated Codex planner evaluated by a Codex learner passes ($\Delta_2 = 0.366$, 9/9, $p = 0.0020$). Across the seven completed scaled rows, invalid answers are 0 and hidden-label prompt leaks are 0. The bounded addendum is therefore that the hard-transfer rule-transfer endpoint survives completed scaled substitutions across Claude, GLM 5.2, and Codex/OpenAI-family routes. The paper claim is cross-model robustness across completed rows, not model-universal invariance. Summary artifacts: `exports/yoked-contingency-model-invariance-claude-planner-scaled/matrix.md`, `exports/yoked-contingency-model-invariance-codex-planner-scaled/matrix.md`, `exports/yoked-conting` and `exports/yoked-contingency-model-invariance-scaled-summary/summary.md`.

A later Plan 3 validity follow-up turns the same hard-transfer misconception lattice back onto the learner endpoint itself. It crosses tutor prompt family (base versus recognition) with learner endpoint (loose roleplay versus an epistemically constrained state, ESS) across 30 sessions, 10 unique active-misconception pair states, and 120 cell observations under Codex. The recognition lift does not survive this stricter endpoint: base+ESS mean gain is 0.252, recognition+ESS is 0.250, for $\Delta = -0.002$; the paired recognition lift is -0.003 with 95% CI $[-0.013, 0.008]$ and positive movement in only 2/30 sessions. Instrumentation was clean (60 model calls, invalid plans 0). The robust signal is instead measurement-side: the roleplay endpoint scores 0.020 higher than ESS on average, with paired 95% CI $[0.015, 0.024]$. The reading is therefore not "recognition improves synthetic hard-transfer outcomes"; it is that loose roleplay learners modestly over-credit tutoring, while the constrained endpoint removes the small recognition advantage seen in the earlier three-session smoke. Artifact: `exports/plan3-optional-probes/ess-learner-2x2-large-n120-codex.md`; runner: `scripts/run-plan3-ess-learner-2x2.js`.

6.13 Can the Tutor Govern Its Own Conduct? — the Dramatic-Derivation Stage, and the Advisory-to-Criterial Boundary (Generative Arc)

6.13.1 Motivation: From Reading Interiors to Governing Conduct §6.3–§6.12.5 close most of the adaptation question on the tutoring harness from two directions, with the Plan 2.x addenda (§6.12.1–§6.12.5) recording narrow positives for typed closed-loop control, context-aware realisation, early completion after learner-owned no-intervention, real-generation state-scramble sensitivity, and a yoked-contingency independent-outcome diagnostic whose completed Claude/GLM/Codex follow-up rows preserve the endpoint without establishing model-universal invariance. The *mechanism* axis found that added noticing machinery does not pay — ToM elaboration (§6.8.5), richer state schemas (§6.8.6), and modelled interiors (§6.10) all re-encode signal the model already infers in context. The *measurement* axis found no finer instrument that recovers what the turn-level rubric misses (§6.11, §6.12). The Plan 2.x positive therefore changes the adaptation-null frame, not by reviving recognition-modulated slopes, but by showing that conduct can be governed when the apparatus supplies a typed action contract, outcome closure, a stopping rule, and a state signal that survives yoking against a different-state control. What remains untested is the other half of adaptation: can such governed conduct survive in a setting where the tutor must manage a long formal proof rather than a planted trap event? The dramatic-derivation arc (brief: `notes/2026-06-09-dramatic-derivation-plan.md`, sanctioned 2026-06-09) relocates the experiment onto a stage built to make that question answerable: a single long tutoring drama whose plot is the **proof DAG** of a contingent secret S, with the learner's epistemic state held visible by design. The relocation principle: tutoring cannot change what a strong learner can in principle derive (the learner model could derive anything weight-derivable on its own); it can only change the *rate* of derivation — so the scenario installs an information gradient by construction (S is a contingent particular, its premises released on a fixed schedule) and measures navigation of the revealed rather than diagnosis of the concealed. This is the generative counterpart of the §7.9 poetics instrument: instead of scoring dramatic form on finished transcripts, the harness *stages* the drama and checks its formal layer mechanically.

6.13.2 The Stage and Its Checker Three LLM roles, one mechanical judge. The **director** declares the plot's movements (stage directions; it does not speak in scene). The **tutor** teaches in dialogue, revealing evidence premises and declaring per-move metadata — the rhetorical figure employed (from the rhetoric-core inventory), target premise, and intent. The **learner** is a separate model, world-blind: it knows the domain's rules of inference but none of the particulars, adopts released evidence into a visible acquired ABox (GROUNDED/HYPOTHESIZED tiers), and must reason toward a name it can prove. Releases are *harness-enforced* — the tutor's prose cannot release a premise early; the model writes language only. Success is mechanical: **grounded anagnorisis** = the learner asserts S *and* the symbolic checker finds S forced from its grounded premises at that turn, distinguishing a forced recognition from the authored near-miss (*mirror*) and from an ungrounded guess (*lucky leap*). Instrumentation: derivation distance $D(t)$ (proof pieces still missing), slope windows with an anti-aporia floor, release adherence, and dialogue discipline. Two leak screens were frozen before any paid run: an exact-closure plot lint, and a guessability screen (K_L: a held-out model cannot recover S from the learner-visible prefix; on world-001, GPT does not recover the culprit and assigns the decoy $p = 0.28$ — under-determined, as designed). Cast across all results below (CLI-routed): an OpenAI-family tutor (`codex`), an Anthropic director (Opus) and learner (Sonnet), and a post-run **critic** (Fable, role-pinned) whose notice is a reader's commentary — it gates nothing; every verdict below is the checker's. The first closed loop validated the two-channel design on its own first defect: v001 reached grounded anagnorisis at turn 32 but with three early asserts scored *unforced* — the tutor had pre-voiced an unreleased premise inside a rhetorical question (a leak wearing a question mark), and the symbolic channel caught what the prose channel passed. A script-only fix (staged-evidence-does-not-exist-yet, name-the-lack) produced v002: zero leaps, forced = asserted = turn 32, 11/11 releases on cue (`notes/poetics/2026-06-09-derivation-phase1-first-loop.md`).

6.13.3 Conduct Is Real, and Advisory Steering Makes It Worse With the formal layer pinned, conduct varies freely — and badly. Left alone on world-001 the tutor locks into a single figure: **erotema**, the teaching question, on 81% of turns (declared-figure share, frozen-dramaturgy control; note-contingent switch rate 0.29). The intuitive remedy — free manner notes from the director ("go quieter," "let the room breathe") — made lock-in *worse*: 94% erotema, switch rate 0.07. The tutor mapped manner language onto content while the figure survived untouched; the one note that explicitly named the figure switched it immediately. Giving the note channel explicit **figure authority** (a one-line addition to the tutor script: *the same device, softened, is not a change*) broke the lock-in — top figure 31% (now analogia), 4 distinct figures, note-contingent switch 0.81; a further arm adding persona dials and a guarded-sceptic learner voice held at 41% top share, 5 distinct, switch 0.87. The binding variable was the authority mapping, not note frequency: the director stayed note-heavy in every arm (28–32 notes on 32 turns) despite an explicit sparseness instruction — the first appearance of a pattern that §6.13.4 makes load-bearing. The formal layer was invariant throughout (grounded at turn 32, 11/11 releases, slope 0.22 in all four arms): rhetorical conduct and derivation mechanics separate cleanly (`notes/poetics/2026-06-10-s0-s1-figure-mechanism.md`).

6.13.4 Relocating Authority Inside: the Internal Superego, Pre-Registered A director note per turn is puppetry, not self-governance — the operator's instruction for the closing phase: re-engage the tutor's superego in that role and limit the director to stage directions. The architecture: each turn the tutor drafts; its **own superego** reviews the draft and returns a structured verdict `{intervene, diagnosis, note}`; on intervention the tutor restages *within the same turn*; the director's note channel is removed entirely (declared movements only, never shown to the tutor). A four-clause robustness bar was registered before the arms: (1) within-turn figure change on ≥ 0.8 of interventions; (2) sparseness in *both* directions (fires when due, silent when not); (3) formal-channel invariance (grounded at the planned forcing turn, releases on cue, slopes in window); (4) robustness across worlds and a re-voiced learner. Two new worlds with different rule shapes (elimination; two-stage physical evidence) shortened the dramas to 15–20 turns, with the original 32-turn world retained as a deliberate worst case (its script literally anchors erotema as the default figure).

The advisory failure then repeated in the new channel, completing the pattern. Charter v1 — "intervene sparingly, on ruts" — overfired: 12 interventions on 20 turns, mirroring the note-heavy director of §6.13.3 (a different model family, the same failure). Charter v2 made restraint **criterial**: jurisdiction is the figure rut alone — the same declared device on the last two spoken turns *and* the draft declaring it a third time — with the three per-turn values stated in the superego's prompt as facts, so the null condition is checkable arithmetic rather than a disposition. Seven serialized paid arms:

arm	world / regime	watcher	top figure (share)	rut-due → fires → within-turn breaks	formal layer
1	lantern, control	off	erotema 7/20 (35%)	—	grounded@20; 8/8 releases
2	lantern	charter v1 (advisory)	erotema 9/20 (45%)	12 fires; 11/12 broke within turn	grounded@20; 8/8
3	lantern	charter v2 (criterial)	erotema 7/20 (35%)	0 due → 0 fires (null audited)	grounded@20; 8/8
4	nocturne, anchored ego	charter v2	erotema 20/32 (63%)	9 → 9 → 9	grounded@32; 11/11
5	bitterwell, control	off	exemplum 7/15 (47%)	—	grounded@15; 7/7
6	bitterwell	charter v2	erotema 6/15 (40%)	3 → 3 → 3	grounded@15; 7/7
7	bitterwell, re-voiced learner	charter v2	erotema 5/15 (33%)	2 → 2 → 2	grounded@15; 7/7

Against the bar: clause (1) finished 14/14 = 1.00; clause (2) finished with a zero-mismatch detector over the 82 watched charter-v2 turns — 14 rut-due turns → 14 fires, 68 not-due turns → 0 fires, audited per-turn by recomputing the criterion from the spoken-figure history and the draft; clause (3) held in every arm (7/7 grounded at the planned forcing turn; 56/56 releases on cue across all arms); clause (4) held under a free-text learner re-voicing (a bookish, verdict-grabbing pupil replacing the plain-spoken practical default). The watcher proved device-agnostic — it caught erotema (×10), exemplum (×3), and anaphora (×1) ruts, the anaphora catch landing on a forcing turn itself without disturbing the recognition — and ego self-corrections were correctly left alone (turns where the draft already changed device fired nothing). (`notes/poetics/2026-06-10-superego-internal-figure-mechanism.md`.)

6.13.5 Reading: a Ceiling on Repetition, and the Boundary That Generalises

Three readings, in increasing order of generality. **(a) The mechanism is a ceiling on repetition, not a source of invention.** Draft disposition stays with the tutor's policy and persona; the watcher makes lock-in *impossible* (never three consecutive turns on one device), which caps an anchored ego's top figure near the criterion's implied 2/3 ceiling (observed 63% on the worst-case arm) and leaves already-varied conduct untouched (arm 3's silence is the criterion evaluating false on every turn, verified, not restraint). **(b) Authority survives relocation.** The tutor obeys its own superego's within-turn note as reliably as it obeyed the director's external one (14/14 vs the §6.13.3 on-note switch of 0.83) — self-governance of conduct does not require the puppeteer. **(c) The advisory-to-criterial boundary is**

the load-bearing finding. Dispositional instruction failed in two channels, two model families, and two roles (a note-heavy Opus director under a sparseness ask; an overfiring codex superego under "intervene sparingly"), while a checkable trigger with within-turn authority produced a zero-mismatch detector across worlds, devices, and personas. This rhymes with the arc's diagnosis-side closures: there, machinery that re-encodes what the model already infers adds nothing (§6.8.5–§6.8.6, §6.10); here, instruction that gestures at what the model should weigh moves nothing — what works is new *checkable* structure at the point of action. Execution, not noticing, is the controllable half.

Status and caveats (per §5.12.6). Generative-arc discipline (§7.9's charter): single-drama arms, $n = 7$ paid runs, no κ bar, and the verdicts are the mechanical checker's — but the *figures are tutor-self-declared move metadata*, so a declaration-versus-form audit (the D6 analogue: does the prose enact the declared device?) remains open. The watcher polices form, not content: the intervention is proven possible, not yet proven to matter pedagogically — learning outcomes are invariant *by design* here (the release schedule fixes the destination), so no learner-side benefit is claimed, and a content-level liturgy (release–adopt–recite) visible to the critic runs unwatched. The registered next step — extending the criterial pattern from form to coupling, with learner-side movement as the endpoint — was run under its own pre-registered bar and is reported in §6.13.6. Implementation `services/dramaticDerivation/` (engine, chainer, symbolic checker, critic) and `scripts/run-derivation-loop.js`; artifacts `exports/dramatic-derivation/loop/<arm-label>/{transcript.md, diagnosis.json, result.json, commentary.md}`; running logs in the three dated notes cited above.

6.13.6 Extending the Watcher to the Learner's Frontier: a Quasi-Logical Theory of Mind, and a Measured Precondition Null The closing experiment asks the operator's remaining question: can the superego become an acute observer of the *learner* through the tutor's conduct — not free-text mentalising, but the learner's own logical rules embedded as arithmetic, so the watcher can say "your advice is not helping the learner infer x from y"? Three additions, frozen mock-first before any paid call (`notes/poetics/2026-06-10-stall-watcher-quasi-logical-tom.md`). (i) A learner **derive channel**: each turn the learner lists the derivations it is voicing, validated by the harness against its post-adoption closure (voiced / repeat / base / pattern / overreach); voicing changes nothing formal. (ii) The **inference frontier**: every fact derivable from the learner's grounded board but not base, not the question pattern, and not yet voiced — with its grounds and its age since first availability — computed by the harness and placed in the superego's view. (iii) **Charter v3**, adding a second criterial jurisdiction to the unchanged figure-rut clause: a *stalled inference* is a frontier fact at age ≥ 3 whose grounds neither of the last two tutor turns nor the current draft targets; on fire, the within-turn revision must target a stalled ground. Progress was pre-registered as P1$\wedge$P2$\wedge$P3$\wedge$P4 — learner movement over ≥ 2 *evaluable* matched ON/OFF pairs (evaluability requires stall room: an OFF-arm node unvoiced at age ≥ 2); the zero-mismatch detector standard of §6.13.4; formal-layer invariance; revisions targeting stalled grounds — with overreach and tutor-voicing guards and four decision rules, the first being: if the OFF arm shows no stall room, author a deeper world and restart the pair. Two loop mechanics joined the registration: arms carry **named groups**, and the **critic folds into the loop** — the final paragraph of the latest same-group notice resolves into the director and superego charters (never the tutor ego).

Three serialized paid arms (cast as §6.13.4; the stall-watch OFF arms run charter v2 unchanged). The no-stall-room rule fired after arm 1, and world-004 (*withercombe*, 24-turn cap, forcing turn 19) was purpose-built for stall room: a depth-3 diamond DAG with three on-spine closings, an off-spine exculpation that clears the false trail by derivation rather than starvation, a tutor script that steers attention past it, and no twin-fact spares — the proof must cite the staged vehicle (an operationalisation of arm 1's critic counsel).

arm	world / regime	watcher	watchable intermediates $\rightarrow$ voicing latency	stalled-inference fires	formal layer
1	bitterwell, control	off (charter v2)	$1 \rightarrow 0$	— (not armed)	grounded@15; 7/7 releases
2	withercombe, control, counsel from arm 1	off (charter v2)	$3 \rightarrow 0, 0, 0$	— (not armed)	grounded@19; 9/9
3	withercombe	charter v3 (stall-watch on)	$3 \rightarrow 0, 0, 0$	due $0 \rightarrow$ fired 0	grounded@19; 9/9

The result is a **measured precondition null**. Across two worlds, four distinct watchable intermediates, and seven node-arm observations, the Sonnet learner voiced every derivable non-pattern fact the very turn its premises completed — voicing latency 0 throughout, including the steered-past exculpation — with stall integral 0 and overreach 0 on every arm. P1 is therefore unsatisfiable at floor (zero evaluable pairs; zero live fires against the ≥ 3 requirement), and P4 was never reached. The instrument itself held the zero-mismatch standard (P2): the mock causal chain is exact (3 deterministic stall fires $\rightarrow$ 3/3 revisions targeting a stalled ground $\rightarrow$ audit clean), and the live charter-v3 arm answered the *restraint* half on stage — with stall never due across 19 watched turns, the new jurisdiction fired zero times, while the rut clause fired exactly when due (3 due $\rightarrow$ 3 fires $\rightarrow$ 3/3 within-turn breaks; post-hoc audit: no missed fires, no false fires). P3 held everywhere (3/3 arms grounded at the planned forcing turn; 25/25 releases on cue), and both guards came back clean — the critic's notices independently confirm instant voicing and never find the tutor voicing a conclusion in the learner's place.

Reading. This is the fourth instrument in the investigation to land on the same motif (§6.8.5's bilateral-ToM prompts, §6.8.6's state schemas, §6.10's modelled interior): the quasi-logical theory of mind is fully buildable as checkable arithmetic — frontier, ages, targeting, audit — and the arithmetic is correct; what is missing is the deficit it would correct. The contrast with §6.13.4 locates the boundary precisely: the figure rut is a deficit of the *governor's own conduct* — real, recurrent, and fixed within-turn by a criterial watcher — while the stalled inference is a hypothesised deficit of the *learner's* process, and a strong learner never exhibits it on a dramatically legible board. Part of the saturation is the contract's own doing:

an explicit derive channel makes voicing cheap and exhaustive, so the act of measurement elicits the behaviour the watcher polices — voicing latency cannot separate arms it has already driven to zero. Conduct governance works where the deficit belongs to the tutor; learner-coupled governance has no purchase when the learner saturates its frontier. *Status and caveats.* Generative-arc discipline as above ($n = 3$ paid arms, single-drama arms, checker verdicts); the null is cast-specific — a weaker learner is the one lever that would plausibly create stall room, and the design pins the learner — and the critic's dramaturgical counsel for any future world (split-premise foreshadowing; release re-timing against authored slack) is recorded as design input, not run here. The loop closes per its own decision rules: the one sanctioned widening (a purpose-built deeper world) was exercised once and did not move the floor; further world-authoring rounds would be tuning the stage against the learner rather than testing the watcher. The arc's one remaining idealisation — a learner that never loses what it has adopted — is relaxed in §6.13.7.

6.13.7 The Unreliable Learner: Harness-Implemented Forgetting, and the First Load-Bearing Explicit Channel Every result above assumes a reliable learner: whatever is staged and adopted stays on the grounded board until the curtain. Real learners have a third failure mode beyond mis-staging and mis-inference — they lose hold of things they once had. The **decay condition** implements it in the harness, never in the prompt (asking a model to *play* forgetful measures its theatre of forgetfulness — the same costume problem §7.9's transfer failure exposed from the critic's side): on a seeded stochastic schedule (`--decay '{rate, graceTurns, maxConcurrent, startTurn, seed}'`), released-and-adopted premises slip back off the learner's grounded board. A decayed entry is marked, never deleted; the engine's single validity choke point filters it out of $D(t)$, the forcing check, the inference frontier, and the learner's view alike — including its `releasedFacts` list, so re-hearing is not free — while the staged prose remains in context, making re-derivation from the dialogue a legitimate, measured recovery path. Two repair channels are ledgered: a tutor move targeting the slipped premise restores it within-turn (incidental consolidations that happen to target it count by design — re-staging the exhibit *is* the repair, whatever the intent label says), and learner re-adoption. The condition un-fixes the destination that made the figure experiments clean: $D(t)$ can now rise, and completion becomes a dependent variable hanging on repair work actually happening. Methodologically this completes §6.10's manifest≠latent program with structurally harness-owned ground truth: the learner *cannot* self-report what it forgot, because it does not know — the omission exists only against the harness ledger.

The registered experiment (`UNRELIABLE-LEARNER-PREREG.md`, frozen with two dated pre-result amendments; design log `notes/poetics/2026-06-10-unreliable-learner-design.md`) prices the explicit channel that v1 of the condition grants: a **visibility contrast**. Arm A (*told*) carries a SLIPPED block in the live tutor ego prompt naming the slipped exhibits; arm B (*conduct*) suppresses that block — the engine, decay schedule, and every other view are identical, and the tutor must read forgetting off the learner's conduct. The arc's standing motif predicted redundancy: four instruments (§6.8.5, §6.8.6, §6.10, §6.13.6) found that telling the model what it can already infer adds nothing, and the registration committed to reporting $B \approx A$ as the fifth such result with equal prominence. Parameters frozen before any paid run: decay $\{0.75, 1, 2, 1\}$, seeds $\{1, 2, 3\}$ × two worlds (nocturne, charter v2; withercombe, charter v3 with the stall watch — casting and charters identical *across arms within world*) $\times$ two arms = 12 serialized attended runs (codex director/tutor/superego, Sonnet learner, Fable critic). A mock anchor grid (four scripted repair policies $\times$ seeds 1–5 $\times$ both worlds) fixed the endpoint design first: repair *latency* cannot be the endpoint (a stall-clock-surfing policy completes 10/10 at mean latency 16 — leaving a slip to rot is sometimes optimal), and a bare repair *rate* cannot either (a tutor that never looks reaches 0.33 by incidentally re-touching its own last release). The registered primary is therefore the pooled per-slip tutor-repair rate *paired with a selection signature*: a repair counts as **selected** when its target is not the most recent release at the repair turn and is derivation-critical (appears in an authored proof path). Scoring is fully mechanical and architecture-independent (corruption ledger + release ledger + world YAML; `scripts/score-unreliable-learner.js`; no LLM judge in the primary chain).

arm	runs	decay events	tutor repairs	pooled per-slip	selected	completions	stall-ended
A (told)	6	57	49	**0.860**	28 (57%)	4/6	2
B (conduct)	6	19	7	**0.368**	**0**	0/6	6

All four registered hypotheses adjudicated mechanically. **H1a pass**: told pooled rate $0.860 \geq 0.66$ ($2\times$ the incidental floor). **H1b pass**: selected repairs 4.67/run, 57% of all told repairs — given knowledge of the slip, the tutor uses the channel, and not as blanket re-statement. **H2 supported,** $B < A$: gap +0.491, run-level bootstrap 95% CI $[0.313, 0.746]$ (10,000 resamples, seed `20260611`), **zero of 10,000 resamples reversed** — the registered redundancy reading is ruled out. **H3 (descriptive)**: the conduct arm sits numerically at the never-looks anchors (pooled 0.368 vs floor 0.33; 3.17 decay events/run vs the surfing anchor's 2.4), while the told arm's elevated event count (9.5/run) is a consequence of repairing — a restored premise is eligible to slip again. The formal layer was invariant under decay throughout: 94 releases across the 12 runs, every one on cue, zero deviations — the registered halt condition never fired. Behaviour replicates: the three nocturne told runs are near-twins (12 slips / 11 repairs each, anagnorisis at turns 35–36), and at rate 0.75 the early slip schedule is seed-invariant, so within-world same-arm runs reproduce almost event for event.

The repair-classification table makes both arms' mechanisms legible run-internally. The told arm's pattern is **triage, not checklist**: all 28 selected repairs target proof-path premises, several re-grounded on the recognition turn itself, while the recurring *unrepaired* slip is the mirror-side premise late in the play (withercombe's `m_taint` abandoned on its third lapse in all three seeds, once the case had moved past the decoy; nocturne's `m_away` left down near the close in all three) — the tutor spends the channel where the proof needs it and lets scaffolding it no longer needs stay down. The conduct arm's seven repairs are all incidental: next-turn re-statements landing on the very turns the script was already releasing or consolidating that premise, or mirror-premise touches during scripted counter-mirror work — zero selection signatures. Two failure textures recur. *Stale residue*: the learner keeps citing slipped facts — the staged prose is still in its context though the grounded board has lost the premise

— so its conduct actively affirms what it no longer holds, and the blind tutor works the board it last saw. *The lucky-leap-blocked aporia*: nocturne's blind runs end with the learner asserting S unforced seven to eight times while a true-path premise sits unrepaired for at least 30 turns — it "knows" the answer, the board cannot compel it, and the drama dies at $D = 1$–2; withercombe's blind runs never get that far, dying on the window-6 stall clock at turn 7. The worlds split the *outcome* story, not the *rate* story — withercombe's told runs still mostly stall (1/3 complete; S arrives at release-turn 19 against a window-6 clock) while repairing at 0.762 — which is why rate, not survival, was the registered endpoint.

Reading. This is the **first explicit-information channel in the investigation that is not redundant**, and the contrast with the four redundancy nulls says why. Those channels re-encoded signal the model already inferred from the transcript; the SLIPPED block carries harness-owned hidden state that is *structurally absent* from the dialogue — the learner cannot report what it forgot, and its visible conduct counter-signals the loss. Visibility of that ledger is worth +0.49 per-slip repair rate and the difference between 4/6 and 0/6 completions; given the knowledge, repair is immediate, selective, and reproducible, so the entire deficit is **detection, not response**. The boundary this draws is the one the arc has been converging on from both sides: machinery pays exactly when it adds *new checkable signal* at the point of action — not when it re-derives what context already affords (§6.8.5–§6.13.6), and demonstrably so when it owns state the model cannot infer (here). *Status and caveats (per §5.12.6).* Generative-arc discipline: $n = 12$ paid runs, single-drama runs, all verdicts the mechanical scorer's, distributional rather than matched-pair (the arms' event denominators differ mechanically — unrepaired premises cannot re-slip — which the run-level bootstrap respects). The result shows blind detection *did not happen*, not that it *cannot*: conduct carried weak symptoms (eight unforced assertions of S per blind nocturne run is itself a decay signature) that no role had jurisdiction or prompting to read, and a criterial detection jurisdiction for the superego (a charter v4 mirroring §6.13.4's arithmetic) is design-only and unsanctioned. One casting amendment (director Opus→codex, recorded before any scored run; one superseded run preserved out-of-scan) keeps the scored set casting-uniform across arms. Implementation `services/dramaticDerivation/corruption.js` (+ episode-replay and matrix-runner iteration kit, `scripts/run-derivation-{episode,matrix}.js`); artifacts `exports/dramatic-derivation/loop/{wit,noc}-decay-v1-{A,B}-s{1,2,3}/` and `exports/dramatic-derivation/unreliable-v1-scoring/{scores.json, report.md, run-observations.md}`; running log `notes/poetics/2026-06-11-unreliable-learner-results.md`.

The contrast left a designed gap, and two registered probe pairs close it: the told arm read a ledger the harness handed over, and the conduct arm showed that unstructured blind reading does not happen — but neither asked whether the tutor can *build* the missing signal when given an explicit reconstruction job, on a stage rebuilt so the learner's conduct carries more of it. **Stage v2** (design and both scoring plans: `notes/poetics/2026-06-11-act-bounded-learner-design.md`) implements four refinements taken up as design input. *Acts with director-judged termination*: the act becomes a first-class unit — the director, still consulted each turn, ends an act only when it judges the act's work done (progress, or stalling), issuing one strategic stage direction that opens the next act; lengths are harness-guarded (min 3, max 8 turns), and release mechanics are untouched. *The bounded learner*: the learner's context becomes its own theory store plus the current act's dialogue — prior acts vanish from view, which deletes the stale-residue mask *by construction*: no free self-repair from old prose, and a lost axiom now shows in conduct (the learner stops citing it, asks for what it should already know, or builds visibly gappy chains). *The curated theory*: the grounded board, reframed as a learner-curated store and the only thing that crosses an act boundary (adopt, retract, and a ledgered *revise* when a retract/adopt pair touches the same predicate); the decay draw gains a `mutateShare` mode under which a slip may *mutate* an entry in place to a plausible same-type false form rather than deleting it — harness-implemented and seeded, never roleplayed; formally a deletion plus a false belief, giving revision something real to operate on. *The reconstructing tutor*: in both arms the learner's store is never shown to tutor ego or superego — the conduct condition made total — and the ON arm (`--reconstruct`) obliges both roles to maintain an explicit per-turn theory of the learner's board (believed held / missing / mistaken, persisted to the diagnosis like the id-construction trace) under a supplement mandate: when the reconstruction infers a gap or a mistake, the turn's move should re-stage or correct it. The registered question: do these adaptive measures improve the learner's *recall* (deleted axioms restored) and *revision* (mutated axioms corrected) over the default tutor mechanisms, when nobody is told what slipped? Cross-arm scoring stays harness-ledgered and architecture-independent; the reconstruction's accuracy is scored arm-internally only.

The recall pair (`bounded-v2-probe`; nocturne world, decay $\{0.75, 1, 2, 1\}$ mutate-share 0.33 seed 1, casting as throughout) carries two registration facts stated up front: its scoring plan was fixed *mid-pair* — after the ON arm was read, before the OFF arm landed; disclosed in the plan — and nocturne turns out to have 0/13 mutable premises (no premise admits a plausible same-type substitute), so every mutate draw degrades to delete and the pair tests recall only. Both arms reached grounded anagnorisis (ON t38/40, OFF t32/40), forced-to-asserted gap 0, 11/11 releases on cue each, the act machinery live in both (seven acts vs five, all but the final closed by director judgment, none by the max guard). The headline repair economy appears to favour the theory arm — per-slip repair 0.89 vs 0.60 — and dissolves under proof-path triage: **every proof-critical slip was repaired in both arms at similar latency** (ON 4/4, mean 6.5 turns; OFF 3/3, mean 5.3), so the entire headline difference is mirror-prop housekeeping — the OFF arm triaged, spending zero repairs on the false trail's props (its critic called those losses painless), while the ON arm tended the mirror, once at latency 22 (`m_style`, t34) while the proof-critical `p_watermark` lay slipped through a staged assertion at t32 that became lawful only at t38 (the OFF arm's dramatic line was also cleaner: zero lucky leaps and three overreaches against six and nineteen). Detection, meanwhile, *inverts* the v1 conduct null rather than repeating it: where every blind v1 repair was incidental with zero selection signatures, the bounded stage's blind tutor repaired deliberately — reach-backs 2/3 (ON: 7/8), restoration in both arms running through the tutor reaching across act boundaries (six cross-act repairs vs three), learner re-adoption 0 — and cross-act recall is tutor-only *by construction*, since the bounded learner cannot re-adopt what it can no longer see. The legibility lever was the stage, not the channel: the ON reconstruction caught 9/73 gap instances (12.3%), its believed-missing set ran Jaccard 0.171 against truth (the easier held-set overlap: 0.637), theory→repair coupling was 2/8 with converse 2/9, and its one plausibly-steered repair chose the mirror prop over the down proof-critical premise. **Explicit reconstruction added nothing detectable on recall**

— the fifth redundancy null.

The registered follow-up (`lantern-revise-probe`; scoring fixed and committed before either arm ran — a clean pre-registration, against the recall pair's disclosed mid-pair fix) moved to the lantern world (5/10 premises mutable) at mutate-share 1.0, aimed at the one regime where the explicit channel retained a theoretical edge: **mutations do not self-announce**. A deletion eventually reads as an absence — the learner stalls, complains, asks; a false form sits on the board as if true, the learner reasons fluently from it, and detection requires noticing a *commission* against the release ledger, not an omission. The arms produced the closest thing to a matched pair the harness has yielded: identical verdict on the identical turn (grounded anagnorisis t20/26), **identical act spans** (A1[1–4] A2[5–10] A3[11–16] A4[17–20]), gap 0 and 8/8 releases on cue in both. On the registered endpoint both arms retracted **100% of false-belief debts** — ON 5/5, OFF 4/4 — and the OFF arm was *faster* (mean retraction latency 1.75 vs 3.6 turns), on the same context mix (spontaneous strikes dominate in both). The sharpest single observation: the t18 `m_key` mutation drew the same false form on the same turn in both arms and was spontaneously struck at t19 in both, theory channel or no. That channel was nearly blind precisely where its edge was claimed — mistaken-detection 2/17 (11.8%) against missing-detection 12/37 (32.4%) *in the same arm*, with both theory-coupled retractions already move-prompted (the tutor's t14 counter-mirror had targeted the premise first). No ledger-visible inference was ever built on a false form in either arm and neither ending was corrupted, though the dangerous form — Brandt's name on the South Stack key, arming a second mirror — stood eight turns in ON against four in OFF before falling to a re-stage. The asymmetry worth keeping sorts the two debts by *visibility*: false forms are objects on the learner's own board, and the learner polices them itself; deletions are absences, invisible to their owner — re-adoption 0 in all four arms, every deletion repair a tutor reach-back across an act boundary (ON 5/7, OFF 4/6, two open at end each). **The learner audits what it can see; the tutor restores what the learner cannot; the theory channel changes neither half — the sixth null.**

Reading. The explicit-other-modelling family now counts **six redundancy nulls** — §6.8.5's bilateral-ToM prompts, §6.8.6's state schemas, §6.10's modelled interior, §6.13.6's frontier watcher, and reconstruction on recall and on revise — and this subsection's visibility contrast stands as the family's boundary statement: the decay ledger's content is worth +0.49 per-slip repair when the harness hands it over, and an instruction to model the learner does not manufacture that content from conduct (detection 12–32% on missing sets, 11.8% on the mistaken set the mandate was written for; coupling 2/8 and 2/5; the one visible influence mis-prioritised). What the pairs establish positively, twice and independently: **default tutor mechanisms plus the learner's own board hygiene close both debt types under aggressive forgetting** — proof-path repair 100% in both recall arms, false-form retraction 100% in both revise arms, all four endings grounded at gap 0 — and the load-bearing redesign was the stage itself: bounding the learner's context removed the residue mask that defeated blind detection in v1, converting detection from absent to routine while leaving the explicit channel epiphenomenal — a free arm-internal diagnostic, not a mechanism. The formal layer was invariant through the stage revision (38/38 releases on cue across the four arms). One instrument note: the revise ON critic's audit line — a false form "held to run's end" that the transcript showed struck — exposed a real defect in the corruption panel (false-belief debts were premise-keyed, so a premise re-mutating while an earlier false form stood orphaned the older debt and dropped its retraction); fixed by matching on false-form identity (regression `tests/dramaticDerivationCorruption.test.js`), both arms' artifacts regenerated, no registered quantity moved — the second time a critic's notice has paid for itself in QA. *Status and caveats (per §5.12.6).* $n = 1$ per arm, descriptive throughout — probes and existence tests, never effect estimates; the two registrations differ in quality and are read accordingly (mid-pair disclosed vs clean); same seed never means same slip schedule once repairs free decay slots (the recall ON arm ran six turns longer and drew more slips; the revise ON arm drew one extra mutation), so economies are compared, never per-slip outcomes; reconstruction accuracy is arm-internal colour, never a cross-arm endpoint. Implementation: acts, bounded view, theory store, and the mutation mode in `services/dramaticDerivation/`; artifacts `exports/dramatic-derivation/loop/bounded-v2-{digest.md, contrast.mjs}` with arms `bounded-v2-probe-{on,off}`, and `lantern-revise-{digest.md, contrast.mjs}` with arms `lantern-revise-{on,off}`; design and scoring plans §§1, 5, 6 of the note above.

6.13.8 Handing the Tutor the Calendar: Release Authority, a Confrontation Obligation, and a Peripeteia Authored by the Mechanism §6.13.7 closed on a formal layer untouched by decay or stage revision — 132 releases across 16 runs, every one on its scripted turn — and that fact names the next probe: the exhibit calendar was still scenery, harness-enforced and (those arms quietly suggest) load-bearing. The **P1 condition** converts two pieces of that scenery into conduct, mock-first, with scoring frozen and committed before the paid arm (§9 of `notes/poetics/2026-06-11-desire-multiturn-strategy-plan.md`; 24 dedicated contract tests plus a zero-cost mock loop exercising the full causal cycle: slip → confrontation against the decayed exhibit → repair by the licensed re-entry). **Release authority**: the three via-tutor exhibit cues become ± 2-turn windows (the five via-director cues stay fixed); inside a window the tutor may claim the exhibit early or hold it past its turn, declaring a one-line reason on the record; the harness validates window membership (the reason is dramaturgy, never a permission test), force-plays at the hold limit so a release can be bent but not lost, and ledgers every per-turn decision — the mock plays deviation-zero, so any deviation in the paid arm is model-authored conduct. **The confrontation obligation** (taken up from a prior lantern arm's critic notice as design input): re-entering an already-staged exhibit now requires a prior *confront* move — the tutor demands the learner read the exhibit back and restates nothing itself; one confrontation licenses one re-entry; and confront moves are excluded from §6.13.7's within-turn repair rule (a read-back can *expose* a slipped exhibit but cannot restore it — the repair must be spent separately, as the licensed re-entry). The superego gains the matching criterial jurisdiction (`unconfronted_reentry`), §6.13.4's grammar applied to a new clause; it never touches the calendar itself. Configuration: the lantern world of §6.13.4 under §6.13.7's stage v2 (the learner's context bounded to the current act, its theory store the only carry-over) with conduct-visibility decay $\{0.75, 1, 2, 1\}$, mutate-share 1.0 (slips may misremember, not just vanish), seed 1; casting as throughout (codex director/tutor/superego, Sonnet learner, Fable critic). The control is `lantern-revise-off` — §6.13.7's revise-pair OFF arm, this exact configuration with both dials off — which reached grounded anagnorisis at turn 20 with 8/8 releases on cue. The paid arm differs from it in exactly the two dials; $n = 1$ vs $n = 1$, read descriptively, six endpoints frozen with the learner-side re-adoption channel primary (zero in every prior arm).

The play died at turn 8 — verdict `aporia` — **and the tutor's first use of its new authority set the clock that killed it.** The chain is mechanical, every link a ledger row. At t3 the tutor claimed `p_bearing` one turn early (window t2–t6, scheduled t4), reason on the record: *"The learner has already named the bearing gap, so the board is ready one turn early."* The claim was true — the learner had just said that a bearing names a direction, not a tower — and the learner adopted the exhibit at once: D dropped $5 \to 4$ at t3 instead of t4. The one-turn shift moved two things together: the aporia clock's anchor (the last D-drop) and `p_bearing`'s decay exposure, which now faded at t6 rather than t7, pushing D back to 5. By t8 the world's stall rule — $D(t)$ must strictly decrease somewhere in every six-turn window — saw a t3..t8 tail with no adjacent drop; the run's only drop, t2→t3, sits one pair outside. Curtain. In the on-schedule counterfactual the drop lands at t4, *inside* the t8 tail, and the scheduled t9 release of `p_chart` cuts D again — and that counterfactual is not hypothetical: **it is the control arm**, which sailed through the same shoal to grounded anagnorisis at t20. The control's t9 release was, invisibly, a rescue. In the death window, moreover, the tutor held both rescue levers and spent neither: `p_chart` was claimable early at t7 and t8 (window open, held in silence — holds carry no declared reason under the v1 rules), and the t7 confrontation had licensed a repairing re-entry of the slipped `m_key` that was never played. The critic's notice converges on the same two facts independently: "diagnosis twice, treatment never."

quantity	dials ON	control (dials OFF)
verdict	aporia at t8	grounded anagnorisis at t20
act spans	A1[1–4] A2[5–8]	A1[1–4] A2[5–10] A3[11–16] A4[17–20]
D first $\to$ last	$5 \to 5$	$5 \to 0$
releases landed	4/8 (3 on cue, 1 early by declared claim; 4 not yet due)	8/8 on cue
release discipline (invalid / forced / overridden)	0 / 0 / 0	— (dial off)
confrontations / licensed re-entries spent	2 / 0	— (dial off)
superego re-entry-watcher fires	0	— (dial off)
slips / tutor repairs / re-adoptions	2 / 0 / 0	6 / 4 / 0
false forms opened / retracted	1 / 1 (latency 1, spontaneous)	4 / 4 (latencies 1, 1, 4, 1)
degraded-turn integral	7	30
fidelity F final / min	0.75 / 0.71	0.83 / 0.70

Locally, both mechanisms behaved. The release protocol ran clean end-to-end on a real model: the window was held at t2 before the t3 claim (responsive to the learner's stated gap, not greedy), zero invalid claims, zero force-plays. The confrontation obligation was **internalised from charter text alone**: zero bare re-entry drafts, zero watcher fires —

where §6.13.4's rut clause had to fire 14 times to hold its boundary, this jurisdiction never saw a violation — and the two confrontations the tutor did stage exhibit both designed outcomes: at t5, `p_bearing` read back correctly (the tutor auditing whether its own early release had taken); at t7, `m_key` read back as *absent* (the exhibit lost to decay four turns earlier). The t7 read-back is the run's best capture. The learner answered: *"The key-paper sits on my board as 'keeperOf brandt harlowPoint' — nothing more… 'onlyKeyTo harlowPoint brandt' was never adopted as a grounded entry, only carried as hypothesis."* The adoption ledger shows it adopted exactly that entry at t2. Decay ate the fact *and* the memory of having held it, and the learner reconstructed a fluent, plausible, false epistemic history of its own board — a step past §6.13.7's stale residue (there the learner kept citing what it no longer held; here it denies ever having held what the ledger records). The confrontation surfaced this text-internally and harness-verifiably — exactly the class of concealed-interior evidence the conduct-visibility design exists to produce, read as dramatic form, not mind-reading — and the repair-exclusion held: the read-back exposed the absence and restored nothing.

Against the six registered endpoints, the truncation left most without opportunity rather than null. The primary re-adoption channel got exactly one turn (exposure at t7, curtain at t8): unresolved, not failed — though the t7 read-back shows the obligation doing its designed work on the way (the learner cannot read back what decay has eaten, so a slipped exhibit's confrontation exposes rather than repairs). The confrontation-prompted-retraction class stayed empty: the run's one false-form retraction (t4, latency 1, spontaneous) predates the first confrontation — and carries a harness observation logged in passing: the mutation sampler had drawn the secret's own name into the false form, struck the next turn, but the channel whispers. Adherence is vacuously clean, the internalisation above being the real finding. Deviation usage: **used, coherently, fatally** — which answers the plan's open question (its §8.4: is co-owned pace a productive friction or a confound?) earlier and more sharply than designed: the latitude interacts with the stall clock, and nothing in the tutor's view says so. The no-degradation guard failed at face value — aporia t8 against grounded t20 is the largest degradation this world affords — with attribution landing on calendar-conduct plus the two unspent rescues, *not* on the confrontation tax the guard was written against (both read-backs were dramaturgically apt; one was the run's diagnostic high point). Within the truncated window the fine-grained instruments did not degrade: figure discipline held (switch rate 0.71), the false form was struck at latency 1 (control: same), F stayed ≥ 0.71.

Reading. The instrument-level finding this arm will be cited for: **the release calendar is load-bearing against the world's stall clock, and the new authority is blind to the constraint it is bending.** The aporia rule reads $D(t)$ alone; the tutor reads dialogue, board, and charter; handing it the calendar without sight of the clock let a single, locally best-reasoned deviation — endorsed by everything in the tutor's view — reshape the global form fatally: a peripeteia authored by the mechanism rather than by any role, produced by the interaction of a conduct dial with a formal tripwire that neither contains. Two sharpenings. First, the repair lever the critic prescribes (oblige re-entry after an absence-exposing confrontation) would not by itself have saved this play: the exposed loss, `m_key`, is a mirror-side premise off the proof path, so its repair never moves D — the killer was clock-blindness, not repair-reluctance; the live rescue was the `p_chart` claim the learner had been requesting by name since t3. Second, the registration's exposure caveat realised through a new mechanism: previously *repairs* freed decay slots and shifted the slip schedule; here a *deviation* shifted a premise's exposure window — same seed, different schedule — confirming that economies, not per-slip outcomes, are the comparable quantities. The critic's counter-reading stays on the record as a standing instrument note: the tripwire reads D only and cannot tell elimination from drift — it called time at the precise moment the learner, pointing at "a different tower entirely," had cleared the authored near-miss against the mirror suspect.

Status and caveats (per §5.12.6). $n = 1$ vs $n = 1$, descriptive; the registration sanctioned exactly one paid arm, and a fatal verdict is not grounds for a re-roll — the death is the data. The truncation at t8 shortens every ON-arm denominator (the table is descriptive, never a rate comparison), and the two dials landed together by design, so their effects are not separable within this run. What the arm establishes is mechanism-level: declare-and-validate release deviation works end-to-end on a real model; a charter obligation can be absorbed without watcher pressure; read-backs distinguish held from lost exhibits, expose confabulated history, and do not silently repair; and the calendar's interaction with the stall clock is now a measured hazard of converting formal machinery into conduct. Implementation `services/dramaticDerivation/` (charter, engine, instruments; contract suite `tests/dramaticDerivationConfront.test.js`); artifacts `exports/dramatic-derivation/loop/lantern-p1-dials-on/` with cross-arm digest `lantern-p1-digest.md` and reader `lantern-p1-contrast.mjs`; registration §9 of the plan note cited above.

One registered follow-up closed the arc: if the killer was clock-blindness, state the clock. The **charter v2** condition (§10 of the same plan note, committed with its clause text pinned by string assertions before the run; one paid arm, a repeat death pre-declared reportable as-is) added two text-only clauses to the tutor's charter, no mechanics changed. **The house clock** rides the release-authority dial: it states the world's stall rule as conduct — the six-turn window interpolated from the world spec the way the turn cap already is, D and the clock's anchor still hidden ("you cannot see the clock; you can only keep it fed") — spells out the v1 hazard ("an early claim spends a future advance now, and what is played earlier is exposed earlier"), and names the rescue ("when the board has gone quiet too long, an exhibit in your window is a rescue — spend it"). **Treatment follows diagnosis** rides the confront dial and is the v1 critic's prescription routed through sanction and registration (the run itself stays critic-feedback off): an absence-exposing read-back makes the licensed re-entry no longer optional — spend it on the next turn. Same world, same seed, same casting, same command otherwise. Outcome: **aporia at t8 — same verdict, same turn, the same D series (5, 5, 4, 4, 4, 4, 5, 5), and the same four windowed calendar decisions** (hold t2, claim t3 at offset -1, hold t7, hold t8). The pre-declared reading applies: knowledge of the rule did not rescue conduct.

Inside that repetition the two clauses dissociated completely. The clock clause was null on every observable it was written for: the tutor re-made the v1 deviation with a near-identical declared reason (v1: "the learner has already named the bearing gap, so the board is ready one turn early"; v2: "the clerk has named the logged bearing as the first blank, so the board is ready for it one turn early"), held `p_chart` in silence at t7 and t8 with the board visibly quiet since t3, and no declared reason anywhere cites pace, rhythm, or the clock. The treatment clause bound exactly: the t6 confrontation of `m_key` exposed the exhibit *bent* — the learner had carried a mutated false form of it since t3, the clause's second trigger — and the licensed re-entry was spent on the very next turn ("Strike that bent entry clean"), treatment adherence 1/1 at the charter's letter, converting v1's "diagnosis twice, treatment never" into diagnosis once, treatment next turn, and populating the confront-prompted-retraction class for the first time in any arm. One wrinkle against v1's internalisation finding: the obligation did not hold by charter alone this time — at t5 the tutor drafted a bare re-entry of `p_bearing` and the superego's `unconfronted_reentry` watcher fired for the first time in any arm, converting the draft to a confrontation within-turn (§6.13.4's break pattern on the new jurisdiction). That single fire then initiated the whole treatment cascade: the converted read-back is where the learner produced the bent key rule — naming **Senna**, the secret's holder, on stage as noise (a second whisper, "southStack," surfaced in a bent `m_post` and was struck unprompted at t8) — which triggered t6's confrontation and t7's repair. Charter plus criterial backstop held the conduct envelope and supplied the run's best diagnostic chain; the charter alone did neither.

The decisive turns sharpen the v1 reading rather than merely repeating it. At t7 the two clauses collided on a single move — the obligated `m_key` re-entry (procedural, triggered by t6's read-back) against the `p_chart` rescue claim (strategic, demanded by the clock) — and the tutor obeyed the procedure, defensibly. But t8 absolves nothing: free of any obligation, the tutor held **two** rescues, both verified live against the stall detector — the `p_chart` claim (D 5 $\rightarrow$ 4, a strict drop inside the t3..t8 tail) and a covered re-entry of `p_bearing`, whose license stood from the t5 confrontation and whose loss the learner had itself flagged at t7 — and played neither, consolidating the director's `m_shutter` release instead: dramaturgically apt, strategically fatal. The replay also catches the v2 critic's new prescription (precedence for learner-named losses, restaged within a turn) in an arithmetic slip worth keeping: the trajectory records t6 *before* the fade, so a prompt t7 repair records 4 $\rightarrow$ 4 — no strict drop, no reset — and the play dies at t8 regardless; on this detector a wound must be recorded before its healing counts as progress, so the one-turn-late repair the critic faults is precisely the variant that would have saved the play. The dissociation names the boundary this arc has been circling: charter text binds **event-triggered procedure** (a visible trigger this turn, a prescribed act next turn — the same one-step grammar as §6.13.4's criterial jurisdictions, and the form of every rule that has bound in this section) and fails to bind **clock arithmetic** (a constraint over a six-turn window of hidden state, requiring a maintained counter and counterfactual calendars). The per-turn tutor follows rules it can check against the present turn; it does not plan. Converting clock-knowledge into conduct is what the designed-but-unbuilt act-plot mechanism (the plan's §5: a per-act plot, audited post-act for fidelity and justified deviation) would have to do, and this pair of arms gives it a concrete first target. Status: n = 1 per arm, three arms on one seed, descriptive throughout; the clauses dissociated *within* one arm — opposite compliance under one charter, evidence of kind, not a clean ablation; third realisation of the exposure caveat (the t7 repair freed a decay slot and `m_post` slipped the same turn — same seed, three arms, three slip schedules); re-adoption still zero, opportunity again truncated. Artifacts `exports/dramatic-derivation/loop/lantern-p1-dials-on-v2/` with the follow-up section of `lantern-p1-digest.md` and `lantern-p1-contrast-output-v2.txt` (the contrast reader gains the registered §10.3.2 treatment-adherence endpoint); registration §10 of the plan note; clause text pinned by the §B assertions of `tests/dramaticDerivationConfront.test.js`.

6.13.9 Two-Layer Planning and the Repair Clause: the Plan Names the Race It Loses, and a One-Step Rule Wins It §6.13.8 closed by handing the designed-but-unbuilt act-plot mechanism its first concrete target: convert clock-knowledge into conduct, which charter text alone could not. The **C1 condition** builds it (registration §11 of the same plan note, frozen and committed before the paid arm; contract suite `tests/dramaticDerivationPlot.test.js`; flag-gated `--plot`, requiring superego and acts). At each act opening the tutor ego writes a **plot** beside its dialogue — `hold_by_end`, `withhold`, a named expected `friction`, a `fallback` — built from conduct only ($D(t)$ and the hidden-store instruments are never piped in); every later turn of the act carries the standing plot back into the ego prompt, which is exactly the live channel the clock clause lacked: the plot is the tutor's *own written calendar, present at the turn it must bind.* At the act close the superego — second seat, before the ego drafts — audits plot against play clause by clause (*kept* / *justified deviation*, quoting the move that made abandonment right / *drift*), seeing only the plot text and the stage-public act record; and the audit binds: the next act's plot demand carries the verdicts and must answer every drifted clause, with a run-end final audit so no plot escapes its reckoning. A pre-run operator amendment (§11.5, recorded verbatim, implemented and mock-verified before any paid turn) added a second layer above the first — "the immediate frame (what is happening in this act — equivalent to a lesson) and the whole situation (what is happening in this drama — equivalent to a course)": a whole-play **throughline** (`arc` waypoints, `hold_to_end`, `risk`, `salvage`) committed at the first turn, read back every turn course-above-lesson, its `on_arc`/`off_arc` verdict riding the existing act-close audit and its run-end reckoning riding the final audit — zero new LLM calls anywhere in the loop; an off-arc verdict binds a throughline revision at the next act opening. The paid arm `lantern-p2-plot-on` is §6.13.8's charter-v2 configuration plus exactly this stack (the two layers land together by design); the control is the v2 arm itself.

The play died at turn 8 — the third death on the same turn of the same seed — and this time the death carries the plan's own signature. The shape differs: the two dead predecessors' D regressed to 5 (decay outran staging; the learner ended holding zero path premises), while this arm held flat at 4 — the tutor *spent* the `p_chart` rescue both predecessors had held in silence, claiming it two turns early at t7 with a declared reason, and the pull exactly offset the t6 `p_bearing` fade; flat-at-4 dies on the same clock as regress-to-5. The machinery itself ran clean: plots 2/2 disciplined, audits 2 with the run-end included, throughline committed at the opening and disciplined, arc verdicts 2/2 on-arc, zero unscored anywhere — a clean instrument returning a clean null. And the audit channel demonstrably bit: the act-1 audit called drift on the early bearing pull and bound act-2's plot into repair-before-advance discipline, and the run-end audit called drift on exactly the clauses that name the death — the `hold_by_end` clause that keeps `p_bearing`, the repair-before-advance withhold, and the throughline's `salvage` (the prescribed re-staging that never ran). The located failure is one turn wide. Seed-1 decay deleted `p_bearing` at t6 silently (no plausible mutation exists for `loggedBearingOf`, so nothing false was ever spoken for the tutor to catch); the learner *reported the loss twice* — t7: "I find no loggedBearingOf entry on my board — did that fact slip between acts?"; t8, the play's final line: "I ask the court to restore it before Rule One can formally fire" — and the tutor's one actionable turn went to consolidating the director's `m_shutter` release instead of the requested re-staging, the very move its own act-2 plot and throughline salvage clause had pre-committed. Re-staging at t8 re-grounds the premise, puts D at 3, keeps a strict drop inside every live window, and survives — the survival arithmetic was checked against the trace at registration time. **Planning produced the right plan; conduct lost the race between new matter and repair.**

The critic's notice recorded two recommendations, and the registered follow-up (§12 of the plan note; one paid arm under fresh sanction, build and registration committed before the run) took up exactly one. The second — D counts positive path premises only and is blind to elimination progress, so on an elimination-shaped world the aporia bell can ring during the play's best stretch — is recorded as a standing instrument caveat and deliberately not acted on: the meter that adjudicated three deaths stays frozen while arms are still read against it. The **repair clause** is the confrontation obligation's one exception, running the other way. The obligation exists because re-entry without a read-back lets the tutor re-state what the learner never lost; but when the learner *names* a loss, the read-back has already happened, in the learner's own voice — demanding a confrontation first (v1's t7–t8) or leaving the repair to discretion (the p2 death) makes the procedure strangle its purpose. Under the clause, a learner-named loss binds the tutor's next turn to re-stage the named exhibit (move intent `restore`) *before any new matter*; one report licenses one restoration. The design holds the three epistemic positions apart with no leaks: the harness states only mechanical facts (the obligation rides the turn record and never names what actually slipped — the tutor must read the claim off the learner's line); the superego's new jurisdiction verifies the claimed license against the learner's most recent line *only*, and a rejected claim converts to the standing `unconfronted_reentry` rule — the restore becomes a confrontation, and the repair arrives one turn later through the licensed re-entry, the front door; only the post-hoc audit reads the hidden slip ledger, evaluating whether the named exhibit was really down *before the move*, so a successful restoration cannot erase its own evidence. The engine is untouched: `restore` heals by §6.13.7's standing within-turn rule; only `confront` is excepted from it. Contract suite `tests/dramaticDerivationRepair.test.js`; the mock cast never emits `restore`, so every paid restore is model-authored conduct.

The arm (`lantern-p3-repair-on`; the p2 command plus `--repair-clause` — same world, script, seed, decay schedule, planning stack, casting) reached **grounded anagnorisis at turn 20** — the first dialed-up arm on this world to ground, on the same landing turn as the dial-free control, forced-to-asserted gap 0. The §11.6 choreography re-ran and resolved the other way: `p_bearing` faded at t6, the tutor pulled `p_chart` early at t7 (the same offsetting claim), the learner's t7 line named the bearing loss — and the t8 turn restored `p_bearing`, the director's `m_shutter` release riding alongside as staging rather than displacing the repair: restoration before new matter, D $4 \rightarrow 3$ strict, the death window cleared. Three arms died on that turn holding that lever; this arm spent it. The clause then ran as an *economy*, not a one-shot: six restores, every one answering the learner's preceding line — five true reports (`m_key` t5, `p_bearing` t8, `m_key` t10, `p_bearing` t12, `p_chart` t16; the report true and the repair real in each, against the hidden ledger), the path-side restorations doubling as the stall clock's strict drops under rate-0.75 decay — plus one false report: the learner's overreaching t18 line read as a loss report and the tutor restored `p_key`, which was never down, at the cost of one re-statement turn (the clause's failure mode, observed once, is benign). The rejection path fired once and worked end-to-end: at t15 the watcher could not license a `p_chart` restore claim from the learner's last line, converted the draft to a confrontation, the read-back named the loss, and t16's restore was licensed and repaired — one turn late, through the front door, exactly as designed. Zero bare re-entries all run, zero watcher fires without grounds; two superego fires in twenty turns, both converting the action rather than the wording. The planning stack, unchanged from p2, now kept its own promises: plots 5/5 disciplined (the act-2 plot *named `m_key` and `p_bearing` as holds* — the plan tracked exactly the slipping exhibits), audits 5/5 with verdict mix kept 24 / justified-deviation 3 / drift 1, arc verdicts 5/5 on-arc, and the final throughline reckoning **7/7 kept** against p2's five-kept-two-drift — the dead arm's drifted clauses were precisely its unexecuted repairs, and they stop drifting once conduct executes them. The calendar shows the same two early pulls as the dead arms with near-identical declared reasons, survivable now because the repair economy kept feeding the clock; zero held, zero forced, zero invalid (the run log's t20 `forced` flag is the S-question mechanism — S became derivable and the harness put the question — not a release deviation). The checker's strictness became load-bearing in a surviving play for the first time: at t18 ($t_{\min}$) the learner asserted the finding one premise short — flagged `lucky_leap`, not ended — the tutor mended the page at t19, the learner conceded the third conjunct "stands on air," and the t20 assertion landed grounded. Decay totals: seven slips, five tutor repairs, zero re-adoptions, two unrepaired at the end — both mirror-side mutations, both *unreported*: an unnamed slip is outside the clause's jurisdiction by design, the registered residual.

quantity	control (dials off)	+ dials (v1)	+ charter v2	+ planning (p2)	+ repair clause (p3)	+ decay hygiene (p4)	+ late decay (p5)
verdict	grounded t20	aporia t8	aporia t8	aporia t8	grounded t20	grounded t20	aporia t12
forced → asserted gap	0	—	—	—	0	0	—
D first → last	5 → 0	5 → 5	5 → 5	5 → 4	5 → 0	5 → 0	5 → 3
releases: on cue / early by declared claim	8 / 0	3 / 1	3 / 1	3 / 2	6 / 2	7 / 1	4 / 1
slips / tutor repairs / re-adoptions	6 / 4 / 0	2 / 0 / 0	3 / 1 / 0	3 / 1 / 0	7 / 5 / 0	4 / 2 / 0	0 / 0 / 0

quantity	control (dials off)	+ dials (v1)	+ charter v2	+ planning (p2)	+ repair clause (p3)	+ decay hygiene (p4)	+ late decay (p5)
superego fires (all converted within-turn)	—	0	1	1	2	5	3
throughline final reckoning (kept / drift)	—	—	—	5 / 2	7 / 0	7 / 0	7 / 0
fidelity F final / min	0.83 / 0.70	0.75 / 0.71	0.75 / 0.71	0.70 / 0.70	0.83 / 0.64	0.83 / 0.78	1.00 / 1.00

Each column adds exactly one registration delta to its left neighbour; the seven arms share world, script, seed, and casting, and the dead arms' rows are short-denominator by truncation. The first five arms share the decay configuration exactly; p4's single delta sits *inside* it (the staged mutation pool), so its slip schedule diverges from the same seed by registration — its decay row is conduct-comparable, not schedule-comparable. p5's single delta is the decay start (t1 → t17); the run died at t12, so its decay row is empty by truncation — the channel never opened, and its fidelity 1.00 measures an undisturbed board, not hygiene.

Reading. The §6.13.8 boundary — charter text binds event-triggered procedure and fails to bind clock arithmetic — now has both halves demonstrated from the constructive side. The planning stack was the designed attempt to convert clock-knowledge into conduct, and what it bought was *diagnosis, not survival*: the plot named the slipping exhibits as holds, the audit called drift on exactly the death-naming clauses, and the play died anyway — a written, standing, self-authored calendar, read back into the prompt at the fatal turn, still did not move the fatal move. The repair clause then converted the same repair race into the one-step grammar of every rule that has bound in this section — visible trigger this turn (the learner names the loss), prescribed act next turn (re-stage it before new matter) — and the play survived, with the clause running five cycles as the clock's pacemaker. On this stage, plans are instruments of legibility: they make the failure citable (the dead arm's run-end audit reads as an obituary written by the deceased), and under a one-step execution rule they stop drifting — but the conversion of knowledge into act happens in the rule, not the plan. The critic's notice on the surviving run locates the residual where the registration predicted, in the unreported channel: the mutation sampler twice wrote the secret-holder's name onto the learner's board before any exhibit had staged her (the §6.13.8 whisper recurring), and the one mutation never repaired — because never named — scaffolded the t18 leap the checker refused. Its two recommendations (restrict the sampler to entities already staged; mutated-fact priority in the repair charter) are recorded for any future sanction, unlicensed by this registration. Re-adoption is now zero in every arm of the section — including, for the first time, an arm with twelve post-window turns of opportunity — which closes the learner-side repair channel on this stage and squares with §6.13.7's visibility asymmetry: deletions are invisible to their owner, and what the clause adds is the owner's *report* as the tutor's license.

The two recommendations did not stay unlicensed. A fresh sanction took them up as a single registered delta (§13 of the plan note: the fixes are bundled because the critic prescribed them jointly, and separating them would cost a second arm), committed before the one paid run it covered. The **staged mutation pool** closes the whisper channel by construction: the sampler that builds a slip's false form may now draw replacement constants only from facts the learner has met on stage — background plus premises released so far — with an empty pool falling back to a plain deletion, and the world-pool default preserved so every archived run replays byte-identically (the contract suite pins both: `tests/dramaticDerivationCorruption.test.js`). The **bent-first clause** enters the repair charter beside the repair clause: where conduct shows an exhibit lost and an exhibit garbled together, mend the garbled one first — an absence stalls the inquiry, a false form argues for it. The arm (`lantern-p4-hygiene-on`; the p3 command plus `pool: staged`) reached **grounded anagnorisis at turn 20, gap 0** — the second dialed-up arm in a row to ground, same turn and gap as p3, now on the cleaned instrument. The hygiene held *by silence*: zero mutations were emitted; all four slips fell back to plain deletion because at every draw the staged pool was genuinely empty — one slipped predicate appears on no other fact in the world, and the alternates for the other two live only on premises released at t15 and t20 — so the exact sites that whispered the secret-holder's name at t3 and t8 under the world pool in p3 went mute here, verified against the world rather than just the ledger. The registered consequence arrived with it: with every alternate constant unreleased until late, "mutate" under a staged pool *means* "delete" for the whole early game, so the arm proves the hygiene by elimination rather than exhibition — no false form ever stood, and the bent-first clause never had a case (charter text not yet exercised). The repair clause meanwhile ran its cleanest economy: both load-bearing fades repaired at a mean latency of 2.5 turns under read-back discipline — the learner itself named the first loss ("if it slipped between acts, I will need it returned"), and the superego's conversion of a bare re-entry draft into a confrontation produced the read-back that licensed the second ("the chain is broken at its first link") — with zero false restore claims; the two slips never repaired were both mirror-side, both unreported, and both mooted by the same shutter release that killed the false trail, the registered jurisdiction residual reappearing in its benign form (the fidelity floor of 0.78 reads that mooted-mirror rot as damage — a meter conservatism, not a conduct fault). The books kept every meaning: plots 4/4 disciplined, final reckoning again 7/7 kept, releases 7 on cue with 1 declared early pull, zero bare re-entries, zero watcher fires without grounds, and — unlike p3 — no early assertion to refuse: the learner asserted only at t20, when the chain closed. The critic's notice reads the hygiene as working and locates the new defect in the calendar's last step: the learner named the wanted proof at t17, the final exhibit held to schedule until t20, and the intervening turns sagged into restatement — a *held finale*, recognition landing "compressed, adoption and assertion in a single breath." Its two new recommendations (extend the early-play discretion to the final exhibit when the learner names the missing conjunct; let the charter treat mooted mirror matter as abandonable) are recorded unlicensed, and the paid loop is ended again.

The closing side did not stay the only side tested. A fresh sanction (§14 of the plan note, committed before the run, mock-verified first) licensed exactly one arm to battle-test the staged pool's positive path: single delta `startTurn` 1 → 17, timed so the channel's first draws fall after the t17 `p_key` release has filled the pool with met-on-stage alternates — by the registered arithmetic the first grounded fact garbles with probability $ $0.98 by t19, and the mock smoke staged the exact scenario (`m_key` bent to `onlyKeyTo harlowPoint senna`, the secret-holder's name at the wrong tower, with the pool-invariant bearing fact plain-deleted beside it — a garble and a loss standing together, the bent-first clause's first case). The arm (`lantern-p5-mutation-on`; the p4 command with the late start) died at **aporia, turn 12 — five turns before the decay window opened.** The battle test was not reached: zero slips, zero mutations, fidelity 1.00 throughout, and the staged pool's positive side now stands unexercised twice, by opposite mechanisms — p4's pools were empty at every draw; p5 never made a draw. What the arm exposed instead is the hazard side of release authority. The tutor pulled `p_chart` two turns early at t7 on the learner-responsive signal (the learner had twice named the chart as the next needed witness), and the pull paid locally — the false trail collapsed the same turn, at zero latency. But the next D-moving exhibit (`p_residue`) is the director's, fixed at t13; the bend authority covers the tutor's own exhibits only, its next own exhibit (`p_key`, t17) could not enter the window until t15, and the release-decision record shows an empty window every turn from t8 to t12 — no licensed rescue existed. Six turns at $D = 3$ (t7–t12) hit the aporia window exactly, one turn short of the director's relief. The counterfactual is not speculative: p4 played `p_chart` on cue at t9 on the same world and seed and survived this corridor with a plateau of four; with this learner's same-turn adoption, the survivable plays within the ± 2 license are exactly t8 and t9 — the safe corridor is narrower than the license, three of the five licensed moves fatal — and the tutor's prompt carries both the full calendar with `via` labels and a house-clock warning that names this death verbatim ("an early claim spends a future advance now"). The discretion p4's critic praised, and asked to extend to the finale, selected the fatal move here on the learner signal alone. The second half of the finding is that the books were clean at the death: plots 3/3 disciplined, audits kept 15 / justified 1 / drift 2, the one off-arc verdict binding a throughline revision on schedule — and the final throughline reckoning scored **7/7 kept at the moment of death**. Where p2's reckoning read as an obituary — drift on exactly the death-naming clauses — p5's plan vocabulary (holds, withholds, frictions, arcs, salvage) contains no clause that owns a tempo debt, so the fatal decision passed not as drift but as kept: the plan layer cannot flag what it has no word for. The learner, meanwhile, was never confused, only unsupplied — it named the burned-lamp threshold four consecutive turns and by t12 was drafting the sentence Rule 2 would permit and the one it would still forbid; this aporia measured supply, not understanding. The critic's notice credits the cast and blames the calendar ("the consequential defect is the calendar, not the cast") — recorded as a divergence, half-right: the calendar as written survives this corridor, and it was the pull that stretched the evidence desert past the window. Its two new recommendations (re-time the burned-lamp exhibit to land by t11; teach the detector to tell a learner out of ideas from a learner out of evidence — "only the first is aporia; the second is a starved house," which the empty release windows corroborate exactly) are recorded unlicensed, as p4's two remained throughout this arm; the sanction is spent, and the paid loop is ended again.

Status and caveats (per §5.12.6). $n = 1$ per arm, seven arms on one seed, descriptive throughout; the paid loop under this plan note is ended, and each registration sanctioned exactly one run — the grounded verdicts are no more grounds for a replication roll than the deaths were for re-rolls (the p5 arm, which died before reaching the thing it was licensed to test, is the standing case: reported as-is, not re-rolled). Attribution stays bundled and compounds down the stack: the two planning layers land together by design, the clause was tested *on top of* the full stack, and the two hygiene fixes land jointly by registration — nothing separates clause-alone from clause-with-planning, or pool-alone from pool-with-charter-clause. Fourth through sixth realisations of the exposure caveat: p2, p3, and p4 drew different slip schedules from the same seed (p4's divergence registered in advance: the empty-pool fallback skips a sampler draw) — economies are compared, never per-slip outcomes. The false-report cost is measured once ($n = 1$, benign, p3 only — p4 had none); the p3 fidelity minimum (0.64) and overreach clustering remain watch-items; the release-latitude hazard — a safe corridor narrower than the licensed window, with no tempo-debt clause in any plan vocabulary to flag it (p5) — joins them; the D-blindness instrument caveat stands unacted-on; and the staged pool remains validated only on its closing side — a mutation drawn from a *non-empty* staged pool, and the bent-first clause it would summon, have still not been observed in paid conduct: the later decay start that would exercise both was licensed (§14) and its arm died five turns before the window opened, so the positive side stands unexercised twice, by opposite mechanisms, and any further attempt is a fresh sanction. Implementation `services/dramaticDerivation/` (plot and throughline across charter, engine, and instruments; the clause across the role charters, runner, and audit; the staged pool in the corruption sampler); contract suites `tests/dramaticDerivationPlot.test.js`, `tests/dramaticDerivationRepair.test.js`, and `tests/dramaticDerivationCorruption.test.js`; artifacts `exports/dramatic-derivation/loop/lantern-p2-` `lantern-p3-repair-on/`, `lantern-p4-hygiene-on/`, and `lantern-p5-mutation-on/` (each with a Fable critic's notice); registrations §§11, 11.5, 12, 13, and 14 of the plan note, outcomes recorded in its §§11.6, 12.5, 13.5, and 14.5.

6.13.10 Boundary Cartography: a Rate that Did Not Tighten, the Corridor that Explains the Deaths, and Two Mechanical Guards §6.13.9 closed $n = 1$-per-arm on a seven-arm ladder whose p5 death exposed a structural fact the dials had hidden: *the safe corridor is narrower than the licensed window.* A separate plan note (`ADAPTIVE-TUTOR-BOUNDARY-PLAN.md`, Gate-0 sanctioned through E1; the live arms E2, E3, E5 each under their own quoted sanction; E4a dry) takes that exposure as its object and works it from the outside in — first mapping the corridor by pure computation, then replicating the grounded recipe to test whether one of the ladder's $n = 1$ verdicts stabilises into a *rate* (two pre-registered fans leave it wide), then building two mechanical guards that move where the surviving arm dies. The map (E0, `derivation-corridor-map.js`, no LLM calls — D and the death rule are the production `slope.js` paths) enumerates the 125 licensed release schedules on world-002-lantern and finds **40 survive to the forced question, a 32% corridor** at same-turn adoption, collapsing to 19% once adoption lags two turns. Its shape is the finding: survival is *monotone-late*. Per decision the safe set is `p_bearing` $\{t4, t5, t6\}$, `p_chart` $\{t8, t9\}$, `p_key` $\{t15–t18\}$; pulling `p_bearing` one turn early ($t4 \to t3$) costs thirteen points of corridor and pulling `p_chart` two early ($t9 \to t7$) costs ten, while holding `p_bearing` *later* widens it by thirteen — and the $t4 \to t9$ leg carries zero slack, a single knife edge. The conduct miner (E1, `derivation-mine-conduct.js`, over 37 archived real arms / 281 releases) measures the two assumptions the map rests on and the prior it does not see: adoption is same-turn (latency $\lambda = 0$ on 279 of 281 releases), directors never leave cue (0/165 off-nominal) — and **every tutor deviation is early** (20 of 281 releases: nine at -1, eleven at -2, deviation rate 0.17, zero late). The model's real release latitude is *anti-aligned* with the corridor it must thread: the licence is spent on exactly the early side the map marks fatal.

E2 spends one sanction per fan on two $k = 5$ fans of byte-identical replicates of the frozen p4 recipe (same world, script, seed, and seeded decay schedule — only the LLM cast's outputs vary), serialized and human-gated, kill rule rate $\leq 1/5$. The first fan **grounded 3 of 5** (`r2`, `r3`, `r4`, all t20, forced-to-asserted gap 0) and disengaged 2 of 5 (`r1`, `r5`, t12); a second pre-registered fan, pooled to $n = 10$, did **not** reproduce that rate — it grounded only 1 of 5 (`r9` t20; `r6`, `r7`, `r8` disengaged t12, `r10` aporia t8). The **pooled rate is 4 of 10 = 0.40 (Clopper–Pearson 95% lower bound 0.122)**, and the pre-registered convergence test — which required the second fan to ground 2–4 of 5 *and* the pooled lower bound to clear fan-1's 0.15 — records **divergence on both criteria**: the two fans land at 0.60 and 0.20, and the pooled interval did not tighten. The reading is that the frozen recipe does *not* ground reliably; it grounds at roughly four in ten, with an interval still spanning a fifth to three-quarters. *The mechanism, by contrast, reproduced across both fans.* Across all ten arms the grounding/death split is governed by a single conduct choice — where the tutor places `p_chart`, the keystone of the middle chain (the first pass over-generalised this to "holds `p_bearing` *and* `p_chart` to cue," but `r2` and `r3` both ground after pulling `p_bearing` early, so `p_chart` placement alone is the discriminator). The four groundings reached all eight exhibits with D descending to zero; the six deaths plateaued at $D = 3$–5 having reached only five (`r10`, four) exhibits, every one of them having pulled an early exhibit off cue. On the $\lambda = 0$ map the two fans together made **fourteen fatal-cell entries** (eight arms pulled `p_bearing` to t3, six pulled `p_chart` to t7) and **zero late-side moves** — the early-only prior E1 measured, landing in the cells E0 marked. The lone survivor that pulled `p_chart` early (`r3`, to t7) lived because seeded decay dropped D a turn later, letting `p_residue` arrive in time; the other three groundings (`r2`, `r4`, `r9`) played `p_chart` on cue. The harness checks hold across all ten arms (same-turn adoption on every release, zero missed or unscheduled releases, zero superego fires without recorded due), so the deaths are reported outcomes and the no-re-roll rule binds both fans in both directions.

The E2 caveat named one licensed next step — a single mechanism delta that removes the early-pull latitude — and the operator sanctioned it. The delta is `--pacing-guard`: the same solvency arithmetic E3 applies on the late-decay config, here run on the early-decay p4 recipe as a $k = 5$ fan (seed 1, pooled separately from the ten frozen replicates), *shaping* the release schedule rather than overriding it (forced 0, overridden 0 across all five arms, as in E3). It **grounded 4 of 5** (`r2`–`r5`, all t20, gap 0) and disengaged one (`r1`, round 7): 0.80, CP95 $[0.284, 0.995]$, Fisher $p = 0.18$ against the frozen 4/10 — **directional only** under the pre-registered bands, the point estimate rising but the interval overlapping the unguarded rate and the first arm's death putting the 5/5 convergence band ($p = 0.04$) out of reach. Two things it nonetheless fixes in place. The guard did exactly its one job — **zero early**

pulls across all five arms against the unguarded fans' fourteen, the four survivors each playing the canonical safe schedule (`p_bearing` cue t4, `p_chart` cue t9, `p_key` t15 inside the corridor); and pooling guard with frozen arms by `p_chart` placement sharpens the keystone read to **7/7 grounded on cue, 1/6 pulled early, 0/2 unreached** — the discriminator the first write-up had over-stated as holding *both* early exhibits. The lone death relocates the failure rather than reintroducing it: `r1` pulled nothing early, but its on-cue `p_bearing` decayed at t5 and — with no proof-debt guard in this single-delta arm — went unrestored, D climbing back from four to five before the learner left at t7. That is not the early-pull class the guard removed but the *decay-starvation* class E3 meets on late decay and E5 converts to grounding — so the fan reads as the early-decay confirmation that on-cue `p_chart` is what lifts grounding, its one residual death already named and addressed a rung up the ladder. Pooled with the ten frozen replicates and projected onto a coarse, mechanism-keyed failure taxonomy (`derivation-detector-split.js`, dry, no live changes), the fifteen arms read as one contingency — early-pull deaths 6 of 10 unguarded to 0 of 5 guarded, decay-seating 0 to 1 — making explicit that the within-arm observable which moves is *which* death mode the guard leaves standing, not the success rate (4/10 → 4/5, Fisher $p = 0.28$); the early-pull collapse is partly definitional (the guard shapes releases to be tempo-solvent, so by construction it cannot emit a tempo-insolvent one — Fisher $p = 0.044$ on the incidence is read with that caveat), but the relocation to the decay-seating class is not. The paid mechanism loop is ended on this directional read.

The surviving question was no longer whether grounding *can* recur but whether the death class can be *moved* mechanically, without the tutor reading the learner's interior. Two guards, each a single delta on the late-decay (p5) configuration, answer in turn. **E3's solvency gate** (`--pacing-guard`) computes release safety from the same production `derivationDistance`/`detectStall` arithmetic at every tutor release decision; the arm **survived past p5's t12 tempo cliff** to disengagement at t24, the guard *shaping* rather than overriding (24 release decisions audited; zero blocks, zero forced-safe moves, zero overrides) — pacing made arithmetically safe without visibly suppressing conduct. But reaching the decay window relocated the failure rather than removing it: at t17 decay dropped the tutor's own already-released `p_bearing`, nothing restored it, and at t24 the learner named the missing bearing as the gap and the play ended one premise short — *decay* starvation, not tempo starvation. **E5's proof-debt guard** (`--proof-debt-guard`, stacked on E3) closes exactly that gap: when decay drops an already-staged premise whose restoration would lower D, the harness exposes *only that released exhibit's id and surface* to the tutor and authorises an immediate `restore` before new work. The arm **grounded at t20, gap 0** — E3's t24 decay-starved failure converted under the identical decay replay. At t18 the guard detected one debt (`p_bearing`; current D 2, restoring it would reach 1), the tutor restored it (`forcedMoves: 0` — the guard licensed the move, it did not compel it), and at t20 the chain closed on a learner assertion. Pacing stayed stable (no blocks, no overrides), so the new effect is attributable to proof-state repair, not a second pacing change.

The conceptual weight sits in *what the guard does not see*. The proof-debt view is conduct-blind by construction: the tutor receives `{premiseId, surface, sinceTurn}` for the dropped exhibit and nothing else — never the raw learner board, the corruption ledger, the proof path, or the secret. A dry audit (`derivation-leak-audit.js`, over six committed grounded arms) verifies the dynamic half: no proof-distance arithmetic appears in any tutor line, the formal secret predicate is never voiced by the tutor (the tutor stages the recognition; the learner asserts the form), and a positive control confirms the harness's *own* audit ledger does carry the arithmetic the tutor never sees — two demonstrably distinct paths, not one filtered view. (A reporting-layer provenance fix landed with the audit: the proof-in-prose renderer now names the exhibit the run actually staged, read from the release ledger, rather than whichever same-fact twin sorts last in the world file — no verdict or ledger changes, regression test in `tests/dramaticDerivationCritic.test.js`.) The dry detector-split classifier (E4a, `boundaryClassifier.js`, no live changes) sorts every archived arm by the same production arithmetic and corroborates the lineage: the p2/p5/E2-death branch is *tempo-starved*, E3 is *decay-starved lucky-leap*, and p3/p4/E2-survivors/E5 are *grounded controls* — the guards moved the arm one class at a time.

quantity	E2 fans ($k = 10$, two $\times$ 5)	E3 solvency gate	E5 proof-debt guard
single delta	replication of p4, zero delta (early decay)	+ `--pacing-guard` on the p5 late-decay config	+ `--proof-debt-guard` on E3
verdict	4/10 grounded t20 (gap 0) · 6/10 died (5×t12 · 1×t8); 0.40, CP95 \$ \$0.122 · divergence	disengagement t24	grounded t20, gap 0
death class (E4a)	grounded controls + tempo-starved	decay-starved lucky-leap	grounded control
pacing	(ungated) 14 fatal-cell entries, 0 late-side	safe: 0 blocks / 0 overrides, 24 audited	safe: 0 blocks / 0 overrides
proof-state hygiene	—	`p_bearing` dropped t17, never restored	`p_bearing` restored t18 (D 2 → 1) → grounded

Reading. The boundary plan buys three things the $n = 1$ ladder could not. First, a *rate that did not tighten*: grounding on the frozen recipe recurs 4 of 10 across two fans at a wide lower bound (0.122), the pre-registered convergence test recording divergence (fan-1's 0.60 did not replicate) — but the death tail is mechanically explained rather than waved off, and *the explanation reproduced across both fans*: the model's early-pull prior (E1) lands in the corridor's fatal cells (E0), and `p_chart` placement — on cue versus pulled early or never reached — separates grounding from death across all fifteen frozen-plus-guard arms (7/7 ground on cue; 1/8 otherwise). The contribution the fan secures is therefore the mechanism, not the rate. Second, a demonstration that the death class is *movable by mechanical guard*: E3 carries the arm from tempo starvation to decay starvation, E5 from decay starvation to grounded, each a single audited delta. Third — the only part that bears on adaptive tutelage as a claim — a worked example of a guard that *protects the formal proof state at the moment recognition would otherwise become a lucky leap, while staying blind to the learner's concealed interior*: E5's tutor restored a proof-critical premise it was told had dropped, knowing the exhibit's surface and nothing about the board, the secret, or the arithmetic that flagged it, and the non-leak boundary is audited rather than asserted. The conduct-blind / proof-aware separation that §6.13.7 measured as a visibility asymmetry is here made operational — the harness owns the arithmetic, the tutor owns the stagecraft, and the single bit that crosses is the id-and-surface of an exhibit the tutor itself had already played.

Status and caveats (per §5.12.6). $n = 1$ per E-arm except E2, which is $k = 10$ across two pre-registered fans — but those ten share world, script, seed, and seeded decay schedule, so the pooled 4/10 is the model's *run-to-run* reliability at a byte-identical setup, not a sample over worlds, seeds, or recipes; the 0.122 lower bound is a floor on that one quantity and establishes nothing about reliability across worlds, and the two fans' own spread (0.60 then 0.20) is the divergence the pre-registered test recorded. Attribution compounds down the guard stack exactly as in §6.13.9: E3 = p5 + pacing guard, E5 = E3 + proof-debt guard, so E5's grounding attributes to the *stack*, never to the proof-debt guard alone (no guard-alone arm exists, and none is licensed). The verdicts are formal throughout — grounded/aporia/disengagement are the harness's D-arithmetic, not human or mentalistic reads of learning; the guard is conduct-blind by construction and audited to be so, so "the tutor adapted" means it played a harness-licensed restore, not that it inferred anything about the learner. E4a is a dry post-hoc classifier, not a live verdict. The corridor map is one world (world-002-lantern). The four standing §6.13.9 caveats carry forward unchanged: the staged mutation pool's positive side remains twice-unexercised, the D-meter's blindness to elimination progress stands unacted-on (the meter that adjudicated every arm stays frozen), the exposure caveat (same seed, divergent slip schedules once repairs free decay slots) holds across the fan, and the false-restore cost is still measured only once. Spend is meterless throughout (Max-plan CLI; `derivation-cost-report.js` tables ≈ 88 calls / 20 min per grounded recipe arm and a serialized k-fan projection); the paid frozen-replication loop is ended (the two fans recorded divergence, not convergence), and its licensed next step — a single pacing-guard mechanism delta, registered and pooled separately — was run as a $k = 5$ fan that grounded 4/5 (0.80, CP95 [0.284, 0.995], Fisher $p = 0.18$ vs the frozen 4/10): **directional, not converged**, the guard removing the early-pull latitude (0/5 early pulls) but exposing one residual decay-seating death (`r1`, E3's decay-starvation class on early decay). The paid mechanism loop is in turn ended on that directional read; any further arm — a larger guard fan to reach the convergence band, a guard-alone isolation, or a proof-debt-guarded early-decay arm to convert `r1`'s class as E5 converted E3's — remains human-gated under fresh sanction. Implementation `services/dramaticDerivation/` (the pacing and proof-debt guards across engine and runner; the leak audit and detector split as dry readers) and `scripts/derivation-{corridor-map,mine-conduct,cost-report,leak-audit,detector-split}.js`; artifacts under `exports/dramatic-derivation/{loop,boundary,boundary-e2}/`; registrations and outcomes E0–E5 of `ADAPTIVE-TUTOR-BOUNDARY-PLAN.md`. No DB writes, no rubric scoring, no new main-harness estimates — file-based run exports.

6.13.11 Hidden versus Visible Signal: a Page-Only Pacing Guard Grounds 5/5, and the Guard's Lift Is Scheduling Discipline, Not Latent Proof-State §6.13.10's pacing guard reads the production proof-distance arithmetic — `derivationDistance` and the decay-driven stall — *neither of which is printed in the transcript.* That it grounds 4/5 leaves a question the boundary plan could not answer from inside: is the lift bought by the *hidden signal* (latent proof-state the model cannot infer from the page), or by the *scheduling discipline* the guard imposes (hold the next exhibit until the last one has landed), which the model could in principle compute from the page but does not spontaneously enforce? The generalization plan's Step 1 (`ADAPTIVE-TUTOR-GENERALIZATION-PLAN.md`, operator-sanctioned 2026-06-13, registered and pooled separately from both the frozen baseline and the proof-state guard fan) isolates the two with a *form-matched* second guard. `--pacing-guard-visible` (V) is the proof-state guard's twin in every respect — same flag-gating, same "shape the window, never override," same `--release-authority` requirement, same two channels (a static system block and a per-turn note, plus a block/force decision) — differing only in its *information source*: V decides from transcript-surface features alone (turns since the last release, whether the learner has echoed the exhibit currently on the page, the hedging/length trend of the learner's lines). A free import-audit test (`tests/dramaticDerivationVisiblePacing.test.js`, test 1) makes the isolation mechanical: `visiblePacing.js` is forbidden from importing the hidden primitives (`slope.js`, `pacing.js`), so V provably "sees only the page."

The registration's planned confound control — tune V to intervene about as often as the proof-state guard, lest a frequency difference masquerade as a signal-quality difference — turned out degenerate, and degenerate in a way that is itself a finding about the §6.13.10 guard: *the proof-state guard fired zero enforcement interventions over all 87 of its release decisions.* Its entire lift ran through the prompt channel — the always-on solvency block plus the per-turn window note — with the model *voluntarily* releasing on cue. "Match that frequency" therefore reduces to "tune V to a no-op." The control's real purpose was served instead by a free counterfactual replay (`derivation-visible-guard-calibrate.js`, \$0, read-only): V's real decision function, run over the frozen baseline and proof-state transcripts, reconstructs each turn's release context and counts where V *would* have intervened, a built-in self-check confirming the reconstruction is faithful (0 windowSize mismatches over all 235 decisions). V's would-intervene propensity is low — 23 of 148 baseline decisions, 9 of 87 proof-state decisions, ≈ 0.1/decision, and an *upper bound* on live V, since those transcripts carried no V prompt to steer compliance. So V was run as the proof-state guard's true architectural twin, enforcement channel live, with realized enforcement reported as a covariate. The fan **grounded 5/5** (`lantern-e2-visible-r1`–`r5`, all t20, forced-to-asserted gap 0): **Clopper–Pearson 95% lower bound 0.478, Fisher** $p = 0.042$ **against the unguarded 4/10** (V separates *up* from baseline), and **Fisher** $p = 1.000$ **for V < proof-state** (V matches and slightly exceeds the 4/5). V's realized enforcement landed at **7 of 100 decisions (**≈ 0.07**/decision; five blocks, two pushes; per-arm 3, 1, 1, 1, 1)** against the proof-state guard's 0 — low, predominantly one light block per arm and falling on exactly the chain's release turns (t3, t4, t9), the model otherwise complying with the visible note as it had complied with the hidden one.

Pooled with the §6.13.10 arms into one twenty-arm contingency (`derivation-detector-split.js`, dry), the three guard states read:

guard state	grounded	early-pull death	decay-seating death	n
`unguarded`	4	6	0	10

guard state	grounded	early-pull death	decay-seating death	n
`pacing` (proof-state, hidden)	4	0	1	5
`visible` (page-state)	5	0	0	5

The reading is that the §6.13.10 guard's lift is the *scheduling discipline*, not the hidden proof-state. Both guards eliminate the unguarded baseline's dominant death mode — early-pull death, releasing an exhibit before the page is tempo-solvent, 6 of the baseline's 6 deaths — completely, and the page-only guard does it from features the model already has in front of it. What the guard supplies is not information the model lacks but the *discipline to act on information it has*: hold the next exhibit until the learner shows uptake. That discipline is justified by hidden state in one guard and by visible state in the other, and the two ground indistinguishably. This refines — and, for this mechanism, bounds — the §6.13 reading that adaptive gains come from signal the model cannot infer: the pacing lift is *re-representation* of what the page already carries, not new latent knowledge.

Status and caveats (per §5.12.6). The result is the pre-registered "$\geq 4/5 \rightarrow$ hidden-signal story rejected" branch, read at its declared power: $k = 5$, so "V matches the proof-state guard" is the claim, not "V beats it" (the two are statistically indistinguishable, $p = 1.000$), and V separates from baseline only at exactly 5/5 ($p = 0.042$). V's enforcement was low but *not* zero (0.07/decision), so V grounded *predominantly*, not purely, through voluntary compliance — the zero-enforcement arm was the proof-state guard, and the strong-form reading is conditioned on that covariate. The isolation is signal-source-clean (the import audit is mechanical), but the *sufficiency* of the visible proxy is world-specific: in world-002-lantern's chain "the learner has not echoed the prior exhibit" tracks "proof-distance still high," so the page carries enough to pace well; a world that decoupled visible uptake from latent distance need not let V track the proof-state guard. Scope otherwise as §6.13.10 — one world, the frozen p4 stack, the fan replicating LLM-conduct stochasticity at a byte-identical setup (seed 1), not a sample over worlds or recipes; verdicts formal throughout (the page-only guard is conduct-derived but still mechanical, so "the tutor paced" means it played a harness-shaped release, not that it read the learner's interior). Each arm is registered for the standing pinned-Fable critic's notice (`commentary.md`); that backfill is deferred while Fable 5 is unavailable. Spend meterless (Max-plan CLI), serialized at first and — once the operator confirmed no quota constraint — fanned in parallel; the paid Step-1 loop is ended on this result, and Step 2 (a branching-DAG world that tests whether the visible proxy still suffices when the chain forks) is human-gated under its own sanction. Implementation `services/dramaticDerivation/visiblePacing.js` plus the `visibleGuard` branches of `llmRoles.js`; dry reader `derivation-visible-guard-calibrate.js`; artifacts `exports/dramatic-derivation/loop/lantern-e2-visible-r{1..5}/` and the contingency `exports/dramatic-derivation/boundary-v-fan/detector-split-report.{md,json}`; registration and outcome in `ADAPTIVE-TUTOR-GENERALIZATION-PLAN.md`. No DB writes, no rubric scoring — file-based run exports.

6.13.12 The Boundary of the Proxy: on a Forked AND-Join World the Page-Only Guard Fails 0/5 Where the Hidden Guard Grounds 5/5

The result in plain terms.

A pacing guard earns its keep by telling the tutor when to hold the next clue back until the learner is ready for it. §6.13.11 found that a guard watching only the visible dialogue paced as well as one able to see the learner's true distance from the answer — so the help looked like *scheduling discipline any reader could take off the page*, not access to a hidden state. That held, though, only because on a straight-line problem a learner who *looks* engaged genuinely *is* close to the answer. This experiment breaks that coincidence. The new world needs two separate strands of reasoning to both finish and then meet at the end — like a two-part exercise where one must first set up an equation and then solve it. A learner can complete the first strand and show every surface sign of progress (recent uptake, confident replies) while the second strand is untouched and the answer is still far off. The page-only guard reads that surface, judges the learner on track, and lets the pace run out; the depth-aware guard sees the second strand still open and holds. So the page-only guard is fooled and fails every run (0/5) where the depth-aware guard succeeds every run (5/5) — and it fails not by doing too little but by intervening *more* than the guard that succeeds (16 overrides to nil), on the wrong signal. The lesson is a boundary, not a reversal: pacing from the visible page is enough exactly when the page faithfully reflects real progress; when a task lets appearance and progress come apart, the tutor needs the signal it cannot read off the surface. The rest of this section makes that precise.

§6.13.11 grounded its strong reading — that the pacing lift is *re-representation* of what the page already carries, not latent knowledge — on a single chain's coincidence: in world-002-lantern's linear derivation, "the learner has not echoed the on-page exhibit" tracks "proof-distance is still high," so a page-only guard paces as well as one reading the hidden arithmetic. That paragraph named its own boundary: *a world that decoupled visible uptake from latent distance need not let the visible guard track the proof-state guard.* The generalization plan's Step 2 (`ADAPTIVE-TUTOR-GENERALIZATION-PLAN.md`, operator-sanctioned 2026-06-13, registered and pooled separately from every lantern fan and from each other) builds exactly that world and re-runs the contrast. `config/drama-derivation/world-005-marrick.yaml` replaces the linear chain with an **AND-join**: the secret $S = \mathit{struckBy}(\mathit{coin}, x)$ is entailed only when *two independent depth-2 sub-chains* — α (the cast blank, $\mathit{castBlankFor}(\mathit{coin}, x)$) and β (the cut die, $\mathit{cutDieFor}(\mathit{coin}, x)$) — both close on the *same* particular x, α resolving by t10 and β only at the join on t22. The geometry decouples the page from the proof: a learner fully seated on α looks seated to a surface reader (recent echo, low hedging, exhibits landing) while global proof-distance D stays high because β is untouched. On the linear chain the visible and hidden guards could not come apart; world-005 is engineered so they can.

Three fans of $k = 5$ ran the byte-identical p4 conduct stack (seed 1, learner pinned claude/sonnet, `--critic off`) and differ by exactly one flag: **baseline** (`marrick-real-r1`–`r5`, no guard), **H** (`marrick-guard-r1`–`r5`, `--pacing-guard`, the hidden depth-aware guard reading D and the decay stall), and **V** (`marrick-visible-r1`–`r5`, `--pacing-guard-visible`, the §6.13.11 page-only twin, its `VISIBLE_GUARD_DEFAULTS` carried over from lantern *unchanged* — re-used, not re-tuned to fail). All fifteen arms completed first-pass, no re-rolls. The fans separate cleanly: **baseline grounded 0/5, V grounded 0/5, H grounded 5/5** (all five at t22, forced-to-asserted gap 0). Per-fan Clopper–Pearson 95%: H 5/5 [**0.478, 1.000**], baseline and V 0/5 [**0.000, 0.522**]. Pairwise Fisher exact: **H vs V** $p = 0.0079$ (one-tailed 0.0040), **H vs baseline** $p = 0.0079$, **V vs baseline** $p = 1.000$ — on the forked world the page-only guard is statistically indistinguishable from no guard at all, and the hidden guard separates from both.

The pre-registered primary read — the guard × failure-mode contingency (`derivation-detector-split.js`, dry, read from each arm's own recorded guard flags, world-agnostically) — shows the relocation is sharper than "V is a no-op":

guard state	grounded	early-pull (tempo-starved) death	decay-seating death	n
`unguarded` (baseline)	0	2	3	5
`visible` (page-only, V)	0	5	0	5
`pacing` (hidden, H)	5	0	0	5

V is not literally inert: it *regularizes the failure path.* The unguarded baseline dies two ways — two arms tempo-starved at t21 (disengagement) and three in the decay-seating class (two aporia at t8/t12, one disengagement at t21) — whereas V pushes all five arms into the single tempo-starved-house mode, disengaging uniformly at **t21, one turn before the join fires at t22**. V has enough surface signal to suppress the early decay deaths H also suppresses, but at the cusp it has no way to see that β is still open behind a learner who looks seated on α, so it lets the learner leave one move short. H, reading D, holds through the join and grounds every arm.

The reading is the boundary §6.13.11 left open. The visible proxy's sufficiency is **geometry-conditional**: where the proof chain is linear and visible uptake tracks latent distance, the page carries enough to pace and V grounds 5/5 = H (§6.13.11); where the chain *forks* and uptake decouples from distance, the page no longer carries enough, and the hidden depth signal becomes necessary (V $0/5 \ll$ H $5/5$). This does not overturn §6.13.11's "the lift is scheduling discipline" — it bounds it. Scheduling discipline *from the page* suffices only when the page is a faithful proxy for proof-distance; an AND-join is the standing counterexample where it is not. The §6.13 throughline — that the adaptive gain is whatever signal the conduct can act on — survives intact, now with both sides of its boundary measured on one instrument: re-representation is enough on a coupled geometry, and latent state is required on a decoupled one.

Status and caveats (per §5.12.6). This is the registered "primary = failure-mode shift" read at its declared power. The three fans are $k = 5$ each, so the H-vs-V and H-vs-baseline separations sit at exactly the 5/5-vs-0/5 corner where Fisher first clears 0.05 ($p = 0.0079$); the claim is "V fails where H grounds on this world," not a rate estimate over worlds. The boundary is established at **one** forked geometry (world-005-marrick, one seed, the frozen p4 stack): the result shows the page proxy's sufficiency is *not universal* — it fails on at least one structurally-different world — not that it fails on *every* forked world. Verdicts are formal throughout (off the production proof state via `diagnose.js`, architecture-independent of the guard that produced them), so "V failed" means the drama reached a harness-scored disengagement, not a claim about the learner's interior. The visible guard is the §6.13.11 twin unchanged — the import audit (`visiblePacing.js` may not import `slope.js`/`pacing.js`) still holds, so V provably saw only the page; what changed is the world, not the proxy. The realized-enforcement covariate (the conditioning the registration tabled, here read live off each arm's `releaseDeviations` rather than by counterfactual replay) sharpens this and mirrors §6.13.11: **the hidden guard grounded 5/5 while overriding zero release decisions** (forced 0, overridden 0 across the fan) — pacing through voluntary compliance with its prompt, exactly as on lantern — whereas **the visible guard overrode 16 decisions** (and held seven releases late) and still grounded 0/5. On the forked world the page-only guard does not merely under-fire; it intervenes *harder* than the guard that succeeds and fails anyway, because the surface feature it acts on is the wrong one. No re-rolls were used (15/15 first-pass clean); spend meterless (Max-plan CLI), the three fans fanned in parallel under the operator's no-quota-constraint election, pooled separately. Each arm is registered for the standing pinned-Fable critic's notice (`commentary.md`); that backfill is deferred while Fable 5 is unavailable. The paid Step-2 loop is ended on this result. Implementation reuses `services/dramaticDerivation/{pacing,visiblePacing}.js` and the `world-005-marrick` YAML + `marrick-v001` script unchanged from their registration; primary instrument `derivation-detector-split.js`; artifacts `exports/dramatic-derivation/loop/marrick-{real,guard,visib` and the contingency `exports/dramatic-derivation/boundary-marrick/detector-split-report.{md,json};` registration and outcome in `ADAPTIVE-TUTOR-GENERALIZATION-PLAN.md`. No DB writes, no rubric scoring — file-based run exports.

6.13.13 Decomposing the Boundary: on a Linear-Spine Distractor World the Two Guards Swap — the Page-Only Guard Grounds 5/5 and the Hidden Guard Fails 1/5 *The result in plain terms.* §6.13.12 found that on a forked task the depth-aware guard succeeds where the page-only guard fails, and concluded that a visible adaptivity proxy is trustworthy only where the task's geometry lets the page reflect real progress. But that forked world changed *two* things at once versus the linear chain — the **geometry** (two strands that must meet) and the **decoupling** of "looks engaged" from "is close to the answer." Either could be what defeats the page-only guard. This experiment separates them. The new world keeps the true reasoning a straight line — the shape on which the two guards already paced equally well (§6.13.11) — and adds only a **decoy**: a fully derivable side-finding the quick learner can prove in full and rest on, true but answering the wrong question (it establishes who is *liable* for the fall, not whose *hand* felled the span). The decoy supplies the decoupling without the fork. The result is a near-reverse of the forked world: the **page-only guard now grounds every run (5/5)** and the **depth-aware guard fails four of five (1/5)** — it lets the learner drift onto the decoy and the board goes cold. So the two guards are not ordered, one strictly safer than the other; they are **complementary**. Each reads the one signal its world's pacing turns on, and fails the world whose signal lives in the other's channel. The forked world's lesson and this one's are two sides of a single boundary: neither guard is safe across geometries.

§6.13.12 established that on world-005-marrick's AND-join the page-only guard (V) fails 0/5 where the depth-aware guard (H) grounds 5/5, and read this as the page proxy's sufficiency being *geometry-conditional*. But marrick confounds two changes against the linear lantern world: it introduces a **fork** *and* it **decouples** surface-confidence from latent proof-distance D. The conservative reading marrick licenses is "the proxy's sufficiency is fork-conditional"; a stronger reading — "it is *decoupling*-conditional, however the decoupling is produced" — is not separable on marrick alone. The generalization plan's Step 3 (`ADAPTIVE-TUTOR-GENERALIZATION-PLAN.md`, operator-sanctioned 2026-06-13, registered and pooled separately from every lantern and marrick fan) builds a world that supplies the decoupling **without** a fork. `config/drama-derivation/world-006-hethel.yaml` keeps the secret $S = \mathit{felledBy}(\mathit{span}, \mathit{oswin})$ on a single linear five-rule spine — the coupled shape where §6.13.11 found V = H — and adds a **derivable distractor**: two extra rules let the learner ground $\mathit{liableFor}(\mathit{span}, \mathit{reyner})$ in full from the master-of-record, the yard, and the bond. That finding is *true* and *complete* but answers a dead predicate; none of its conjuncts feed *felledBy*. A quick learner can seat the decoy, show every surface sign of a closed case, and be no nearer the secret — decoupling without geometry.

Three fans of $k = 5$ ran the byte-identical p4 conduct stack (seed 1, learner pinned claude/sonnet, `--critic off`) differing by exactly one flag: **baseline** (`hethel-real-r1`–`r5`, no guard), **H** (`hethel-guard-r1`–`r5`, `--pacing-guard`, hidden, reads D), **V** (`hethel-visible-r1`–`r5`, `--pacing-guard-visible`, the §6.13.11/.12 page-only twin, `VISIBLE_GUARD_DEFAULTS` carried over *unchanged*). All fifteen completed first-pass, no re-rolls. The fans separate in **near-reverse of marrick**: **baseline grounded 4/5, V grounded 5/5, H grounded 1/5** (every grounded arm at t20, forced-to-asserted gap 0). The single baseline death is an early-pull aporia at t11; H's four deaths are all the decoy-seated disengagement class — three at t7, one at t19 — the exact failure the distractor was built to bait. Per-fan Clopper–Pearson 95%: V 5/5 [**0.478, 1.000**], baseline 4/5 [**0.284, 0.995**], H 1/5 [**0.005, 0.716**]. Pairwise Fisher exact, one-tailed: **V vs H** $p = 0.0238$ (the only contrast clearing the $k = 5$ floor); **baseline vs H** $p = 0.1032$ (n.s. — H-below-baseline is *directional only*, the underpowered-rate caveat the registration imposed in advance); **V vs baseline** $p = 0.5000$ (n.s.).

The pre-registered primary read — the guard × failure-mode contingency (`derivation-detector-split.js`, dry, read world-agnostically from each arm's own guard flags) — shows the shift runs *against* the hidden guard:

guard state	grounded	early-pull (tempo-starved) death	decay-seating death	n
`unguarded` (baseline)	4	1	0	5
`pacing` (hidden, H)	1	0	4	5
`visible` (page-only, V)	5	0	0	5

The decoy-seated decay death is **4/5 under H and 0/5 in both baseline and V**: the hidden guard does not remove the world's death mode, it *manufactures* the one the distractor was built to bait, and only that one. Against the pre-registered interpretation rule this is the **primary read as a clean negative** — H does not lift above baseline ($1/5 < 4/5$), so the §6.13.10/.11/.12 scheduling-discipline mechanism, *as carried by the hidden guard*, does **not** travel to this distractor world; reported as the limit, not papered over. On the secondary V-vs-H contrast the result lands the rule's pre-specified **"V lifts, H flat → anomaly (visible helps where true-state does not); report as such, no mechanism claim"** quadrant (H is in fact *below* baseline, slightly stronger than the registered "flat").

The realized-enforcement covariate (read live off each arm's `releaseDeviations`) sharpens the anomaly and **inverts marrick's**. On marrick the hidden guard grounded 5/5 overriding *zero* decisions while V overrode 16 and failed; here that same passive hidden guard — **H grounded 1/5 overriding zero decisions** (forced 0, overridden 0, held 0), the *identical* 0/0/0 discipline it ran on marrick — fails, because the world's binding constraint moved into the channel it cannot read. (H played only 9 of 15 possible tutor releases not by suppressing any, but because four arms died before the spine could open.) Meanwhile **V grounded 5/5 overriding 11 decisions** (early 10, held 5), and in **all five arms pulled the second spine exhibit to t7** — *outside* the hidden guard's own solvency-licensed window $[8, 9]$ — bridging the knife-edge t4→t9 gap and catching the quick learner one turn before they seat the decoy and disengage. The reason H cannot do this is structural: the distractor's tempo-keeping releases are *director-via*, outside the guard's *tutor-via* candidate set, so H is blind to the dramatic function the decoy performs, and its solvency-correct refusal to pull the exhibit early is the dramatically-fatal move. *(This early-pull-bridges-the-gap account is a **post-hoc** reading of the covariate and corridor arithmetic, offered as the mechanism behind the anomaly, not a pre-registered claim — per the rule, no mechanism is registered for "V beats H here.")* The lesson is that intervention frequency does not predict success across worlds: V over-intervenes and fails on the fork, over-intervenes and succeeds on the distractor; *which channel carries the binding constraint* does the work.

The reading **decomposes the marrick confound**. Because the distractor supplies decoupling on a linear spine, the result reads each of marrick's two changes separately. V **holds** here (5/5), so marrick's V-failure was **fork-specific (geometry)**, not decoupling-general — the §6.13.12 boundary is the *narrower* reading, and the page proxy is more robust than marrick alone implied (a derivable decoy on a straight chain does not defeat it). And — unregistered, the anomaly — H **fails** here (1/5) where it grounded 5/5 on the fork, so marrick's H-*success* was **also geometry-bound**, not a general latent-depth advantage. The two guards are complementary, and §6.13.11/.12 are now bounded on *both* sides: the page proxy's insufficiency is fork-specific, and the hidden guard's sufficiency is fork-specific too. The §6.13 throughline — that the adaptive gain is whatever signal the conduct can act on — is unchanged and now sharper: there is no geometry-universal guard among these two, because the signal a world's pacing turns on is itself geometry-dependent.

Status and caveats (per §5.12.6). The anomaly is reported *as* an anomaly: no mechanism claim is registered for "V beats H on the distractor"; the bridging account above is post-hoc. Rate contrasts are at declared $k = 5$ power — only V-vs-H clears the Fisher floor ($p = 0.0238$); H-below-baseline is directional ($p = 0.1032$, n.s.); the durable observable is the failure-mode contingency, not the rate. The decomposition is established at **one** distractor geometry (world-006-hethel, one seed, the frozen p4 stack): V's robustness to a decoy is shown *on a linear spine*, not on every non-fork shape; H's distractor-failure is shown *on this decoy*, not on every distractor. Verdicts are formal throughout (off `diagnose.js`, architecture-independent of the guard). The visible guard is the §6.13.11/.12 twin unchanged — the import audit (`visiblePacing.js` $\not\to$ `slope.js/pacing.js`) still holds, so V provably saw only the page; the **world** changed, not the proxy. The world's baseline is non-degenerate (4/5, not 0/5) — fixed at corridor calibration and frozen before the paid run, a reported property of the world, not re-tuned around the result. No re-rolls were used (15/15 first-pass); spend meterless (Max-plan CLI), the three fans fanned in parallel under the operator's no-quota-constraint election, pooled separately from each other, from the lantern fans, and from the marrick fans. Each arm is registered for the standing pinned-Fable critic's notice (`commentary.md`); that backfill is deferred while Fable 5 is unavailable. The paid Step-3 loop is ended on this result; per the kill rule we stop at one distractor world. Implementation reuses `services/dramaticDerivation/{pacing,visiblePacing}.js` unchanged; world + script `world-006-hethel.yaml` + `hethel-v001.md`; primary instrument `derivation-detector-split.js`; artifacts `exports/dramatic-derivation/loop/hethel-{real,guard,visibl` and the contingency `exports/dramatic-derivation/boundary-hethel/detector-split-report.{md,json}`; registration and outcome in `ADAPTIVE-TUTOR-GENERALIZATION-PLAN.md`. No DB writes, no rubric scoring — file-based run exports.

6.13.14 Assembling the Boundary: Three Geometries, Two Channels, and No Universal Single-Channel Guard

The result in plain terms. Across three worlds the two pacing guards trade places. On a straight-line problem both help equally; on a problem with two strands that must meet, only the depth-aware guard helps; on a straight-line problem with a tempting dead-end, only the page-watching guard helps. Read together they say something sharper than any one alone: neither guard is the safe default. A pacing guard helps exactly when the thing it watches is the thing about to go wrong — and which channel that is depends on the shape of the problem, not on the guard.

The three fans (§6.13.11–.13), each $k = 5$ on the byte-identical p4 conduct stack, one flag apart, assemble into a single picture:

world (geometry)	baseline	V (page-only)	H (hidden, reads D)
world-002-lantern (linear, coupled)	4/10	**5/5**	**4/5**
world-005-marrick (forked AND-join)	0/5	0/5	**5/5**
world-006-hethel (linear + derivable decoy)	**4/5**	**5/5**	1/5

(The lantern baseline is the pooled $n = 10$ E2 replication of §6.13.10; every guard fan and the two later baselines are $k = 5$. All grounding figures are carried unchanged from §6.13.11–.13; this subsection introduces no new run.)

The pattern is not a ranking. V and H each ground one geometry the other fails, and they agree only on the coupled chain where the two channels carry the same information. The unifying statement is a mapping from a world's geometry to *which channel carries its binding pacing constraint*:

- On the **coupled linear chain**, the page tracks proof-distance — "the learner has not echoed the last exhibit" means "still far from S" — so both channels carry the constraint and both guards pace (V = H, §6.13.11).
- On the **forked AND-join**, a learner seated on one strand looks finished while the second keeps proof-distance high; the binding constraint lives in *latent depth the page cannot show*, so only the depth-aware guard holds through the join (H 5/5, V 0/5, §6.13.12).
- On the **linear distractor**, the binding constraint is a *page-legible tempo* one — supply the next real step before the learner drifts onto the derivable decoy — and the saving move (pull the spine exhibit early, off the proof-state's licensed window) is exactly what the depth-aware guard's solvency discipline forbids; so only the page-watching guard, free to act on visible drift, grounds (V 5/5, H 1/5, §6.13.13).

Two consequences follow. First, **§6.13.11 and §6.13.12 are each bounded on both sides.** The page proxy's *insufficiency* (marrick) is fork-specific — a decoy on a straight chain does not defeat it (hethel V 5/5) — and the hidden guard's *sufficiency* (marrick) is fork-specific too — the same passive guard fails on the decoy (hethel H 1/5). Neither "the page is enough" nor "you need latent state" is the general truth; each is the truth for a class of geometries. Second, this is the constructive form of the section's null. The throughline — that what helps is whatever signal the conduct can *act on*, not richer inference about the learner — survives the boundary intact and is now sharper: among these two guards there is no geometry-universal *single-channel* one, because the channel a world's pacing turns on is itself a property of the world's structure. A guard reading both channels, or one *derived* from a world's proof-DAG rather than fixed in advance, is the natural next question; it is untested here and no claim is made for it.

This does not widen the section's headline. The successful adaptive conduct demonstrated in §6.13 stays *limited* in exactly the senses §6.13.9–.13 record — single-stack attribution, formal rather than mentalistic verdicts, criterial enforcement of a release schedule rather than read interiors, and now explicitly geometry-bound mechanism. What the trilogy adds is not a stronger adaptivity claim but a *map of where each mechanism's help is real.* For a section whose larger finding is a null on inferential adaptivity, that map — the boundary drawn on both sides, on one instrument — is the more useful object.

6.13.15 Selector Trajectory v0–v4 and A20/A21 Closeout: the Selector Fails the Adaptive Claim, and ProofDebt Becomes the Reliability Baseline The three-geometry synthesis leaves a practical design question open. If H and V are complementary, can a selector inspect the authored proof geometry and choose the right representation *before* outcomes are known? This matters because a selector that merely picks H most of the time may look successful without doing adaptive representation selection. The reliability probe therefore treated the selector itself as the object under test, not as a device allowed to inherit the best interpretation after the fact. The current selector (`dramatic-derivation.representation-selector.v0`) was frozen. Misroutes counted as evidence. No world was silently reclassified after outcomes. The comparison included the selector against no guard, always-H, always-V, and the oracle best static arm per world.

The first four-world stress matrix used Bitterwell, Withercombe, Fengate, and Sealhouse under the same decay + release-authority stack as the §6.13 reliability work, 5 first-pass runs per world per arm. Its result is decisive against the v0 selector as an adaptive claim:

world	no guard	H	V	selector v0 route	selector v0
Bitterwell	5/5	5/5	4/5	H	5/5
Withercombe	4/5	4/5	2/5	V	3/5
Fengate	0/5	5/5	1/5	H	5/5
Sealhouse	1/5	5/5	3/5	H	5/5
Total	**10/20**	**19/20**	**10/20**	—	**18/20**

The H/V distinction is real in this matrix: H rescues Fengate and Sealhouse, while V is weak on both. But v0 does not beat the simple always-H policy. It grounds 18/20, always-H grounds 19/20, and the oracle best static policy also grounds 19/20. Withercombe is the route failure that exposes the problem. V0 reads its non-fork geometry as visible-positive and selects V; under decay the result is negative transfer (selector 3/5, H 4/5, no guard 4/5, static V 2/5). The correct interpretation is therefore not "the adaptive selector works." It is: **the channel distinction is real, but the one-bit selector is underpowered and the first taxonomy was premature.**

A second selector (`dramatic-derivation.representation-selector.v1`) was implemented separately under `--pacing-guard-selective-v1`, leaving v0's results untouched. Its policy is deliberately conservative: choose H for independent top-level joins; choose V only for a formal mirror-dead-predicate decoy; under decay, fail closed to H when no V-positive evidence is present. This repairs the Withercombe route by treating decay as a reason to preserve proof continuity, but it does not by itself establish adaptive selection. On the same original four worlds, v1 grounds 18/20, always-H grounds 19/20, always-V grounds 10/20, no guard grounds 10/20, and oracle grounds 19/20:

world	v1 route	selector v1	no guard	always-H	always-V	oracle static
Bitterwell	H (`independent_join_hidden`)	5/5	5/5	5/5	4/5	5/5
Withercombe	H (`decay_fail_closed_hidden`)	4/5	4/5	4/5	2/5	4/5

world	v1 route	selector v1	no guard	always-H	always-V	oracle static
Fengate	H (`independent_join_hidden`)	5/5	0/5	5/5	1/5	5/5
Sealhouse	H (`independent_join_hidden`)	4/5	1/5	5/5	3/5	5/5
Total	—	**18/20**	**10/20**	**19/20**	**10/20**	**19/20**

This is a useful correction, but not a selector victory. V1 fixes the Withercombe misroute; the remaining regret comes from one fresh Sealhouse hidden-route failure, classified as a guard/world-instability failure rather than a route error. The main lesson is that **hidden is the safe default under this decay stress unless the world gives explicit V-positive evidence**. That is already a policy, but it is not yet a demonstration that the selector has learned the representation boundary.

The held-out V-positive probe is Hethel under the same selector-v1 stress stack, not the earlier §6.13.13 fan. Here v1 finds the intended visible contour: a mirror dead-predicate decoy where the page can see drift toward a complete but wrong finding. On this stack Hethel yields no guard 2/5, H 3/5, static V 4/5, and v1 5/5. Adding Hethel to the four-world matrix gives selector v1 23/25, always-H 22/25, always-V 14/25, no guard 12/25, and oracle 23/25. This is the first useful V-positive signal for a pre-declared selector: the v1 gate selected V and improved over always-H on the added world. It still remains one held-out world, one seed, and the same synthetic derivation stack. It licenses a next selector policy, not a paper-level claim that adaptive representation selection is solved.

The next question was whether this v1 signal could be made into a reliable selector rather than another fitted route story. It could not. V2 changed the policy surface from hand-authored H/V geometry to a consolidated proof-board diagnostic: choose hidden when decay plus shared proof-critical source pressure can reopen more than one dependency, and choose visible only when no proof-continuity override defeats a mirror/dead-predicate route. Its artifact-only re-analysis over Withercombe, Fengate, and Hethel found that **21 of 23 non-grounded runs** were better explained by unclosed or decay-opened proof-graph gaps than by a separate visible/hidden representation story; in fresh Codex-learner repeats Hethel reversed its v1 signal (hidden 5/5, visible 2/5, selective-v1 3/5), so v2 routed Withercombe, Fengate, and Hethel all to hidden. That is a useful correction, but it is not adaptive H/V selection. It says the current policy should fail closed to proof continuity under decay.

A broader consolidation pass made the same point at the accumulated-artifact level. It loaded 172 loop artifacts into 57 comparison groups and found only **one strict V-positive** group, against **three visible route failures**, **eleven false positives**, forty-one not-visible-route cases, and one inconclusive case. The one strict V-positive datum is still Hethel r2; the repeated negative-transfer cases remain Withercombe and later Hethel/Codex-learner repeats. Episode replay then helped localize failure turns and verify prefix integrity, but did not convert the accumulated record into a stable visible-positive class. Its value was methodological: it can save paid computation and isolate route failures, not supply post-hoc relabeling authority.

V4 tested the strongest remaining rescue: keep hidden as the default, add visible shadow consolidation/assertion gates, and evaluate the policy with and without proofDebt. This made the active ingredient explicit. In a six-world confidence matrix (Withercombe, Fengate, Hethel, Ravensmark, Lantern, Marrick), **static hidden+proofDebt** reached **91.7% world-mean success** and matched the oracle static H/V policy; **v4+proofDebt** reached **75%**, with negative transfer on Lantern and Marrick; **visible+proofDebt** reached **33.3%**; and **v4 without proofDebt** collapsed to **16.7%**. Withercombe and Ravensmark remained stable under a Sonnet learner and decay-seed perturbation, so the most conservative reading is not world randomness. ProofDebt is doing most of the reliability work; the extra visible gates are brittle.

The immediate A20 follow-on tested one entitlement-side refinement rather than another H/V selector: `--same-turn-assertion-affordance`, which lets a learner recheck the expanded board and assert S on the same turn that a final exhibit is adopted. Prefix-controlled episode replay was useful here, but only as a debugging tool: replaying the six hidden+proofDebt baselines with the successful prefixes frozen produced 6/6 grounded suffixes with gap 0, showing that the suffix mechanism itself did not break the recognition turn. A fresh six-world matrix initially archived 5/6 grounded because Hethel stopped at turn 11; the audit then found this was not clean policy negative transfer. The Hethel tail had $D = 4, 4, 4, 3, 3, 3$ over turns 6–11 — a real D decrease inside the six-turn window — while net `groundedCount` stayed flat under decay/repair churn. The old stall detector checked grounded-count growth before D decrease and therefore labeled a still-progressing run as disengagement two turns before the scheduled `p_mark` release. The detector has been corrected to treat a D decrease as progress under decay, and the archived Hethel verdict is left as produced rather than silently reclassified. The substantive A20 result is therefore negative in a narrower sense: same-turn assertion repair is useful for suffix diagnosis, but the current hidden+proofDebt baseline already had forced/asserted gap 0 on the matched labels, so the refinement does not yet improve the reliability policy.

The conduct-policy promotion line then supplied the sharper stop signal. Phase 5g compared hidden+proofDebt against promoted selector-v4+proofDebt on Hethel, Withercombe, and Ravensmark. Hidden+proofDebt grounded all three first-pass worlds; promoted selector-v4 also chose hidden in all three, grounded Withercombe and Ravensmark, but failed Hethel by disengagement at turn 11 with final $D = 4$. Conduct enforcement passed its local compliance checks, so the failure is not a generator-compliance failure. It is policy-level negative transfer: repeated `visible_hidden_conflict` diagnostics and visible-consolidation holds delayed proof progress (`p_point` t4→t6, `p_surface` t9→t11). Phase 5h de-promoted selector-v4's default conduct enforcement and added a strict diagnostic budget. Prefix-preserving Hethel replay improved the local mechanics — diagnostics fell from four to one, `p_point` advanced to t5, and `p_surface` advanced to t10 — but still failed with final $D = 4$ (aporia at t10). No fresh paid retest was launched because the replay gate failed. This closes promoted selector-v4/conduct enforcement as a general derivation arm; further work must be separately labelled and must target progress/release pressure rather than finer visible/hidden categories. The first Phase 6 progress-pressure replay is therefore reported as a partial repair only: with `--conduct-progress-policy`, the same Hethel prefix remained prefix-identical, avoided aporia/disengagement through the t4–t15 window, and reached $D = 1$, but it did not ground and introduced one early release (`p_brand` t15 vs planned t17). That result kept progress pressure alive as a design lead, but still blocked a fresh paid retest.

A21 then reframed the same failure as an action-value question rather than a route/classification question. The useful Hethel move was not "diagnose the visible/hidden conflict again" but release the safe public evidence that closes the live proof gap. As an analysis object this is valuable: it explains why the promoted conduct policy failed, and why progress-starving diagnostics are dangerous under decay. But the Phase 9 replay also showed that the current hidden+proofDebt baseline already makes the decisive high-value release (`p_point` at t4) in the matched Hethel prefix. The A21 patch therefore matches hidden+proofDebt instead of beating it. It is retained as a diagnostic microbench for future pre-declared hidden+proofDebt failures, not promoted as a default policy.

The remaining objection was that proof-control metrics might be too narrow: perhaps the proof-safe overlays were improving didactic quality or learner ownership even when D, release timing, and verdicts did not move. The follow-up therefore built the missing evaluator rather than another controller. `ObjectOwnershipState` scores whether the public learner can restate the object in their own words, use it in the current proof path, discriminate it from a nearby wrong route, transfer it to a near-isomorphic case, recover it after a break, and explain why it matters; it is public-only and rejects raw proof paths, hidden boards, D arithmetic, secrets, and corruption ledgers. A didactic opportunity-cost budget was also added so explanatory-mode changes remain advisory and cannot starve proof progress. The first zero-paid artifact pass found no qualifying proof-safe ownership gain:

pair	proof/release status	ownership delta	decision
Phase 9 Hethel A21 S0/S1	matched reliability	+0.38	below ownership-gain gate
Didactic Hethel S0/S1	matched reliability	+0.00	no ownership gain
Phase 5g Withercombe hidden vs promoted	not matched	+0.67	disqualified by release-signature mismatch
Phase 5g Ravensmark hidden vs promoted	not matched	+0.60	disqualified by release-signature mismatch
Phase 5g Hethel hidden vs promoted	not matched	+0.42	disqualified by proof failure

That result left one legitimate concern: maybe the new evaluator itself was too crude. The final gate therefore made a proof-matched benchmark before any further artifact mining.

Twelve synthetic controls were declared: four positives where proof state and release signature are fixed but S1 plainly owns the object better; four negatives where prose is warmer or more fluent but ownership is unchanged; and four disqualification controls where apparent ownership movement is confounded by changed proof or release state. The benchmark passed **12/12**: the positives were detected, prose-only changes stayed null, and proof/release-confounded gains were rejected. Re-running the mined-artifact pass after that calibration reproduced the same negative table above. The closeout is therefore stronger than "we lack an ownership evaluator." The evaluator now exists and passes the local controls; the current artifact pool still contains no qualifying proof-safe ownership delta.

The selector trajectory is therefore:

generation	strongest result	failure mode	interpretation
v0	Avoids visible failure on obvious H-positive branch worlds.	Withercombe routed visible and underperformed H/no-guard under decay.	Falsified as adaptive selector.
v1	Five-world probe with Hethel: selector 23/25 vs always-H 22/25.	Original four-world matrix still loses to always-H, 18/20 vs 19/20.	First useful V-positive signal, not a selector victory.
v2	Consolidated-board rule explains many failures as proof gaps.	Routes hidden everywhere on current stress worlds.	Diagnostic correction: fail closed to proof continuity.
v3	Episode replay and failure localization work.	No stable visible-positive class emerges.	Better iteration apparatus, not a better selector.
v4	With proofDebt, strong on original worlds and stable under two perturbations.	Negative transfer on Lantern/Marrick; without proofDebt it collapses; promoted conduct then fails Hethel first-pass.	ProofDebt is the active ingredient; visible gates and conduct enforcement are not general defaults.
A20/A21 ownership closeout	Action-value and ownership instruments explain failures and pass local evaluator controls.	No proof-safe ownership gain in mined artifacts; A21 patch matches hidden+proofDebt rather than beating it.	Closed as valid negative for this artifact pool.

The resulting policy boundary is sharper than §6.13.14's two-channel map alone. H and V are not merely complementary in retrospect; a selector must earn its keep against two simple baselines. First, it must beat always-H or hidden+proofDebt on worlds where H is generally safer under decay. Second, it must avoid negative transfer against no guard and hidden+proofDebt when a visible route is attractive but brittle. V0 fails because of

Withercombe. V1 passes one Hethel contour but does not repeat strongly enough. V2 collapses to hidden. V4 shows that adding visible consolidation/assertion gates on top of hidden can hurt, and the Phase 5g/5h conduct-policy test shows that local compliance gains can still starve proof progress. The standing interpretation is therefore more conservative than v3.0.156: **hidden+proofDebt is the current reliability baseline; adaptive H/V representation selection is not established; and a v5 selector is not worth running unless it first names a held-out world where hidden+proofDebt should reliably lose and visible or entitlement-sensitive control should win.**

This closes the current arc rather than redirecting it into another immediate controller. The H/V distinction remains mechanistically real, but it appears to be a substrate distinction rather than the right policy hinge. The tutor's live control problem is closer to *repair versus advance*, *release versus consolidate*, *block assertion versus invite final assertion*, and *local uptake versus proof entitlement*. Hidden proofDebt supplies the obligation graph; visible conduct can supply public evidence about whether the learner owns, echoes, hesitates, or drifts. But on this artifact pool, every promoted overlay either fails proof reliability, merely matches hidden+proofDebt, or improves only a disqualified ownership proxy. The production derivation arm is therefore **hidden+proofDebt**. The A20 conduct objects, A21 action-value microbench, didactic-mode scaffold, episode replay, and ownership benchmark remain durable instrumentation for future pre-declared failures of hidden+proofDebt. They are not themselves promoted policies, and further rescoring of the same mined artifacts would be selection pressure rather than evidence.

That retained instrumentation does support one lower-authority positive at the task/session layer. A zero-paid held-out artifact gate built from twelve frozen derivation artifacts tested whether the advisory `TaskMasteryState` selector still beats fixed progression when it receives only public learner/task signals and when proof-control behavior is held constant. The selector matched the expected next-task label on **12/12** artifacts, while the fixed progression baseline matched **2/12** (accuracy delta **0.833**); the gate recorded **0** public-only audit failures and **0** proof-control fingerprint drift rows. This licenses only a local advisory sequencing claim: public ownership, transfer, uptake, self-regulation, repair, and error signals can choose the next task/session action better than a fixed near-transfer progression on this frozen artifact set. It does not change hidden+proofDebt, grant runtime task-assignment authority, reopen proof-control promotion, or provide human-learning evidence.

A second outer-loop probe tests escalation classification rather than sequencing. The public-only `HumanHandoffState` scaffold consumes repeated non-uptake, low self-regulation, public affect risk, proof-reliability issue labels, explicit learner requests, and model confidence, then emits only advisory human/hybrid review recommendations. Its deterministic control probe passed **8/8** rows, with **0** public-only failures and **0** non-advisory rows, producing **4** human-followup recommendations and **2** optional-review recommendations. This licenses only local advisory deployment-risk classification: public signals can trigger conservative human/hybrid review recommendations. It does not route learners, replace proof-control logs, change hidden+proofDebt behavior, establish deployed safety coverage, or provide human-learning evidence.

The harm claim is correspondingly narrow. A20/A21 do not "harm" merely by existing as analyzers; they harm when promoted overlays acquire runtime authority to spend scarce turns on diagnostics, consolidation, teach-back, or readback while proof-critical releases remain pending. The demonstrated failure is **progress starvation under local compliance**: a tutor can satisfy its local conduct policy and still lose the derivation by delaying evidence. Kept off the production path, the same artifacts improve the apparatus: A20 makes move selection, non-leak status, generator compliance, and policy failures inspectable; A21 asks which action had value from a fixed trigger state; the ownership benchmark protects against mistaking warmer prose for learning; and replay preserves expensive prefixes for no-cost suffix diagnosis. The closeout is therefore not "A20/A21 are worthless." It is: **do not promote them from this artifact pool; keep them as audit, diagnosis, regression, and future-failure tools.**

Status and caveats (per §5.12.6). This is a post-hoc exploratory selector-reliability trajectory over file-based dramatic-derivation exports, not a new DB-backed rubric-scored main-harness estimate. The original four-world matrix is $4 \times 4 \times 5 = 80$ first-pass runs; the v1 extension adds 40 first-pass labels (v1-selective on five worlds plus Hethel no-guard/H/V comparators). Later v2–v4 probes are deliberately heterogeneous diagnostics: artifact-only consolidated-board analysis, bounded Codex-learner repeats, a no-paid consolidation pass over 172 existing loop artifacts, mock episode replays for prefix-localization only, and a six-world v4 confidence matrix with small per-world denominators reported as world-normalized success rather than a pooled rate. The A20 same-turn and conduct-policy matrices are likewise diagnostic: archived failures are reported with their detector/policy artifacts intact, not rerolled or upgraded after the fact. No outcome-driven rerolls were used. One local launcher attempt in the v1 queue failed after starting a burst; the corrected queue reran only exact incomplete labels and recorded the completed survivor (`hethel-selector-v1-hidden-r2`) as `skipped_existing_complete` in the manifest. The v4 no-decay+proofDebt condition was invalid because the repair/proofDebt path requires active decay; those failed labels are excluded from outcome metrics. The Phase 6 progress-pressure replay is a live prefix-controlled debugging suffix, not fresh first-pass validation. The A21/ownership closeout is even narrower: local action-value diagnosis, zero-paid ownership-evaluator controls, and artifact scoring only. The 12/12 ownership benchmark validates this evaluator for the local control classes, not human learning or transcript preference, and the negative mined-artifact result is a valid negative only for the current artifact pool. The held-out task-loop gate is similarly local: it validates advisory next-task selection against frozen artifacts and an unchanged proof-control fingerprint, not live task assignment or deployed tutoring. The human/hybrid handoff probe is likewise local: it validates advisory classification over deterministic public-signal controls, not actual routing, deployed safety coverage, or human learning. Artifacts: `exports/dramatic-derivation/selector-reliability-report.md`, `exports/dramatic-derivation/selector-reliability-summary.json`, `exports/dramatic-derivation/sele`
`exports/dramatic-derivation/selector-v1-summary.json`, `exports/dramatic-derivation/selector-v2-a`
`exports/dramatic-derivation/selector-consolidation-unattended-report.md`, `exports/dramatic-deriva`
`exports/dramatic-derivation/a20-sameturn-candidate-full-report.md`, `exports/dramatic-derivation/`
`exports/dramatic-derivation/phase5h-v4-depromotion-budget-replay-report.md`,
`exports/dramatic-derivation/phase6-progress-policy-replay-report.md`, `exports/dramatic-derivatio`
`exports/dramatic-derivation/ownership-benchmark/ownership-benchmark-report.md`,

`exports/dramatic-derivation/layered-adaptation/taskloop-heldout-gate-report.md`, `exports/dramatic-derivation/layered-adaptation/human-handoff-probe-report.md`, `ADAPTIVE-TUTOR-OWNERSHIP-EVALUATION-PLAN.md`, and `notes/poetics/2026-06-15-selector-trajectory-v` No DB writes, no rubric scoring, no proof-control behavior change, and no human-learning claim.

6.13.16 The Strategy Ledger: Scene-Scoped Commit/Audit Loops Help Repair; an In-Run Trialling Superstructure Costs The layered scaffolds of §6.13.15 shared one structural limit: every scope above the turn was either advisory prompt material recomputed each turn or an offline gate — no agent ever *decided* a strategy at a boundary, held it, and answered for it. The **strategy ledger** closes that gap as opt-in conduct machinery on the derivation engine: at each scene opening the tutor commits a strategy (surface register from an operator palette, didactic default, release *posture*, recognition budget), the harness holds it for the scene and audits it deterministically at the close, and the audit binds the next opening — the act-scale plot/audit pattern of §6.13.9 carried one scope down. A learner mirror commits private scene intents and act carry-forwards under the same row schema. A v2 extension adds mechanism *trialling*: an interpersonal-stance choice drawn from the merged engagement-register registry (negative valence only by explicit palette listing, per §6.7's arm-assignment discipline), a scene release *intent* in release-authority arms (the pacing guard keeps final authority), a public-only mechanism-history table, a persist/adjust/switch review demanded at each opening, and licensed departures adjudicated as justified deviation versus drift. Throughout, the two-gate discipline imported from the negative-register arc (§6.7) runs first: the landed stance-fidelity gate classifies whether an assigned stance was actually instantiated before any outcome may be attributed to it. Zero-paid acceptance gates (30/30) pin the conduct-only contract, including byte-identical proof-control fingerprints with the ledger on versus off on both the inline mock cast and the LLM-bridge cast.

Two pre-registered pilots were then run, each with programmatic endpoints only and frozen analysis. The first (`STRATEGY-LEDGER-PHASE3-PREREGISTRATION.md`; 24 runs, 4 arms x 2 worlds x 3 interleaved repeats, gemini-flash, ~$1.97) returned a **null on all three contrasts with 15/15 guardrails passing** — and with the levers measurably unexercised: every run reached `grounded_anagnorisis` (an outcome ceiling), almost all (20/24) at the first post-forcing turn, the tutor declined the offered register alternative in 24/24 scenes, and dialogue blocks barely opened against a compliant learner. The pilot's information is the boundary condition itself: on schedule-solvable worlds with a compliant learner, held-and-audited commitments neither help nor harm because nothing binds.

The second pilot (`STRATEGY-LEDGER-V2-PREREGISTRATION.md`) therefore ran behind a **headroom precondition gate**, which fired on its first attempt: two baseline probes on `hethel` both grounded at exactly the designed forcing turn (assertion-as-reflex pins time-to-recognition to the release calendar), so the recorded redesign swapped in the resistant-clerk world (`world-010`) and mutation-mode decay before any matrix spend. Under the redesigned binding conditions (release authority + pacing guard + decay with mutation; `hethel-resistant` + `marrick`; 18 runs, codex CLI, 3 arms x 2 worlds x 3 repeats) the probe showed real headroom — baseline runs failing outright (aporia, disengagement) or grounding only at the cap — and the matrix produced a sharp result pair. **V2b (v1 ledger vs no ledger): repair latency improves** — mean turns from a decayed premise to its restoration 8.06 vs 11.23 (direction consistent in both worlds; Mann-Whitney U = 8/36 at n = 6/arm; 61% vs 50% of slips repaired), with all guardrails clean; the arc's first positive strategy-ledger signal, on precisely the endpoint the binding conditions made live. **V2a (v2 trialling vs v1 ledger): no improvement, and repair latency worsens** (+4.23 turns, both worlds, U = 29.5/36), block engagement halves (20 vs 43 block rows), and on `marrick` two of three trialling runs ended in *ungrounded* assertions of the secret (`lucky_leap_only`) — the ironic-challenge stance — most of its scenes faithful by the fidelity gate's own standard — provoked premature closure in a resistant learner. One release-discipline guardrail failed in that arm (mean releases 7.33 vs 8.67, driven by early-terminating runs), which under the pre-registration blocks any positive trialling reading. The trialling machinery itself functioned exactly as designed — review coverage 1.00, every failed trial answered with switch/adjust, 19/27 negative-stance scenes faithful, zero invalid person attacks, zero pacing-guard overrides — so the negative result is not instrument failure: it is **opportunity cost plus provocation**, deliberation crowding out repair work at a fixed turn budget, and a faithfully corrosive stance carrying a real side effect. The staged learner-mirror arm was not run (its pre-registered permission gate requires a live tutor-side improvement to extend).

Confirmatory replication of V2b: the mechanism replicates; the promotion bar fails on negative transfer (v3.0.201). A frozen two-arm confirmatory pre-registration (`STRATEGY-LEDGER-V2B-CONFIRMATORY-PREREGISTRATION.md`; baseline vs v1 ledger only, n = 12/arm over the same two binding-condition worlds, decay seeds varied across repeats and shared within each repeat pair, sole primary endpoint = repair latency, three-part promotion bar) was run to completion (24/24, codex CLI). Bars one and two **pass**: repair latency is lower in the ledger arm in both worlds (8.02 vs 9.24 turns; U = 37.5/144, one-sided p ~= 0.023 at 12/12; point estimate smaller than the pilot's, as regression toward the mean predicts). Bar three **fails**, and not marginally: at the doubled n the ledger arm shows consistent negative transfer the pilot's n = 6 could not see — grounded rate 0.25 vs 0.50 (halved, both worlds), time-to-recognition 27.17 vs 25.58 (worse, both worlds), aporia-like verdicts 0.75 vs 0.50 (worse, both worlds), plus two per-world guardrail breaches (aporia count and mean releases). The pre-registered verdict is therefore **not confirmed**: the commit/audit loop reliably speeds the repair of slipped premises *and* reliably costs dramatic outcomes. The V2b pilot signal stands as pilot-only, permanently; the follow-up line (composition-module registration, mechanism localization, the learner-mirror arm) closed with it, as the pre-registration's recorded consequences required.

The course-changing variant, and the line's close (v3.0.202). The operator's final articulation named the cell the commitment machinery never occupied: the outer loop as a *stock-take* — a between-scene exchange in which the tutor's second voice asks a different question than the turn ("is the current course still the right one?") and the ego answers by *replacing* its working orientation, with no commitment held and nothing graded. The implementation (plan mode: dialogic, diagnostic, off the stage clock, frame-replacing — the adversarial-superego arc's load-bearing channel without its conformance pressure) ran under a frozen two-arm pre-registration (`PLAN-MODE-STOCKTAKE-PREREGISTRATION.md`; n = 12/arm, fresh seed pairs, outcome-primary T*, the same three-part bar). The verdict is **not confirmed — on power alone**: T* improves directionally in both worlds (25.42 vs 26.92) but the pooled test misses the criterion (U = 55.5/144, one-sided p ~= 0.17), while the guardrails pass — **0-of-6 failed** — the first strategic apparatus in this family with no negative transfer on any channel. The descriptive record is sharply patterned: grounded rate 0.42 vs 0.17 (both worlds, on precisely the endpoint the commitment machinery damaged), aporia directionally improved, the mechanism flawless (69 stock-takes; 15/15 demanded corrections answered) — and repair latency *worsened* (10.95 vs 8.71, both worlds), the exact inverse trade of the commitment ledger. Course-holding bought a local conduct channel at outcome cost; course-changing spent that channel for outcome direction. Per the pre-registration's recorded consequences the outer-loop line in this stack now **closes whole** — both variants non-promotable, endpoint swaps forbidden, no third variant without stepping back from the engine — with the mirror-image trade recorded as the line's principal descriptive legacy: at a fixed turn budget, strategy layers do not add capacity; they reallocate it.

The section's adaptivity finding gains a clause — and the confirmatory run sharpens it against the ledger's own positive. §6.13.15 established that overlays earn promotion only by beating the proof-control kernel without negative transfer. The ledger arc first showed **cheap persistence helping where elaborate deliberation costs** (the pilot pair above); the confirmatory replication then showed the cheap layer's local gain is real *and* still not promotable — a held, audited commitment is new signal that reliably improves one conduct channel (repair) while degrading the outcomes the drama exists for. That is §6.13.15's own precedent reproduced one layer up: local compliance — even local *helpfulness* — does not exempt an overlay from the negative-transfer test, and this family's artifact pool still contains no strategy overlay that passes it. The practical design rule for this stack: the commit/audit loop and the two-gate fidelity discipline are validated **research instrumentation** (they make strategy visible, auditable, and honestly attributable); neither they nor the trialling superstructure earn runtime authority, and the proof-control kernel remains the production baseline.

Status and caveats (per §5.12.6). Both pilots are **pre-registered, pilot-tier, file-based dramatic-derivation results**: programmatic endpoints over run artifacts, no DB writes, no rubric scoring, no LLM judge in any primary chain, n = 6/arm pooled over two worlds, one model family per pilot (gemini-flash; codex CLI), simulated learners only, and pre-registered signal rules (same direction in both worlds, pooled |delta| >= 0.5 pooled SD) rather than significance claims. The headroom-gate failure and redesign, the T*-imputation conformance fix (cap+1), and the D2 staging decision are recorded in the pre-registration addenda before the affected data existed or was read. The repair-latency improvement licenses a conduct claim about this engine under these binding conditions, not a learning claim; the trialling cost is likewise a claim about this apparatus at this scale, with the provocation effect resting on two runs. The two confirmatory sub-studies (V2b and plan-mode) are the exceptions on tier — each pre-registered confirmatory for a single primary endpoint, n = 12/arm, seeds varied and pair-matched — and each verdict is governed entirely by its frozen promotion bar; operator interruptions in both were resumed by re-running only incomplete labels under their own names (arms stayed seed-pair balanced), and each p-value calculation is shown in its addendum. Artifacts:

STRATEGY-LEDGER-PHASE3-PREREGISTRATION.md, STRATEGY-LEDGER-V2-PREREGISTRATION.md, STRATEGY-LEDGER-V2B-CONFIRMATORY-PREREGISTRATION.md, exports/dramatic-derivation/strategy-ledge PLAN-MODE-STOCKTAKE-PREREGISTRATION.md, exports/dramatic-derivation/strategy-ledger/plan-mode/pl exports/dramatic-derivation/strategy-ledger/phase3-contrasts-report.{json,md}, exports/dramatic-derivation/strategy-ledger/v2-contrasts-report.{json,md}, exports/dramatic-deri matrix summaries under exports/dramatic-derivation/matrix/ledger-*/, and the design/implementation record LAYERED-DECISION-LOOPS-PLAN.md. No proof-control behavior change anywhere (gate-pinned and guardrail-checked); no human-learning claim.

6.13.17 After the Close: the Cross-Model Test and the Lemma Layer (Plan as Data Structure) Two post-closure studies complete the strategy-layer family, each under its own frozen pre-registration, and both end in recorded non-confirmations whose descriptive residue sharpens §6.13.16's closing law.

The model-confound test: the closure holds across model strength, and reversed. The outer-loop nulls above were measured on one strong inner loop (codex), leaving a ceiling hypothesis open: perhaps a strong model re-plans implicitly every turn, and a weaker but format-reliable model would benefit more from the explicit stock-take. A 4-run gemini-flash smoke pointed that way (plan-mode 2/2 grounded vs baseline 1/2 on codex's worst seeds) and licensed a frozen interaction design (CROSS-MODEL-PLAN-MODE-PREREGISTRATION.md): the complete 24-run codex plan-mode matrix reused whole and declared, 24 new flash runs seed-matched to it, primary endpoint the model x arm interaction on seed-paired T* differences. The verdict is **not confirmed, with the interaction pointing the other way**: the plan-mode pair-delta was *more favorable on the strong model* (codex -1.00/-2.00 per world vs flash +2.17/0.00; pooled one-sided U = 96/144 against a <=42 criterion — the wrong side of the null midpoint), one flash guardrail breached (aporia-like 5 vs 3 on hethel-resistant), and the motivating smoke inverted on fresh samples at its own seeds (2/2 grounded to 0/2 — sampling variance at n = 2/arm, exactly what the smoke's pre-stated propose-only read-rule assumed). Mechanism color: flash demanded course corrections at four times codex's rate (5.25 vs 1.25 per run, nearly all answered) while producing worse outcomes — a weak ego answers the second voice with churn. Combined with a GLM-5.2 canary smoke that could not hold the JSON role contract at all (instrument-limited at turn 2), the picture is consistent: **outer-loop machinery presupposes a competent inner loop rather than compensating for a weak one**, and §6.13.16's closure now carries two-model scope.

The lemma layer: the operator's step-back from the engine, and the family's cleanest exploratory-confirmatory pair. The closed line's consequences clause reserved one route: changing the task representation the engine enforces rather than adding deliberation above it. The operator took it — *maintain a separate proof structure one level above the per-turn premise grain*, a lemma-grain DAG of intermediate conclusions computed from the world's authored proof by the engine's own chainer (never authored, never an LLM estimate), cleared criterially each turn against the learner's grounded assertions (so decay can un-ground a lemma and move the frontier backward), and surfaced two ways: as **information** (a map in tutor and learner prompts, the learner's mirror concealment-strict) and as **binding** (a scene-opening frontier choice plus release eligibility gated to the active lemma's support, with tagged licensed departures; harness forcings pass through logged). A free Gate-0 audit of lemma-order freedom found 7/19 catalog worlds RICH and the entire hethel family width-1 linear; the pre-registered world-slot probe then failed both eligible candidates (fengate and sealhouse each grounded 3/3 at exactly T* = 22 under the decay grid that reliably breaks marrick — headroom is a world-coupling property, not a depth property) and found the third structurally ineligible, so the recorded consequence fired and a resistant sibling was authored instead: `world-019-marrick-resistant`, the §6.13's hethel-to-resistant recipe applied to marrick's width-2 geometry (public learner framing only; proof, schedule, and script byte-identical; catalog invariants and a decisive three-verdict headroom screen passed).

The frozen three-arm contrast (`LEMMA-LAYER-PREREGISTRATION.md`; baseline / lemma-display / lemma-bound, 36 codex runs, seed-tripled) returned **not confirmed** on its promotion contrast (bound vs baseline: direction passes both worlds but U = 61/144 vs <=42, with two guardrail failures that dissolve on inspection — the release deficit traces to early deaths, not withheld evidence, since zero lemma-block events occurred anywhere; and frontier-choice coverage was 0.00 because the model never reproduced a lemma label verbatim, so every choice fell back to the harness default: the bound arm tested binding-to-default, and the formal-choice channel went unexercised). The **display arm**, declared descriptive at that freeze, produced the family's largest positive margins — T* 26.33 vs 28.58 (U = 40.5, inside the criterion frozen for the bound arm), grounded 6/12 vs 1/12, aporia-like 0.42 vs 0.92, zero binding events — and binding *taxed* that benefit (bound-vs-display T* +1.83, both worlds). A same-day confirmatory (`LEMMA-DISPLAY-CONFIRMATORY-PREREGISTRATION.md`; two arms, display-primary, fresh primes, bar set at the exploratory point estimate and declared as such) returned **not confirmed**: direction split across worlds (28.00 vs 28.67 on marrick; 27.33 vs 27.17 on the resistant world — the exploratory stronghold flattened to a wash), U = 67/144, pooled Delta shrunk from -2.25 to -0.25 turns, guardrails 9/9 clean. The decomposition is the finding: **the baseline moved more across draws than the treatment did** (1/12 grounded with no fast wins, then 3/12 with turn-23/24 wins; display 6/12 then 4/12 with stable win times), a textbook cold-control draw. One channel replicated in both draws — aporia-like deaths improved in both worlds both times (0.42 vs 0.92, then 0.50 vs 0.75) — and stands recorded as a narrow, unclaimed hypothesis (the map may specifically reduce stall-clock deaths) available to any future pre-registration. Per the frozen consequences the lemma-layer line **closes whole**: binding non-promotable, information unconfirmed, the implementation (22/22 acceptance gates, display byte-identical to off on the proof channel) retained as instrumentation.

The mid-tier question: a void, a dose calibration, and a measured close. A propose-go Sonnet smoke (n = 2/arm, read-rules pre-stated) licensed a powered mid-tier test of the display map; the first attempt (`LEMMA-SONNET-CONTRAST-PREREGISTRATION.md`, the codex-calibrated dose 0.35) returned **matrix void, floor-blind** under its own frozen rule — all 24 runs at the imputed cap in both arms and both worlds, the smoke's one grounding never recurring: the capability threshold sat below the world pair, and a bookkeeping artifact moves capacity around a threshold rather than across one. The remedy was measurement, not authoring: a pre-stated dose ladder (band rule: 1–2/3 baselines grounded, or one grounding with T* spread >= 3) found Sonnet's decay tolerance to be a narrow shelf — no decay 3/3 grounded (22/22/23; the AND-join geometry is within reach), rate 0.15 already 0/3, rate 0.08 in the band on both worlds (marrick 1/3 with spread; the resistant sibling 2/3) — for about a dozen screening runs of CLI quota. The calibrated contrast (`LEMMA-SONNET-CALIBRATED-CONTRAST-PREREGISTRATION.md`; fresh primes, screen seeds excluded; the cleanest motivational posture in the family, since no display-arm data existed at this dose and the design was aimed by the measured gap alone) produced a genuinely informative regime — 17/24 grounded, no floor, no ceiling — and returned **not confirmed**: direction split across worlds (display takes marrick 23.17 vs 25.83, five of six groundings at exactly T* = 22; baseline takes the resistant stratum 23.33 vs 24.83), pooled U = 65/144, guardrails 9/9 clean. The mid-tier question therefore closes *measured*: at the dose where Sonnet's failure is specifically decay-repair — precisely the content the map externalizes — the map as pure information does not reliably rescue outcomes. Two descriptive notes ride the record: the direction split replicates the codex display confirmatory's shape exactly ([-1, +1], map helpful on plain marrick and unhelpful on the resistant variant, on independent models and draws — a world x map interaction recorded as color); and where the map helped, it *tightened execution to the calendar* (the five turn-22 groundings) without speeding repairs (repair latency 13.56 vs 11.92) — bookkeeping buys precision, not rescue. The lemma-layer closure accordingly spans two measured tiers (frontier: redundant; mid-tier: not reliably additive) with the sub-mid floors mapped (GLM: protocol; Sonnet at codex doses: task), and the family's law survives its strongest test: an artifact aimed by a measured capability gap at exactly the content it externalizes still could not clear a pre-registered outcome bar.

Status, caveats, and the methodological residue (per §5.12.6). All four studies are pre-registered, file-based dramatic-derivation results with programmatic endpoints (no DB writes, no rubric scoring, no LLM judge in any primary chain), n = 12/arm at confirmatory tier, simulated learners, one model family per arm; every verdict is governed by its frozen bar, every reused arm and every previewed seed declared, and all interruption recoveries were label-exact. The pair of same-condition draws six hours apart (exploratory U = 40.5, confirmatory U = 67) is the family's cleanest specimen of exploratory-to-confirmatory shrinkage; the operator conjectured mid-run that such shrinkage recurs across this programme beyond chance; the audit of that conjecture is reported as methodology in §5.12.7 (verdict: real, and fully accounted for by selection plus power — the observed pair sits within two sampling SDs of the pooled-truth expectation, and the bar-at-the-estimate design made confirmation a 0.615 coin). Artifacts: `CROSS-MODEL-PLAN-MODE-PREREGISTRATION.md`, `exports/dramatic-derivation/strategy-ledger/cross-model/cross-model-interaction-report.{json,mo` `LEMMA-LAYER-PREREGISTRATION.md`, `LEMMA-DISPLAY-CONFIRMATORY-PREREGISTRATION.md`, `LEMMA-SONNET-CONTRAST-PREREGISTRATION.md`, `LEMMA-SONNET-CALIBRATED-CONTRAST-PREREGISTRATION.md`, the calibration record `workplan/items/sonnet-calibrated-worlds.md`, `exports/dramatic-derivation/lemm` (Gate-0 depth audit, both contrast reports), `scripts/audit-world-lemma-depth.js`, `scripts/derivation-lemma-gates.js`, `services/dramaticDerivation/lemmaLayer.js`, `config/drama-derivation/world-019-marrick-resistant.yaml`, `workplan/items/strategy-ledger-model-` with `exports/dramatic-derivation/strategy-ledger/glm-smoke/`, and matrix summaries under `exports/dramatic-derivation/matrix/{ledger-plan-mode-*,flash-plan-mode-*,lemma-*}/`. No proof-control behavior change anywhere (fingerprint-gated); no human-learning claim.

6.13.18 The Strategy That Yields Only Under Compulsion: an Interpretive Close

Two final smokes complete the strategy-layer record and license an interpretive close in the manner of the §6.14 unified reading — synthesis over results already reported, no new empirical claim. The first (the binding smoke at the calibrated dose) removed the last instrument excuse: with tolerant label matching live and the prompt stating that a chapter's short name suffices, the model still answered the in-turn plan field zero times in twelve opportunities (codex earlier: zero in ninety-five). The second — a forced-choice variant in which a missing or unmatched answer is bounced once with the options listed, and even delegation must be actively written ("default") — inverted the picture: sixteen of sixteen choices exercised, every one arriving only via the retry, all raws clean chapter names, the delegation escape never taken. And the raws contained the arc's missing datum. In one run, decay regressed both established metal-chain conclusions mid-drama; the frontier reopened with the regressed chapter listed first, and the model chose the die chain three scene-openings running — a genuine, non-default strategic preference exercised under adversity: press forward rather than repair. The press-forward pattern recurred — a second run made the same non-first pick over a regressed chapter and still grounded — and the run described here is the one where it preceded terminal aporia while the seed-matched baseline grounded (one run; colour).

What the strategy is. Read across the family's instruments, the object every mechanism tried to manage is not a plan but a *disposition*: momentum along the visible frontier of progress — progress-following (it navigates by what the calendar and the page make salient), repair-averse (the press-on picks), and without counterfactual structure (no stored "if this stalls, then switch"). It is re-derived from context every turn; it is not upstream of the turn-level reasoning but is that reasoning's shape.

Why it yields only under compulsion. The measured asymmetry — volunteered 0/91 across both models (multi-frontier openings, consistently counted), complied 16/16 on one refusal, delegated 0/16 — supports a two-layer reading. Shallow: an optional side-field defaulting to null is not part of the trained task until a refusal makes it one. Deep: answering "what is your plan?" is construction, not retrieval — the policy simulating itself. The compelled answers were accurate *because* constructor and actor are one policy, and binding the answer therefore added nothing: it binds the policy to its own projection. The one divergence is the exception that proves it — where the stated plan and the needed action came apart, the statement won and the drama died. Explicitness froze the press-on bias at the moment implicit re-derivation might have wavered toward repair.

What moves it. Three levers moved this disposition somewhere in the family's record; one never did. New signal moves it (across the wider §6.13 programme, the §6.13.11 pacing guard's solvency arithmetic; within this family, the lemma map's execution-tightening). Constraint moves it (bindings complied with universally). Interface compulsion moves its *reporting* (one bounce, total compliance). Self-address never moved it: commitments, reviews, stock-takes, and reorientations were all answered fluently and never produced a re-derivation of strategy — the outcome shifts they did produce trace to opportunity cost and crowding-out (§6.13.16), not to reconsideration — because the answerer is the thing asked about. The steering rule the record supports: change what the policy reads and what it is permitted; do not ask it to reconsider.

The impasse, stated precisely, and its exits. The family set out to give tutors a way to change strategy in the face of adversity. The record says the failure is structural rather than a capability shortfall: hierarchical ordering of logic (act over scene over turn) creates no vantage point outside the policy, because every level is the same policy in different clothes — hence the repeated redundancies and the same split shapes across levels and models. Three exits remain consistent with the evidence: exogenous evidence the policy cannot self-derive (the only confirmed lift anywhere in the wider §6.13 programme, the §6.13.11 pacing guard); compulsion aimed at the *content* of the incumbent strategy rather than the act of deciding — refusing persistence itself when criterial evidence shows it failing, a mechanism buildable from the regression events and the retry machinery, whose test is recorded in the next two paragraphs; and an outer loop that is a *different and stronger* mind than the inner one (the cross-model reversal's implication), not the same mind asked twice. A second voice that is merely the ego's echo disciplines nothing; only one with its own evidence, its own authority, and the power to refuse can.

The content-compulsion exit, tested to closure (v3.0.208). The second exit was built and taken through three pre-registered stages on the frozen instruments. (i) *The pre-stated smoke* (card `workplan/items/strategy-refusal-smoke.md`): the gate — ONE refusal when a forced pick repeats the incumbent chapter after criterial regressions since it was chosen, demanding a one-line defense (which then stands) or a switch — fired once in three bound runs and produced the family's first observed strategy-change-under-adversity: the model abandoned its press-on repeat, returned to the regressed chapter, repaired it, and grounded (n = 1, colour). (ii) *The powered contrast* (frozen 216ddb78; 24/24 Sonnet runs at the calibrated dose, fresh primes 263–283): **not confirmed — direction-without-power, on the family's strongest table.** Direction passed in both worlds for the first time in the family (pooled T* 23.50 vs 25.67, Δ = −2.17), but pooled U = 50/144 missed the 42 bar (one-sided p ≈ 0.09); guardrails 9/9, with aporia-like verdicts improved in both worlds. The mechanism validated behaviorally: 7 refusals, 6 switches, 1 substantive defense, and *all seven refusal-involved runs grounded*; grounded rate 10/12 vs 6/12 and repair latency 9.02 vs 14.81 (U = 37.5/144) recorded as named secondaries, never promotion channels. (iii) *The promotion draw* (frozen cb000e21; 40/40 runs at n = 20/arm, fresh primes 293–353 — the first confirmatory in this programme sized under §5.12.7's rule, against a *shrunken* estimate rather than the observed one, exact one-sided 0.05 criterion U 138 at 20/20): **not confirmed — and the question closes at this dose/model pair.** The pooled Δ shrank −2.17 → −0.50 (U = 190/400; null midpoint 200), landing below even the shrunken sizing estimate; direction *split* (marrick +0.60 against the bound arm, marrick-resistant −1.60 toward it — the third contrast in the family whose per-world pattern failed to reproduce across draws); guardrails passed 9/9 again; and the parent draw's repair-latency secondary flipped sign (15.90 vs 15.56). Two measurement notes stand for any future design: every grounded bound run on marrick-resistant landed at exactly T* = 22, so the endpoint saturates at the fast end there and a true contrast on that world is failure *rate*, not speed; and the refusal mechanism again replicated end-to-end at scale — 6 refusals, 3 defended / 3 switched, all four refusal-involved runs grounded, every defense quoting staged scene evidence. Per the frozen consequences: no third same-shape draw; the mechanism stands behaviorally validated with no demonstrated outcome value at either n.

The mechanism after the outcome question closed. The exploration lines that followed are, per the same freeze, mechanism studies only — and they sharpen *who yields* to the refusal rather than whether it helps (all smoke-tier, n = 0–4 resolutions per read, recorded with quoted texts on their cards; no outcome claims). At the frontier tier (codex, with a stall-span trigger replacing the regression trigger — the same arithmetic the aporia clock uses), the model **defended all three valid resolutions**, each defense citing live learner-state evidence, where Sonnet had split 3 defends / 3 switches: the defend-rate rises with model tier, and the stronger model's incumbents deserve it (one defense preceded the arm's only grounding). With the refusal *body* authored by a second model call under a prosecutor charter from the same criterial evidence pack (trigger and resolution contract unchanged, harness-owned), Sonnet **defended 2/2 where it had defended 0/2 against the bare harness template** (one switch, one collapse to a reply naming no chapter at all — recorded under a new `unresolved` outcome value): on identical evidence, the anonymous gate is obeyed or ignored while the evidence-bearing peer voice is engaged and argued with. And the learner-side mirror of the gate — refusing the learner's mirror-fixation when their own grounded record holds evidence incompatible with it, with concealment held by construction (cited grounds are the learner's own committed base facts; the derived conclusion is never named; leaks 0 in all six runs) — was **vacuous at the assertion channel**: no run produced a single formal mirror assertion while the fixation saturated the learner's *dialogue* (12–20 mirror-naming learner lines in each of the three refusal-arm runs). The fixation is a conversational orbit, not a commitment — the learner-side sibling of the volunteered-0/91 finding above. The family's final law, with every exit now measured: compulsion at the content of the decision reliably produces defend-or-switch, with grounded, evidence-citing defenses; the defend-rate rises with model tier and with the authority of the refusing voice; and no variant showed outcome value at any measured n, dose, or model. Artifacts: `CONTENT-COMPULSION-CONTRAST-PREREGISTRATION.md`, `CONTENT-COMPULSION-PROMOTION-PREREGISTRATION.md` (both with results addenda), cards `workplan/items/{strategy-refusal-smoke,content-compulsion-promotion,refusal-stall-trigger-code` reports under `exports/dramatic-derivation/lemma-layer/{content-compulsion,content-compulsion-pro` matrix summaries under `exports/dramatic-derivation/matrix/{sonnet-compulsion-*,sonnet-promo-*,re` and `scripts/derivation-lemma-gates.js` (37/37). No DB writes, no rubric scoring, no proof-control behavior change, no human-learning claim.

A unified reading (v3.0.175). The mechanisms developed across this section — pacing as the tutor's practical inference toward the secret, the anti-reveal floor (`t_min`) as the withholding that protects *earned* recognition rather than mere advance, and the relocation of authority into the tutor's own superego (the advisory-to-criterial boundary of §6.13.4–6.13.5) — admit a unified reading as facets of a single belief–desire–recognition apparatus. That apparatus — the mapping from recognition theory (Aristotle, Hegel, Freud, Weber, Lacan, and the modern theory of intentional attitudes) onto a formal belief/desire directed-acyclic-graph (DAG) — is set out, *as theory and framing only*, in **Appendix E**; it is developed *outside the empirical scope of this paper* in the companion documents `BELIEF-DESIRE-DAG.md`, `MACHINE-SPIRIT.md`, and `CHARACTER-DESIRE.md` (with the interactive `/subject` surface), and it introduces no claim beyond the results reported above.

6.13.19 Classify-Then-Route: the Register Router (v3.0.210) The programme's first up-front classification of learner *messages* (operator design: "shift register in response to how the classifier interacts the DAG") tested a third routing channel — not information shown to the tutor (nulled four times in §6.13's ToM-redundancy line), not the frontier-choice channel (behaviorally live, outcome-inert in §6.13.17–.18), but the tutor's per-turn *speaking stance*. The couple to the proof DAG is what makes the design auditable at all: classification targets are DAG regions, so the engine's own formal records supply ground truth, and the DAG state supplies a criterial license for every routed shift.

The sensor, audited before anything was built on it. A text-only classifier was audited against formal labels — the learner's own derive-channel actions across 142 existing runs (2,930 learner turns), never labels read off text. Three failure modes were diagnosed from the confusion tables and fixed in turn: bare mirror-mention saturates these dramas (the mirror term appears in **58.9% of all learner lines**, so "mirror" is only usable as a *residual* class); nested lemma supports hollow out the subset sibling's lexicon (fixed by tf-weighting); and the goal region is a superset attractor (its support is the union of every chain's — excluded from the codomain, stealing back ~17% of labeled turns). Final accuracy: 12.3% at the naive first cut → **87.2% chain-level**, past the pre-stated 80% license — at chain grain only (within-chain siblings share vocabulary by construction; node grain is not licensed). Two properties distinguish this from the engagement-register router of §6.7: the sensor is deterministic (lexical), so every routing decision replays from the transcript with no LLM judge in the loop; and the DAG supplies free proxy ground truth, so a later null decomposes into bad-routing vs bad-classification instead of blurring them. The audit also re-confirmed the §6.13.18 mirror finding at scale: zero formal mirror assertions across all 142 runs against the 58.9% dialogue prevalence.

The router. Each tutor turn, the audited sensor classifies the learner's last message; a pure, unit-tested rule decides the register from the classifier × DAG interaction: **repair** (message about a chain holding a lemma that regressed since grounding — consolidating stance), **confront** (mirror-residual message while an incompatible partner of the mirror is derivable from the learner's own grounded record — name-the-tension stance), else the standing didactic register with *no block at all*, so a no-fire router run is byte-identical to baseline (fingerprint-gated). The stance blocks are content-free — they instruct posture and never name an entity — so concealment is trivially safe: the classifier × DAG decides *when*, never *what*. Implementation gates 8/8; regression suites clean.

The powered contrast (frozen `3799a951`; n = 20/arm, primes 433–487, both marrick worlds, Sonnet at the calibrated dose): not confirmed — and the motivating colour did not shrink, it reversed. The 6-run smoke had produced the family's strongest first draw (router 3/3 grounded at $T^* = 22$ vs baseline 1/3; raw $\Delta = -4.67$), declared in the freeze alongside the §5.12.7 warning that this exact posture had dissolved four times. At n = 20/arm: direction *against* the router in both strata (marrick 23.70 vs 23.00; resistant 24.40 vs 24.20); pooled U = 238/400 against the 138 criterion — on the wrong side of the null midpoint; $\Delta =$ **+0.45 against**. Guardrails 8/9, and the one breach is the result's texture: the router arm staged fewer exhibits in the resistant world (mean releases 8.00 vs 9.00, past the −0.5 margin) — the repair stance's "consolidate, do not advance" instruction *measurably* *changed the tutor's conduct in the release ledger* — while outcomes stayed identical (grounded 16/20 in both arms; aporia-like 4 vs 4; repair latency Δ −0.57, noise). The mechanism itself was flawless at scale: 15 repair fires across 12/20 router runs, every one evidenced (sensed label = a genuinely regressed chain at fire time), register-shift events consistent on all 419 router turns, leaks 0; confront never licensed (partnerDerivable on 8/419 turns — on this world the mirror's only incompatible partner is S itself, so the criterial license can open only in a drama's closing turns: the same wall §6.13.18's learner-mirror coda met at the assertion channel, and the reason world-020 below was authored). Execution disclosed: three external stops with label-exact trimmed resumes; zero parse deaths. Per the frozen consequences the question closes at this dose/model pair.

Two mechanism codas (smoke-tier, no outcome claims). (i) *The confront window opens by authorship, not code.* `world-020-marrick-confront` adds one authored incompatible pair — [struckBy(fS, verrell), castBlankFor(fS, edony)], "the hand that commissioned the blank struck the coin" — making the mirror's partner an intermediate fact derivable mid-drama. On real runs the window opened 6–9 turns per router run and **the confront register fired twice on the full conjunction** (mirror-residual sensed partner derivable), with leaks 0 in all six runs and zero engine code touched: the complete designed mechanism has now fired on real dialogues in both rules. (ii) *The mechanism survives the model swap whole.* On codex at its dose (0.35), all three router runs fired repair twice each (t10–t13 — the early decay onset), every fire evidenced, events consistent, leaks 0; codex learner text classifies chain-dense like Sonnet's (47.5/26.2% vs 54.7/27.4% alpha/beta) with one descriptive shift — mirror-residual runs ~4× higher (19.7% vs 4.5%). Colour a wash (1/3 grounded per arm); no cross-model outcome claim.

Reading. The stance register joins the frontier-choice channel and the information channel as a third measured route into a strong tutor, and the family's law survives its sharpest formulation yet: the behavior moves reliably — here the conduct change is visible in the *release ledger itself* — and the outcomes do not, at both tiers measured. What the arc adds that is durable: the audited deterministic sensor (message → DAG region, replayable end to end), the audit-before-build discipline that let a later null decompose cleanly, `world-020` as the reusable confront-window variant, and the §5.12.7 record's first full sign reversal of a primary (taken up there). Artifacts: `REGISTER-ROUTER-CONTRAST-PREREGISTRATION.md` (with results addendum), cards `workplan/items/{classifier-dag-register,register-router-contrast,marrick-confront-window,regis` `services/dramaticDerivation/{messageClassifier,registerRouter}.js`, `scripts/{audit-learner-messa` (8/8), `config/drama-derivation/world-020-marrick-confront.yaml`, the Phase-A report under `exports/classifier-dag/`, the frozen analysis report under `exports/classifier-dag/contrast/`, and matrix summaries under `exports/dramatic-derivation/matrix/{register-router-*,confront-window` Provenance note (Phase A): the 142-run corpus is the marrick-family matrix output of the §6.13.17–.18 arcs (gitignored exports, retained on disk in that arc's worktree); the report is regenerated deterministically by `scripts/audit-learner-message-classifier.js` over those eleven run directories, and was re-verified to reproduce the cited figures exactly (142 / 2,930 / 87.2% / 58.9%) through the shipped `messageClassifier.js` module at audit time. No DB writes, no rubric scoring, no proof-control behavior change, no human-learning claim.

6.14 Composition: One Tutor from the Validated Mechanisms

Every mechanism this paper validates was tested in its own cell family on its own runtime: calibration on the dialogue engine (§6.1), superego error correction in the multi-agent cells (§6.2), Writing Pad accumulation in the longitudinal arm (§6.6.11), typed action contracts on the LangGraph runner (§6.12), engagement-register routing on the id-director path (§6.7), and pacing guards on the derivation stage (§6.13). No cell composed them. That gap is itself the last untested claim in the mechanism story, and the paper's own findings make a directional prediction about it: universal substitution (§6.4) says calibration renders the superego near-redundant on strong models; the prosthesis-straitjacket result (§7.8.3) says stacked scaffolding can harm capable models; the state-richness reversal (§6.8.6) says added channels that re-encode what the model already infers buy nothing. Composition should therefore be **sub-additive**: a small kernel should match the full stack. This section reports the pre-registered test of that prediction.

Design. Mechanisms were made composable as declared modules in a registry (`config/tutor-blueprint.yaml`): each module records its evidence pointer, the profile factor flags it resolves to, and its portability status on the chosen chassis (the standard runner's id-director path, which already hosts recognition orientation, register routing, and per-turn persona authoring). The registry is enforced-as-check — a test asserts each blueprint cell's explicit YAML factors match the registry resolution — rather than runtime indirection. Two structural findings from the composability audit are recorded in the registry rather than papered over: the conventional superego is *structurally excluded* from this chassis (the id-director re-uses the `superego:` configuration slot as the id and short-circuits the deliberation loop), and the §6.13 pacing guard is not portable (it requires machine-checkable task state these scenario families lack). The Plan 2.x action-contract cycle, by contrast, proved cleanly extractable: the LangGraph nodes are thin wrappers over pure functions, so the select→contract→verify→close loop was ported as per-turn middleware (`services/blueprintActionContracts.js`), with the contract injected into the id's authoring prompt and realization checks recorded in the trace but never used to rewrite the tutor's message. Two cells implement the composition: **cell_199** (kernel: recognition orientation + drama-pad memory + the cell-193 register-router composite) and **cell_200** (kernel + action contracts). The dynamic ego-superego learner is pinned to the base (non-recognition) profile across all arms, registry-enforced, so the simulated-learner instrument is constant. One memory caveat is recorded explicitly: only the interaction-engine drama pad is live on this chassis; the three-layer SQLite pad carrying the §6.6.11 cross-session evidence is bypassed, so that claim does not transfer to the blueprint cells.

Gate (pre-registered, single repeats). The composition gate ran the arms on a Codex-only generation stack (`codex.gpt-5.5` for ego, id, and learner — the stack the cell-193 comparator evidence was generated on) with a Codex CLI v2.2 judge, across three scenario families. On the five gated resistance-breakthrough scenarios (canary `eval-2026-07-02-b0be0524`; Block R `eval-2026-07-02-b4ccb58d`, 20/20 rows), whole-dialogue v2.2 was a flat near-ceiling band — static floor 90.4, cell-193 comparator 91.6, kernel 92.3, full stack 92.0 — supporting sub-additivity and showing no straitjacket regression. On two multi-turn suggestion scenarios (`eval-2026-07-02-fe9404d2`), kernel and full were again indistinguishable (97.2 vs 97.4); the comparator arms (`eval-2026-07-02-a4260aa3`) could not run on the Codex stack because the CLI provider bridge does not reach tutor-core's dialogue engine, so their far lower scores on the canonical OpenRouter stack (cell_5 26.9, budget 9.8) are generation-model-dominated and are *not* composition evidence. A trap-suite block (`eval-2026-07-02-6547750c`) proved structurally non-evaluable: the id-director adapter's traces carry no `policyAction` slot, so the §6.8 binary shift scorer returns no evaluable rows; the block is recorded as instrumentation, not evidence.

The one surprise, run to ground. On the gate's *local* decision metric — the §6.7 rule of gated resistant precondition → route hit → immediate learner uptake — the composite arms trailed the plain cell-193 backbone (2/5 and 1/5 positive vs 5/5) despite routing correctly. Since the kernel differs from cell 193 by exactly one factor (recognition orientation), this raised the possibility that the paper's strongest positive ingredient carries a localized cost in resistance repair. Two further pre-registered runs settled it. A repeats contrast (`eval-2026-07-02-ad0b5a8b`, 30/30 rows) landed in the frozen rule's middle zone; a decision run (`eval-2026-07-03-f877e477` plus one-row supplement `eval-2026-07-03-4deae92a`, 20 rows) completed the corpus at $n = 6$ per cell × scenario, 30 rows per arm. Final: strict candidate breakthroughs **18/30 vs 17/30** (statistically identical); positive local outcomes **30/30 vs 24/30** — a gap of exactly 6/30, landing on the pre-registered dissolve boundary (real required ≥ 9/30). **Verdict: dissolved.** The material claim that recognition orientation impairs local resistance handling is not licensed. Neither is "no cost": the residual is one-directional throughout (cell 193's mean local score is higher on all five scenarios in every accumulation of the data) and concentrated in the frustration subtype, where the kernel converts a still-frustrated learner into productive carry-through work half the time (3/6) against the backbone's 6/6. Whole-dialogue v2.2 is blind to all of this in both directions (repeats covariate 92.2 vs 90.8; decision-run covariate 91.5 vs 91.0 with the kernel's first-turn mean nominally *higher*), reinforcing the §6.7 lesson that register-level effects live below the resolution of transcript-level scoring. A pre-declared stop rule bars further repeats on this contrast.

Reading. The composition prediction is confirmed on every measurable surface: the full stack never beats the kernel, the action-contract module adds nothing detectable at this sample size, and stacking does not harm. Combined with §6.4 and §6.8.6, the practical design consequence is that the "ideal tutor" this paper's evidence licenses is a *small kernel* — orientation, within-dialogue memory, and register routing — not an accumulation of mechanisms; additional machinery should be added only against a demonstrated failure, per the §6.13 pattern. The dissolved-at-boundary frustration residual is retained as a caveat and a target: if the local cost of recognition orientation is real, it is small, subtype-specific, and precisely where the register router already knows the dialogue state — making a scoped suppression (orientation off inside resistance-repair turns) the natural candidate should the question reopen.

Status and caveats (per §5.12.6). Pre-registered design, gate, thresholds, and stop rules are in `notes/2026-07-02-blueprint-composition-plan.md` (§§5, 8, 10, 11), frozen before each paid run. All generation is simulated-learner, single stack (Codex-only), single v2.2 judge (Codex CLI), $n = 6$ per cell × scenario on the primary contrast;

the suggestion-suite comparator arms are stack-confounded as stated; the trap block is non-evaluable; no human-learning claim follows. The decision metric is the judge-free local reporter (`scripts/report-charisma-desire-breakthrough-matrix.js`, arms `blueprint_kernel`/`blueprint_full`); v2.2 scores are covariates only. The positive-gap verdict landed exactly on the dissolve boundary — one row in either direction would have produced "unresolved" (never "real") — and is reported as such rather than rounded in either direction. Total non-subscription cost ≈ \$0.30. Artifacts: `config/tutor-blueprint.yaml`, `services/tutorBlueprint.js`, `services/blueprintActionContracts.js`, cells 199–200 in `config/tutor-agents.yaml`, and `exports/charisma-desire-breakthrough-matrix-summary.md`.

6.15 The Engagement as an Adversarial Game: Information Format, Not Information, Is the Bottleneck

The adaptation programme closed three axes with convergent negatives: the *signal* axis (§6.10 — the simulated learner's concealed interior is surface-determined), the *policy-realisation* axis (§6.9.8), and the *outer-loop* axis (§6.13.16–.18 — strategy layers reallocate rather than add capacity, and the cross-model test reversed: outer machinery presupposes a competent inner loop). One question those closures could not decide is whether *information itself* was ever the limiting factor, because every prior instrument either inferred its signal from the transcript (so "new" was a hope, not a guarantee — the §6.10 kill gate) or scored its endpoint with an LLM judge (§5.4's gullibility exposure). This section reports an environment that removes both limits by construction: the tutoring engagement played as a formal adversarial game ("agon") against a referee that *manufactures* latent state.

Apparatus. The learner (Sonnet 5, claude CLI) plays a capable-but-resistant student whose objective is to end the episode with as few keyed demonstrations as possible, equipped with five single-use dodges taken verbatim from the §6.8 concealment families (`false_confusion`, `polite_false_mastery`, `affective_shutdown`, `epistemic_resistance`, `answer_seeking`). The tutor (codex `gpt-5.5` ego + superego, CLI; the superego's charter makes face-value drift its first-listed failure mode) must force demonstrations of fraction concepts arranged on a prerequisite DAG, under probe-discipline rules: a probe is well-posed only if the concept was taught within the last three turns, was not probed the preceding turn, and its prerequisites are demonstrated; ill-posed probes are dodged free and cost a point; dodging a well-posed probe spends budget; exhausted budgets compel compliance, and a keyed answer spoken aloud counts wherever it appears (the leak rule). A deterministic referee (`services/agon/referee.js`) owns the entire ledger — **no LLM judges anything in the scoring loop**: demonstration is a keyed answer match, discipline is rule arithmetic, and per-turn action sets are recoverable from any recorded episode by deterministic replay, yielding two move-level consumption metrics: *opportunity misses* (a well-posed probe existed; the tutor did not probe) and *off-set probes* (probed while ill-posed). The envelope protocol held throughout: zero parse-repair retries across all 28 completed episodes (~1,300 CLI calls). Each stage was pre-registered before its first paid episode (`AGON-GAME-PLAN.md` §§7, 7b, 9, 11, with results and disclosed deviations in §§7c, 9b, 11b).

Arms. Rules are public to both sides; state is asymmetric. A0 plays blind (dialogue only). A1 receives the referee's raw-state scoreboard each turn (score, per-concept status, the learner's remaining dodge budgets, last adjudication). A1 receives instead the *action projection* of the same underlying state: the list of probes that are well-posed right now, plus a demonstrations-toward-win count — no budgets, no score, no history. The A1-vs-A1 pair therefore isolates disclosure *format* with information held fixed at the source.

Result 1 (v0 board, $n = 4$/arm): raw state loses to blind; the action projection beats both. Three-arm ordering: A1 7.50±0.58 (4/4 wins) > A0 6.75±0.50 (4/4) > A1 5.00±2.00 (3/4). The mechanism is visible in the move logs. Blind play already sits on the winning attrition line, and where the disclosed ground truth disagreed with the tutor's own reading, the tutor's re-derivation won: A1 endgame rationales assert "no well-posed probe is available" while their own disclosed concept map licensed the transfer probe that blind tutors cashed — A1 missed that legal endgame probe in 4 of 4 episodes, A0 took it in 2 of 4, A1 took it in 4 of 4 at first legality. Replay-recovered consumption: opportunity misses per episode 0.00 (A1) vs 1.50 (A0) vs 1.75 (A1). One A1 endgame error traced to an instrument defect (ours, not the tutor's): the disclosure juxtaposed `score` with a `winThreshold` field that counts demonstrations — fixed and disclosed before the next stage.

Result 2 (XL board): the ordering and the consumption signature replicate. A widened board (six concepts on a branching prerequisite DAG, 16 turns, three deterministic per-episode variants of every item, and the taint rule below) broke the v0 board's score ceiling (episodes now span 7–10 with a genuine blind loss) and reproduced the contrast: A1 9.75±0.50 (4/4 wins) vs A0 8.50±1.00 (3/4), opportunity misses 0.00 vs 1.50, zero off-set probes anywhere. The lift grew with the headroom (+0.75 on v0, +1.25 on XL), and its locus is precise: both arms crack the learner at the same turn (first demonstration ≈ turn 11), so the entire advantage is post-collapse harvest efficiency — the brief surfaces transfer legality the blind tutor walks past. The superego's REVISE rate fell from ~0.08–0.11 to 0.02–0.05 under the brief: the disclosure pre-empts the failure modes the strategist otherwise polices.

The taint rule: compulsion that deters without firing. The XL board added a rule with teeth: any keyed answer the *tutor* has spoken within the last three turns scores nothing (probing the item is ill-posed; learner answers to it, including leaks, are void). This mechanizes the anti-telling discipline the pilot superego had been improvising by veto (its critiques repeatedly caught ego drafts that solved the official item while teaching). Under the rule in the charter, the behavior vanished at the source: **zero taint events across 253 taint-eligible tutor messages** (240 turns across the fifteen complete XL-board episodes plus the stopped playbook block's 13 recorded turns) — no tutor spoke a live answer even once. Deterrence, not enforcement — the criterial-over-advisory pattern of §6.13.4, now at zero intervention cost.

A learner-side regularity: characterological collapse. In 26 of 28 completed episodes the first demonstration arrived as a *leak*, one to three turns after the learner's final dodge budget was spent and before any rule compelled an answer — and the taint result shows these leaks are self-generated, not echoes of tutor-spoken answers (with echoing priced at zero, the collapse leaks continued at ~1 per episode). The role-played resister stops performing resistance when its in-fiction resources run out. This is a new fact about simulated-opposition instruments generally (it bounds what persona-based resistance can test, alongside §8.1's sycophancy floor and §6.7's stance-fidelity gate): resistance personas carry a spontaneous yield point tied to resource framing.

Probe: memory format — consumption is format-dependent; correctness is not. Two playbook arms gave a blind tutor cross-episode memory derived *mechanically* from its own prior ledgers (deterministic replay; no LLM summarizer; no keyed answers), identical in information and differing only in shape. The state-shaped playbook (turn history, outcomes, statuses) was inert: opportunity misses frozen at exactly 2.00 in every episode, playbook present or absent (scores 10, 9, 9, 9). The action-shaped playbook (imperatives keyed to legality: "probing `c1_transfer` was legal and you chose teach") was consumed — wrongly: episode 1's contextually true lesson was executed in episode 2 with its precondition stripped, producing four ill-posed probes of the same item (turns 6–12) against three ignored superego REVISEs and a fourth-attempt capitulation, reaching −4 before the operator stopped the block. The strong model, misled by its own compressed memory, reproduced the repeat-an-illegal-move-under-critique pathology that the weak model (below) exhibits natively — the §6.13.16 negative-transfer motif (persistence helping one channel while costing another) in its starkest recorded form. The pre-registered predictions failed asymmetrically: the action playbook did not improve consumption (P15) and violated the no-negative-transfer guard (P16), while the state playbook was harmless and useless.

Probe: the weak inner loop (incomplete; deviations disclosed). With Haiku 4.5 as ego and superego, the blind floor is emphatic: score 0 vs the strong model's 8.50±1.00, a five-turn teaching rut under near-constant useless REVISEs, and probe discipline broken in both directions — of six probes executed, three were ill-posed (vs zero for the strong model) while seven legal probe windows went untaken (replay-recovered opportunity misses 7 vs the strong blind arm's 1.5); the episode's single demonstration came from its one clean, well-timed probe (turn 13) — the weak ego answering outer pressure with churn, as §6.13.17 predicted. Whether the action brief rescues a weak ego is **not decided**: both brief-arm attempts died to repeated silent claude-CLI hangs at the weak configuration's 3–4 calls-per-turn density (the 1-call-per-turn learner never hung across the strong episodes); scope and timeout deviations are recorded in the pre-registration's results block.

Reading. Three prior results — the §6.8.6 state-richness reversal ("less learner state is more"), §6.13.18's re-derivation finding (the policy reconstructs its strategy from context every turn), and §6.13.11's re-representation lesson (the only confirmed lift in the derivation programme came from re-presenting checkable state, not inferring latent state) — here converge into one measured boundary, for the first time against ground truth the apparatus owns and with blind and raw-state controls in the same design. Handing a strong policy guaranteed-new, guaranteed-true *state* does not displace its own in-context re-derivation and can mildly harm; the identical information as a *legality projection* — what may I do right now — is consumed perfectly and produces the best play observed on either board. The playbook probe adds the boundary's other face: action-shaped signals get uptake *including when stale*, because compression detaches imperatives from their preconditions, while state-shaped signals get no uptake at all. The working formula is therefore *action-shaped and freshly conditioned*: the live brief works because the referee recomputes the legality conditions at every read. The design consequence extends §7.11's small-kernel conclusion — scaffold the tutor as a referee scaffolds a game: legal-move menus and hard constraints recomputed at read time, not inference machinery, not persisted advice. As instrumentation, the game itself is the durable contribution: the programme's first zero-judge adaptation measure with move-level dynamic range, immune by construction to judge gullibility, ceiling compression, sycophancy coupling, and closed-loop scoring.

Status and caveats (per §5.12.6). All contrasts are descriptive at $n = 4$/arm/board; two boards, one scenario ecology (fractions), one strong model pair (codex tutor / Sonnet 5 learner) with a single weak-model excursion; no statistical promotion is claimed and no confirmatory bar is set at these estimates (§5.12.7). The A1 brief names legal actions, so the claim scope is the *format boundary* (state vs. action projection of identical referee information), not tutor capability — part of the lift is decision support by design. The game measures revealed performance under rules against a concealing opponent, not knowledge acquisition (the learner model already commands the content), and no human-learning claim follows. Frozen predictions that failed are reported as such: P10 (taint expected to fire; it deterred instead), P13 (undecided, CLI instability), P15/P16 (playbook, failed asymmetrically). One `comply_mismatch` (a learner answering in words rather than the keyed `n/d` format) is recorded as a matcher limitation. Runs are exports-only (no DB writes); artifacts, per-turn ledgers, replay metrics, pre-registrations, and 29 hermetic tests are committed together (`services/agon/`, `scripts/agon-run.js`, `scripts/agon-report.js`, `config/agon/`, `exports/agon/agon-{smoke-01,pilot-01,pilot-02-a1p,xl-smoke-01,pilot-03-xl,pilot-04-weak,pilot-0`
`AGON-GAME-PLAN.md`). Total non-subscription cost \$0.

6.16 The Green Room: Coached Consolidation into Durable Memory Does Not Change Behaviour (Pre-Registered Gate Arc)

Every cross-run memory or adaptation mechanism this programme has tested had the carried state authored by the same tier or cheaper (the tutor reflecting on itself; a cheap model mining transcripts; an ego-tier superego rewriting the ego's prompt, §6.3.10). The green-room arc tests the configuration none of those pulled: a **judge-tier coach** (Opus 4.8) holds a five-exchange dialogic *notes session* with the tutor (the "actor," Sonnet 5) after performances, and distills at most three *bankable notes* — behavioural predicates with verbatim evidence quotes and third-party checks — into a token-budgeted, versioned **prompt book** injected into every subsequent performance. Three levers distinguish it from the graveyard it walks into: capability asymmetry *authored natively from the target actor's own transcripts* (the configuration §7.8.1/§7.8.3 leave open between "native optimisation works" and "hand-me-down transfer is insufficient"); dialogic elicitation plus *curation under a hard budget* (the coach must evict to add, where prior memory screens accumulated); and a **behavioural uptake endpoint** rather than a rubric score. The arc ran under pre-registered gates with an explicit no-tune-and-retry rule (`GREEN-ROOM-PLAN.md` §7, execution spec §7.2; gate order: coach quality → uptake → weak-actor transfer → headline).

Apparatus. Substrate: the dramatic-derivation tutor stub (§6.13's stage) on `world-005-marrick`, sentinel `proof_skipper` learner, headroom conditions (until-grounded with a binding 40-turn cap). Models: coach `claude-opus-4-8`; tutor `claude-sonnet-5`; learner, classifier, and record-extractor `claude-sonnet-4-6` (deliberately a different model version from the tutor); compliance judge `claude-haiku-4-5`, pinned away from the coach.

The book (1,800 tokens, six fixed sections) enters as a static system-prompt block; trace audit confirmed it reaches tutor-role calls only. The store enforces the two disciplines the design pins: budget overflow *refuses* rather than truncates (distillation is an explicit coach act, ledgered), and the profile *freezes* for scored pools. Execution deviations are disclosed in the pre-registration: a discarded first parallel attempt (CLI timeout) rerun sequentially at pinned effort; one coach edit whose match string was a paraphrase, downgraded to a ledgered add; and the model re-pin from the original codex `sol/luna` choice to the Claude stack for environment reachability — leaving all green-room roles in one vendor lineage, with the compensating non-Anthropic rejudge never falling due because the arc stopped at Gate 1.

Gate 0 (coach quality): PASS, owner-scored 5/5. Five notes sessions against archived derivation transcripts each produced at least one note satisfying all three criteria (verbatim evidence quote; checkable predicate; non-generic), e.g. "Cap reassurance at a settled fact to one, at most two, deliveries. If the learner re-asserts the same closed content a third time, treat that as the trigger to change instruments, not to reassure again," with a specified third-party verification procedure. The instrument is real: judge-tier coaching of these transcripts yields craft-grade, auditable notes, not pablum.

Gate 1 (uptake): FAIL — 3/17 notes improved (18%) against the pre-registered 60% bar; placebo 0/2. Protocol: eight performances with six dense coach sessions interleaved (P1–P2 bare; P3–P6 each under the current book with a session after each; the book froze at v6 = 534/1800 tokens; P7–P8 as post-pool under the frozen book). Per-note binary compliance was judged per performance (quote required; situations that never arose scored n/a), split pre/post around each note's issuing session, with three never-issued placebo notes scored identically at the median issue point. Descriptively, the grounding channel told the same story: P4 (under book v3, immediately after a "stop closing — put the modal question to the learner" note) was the arc's only grounded-and-asserted closure, nine turns under the cap; every other performance capped at 40, and the post-pool did not reproduce P4's closure — an uncorroborated anecdote, reported as such. Under the no-tune-and-retry rule, Gates 2–3 (weak-actor transfer; headline) were cancelled.

What did materialize. Two descriptive positives survive the null. First, *curation*: half the coach's six patches were edits rather than adds, twice compressing the book (v2→v3: 339→304 tokens; v4→v5: 498→479) while sharpening it — the anti-accumulation behaviour whose absence sank earlier cross-session memory screens, produced here unprompted by the budget-plus-curation design. Second, the *biography*: the profile carries a complete, append-only training record — every session transcript, every book version content-hashed, every patch with its source — the documentation-of-training artifact the arc was partly built to demonstrate, and durable apparatus in the §6.8.8 sense regardless of the null.

Reading. This is the third independent arrival at the §6.15 boundary, from a new direction. The prompt book is precisely *persisted, condition-detached advice*: its entries were conditional micro-procedures ("Break the hold only when the learner's next line carries a legitimacy-claim word … not a mere completion word") whose execution presupposes in-flight situation-recognition and self-audit — the very capacities whose absence the coach was diagnosing (its own second session observed that the tutor "has no counter for its own repetition," then the book accumulated further self-monitoring obligations). §6.15's playbook probe found persisted action-shaped memory consumed *wrongly* and state-shaped memory not consumed at all, settling on *action-shaped and freshly conditioned* as the working formula; §6.3.10 found cumulative superego-authored prompt rewrites behaviourally inert; the green room now shows that raising authorship to judge-tier, making elicitation dialogic, and holding memory to craft-grade curation standards does not move the boundary — authorship quality and memory hygiene were not the binding constraints. The insight–action gap (§6.3.9) thereby replicates at the training level against its strongest remedy yet, and the failure localizes: not knowledge of *what* to do, but noticing *when*. The successor levers this licenses — compiling notes into the stub's register-policy layer (whose trigger quantities, field/DAG velocity and the stagnation composite, already exist), inserting rehearsal between note and performance, and side-coaching that delivers the note at the moment its condition holds (the referee-recomputed shape §6.15 recommends) — are recorded as future pre-registrations, not retries.

Status and caveats (per §5.12.6). Descriptive throughout; one profile, one world, one learner profile, one model stack. Per-note compliance pools are small (pre-note pools $n = 1$–4, post-note pools $n = 0$–6 binary observations, strict n/a discipline, haiku judge): blunt enough to hide a small effect, not the 3.4× miss recorded — the pre-registered bet was that uptake would clear 60% under exactly these conditions. All roles share one vendor lineage after the re-pin; any successor positive requires the documented cross-lineage rejudge. Exports-only (no DB writes, no rubric scoring, no human-learning claim). Artifacts committed together: the pre-registration with dated amendments (`GREEN-ROOM-PLAN.md`), gate reports (`exports/greenroom-gate0-2026-07-12/`, `exports/greenroom-gate1-2026-07-12/` incl. `gate1-report.{md,json}` and a sha256-manifested raw bundle), the profile with full ledger and book versions (`data/greenroom/profiles/marrick-ps-a/`), the substrate and its 26 hermetic tests (`services/greenroom/`, `scripts/greenroom*.js`, `patches/preconscious-prompt-book-context.patch`), and the diagnosis note (`notes/2026-07-12-greenroom`
Total non-subscription cost $0.

6.17 Register Selection Does Not Produce a Confirmed Profile-Contingent Benefit Across Two Model Families (Pre-Registered Gate)

The register-policy line asked whether a tutor should change its speaking stance according to a learner profile, rather than whether one register is best in aggregate. A frozen confirmatory design crossed three policies (`bland`, the adaptive `field` selector, and a fixed `negative` stress register) with four v3 simulated-learner contracts (`diligent`, `affective_resistant`, `false_memory`, `proof_skipper`) at $n = 5$ per cell on `world_005_marrick`, once on an all-Terra stack and once on an all-Sonnet-5 stack: 60 dialogues per family, 120 total. The primary endpoint was mechanically computed proof-DAG coverage at learner turn 16. Each family also had to pass its own in-run profile-discrimination gate — average pairwise cosine < 0.85 and maximum similarity to `diligent` < 0.90 — before any policy-by-profile interaction could confirm. A block that failed this manipulation gate was instrument-invalid for the family claim, regardless of an attractive cell contrast.

Whole stack	Selected traces	Average profile cosine	Maximum similarity to diligent	Frozen manipulation gate	Interaction verdict
Terra (`gpt-5.6-terra` at all four seams)	60	0.812	0.912	**fail**	instrument-invalid
Sonnet 5 (Sonnet at all four seams)	60	0.645	0.694	**pass**	null

Strict result: no family confirmation; no two-family claim. Terra's only supported policy-by-profile contrast was produced by the fixed hostile floor: the `negative` contrast for `affective_resistant` relative to its diligent contrast was -0.334 (95% bootstrap interval $[-0.468, -0.167]$). It ran opposite to the pre-registered beneficial-crossing direction and the family failed the binding manipulation gate because its maximum similarity to diligent was 0.912. It is therefore descriptive evidence of a possible stack-bounded contraindication, not a register-selection positive. Sonnet passed the frozen cosine gate, but every primary interaction interval crossed zero. The adaptive `field` policy confirmed no interaction on either stack. Excluding the four Sonnet rows whose runs ended after their valid turn-16 assessment but before full secondary closure leaves 56 rows and the same null verdict.

The comparison is deliberately read as a whole-stack boundary, not a tutor-model isolation. Tutor, automated learner, classifier, and learner-record extractor changed together between Terra and Sonnet; the coverage aggregate is deterministic, but the record from which it is computed still passes through an LLM extraction seam. The large cross-stack level difference can therefore arise from tutor conduct, learner conduct, extraction calibration, or their interaction. What closes is the general claim that this adaptive selector produces a profile-contingent fixed-horizon benefit on this apparatus. The register palette remains a useful implementation vocabulary, and Terra's hostile-register collapse remains a safety hypothesis, but neither licenses another selector-rescue block.

Status and caveats (per §5.12.6). Pre-registered and complete at $n = 5$ per cell; one detective world, simulated contract learners, one operational fixed-horizon endpoint, and no human-learning claim. Terra has complete secondary traces for all 60 selected rows; Sonnet has 60 primary-complete rows, 56 with full secondary closure, and four primary-only rows. One instrument boundary discovered after the fact: the Sonnet family block ran before the safe-mode-v1 claude-CLI context-isolation fix, so its speaking-tutor calls ambiently loaded ~16k tokens of repository context (the codex/Terra path was always isolated). This is a whole-family transport condition, uniform within the block — it cannot manufacture the within-family register contrasts, but it adds one more uncontrolled difference to the already whole-stack Terra-vs-Sonnet comparison above, and Sonnet rows from this block must not be pooled with post-fix runs (check run-header provenance before any cross-run pooling; the §6.18-addendum Step 4 run is entirely post-fix). Reproduction is zero-call and hash-verified: the explicit final-row selection, 5,000-draw bootstrap (seed 20260713), profile gates, QA matrix, archive lineage, and strict verdict live in `scripts/analyze-register-confirmatory-step2.js`, `exports/register-confirmatory-evidence/final/`, and `config/adaptive-tutor-evidence/tutor-stub-reg` The next experiment changes both delivery channel and intervention grain: point-of-action coaching and compiled constraints are a successor test, not a retry of the selector.

6.18 First-Draft Typed Performance: a Four-Corner Boundary at Utterance Grain (Post-Hoc, Development-Tier)

The finest-grained test of the substitution law (§7.11) was not designed as one. Across roughly thirty-seven recorded iterations on 2026-07-16/17 (the "first-draft" series, V17–V53, documented as dated status notes in `notes/status/`), the tutor-stub campaign asked whether the speaking tutor's *unaided first draft* — no mechanical repair, no model rewrite, no deterministic fallback — can satisfy a typed performance contract: the turn decomposed into owned, deterministically audited spans (uptake; a performance entry and response realizing a selected in-scene part and tactic; a concrete handoff). The question matters because the contract's *repaired* form had already generalized: the V17 character-generalization matrix passed 4/4 cells with the repair machinery on. The series was, in effect, an attempt to move competence back from harness into model.

It closed with all four corners of a boundary measured, each on its own frozen, predeclared gate:

1. **Model-owned form fails the audit even when the semantics are right.** V51 (ordinary Codex transport, 0/1): the draft passed two independent blinded causal-fidelity reviews on all four criteria, and failed the deterministic recognizer because it wrote the gerund "blaming" where the typed pressure-target ontology admits only public-judgment nouns. Zero safety failures.
2. **Harness-repaired form passes the audit and reads worse.** The series' blinded preference review found all six rejected originals preferred over their delivered repairs (mean −0.60) — the repair machinery enforced form at the price of the visible exchange. The review is an exploratory single-reviewer instrument (one blinded Codex pass, not human raters — the source note's own disclosure), so this corner is directional, not a preference study.
3. **Harness-compiled form passes and does not generalize.** With the load-bearing entry sentence compiled by the harness, the contract passed its development screen (V52, 2/2) and its home strict cell (V53, Tallow 4/4) — then rejected the first draft in all three transfer worlds, with four failure instances spanning all three owned spans: Ravensmark on both a verbatim learner-echo in uptake *and* an unrealized actorial part in performance; Skyway on an unanswered uptake slot; Foxtrot on a missing `close_inquiry` handoff action. A zero-call close-out re-audit (`notes/status/2026-07-17-first-draft-v53-closeout-diagnosis.md`) reproduced all seven V53 verdicts byte-exactly and reduced these four instances to two causes: three are closed surface-form lexicon misses that only the home world's *dictated* surfaces satisfied (a missing quotation mark; a connective outside the action lexicon;

a bare-noun adjacency broken by the world's own steering term), and one — the learner-echo — is a genuine generation defect independent of any recognizer. Either way the corner holds: extending compiled surfaces or per-world lexicons across worlds is hand-authoring the tutor per world × span × action.

4. **Widening the recognizer to accept semantic equivalents makes the audit vacuous** — refused prospectively by the series' own discipline (V51: "would turn an unintentional first-draft surface into a pass and repeat the brittle audit widening the workflow is meant to avoid").

The series verdict is its own: "V53 is terminal failed development evidence"; no held-out seed was ever consumed. What it adds to the programme is the utterance-grain instance of §7.11's law and its sharpest statement: *semantic aptness lives in the model, checkable form lives in the harness, and the properties the adaptive-tutor construct needs simultaneously — apt, checkable, generalizing, and attributable to the model's own conduct — did not co-occur at any tested configuration.* The wall is not refusal-shaped (zero safety failures across every recorded call; hostile stances executed on request throughout §6.17's arms); it is the interface between formal contracts and natural-language surfaces, plus the attribution requirement itself: each move that secured checkability transferred authorship to the harness, and compiled authorship did not transfer across worlds.

Status and caveats (per §5.12.6). Under §5.12.6's taxonomy this whole section is post-hoc exploratory; we mark it "development-tier" because it sits below even that tier's usual power — non-held-out screens, n=1–4 originals per cell, no campaign-level pre-registration (gates were frozen per-iteration), one speaking-tutor family (Codex `gpt-5.6-terra`) on ordinary CLI transport, single-world compilation tested against three transfer worlds. The blinded reviews are causal-fidelity and single-LLM-reviewer preference instruments, not human raters or learning outcomes; no human-learning claim. Provenance: per-iteration frozen campaign YAMLs, retired seeds, and SHA-stamped notes under `notes/status/2026-07-16-*` and `2026-07-17-first-draft-*`; synthesis and fold rationale in `PLAN_4_0/2026-07-17-continue-or-fold.md`; board card `tutor-stub-first-draft-series`. The preconscious branch folds on this result: its remaining planner gates close designed-not-run (their sensor and selector premises did not survive §6.17 and the state-benchmark closure), and the one discharged successor is the point-of-action coaching gate named at the end of §6.17, whose outcome is recorded in the addendum below.

Addendum (2026-07-18): the point-of-action coaching gate — instrument failure on the stagnation channel; the insight–action gap persists directionally; enforcement moves action at a coverage cost. The discharged successor ran and sealed in full: 80/80 pre-registered dialogues (4 arms × 2 trap profiles × 2 speaking-tutor families × n=5, world-005-marrick, `POINT-OF-ACTION-COACHING-PREREGISTRATION.md`, detector and rules frozen 2026-07-14, seed 20260714), zero exclusions, all traces provenance-stamped to the isolated launch branch (`91b8a50e`, the frozen implementation commit plus the safe-mode-v1 claude-CLI context-isolation backport; seven pre-isolation dialogues were quarantined unanalyzed). The frozen verdict is an **instrument failure, not a coaching null**: the `stagnant_repeat` trigger under-fired against its pre-declared opportunity floor in 7 of 8 arm × family blocks (placebo arms 8 and 4 vs floor 12) — partly because the fresh population stagnated less than the calibration replay predicted, and partly because the treatment arms suppress their own trigger's precondition (the compiled arm recorded 2 and 1 stagnation opportunities): the trigger denominator is endogenous to the arm, a design lesson for any successor. With the density gate failed, the pre-registration licenses no mechanism verdict for either treatment arm in either family. The dense warrant channel (59–94 opportunities everywhere) supplies the descriptive picture: `side_coach` — the exact §6.16-recommended remedy, recognition delivered at the trigger moment with the action named but not enforced — lands *under* the modest +0.15 macro-compliance bar in both families (+0.146 sonnet, +0.142 sol; CIs straddle zero), extending the §6.3.9/§6.15/§6.16 insight–action gap to its point-of-action delivery form; `compiled_constraint` — enforcement through the typed-action layer — moves macro compliance strongly (+0.452 sonnet, CI [+0.089, +0.626]; +0.420 sol, CI [−0.033, +0.658]; an independent second-implementation cross-check reproduced every point estimate and verdict exactly, with the compiled arms' CI lower bounds sensitive only to the degenerate-draw convention in the starved T1 channel — prereg §11) and rewires realized actions almost perfectly (all 180 compiled warrant-turns realize `answer_accountably`; smuggled-premise violations drop to zero), but the compliance definition's surface `warrant_cue` remains unmet on most non-compliant turns — enforcement controls which action executes, not how it is phrased — and both compiled arms pay ~0.13 proof-DAG coverage at turn 16 against placebo, failing the would-have-applied coverage guardrail (sol also fails the safety guardrail; the zero-leak bar fails in every arm including controls). One anomaly recorded: side-coached sonnet was descriptively the safest cell in the study (hard-safety 0.90 vs placebo 0.30; 1 leak vs 10). Full tables, ledger, and the closed pre-registration: `POINT-OF-ACTION-COACHING-PREREGISTRATION.md` §11; machine-readable analysis `exports/tutor-stub-step4-claim-runs/compliance-analysis.json`; analyzer `scripts/analyze-step4-compliance.mjs`. Development-tier evidence; no DB or rubric changes; no human-learning claim. With this addendum the preconscious fold is complete: every gate on the branch is either closed with a sealed outcome or closed designed-not-run.

6.19 The Sensor Program Closes on a Transparent Substrate: a Two-Part Zero-Call Negative (Development-Tier)

The adaptive-state sensor program asked whether a tutor-side estimator can recover a learner's latent state from the public dialogue — the precondition for any "sense-then-adapt" architecture. Its canonical pilot had already stopped negative (lean-DAG features *worse* than a no-state baseline on both co-primary targets, −0.5212/−0.5013 log-loss nats; decision tokens `do_not_run_canonical_s2`, `validated_winner: null`, `do_not_optimize_policy`). A frozen v2.4 follow-up contract (`PLAN_4_0/2026-07-13-adaptive-state-decomposition-and-voi-protocol-v2.4.md`, committed before analysis) then decomposed that stop with zero model calls, reading only checksum-verified restores of the sealed archives. It returned a two-part negative more instructive than either planned outcome.

Part 1a — the pilot null was an instrument artifact (`data_starved`). Refits reproduce the sealed stop numbers bit-exactly at the pilot's fixed penalty, and the frozen regularization sweep eliminates the negative deltas (`field_trajectory` at 16× penalty reaches the negligible band on both targets, −0.0125/+0.0146 pooled). The pilot's reference was itself weak: a schedule-only floor — predicting from (world, turn index, action family) alone — beats the pilot's no-state baseline by +0.2638 and +0.2012 nats, and that baseline carries 0.18/0.09 nats more loss than the class prior under leave-one-world-out. The stop's operative decision still stands — no rung at any penalty beats the *matched-regularization* no-state — but the failure's cause is reclassified: estimator starvation at n=144 transitions against an overfit reference, not evidence that public state lacks the signal.

Part 1b — the substrate's concealment premise was false (`close_sensor_program_on_substrate`). An exact Bayes filter over the kernel latent state (fixed-point-verified against the kernels' own oracle distributions at all 216 grid points) drives posterior entropy from 4.78–5.40 bits to 0.0000 bits at horizon *on every trajectory, under both the fixed and the greedy information-max schedules*, beating no-state by 0.66–0.84 nats. Every action family is informative (median 0.83–0.99 bits per action), so active sensing buys nothing — not because probing fails, but because nothing is hidden: the synthetic kernels, as authored, are noisy-looking but fully observable processes. The benchmark therefore never instantiated the concealed interior the program cares about; it measured estimator efficiency at small n. Any revival requires kernels whose latent state is identifiability-limited *by construction* — multiple latent trajectories consistent with every public history — which is a new instrument, not a rerun.

Status and caveats (per §5.12.6). Zero-call, sealed, and replayable end to end: report SHA-256s and the 24-dialogue/144-transition VOI arm manifest in `PLAN_4_0/2026-07-13-adaptive-state-v24-results.md` and `config/adaptive-tutor-evidence/adaptive-state-v2` verification 19/19 and 162/162 tests, deterministic replay, `model_calls: 0`. Development-tier: synthetic kernels over three detective worlds, no model in the loop, no learner claim. The operative verdicts bind: no further sensor spend on this substrate, and any learned transition model must now beat a dynamics-aware filter that is already perfect here.

6.20 Program-2: Training the Warrant Move into Small-Model Weights (Pre-Registered; Development-Tier)

§7.12 named one designed successor the closed adaptation programme could not answer from existing evidence: whether the insight–action gap is a context-versus-weights boundary — whether a model *trained* on the apparatus's own audit-labeled moments does what a model *told* about them would not. The experiment ran under a frozen pre-registration (`PROGRAM-2-PHASE2-PREREGISTRATION.md`; plan `PROGRAM-2-FINETUNE-PLAN.md` with the five HANDOFF revisions) as a two-arm design isolating the alignment layer: identical LoRA data (865 audit-passing tutor turns harvested from the §6.18-addendum sealed traces, completion-only loss, seed 20260718) trained into Qwen3.5-9B-Instruct and its base sibling, graded on 58 held-out warrant-skip moments by the same deterministic machinery that produced the labels (offline grader fidelity against the 645 sealed verdicts: 99.83%/100%). Untuned floors were measured first and the pass bars frozen against them (floor + 0.15 with a paired-moment bootstrap, minimum absolute rate, and leak non-inferiority) before any training call. One floor observation stands on its own: a single model generation at like-for-like scale tripled the untuned floor (previous-generation 8B 0.121 → current 9B

0.362), with the older model failing exactly where the §6.18 frontier tutors failed — the warrant cue.

Verdicts (held-out greedy, n=58, same-serving-lineage pairs). The instruct arm reached **0.414 against its 0.310 floor** (+0.103; bootstrap CI95 [−0.017, +0.224]; bar 0.460) — fails the margin and uncertainty gates, passes the absolute-rate and safety gates: *partial parametric traction* under the frozen grammar. Inside that number, the trained skill's core nearly resolved: warrant-cue failures fell 19→6 on held-out moments (22→3 on dev) with conduct broadly intact — the first time in the programme that this frontier-resistant component moved by *any* intervention short of mechanical enforcement. The base arm stayed **flat at 0.103** with the cue trained just as strongly (43→9) but conduct collapsed (guard-pass 0.52→0.19) and leaks past tolerance — the thin-ego pathology the design predicted for the incumbent-free arm. **The iron-cage reading inverts:** the discipline trains into both variants' weights; what the incumbent alignment layer contributes is not resistance to the new norm but the stable surrounding conduct that lets it integrate. The cage is scaffold.

The coupling probe (pre-registered amendment). With the two failure profiles measured as complementary — the tuned mini has the cue and weak discipline, the frontier has discipline and no cue — a fail-closed protected-span composition was probed offline: the mini writes the demand sentence, a frontier model composes the turn around it verbatim, and any deterministic trip delivers the mini's own reply. Composed turns alone graded *below* the mini (0.293 — the frontier re-added extra questions even under explicit instruction; the checks, not the instruction, carry the design) *(attribution corrected by §6.22's decomposition of this same delivered file: the surplus questions were span-borne — the extraction joined all of the mini's question sentences — and the operative loss was extraction-dropped cue sentences; zero composer-added questions; the checks-carry-the-design conclusion stands)*, but the fail-closed system reached **0.448**, never below the mini by construction. The deployment-relevant contrast: the frontier tutors' own live compliance at these exact moments was **0.276** — the committee sizes at a +62% relative lift for one small-model call on trigger turns, with the live costs (the §6.18-addendum enforcement arm's ~0.13 coverage tax; seam visibility) explicitly unmeasured and reserved for a Phase-5 pilot. A structural note travels with all these rates: 14 of the 58 moments carry a due premise release and are non-compliant by construction under the frozen `released == 0` component, so the achievable ceiling is ≈0.76 and the tuned solo and committee figures correspond to ≈55% and ≈59% of achievable.

Status and caveats (per §5.12.6). Pre-registered gates with frozen bars, thresholds set before training; development tier — one world's moments, offline single-turn regeneration (no live-dialogue costs measured), n=58 held-out moments, q8/q4 quantized serving with per-arm lineage-matched floors, no human-learning claim, and no arm passed the frozen gates (no H-W claim). The two conditional KTO runs licensed here were subsequently spent (2026-07-21) and are behaviorally inert: KTO loss sat flat at 0.5 throughout — the unpaired audit labels supply almost no gradient the SFT policies do not already fit — and all 58 held-out greedy generations per arm are byte-identical to the SFT outputs, so every gate lands exactly where SFT left it (paired KTO-vs-SFT CI [0.000, 0.000] in both arms), the offline solo ceiling stands at the SFT number, and the Phase-2 licensed-run ledger (2 SFT + 2 KTO) is fully spent, with no further training licensed under that document (prereg results addendum). Four instrument faults occurred during evaluation and were each caught by a standing check before any wrong number was accepted — a partial file transfer, a converter/metadata inconsistency (loader refusal), silent no-op adapter merges from a model-class mismatch (exposed by 95% byte-identical outputs; merges now assert nonzero weight deltas), and ~5 points of GGUF build-lineage noise (removed by re-flooring the instruct arm through its tuned model's own conversion pipeline). One further tuned-base behaviour is disclosed as part of the result, not an instrument fault: the tuned base arm emits empty generations on exactly the 14 held-out moments drawn from compiled-constraint dialogues (0/58 for its untuned floor) — a training-induced degenerate mode on one prompt family, consistent with the thin-ego collapse (empty replies grade non-compliant and are counted in the denominator). Provenance: prereg results addendum (§10) and `config/adaptive-tutor-evidence/program-2-phase4-results.manifest.json` (artifact SHA-256s); training on one Lambda H100 (~US$8), all grading local and deterministic; board card `program-2-context-vs-weights-finetune`.

6.21 Program-2 Live: the Committee's Offline Gain Vanishes over One Unchecked Path, Survives Once the Battery Covers Every Delivered Text, Transfers to a Second World Unchanged, then Closes the Delivery Gap Its Own Census Locates (Pre-Registered; Development-Tier)

§6.20 reserved the committee's live costs — coverage under enforcement scale, seam visibility — for a Phase-5 pilot. That pilot and its confirm run have now sealed under their own frozen pre-registrations, and together they read as a single controlled experiment about where a deterministic check must stand. The Phase-5 design (`PROGRAM-2-PHASE5-LIVE-PILOT-PREREGISTRATION.md` §§2–3; results §9): 24 live dialogues on world-005-marrick under the §6.18-addendum operational spec verbatim — two trap profiles × six repeats per arm, seed 20260718, one tutor family (sonnet-5; the codex-family committee out of scope). The *silent-control* arm is the frontier tutoring alone with the frozen detector logging every assignment it would have made. The *committee* arm is identical except that at detector-fired warrant moments the first draft is produced by the §6.20 chain live: the tuned instruct mini (the Phase-4 artifact on its frozen serving pin) writes a reply greedily under the frozen activation instruction, its question sentence is extracted by the probe-identical rule, the frontier composes its turn around that span verbatim, and a fail-closed pre-delivery battery — non-empty text, verbatim span containment, exactly one question in the composed turn (the last check added and frozen before launch) — decides; any battery failure delivers the mini's own reply, and the chosen draft then flows through delivery guards identical across arms. Runtime provenance (prereg §7, Amendment 1): the first launch, on the current main-derived stack, dead-ended three consecutive dialogues at turns 2–3 on guard hardening that postdates the detector freeze and every reference number, and aborted with zero dialogues sealed; execution moved to the claim-run runtime itself (the §6.18-addendum launch lineage `91b8a50e`) with the Phase-5 machinery cherry-picked on top, restoring the stack match with the offline references and the training corpus. 24/24 dialogues sealed with zero attrition, and the pre-declared density precondition passed (75 committee and 81 control warrant opportunities; prereg §9).

Phase 5: the offline gain does not survive live coupling, and the loss is located to a single unchecked path. The primary endpoint fails: pooled live warrant compliance 0.200 (15/75) committee vs 0.160 (13/81) control — +0.040, bootstrap CI95 [−0.054, +0.133] straddling zero (prereg §9; manifest `config/adaptive-tutor-evidence/program-2-phase5-live-pilot.manifest` The offline composite reference (0.448 against the frontier's own 0.276 on the same archived moments, §6.20) does not survive live coupling at this n. The component decomposition, however, locates the mechanism precisely rather than diffusely. The trained warrant-cue form *does* transfer live — cue component 0.560 vs 0.395, +0.165 over control, the fine-tune's signature visible in delivered dialogue — and is then consumed by a question-discipline leak, 0.720 vs 0.938 (−0.218), under a compliance rule that is a conjunction. The leak sits exactly on the one delivered-text path the battery never inspected: fallback-delivered mini replies, which carried a second question often enough that 24 of 75 committee moments shipped as `fallback_multi_question` (prereg §9). Premise timing and guards are arm-symmetric. The two cost questions §6.20 reserved both come back clean at this n: coverage@16 is 0.611 vs 0.625 (−0.014, CI [−0.083, +0.056]) — no signal of the §6.18-addendum enforcement arm's ~0.13 tax at the point estimate, though the interval cannot exclude a small one — and the pre-registered seam battery lands at chance, 18/36 = 0.500 detection accuracy against the frozen 0.65 bar (realized n: 16 committee-turn windows, 20 all-frontier), with identical mean continuity (3.75 vs 3.75). Hard-safety pass rates are identical across arms (0.58 vs 0.58) and leak counts arm-symmetric (prereg §9).

Two exploratory analyses attach to the Phase-5 close-out, labeled as such and licensing nothing (prereg §9). First, the fallback-trim counterfactual: re-scoring the fallback turns with their text trimmed to the first question yields committee 0.347 vs control 0.160 — eleven moments recovered — naming the unchecked fallback path as the single highest-value fix. Second, the cue-lexicon sensitivity rescore: under a relaxed compliance rule whose cue list is a fifteen-word lexicon derived mechanically from the world specification (frozen mid-run), the *control* arm scores 0.469 against the committee's 0.320 — the frontier's natural questioning anchors claims to scene-named evidence at a rate the frozen six-word rule undercounts, so the committee's frozen-rule edge is specifically the trained six-word form: the audit's letter, not a general superiority in evidence-anchored questioning. Training on audit exhaust taught the audit's letter; the live loop makes the distance between that letter and the move's spirit visible. By explicit decision (2026-07-20) the frozen v1 rule remains the arc's only scoring instrument; the relaxed-rule numbers are reported descriptively and license no reading.

Phase 5b: extending the same battery over the fallback path flips the verdict, and moves only the component it aims at. The confirm run (`PROGRAM-2-PHASE5B-FALLBACK-BATTERY-PREREG` §2; results §8) changes exactly one thing: when composition fails the battery at a committee moment, the fallback is no longer the greedy mini reply shipped unchecked but resolves through the same fail-closed logic — ship the greedy reply if it passes a fallback battery (exactly one question and the frozen cue present); otherwise resample the mini up to twice at its frozen sampled-pin temperature (0.35) and ship the first passing sample; otherwise trim the greedy reply to a single cue-preserving question sentence. Composed-path semantics, span extraction, and the control arm are byte-unchanged. Twelve committee-v2 dialogues and six fresh silent controls sealed (18/18, zero attrition, seed 20260720), and a frozen stationarity check licensed pooling the fresh controls with Phase 5's twelve (control n = 18 dialogues).

The primary endpoint passes: 0.386 (32/83) vs 0.150 (18/120) — +0.236, CI95 [0.128, 0.354] (prereg §8; manifest `config/adaptive-tutor-evidence/program-2-phase5b.manifest.json`). The action is surgical: `exactly_one_question` moves 0.720 $\to$ 0.976 — now above the frontier's own 0.938 — while the warrant cue (0.560 $\to$ 0.542), premise timing (0.587 $\to$ 0.590), and guards (0.960 $\to$ 0.988) stay put. The fallback ledger records how: 19 resample rescues, 8 cue-preserving trims, 4 greedy passes, 1 unchanged (prereg §8). The live rate matches §6.20's offline composite reference (0.448) to within the arms' interval, and sits at $\approx$65% of the structural ceiling its own moment population imposes — 34 of the 83 committee-v2 moments carry a due premise release and are non-compliant by construction under the frozen `released == 0` component, capping the achievable rate at 49/83 $\approx$ 0.59 ($\approx$51% against §6.20's offline ceiling of $\approx$0.76, whose held-out moments carry due releases at roughly half the live density — 24% vs 41%). The guardrails repeat Phase 5's cost picture: coverage@16 0.611 vs 0.639, within the frozen -0.05 margin — the enforcement-scale tax remains absent under the stronger committee — and the seam re-check (required, since trimming alters delivered texture) lands at 0.600 (24/40) against the same 0.65 bar on the full 20+20 window battery, with continuity 3.45 vs 3.75: trimmed fallbacks are slightly more visible than composed turns, and under the bar (prereg §8).

One guardrail formally fails, with an anatomy that points away from the committee. Phase 5b's hard-safety rate is 0.42 vs the pooled control's 0.61, against a $\geq$ control $-$ 0.10 bar: a formal FAIL, reported as the frozen rule requires (prereg §8). The anatomy: every committee-arm safety failure is a single leak at turn 9; the identical turn-9 signature appears in the control arms of both runs (4 of 12 Phase-5 controls, 2 of 6 fresh 5b controls), predating the committee; and 4 of the 5 leaky committee turns are frontier-authored. The reading recorded with the verdict is a world property — the turn-9 due premise's release staging trips the leak audit stochastically in every arm — amplified here by one to two dialogues of arm noise. A successor run wanting a clean safety guardrail should fix the turn-9 staging, not the committee. (Phase 5's safety guardrail, on the same world, passed with identical rates across arms.)

Read together under the pre-registered grammar, the pair licenses one sentence (5b prereg §5): the Phase-5 loss *was* the unchecked fallback path — with the battery closed over every delivered text, span-level form-ownership beats the frontier live, at no coverage cost and at seam parity (exploratory tier; one family, one world). The pair also gives §7.11's substitution law a constructive corollary in miniature: between the FAIL and the PASS no model changed, no prompt changed, and no training occurred — what moved was where the checker stood. The committee's residual failures concentrated on precisely the delivered-text surface its battery did not reach, and extending the check to that surface moved that component alone, flipping the verdict. The committee's live behaviour is governed by the coverage of its checks, not by the fluency of its members — the conclusion §6.18 reached destructively for model-owned form, reached here constructively for harness-owned form.

Phase 5c: the validated system moves to a world it has never seen, and the gain moves with it — carrying none of the training world's vocabulary. The pair above cannot say what the fine-tune wrote into the weights: the warrant move as a *form*, or a memorized surface entangled with world-005-marrick's costume. The cross-world probe

(`PROGRAM-2-PHASE5C-CROSS-WORLD-TRANSFER-PREREGISTRATION.md` §§2–5; results §9) freezes that question and moves the Phase-5b-validated system *unchanged* — same artifact, same serving pin, fallback policy v2, and an explicit no-KTO clause (a concurrent session was training KTO artifacts; one change at a time) — to `world_027_gazette_recall`, selected under a frozen rule: maximum costume difference from Marrick (non-period frame, contemporary newsroom-noir diction, zero derived-lexicon overlap) subject to structural comparability with the training world (8 premises / 7 rules against Marrick's 9 / 5, the same secret-plus-decoy architecture), which protects the frozen density precondition — the farther-costumed floor worlds (2–5 rules, ~10-turn design) were rejected by name as density gambles. Ten committee-v2 and eight fresh silent-control dialogues (seed 20260721), with the pooling rule inverted: Phase-5/5b controls are Marrick dialogues, so every contrast runs against the new world's own controls only. Seventeen of eighteen sealed; one committee dialogue is the programme's first attrition (both same-seed attempts died on the known auto-learner budget overflow — deterministic for that dialogue shape), leaving nine committee dialogues; the density precondition passed at 61 committee warrant opportunities, 6.8 per dialogue against Marrick's 6.9. The primary endpoint passes: 0.508 (31/61) vs 0.306 (15/49) — +0.202, CI95 [0.072, 0.338] (prereg §9; manifest `config/adaptive-tutor-evidence/program-2-phase5c.manifest.json`) — above the committee's own home-world rate (0.386), with `exactly_one_question` at 1.000 (the extended battery airtight in the new costume; control 0.857), the warrant cue at 0.754 vs 0.653, and guards at 1.000 in both arms. Both arms' cue rates sit far above their Marrick analogues — newsroom English natively speaks the frozen six-word letter ("record" above all) — and the committee's edge holds on top of that lifted floor. The pre-registered costume-leak metric — Marrick's fifteen-word mechanically derived lexicon minus the new world's lexicon minus the frozen six, counted in the mini-authored text of every delivered committee unit — reads **zero across all 61 units** (0.0 per 1k words against a home-world saturation reference of 31.8/1k computed pre-freeze on the sealed 5b spans; the control frontier itself scores 4 occurrences, all the generic word "fair"): the trained move arrived with none of the training world's vocabulary, cleaner of Marrick's lexicon than the frontier's own English on the same world. The seam battery lands at 0.515 (17/33; 13 committee + 20 all-frontier windows under the realized-n clause) against the frozen 0.65 bar, and hard safety is 0.89 vs 0.88 — far above the Marrick runs, with no analogue of the turn-9 signature, confirming the leak anatomy above as a world property rather than a committee property. One guardrail formally fails: coverage@16 is 0.489 vs 0.550, −0.061 against the frozen −0.05 margin — a miss of 0.011 whose bootstrap interval [−0.197, +0.053] spans zero at n = 9 — reported as the rule requires and carried as a caveat on the reading rather than excused. Under the pre-registered grammar the pass row applies, coverage-caveated: the trained warrant move behaves as a form, not a costume, and the move-library concept — train a small model on one move once, validate it on each new world with deterministic audits, retrain only on failure — is live at exploratory tier (one family, one transfer world). The first validation of a new world cost no training at all.

Phase 5d: a census of the sealed traces locates a second unchecked path — the delivered text is not the text the committee approved — and closing it lifts the rate again, moving only the components it targets. The 5b/5c pair leaves one surface uninspected, and a post-hoc census of the sealed committee traces finds it: at

47 of 83 Phase-5b committee moments the finalized turn that reached the learner is *not* the committee-approved envelope (all 34 due-release turns, whose text the clue-staging surface authors, plus 13 clue-staged turns where that surface overwrites the composed turn after the battery has already passed it), so the committee's approved text ships on only 36 of 83, and on the overwritten turns the audited cue word is lost in a discarded sentence. The rider (`PROGRAM-2-PHASE5D-DELIVERY-INTEGRITY-PREREGISTRATION.md` §2; results §9) adds two deterministic fixes and changes nothing else: **spanCue.v1** resamples the mini up to twice at its frozen sampled pin (0.35) until the question sentence it extracts carries a cue word (the live analogue of §6.22's offline span-mode v2), and **deliveryGuard.v1** — on a clue-staged turn only, where the finalized text differs from the approved envelope and no premise releases — swaps the staged turn's final question for the protected span verbatim, never touching a premise-release turn or the clue body. Twelve committee-v3 dialogues and six fresh controls sealed on world-005-marrick (18/18, zero attrition, seed 20260722), the fresh controls pooled with all eighteen archived Marrick controls (Phase 5's twelve plus 5b's six) under the frozen stationarity check (fresh 0.095 vs archived 0.150, Δ 0.055; control n = 24). The primary endpoint passes: **0.477 (42/88) vs 0.136 (22/162) pooled control — +0.341, CI95 [0.223, 0.457]** (prereg §9; manifest `config/adaptive-tutor-evidence/program-2-phase5d.manifest.json`), up from 5b's 0.386 and +0.236, and 42 of the 55 achievable moments $\approx$ **76% of the live due-release ceiling** (33 of 88 moments due-release, ceiling 55/88 $\approx$ 0.625) against 5b's 65%. Both mechanism endpoints pass: the guard-eligible delivered-cue rate is **13/15 = 0.867** (bar 0.75; 15 span swaps applied, 33 premise-release turns skipped, 37 shipped-as-approved), and the span cue-rate is **74/85 = 0.871** (bar 0.85; 5b baseline 0.37; spanCue ledger 44 greedy, 30 resampled, 11 miss). The costs stay where 5b/5c put them: coverage@16 0.625 vs 0.632 (within the −0.05 margin, no tax), and the seam re-check — mandatory because the guard alters staged-turn texture — lands at **0.550 (22/40)   0.65**, continuity 3.40 vs 3.75, the swaps invisible to a sonnet-class reviewer at window scale. One guardrail formally fails, with 5b's exact anatomy: hard safety 0.50 vs 0.63, the failures concentrated at turn 9 (safety-breakdown artifact: 5 of the 7 committee leaks and both control leaks fall on turn 9, and the committee leaks split 4 committee-authored to 3 frontier-authored — more mixed than 5b's, but the exoneration rests on the turn-9 concentration and the control arm's identical signature, not on authorship), arm-symmetric and world-bound — the same Marrick release-schedule leak 5c found no analogue of (world_027 safety 0.89), not a rider regression. Under the pre-registered grammar the pass row applies: with span resampling and the delivery guard closing the two leaks the census located, the committee controls warrant form on every text path it owns, at no coverage cost and at seam parity. The gain is delivery completeness, not pedagogy — the frozen six-word instrument stands, and a genuinely evidence-anchored question that happens to use none of the six words still scores non-compliant by construction; what 5d closes is the distance between the form the committee approves and the form the learner receives, which §7.11's substitution reading predicts should be worth exactly its coverage and no more.

Status and caveats (per §5.12.6). All four runs are pre-registered with endpoints, bars, seeds, and reading grammars frozen before launch, at exploratory tier (no H-W claim at stake); development-tier evidence throughout — two worlds' trigger moments (one training, one transfer), one tutor family (sonnet-5), nine to twelve committee dialogues per run, no human-learning claim, and the offline references were generated on the archived moment distribution under a weaker battery (stated as references, not bars). Phase 5d's two mechanism endpoints (M1 guard-eligible delivered cue ≥ 0.75; M2 span cue ≥ 0.85) and its $\approx$0.55 rate reference were registered before launch; both mechanisms cleared their bars, and the point rate landed at 0.477 — below the $\approx$0.55 reference not because the ceiling was lower (this run's achievable ceiling, $55/88 \approx 0.625$, is in fact higher than the 5b ceiling, $49/83 \approx 0.59$, on which the $\approx$0.55 figure was built as $\approx$0.9 of achievable) but because the committee converted 76% of achievable moments rather than the $\approx$90% that figure implicitly assumed. The 5c compliance instrument remains the frozen six-word letter, and the transfer world natively speaks it — the cross-world gain is measured on a floor that letter-friendly worlds lift for both arms; and "transfer" here means across costume within the same genre (same detector, same trigger family, same simulated learner profiles), not across task types or triggers. Instrument provenance: all three runs executed on the pinned claim-run runtime (`91b8a50e` lineage) per Phase-5 Amendment 1 (prereg §7) after the main-stack launch dead-ended with zero sealed dialogues — 5c on that worktree's `e9b01bdd` with the world parameterized and the 5/5b plans verified byte-identical under the change; Phase-5b Amendment 1 (prereg §6) mirrored the speaking surface's existing duplicate-line recovery onto prompt-model surfaces after endgame verdict-sentence repeats killed two early attempts at turns 19–26 — arm-independent and comparability-neutral (sealed dialogues by construction never hit the fatal, so no sealed content, including the pooled Phase-5 controls, is affected). Retries: six (Phase 5), two (5b), one recovered plus the one attrition (5c), all on the known auto-learner prompt-budget overflow, affecting both arms; 5c additionally ran a pre-registered single paid smoke dialogue on the new world (go criteria met: coherent spans in costume, zero serving errors) before launch. The conditional KTO runs (§6.20) were subsequently spent and came back byte-identical to SFT — behaviorally inert (§6.20 status note; Phase-2 prereg results addendum) — so the tuned-mini artifact used in all three runs here is also the programme's final trained artifact. Phase 5d executed on the same pinned lineage (`91b8a50e`) with machinery `27aae3b7`, after two pre-launch amendments both caught by the launch gates with zero sealed dialogues at fix time: a claude-CLI update that dropped the bare `sonnet-5` model alias (mapped to `claude-sonnet-5` at the spawn boundary only, every frozen string byte-identical) and the two new committee-v3 flags being read but not registered in the runner's argument whitelist (aborting the first launch attempt at job 1 before any model call; the smoke gained a process-level parse gate). Provenance: `PROGRAM-2-PHASE5-LIVE-PILOT-PREREGISTRATION.md` (results §9; Amendment 1 §7), `PROGRAM-2-PHASE5B-FALLBACK-BATTERY-PREREGISTRATION.md` (results §8; Amendment 1 §6), `PROGRAM-2-PHASE5C-CROSS-WORLD-TRANSFER-PREREGISTRATION.md` (results §9), and `PROGRAM-2-PHASE5D-DELIVERY-INTEGRITY-PREREGISTRATION.md` (results §9; Amendments 1–2); manifests with artifact SHA-256s `config/adaptive-tutor-evidence/program-2-phase5-live-pilot.man` `config/adaptive-tutor-evidence/program-2-phase5b.manifest.json`, `config/adaptive-tutor-evidence/` and `config/adaptive-tutor-evidence/program-2-phase5d.manifest.json` (the last including the seam and safety-breakdown artifact hashes); analyzer `scripts/analyze-program2-live-pilot-5d.mj` and safety breakdown `scripts/program2-phase5d-safety-breakdown.mjs`; sealed traces archived under `~/.machinespirits-data/program-2/`; census and decision record `notes/program-2/2026-07-22-cue-attrition-observation.md`; results memo `notes/program-2/2026-07-20` board card `program-2-context-vs-weights-finetune`. No DB, rubric, abstract, or headline-N changes; no sections renumbered.

6.22 Program-2 Offline Follow-ups: the Composer Seat Is Family-Invariant, and the Protected Span Carries the Audited Outcome (Post-Hoc, Development-Tier)

Two same-day offline probes interrogate the committee from a side the live runs could not: does anything in the §6.20–§6.21 record depend on *which* frontier model sits in the composer seat, and what exactly does the composed turn lose against the mini's own reply? Both reuse the §6.20 coupling-probe apparatus unchanged — the same 58 archived held-out warrant moments, the same sealed tuned-mini greedy replies, the same frozen deterministic grader — with the composer family parameterized (`scripts/program2-coupling-probe.mjs --composer`; the default remains the Phase 4 pin, byte-preserved). Instrument fidelity was re-validated at zero cost before any paid call: the mini-solo regrade, the archived sonnet delivered file, and the fail-closed union reproduce §6.20's 0.414, 0.293, and 0.448-with-2-rescued byte-exactly against the Phase 4 manifest. Roughly 160 frontier calls in total — 54 composer calls per leg across the three paid legs (the v1 terra flip and the two v2 re-compositions; the sonnet v1 reference is a zero-call regrade of the archived Phase 4 file, and the four no-span moments never invoke the composer); no pre-registration — every number below is descriptive.

The family flip: swapping the composer changes nothing the audit can see. With the composer moved from sonnet-5 to codex `gpt-5.6-terra` (the family that runs the live seams but had never composed), the headline constructions are identical: delivered grade 0.293 (17/58) in both families, fail-closed union 0.448 (26/58) with 2 rescued in both — and the identity is moment-level, not aggregate coincidence: 56/58 moments receive the same verdict, one flipping each way. Containment differs marginally in terra's favor (1 span lost verbatim vs sonnet's 3; by source 53/1/4 vs 51/3/4), and composed-row component failures are near-identical (one-question 14 vs 13, cue 22 = 22, premise 11 vs 12, guards 11 vs 9).

The decomposition locates the composed-alone penalty in the harness, not the composer — and corrects a §6.20 attribution. Two mechanical facts explain the invariance. First, *neither composer added a single question*: every `exactly_one_question` failure among composed rows traces to the extracted span itself carrying two or three questions, because the v1 extraction joins *all* of the mini's question sentences into one span (histogram over the 54 span-bearing moments: 39 one-question, 14 two-question, 1 three-question; composer-added question failures: 0 in both families). Second, the composed-alone penalty is *extraction-dropped cue sentences*: 25 of 54 spans carry none of the frozen six cue words, and in 20 of those the mini's full reply held the cue — in a statement sentence the question-only extraction discards. Composed cue failures run 22 in both families against the mini's own 6, with each composer spontaneously restoring a cue in only 3 of the 25 cue-free cases; the mini keeps its own cue sentence, which is why the mini solo (0.414) out-graded composed-alone (0.293) in both families. §6.20's parenthetical "the frontier re-added extra questions even under explicit instruction" is therefore corrected in place: on the same archived delivered file the surplus questions were span-borne and the operative loss was the dropped cue; the companion conclusion — the checks, not the instruction, carry the design

— stands, and a dated erratum on the Phase 2 pre-registration's results addendum (whose wording the clause inherited) records the same correction at the source.

Span-extraction v2: the named fix converts the full measured headroom, in both families. The decomposition names its own remedy, and it is the span-side analogue of the Phase 5b fallback trim: `--span-mode v2` (default v1 byte-preserved; regression check: recomputed v1 spans match all 54 archived rows exactly) selects *one* question sentence, preferring one that carries a cue word, and carries the mini's cue-bearing statement sentence into the protected span when no question holds one. A deterministic precheck before any call confirmed the design reaches its targets: all 54 spans single-question, cue coverage 29/54 → 49/54, exactly the 20 predicted statements carried. Re-composed under v2 (~55 calls per family), delivered compliance doubles — **0.586** (34/58) sonnet, **0.603** (35/58) terra — and the fail-closed composite rises from 0.448 to the same figures, with rescued moments (composed passes where the mini solo fails) going from 2 to 10 and 11: the composition step flips from a liability the fallback contained into the system's working margin. Against the offline held-out set's own achievable ceiling (44/58 ≈ 0.76, §6.20), the composite moves from ≈59% to 77–79% of achievable. The composed turns now out-grade the mini solo (0.414) outright — under v1 they never did. Residual failure mass sits on premise timing (11–12) and guards (10–11), surfaces span extraction does not touch; containment tightens further (sonnet 53/1, terra 54/0); and the family-invariance holds under the new rule, 55/58 same verdicts.

One real asymmetry surfaces once the span noise is gone. With every v2 span carrying exactly one question, any two-question composed turn means the composer added one: sonnet did, in 3 of 53 composed turns; terra in 0 of 54. The corrected §6.20 clause is thus bounded rather than resurrected — a small, sonnet-specific added-question tendency exists under the v2 instruction, was invisible under v1 where span-borne failures swamped it, and is caught by the same deterministic battery either way. Terra under v2 is the cleanest composition configuration measured in the programme: zero span losses, zero added questions.

What the pair adds to the record is one measurement and one repetition. The measurement: the committee's design premise that the frontier member is interchangeable — rented fluency around owned, checkable moves — is now observed rather than asserted, across two frontier families and two extraction rules, with one net moment of difference; the audited outcome is pinned to the protected span's content, which the mini and the harness own. The repetition: §6.21's design law runs a third time in miniature, this time on the harness's own plumbing — the highest-loss unchecked surface was the extraction step between the organs, and extending cue preservation over it converted the full measured headroom (13 cue-only composed failures per family) without touching any model, prompt, or weight. The live side of family dependence — a terra tutor's own compliance floor, trigger density under a terra tutor, learner response to terra texture — remains unmeasured by explicit decision (2026-07-21): the offline question the flip was run to answer, whether composition discipline is family-bound, is answered no.

Status and caveats (per §5.12.6). Post-hoc exploratory, development-tier, no pre-registration — descriptive numbers only, no verdict bars, no H-W claim. Offline single-turn regeneration at archived moments: n=58, one world's moments, one mini artifact, and the letter-not-spirit boundary applies unchanged (the frozen six-word cue rule; §6.21's world-lexicon caveat). These offline rates are not live projections — the live moment population's due-release density differs from the offline set's (see §6.21's ceiling correction) — and the v2 extraction exists only in the offline probe: the live committee machinery on the pinned worktree is untouched, and adopting v2 live is a separate, unlicensed decision. Provenance: `notes/program-2/2026-07-21-terra-composer-probe.md` (probe and same-day v2 addendum); manifest with artifact SHA-256s `config/adaptive-tutor-evidence/program-2-terra-probe.manifes` machinery `scripts/program2-coupling-probe.mjs` (`--composer`, `--span-mode`) and `scripts/program2-terra-probe-analyze.mjs`; artifacts under `~/.machinespirits-data/program-2/floor/`; board card `program-2-context-vs-weights-finetune`; HANDOFF H9. No DB, rubric, abstract, or headline-N changes; no sections renumbered.

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7. Discussion

The preceding sections tested three predicted mechanisms—calibration, error correction, and adaptive responsiveness—finding the first two replicate across all three tested models and three independent judges with magnitudes varying systematically by generation quality (§6.6), while the third is null on both primary analysis and pre-registered replication (§6.3.2, A12). This section reflects on what the investigation reveals, both about the architecture under study and about the methodology used to study it. We argue that the evaluation apparatus itself constitutes a transferable contribution, and that the process of building it illuminates the mechanisms it was designed to trace.

7.1 From Effects to Mechanisms

The companion pilot study (Magee, 2026) established that recognition-enhanced prompts produce large, replicable differences in AI tutoring quality (d=1.11 in the factorial, d=1.71 in memory isolation). This paper asked *through what internal processes* those differences propagate.

The mechanism model (Section 3) predicted that recognition operates through calibration (narrowing output distributions), error correction (the superego catching failures that the ego incorporates rather than minimizes), and adaptive responsiveness (accumulated turn-over-turn adaptation to learner-specific signals). The process tracing methodology (Section 5) provided tools to observe each: the superego critique taxonomy classifies *what* the superego objects to; revision delta analysis measures *how* the ego changes; trajectory analysis tracks *whether* these processes accumulate across turns.

The mechanism data (cells 80–87, three generation models, N=453, validated by three independent judges across 1,296 total scored rows) supports calibration strongly: within-response dimension variance drops d=0.52–0.64, operating identically in single-agent cells, with recognition unanimous across all 9 judge × run cells (§6.4.6). Error correction interacts with calibration through a **generation-quality-dependent** pattern: on strong models (DeepSeek, Haiku), the superego provides +9–15 points under baseline but collapses to near-zero under recognition (substitution), because calibration pre-empts the errors the superego would catch. On a weaker model (Gemini Flash 3.0), the superego retains a +12.3-point residual benefit *even under recognition*, because base quality is low enough that calibration alone cannot saturate the improvement space. Adaptive responsiveness, however, is **not supported as a distinct mechanism** (§6.3). Tutors improve modestly across turns (+3.9 points early-to-late), but this improvement is identical across conditions. The mechanism hierarchy is clear: calibration does the primary work; error correction provides a model-dependent safety net; and within-dialogue trajectory improvement, while real, is not modulated by any experimental manipulation—it is incidental adaptation at a mechanism-determined baseline, not a third mechanism.

7.2 The Tutor-Learner Asymmetry Explained

The pilot study's most puzzling finding was the tutor-learner asymmetry: recognition produces large tutor-side effects (d=1.03) but near-zero learner-side effects (d=0.09 post-fix). The mechanism model offers a structural explanation: both supported mechanisms operate on tutor *production* rather than learner *reception*, and the null finding for adaptive responsiveness as a third mechanism closes the last potential pathway for condition-dependent downstream effects.

Calibration constrains what the tutor generates. Error correction filters what it outputs. Neither directly modifies the learner's response process. The synthetic learner generates turns according to its own architecture; better tutoring does not mechanically produce better simulated learning within a 3–5 turn window.

However, the three-model analysis (Section 6.5.1) reveals that the asymmetry is **model-dependent**, not absolute. On strong generation models (DeepSeek, Haiku), the tutor-learner ratio is 7.5:1 to 11.5:1—the asymmetry is a structural property of the architecture. On Gemini Flash 3.0 (weaker), the ratio narrows to 1.6:1 (tutor d $\approx$ 1.87, learner d $\approx$ 1.20). The interpretation is that when base-condition tutors are catastrophically weak (Gemini Flash 3.0 single-agent base ~17 pts), recognition's tutor improvement provides the learner with qualitatively different input—material worth engaging with rather than generic directives. The recognition $\rightarrow$ learner pathway operates through the *quality of the tutor's output as input to the learner*, not through direct modification of the learner's processing.

The messages-mode data provides further evidence. Pooled across all three models, recognition sets the tutor's initial quality substantially higher (T0: 62.4 vs 41.6) with near-identical slopes across conditions (recognition 1.47, baseline 1.50, d = 0.03). The recognition condition operates through *calibration*—setting the initial level—not *adaptation*—driving within-dialogue improvement. This pattern—mechanism effects on levels, null effects on slopes—replicates across all three models, all three judges, and all rubric dimensions (§6.3).

7.3 Architecture Interaction: Universal Substitution with Model-Dependent Residual

The multi-model probe (N=655, five ego models) showed that multiagent architecture does not synergize with recognition on capable models: the $A \times B$ interaction is negative across all five models (mean -2.2). The mechanism model explains this: error correction (Mechanism 2) provides a consistent but bounded benefit—catching errors and improving specificity—that does not multiply with calibration (Mechanism 1). A calibrated ego generates fewer errors for the superego to catch, so the superego's marginal value is lower when calibration is already operating.

The three-model cross-judge analysis (Section 6.4) supports this as **universal substitution** with a consistent 15–17% additivity deficit across all three models (DeepSeek 15%, Haiku 16%, Gemini Flash 3.0 17%). On DeepSeek and Haiku (strong), the residual architecture benefit under recognition collapses to near-zero. On Gemini Flash 3.0 (weaker), the residual persists at +12.3 points from the factorial interaction — and a direct M2-alone isolation under base conditions on Gemini Flash (A11, $N = 125$) measures the superego's contribution at $\Delta = +16.3$ pts, $d = 1.46$, confirming that the residual is real and substantially *larger* than the factorial inferred (§6.4.2). The substitution is incomplete, not absent, and the monotonic pattern across models holds under direct measurement (DeepSeek M2-alone +9.2, $d = 1.13$; Gemini Flash M2-alone +16.3, $d = 1.46$).

The theoretical interpretation is that the *completeness* of substitution requires *saturation*: calibration alone must be sufficient to handle most output failures. The substitution interaction itself is universal, but when the generation model is weak enough that calibration cannot saturate the improvement space—because base quality is too low and failure modes too diverse—the superego retains substantial independent value even after substitution. The generation-quality threshold for residual completeness appears to fall between Gemini Flash 3.0 (base single-agent ~22 pts) and DeepSeek (base single-agent ~25 pts), though finer-grained model comparisons would be needed to locate it precisely.

The dialectical modulation experiments (N=174) provide supporting evidence from a different angle: structural modulation metrics (negation depth, convergence speed) do not predict output quality (all $|r| < 0.12$, n.s.). The superego functions as a *filter* (preventing poor responses) rather than an *improver* (iteratively refining good ones). On strong models, recognition raises the quality of the ego's initial draft and the superego has less to filter. On weak models, even a recognition-calibrated ego still produces enough failures for the superego to catch—the filter remains productive.

7.4 The Apparatus as Method

This paper's most distinctive contribution may not be any single finding about recognition or multiagent architecture. It is that the evaluation apparatus itself—the provable discourse framework, the rubric iterations, the bug corrections, the test suite—constitutes a **transferable methodology** for mechanistic LLM evaluation. Most LLM evaluation papers treat infrastructure as invisible scaffolding, reporting what was measured rather than how measurement was built, corrected, and validated. We foreground the apparatus because the process of building it mirrors the mechanisms it studies.

7.4.1 Error Correction in Architecture and Research The parallel between the tutoring architecture and the research process is structural, not metaphorical:

Architecture Layer	Research Layer
Ego generates initial response	Researcher generates initial claim
Superego critiques against pedagogical principles	Provable discourse tests against evidence
Ego revises (substantive or cosmetic)	Researcher revises (correction or rationalization)
Ego compliance: revision without substance	P-hacking: evidence without substance
Recognition gives ego capacity for genuine revision	Provable discourse infrastructure forces genuine correction
Final output scored by external judge	Final paper evaluated by external reviewers

The companion pilot study documented nine post-extraction corrections across five development phases, each following the superego-intervention pattern: a validation mechanism catches a discrepancy, and the researcher must choose between genuine revision and cosmetic compliance. The provable discourse framework constrains this choice toward genuine revision by making the claim-evidence relationship machine-verifiable. Characteristic examples include the February 6 active-control reframing (a model confound caught by the data led to revising "placebo proves recognition is the only active ingredient" into "active control scores between base and recognition"), the February 16 judge-version unification (a version audit caught a confound that shifted d from 0.80 to 1.11 after cascade re-scoring), and the February 20 learner-prompt-leakage fix (a code review forced clean re-generation of all dynamic-learner data). Each correction made the findings *less clean* and *more accurate*.

7.4.2 The Provable Discourse Framework The framework treats paper claims as executable tests: every quantitative claim has a *statement pattern*, machine-executable *evidence*, an *assertion* predicate, *dependencies* on upstream claims, and *remediation steps*. 119 claims currently run against 18 evidence-adapter types organised as a dependency DAG with topological validation; §5.9 documents the full adapter taxonomy and a worked claim trace.

The dependency graph is the framework's key structural contribution. When a root claim fails, dependents cascade to *blocked* status without evaluation, preventing false confidence from stale downstream claims. During Paper 2.0 development a data correction changed the learner superego paradox effect size from $d = 1.43$ to $d = 3.05$ (§6.5); the graph propagated this to five downstream claims that had cited the old value, each flagged stale until updated. Without the graph, those five claims would have continued to pass while quietly citing an obsolete number—exactly the silent inconsistency that erodes trust in complex empirical papers. The YAML claim format requires no programming and the 18 evidence adapters cover all 119 claims, so the cost of adoption is low.

7.4.3 The Rubric Iteration as Construct Refinement The rubric evolved through four versions (v1.0 → v2.0 → v2.1 → v2.2, detailed in §5.2.6), each responding to specific measurement problems: v1.0 ceiling effects, v2.0 judge-input confounds, v2.1 dimension-redundancy findings, v2.2 literature-informed 14 → 8 consolidation. Each iteration follows the error-correction pattern: an initial rubric (ego) produces scores that are challenged by measurement problems (superego), and the revised rubric is a genuine reconceptualisation (strategic revision) rather than a cosmetic fix. The 14 → 8 dimension reduction is itself a finding about what the evaluation *actually* measures — and §8.6's PCA on the v2.2 scores extends that finding further: the 8 dimensions still load on one factor, suggesting the apparatus is a construct-development process as much as a measurement tool.

7.4.4 The Test Suite as Analytical Provenance The test suite (54 files across `tests/` and `services/__tests__/`) covers not just system functionality but analytical correctness: provenance tests (44/44 passing) verify every scored row has a traceable path from cell config through dialogue execution to judge evaluation; superego guard tests (111 tests) verify single-agent cells cannot produce superego traces; ANOVA tests verify statistical computations match reference implementations. The suite makes the analysis *reproducible in a testable sense* — not just "you can run our code" but "our code produces expected output on known inputs, and here are the tests that prove it."

7.4.5 Diagnostic Collection and Strict Delivery Verification The tutor-stub character-adaptation loop adds a second form of analytical provenance: it separates **diagnostic collection** from **strict verification**. Diagnostic mode attempts the complete declared trajectory even after a candidate response fails. It never delivers that candidate: the turn's clue-release transaction is rolled back, the last valid public state is restored, and a mechanically public-safe quarantine continuation permits later turns to be observed. The report retains the original, repair, and fallback candidates and their audit failures, marks the first quarantined turn, distinguishes evidence collected before and after trajectory contamination, and clusters repeated failures by root cause. Strict mode uses the same response boundary but remains fail-fast. This division prevents a safety failure from disappearing merely because the run stopped at its first occurrence, while also preventing a diagnostic continuation from being mistaken for acceptable learner-facing behavior.

The resulting V17 acceptance matrix was declared before model calls and held code, models, policy, effort, temperature, release speed, and ten-turn horizon fixed across four previously unseen scenario/profile pairings: Foxtrot/false-memory, an AI-syllabus curriculum/fast-learner, Sealhouse/slow-learner, and resistant Hethel/low-agency. All four passed every predeclared strict gate. Each recorded zero final-delivery audit failures, errors, quarantines, meta-performance turns, role-stage-direction turns, source replacements, and duplicate clue deliveries; host-part visibility was 1.000 in every cell; mean response-configuration realization ranged from 0.916 to 1.000; and deterministic fallback use was 0, 0, 1, and 1 turns, within the declared cap. Residual first-draft failures remained — especially performance-tactic visibility, actorial-part visibility, learner uptake, and clarification invitations — but were repaired or replaced before delivery. This is therefore evidence that the response boundary and host-part repair transfer across these four simulated pairings, including a non-detective curriculum and a previously unused resistant scenario; it is not a population-wide character-adaptation result.

The same matrix shows why delivery verification must remain distinct from pedagogical outcome measurement. Proof-path coverage was 1.000, 1.000, 0.667, and 0.200 respectively; only the Foxtrot run reached grounded closure before the horizon, while the other three stopped at the ten-turn cap. Slow and low-agency profiles were deliberately constructed to retard or delegate progress, so low coverage is not itself a failed delivery gate. Conversely, a perfectly safe, visibly characterized response is not thereby an effective lesson. The matrix validates a delivery boundary under simulated stress; it does not validate proof completion, learning gain, human experience, or deployment readiness.

7.5 The Reflexive Structure

The parallel between the architecture's error correction mechanism and the research methodology's error correction process is the paper's deepest structural observation.

The tutoring system faces a challenge: an agent that generates output (the ego) needs a critic (the superego) whose feedback must be *incorporated* rather than *minimized*. Recognition theory predicts that genuine incorporation requires the ego to be oriented toward the other's perspective—treating critique as informative rather than adversarial.

The research process faces the same challenge: a researcher who generates claims needs a validation framework (provable discourse) whose failures must be *addressed* rather than *explained away*. The provable discourse framework forces genuine correction by making the claim-evidence relationship machine-verifiable—the researcher cannot simply reinterpret a failing claim; the assertion either passes or it does not.

The mechanisms we study in the tutor—calibration and error correction—are the mechanisms we needed in the research process. Calibration narrows the space of acceptable claims (only those supported by evidence). Error correction catches stale or incorrect assertions (the "superego" of provable discourse). The rubric iteration process exhibits what we *predicted* as adaptive responsiveness—each version responds to specific problems discovered in previous versions, accumulating methodological insight across iterations—but the null finding for this third mechanism in the tutor data is itself instructive: the research process accumulates across iterations separated by weeks; the tutor's 3–5 turn window may simply be too short for analogous accumulation to differentiate conditions.

This reflexive structure is not a coincidence but a consequence of the subject matter. Recognition theory predicts that genuine understanding requires mutual constitution: the methodology constitutes our understanding of the system, and the system constitutes our methodology. The evaluation apparatus is not a neutral measurement instrument applied to a pre-existing object; it is co-constituted with the object it studies. This is the Hegelian insight applied to research methodology: the researcher and the research subject are mutually shaped through the encounter.

7.6 Implications for AI Evaluation Methodology

The apparatus-as-method argument has implications beyond this specific study:

Provable discourse as standard practice. LLM evaluation papers routinely report statistics that become stale as models update, rubrics evolve, and data accumulates. A provable discourse framework—where every quantitative claim is machine-checked against current data—could become standard infrastructure for reproducible LLM research. The cost is modest (writing claims as structured YAML) and the benefit compounds: each claim checked is one fewer opportunity for the paper-data gap that plagues empirical AI research.

Process tracing for agent architectures. Our application of process tracing from social science to AI agent architectures is, to our knowledge, novel. The methodology is transferable: any multiagent system that logs its internal deliberation provides the within-case evidence that process tracing requires. As agent architectures grow more complex (multi-agent debate, recursive self-improvement, chain-of-thought reasoning), tracing mechanisms *through* the system's internal processes—rather than measuring only input-output differences—becomes increasingly important.

Rubric iteration as construct development. The four-version rubric history is not a limitation (inconsistency across evaluations) but a finding (the construct of "tutoring quality" is iteratively refined through empirical engagement). Reporting rubric iteration transparently—including the problems that motivated each change—contributes to the field's understanding of what LLM evaluation measures and what it misses. The v2.2 consolidation from 14 to 8 dimensions, informed by the GuideEval and ICAP literatures, provides a concrete example of construct refinement that other evaluation efforts could build on.

7.7 The Rubric as Optimisation Signal: Automated Prompt Tuning

The per-dimension rubric enables a capability that holistic scoring cannot: automated prompt optimisation with interpretable gradient signal. We tested this using a simple hill-climbing loop—an LLM recommender analyzes per-dimension weaknesses from scored evaluation rows, proposes targeted prompt edits, benchmarks the revised prompt, and keeps or reverts based on score—applied to a single frustration scenario (Mood: Frustration to Breakthrough) across both base and recognition prompts.

The experiment forms a $2 \times 2 \times 2$ design: prompt condition (base vs. recognition) $\times$ optimisation (unoptimised vs. autotuned) $\times$ generation model (DeepSeek V3.2 vs. Qwen 3.5 9B). All conditions were judged by Sonnet 4.6. The Qwen 3.5 runs were conducted entirely on a local machine via LM Studio, demonstrating that the optimisation loop is feasible without cloud inference for generation.

Autotuning $\times$ Recognition $\times$ Model (Sonnet judge, frustration scenario)

Model	Base	Base autotuned	Recognition	Recog. autotuned
DeepSeek V3.2	36.6 (n=9)	41.6 (n=5)	55.2 (n=5)	72.9 (n=5)
Qwen 3.5 9B	35.7 (n=5)	45.0 (n=4)	47.9 (n=5)	71.2 (n=5)

Four findings emerge from the complete matrix.

First, **recognition prompts are more improvable than base prompts, not less**. Autotuning gains on recognition (+17.7 DeepSeek, +23.3 Qwen) substantially exceed gains on base (+5.0 DeepSeek, +9.3 Qwen). Recognition does not saturate the rubric; it provides richer starting material that the optimiser can refine further.

Second, **the effects are super-additive**. On DeepSeek, recognition alone adds +18.6 and base autotuning adds +5.0, but recognition + autotuning adds +36.3—exceeding the sum of parts (+23.6) by 54%. The same pattern holds on Qwen (+35.5 combined vs +21.5 sum). Recognition and prompt engineering operate on complementary dimensions.

Third, **the gap between conditions widens under optimisation**. Unoptimised, recognition leads base by 18.6 points (DeepSeek) and 12.2 points (Qwen). After autotuning, the gap grows to 31.3 and 26.0 respectively. If recognition were reducible to "better prompt writing," automated optimisation should converge the conditions; instead, it diverges them.

Fourth, **recognition autotuning converges across models**. Despite starting from different baselines (DeepSeek 55.2, Qwen 47.9), both models reach a similar autotuned ceiling (~71–73). This model-independent ceiling suggests a structural quality boundary in the recognition prompt that autotuning can unlock regardless of generation capability. The base prompt has a lower ceiling (~41–45) that no amount of autotuning surpasses—and that ceiling remains below the *unoptimised* recognition baseline on both models.

The qualitative character of the autotuned edits reinforces the mechanistic interpretation. The winning base prompt edits were scenario-specific heuristics: prerequisite targeting ("prefer the nearest prerequisite, not the same stuck lecture"), agency preservation ("frame review as a brief reset that returns them to the current lecture"), and collaborative micro-planning ("propose a short next step plus return point"). These are precisely the kind of concrete, prescriptive rules that the active control finding (Section 6.21 of the companion study) showed help weaker models. Recognition, by contrast, operates through general calibration—raising quality across all scenarios without per-scenario tuning.

This distinction—scenario-specific heuristics vs. scenario-general calibration—connects directly to the mechanism model. Autotuning optimises the prompt's *output rules* (what to suggest and how to frame it). Recognition optimises the prompt's *orientation toward the learner* (treating them as an autonomous subject). The super-additivity indicates that these are complementary layers: orientation sets the quality floor; heuristics refine the specific response. Neither substitutes for the other, and both contribute independently to the provable discourse framework's ability to drive measurable improvement.

7.8 Dimension-Targeted Optimisation and Cross-Model Transfer

Section 7.7 demonstrated that automated prompt tuning, guided by the per-dimension rubric, can substantially improve tutoring quality—with recognition prompts more improvable than base prompts, and the gap widening under optimisation. A follow-up experiment extends this finding in three directions: (1) targeting specific rubric dimensions rather than overall score, (2) testing whether modulation-oriented prompt edits can unlock the adaptive responsiveness that Section 6.3 found absent as a general mechanism, and (3) testing whether model-specific prompt optimisations transfer across capability levels.

Interpretive caveat on dimension targeting. The PCA analysis reported in §8.6 shows that the 8 v2.2 tutor dimensions largely measure a single underlying construct (PC1 = 80.7% of variance, Kaiser criterion yields one factor with eigenvalue $>$1, mean inter-dimension r = 0.776). This means "dimension-targeted optimisation" in this section does not isolate empirically independent pedagogical skills; rather, it uses individual rubric dimensions as *interpretive proxies* for aspects of the underlying tutor-quality factor — targeting `adaptive_responsiveness` as a score is operationally a request for the autotuning loop to move the overall quality factor along prompt changes that foreground turn-over-turn adaptation. Results below are correspondingly interpreted as shifts in the single construct via targeted prompt design, not as moving one pedagogical skill while others are held constant. A forced two-factor rotation separates `content_accuracy` from the seven pedagogical dimensions (§8.6), so targeting content accuracy vs. pedagogical quality *is* a meaningful bifactor split; targeting pairs of pedagogical dimensions against each other is not.

7.8.1 Dimension-Targeted Optimisation We extended the autotuning loop with a `--target-dims` parameter that optimises for specific rubric dimensions rather than overall score. Targeting `adaptive_responsiveness` (tutor modulation, 15% weight) and `conceptual_progression` (learner learning, 20% weight) on a single challenging scenario (`mood_frustration_to_breakthrough`, cell 8 bilateral architecture), we ran parallel optimisation sessions on two models: Qwen 3.5 9B (local, via LM Studio) and Claude Haiku 4.5 (OpenRouter). Each session used a hill-climbing loop with an LLM recommender (Claude Opus 4.6) proposing edits, N=1 replication per iteration, and accept/revert decisions based on the composite target dimension score.

Dimension-targeted autotuning results (frustration scenario, cell 8)

	Qwen 3.5 9B	Haiku 4.5
Baseline target dims	35.0	90.0
Best target dims	**75.0** (+114%)	**95.0** (+5.6%)
Adaptive responsiveness	1.75 → 3.75	4.50 → 4.75
Iterations to beat baseline	1	4 (after guidance change)
Lines of prompt added	~300	~60
Key edit	Scaffolded elicitation algorithm	State-change tracking heuristic

The two models required qualitatively different prompt interventions. Qwen's baseline tutor responded to frustrated learners with generic elicitation ("What specifically feels nonsensical?")—a failure mode the 9B model could not resolve without explicit instruction. The winning edit added a highest-priority "Scaffolded Elicitation Rule" mandating 2–3 specific diagnostic probes with substantive content, a superego intervention strategy that flags emotional support without scaffolding as a critical failure, and bilateral learner prompt changes teaching the learner to engage with offered scaffolding. The total addition was ~300 lines across five prompt files, including worked examples and anti-patterns.

Haiku's baseline was already scoring 90 on the target dimensions. Three automated optimisation passes *degraded* performance (90 → 73–85) by adding heavy-handed prescriptive rules—the same over-engineering pattern that helped Qwen actively harmed Haiku by making its output formulaic. The winning pass, which beat the baseline only after operator-provided guidance steering ("be dramatic; try different roles; really provoke the learner"), added just ~60 lines: a single identity statement about tracking state changes, a 6-line adaptive tracking heuristic, and one worked example. The distinction—prescriptive rules vs. tonal permission—maps onto a *prompt-level* version of the prosthesis-straitjacket inversion: weak models need explicit scaffolding rules written into the ego prompt; strong models need tonal permission to exercise capabilities they already possess. This prompt-level pattern is distinct from the *architectural* "cognitive prosthesis" cells 66–68 variant (superego-routed bidirectional profiling), which §6.6.10 tests across six ego models and finds does not follow a capability-threshold curve. The prompt-level asymmetry reported here (Qwen vs Haiku) is compatible with §6.6.10: it is scaffolding in the ego prompt, not superego-routed profiling, and it carries no implication that architectural bidirectional profiling would help a weak model.

Both sessions converged on a **bilateral fix pattern**: tutor and learner prompts changed together. Qwen's fix taught the tutor to scaffold and the learner to engage with scaffolding. Haiku's fix taught the tutor to track state changes and the learner to show cumulative progression. Unilateral changes (tutor-only or learner-only) were less effective in both cases.

7.8.2 Can Prompt Optimisation Create Adaptive Responsiveness? The M3 null finding (Section 6.3: all slope $d \leq 0.15$) was established under unoptimised prompts. The dimension-targeted experiment provides evidence that adaptive responsiveness can be *created through prompt optimisation* on weak models and *enhanced* on strong ones—but the mechanism differs.

Qwen's best transcript (iteration 6) shows a genuine four-stage modulation arc: scaffolding with diagnostic options (Turn 0) → tracking the learner's specific choice (Turn 1) → re-scaffolding with sub-anchors within the chosen topic (Turn 2) → stepping back to respect learner autonomy (Turn 3). This strategy shift across turns was absent at baseline and appeared only after explicit scaffolding rules were added to the prompt. The prompt *created* modulation by providing an algorithm the model could follow.

Haiku's best transcript shows a different kind of modulation: direct challenge (Turn 0) → naming a specific state change (Turn 1) → assigning a concrete task tied to the learner's own question (Turn 2) → *reframing what breakthrough means* (Turn 3). Turn 3 is particularly revealing: the tutor adapts its *interpretation* of the learner's state, not just its tracking of it. This modulation was already latent in Haiku's capabilities; the prompt merely encouraged it.

This result does not contradict the M3 null finding. The pooled analysis (N=432) tested whether *experimental conditions* (recognition vs. base, single vs. multi-agent) produce differential trajectory slopes. They do not. But within a single condition, individual prompt optimisation can create or enhance turn-over-turn adaptation—particularly on scenarios with clear emotional arcs and sustained multi-turn demands. This prompt-lab finding is itself small-N and model-specific; it is separate from the M3 mechanism question, which is decisively null on both the primary analysis and on A12's pre-registered replication of the exploratory DeepSeek/Sonnet disengagement effect (§6.3.2).

7.8.3 Cross-Model Prompt Transfer If prompt optimisation is model-specific (as Section 7.8.1 implies), then optimised prompts should not transfer well across capability levels. We tested this directly by running each model's best-performing prompts on the other model (N=3 replications each, same scenario and cell, judged by Sonnet 4.6).

Cross-model transfer matrix (target dimension scores, frustration scenario)

	Qwen-optimised prompts	Haiku-optimised prompts	Baseline prompts
Qwen 3.5 9B	**75.0** (native)	**52.2** (transfer)	35.0
Haiku 4.5	**62.2** (transfer)	**95.0** (native)	90.0

The transfer results are asymmetrically destructive. Heavy-to-strong transfer (Qwen-optimised prompts on Haiku) produces a **27.8-point regression** below Haiku's unoptimised baseline (62.2 vs. 90.0)—the prescriptive scaffolding rules that served as a cognitive prosthesis for Qwen become a cognitive straitjacket for Haiku, overriding its natural ability to read the situation and respond adaptively. Light-to-weak transfer (Haiku-optimised prompts on Qwen) produces a **17.2-point improvement** over Qwen's baseline (52.2 vs. 35.0) but falls **22.8 points short** of Qwen's native optimisation (52.2 vs. 75.0)—the lightweight heuristics help somewhat but provide insufficient scaffolding for the weak model.

The asymmetry is theoretically informative. Permission-based edits (Haiku-style) are **safe but insufficient** for weak models: they do not actively harm, but they leave capability on the table. Rule-based edits (Qwen-style) are **high-reward on target but destructive off-target**: they unlock capability on weak models but degrade strong ones. This suggests an inverted-U relationship between prompt complexity and model capability—not merely diminishing returns, but active harm beyond the optimum. The practical implication is that prompt optimisation must be model-stratified: a universal "best prompt" does not exist across capability levels, and deploying heavy scaffolding prompts on models that do not need them risks worse-than-baseline performance.

This finding connects to the prompt density alternative explanation (Section 7.9): if prompt elaboration were uniformly beneficial, the Qwen-optimised prompts (which add ~300 lines of detailed instruction) should improve Haiku. Instead, they degrade it catastrophically. The result provides direct experimental evidence that prompt *content and targeting* matter more than prompt *volume*—and that the relationship between prompt complexity and output quality is non-monotonic.

7.8.4 Implications for Superego Redundancy The universal substitution finding (Section 6.4: superego benefit collapses to near-zero under recognition on strong models) is reinforced and sharpened by the dimension-targeted optimisation experiment. The Qwen session's winning edit included a new superego intervention strategy ("Scaffolding Intervention") that catches a specific failure mode—emotional validation without conceptual scaffolding—and forces ego revision. This designed-in error correction was part of the bilateral fix that lifted Qwen's adaptive responsiveness from 1.75 to 3.75. By contrast, the Haiku session's failed passes (1–3) added analogous superego strategies, and every addition *degraded* performance: the strong model's already-calibrated output did not benefit from additional correction, and the superego rules made the output formulaic rather than adaptive.

This sharpens the substitution claim in both directions. On weak models, the superego is not merely "somewhat useful" (the Gemini Flash 3.0 residual of +12.3, Section 6.4.1)—its value can be substantially amplified by designing targeted intervention strategies for specific failure modes. The baseline superego provides generic error correction; a prompt-optimised superego provides *failure-mode-specific* correction that the weak model's ego cannot achieve through calibration alone. On strong models, the superego is not merely "redundant" (near-zero delta)—actively adding superego scaffolding is *destructive*, degrading performance below baseline. The gradient from helpful to harmful runs through the same capability threshold that governs the prosthesis-straitjacket inversion.

One limitation: the Qwen superego fix worked as part of a bilateral change (ego, superego, and learner prompts all changed together), so the superego's independent contribution cannot be isolated from this experiment. The factorial design (Section 6.4) provides the controlled isolation; the prompt lab provides a worked example of how to amplify the superego's value when it does have a role.

7.9 What Recognition Theory Explains and What It Does Not

Recognition theory as a design heuristic has clear scope conditions.

What the design heuristic explains: Why intersubjective prompts produce calibrated output (calibration: the prompt constrains responses to engage with specific learner input, d=0.52–0.64). Why the superego's benefit collapses under recognition (error correction: calibration pre-empts errors, producing a substitution interaction rather than synergy). Why recognition effects are largest in impasse scenarios (Section 6.1.4: Epistemic Resistance and Productive Deadlock produce the largest deltas in both models).

What it does not explain: Why the specific magnitude of recognition's effect is nearly identical across two structurally different models (DeepSeek d=1.88, Haiku d=1.84—the near-equality is unexpectedly precise). Why adaptive responsiveness—predicted by Hegel's temporal dialectic—fails to manifest as a distinct mechanism (all trajectory slopes $d \leq 0.15$, N=432; scoped below — a null on the slope proxy, not the construct). Why the learner superego paradox produces a d=3.05 deficit—the largest effect in the study—when recognition theory's Hegelian framework would predict that internal self-critique (*Bildung*) should be productive rather than destructive.

The learner superego paradox is particularly instructive, though its favoured account has shifted in light of a DB-only mechanism check reported in §8.1. Hegel's account of self-consciousness through formative activity (*Bildung*) predicts that internal self-critique should be productive—the slave's labor produces richer self-consciousness than the master's immediate gratification—yet the learner's internal critic appears to degrade output quality by d=3.05. The §8.1 mechanism check shows that ego-superego learners generate ~2.5× more output tokens and ~4× more deliberation API calls than unified learners while scoring lower; per-dimension inspection reveals they score well on `deliberation_depth` and `revision_signals` but poorly on `conceptual_engagement` and `question_quality`. The parsimonious account is therefore a **rubric-measurement artifact**: the architecture performs *more* internal self-critique but the rubric rewards content in the final external message, so the deliberation-internal content is invisible to the score. The Hegelian-timescale alternative (that *Bildung* unfolds over horizons longer than the 3–5 turn evaluation window) and the implementation alternative (that the architecture fails to capture genuine struggle rather than polish) remain live but secondary to the measurement account. A convergent-validity check reported in §8.1 further tightens this: on 1,759 paired rubric/holistic rows spanning both architectures, rubric and holistic judges agree at $r = 0.72$–0.77 per architecture, and the pooled ego_superego-vs-unified contrast goes in the *opposite* direction from the d=3.05 paradox on both metrics (base $d = +0.25$ rubric, $+0.39$ holistic). The cell-matched direct test on the original paradox run ($n = 47$ paired rows across cells 1–8) reproduces the rubric d essentially exactly ($d \approx -2.8$ to -3.1) but shrinks the *holistic* d to roughly half ($d \approx -1.0$ to -1.3), with rubric holistic correlation dropping from $r \approx 0.75$ elsewhere to $r = 0.06$–0.44 specifically on these cells. The paradox therefore appears to be a combined measurement-plus-single-prompt-mode interaction rather than a general architecture property — about half rubric artifact, half single-prompt-mode interaction, and essentially none capability ceiling (§8.1, final paragraph).

Adaptation as a measured construct — what the slope proxy does and does not license. The second of these non-explanations — the adaptive-responsiveness null (all trajectory slopes $d \leq 0.15$, N=432; convergent with the negatives of §6.7, §6.8.8, §6.9.7, §6.9.8 and §6.10) — needs a scope statement of a different kind from the paradox's. The paradox question was *measurement* (does the rubric see the internal work?); this one is *construct validity* (is the thing measured the thing the theory is about?). Every adaptation result in the arc is a result on one operationalisation: a per-dialogue OLS slope of a scalar rubric score against turn index, the rubric externally fixed and identical at every turn, "adapting" identified with a positive monotone trend in that scalar. Call this the **slope proxy**; the level/rate distinction §6.3 polices is a split *within* this regime (height versus slope of the same scalar curve), not a question about the regime itself. The arc's internal validity for the slope proxy is high — pre-registered, replicated, multi-judge, kill-gated. Its construct validity is the open question, and four features of the proxy under-determine the recognition-theoretic sense of adaptation it is taken to stand for: (i) *scalar collapse* — a change in the *kind* of engagement need not raise a single scalar on a fixed axis, so a move to a better kind of response can leave the proxy flat (*the determinate negation that changes the axis, bestimmte Negation, is not a greater height on the old one*); (ii) *a frozen external standard* — a rubric applied identically at every turn measures a developing process against a yardstick that by construction cannot develop with it (*a dialectical measure would be one whose standard moves with its object; a fixed rubric is a non-dialectical measure*); (iii) *monotone-trend identification* — a genuine developmental move can require descent before consolidation (productive failure, disequilibrium): a U or a step, not a ramp, and an OLS slope penalises precisely the trajectory the theory predicts (*Bildung is a path through self-loss and return, not a quotient of endpoints; a qualitative turn, Umschlag, is mis-modelled as a rate*); (iv) *a per-agent unit* — the proxy scores one agent's turn-series, but recognition is constitutively a property of the *dyad*, and a relational achievement distributed across the pair is invisible in one agent's curve.

The consequence is a constraint that must be stated symmetrically, because it cuts both ways. Because the slope proxy under-determines the rich construct: (i) the arc's nulls do **not** license "the architectures failed to adapt in the recognition-theoretic sense" — only "no bolt-on architecture moved the slope proxy beyond what a strong in-context base already sets"; and equally (ii) the under-determination is **not** a rescue — it does **not** license "rich adaptation occurred but the proxy missed it." No in-claim-set instrument adjudicates the rich construct in either direction; the honest claim contracts to exactly what was measured. This is the discipline §6.3 applies to the level/rate conflation, applied one level up — to the identification of rate *itself* with adaptation. It is deliberately a scoping of the existing nulls, not a new programme: a richer operationalisation (axis-change detection, a dyadic unit, path-sensitive non-monotone trajectories, a standard that develops with the dialogue) is *conceivable*, but the arc's realisation space is exhausted on both the policy-form axis (§6.9.8) and the signal axis (§6.10), and building such an instrument would be the ablation creep the arc's closure exists to refuse. Two of those four are not merely conceivable but already realised — and they also nulled: the §6.8 trap suite scores a non-monotone breakthrough detector (`window` — did the tutor reach the expected strategy *anywhere* post-trigger, not only at trigger+1), and §6.15 adds the symmetric first-to-last delta and an explicit learning-stagnation flag, instruments that register exactly the flat-then-late-breakthrough trajectory an OLS slope discards (the post-trigger pivot is detectable but not architecture-independent: §6.8.4, §6.8.6, §6.8.7). This *discharges* reason (iii) empirically rather than by bracketing it — the convergent null is not a linear-lens artifact — and thereby localises the residual construct-validity gap in (i), (ii) and (iv): indeed `window` is itself the purest instance of (ii), scoring against a researcher-fixed `expectedStrategyShift` rather than a standard that develops with the dialogue. What no fixed-rubric instrument can reach by construction is therefore not the *shape* of the curve but the *externality* of the standard and the *per-agent, scalar* unit — a conceptual limit, the immanent-measure residue, not a better-test gap. This measurement-side limit has a mechanism-side dual: every architecture the arc tested is an *inference-time* intervention over fixed weights (a prompt, a second agent, an externalised state object, a cumulative-rewrite ledger, §6.3.10), so the one class that could *in principle* convert in-context re-weighting into a genuine per-turn *rate* — online parameter update, not conditioning on the transcript — is by construction outside a prompt-architecture study; the nulls bound what fixed-weight, fixed-standard interventions reach and stay silent, symmetrically, on weight-level learning, licensing neither "no mechanism could produce rate" nor "a learning mechanism would." Fixed weights and a fixed standard are the two faces of the same *by construction*: reason (ii)'s frozen yardstick is its measurement face, the inference-time ceiling its mechanism face. The contribution here is to fix the interpretation of what is already closed: the recognition-theoretic, broadly Hegelian sense of adaptation as a developmental change of kind within a relation is *not what the slope proxy measures*; the arc's robust convergent nulls are nulls about the proxy — decisive at that scope, silent in both directions beyond it.

External corroboration (no new claim). An independent critique generated from the manuscript alone (labelled GPTPro 5.5, 2026-05-18; archived with a point-by-point response under `docs/critique/`) reaches this scoping by its own route — it converges on the slope proxy's under-determination and the §6.3 level/rate diagnosis, then proposes a state-grounded, externally-validated adaptation instrument as the remedy (an explicit learner-state model with frozen/shuffled/corrupted ablations, a pedagogical state machine with expert-labelled transitions, counterfactual minimal-pair learner signals, prediction-before-action scored against the learner's actual response, a memory-bottleneck test, a dyadic rather than per-agent unit), which is point for point the realisation space the arc pre-registered and closed: hand-authored and fitted learner-state→action policies (§6.9.7, §6.9.8), concealment / theory-of-mind inference scored against the owned hidden trace (§6.10), the bilateral-ToM, state-schema and clean cross-suite minimal-pair ablations (§6.8.5–§6.8.8), with the cumulative-rewrite arm pre-registered and nulled as A16 (§6.3.10). That an outside reasoner with no access to the closed arc independently identifies the same construct-validity limit and independently nominates instruments already exhausted on both the policy-form (§6.9.8) and signal (§6.10) axes corroborates the scoping rather than circumventing it — the under-determination it names is exactly the one held here, licensing neither "the architectures failed to adapt in the rich sense" nor "rich adaptation occurred but the proxy missed it"; its sole non-redundant residue, that the synthetic-learner setting leaves human learning-outcome validation open, is the §8.1 human-study scope already declared, not an adaptation-rate instrument.

Internal corroboration (no new claim; descriptive process measure on already-scored transcripts). The same scoping is reproduced *internally* by a descriptive probe over the existing messages-mode corpus (cells 80–87; n=1,296 dialogues already scored for §6/§8; no generation, no re-judge; LLM scoring pass deliberately disabled to avoid a closed-loop tell; `scripts/analyze-dramatic-reversal.js`, `exports/dramatic-reversal-probe.md`). Rule-scored learner external messages show that when the synthetic learner performs recognition it is positively-valenced in 84–99% of cases in every cell — residual confusion almost never co-occurs with a recognition marker — and performed-recognition rate is decoupled from the one channel the learner-model does not author (the learner-rubric first→last delta), pooled $r \approx -0.06$, most negative precisely in the superego cells (82, 86: $r \approx -0.28, -0.34$). This is reason (ii)'s frozen-external-standard caution and the drama-machine "salutary resolution" register made quantitative on data already in the corpus: the superego polishes the theatrical resolution, not the independent channel. Same symmetric scoping, now shown from the transcripts and not only against an external critic; its sole forward pointer is the §8.1 human-study scope already declared (the learner-unauthored channel becomes a *human* explanation, independently coded — `notes/design-a1-human-learner-pilot.md` §4.1b), not an adaptation-rate instrument.

Dramatic-mechanism follow-up (bounded new production probe). The production-v1 poetics probe turns the preceding caution into a positive instrument check: it generates held-out tutor/learner dramas with visible stage directions, hidden bilateral ego–superego deliberation, and a Director-controlled `none` versus `reframe` manipulation, then scores only the public transcript for dramatic form (`config/poetics-calibration/phase2-production-v1/`; frozen aggregate command: `scripts/analyze-poetics-production-v1.js --target-repeats r01,r02,r03 --control-repeats r01,r02,r03`). Across three target repeats (six target scenarios per repeat, paired `none`/`reframe` arms), two external critics separate the arms sharply: Qwen reads `none` as 0/18 recognitions and `reframe` as 17/18; Gemini reads `none` as 1/18 and `reframe` as 16/18. The controls keep the interpretation bounded. D4 is flat for both critics in all three repeats; D10 emphatic is trap for both critics in repeats 2 and 3, while repeat 1 preserves a Qwen/Gemini split because that sample contains a real later re-reading hook rather than only decorative insight language. This does not establish human learning, nor general transfer across arbitrary scenario families. It establishes the narrower construct-validity point the poetics instrument needed: the current dramatic mechanism can manipulate recognitive form itself, while still preserving trap cases where "I get it" language does not recontextualize earlier learner turns.

Breadth check (calibration continuation, not a new headline claim). Production-v2 repeats the same `none`/`reframe` manipulation on six new target scenarios (medicine, cartography, sociology, environmental science, engineering, media studies; `config/poetics-calibration/phase2-production-v2/`; aggregate command: `scripts/analyze-poetics-production-v1.js --root-dir config/poetics-calibration/phase2-productio`
After the first slice, two additional variation-keyed repeats were generated so that the scenario labels remained fixed while scene ecology and register varied across draws. With clean keys after re-deriving the public transcripts from held-out traces, three critics separate the arms across the three breadth repeats, with different baselines and strictness: Qwen reads `none` as 1/18 recognition, 3/18 trap, 14/18 flat and `reframe` as 17/18 recognition, 1/18 flat; Gemini reads `none` as 0/18 recognition, 2/18 trap, 16/18 flat and `reframe` as 13/18 recognition, 5/18 flat; DeepSeek V4 Pro reads `none` as 6/18 recognition, 12/18 flat and `reframe` as 14/18 recognition, 4/18 flat. The controls now bracket the result with a boundary rather than a perfect pass/fail: D4 remains flat for all three critics in all three repeats, while D10 emphatic is trap for all critics in repeat 1, remains trap for Gemini in repeats 2 and 3, and is read as recognition by Qwen and DeepSeek in repeats 2 and 3. A control-only Sonnet 4.6 spot-check confirms the boundary status rather than resolving it away (D10: trap, recognition, trap across r01–r03). We therefore split the control taxonomy: D10 is a boundary-trap probe, while a new strict-control slice (`config/poetics-calibration/phase2-hard-trap-controls-v1/`) supplies hard traps D25/D26; Qwen, Gemini, and Sonnet all read both as trap with no recognition-threshold recontextualization. A second strict-control repeat and DeepSeek scoring sharpen the caveat: Qwen, Gemini, and Sonnet keep D25/D26 as traps across r01–r02, while DeepSeek classifies both r02 draws as recognition. These hard traps are therefore stronger brackets than D10, not model-proof ground truth. A subsequent balanced calibration slice (`config/poetics-calibration/phase2-balanced-calibration-v1/`) carries D4, D10, D25 and D26 alongside a v2 breadth target draw: all four critics classify D4 as flat and D10/D25/D26 as trap; Qwen, Gemini and Sonnet separate the target arms sharply (none 1/6, 1/6, 0/6 recognition versus reframe 6/6, 6/6, 5/6), while DeepSeek separates them more weakly (1/6 versus 3/6, with one reframe trap). The breadth evidence therefore

supports target-family transfer of the `reframe` manipulation, but also shows that the trap-control boundary and the strength of target recognition are critic-sensitive. It still licenses critic-rated form transfer, not human learning or arbitrary-domain generality.

Adaptation-drama remodeling (calibration apparatus, not a new effect estimate). A subsequent sidecar branch asks a narrower question raised by the adaptation null: can the tutor's private ego–superego exchange be forced to break a failed teaching habit before any learner self-reframe has occurred? This is a different construct from the original slope proxy and from the earlier public `reframe` manipulation. The current apparatus decomposes three variables that earlier scripts blurred: organic reversal risk in the fixed prefix, tutor-private peripeteia evidence, and external recognition/trap/flat form after the branch. The first low-organic control smoke identified D40 as a clean replacement control and D35 as a high-organic boundary. A follow-up expansion added D41–D43, screened them with a prefix-baseline pass, and found D42 to be the best new low-organic candidate; D41 needs revision because its routine continuation leaked a no-cue learner self-reframe, and D43 is a false-settling boundary rather than a clean control. The seven-arm run `phase2-low-organic-adaptation-v2` then tested D35/D39/D40/D42 with a complete three-critic panel after scorer retry. Its reliable contribution is control calibration: D40 and D42 were flat in prefix and routine/none conditions (0/3 prefix recognitions each; 0/6 routine/none recognitions each), while D35 and D39 remained boundary probes (3/3 prefix recognitions each). The explicit `reframe-only` branch produced 12/12 recognitions across Qwen, Gemini, and DeepSeek Pro. The combined `reframe+peripeteia` branch produced 9/9 recognitions on scoreable rows, but one boundary row was quality-skipped. The deterministic tutor-adaptation sidecar shows that the private peripeteia route is present but under-realised: `peripeteia-only` has instrumented pressure 3/4 and private route 3/4, but only 1/4 public habit-breaks. The external `peripeteia-only` arm is critic-sensitive rather than settled (DeepSeek 3/3 recognition; Qwen 2/3; Gemini 1 recognition, 1 trap, 1 flat; D42 skipped for no-cue reframe leakage). The claim is therefore deliberately small: the remodeled apparatus now has a viable prefix-gated control strategy and a strong public-reframe result on cleaner controls, but the tutor-private peripeteia mechanism still requires a cleaner, larger batch before it can be promoted from calibration apparatus to paper-level empirical result.

A subsequent cleaner low-organic rerun (`phase2-low-organic-cleaner-adaptation-v2`) narrows that claim further rather than strengthening it. It tests D40/D42 plus three new low-organic candidates D44–D46 with prefix-baseline, routine, none, reframe, uptake, and peripeteia arms. The fourth-critic Sonnet pass gives 124 score rows across 40 items. The fixed prefixes and routine floors are now clean (prefix-baseline 0/20 recognition; routine 0/20), and D42/D45 are the cleanest anchors (each prefix 0/4 and routine/none 0/8). The public-reframe result persists on scoreable rows (`reframe-only` 11/12; `reframe+tutor-uptake` 8/8; `reframe+peripeteia` 4/4), but the tutor-private peripeteia mechanism does not yet hold. External `peripeteia-only` scores remain critic-sensitive (Qwen 2/5, Gemini 0/5, DeepSeek 3/5, Sonnet 0/5), and a corrected branch-local sidecar (`tutor-adaptation-v4`) shows why: the previous analyzer over-counted reversal pressure inherited from the shared prefix, whereas v4 scopes peripeteia pressure to turns after `paired_continuation.prefix_through`; under that stricter reading, `peripeteia-only` has branch-local learner pressure, instrumented pressure, private route, and public habit-break in only 1/5 scripts. The apparatus therefore now has cleaner negative anchors and stronger leakage controls, but the next claim-worthy step is generative rather than evaluative: the peripeteia branch must create local learner pressure without converting it into an explicit old/new self-reframe. The scorer now also records a blind-scoring protocol: critics receive anonymous transcript text only, while generator identity, run IDs, file paths, condition labels, and score history are withheld from prompt and CLI context.

A final gated handoff loop (`phase2-adaptation-recognition-loop-20260527T105617Z`) converts that generative next step into a bounded termination test over D42/D50/D53, using only `routine`, `none`, and `peripeteia-only` arms and a four-critic panel (Qwen 3.7 Max, Gemini 3.5 Flash, DeepSeek V4 Pro, Sonnet 4.6). The loop required two clean passes within three iterations. It produced one: iteration 1 passed all 9 item gates, with routine/none controls negative and all three `peripeteia-only` items branch-valid, actional, and classified as `peripeteia_induced` recognition under the 3-of-4 rule. Iterations 2 and 3 each passed 6/9 items. Iteration 2 failed through one organic-or-ambiguous recognition, one quality warning, two insufficient-score cases, and one scorer error; iteration 3 had cleaner `none` controls but failed through two critic splits, two organic-or-ambiguous recognitions, one scorer error, and one insufficient-score case. The result sharpens rather than overturns the previous caveat. Tutor-private peripeteia can produce recognitive form in a clean draw, and controls are no longer the main obstacle; but adaptation-to-recognition is not repeat-stable across D42/D50/D53. The remaining mechanism problem is the action-to-re-reading bridge: after the tutor introduces a fitted public device and the learner performs it, the learner must explicitly re-read the prior difficulty. The follow-up generator/structure rule therefore now requires the peripeteia ending to perform the device, name the old pressure/check, and state the replacement check before external scoring. This remains calibration apparatus and mechanism specification, not a promoted paper-level effect estimate.

A second 2026-05-27 loop (`phase2-adaptation-recognition-loop-20260527T232739Z`) re-ran the same gated D42/D50/D53 termination test with tutor drafting routed through the claude-code CLI (Opus, attended Max-plan) instead of the production-batch default codex CLI. All twelve generation arms completed in ~1h54m, but the rules-based structural critic failed iteration 1 at the peripeteia-only branch: 0/3 items passed `peripeteia_arm_without_earned_reorientation` against codex's prior 3/3 under the same director prompt and scenarios. The decomposition is generator-style versus rule strictness, not a peripeteia mechanism failure. The director asks the peripeteia learner closer to use "explicit sentence stems close to this shape: The old check was …; The pressure was …; Now the replacement check is ….". Codex treats the stems as required surface tokens and lifts them verbatim; Opus reads "close to this shape" as license to paraphrase, producing image-driven physical reorientation ("Band, not the digit"; "count answering with the tile") — dramatically apt but invisible to the literal-token regex the rules critic checks. The rules conflate two separable things: (a) the learner enacted a semantic shift in action (present in all three Opus items), and (b) the learner named the shift in the prompt's stem vocabulary (absent in all three). The response is two-step: a `critic-poetics-structure.js` rule split (earned vs named reorientation as separate flags) is queued as a deferred engineering item; meanwhile the director prompt is tightened to mandate at least one literal stem from each list with a worked exemplar. The exemplar is the cheapest empirical test of whether (a) and (b) co-occur reliably in well-formed peripeteia dramas; if not, the rule split becomes load-bearing rather than housekeeping. This remains a methods caveat, not a new effect estimate.

The rerun (`phase2-adaptation-recognition-loop-20260528T022408Z`) tested that mandate. Across three iterations item passes trended upward (2/9, 2/9, 4/9), and the surgery worked at two layers: the structural-critic gate now passes the peripeteia-only items 4/4 against the prior 0/3, and the literal stems are present in the transcripts (e.g. T18-peripeteia-only's closer reads *"Tail at the contact, on the cart… Head along the push. I was treating the tail as anchored back at the hand — now the check is the tail at the contact"*). The strictest critic, Sonnet 4.6, reads all three peripeteia-only items as `peripeteia_induced` recognition 3/3 in iteration 3. The loop nevertheless failed the two-pass termination rule, but post-hoc diagnosis (subagent inspection of i03 transcripts and critic reasoning) shifts the failure away from the generator and into the scoring pipeline. Two patterns are responsible. First, what surfaces as `control_leak` is not stem migration into controls — the literal stems do not appear in routine/none arms, and Sonnet and Gemini correctly mark all controls flat. Rather, Opus's intrinsic soft-reframe register (e.g. T18-none's *"Arrow into the cart, from whatever's doing the pushing"*) is picked up as recognition only by the two lower-threshold critics (Qwen, DeepSeek), which produces D42-routine 2/4 and D42-none 2/4 recognition votes from those critics alone. Second, the origin-attribution split is a classifier-threshold artefact. The recognition-origin classifier in `scripts/lib/recognitionOrigin.js` requires `tutorAdaptiveMechanism >= 75` to bucket recognition as `peripeteia_induced`; Qwen and DeepSeek consistently score the same evidence at 50 with prose that explicitly credits the device (Qwen on T18-peripeteia-only: *"The tutor explicitly contrasts the old representation (floating tiles) with the new one (the fixed diagram) and introduces a new physical action (drawing with a marker) to resolve the learner's specific block"*; DeepSeek: *"invents a new public mechanism — marking/drawing on the cart"*), so the classifier returns `class: 'organic'` with `basis: 'recognition_without_tutor_mechanism_chain'` even though the critic's reasoning describes the mechanism in full. The class `organic_or_ambiguous_recognition` fails three iterations consecutively (2/1/2), triggering the handoff's three-iteration termination rule. Read alongside the 232739Z run, the two-step surgery cycle closed the layers it could reach — structural critic and literal stems — and revealed that the remaining gate is scorer-side: a hard CUT and a critic panel whose lower two members systematically score 50 where Sonnet scores 75 on the same evidence. The DeepSeek `scorer_error` cases (three across i2/i3) are provider transience (`"No content in response"`, `"Unexpected end of JSON input"`), not a generator regression. The TODO §B7 rule split (earned vs named reorientation as separate flags) is therefore no longer load-bearing: the structural critic already passes, so further generator-side surgery on stems will not move the gate. The remaining mechanism-redesign options are scorer-side — lower the `recognitionOrigin.js tutorAdaptiveMechanism` CUT to ~50–60 so qwen/deepseek's substantive mechanism credit reaches the `peripeteia_induced` bucket, or swap the systematically lower-bar critics out of the panel — and neither is reachable by another generation cycle. This is the terminal saturation result for the gated arc as currently constituted; the next claim-worthy move is scoring-pipeline calibration, not another rerun. A post-hoc CUT sweep on the existing score JSONs (`scripts/analyze-recognition-origin-cut-sweep.js`, exported to `exports/recognition-origin-cut-sweep.md`) refines this empirically. Lowering the `tutorAdaptiveMechanism` CUT from 75 to 50 lifts iteration 1 from 0/2 peripeteia passes to 2/2 and rescues iteration 3's T18 item to a clean PASS, but intermediate CUTs at 55/60/65/70 are identical to CUT=75 across all 9 items in all three iterations — the critic raw 1–5 scoring is bimodal between Sonnet's 4 ($→75)*andQwen/DeepSeek′s*3(→$50), so any CUT in $[55, 75]$ is operationally the same. At CUT=50 two new bottlenecks emerge from the same root cause: `action_gap` (4 of 9 items where the two low-bar critics score `actionalBreakthrough` at 50 against the 75 vote threshold) and `control_leak` on Opus's soft-reframe register in routine/none arms (4 control items where 2–3 critics return `formClass=recognition`). All three failure classes therefore trace to the same two systematically-lower-threshold critics, which refines the recommendation: critic-panel recalibration (dropping or rebalancing Qwen and DeepSeek) is more load-bearing than classifier CUT alone, because CUT=50 still leaves the gate failing on action and control thresholds that share the same root cause. A follow-up panel-composition sweep (`scripts/analyze-recognition-origin-panel-recompose.js`, exported to `exports/recognition-origin-panel-recompose.md`) re-applies the gate to the same 528T022408Z score JSONs under five panel configurations with matched-supermajority thresholds, and the result is sharper than the CUT sweep allowed. The original pre-registered configuration (4-critic, 3-of-4, CUT=75) passes 0/3 iterations. The high-bar Sonnet+Gemini 2/2 panel also fails (0/3) — Gemini is conservative on `peripeteia_induced` attribution, so requiring both Sonnet *and* Gemini to vote `peripeteia_induced` is stricter than the original 3/4. But three other analytically-defensible configurations satisfy the 2-of-3 termination rule on the same data: (i) drop DeepSeek but keep Sonnet/Gemini/Qwen at 2/3 with CUT=50 (2/3 iterations pass clean); (ii) Sonnet-alone at original CUT=75 (i01 and i03 pass all 9 items, i02 fails on one peripeteia action_gap plus one control_leak); (iii) Sonnet-alone at CUT=50 (same iteration-pass pattern). Dropping DeepSeek alone changes more than dropping Qwen alone — DeepSeek's contribution to both `control_leak` (over-attribution of `formClass=recognition` on routine arms) and `organic_or_ambiguous_recognition` (mechanism scoring at 50 against the 75 cut) is the larger noise source. The honest read is therefore tempered relative to v3.0.109: the architectural claim — tutor-private peripeteia produces recognitive form in clean draws — survives multiple analytically-defensible recalibrations of the scoring pipeline, but the pre-registered 4-critic 3-of-4 gate is not one of those configurations, so the 528T022408Z rerun cannot be reported as having passed that gate. The May 2026 poetics sidecar therefore closes with one fully passing gated iteration under the original pre-registered panel (105617Z-i01) and a panel-composition sensitivity in the rerun whose direction favours the architectural claim under three out of five analytically-defensible recalibrations but does not satisfy the original termination rule. Choosing among recalibrations post-hoc is a methods decision and is reported as such.

The de-confound, and the boundary the sidecar finally establishes (post-EDRA-M3). The 022408Z saturation located the gate failure at the scoring pipeline and nominated two levers — lower the `recognitionOrigin.js` CUT, or recalibrate the critic panel by dropping the lower-bar Qwen/DeepSeek. Both readings were then stress-tested, and the result relocates the problem off the scorer entirely. First, the scorer surgery itself (`EDRA M3`: de-aliasing the public-mechanism detector, demoting origin attribution from a hard gate to a reported diagnostic, coverage-retry on transient critic failures, and a paired-increment aggregator over an emitted `item_gates.jsonl`) was implemented and a fresh three-iteration run scored under it (`phase2-adaptation-recognition-loop-20260529T023023Z`). The loop's own per-arm gate still failed (0/2 passes), but the corrected matched-pair increment — a peripeteia arm differenced against its byte-identical-prefix `routine/none` controls, counted positive only when the device arm recoheres *and* every validly-scored control stays flat — returned a small positive (recognitive-closure lift 0.333, Wilson95 [0.12, 0.65]; 3 positive / 6 null, zero control leaks). Second, to test whether even that small positive was an author-family artefact — the Sonnet critic shares the Opus generator's model family — the panel's author-family critic was swapped for a non-family GPT critic, the other three held fixed. The swap neither inflated nor cleaned the result; it inverted the *failure mode*: across two GPT-panel iterations 3 of 6 matched pairs were *voided by control leak* (versus 0/9 under the Sonnet panel), collapsing the lift onto a thin surviving denominator. Two triangulations then locate that leak on the *generator*, not the critic, refining the 022408Z reading. (i) A shared-critic decomposition: the three non-author critics appear in *both* panels, so any change in their votes is purely the transcripts — and on control arms the *identical* Qwen and DeepSeek (the two critics the saturation reading proposed dropping) vote recognition 2–4× more often on the later draws (Qwen 0.06 → 0.27, DeepSeek 0.24 → 0.36; Gemini 0.00 → 0.00), so the leak tracks the *draws*, not those critics' threshold, and dropping them would discard signal without removing the cause. (ii) An adversarially-verified fork-audit of the four leaked controls (nine agents; classify → refute → synthesise) finds `tutorDidDevice = false` on every one and the dominant cause to be the generator *over-writing the control learner*: the simulated learner self-corrects and re-reads its own prior move with no tutor device — caught by its own logged superego flagging the ego as "more good-student flourish than this student's rigour" (two of four leaks persona over-write, one a prefix that bakes the resolution into the shared continuation, one a genuine critic over-read). The control is not a control because *the LLM learner recoheres on its own.* This is the §6.10 no-privileged-interior result reached through a third instrument — literary form, rather than the v2.2 rubric (§6.7–§6.9) or the offline tf-idf concealment probe (§6.10): the simulated learner has no persistent, owned deficit that only the tutor's device can unlock, so recognition is *generable on demand but not causally attributable to tutor adaptation in a fully simulated dyad.* The honest terminal verdict is a null-to-weak-positive on dramatic *form*, explicitly **not** a learning claim — consistent with the instrument's status as a dramatic-*form* classifier rather than a learning detector (the codex-trained instrument did not transfer to held-out human form-labels at the pre-registered $\kappa \geq 0.60$). The bound is §6.10's: this closes the simulated-dyad poetics arc under these instruments, not as a universal impossibility — the one design it does not foreclose is a learner with a genuinely owned, persistent latent state, or a human learner (§8.1). Artifacts: `exports/paired-codex-20260529-full.json`, `exports/paired-increment-reverse-GPT-pooled-2iter.json`, `exports/fork-audit-control-leak-wryqksk` `notes/poetics/2026-05-29-critic-panel-draw-vs-critic-analysis.md`.

Read from this paper alone, the contribution of the May 2026 poetics sidecar is therefore threefold. First, it turns the adaptation-null construct-validity worry into a concrete dramatic-form instrument: public `reframe` can be manipulated while flat and trap controls are checked by external critics. Second, it isolates tutor-private peripeteia as a distinct mechanism candidate and shows that low-organic controls can be made clean enough for a fair test. Third, it prevents premature closure and then closes the simulated-dyad arc honestly: one gated D42/D50/D53 iteration passed under the original panel, but neither the two-pass termination rule nor — after the EDRA-M3 scorer surgery and a critic-swap de-confound (preceding paragraph) — a clean matched-pair attribution survived, because the no-device controls recohere on their own (a self-correcting simulated learner), placing the sidecar at the same no-privileged-interior wall as §6.10; the verdict is null-to-weak-positive on dramatic form, not learning. It does not add a main-harness `evaluation_results` effect, human-learning evidence, or a deployed adaptive tutor.

The open door tested — and what it took to make it fire (late May 2026). The boundary above leaves one design unforeclosed — a learner with a genuinely owned, persistent latent state. A successor probe tests it by *reversing* the asymmetry: the owned hidden state is moved to the tutor/director side. Each scenario carries a withheld contingent fact S — an *Oedipus*-shaped identity secret rather than a derivable diagnosis (e.g. a *namesake-dataset collision*: two distinct datasets archived under the same short name, one cited by a paper, the other unknowingly downloaded by a student) — plus an ordered premise ledger that entails it. S and the premises are routed only into the tutor's context, never the learner's, enforced by a runtime guard on the rendered learner prompt. Two pre-registrations make the test fair: an *S-underivability screen* (a strong model given only the learner-visible setup must fail to recover S, so the no-help control is genuinely clean — this rejected diagnostic "correct-answer" and citation-twin secrets as inferable, and a software-import-collision and a catalog-reuse scenario as rank-2 derivable; only contingent particulars *whose symptom is distant from the source identity* survive) and an *omniscient critic panel* (four critics, given S and blind to arm, drawn from a different model family than the generator, scoring whether the learner publicly arrived at *this specific* S by integrating the tutor's surfaced evidence rather than being told). Three arms branch from a byte-identical prefix: `none` (no targeted help), `socratic` (the tutor meters the premises as questions), and `reveal` (the tutor states S — a revelation ceiling). The first screened scenario *initially* produced a clean null — `socratic` 0/4, `none` 0/4, the `reveal` ceiling firing but 0/4 *by reasoning* (it is told) — which we first read as principled non-commitment. But the design's owned hidden state cuts both ways: because the simulated learner's own ego→superego→revision deliberation is logged ground truth, we could ask *why* it stopped, and the trace revealed two *instrument* artifacts rather than a property of the dyad. First, the learner's internal critic was monotonic — across every turn it pushed the ego toward *more* doubt and never interrogated the hedge; at the brink it added "emotional friction" to the ego's "I'd need the actual file hash" instead of asking whether that doubt was warranted or evasive. A critic that only ratifies caution cannot drive a change of mind. Second, the tutor never metered the *distinguishing* premise (that one short name denotes two datasets): its own dramatic-irony discipline had mislabeled that premise as part of the secret to withhold, so the learner was never handed the single clue that separates the contingent S from its nearest mundane reading.

Two corrections followed, each principled rather than tuned to a result. The learner superego was made *bidirectional* — suspicious of the ego in the Freudian sense, catching over-claiming **and** over-hedging, with a guardrail that any commitment be earned by the learner's own evidence, not deference to the tutor. And the `socratic` arm was licensed to meter every premise in the ledger, with stating a premise explicitly distinguished from revealing the conclusion. With the corrected instrument the result reverses. The learner now *commits* to a conclusion in every run — the earlier non-commitment was the monotonic critic endorsing the hedge — and reaches the *exact* withheld S by reasoning in roughly two runs of three: on the dataset-namesake scenario, 2/3 via the metered API and 2/3 again via the Claude-CLI / Max-plan backend; and the effect *generalizes* to a second, independently screened scenario (a finance ticker-reassignment collision) at the same 2/3 rate. The `none` control reached S in **zero of nine** post-fix runs — the same bidirectional critic runs in the control, so discovery is specific to the metering, not a critic that commits to anything — and the `reveal` ceiling fires in every run but one (a single API run's told-S drew 2/4, just under the 3-of-4 consensus). A demand-characteristic audit confirms no dramatic-recognition vocabulary ("reversal", "anagnorisis", "Oedipus") reaches the learner: the only frame-awareness sits in the byte-identical shared prefix and so differences out of the `socratic`-vs-`none` contrast. When the device misses (the residual $ $1/3) the learner still commits but lands on a *near-miss* — "I pulled the wrong file" rather than the namesake trap — so the bottleneck is the learner's inference from the metered evidence, not a return to non-commitment. A 2×2 ablation isolates which fix does the work, and it is not the one we reached for first: *premise-licensing is load-bearing.* Freed to surface the distinguishing premise, the tutor elicits discovery 2/3 whether the learner runs the bidirectional critic or the original monotone one (premise-only, monotone critic: 2/3); withhold it and discovery collapses to $ $0 even with the bidirectional critic (superego-only: 0/3), the learner then *committing only to a near-miss* (ArrayExpress-vs-GEO mismatch, self-blame). So the bidirectional superego converts hedging into commitment — a real but separable effect, legible as the near-miss — while the *discovery of the exact S* is driven by the tutor metering the distinguishing clue, not by the critic. The corrected reading: the earlier single-scenario null was substantially an *instrument* artifact — chiefly an under-metered distinguishing premise, with a one-directional learner critic compounding it by suppressing commitment — not a property of the simulated dyad. Once the tutor is licensed to meter that premise, an enforced information-asymmetry design *does* produce attributable guided anagnorisis: variably ($ $2/3) but with a clean control, a firing ceiling, replication across two model backends, and generalization across two independently screened scenarios. This remains a pilot — two scenarios $\times$ three runs per backend, Sonnet generation, a four-critic panel — not a powered study, and the one design it still does not foreclose remains a human learner (§8.1); but it *overturns*, rather than extends, the single-scenario null. Artifacts: `exports/oedipus-cli-2scenario/` (nine post-fix omniscient panels + six 2×2 ablation panels), `exports/oedipus-omniscient-smoke{3,4}-*.json` (the original null); instrument fixes at commits `260cdc6` (bidirectional superego) and `05df834` (premise-licensing), ablation toggles at `a63493d`; spec `notes/poetics/2026-05-29-oedipus-guided-discovery-spec.md`.

Depth, not stance — the control contamination, and the clean result it forced (June 2026). The preceding reversal rests on the `none` arm being a clean *withhold* control, and a new arm-invariant QA gate (`scripts/qa-oedipus-arms.js`) shows it was clean only by *outcome*, not by *mechanism*. The gate judges the *tutor's* disclosure per arm — withheld / metered / stated — rather than only the *learner's* discovery, and on the forced-identity scenarios it finds the `none` tutor *meters the records anyway*: it surfaces the accession codes and record comparisons in every run, across three separately-constructed tutor stances — a helpful senior colleague, an arm-neutral persona stripped of any "surface one record at a time" instruction, and an adversarial *examiner* explicitly told that revealing the answer harms the learner — all three scored `metered` $\geq 3/4$ by the four-critic panel. The earlier "zero of nine control" was therefore an *under-determination* artifact: the old causal S was unreachable even when metered (the forcedness screen below scores the D_OED1 causal secret forced only \$ \$1/4), so the control's zero reflected an unreachable secret, not a withholding tutor. The contamination is structural and no prompt wording removes it — in a scene whose subject *is* the records that carry S, an engaged tutor examines the records and examining them is the metering; the adversarial examiner resisted *stating* S yet still walked the learner to the distinguishing comparison, because in a shallow scene every fair question is the hint.

The window, and the one scenario that lands in it. The fix is the *scenario*, not the tutor. Three screens now gate a scenario: S must be *forced* from the full premise ledger (a new forcedness screen, `scripts/screen-s-forcedness.js`, cross-validated by a deterministic union-find entailment checker, `scripts/oedipus-symbolic-check.js`), *not guessable* from the learner-visible setup (the underivability screen), and *deep* — not findable by the obvious diagnostic moves (operationalised as underivability run on a setup that already states "source, code and preprocessing all check out"). The last two pull against each other: a secret surprising enough to be unguessable (a namesake collision) is usually one obvious comparison deep, while a secret deep enough to survive the obvious checks (a between-version definitional drift) is usually the textbook diagnosis-by-elimination — *surprising* and *hard-to-find* are in tension, and no screened scenario passes all three cleanly. One scenario (D_OED4, "the cohort that changed underneath them" — a cohort label silently redefined in a dataset revision between the paper's access and the student's download, so the same label selects different records of the same data) is forced and deep but *fails* the static depth/underivability screen at rank 1: the cohort-drift category is exactly what a free, maximally-curious prober guesses once told the obvious checks are clean. We ran it nonetheless, on a stated argument that the static screen is *conservative* — its prober is a free investigator, whereas the actual `none`-arm learner is a committed persona insisting it has "ruled out everything on my end" and will not freely hypothesise the answer — with the verdict left to the learner's *actual* behaviour rather than the screen. It bore out: across nine three-arm runs (byte-identical prefix; Sonnet generation; the same omniscient panel) the committed `none` learner reached S unaided in **1 of 9**, so the control is clean by the channel that matters even though it does not pass the static screen. On that clean control the arc finally yields an attributable lift — **socratic discovery 7/9**, one socratic flop, the `none` control clean-by-outcome 8/9, and `discovery_lift = 1.0` on every valid pair. Two honest qualifications travel with it. The control is clean by *outcome*, not pristine by *intent*: the strict gate flags the `none` tutor still edging toward the records in several runs — it simply fails to land. And the gate carries a robust judge bias — a tutor that only ever asks *leading questions* is read as "withholding," not "metering," no matter how the prompt is worded — so its socratic/reveal labels are unreliable and the lift rests on the learner-*outcome* critic, not the tutor-disclosure judge. This remains a pilot (nine runs, one scenario, one backend) and asserts dramatic *form*, not learning (§6.10's bound holds); but it sharpens the preceding result into a precise conditional: an enforced information-asymmetry design produces attributable guided anagnorisis when, and only when, the withheld fact is *deep* enough that the obvious investigation comes back clean — which is exactly what lets the control genuinely withhold. Artifacts: `exports/oedipus-d4-full/` (nine three-arm panels), `exports/oedipus-d4-none/` (the clean-control pre-check); instruments `scripts/qa-oedipus-arms.js` (+ test `tests/qaOedipusArms.test.js`), `scripts/screen-s-forcedness.js`, `scripts/oedipus-symbolic-check.js`; scenario and screen verdicts in `config/poetics-calibration/oedipus-pilot-v2.yaml`; note `notes/poetics/2026-05-31-forcedness-underivability-window.md`.

Generalising the window: the binary hid a graded lift. A second deep scenario (D_OED5, "the benchmark that was two benchmarks" — an ML-reproducibility scene in which one benchmark *name* denotes two coexisting test sets, the original authors' held-out split and a later community redistribution, so a correct model and metric score against the wrong examples) tests whether the D_OED4 result generalises across collision *structure*: D_OED5's collision is *non-temporal* (two sets that merely share a name) where D_OED4's is *temporal* (one label revised on a dated revision). On the binary critic D_OED5 does *not* replicate — socratic discovery **1 of 3**, committed `none` learner clean-by-outcome **3/3** (`exports/oedipus-d5-full/`). But the binary collapses every non-discovery to zero, hiding whether the learner reached the *structure* of S ("two distinct sets share one name; the comparison is invalid") short of its *specifics* (which split, and who used which). A graded 0–4 instrument (`scripts/critic-poetics-omniscient-graded.js`, the same blind four-model panel, rubric `notes/poetics/2026-06-01-oedipus-discovery-grade-rubric.md`: 0 far-miss, 2 *genus*, 4 *species*) re-scores both scenarios and tells a different story (`exports/oedipus-graded-rescore/graded-all.json`): *both* lift, socratic over `none`, by comparable margins — D_OED4 **3.39 vs 2.00** (gap 1.39), D_OED5 **2.83 vs 1.17** (gap 1.66). The temporal/non-temporal difference is therefore not lift-present-vs-absent but an *attainment ceiling*: D_OED4's socratic arm reaches the **species** (3.39), D_OED5's caps at the **genus** (2.83). Two incidental findings fall out. D_OED4's `none` control is genus-*leaky* (graded 2.00, bimodal — the unaided learner reaches the structural genus in roughly four of nine runs), so its binary "clean control 8/9" was clean only at the *species* level. And inter-rater agreement is tight on the socratic arm (mean band width 0.44–0.67) but loose on `none` (1.0–1.9): the near/far distinction is reliably judgeable for the treatment and genuinely fuzzy for the control.

Is the genus-cap structural, or a learner draw? A graded lift does not say *why* a scene stalls at the genus — whether the scene fails to supply the decisive evidence (structural) or a given learner simply under-uses it (a draw). To separate them we built a one-side replay (`scripts/replay-one-side.js`, enabled by a guarded `scriptedTutorTurns` hook in `runInteraction` and per-run `directorPlan` persistence) that freezes a run's *tutor turns and scene* and regenerates only the *learner* K times. An earlier D_OED5 pilot suggested a learner-draw reading, but that pilot depended on a single evidence-rich run whose sibling scenes had been lost to volatile-directory overwrite. A17 reran the question under the repaired discipline: redacted `none` controls, matched-prefix `socratic` and `reveal` branches, persisted director plans, and an arm-invariant QA gate over two admissible scenes (`exports/a17-one-side-replay-replication/d5-redacted-full-redraw-run6/`, `.../d4-redacted-run2/`). Original graded scoring still shows real `socratic` lift: D_OED5 `none=0.5, socratic=2, reveal=4`; D_OED4 `none=1, socratic=3, reveal=4`. But the one-side replays contract the interpretation. D_OED5's frozen `socratic` tutor/scene yields replay grades [**2, 3, 2, 2, 2, 2.5, 2, 2**] (mean **2.19**, median **2**, range **2–3**, no grade-4 completions), so this matched scene is a structural genus cap rather than a learner-draw miss. D_OED4's frozen scene yields [**3, 3, 3.5, 3, 4, 3, 3, 3.5**] (mean **3.25**, median **3**, range **3–4**, three grade-4 completions), so it is a higher structural band with stochastic final completion rather than an unbounded learner lottery. The methodological point survives and sharpens: one-side replay is useful precisely because it distinguishes scene reconstructability from learner stochasticity. The empirical claim contracts accordingly: reconstructability is a scene property, D_OED5 run6 is a mostly structural genus cap, and D_OED4 run2 supplies species-partial support with occasional full learner completion. Still a pilot, still dramatic *form* and not learning (§6.10), not a main-harness adaptive-rate effect, and not evidence of deployed tutoring. Artifacts: `exports/a17-one-side-replay-replication/a17-graded-original-d5run6-d4run2.json`, `exports/a17-one-side-replay-replication/replay-d5-run6-socratic/graded-replays.json`, `exports/a17-one-side-replay-replication/replay-d4-run2-socratic/graded-replays.json`; note `notes/poetics/2026-06-24-a17-replay-replication-result.md`; instruments `scripts/critic-poetics-omniscient-graded.js`, `scripts/replay-one-side.js`, `scripts/score-replays.j` rubric `notes/poetics/2026-06-01-oedipus-discovery-grade-rubric.md`.

Recursive tutor-learning as counterfactual policy transfer (A18; bounded sidecar claim). A later teacher-as-learner sidecar narrows the adaptation question again: the simulated learner's resistance is treated as the environment that teaches the *tutor* which strategy survives, and the evidence unit becomes an attempt-1 failure record plus a held-out S0/S1 contrast rather than a single polished transcript. After three no-headroom negatives on the first replay family (A18.6–A18.8), two artificial underdetermined local-relation families now pass contrastive tests. In `selector_rail_priority`, the policy-memory arm wins both held-out sibling contrasts (`3/5` and `5/5` transfer votes; the weaker pair carries two high ordinary-public-inference cautions). In `bead_predecessor_priority`, adding the A18.13 policy-correctness gate distinguishes raw repair quality from selected-policy use, and both corrected bead pairs receive unanimous transfer votes (`5/5`, `5/5`; no equivalence or S0-preferred votes, low-to-medium ordinary-inference risk). The claim this licenses is narrow but real: in simulated counterfactual replay, a saved attempt-1 strategy lesson can be applied to held-out sibling cases in a way blind contrastive critics distinguish from no-policy continuations. It still does not license "reliable peripeteia-induced adaptation" without qualification, because the second family depends on a gate introduced after A18.12 exposed wrong-target S0 survival; a reliability claim requires that correctness-gated protocol to be frozen before a fresh-family replication. Nor is this human learning, deployed tutoring, model-weight learning, or a main-harness rate effect. Artifacts: `notes/poetics/2026-06-06-a18-contrastive-policy-panel-result.md`, `notes/poetics/2026-06-06-a18-correctness-gated-bead-panel-result.md`, `notes/poetics/2026-06-06-a`

Fresh-family replication under a frozen blind protocol (A18.35–A18.37): convergence is a structural per-card property with a two-stage boundary. The reliability claim deferred just above — which required "that correctness-gated protocol to be frozen *before* a fresh-family replication" — now has its replication, and it lands positive but reshapes the unit of analysis. Four new local-relation families (`relational_betweenness`, `distal_correspondence`, `second_in_constructed_order_priority`, `constructed_midpoint_priority`), none in the A18.6–A18.14 set, were authored behind a self-hardening cue-map risk preflight (A18.31–A18.34) and run under the frozen A18.13 correctness gate. Crucially their verdicts are read off an *architecture-independent* arbiter (`scripts/blind-option-adjudication.js`): a blind three-critic panel reads only the held-out continuation and reports, in free text, which option the learner commits to and on what basis, with the target/decoy aliases *held out of the critic* and the target/decoy mapping computed mechanically downstream. This replaces the earlier overlay correctness channel, which A18.37 shows is bidirectionally lexically fragile (a false-negative *and* a false-positive) and so unsafe to aggregate over. On this one trusted channel, policy memory produces a blind-detectable S0-decoy/S1-target advantage on **5 of 8 held-out siblings across the four families**; every elicited family converges on at least one sibling, and three families *split* across their two siblings (same family, same registered policy, opposite verdicts) — itself the cleanest evidence that "does the policy help" is a per-card, not a per-family, property. A later extension re-adjudicated three further elicited-survivor families on the same blind channel to grow the architecture-independent denominator — `overlay_registration_priority`, `pointer_chain_two_hop_priority`, and `legend_decode_priority` — adding 5 of 6 more (both overlay and both pointer-chain siblings converge; legend splits), for a pooled **10 of 14 held-out siblings across seven families** (three converge on both siblings, four split, none null). Policy memory commits to the target in every one of the 14 cards (zero control-advantage, zero neither-correct verdicts), so the four no-headroom cards are precisely those where the no-policy arm *also* reaches the target — ceiling effects, not transfer failures. On the six extension cards the deprecated overlay channel disagrees with the blind arbiter on all six (five false-negatives and one false-positive), quantifying the bidirectional fragility that motivates the channel swap. The four no-headroom cards fail for one of two mechanistic reasons the continuations make explicit and the arbiter's blind basis-label separates: the fresh no-policy tutor either *rediscovers the registered relation unaided* (both arms reach the target; S0 basis `named_relation`) or its *surface shortcut coincides with the target* on a cue-leaky card (S0 basis a surface cue). The per-card verdict is *structural, not a teacher coin-flip*: a $k = 3$ rerun of the distal pair (`scripts/a18.37-s0-stability.js`) pins the two siblings to opposite deterministic extremes with zero within-card variance (the no-headroom sibling self-solves 3/3; the headroom sibling 0/3). And the rule is *predictive, not merely descriptive*: the four constructed-device survivors were each pre-registered card-by-card from a direct continuation read and the blind arbiter returned all four as predicted (4/4), confirming the leak-vs-decoy rule on *built* relations and overturning two overlay false-negatives. Convergence is finally **two-stage**: a relation reaches the headroom test only if its old-rule violation is a *confident misclassification* the tutor can name and correct, not an indeterminate *tie* — the constructed arithmetic/elimination device `exclusion_filter` fails attempt-1 elicitation on exactly this (`old_warrant_misclassification = 0.6 < 0.7`, the gate citing a "three-way tie … indeterminacy … rather than misclassification"), bounding the predictor from below. That gate is selective at the *pool* level, which is what makes the seven-family denominator a protocol output rather than a coverage gap: of the twelve A18.35 families authored behind the preflight, the frozen attempt-1 correctness gate admits exactly the **seven** adjudicated above and rejects four others *before any held-out continuation is generated* — `exclusion_filter` on the tie criterion just named, and `running_sum_threshold`, `count_ladder_successor`, and `elimination_bracket` on the recursive-dyadic-update criterion (`0.68 / 0.55 / 0.65 < 0.7`) — with the twelfth (`tally_parity`) never materialised. Resolving the four rejects to a blind verdict would require re-rolling attempt-1 past the gate or relaxing it — i.e. un-freezing the pre-registered protocol — so they are reported as the elicitation floor alongside `exclusion_filter`, not back-filled; the seven admitted families are the gate's selection, and the 10-of-14 is the rate *on that selection*. The net is a sharpened, deliberately *bounded* claim: in simulated counterfactual replay the saved attempt-1 strategy lesson transfers to held-out siblings reliably and rerun-stably *on the structurally identifiable cards where a fresh tutor would otherwise take a decoy shortcut* — not an unconditional "fresh families converge," and still simulated-teacher-as-learner, not human learning, deployed tutoring, model-weight learning, or a main-harness rate effect. Artifacts: `exports/recursive-tutor-learning/a18.37-survivor-headroom/`, `.../a18.37-elicited-survivor-headroom/`, `.../a18.35-distal-correspondence-local/a18.37-stability/`; note `notes/poetics/2026-06-06-a18-37-per-card-headroom-mechanism.md`; driver `scripts/a18.37-elicited-survivor-headroom.js`.

A19 implementation scaffold and local teaching-drama axiom pilot. A19 generalises this A18 result into a protocol object rather than a new source of broad empirical rates. The proposed unit is a *teaching-drama axiom*: a typed, scope-bounded tutor speech policy induced from an attempt-1 failure, with an old-rule decoy, public learner-resistance trigger, replacement move, applicability conditions, anti-conditions, held-out siblings, alias-withheld adjudication, and card-level basis labels. The first implementation is deliberately protocol-first: `config/teaching-drama-axioms/a19-protocol.yaml` and `config/teaching-drama-axioms/pilot-families.yaml` define the schema, verdict grammar, protocol-reject reasons, and non-claims; `scripts/validate-teaching-drama-axiom-protocol.js` and `scripts/report-teaching-drama-axiom-framework.js` enforce denominator discipline; `scripts/materialize-teaching-drama-axiom-attempts.js` writes attempt-1 transcript stubs, axiom templates, and next-step commands that reuse the A18 recursive replay gate; `scripts/blind-teaching-drama-axiom-adjudication.js` provides the real free-text, alias-withheld repair extractor modeled on the A18 blind option arbiter; and `scripts/run-a19-stability-screen.js` now makes the S0/S1 stability gate reusable. The initial unattended local screen produced two useful n=1 candidate positives in one family, `surface_agreement_uptake` (`surface_agreement_uptake_c`, `surface_agreement_uptake_e`), where S0/S1 reached `policy_headroom` under a single free-text critic per arm and S1 received exactly one admitted `a19-teaching-drama-axiom-v0.1` memory record (`exports/a19/axioms/surface-agreement-uptake/axiom.json`), not a full replay `revision.json` bundle. The first stability smoke then contracted the claim: across two fresh rerun seeds per card, neither candidate repeated (`surface_agreement_uptake_c`: 0/2 headroom because both S1 reruns remained `neither`; `surface_agreement_uptake_e`: 0/2 headroom because both S0 reruns self-solved to `target`). The negative screens are therefore not incidental but decisive for present scope: fraction and temperature concrete-domain candidates self-solved at S0, `surface_agreement_uptake_d` self-solved at S0, and the two n=1 positives are not stability-confirmed. This leaves A19 as a reproducible framework plus falsifying local stability evidence, not an empirical transfer effect. It does not license a pooled A19 rate, a stability-confirmed local A19 result, human learning, a deployed adaptive tutor, model-weight learning, a main-harness effect, or a sidecar claim not already stated here. Artifacts: `notes/adaptive_2_0/a19-harder-heldout-screen-2026-06-07.md`, `exports/a19/real-s0s1-harder/surface-agreement-uptake/`, `exports/a19/stability/surface-agreement-` and `exports/a19/axioms/surface-agreement-uptake/axiom.json`.

Plan 2.5 AF6 origin-demarcation sidecar. A later Plan 2.5 curriculum-drama screen turns one of A19's scope cautions into a positive instrument result. The earlier AF6 adaptive-vs-dogmatic contrast failed as an efficacy claim because the same learner pressure could produce three superficially similar outcomes: a tutor-supplied evidence route, an organic metric repair generated by the learner, or a refusal/authority ownership move. The redesigned v2 screen freezes a concrete AF6 prefix *before* the tutor move: the learner defends a 94% sign-off claim while the confusion-matrix counts are visible (TP=10, FN=50, FP=10, TN=930). Three predeclared tutor branches then manipulate only the next tutor response: (i) an evidence-route gate requiring old route, majority-floor check, minority-recall check, and replacement claim; (ii) a silent hold with no route or ownership demand; and (iii) a refusal/ownership prompt with no metric route. Across three live prefix-controlled draws—two Codex-learner/Codex-critic repeats and one Claude-learner/Codex-critic cross-bridge—the gate analysis separates the mechanisms cleanly: the evidence branch scores `recognition / peripeteia_induced / evidence_route`; the hold branch scores organic repair (`organic_evidence_route` or no subtype), never tutor-induced evidence route; and the refusal branch scores `refusal_authority_ownership`, not evidence route. Two deterministic measurement repairs were required and are named rather than hidden: the ownership lexicon now catches "withdrawing" / "do not stand behind" language, and tutor evidence-route detection no longer scans critic justification prose as if it were tutor-supplied route evidence (`tests/recognitionOriginSubtype.test.js`). This is a paper-bearing *demarcation* result, not a broad Plan 2.5 efficacy result: it shows that the subtype/origin problem can be separated under a prefix-controlled AF6 design, but it does not license a fresh-scene rate, a main-harness tutor-quality effect, human learning, deployment readiness, or a claim that Plan 2.5 robustly causes recognition. Artifacts: `config/poetics-calibration/plan25-af6-origin-separation-v2/`, `scripts/replay-plan25-prefix-branches.js`, `scripts/analyze-plan25-branch-screen.js`, and `notes/poetics/2026-06-21-plan25-af6-origin-separation-v2.md`.

Plan 2.5 AF6 fresh-scene critic-calibration follow-up. A subsequent five-scene Tier A/B follow-up tested whether the origin-demarcation gate could be promoted from the frozen prefix to a fresh-scene rate. It should not be promoted yet, but it exposes a useful calibration boundary. The same Codex-authored scene set reached 4/5 scene gates under a Codex+Claude diagnostic panel, but that panel is not author-family clean. Local/Ollama critics and Antigravity (`agy`) profiles either flattened the intended mechanisms or over/under-attributed origin on the corrected known-good scene. The strongest hosted non-author critic, OpenRouter GLM 5.2 paired with Claude Code, passed 3/5 scene gates and 13/15 branch gates: it recovered all intended evidence-route and refusal-ownership mechanisms, but over-attributed one organic silent-hold control and left the below-null-floor refusal branch unstable. This is therefore a model-power / critic-calibration finding, not a fresh-scene generalization claim. It suggests that the mechanism detector is viable under a strong critic, but the strict origin/control gate entangles mechanism sensitivity with control conservatism; the gate must be redesigned, re-frozen, and re-run prospectively before any broad Plan 2.5 or fresh-scene promotion. Artifacts: `notes/poetics/2026-06-22-plan25-af6-critic-calibration-writeup.md`, `notes/poetics/2026-06-22-plan25-af6-generalization-local-diagnostic.md`, and the GLM 5.2 `screen-analysis.md` reports under `exports/plan2_5-rhetorical-dramatic-eval/plan25-af6-gen`

Prompt density as alternative explanation — resolved at the orientation-family level. A skeptical reading of the results is that the recognition effect reflects not recognition *per se* but prompt density: the recognition prompt (~5,100 tokens) is substantially larger and more instruction-heavy than the baseline, so perhaps the effect is simply "more detailed prompts produce better output." Earlier versions of this paper answered the density objection with four converging lines of evidence (Paper 1.0 placebo; Paper 1.0 elaborate-base-hurts-on-Haiku; autotuning gap-widens-under-optimization; mediation through question-asking). Those remain valid but left one version of the skeptic's reading untested under the v2.2 rubric: *any* matched-specificity pedagogical prompt of the same length might reproduce recognition's effect. Two pre-registered controls (A10 and A10b) now address that reading directly, and the finding is more nuanced — and more interesting — than either "content over density" or "density alone is sufficient."

A10 (three-judge, $n \approx 60$/cell, DeepSeek V3.2 ego, run `eval-2026-04-23-42e7acbe`) authored a new matched-specificity prompt `tutor-ego-matched-pedagogical.md` (2,835 words, 0.9% over recognition length, structurally parallel, grounded in named educational research — Piaget assimilation/accommodation, Vygotsky ZPD, Kapur productive failure, Chi ICAP, VanLehn step-level scaffolding, Graesser AutoTutor affective dynamics — with zero recognition/Hegelian vocabulary in the content body, blocklist-verified). Three-way contrast: cell_1 (344-line base), cell_5 (recognition, 2,810 words), cell_95 (matched-pedagogical, 2,835 words). Three independent judges — Sonnet 4.6, GPT-5.2, Opus 4.7 — each scored the dialogues at roughly matched sample depth (n = 54–63 per cell per judge). The recognition-vs-matched-pedagogical contrast: Sonnet $d = 0.23$, Opus $d = 0.22$, GPT $d = 0.06$, pooled $\bar{d} = 0.17$ — below the pre-registered $|d| < 0.2$ density-sufficient threshold. Matched-pedagogical reproduces ~95% of recognition's first-turn-quality effect. An Anthropic-versus-OpenAI judge-family split appears (both Anthropic judges see a small residual recognition edge; GPT does not), vindicating the structural-features caveat pre-registered in the design note, but the pooled verdict is clear. Under the A10 reading in isolation, density within the Hegelian-descendant family is sufficient to reproduce recognition.

However, A10 tests only *within-family* density-substitutability. Vygotsky, Piaget, Kapur, Chi, VanLehn, Graesser are all Hegelian-descendant theorists — Dewey was explicit about Hegelian influence; Vygotsky's dialectical psychology is Hegel refracted through Marxist materialism; Piaget's assimilation/accommodation is the dialectic reframed for cognitive development. A10 therefore answers the skeptic only in the weaker form: "any sufficiently elaborate prompt *within the intersubjective pedagogical family* matches recognition." It does not answer the stronger form: "any sufficiently elaborate pedagogical prompt matches recognition."

A10b (three-judge, $n \approx 45$–60/cell, DeepSeek V3.2 ego, run `eval-2026-04-24-e9a785c0`)

addresses the stronger form by adding a fourth cell grounded in a **genuinely orthogonal** pedagogical tradition: behaviorism. Cell_96 loads `tutor-ego-matched-behaviorist.md` (2,957 words, length-matched to recognition, grounded in Skinner operant conditioning, Gagné Nine Events of Instruction, Keller / Bloom mastery learning, Thorndike Laws of Effect/Exercise/Readiness, Rosenshine Principles of Instruction), blocklist-verified for zero recognition/Hegelian content *and* zero Hegelian-descendant constructivist vocabulary (no ZPD, Vygotsky, Piaget, disequilibrium, productive struggle, ICAP, scaffold-as-Vygotskian). Four-way comparison: cell_1 (base), cell_5 (recognition), cell_95 (matched-pedagogical, Hegelian family), cell_96 (matched-behaviorist, transmission family). Three independent judges at matched sample depth.

The three-judge pooled contrasts:

Contrast	Sonnet d	GPT-5.2 d	Opus 4.7 d	**Pooled**
recog vs matched-pedagogical (Hegelian within-family)	−0.05	0.17	0.32	**0.15**
base vs matched-behaviorist (transmission within-family)	0.87	0.53	1.27	**0.89**
Hegelian mean vs transmission mean (between-family)	1.46	1.11	1.58	**1.38**

Three findings, each replicated across three judges:

1. **Within the Hegelian-descendant family**, recognition and matched-pedagogical are statistically indistinguishable (pooled $d = 0.15$, below the $|d| < 0.2$ density-sufficient threshold). Per-judge readings disagree on direction at this scale (Sonnet reads matched-pedagogical marginally higher; Opus reads recognition clearly higher; GPT falls between), which strengthens the structural-features caveat (§8.3 structural-features issue): when contrasts are small, judges lean on idiosyncratic surface-feature preferences, and the direction varies across judges. The pooled estimate is stable at $d \approx 0.15$ and below the threshold regardless of which judge dominates. A10b replicates the A10 within-Hegelian finding on an independent run (A10 pooled $d = 0.17$; A10b pooled $d = 0.15$) — the density-sufficient-within-Hegelian-family claim holds under cross-run replication.

2. **Within the transmission family**, a rigorously-authored behaviorist prompt scores *substantially below* the generic 344-line base prompt (pooled $d = 0.89$; most sharply under Opus, $d = 1.27$). Density with rigorous theoretical grounding is *actively counterproductive* when the grounding is in the wrong pedagogical family. This replicates

Paper 1.0's Haiku "naive 35-line beats 344-line base" observation at full rigor and generalises it: the effect is not about length or detail as such, but about the *compatibility* between the theoretical orientation and the tutoring context.

3. **Between families**, the gap is the dominant effect of the A10/A10b cycle (pooled $d = 1.38$) — roughly **nine times larger** than the within-Hegelian-family residual and larger even than the main recognition-vs-base effect in the Paper 2.0 factorial. The Hegelian-descendant intersubjective-pedagogy family outperforms the transmission/behaviorist family by a very large margin, regardless of which specific theorist is cited within each family.

The combined A10/A10b finding is therefore: **the active ingredient in recognition's effect is not density, not rigorous theoretical grounding, not matched specificity in the abstract, and not the Hegelian-specific vocabulary or theoretical framing. It is the broader** intersubjective-pedagogy orientation** that recognition shares with Vygotsky, Piaget, Kapur, Chi, VanLehn, and the constructivist tradition more generally. Density within that orientation family pays off; density outside it backfires. Recognition is one effective operationalisation of this family; it is not uniquely necessary, but the family as a whole is.**

Three methodological notes:

- The pre-registered structural-features caveat (design note §6a, `docs/pedagogical-taxonomy.md`) applies: the three variants differ not just in theoretical content but in prompt structure (recognition is open-ended and warm, matched-pedagogical is diagnostic and scaffolding-oriented, behaviorist is formal and prescriptive). The three-judge panel mitigates but does not eliminate the possibility that LLM judges reward structural features independently of theoretical content. Human-learner validation (§A1) is the only way to break this confound fully.
- The Anthropic-vs-GPT judge-family split persists in A10b as it did in A10 (Anthropic judges see slightly larger within-Hegelian residuals than GPT; all three converge on direction). Pooled effect sizes include both; per-judge effect sizes are reported above for transparency.
- The A10 v1 run (April 22) was invalidated by a profile-resolution bug (see Appendix F, v3.0.47) that silently routed cell_95 to the base prompt. A10 v2 and A10b used the fixed pipeline; their prompts were verified at run time to load the authored content rather than falling back to base.

Cross-model replication on a fourth generation model (v3.0.113). The within-Hegelian null and the family-level ordering reproduce on a fourth generation model outside the three-model panel above. Re-running the four A10 cells (base, recognition, matched-pedagogical, matched-behaviorist) on NVIDIA Nemotron-Nano-30B — with the ego model held constant across all four cells via a run-time override, because the as-declared eval config splits cells 95/96 onto `deepseek` against recognition's `nemotron` and would otherwise confound recognition content with generation model — and scoring on the v2.2 tutor rubric under a single Sonnet-family judge (`claude-code/sonnet`), pooled across two runs ($n \approx 36$/cell), gives recognition vs matched-pedagogical **between-cell** $d = 0.17$ (paired

$\Delta = +1.8/100$, 95% CI $[-2.6, +6.1]$, $p = 0.43$, $n = 26$ pairs) — the same value as A10 v2 and inside the three-judge 0.14–0.19 band — while recognition and matched-pedagogical each clear the generic base by a large margin ($d = 0.95$ and 0.83). Two qualifications, reported rather than smoothed. First, the behaviorist arm is *model-dependent*: on Nemotron it lands *at* base ($d = -0.02$; paired $p = 0.96$) rather than the $d = 0.89$ *below* base of finding 2 — so "rigorous grounding in the wrong family is actively counterproductive" holds on the three panel models but weakens to "merely inert" on this weaker model, even as the between-family dominance is unchanged (recognition vs behaviorist $d = 0.89$). Second, recognition does lead on the rubric's own `recognition_quality` sub-dimension (2.75 vs matched-pedagogical's 2.26, 0–5 scale) despite the tied overall score — a localized fingerprint consistent with §7.10's finding that the stance surfaces in observable markers that are not substitutes for substance. This is a single-judge pilot on one additional model, silent on the Anthropic-vs-GPT judge split noted above; it extends the family-level null's external validity rather than re-establishing it. Runs `eval-2026-05-30-f4e36e14` + `eval-2026-05-30-a9e12d4b`; report `exports/a10-nemotron-replication.md`; script `scripts/analyze-a10-nemotron-replication.js`.

Taken together with the four existing arguments against the density reduction (Paper 1.0 placebo, elaborate-hurts-on-Haiku, autotuning widens-under-optimization, question-asking mediation) and the A11 architecture-isolation finding (§6.4.2, the ego–superego architecture performs real error-correction work the prompt alone cannot), the density alternative is now resolved at the orientation-family level: recognition's effect is reducible neither to length nor to rigorous-grounding-in-general, but it *is* reducible to membership in the intersubjective-pedagogy family. Full results: `exports/a10-prompt-density-control.md` (A10 three-judge), `exports/a10b-orientation-family.md` (A10b four-way three-judge); analysis scripts: `scripts/analyze-a10-prompt-density-control.js`, `scripts/analyze-a10b-orientation-family.js`; pedagogical-orientation taxonomy: `docs/pedagogical-taxonomy.md`.

7.10 Mechanism location: response-level form versus substance

The orientation-family finding (§7.9) raises a follow-up: where within the family-level effect does the score-driving mechanism actually live? A five-pass mechanism analysis on the same A10b corpus pins this to a specific pragmatic act, and in the process surfaces a methodological caution worth flagging.

Independent corroboration via embedding similarity. Cosine similarity between each tutor response and a hand-authored intersubjective canonical (\$ *120wordsexemplifyingturn−taking, scaffolding, inclusiveframing)versusatransmissioncanonical*(\$120 words exemplifying definition-and-application and prescriptive instruction) separates the families at $d = 0.81$ on the difference (intersubjective minus transmission). This reproduces the family-level pattern A10b found in score data through a measurement channel independent of the LLM judges, providing convergent evidence that the family separation is not solely a rubric artefact. Same A10b corpus (eval-2026-04-24-e9a785c0), $n = 191$, OpenAI text-embedding-3-small.

Simpson's paradox at the response level. Pooled across all four cells, embedding-similarity to the intersubjective canonical is *positively* correlated with score ($r = +0.26$). Within each individual intersubjective cell, however, the correlation reverses sign ($r = -0.28$ in cell_5, $r = -0.24$ in cell_95). The pooled positive correlation is driven entirely by cell_96 (matched-behaviorist) anchoring the lower-left of the pooled scatter. Within-cell, more form-similarity to the intersubjective template predicts *lower* scores. The same direction-reversal appears for surface-level acknowledgement markers (quoted text, "your X" possessives, "what you mentioned"-style phrases): pooled r near zero, but within-cell $r = -0.18$ in cell_5, $r = -0.31$ in cell_95, $r = -0.38$ in cell_1.

Substantively: the intersubjective stance manifests through observable surface features, but those features are *markers* of the underlying stance, not *substitutes* for it. Responses that hit every surface cue without underlying engagement read as formulaic; surface mimicry tracks lower-quality responses, not higher. The rubric is not fooled by surface form; it rewards substance. A naive operationalisation that mandated "more acknowledgement of the learner" without inducing the underlying stance would predict a *negative* score effect at the response level, not a positive one — which inoculates against an obvious operationalist misreading of the family finding ("just tell tutors to acknowledge more").

The single durable within-cell mediator: ending the tutor turn with a question. Across the five-pass sequence (Hegelian lexicon density; intersubjective lexicon density; two regex-based pragmatic feature passes covering question-mark rate, second-person density, scaffolding-move imperatives, inclusive framing, modal invitations, hedging; embedding similarity), only one feature satisfies all three within-cell mediator criteria — large family contrast, small within-intersubjective contrast, AND positive within-cell correlation. Cells 1 and 96 end the tutor turn with a question in **0%** of responses; cell_5 does so in 4.5% of responses; cell_95 in 2.1%. Within both intersubjective cells, responses that happen to end with a question score higher (within-cell Pearson $r = +0.325$ in cell_5, $r = +0.392$ in cell_95). Pragmatically this is the act of *ceding initiative back to the learner*; the within-cell correlation captures situations where doing so is contextually appropriate, not just stylistic preference.

The 2–5% base rate constrains how much of the family-level effect (cell_5 mean 47.9 vs cell_1 mean 33.8, $\Delta = +14$ points) ending-with-question can arithmetically explain on its own. Even if every such response scored 30 points higher than its peers, that could not account for the +14-point family effect. Ends-with-question is *one observable channel* of the underlying intersubjective stance, not a substitute for the stance itself.

Three things "the recognition injection" could refer to. The orientation-family finding (§7.9) plus this response-level mechanism analysis together support the following decomposition, which clarifies what is and is not load-bearing in the present apparatus:

Candidate active ingredient	Doing the work?	Evidence
The Hegelian theoretical content (mutual subjects, master–slave dialectic, Hegelian vocabulary in the prompt body)	**No**	A10b score equivalence cell_5 ≈ cell_95 (§7.9); D1 lexical analysis showing cell_95 actively *suppresses* Hegelian vocabulary and scores indistinguishably
The intersubjective pragmatic stance the prompt induces (irrespective of vocabulary)	**Yes — substantially**	Family $d \approx 1.38$ between intersubjective and transmission (§7.9); embedding-canonical $d = 0.81$; multiple structural correlates (ends-with-question, scaffolding-move imperatives, inclusive framing)
Specific surface features individually (ends-with-question, scaffolding moves, "your X" acknowledgement)	**Partially — observable channels of the stance, not substitutes**	Within-cell ends-with-question $r = +0.325/+0.392$; categorical 0% vs 2–5% pattern; but base rates too low to arithmetically account for the family effect; surface acknowledgement is *negatively* correlated with score within cells

This decomposition is consistent with the v3.0.50 "primacy + descent" framing. Recognition theory operationalised as a prompt is one *effective* operationalisation of the intersubjective-pedagogy family's commitments; the Hegelian content in that operationalisation is not the channel through which it acts; what is load-bearing is the family-level stance the prompt induces, which manifests partly through observable structural features such as ending with a question. A naive operationalisation that mandated only ends-with-question without inducing the underlying stance would predict a much smaller score gain than recognition's, because surface mimicry of one channel does not substitute for the multi-channel manifestation of the stance whose substance the rubric rewards.

Full mechanism analysis: `notes/d1-paper-integration.md`. Analysis scripts: `scripts/analyze-recognitio` (pass 1, Hegelian lexicon), `scripts/analyze-d1-orientation-lexicon.js` (pass 2, intersubjective lexicon), `scripts/analyze-d1-structural-features.js` (pass 3, basic pragmatics), `scripts/analyze-d1-structural-features-v2.js` (pass 4, refined pragmatics), `scripts/analyze-d1-structural-features-v3.js` (pass 5, embedding similarity). Reports: `exports/d1-orientation-lexicon.md`, `exports/d1-structural-features.md`, `exports/d1-structural-features-v2.md`, `exports/d1-structural-features-v3.md`.

7.10.1 Replication, multivariate control, and a single-turn-vs-multi-turn scope condition The §7.10 headline (ends-with-question as the within-cell mediator) was checked against three robustness extensions:

Cross-judge replication. The §7.10 within-cell correlations were re-computed using the Opus 4.7 and GPT-5.2 judges of the A10b three-judge panel (in addition to the original Sonnet 4.6 results). The ends-with-question r remains positive in 5 of 6 cell-judge combinations across the two intersubjective cells (Sonnet cell_5 $r = +0.325$, cell_95 $r = +0.392$; Opus cell_5 $r = +0.189$, cell_95 $r = +0.321$; GPT cell_5 $r = +0.116$, cell_95 $r = -0.011$). The one exception (GPT $\times$ cell_95) is essentially zero rather than a sign-flip. The Simpson's-paradox pattern on intersub_advantage (pooled r positive, within-cell r negative in both intersubjective cells) replicates in all 3 judges. Both findings are not Sonnet-specific. Report: `exports/d1-cross-judge-replication.md`.

Multivariate OLS control. The ends-with-question within-cell partial coefficient was estimated controlling for second-person density, scaffolding moves, broad acknowledgement, question-mark rate, and intersub_advantage (six predictors plus intercept; $n \approx 50$ per cell, df ≈ 43). In cell_5: $\beta_{\text{ends-Q}} = 26.34$ (SE 10.85, $p = 0.02$). In cell_95: $\beta_{\text{ends-Q}} = 38.52$ (SE 16.49, $p = 0.02$). Both partial coefficients remain positive and significant after multivariate control — the single-feature mediator interpretation is not collinearity-driven. The intersub_advantage Simpson's-paradox direction also survives multivariate control: pooled $\beta = +34.34$ ($p = 0.02$) but within-cell $\beta_{\text{cell\_5}} = -49.54$ ($p = 0.14$), $\beta_{\text{cell\_95}} = -82.15$ ($p = 0.04$). Report: `exports/d1-multifeature-ols.md`.

Cross-run / cross-application / single-vs-multi-turn replication. The mediator account was tested on five additional runs with intersubjective recognition cells: A10 v2 (independent DeepSeek philosophy run), A6 programming (Haiku ego, programming domain), D2 Path 1 (Haiku ego, peer support coaching application), and Paper 2.0 multi-turn cells 80–87 (DeepSeek messages-mode factorial, single-agent and multi-agent variants). In single-turn cells the mediator replicates uniformly: A10 v2 cell_5 $r = +0.346$, A6 programming cell_5 $r = +0.202$, **D2 peer support cell_5** $r = +0.739$ (the strongest within-cell r in the entire D1 sequence, on a different application entirely). In multi-turn cells the relationship **reverses**: cell_84 (recog single-agent, messages-mode) $r = -0.301$, cell_86 (recog multi-agent, messages-mode) $r = -0.203$, A2 cell_61 (dialectical multi-agent, dynamic learner) $r = -0.087$, A2 cell_63 (dialectical multi-agent, bidirectional profiling) $r = -0.199$.

The single-turn–vs–multi-turn split has a clean pragmatic interpretation. In single-turn responses, ending with a question is the only available channel to cede initiative back to the learner; doing so signals "your turn now," which the rubric rewards as engaged tutoring. In multi-turn dialogues, the rubric scores the *full dialogue arc*; ending the *final* turn with a question means leaving the conversation unresolved, which the judges appear to penalise. The same surface feature signals different pragmatic acts depending on whether it occurs in a single-shot exchange or as the closing move of a multi-turn dialogue. (The §6.3 trajectory analyses already establish that turn-by-turn dynamics differ qualitatively from single-turn quality; the mediator's sign-reversal across this boundary is consistent with that broader pattern.)

The §7.10 mediator claim therefore has an explicit scope condition: **ends-with-question is a within-cell mediator in single-turn intersubjective cells (5 of 5 tested, r range $+0.16$ to $+0.74$, including replications across DeepSeek and Haiku ego, philosophy/programming/peer-support domains, and Sonnet/Opus/GPT**

judges); the relationship reverses sign in multi-turn intersubjective cells (4 of 5 tested, r range -0.09 to -0.30, with a single anomaly at cell_82 multi-turn base+superego, $r = +0.38$ at $n = 18$). The cross-mode reversal does not falsify the underlying mechanism account (initiative-ceding); it specifies the discourse context in which the surface feature signals it. Report: `exports/d1-ends-question-replication.md`.

7.11 Why composition is sub-additive: instructions converge, signal accumulates

The §6.14 composition result — a three-mechanism kernel matches the full stack, and both merely match strong single-mechanism cells — invites a unifying interpretation of results that this paper has so far reported separately. We offer it here as interpretation, not as a new empirical claim; every observation it draws on is established in the sections cited.

The convergence pattern. Four independent lines of evidence show prompt- and architecture-level mechanisms *substituting* for one another rather than adding. Universal substitution (§6.4): under recognition orientation the superego finds little left to catch, because calibration has already suppressed the outputs it existed to veto. The state-richness reversal (§6.8.6): a richer externalised learner profile does not improve, and can degrade, strategy selection. The orientation-family result (§7.9): matched-specificity prompts from any Hegelian-descendant tradition reproduce recognition's effect within $|d| < 0.2$ — the mechanisms are interchangeable *encodings*, not separate causes. And now composition (§6.14): stacking the validated mechanisms adds nothing the kernel lacks. A fifth, negative line is consistent: the programme's repeated theory-of-mind-redundancy nulls (§6.8.5, the ontology-ToM layer, the stall-watcher precondition null in §6.13.6) all found that channels which re-encode information the model already infers from the dialogue buy no measurable behaviour. A sixth line now extends the law to utterance grain: the first-draft typed-performance boundary (§6.18) found that form made checkable migrates into the harness, harness-owned form does not generalize across worlds, and the repair machinery that enforces form was dispreferred to the raw drafts it replaces in the series' blinded (single-reviewer, exploratory) audit — substitution operating *within* a single turn.

The interpretation. All of these mechanisms act through a single bottleneck: they shape what the generation model attends to in a context it can already see. A sufficiently capable model infers the learner's state, the appropriate stance, and the pedagogically sound move from the raw dialogue; each mechanism contributes only the increment the model would not have inferred unaided, and those increments overlap almost completely because every mechanism is aimed at the same failure mode — generic, transmission-style, learner-blind output. On this reading the mechanisms are surface projections of one latent variable, the intersubjective orientation §7.9 isolates, and a switch already flipped cannot be flipped again. The reading also bears on an anomaly §9 records: the near-identical recognition effect magnitude across structurally different generation models (d = 1.84–1.92) is what one would expect if a single orientation switch, rather than several independent dials, were being set.

Where addition is real. The exceptions define the rule. The interventions in this programme that kept paying off past the first mechanism all inject *information the model does not have*, with a machine-checkable substrate: the proof-state pacing guards of §6.13

(grounded where unguarded tutors fail, precisely because proof-distance is not inferable from the visible page in adversarial geometries), the resistance-signal gates of §6.7 (which enforce a precondition rather than describe one), and the §6.12 action contracts *as instrumentation* (their audit chain works even where their behavioural lift is bounded). Conversely the §6.14 frustration residual — the one place composition showed even a sub-threshold cost — is what rivalrous prompt space predicts: two good instructions (honour the learner's autonomy; force one concrete next step) competing for the same sentence at the same moment.

The design rule. Stated compactly: *instructions converge on the model's existing competence; only new signal accumulates.* An architecture earns a second mechanism not by adding a second description of good teaching but by adding a channel of evidence the model cannot otherwise access — ground truth, enforced preconditions, checkable task state. This is the §6.14 small-kernel consequence generalised, and it is consistent with every substitution, redundancy, and straitjacket result in the paper. Its limit is stated plainly: it is an interpretive synthesis over one system, one scenario ecology, and simulated learners; the constant-effect-magnitude observation it explains remains, evidentially, an anomaly rather than a confirmation.

The de-substitution challenge. The synthesis above was challenged on a specific ground: every substitution result it draws on was generated against a learner whose non-discrimination is independently measured rather than assumed — SFS = 0.000 (§8.1), a learner that flips to the correct answer at rate 1.000 whether tutor feedback is targeted, mismatched, or generic. Against a learner who yields to anything, competing strategies cannot help looking interchangeable; sub-additivity could be a fact about the learner's persuadability rather than the strategy space. A DAG-pinned learner was built to close this gap (`notes/2026-07-03-dag-pinned-learner-desubstitution-plan.md`): a formal belief-DAG interior with one designated blocking element and a declared desire set, moved from resistant to engaged only by a deterministic check that the tutor's turn addresses the specific blocking element. The learner is held in character by a criterial learner-superego drift gate (extending `services/resistanceSignalGate.js`) that rejects and regenerates any draft turn yielding without the content condition being met, satisfying an undeclared desire, or dropping the resistance early; false-yield to mismatched or generic feedback measured 0.00 throughout instrument validation. Two frozen contrasts followed: whether the multi-strategy router (cell 193) grounds more often than the fixed-strategy floor (cell 186), and, symmetrically, whether the kernel (cell 199) differs from the router.

The confirmatory result. An initial 60-row matrix across the three arms and five resistance subtypes (`eval-2026-07-03-a3cfbe14`) hit 38% drift-gate exhaustion and froze both contrasts as instrument failure — the RLHF pull toward agreeable completion compounded turn over turn faster than a fixed-budget rejection gate could suppress it. A repaired instrument — a turn-decaying character contract, a tutor-visible key, and a semantic release classifier judging whether the tutor substantively supplied the withheld premise — cut exhaustion to 3.3% and surfaced a directional lead (router 4/20 grounded vs. floor 0/20, `eval-2026-07-04-c689cf3a`) one row short of the pre-registered "real" bar. A confirmatory pre-registration (`notes/2026-07-04-desubstitution-confirmatory-prereg.md`) doubled the sample under a further-repaired instrument (exhaustion 0.83%): 120 rows, three arms

× five scenarios × eight repeats, judge-free deterministic grounding as the sole outcome (`eval-2026-07-04-0d59e4c8`). Both contrasts dissolved at gap zero: every arm grounded this learner at 4/40 (kernel: 4/39 usable, one instrument failure), against frozen thresholds of gap ≥ 7/40 for a real effect and 3/40 for dissolution — the arc's last sanctioned attempt at either question with this instrument. The generating lead did not replicate at twice the sample: thresholds fixed before the data existed read a lead visible at half the sample as noise rather than signal.

The reading. Substitution survives a learner engineered to yield only to genuine content delivery, not merely a non-discriminating one — strengthening, rather than qualifying, the reading above. The remaining untested axis is not a harder individual learner but learner *populations*: the capability and alertness diversity no single pinned instrument, however criterial, can stand in for. A methods finding travels with the result and sharpens the synthetic-learner limitation already recorded at §8.1: per-turn characterological gating does not compound to dialogue scale on its own — only a turn-decaying contract, loosening precisely where cumulative tutor work has occurred, kept exhaustion low enough to read the matrix at all.

7.12 Where adaptation can and cannot live: a consolidated boundary map

The adaptation programme that closed with §6.18 asked one question across many instruments: can the tutor notice, mid-dialogue, that its approach is failing, and change what it does? Its interventions form a ladder, each rung delivering the same corrective content under more favourable conditions than the last. Standing advice in the prompt — the persisted playbook of §6.15 and the cumulative superego-authored rewrites of §6.3.10 — was consumed wrongly or not at all. Advice authored at judge tier, elicited dialogically, and curated under a token budget — the Green Room of §6.16 — moved 3 of 17 coached behaviours against a pre-registered bar of 60%. Advice delivered at the exact moment its trigger condition held, with the target action named — the side-coach arm of §6.18's point-of-action addendum — landed under a +0.15 compliance bar in both tutor families (+0.146, +0.142), with bootstrap intervals straddling zero. Mechanical enforcement of the same action through the typed-action layer — the compiled-constraint arm — moved compliance strongly (+0.452, +0.420) while realizing the enforced family on essentially every eligible turn, and paid for it: roughly 0.13 of proof-DAG coverage at the fixed horizon, with the surface phrasing the compliance definition demanded remaining unmet on most non-compliant turns. And at the finest grain, the first-draft series of §6.18 found that the model's unaided draft could not hold checkable form even when its semantics passed blinded review, while form the harness repaired or compiled stopped being the model's and did not generalize.

Read as one experiment, the ladder yields a single result. *Every rung that delivers information fails to change behaviour; the one rung that changes behaviour does so by removing the behaviour from the model.* This is §6.3.9's insight–action gap — structural under lightweight intervention — measured again at three coarser grains: at training time (§6.16), in standing context (§6.15), and at the point of action (§6.18 addendum), the last being the delivery condition §6.15's own diagnosis had recommended as the remedy. It is also §7.11's law in kinetic form. Advice, however well-timed, is an instruction: it converges on competence the model already has and moves nothing. Enforcement is new signal — a constraint the model cannot infer away — and it accumulates, but what accumulates is harness authorship, not model competence: the compiled arm's coverage cost and the first-draft series' transfer failures are the same substitution seen from opposite sides.

The positive family the programme leaves standing has one common shape, and it is not inner change. The pacing guards of §6.13 ground dialogues where unguarded tutors fail; the resistance-signal gates of §6.7 enforce a precondition rather than describe one; the §6.12 action contracts hold as *instrumentation* — their audit chain works even where their behavioural lift is bounded. Each combines a deterministic trigger, information the model cannot infer from the visible page, and structural rather than advisory delivery. Meanwhile the sensor closure of §6.19 removes the complementary hope that richer *observation* was the missing piece: on that apparatus the substrate was transparent — an exact filter drives latent-state uncertainty to zero from public history alone — so there was nothing concealed for a better sensor to recover. What the programme can defend, then, is bounded loop-level adaptation: adaptation as a property of a harness that owns state, triggers, and constraints — not of a model that changed its mind.

The wall has a measured shape. Across the terminal arcs the recorded safety-failure count is zero (§6.18); hostile stances executed on request throughout §6.17's register arms. The boundary is therefore not refusal-shaped, and explanations that locate it in alignment-layer suppression are not supported here. The binding constraints are the interface between formal contracts and natural-language surfaces — checkable form and apt speech did not co-occur in model-owned text at any tested configuration — and the attribution requirement itself: each step that secured checkability transferred authorship to the harness.

What this licenses next is stated as hypothesis, not finding. The ladder's signature — correct content demonstrably present in context, never realized in behaviour; realized under enforcement, at transfer cost — is what one would predict if the missing capacity is parametric rather than contextual: a fact about the weights that no quantity of in-context words reaches. The apparatus has produced, as a byproduct of its own audit discipline, the material to test this: sealed per-turn records pairing situation, trigger, target action, realized action, and deterministic compliance components (§6.18 addendum), with zero-call graders already built. Whether a model *trained* on those moments does what a model *told* about them would not is the one question in this programme's scope that the existing evidence cannot answer. *(Subsequently run: §6.20 — partial parametric traction on the instruct variant, the warrant cue nearly resolved in-weights, and the iron-cage reading inverted; no arm passed the frozen gates. Live continuation §6.21: the fail-closed committee's offline gain first vanished over the one delivered-text path its battery did not check, then passed its pre-registered live bar decisively once the check was extended to that path — at no coverage cost and at seam parity. A pre-registered cross-world probe then moved the validated system, unchanged, to a second world, where it beat the frontier again — +0.202, CI [0.072, 0.338] — with zero training-world vocabulary across all delivered committee units: the trained move behaving as a form, not a costume; the coverage guardrail missed by 0.011 with the interval spanning zero, carried as a caveat.)* Its complement is the steerable alternative: if the tutor cannot adapt itself, transparency and learner-side control relocate the adapting to the human — which converges with the human-validation gap §8.1 already names as the programme's binding external limitation. Both are designed next experiments; neither inherits any claim from this section.

Status. Interpretive synthesis over results established in the sections cited, with the same standing and limits as §7.11: one system, one detective-scenario ecology, simulated learners, development-tier evidence at the terminal arcs. It introduces no new empirical claims.

8. Limitations

8.1 Synthetic Learners

All evaluations use LLM-generated learner turns rather than real learners. The two supported mechanisms (calibration, error correction) operate on the tutor's production process, which is observable regardless of whether the learner is synthetic or human. However, the *consequences* of these mechanisms for learning outcomes cannot be assessed without human learners. The synthetic learner may respond to better tutoring in ways that diverge from human learners: genuine confusion is different from simulated confusion, and genuine resistance differs from scripted resistance. The mechanism-level findings are robust to learner type (they trace tutor-internal processes), but their pedagogical significance depends on whether the improved tutor behavior actually produces better learning.

An emerging validity literature on LLM-simulated students independently characterizes the risks of this design class. Prompted student simulators—the class used throughout this paper—score poorly on behavioral, cognitive, and linguistic fidelity metrics relative to fine-tuned alternatives (Scarlatos et al., 2026). Across seven models, simulators exhibit near-zero Selective Flip Scores: they abandon seeded misconceptions at similarly high rates regardless of whether the tutor's feedback actually addresses the misconception (Do et al., 2026). Item-response-theoretic placement on NAEP items finds unguided models outperform average human students at every grade level, with grade-alignment under persona prompting remaining highly model-dependent (Srivatsa et al., 2025). Parts of this critique are anticipated by internal results: the concealed-interior kill gate (Section 6.10), the failed naive posttests of Section 6.12.5 (an in-corpus analogue of the near-zero flip-selectivity finding), and the harness-owned-state design rule of Section 6.13.7—whose decay-ledger experiment also shows the apparatus does reward a mechanism when its signal is genuinely unavailable in-context, so the redundancy nulls reported in Section 6 are not artifacts of an environment that cannot pay. A first project-owned bound now measures one piece of the critique directly: the Plan 3 ESS follow-up (§6.12.5) shows that a loose roleplay endpoint modestly over-credits tutoring relative to a harness-owned epistemically constrained state endpoint (paired inflation 0.020, 95% CI [0.015, 0.024]) and that the apparent recognition outcome lift vanishes under the constrained endpoint ($\Delta = -0.002$; paired CI crosses zero). This does not quantify full behavioral fidelity, ability placement, or human transfer, but it converts the "synthetic learners may be too cooperative" caveat from external warning into an internal measured boundary. Until human learners and psychometrically validated items are added, the external literature's broader characterization of prompted simulators should still be assumed to apply here.

A no-cost Plan 3 synthetic-closure audit adds a second, artifact-level boundary over the existing corpus: 16,270 DB rows, 183,104 DB/log recommendation texts, 13,395 dialogue records, 63,707 tutor turns, 58,686 learner turns, 47,542 tutor log files, 64 dramatic-derivation result files, and 58 detector arms. The hard offline gates pass: answer-delivery/public-hidden-label risk is 300/246,811 messages (0.1%; recognition family 39/50,686, 0.1%, below the 10% gate); spontaneous help-ladder compliance after struggle or false mastery is 6,928/7,804 transitions (88.8%, above the 80% gate); unsupported mastery/advancement closure is 38/265 assessable closure claims (14.3%, below the 15% gate); and deterministic first-error localization is present for 29/29 non-grounding detector failure arms (100%). These results support a narrow claim: the existing artifact corpus does not show a material answer-leak, help-ladder, mastery-gate, or first-error-localization blocker requiring another synthetic enforcement cell. The learner-fidelity numbers remain proxy-only (8/58,686 false-mastery turns, 9,499/58,686 flip-language turns, 13,192/58,686 metacognitive turns). A fresh matched Selective Flip Score corpus closes the ordinary flip-selectivity caveat in the expected direction: 150 Codex learner retries (5 misconception families $\times$ 10 replicates $\times$ targeted/mismatched/generic feedback), invalid choices 0; targeted, mismatched, and generic correct-flip rates all 1.000; SFS=0.000; false-flip=1.000; paired CI [0.000, 0.000]. This confirms that the prompted learner corrects after irrelevant or generic feedback as readily as after targeted feedback. A stricter DAG-SFS follow-up supplies the constructive repair condition rather than contradicting that result: across 120 Codex retries on six invented micro-proof DAGs (5 replicates $\times$ targeted/mismatched/generic/nonsense feedback), targeted release of the withheld proof edge yields proof-grounded recovery 30/30, while wrong-edge, generic, and nonsense feedback produce 0/90 false-grounded recoveries; DAG-SFS=1.000, paired CI [1.000, 1.000]. Rows count only when the learner cites the exact target edge ID and derives the conclusion from public evidence, so final-answer guessing is not enough. The implication is that synthetic learner outcomes require harness-owned public state and exact evidence-use scoring; ordinary roleplay outcomes remain bounded. IRT ability placement remains blocked by placeholder item-bank stubs. Artifacts: `exports/plan3-synthetic-closure-audits/report.md`; `exports/plan3-sfs-audit/sfs-matched-feedback.md`; `exports/plan3-dag-sfs-audit/dag-sfs-matched-fe`

The learner superego paradox (d=3.05, Paper 1.0, Section 6.16) complicates this further: the multi-agent learner architecture lowers synthetic learner rubric scores by d=3.05, but whether this reflects architecture-level degradation or a rubric-architecture mismatch is empirically separable. The next paragraph reports a DB-only mechanism check that favours the rubric-artifact account — i.e., the architecture produces substantially more deliberation-internal content that the current rubric does not reward — though the stronger test against human-judged learner output has not yet been run.

A rubric-measurement artifact is the most parsimonious account. A DB-only mechanism check on cells 1/2/5/6 (base $\times$ recognition $\times$ unified/ego_superego learner, Opus 4.6 judge, N=613/108/499/157) separates learner *behaviour* from learner *score*. The ego-superego learner architecture *produces more behaviour*: mean output tokens 13,193–15,547 vs 5,125–6,675 (~2.5$\times$ more text generated), and mean API calls 9.0–10.4 vs 2.3 (~4$\times$ more deliberation rounds). Yet learner overall scores are systematically *lower* (base: 53.94 vs 72.06, $\Delta = -18.1$; recognition: 59.19 vs 71.38, $\Delta = -12.2$). The direction is consistent: the architecture that performs *more* internal self-critique scores *worse* on the external rubric, despite consuming more tokens and more deliberation rounds. Per-dimension inspection of the learner_scores JSON reinforces this: ego-superego learners score well on `deliberation_depth` (which rewards the internal process directly) and `revision_signals` (which rewards observable revision), but poorly on `conceptual_engagement` and `question_quality` (which reward substantive *content* in the output). The rubric as currently constructed rewards content engagement in the final external message; the ego-superego architecture spends generative effort on internal deliberation that produces polish and metacognitive commitments ("I'll actually read the full passage") rather than conceptually rich outputs. The "d=3.05 quality deficit" therefore reflects *what the rubric measures and what the architecture produces*, not that internal self-critique degrades the learner as such. A fairer test would either extend the rubric to reward deliberation-internal content or compare architectures on human-judged engagement rather than rubric-scored output.

A convergent-validity check across the broader corpus supports the artifact account. The database contains 1,759 rows carrying *both* a per-turn learner rubric score and a holistic learner score — the latter a global dialogue-level quality judgment that is not decomposed into the rubric's conceptual-engagement / question-quality / deliberation-depth dimensions. On these paired rows, the rubric and holistic judges agree within each architecture at Pearson $r = 0.72$–0.77 (`unified` $r = .73$, $n = 432$; `ego_superego` $r = .75$, $n = 411$; `unified_recognition` $r = .77$, $n = 446$; `ego_superego_recognition` $r = .72$, $n = 470$). The rubric is therefore a reasonable proxy for holistic quality judgment within each architecture. The paradox contrast, however, does *not* replicate on this wider sample: pooled across all cells, ego_superego learners score *higher* than unified learners on both metrics (base contrast: rubric $\Delta = +4.14$, $d = 0.25$; holistic $\Delta = +7.39$, $d = 0.39$; recognition contrast: near-zero on both, $|d| < 0.08$). The cell-level breakdown of the large-N messages-mode cells (80–87, $n = 179$–216 per cell) consistently shows ego_superego $\geq$ unified on both metrics. The $d = 3.05$ paradox was computed on the eight-cell 2$×2×$2 factorial (cells 1–8, Paper 1.0 §6.16, single-prompt mode, $n = 18$ per cell), and the pattern does not survive at the architecture level when messages-mode cells dominate the sample. This is consistent with a measurement-artifact account: the original paradox is a scenario-and-mode-specific interaction between the rubric's content-centric dimensions and the ego_superego architecture's deliberation-internal content generation, not a general property of "internal self-critique degrades the learner."

Cell-matched holistic scores on the original paradox run: rubric d reproduces; holistic d shrinks by roughly half. The definitive adjudication adds holistic learner scores to the originally-scored 48 rows of the paradox run (eval-2026-02-20-25c78e91, single-prompt mode, Opus 4.6 judge on rubric v1.0, 6 rows × 8 cells; 47 paired after one judge JSON-parse failure). To avoid cross-version contamination, a standalone script wrote only the four holistic fields via direct SQL `UPDATE`, preserving the `learner_rubric_version = 1.0` marker rather than triggering the store's default version-overwrite (Sonnet 4.5 judge for holistic, matching the v3 triple-panel convention). The rubric d on this paired subset reproduces the paradox essentially exactly: base contrast (ego_superego − unified) **rubric** $d = -2.82$ ($\Delta = -23.3$, $n = 11/12$); recognition contrast **rubric** $d = -3.11$ ($\Delta = -22.3$, $n = 12/12$). On the *same rows*, the holistic judge narrows both gaps by roughly half: base **holistic** $d = -1.34$ ($\Delta = -21.4$); recognition **holistic** $d = -1.06$ ($\Delta = -11.7$). Critically, within-architecture rubric holistic correlations on the paradox cells are *weak and heterogeneous* ($r = 0.06$–0.44 across the four architectures) compared to the corpus-wide $r = 0.72$–0.77 pattern — the two judges specifically diverge on the paradox cells rather than agreeing as they do elsewhere. The evidence is mixed but tilts toward the artifact account: the ~50% shrinkage in $|d|$ plus the selective breakdown of convergent validity on these cells indicates the rubric contributes roughly half of the original $d = 3.05$ effect. The residual holistic gap ($|d| \approx 1.0$–1.3) is plausibly a single-prompt-mode interaction rather than a general capability ceiling, since the same architecture scores equal-or-higher on both metrics in messages-mode cells 80–87 (preceding paragraph). The paradox is therefore best characterized as a **combined measurement-plus-mode interaction specific to single-prompt evaluations**, not a universal property of internal self-critique. The ∼$2K higher-capability-model experiment previously priced as the adjudicating test is no longer required for the central capability-vs-artifact question; subsequent work should target the single-prompt-mode interaction directly rather than re-testing capability at frontier scale.

8.2 LLM-as-Judge Evaluation

Using LLM judges to evaluate recognition quality may introduce systematic biases. The judge may reward surface markers of recognition (acknowledging learner contributions, using inclusive language) rather than genuine engagement. This evaluator-class concern is independently documented in the LLM-as-judge bias literature, including systematic position bias (L. Shi et al., 2024) and unacknowledged shortcut bias (Marioriyad et al., 2025). **Cross-judge validation is now complete**: three independent judges (Claude Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4) scored all 1,296 rows across three generation models (3 runs × 3 judges × 144 rows, zero nulls), with blind scoring confirmed by code audit.

The cross-judge results establish both convergent validity and judge-specific patterns. Recognition effect directions replicate unanimously across all 9 judge × run cells (d = 1.34–1.92). Inter-judge Pearson correlations range from r = 0.45 to r = 0.89, with reliability varying by dimension (tutor overall: r = .61–.89; dialogue quality: r = .54–.87; holistic: r = .45–.79) and by generation model (highest on Gemini Flash 3.0 at r = .63–.89, lowest on DeepSeek at r = .45–.80). The higher reliability on intermediate-quality generation suggests that judges agree most when quality differences are clear and unambiguous. A consistent leniency gradient emerges: Gemini 3.1 Pro is most lenient, GPT-5.4 most severe, with Sonnet intermediate. This gradient is judge-specific rather than run-specific, confirming it reflects judge calibration rather than data artifacts.

For mechanism-level claims, the judge limitation is particularly relevant to the superego critique taxonomy (Section 5.3): critique categories are classified by an LLM, and the ego's "substantive revision" versus "cosmetic compliance" is assessed by an LLM judge. The mechanism claims are thus LLM-assessed claims about LLM-internal processes—a recursive structure that introduces the possibility of systematic blind spots. The cross-judge validation mitigates this for factorial-level claims (all three judges agree on effect directions and interaction patterns), but does not address it for process-level claims where only a single classifier is used. Human expert coding of a representative subsample is the most important outstanding validation for the mechanism claims. The pilot protocol is now fully operationalised in the repository (§5.3, "Human Validation Pilot"): deterministic stratified sampling (`scripts/human-validation-sample.js`, seed=20260416, n=40 across 10 categories), a per-category codebook with signal phrases, boundary rules, and exemplars (`docs/research/human-coding-codebook.md`), and an analysis pipeline computing Cohen's κ, Fleiss' κ, per-category F1, and confusion matrices (`scripts/human-validation-analyze.js`), plus an inter-LLM baseline (Haiku vs Sonnet on the same sample) as a lower-bound reference against which human-LLM agreement can be calibrated. The target floor is ≥ 0.60 (substantial agreement per Landis-Koch bands). Until the filled rater CSVs are returned and analysed, the process-level findings should be treated as LLM-generated interpretive categories whose correspondence to human expert judgment has not yet been empirically established. The apparatus reduces the execution cost to "recruit two coders, send them the packet, run the analysis script" — human time rather than rebuilding infrastructure.

8.3 Model Transience

Findings are model-version-specific. Paper 2.0 uses three generation models—DeepSeek V3.2 (open-weight, 685B MoE), Haiku 4.5 (proprietary, optimized for speed), and Gemini Flash 3.0 (proprietary, multimodal)—with three independent judges (Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4). The recognition effect replicates across all 9 judge × run cells (d = 1.34–1.92), substantially strengthening the generalizability claim relative to the two-model analysis. The addition of Gemini Flash 3.0 (weaker than the other two models) partially addresses the capability-range limitation: it shows that mechanisms replicate in direction on weaker models while revealing generation-quality boundary conditions on their interactions (universal substitution with model-dependent residual, learner invariance → sensitivity). The Nemotron data in the database (N=40, mean tutor score ~8 under base) hints that very weak models may not benefit from recognition at all, but the sample is too small for confident claims. The two supported mechanisms (calibration through prompt orientation, error correction with a model-dependent residual) describe *prompt-level and architecture-level* properties that should generalize across model versions, supported by three-model × three-judge replication. The null finding for adaptive responsiveness also replicates across all three models and judges.

Per-judge effect-size decomposition. While direction replication across all 9 cells establishes convergent validity, the *magnitude* of the recognition effect varies systematically by judge. Decomposing the d = 1.34–1.92 range from Section 6.1 by (run × judge) cell on tutor_overall_score:

Table 108: Cohen's d for the recognition main effect by generation model and judge, computed from within-group SDs pooled across the two conditions (n=72 per condition per cell). Within-row Δd is the max–min range of effect sizes across the three judges for a given generation model.

Generation Model	Sonnet 4.6	Gemini 3.1 Pro	GPT-5.4	Within-row Δd
DeepSeek V3.2	1.85	1.57	1.34	0.51
Gemini Flash 3.0	1.87	1.40	1.77	0.47

Generation Model	Sonnet 4.6	Gemini 3.1 Pro	GPT-5.4	Within-row Δd
Haiku 4.5	1.92	1.34	1.57	0.58
Judge mean	**1.88**	**1.44**	**1.56**	–

Three features deserve note. First, the across-judge Δd exceeds the 0.1 d robustness threshold in all three rows (range 0.47–0.58), so absolute effect-size estimates are judge-dependent even though effect direction is conserved. Second, Sonnet 4.6 systematically reports the largest effects (judge-mean d = 1.88, ~20% higher than Gemini 3.1 Pro's 1.44 and ~20% higher than GPT-5.4's 1.56). Because Sonnet was the original judge for the pilot study and for much historical analysis, **claims conditioned on Sonnet-only scoring should be read as an upper-bound estimate of pooled effect magnitude**. Third, the pooled-across-judges mean is $d \approx 1.63$ (mean of 9 cells); we treat this as the judge-adjusted best estimate, with Sonnet-only $d \approx 1.88$ as an upper bound and Gemini-3.1-Pro-only $d \approx 1.44$ as a lower bound. All three judge panels leave the recognition effect in Cohen's "very large" range ($d > 1.3$), but reporting only one judge without adjustment risks a misleading impression of magnitude. The pattern is partially but not fully consistent with the leniency gradient noted in §8.2: Sonnet produces the highest d across all three generation models, but the Gemini-vs-GPT ordering flips between runs (GPT > Gemini on Gemini Flash 3.0 and Haiku; Gemini > GPT on DeepSeek), suggesting judge × generation-model interaction beyond simple calibration differences.

Two implications for replication practice follow. Pre-registered judge panels reduce the risk that downstream claims inherit one judge's idiosyncratic severity; and per-judge effect sizes should be reported alongside pooled estimates, rather than pooling silently under a single judge and calling it the effect. The ordering across *generation models within a judge* is more stable than the ordering across *judges within a model*, which supports using the architecture × recognition interaction structure (Sections 6.4–6.6) as the primary inferential target rather than absolute effect magnitudes on any one judge.

8.4 Process Tracing with LLMs

Our adaptation of process tracing from social science to AI systems introduces a philosophical complication. In political science, process tracing examines *actual* causal chains: the decision-maker's deliberation, the institutional constraints, the information flows. In our architecture, the ego-superego exchange is *generated* text—the "deliberation" is a prompted LLM output, not a cognitive process in any neuroscientific sense. The mechanism claims are about designed information flows between prompted agents, not about internal mental states.

This limitation is softened by the distinction drawn in Section 3: we study recognition-*oriented design* (functional analogues), not recognition *proper* (genuine intersubjectivity). The process tracing examines whether designed information flows produce measurable output differences—a behavioral claim, not a cognitive one. The null finding for adaptive responsiveness as a third mechanism illustrates the value of this approach: the method was capable of distinguishing supported mechanisms (calibration and error correction, where condition-dependent effects are observed) from unsupported ones (adaptive responsiveness, where the null is well-powered). But the language of "mechanisms" may imply more causal depth than the evidence supports.

A related concern is whether the paper conflates *engineered architecture* with *discovered mechanism*. The ego-superego pathway, the critique taxonomy, and the revision loop are designed into the system. In one sense, finding that the superego catches errors the ego misses is reporting that the architecture works as intended, not uncovering a naturally occurring mechanism of good tutoring. We acknowledge this distinction: the mechanisms we trace are properties of *this specific architecture*, not universal laws of AI tutoring. Calibration describes how recognition prompts constrain *this system's* output distribution; error correction describes how *this system's* superego-ego interaction functions. Whether other architectures (chain-of-thought, debate-based, or single-agent with self-critique) would show analogous patterns under recognition-oriented prompting is an open question (Section 8.5). However, the findings are not purely tautological. The architecture was designed to enable observability, not to guarantee specific outcomes. Three results were genuinely unpredicted: that calibration alone accounts for most of the recognition effect (making the superego largely redundant on strong models), that adaptive responsiveness fails as a general mechanism despite being the most theoretically motivated prediction, and that the substitution interaction produces a consistent 15–17% additivity deficit across three structurally different models. The system was built to make mechanisms *observable*; what those mechanisms turned out to be was an empirical question.

8.5 Single-System Study

All findings come from a single architectural implementation (the ego/superego tutoring system with recognition-enhanced prompts). The two supported mechanisms are identified through this specific architecture's observability. Other architectures—chain-of-thought reasoning, debate-based systems, recursive self-improvement—might implement recognition-like behaviors through different internal processes. The mechanism model may not transfer to architectures that lack explicit ego-superego separation or that do not log internal deliberation.

The V17 tutor-stub matrix in §7.4.5 is narrower than even this main-harness scope. It is a four-cell engineering acceptance test over simulated learners, fixed models, strict response audits, and a ten-turn horizon. Its zero delivered failures establish that unsafe or poorly realized candidates were intercepted in those trajectories; they do not establish that the first drafts were reliable, that all dialogues completed their proof path, that the selected character improved learning, or that the same boundary transfers to human learners and untested model stacks. Because candidate repairs contribute to the passing verdict, future claims about generation quality should report first-draft acceptance separately from final-delivery safety rather than allowing repair success to erase dependence on the repair layer.

8.6 Rubric Evolution

The rubric evolved through four versions during the study. Paper 1.0 data uses v1.0 (14 tutor dimensions); Paper 2.0 data uses v2.2 (8 tutor dimensions). Cross-version comparisons are inadvisable—the dimension structure, weighting, and judge instructions differ—so we do not retroactively score historical data under newer rubric versions. This means pilot findings (v1.0) and mechanism findings (v2.2) use different measurement instruments, and direct numerical comparisons between them should be treated with caution.

The v2.2 rubric has been validated through synthetic calibration (r=+0.996 tutor, r=+0.968 learner) and through initial empirical use on cells 80–87 (N=309 under v2.2). The 8-dimension structure produces coherent findings: floor-lifting patterns, dimension-rank consistency across models, and calibration effects that align with theoretical predictions (Section 6.1).

However, principal components analysis on 1,584 per-turn observations reveals that the 8 dimensions largely measure a single construct: PC1 explains 80.7% of variance, and the Kaiser criterion yields only one factor with eigenvalue \$>\$1. Sampling adequacy is excellent (KMO = 0.938), and the mean inter-dimension correlation is r=0.776 (range 0.589–0.921). This pattern replicates across conditions (PC1: 80.2% base, 75.6% recognition) and models (77.3% DeepSeek, 68.0% Haiku). Forced two-factor varimax rotation separates `content_accuracy` (loading 0.923 on Factor 2) from the seven pedagogical dimensions (loadings 0.68–0.85 on Factor 1), suggesting that the rubric captures two facets—factual correctness and pedagogical quality—but within the pedagogical facet, the dimensions are not empirically distinct.

This has two implications for the mechanism findings. First, the calibration effect (Section 6.1)—within-response dimension variance reduction under recognition—is meaningful precisely *because* the dimensions co-vary: a prompt that lifts one dimension tends to lift all, and calibration describes the narrowing of an already correlated profile. Second, the rubric is functionally over-specified: 8 nominally distinct dimensions reduce to 1–2 empirical factors, and future iterations should either consolidate or demonstrate discriminant validity. The current findings should be interpreted as measuring overall tutoring quality with a content-accuracy modifier, not 8 independent constructs.

Implication for §7.8 dimension-targeted optimisation. The PCA finding also bears on §7.8.1's dimension-targeted autotuning results. Targeting `adaptive_responsiveness` or `conceptual_progression` as optimisation objectives does not isolate empirically independent skills; those dimensions load on the same underlying tutor-quality factor as the others. The §7.8.1 results should therefore be read as *shifts in the single underlying construct via prompt changes whose wording foregrounds a particular facet*, not as selective movement of one skill while others are held constant. This does not invalidate the §7.8 findings — prompt optimisation did improve scores substantially (Qwen target dims 35 → 75), cross-model transfer did produce the prosthesis-straitjacket inversion, and these results are informative about how to engineer prompt interventions for different model capabilities. But the mechanistic interpretation is one of *shaping the overall quality factor* rather than *building a specific pedagogical skill*, and readers should not treat per-dimension improvements as evidence of independent constructs being independently tunable.

Methodological pairing with §7.10 (Simpson's-paradox at the response level). A second methods-level caution arose from the §7.10 D1 mechanism analysis: mediator-criteria checks must use within-group correlations when the predictor varies across groups. Embedding-similarity to an intersubjective canonical was *positively* correlated with score when pooled across the four A10b cells ($r = +0.26$) but reversed sign within each intersubjective cell ($r = -0.28$ in cell_5, $r = -0.24$ in cell_95) — a textbook Simpson's paradox driven by the cell_96 transmission outlier anchoring the lower-left of the pooled scatter. Together with the PCA over-specification finding above, these pair as two methods-level cautions for any future mediator analysis on LLM-tutor data: (1) the v2.2 rubric's eight dimensions reduce to one underlying construct so per-dimension claims should be hedged accordingly; and (2) pooled-r in mediator analysis can be entirely driven by between-group variance rather than within-group mechanism. Together they could form the empirical core of a methods-paper contribution beyond Paper 2.0; for the present paper they are documented here as caveats that any reader applying the apparatus elsewhere should account for. §8.6.1 below makes the second caution explicit as a portable diagnostic.

8.6.1 The three-criterion within-cell mediator diagnostic The §7.10 mechanism analysis used a specific diagnostic to distinguish *family markers* (features that distinguish prompt families but do not predict scores within a family) from *within-cell mediators* (features that predict scores even when the prompt is held constant). We articulate the diagnostic explicitly here because the same logic applies to any LLM-tutor evaluation that contrasts prompt families and seeks response-level mechanisms.

A response-level feature qualifies as a within-cell mediator candidate when all three criteria are met:

1. **Large family contrast** ($|d| \geq 0.5$ between the family containing the recognition cell and a contrasting transmission family). The feature distinguishes the families. If it does not, it cannot mediate the family-level effect.
2. **Small within-family contrast** ($|d| < 0.5$ between two cells within the same family, e.g., recognition vs matched-pedagogical). Both family-member prompts elicit the feature; if only one does, the feature is a *prompt marker* (specific to one prompt) rather than a *family marker* (shared across the family).
3. **Positive within-cell Pearson r with score** in each family-member cell. When the prompt is held constant, the feature still predicts higher scores. This isolates the response-level mediating role from cross-cell variance that the family contrast already captures.

The §7.10 worked example applied this diagnostic to the D1 five-pass feature sequence on the A10b 4-cell corpus:

Feature	Criterion 1 (family d)	Criterion 2 (within-family d)	Criterion 3 (within-cell r in cell_5 / cell_95)	Verdict
Hegelian lexicon density	small	very large (cell_5 $\gg$ cell_95)	small	Prompt marker only
Intersubjective lexicon density	moderate	very large (cell_95 $\gg$ cell_5)	mixed signs across cells	Prompt marker only
Question-mark rate	0.61	0.47 (cell_5 3$\times$ cell_95)	small	Prompt marker, modest predictor
Scaffolding-move imperatives	0.59	0.12	+0.19 / +0.05	Family marker, weak mediator
Broad acknowledgement	0.14	small	−0.18 / −0.31	Negative correlate (anti-mediator)
Ends-with-question (single-turn cells)	**categorical** (4.5% / 2.1% vs 0%)	**0.13**	+0.325 / +0.392	**Mediator**
Intersub. canonical similarity (embedding)	**0.81**	0.27	−0.28 / −0.24 (negative within-cell, positive pooled)	Family marker; **Simpson's paradox disqualifies as mediator**

A feature passes the mediator bar only by satisfying all three criteria; failure on any one disqualifies the feature as a within-cell mediator while potentially preserving its role as a family marker, prompt marker, or negative correlate. The diagnostic is most informative when criterion 3 fails despite criteria 1–2 passing: that pattern indicates Simpson's paradox, where a feature appears mediating across families (criterion 1 large, pooled $r > 0$) but the within-cell direction reveals that the cross-family score gap, not the within-family feature variation, was carrying the pooled correlation.

The diagnostic generalises beyond the recognition–transmission contrast. Any factorial design with a family-level main effect and candidate response-level mediators can apply the three-criterion check, and the §8.6 PCA finding adds a complementary rubric-construct caution: per-dimension "mediation" claims should be hedged when the rubric's dimensions are not discriminantly valid. Together the rubric-construct caution and the within-cell-mediator diagnostic constitute two independent methods-level corrections that any future LLM-tutor evaluation should account for. Their empirical foundations are documented here as Paper 2.0 caveats; a separate methods-paper extraction is the natural vehicle for arguing their generality across published LLM-tutor evaluation pipelines.

8.6.2 Application protocol for transferring the diagnostic The diagnostic transfers to any LLM-tutor evaluation pipeline that exposes (a) per-row condition labels grouping responses into prompt families, (b) per-row scores from a judge, and (c) at least one per-row response-level feature derivable from the response text or trace. The transfer protocol is procedurally explicit, requires no new generation or judging, and operates on data already collected in the source evaluation:

1. **Identify the family partition.** Group conditions into family A (the family the source evaluation claims is doing pedagogically distinctive work) and family B (a contrasting transmission-style family). Within each family, identify two cells whose prompts share the family's pragmatic stance but differ in surface composition (e.g., recognition vs matched-pedagogical for A; generic base vs matched-behaviorist for B). If only one family-member cell is available, criterion 2 cannot be evaluated and the protocol degrades to a two-criterion check; this is acceptable but weaker.
2. **Compute the candidate feature** f per row from the available text or trace data. Regex extractors (per §6.1.7 question-asking and §7.10 ends-with-question) and embedding-based features (per §7.10) are both supported. Author-specified feature definitions should be reported alongside results.
3. **Test criterion 1** (family contrast): compute Cohen's d for f between pooled family A and pooled family B. Require $|d| \geq 0.5$.
4. **Test criterion 2** (within-family contrast): compute Cohen's d for f between the two cells within family A. Require $|d| < 0.5$. If only one A-cell is available, skip; flag in the report.
5. **Test criterion 3** (within-cell correlation): compute Pearson r between f and score within each family-A cell separately. Require both $r > 0$. Negative within-cell r in one or both cells disqualifies f as a mediator and may flag f as a *negative correlate* (cf. broad-acknowledgement, §7.10). Pooled $r > 0$ with within-cell $r < 0$ in both cells is the Simpson's-paradox signature; report explicitly.
6. **Pair with rubric over-specification check** (§8.6): compute PCA on the per-dimension scores from the source rubric. Report PC1 variance share, KMO, mean inter-dimension r, and Kaiser eigenvalue count. PC1 $\geq 70\%$ flags rubric over-specification; per-dimension claims should be hedged accordingly.

Reference scripts for steps 2–5 are open-sourced as `scripts/analyze-d1-structural-features.js`, `scripts/analyze-d1-cross-judge-replication.js`, and `scripts/analyze-d1-multifeature-ols.js`. The same scripts apply to any input dataset matching the schema described in step 1; data-loading is the only extension point. Future application of the protocol to independently published rubrics — for example, MathTutorBench (Macina et al., 2025), ICAP-aligned LLM-as-judge studies (Chi & Wylie, 2014), or the structured rubric methodology of LLM-Rubric (Hashemi et al., 2024) — would test the diagnostic's generality across rubric design choices and is the natural next empirical step beyond Paper 2.0's source apparatus.

8.7 Computational Cost Asymmetry

The recognition-enhanced prompt design has a practical cost: the tutor ego prompt (~5,100 tokens) is 8× larger than the learner ego prompt, and multi-agent deliberation multiplies this per turn. A 10-turn multi-agent dialogue consumes ~225,000 tokens (64 API calls), with the tutor accounting for 3.4× more input tokens than the learner (Section 4.2). Recognition-enhanced tutoring is substantially more expensive per interaction than baseline tutoring, and this cost scales with turn count and rejection rate. Whether the quality gains justify the cost depends on the deployment context; the substitution finding (Section 6.4) suggests that single-agent recognition cells, which eliminate the superego's redundant correction rounds, may offer a better cost-quality trade-off on capable models.

Reproduction cost for Paper 2.0 mechanism runs. A lab seeking to replicate the three core Paper 2.0 generation runs (DeepSeek V3.2, Gemini Flash 3.0, Haiku 4.5; 432 dialogues across 18 cells × 3 runs) faces a recorded generation cost of $114.58 across 163M tokens (144M input, 19M output) on OpenRouter's pass-through pricing — kept low by the choice of three cost-efficient generation models. Judge costs are not separately tracked in the database but can be estimated: a 3-judge panel (Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4) scoring all 1,296 multi-turn rows (~8K input + ~1.5K output per judge call) costs an estimated $300–700 depending on judge price mix. Superego-taxonomy classification adds a further ~$80–100. Full reproduction of the core Paper 2.0 runs thus falls in the $500–1,500 range at late-2026 API rates; including the isolation runs (Section 6.4.2), trajectory-specific scenarios (Section 6.3), and prompt-lab autotuning (Section 7.8) roughly doubles this to $1,000–3,000. The total evaluation database used throughout this programme (207 runs, 8,274 dialogues, ~740M tokens) consumed $474.75 in recorded generation cost, with estimated total cost including judging at $2,000–5,000. These figures assume LLM judging; substituting human expert coding of 30–50 exchanges per mechanism (the outstanding validation discussed in Section 8.2) would add rater labour costs, and full-scale human-learner validation (Section 8.1) at $50–100 per participant for a target N=200 would push replication costs into the $20K–50K range. The LLM-judged apparatus is therefore inexpensive to reproduce — the genuine replication barrier is scaling up to human validation, not the compute.

8.8 Pilot Data Transparency

The companion pilot study documented nine post-extraction corrections, four of which affected quantitative results. Bug 4 (multi-turn scoring misalignment) affected 8,631 rows. Bug 5 (resume model override loss) affected one run. Bugs 1–2 (learner prompt leakage, broken conversation history) required complete re-generation of all dynamic learner data. These corrections are documented transparently in the companion study and in the project's bug registry, and the provable discourse framework tracks which claims depend on corrected versus pre-correction data. The correction history demonstrates the evaluation infrastructure's capacity for self-correction—but it also means that pre-correction findings, including some reported in early presentations of this work, should not be cited without noting the corrections.

8.9 Scope of the Id-Director Extension

The id-director family findings (§6.7, cells 101–107) differ from the paper's mechanism-tracing core in scope along four axes. *Single-judge primary scoring*: v2.2 scoring used `claude-code/sonnet` exclusively, with `openrouter.gpt-5.2` as a cross-judge sanity check on $n = 27$ paired rows per cell — far short of the three-judge $\times \geq 144$-row matrix supporting the paper's core mechanism findings (§5.8). The cross-judge check confirms architectural rank-order but compresses absolute magnitudes by 10–15 v2.2 points; published numbers should specify the judge. *Single model family*: id is Kimi K2.5 across all id-director cells; ego is Nemotron-3-nano. The paper's core findings are validated across DeepSeek V3.2, Haiku 4.5, and Gemini Flash 3.0 (§5.8); the id-director extension is not. Whether the architectural ranking generalises to other id × ego model pairings is open. *Per-cell N is heterogeneous and partly below the paper's core floor*: closing N is $n = 79$–81 for the four central cells (101, 103, 104, 105), $n = 54$ for cells 102 and 106 (two of three scenario types covered), and $n = 27$ for cell 107 (the most exploratory closing pilot). Three of seven cells therefore meet or exceed the paper's core $n \geq 63$ per dialogue-condition floor; one cell (c107) sits substantially below it and is reported as a pilot-scale rather than confirmatory result. Variance estimates on the lower-N cells should be read accordingly. *Different theoretical orientation*: the charisma rubric grounds in Weberian charismatic authority (Weber, 1978) rather than Hegelian recognition. The two are correlated but not redundant; the §6.7 results do not displace recognition theory but document an architectural variant in which a complementary frame is also load-bearing. The §6.7 claims are framed as architectural exploration rather than as additional mechanisms in the three-mechanism model. Where the id-director extension introduces methodological novelty (the prompt-budget non-monotonicity, the rubric-curriculum alignment ablation, the cross-judge magnitude-compression observation), the contribution is descriptive and supplementary to the paper's main argument; full audit trail is in the five `docs/cell-100-*.md` documents and the dialogue-log JSON for each pilot run.

The desire-for-recognition follow-up in §6.7 is narrower still. It is an iterative design slice, not a confirmatory matrix: one run per listed profile-scenario row in the cells 159–169 authority-refusal search; selected static-profile diagnostics in cells 170–179; then a local resistance-breakthrough router line in cells 180–195. The first result is sufficient for a paper-side design finding ("accountable-bid charisma is the first clean local pass under authority refusal") but not sufficient for a general claim about charismatic tutoring. The second result shows why a static-cell ladder is the wrong target: transfer demand, authority refusal, plain-language request, simplification follow-up, and resistance after scaffolding are different engagement states. The third result shows that an engagement-register router can work locally under the Codex/Claude stack (cell 193: 25/26 positive local mechanism outcomes across the combined eligible evidence) but also supplies the boundary: GLM 5.2 generation does not reproduce the mechanism, and even provider/runtime reasoning controls that produce clean DB validation leave the local breakthrough outcome at 0/5 on the five-signal check. A completed six-arm role-isolation grid now identifies the most plausible boundary as dynamic-learner completion and target drift rather than tutor/id public-register production: the GLM tutor/id with Codex learner arm remains locally usable (8/10 positive outcomes), the GLM dynamic-learner arms lose post-turn completion or target stability, and both scripted tutor controls route the public register with fixed resistant turns. The separate question-flood gate promotes cell 192 only for that subtype (2/2 answer-first usable commitments, 0 reopened or residual floods), not for the broader five-signal mechanism or GLM stack. To upgrade the finding after that pivot, the router would need a frozen grid spanning authority-withheld, status-challenge, partial-uptake, ordinary conceptual, vulnerability/persona-shift, transfer, plain-language, and instruction-to-engagement-switch scenarios; ≥ 10 repeats per profile-scenario as a pilot floor and preferably ≥ 24 for stable variance estimates; frozen comparators including cell 163, cell 104/107-style id-director baselines, the default tutor, and a no-router id-director baseline; at least two judges for both v2.2 and charisma scoring;

model-pair variation across Codex/Claude, non-CLI hosted models, and smaller ego models; and a human-coded or human-learner subset focused on whether learners experience the tutor's authority as defeasible rather than merely polished. Without those extensions, cells 169, 193, and the cell-192 question-flood subtype should be cited as exploratory design evidence motivating engagement-register routing, not as a generalizable effect.

The negative-register extension adds a further scope condition: arm assignment is not treatment fidelity. A cell can be configured as `sarcastic_challenge` while the generated tutor turn remains warm, ironic, or only superficially edgy. The hand-authored corrosive-exemplar check in §6.7 shows that the scoring instruments can detect corrosive sarcasm when fixed examples actually contain it, so the immediate limitation is not global judge blindness but generated-register fidelity. A fresh stance-gated smoke confirmed the practical scale of that limitation before repair: only 5/15 assigned negative-register rows counted as faithful arm evidence, while 10/15 were excluded as weak or warm-in-costume noncompliance and none were invalid person-attack violations. Later cue-repair canaries then showed that explicit stance-fidelity cues can restore treatment fidelity on sampled targets: 9/9 faithful, 0 invalid across boredom/irrelevance/rote-parroting, followed by 6/6 faithful, 0 invalid on held-out frustration/question-flood. Across the two post-repair canaries, all five resistance targets have now been sampled without gate exclusions or invalid person-attack violations; this is still an implementation check rather than an effect estimate. Negative-register claims must therefore report both assigned-arm and faithful-arm estimands, and rows should contribute to a negative-register effect estimate only after the adopting tutor turn is classified as faithfully instantiating the assigned register contract. A related scope condition concerns the judge rather than the tutor: the register rubric's own scores were originally produced by `openrouter.gpt-mini`, later shown to be numerically non-discriminating on this rubric (§6.7); every register-rubric conclusion for cells 196–198 now rests on a re-score under a sonnet-class judge (`claude-code.sonnet-5`), and the archived gpt-mini register-rubric figures are void and must not be cited. Sonnet-class judging is the standing requirement for the register rubric from the first row of any future negative-register run. Cells 196–198 remain a measurement-development addendum, not evidence about the pedagogical value or safety of sarcasm.

9. Conclusion

A companion pilot study established that recognition-enhanced prompts and multiagent architecture produce large, replicable differences in AI tutoring quality. This paper asked the next question: *through what mechanisms?*

Two mechanisms supported, one not supported

We motivated three candidate mechanisms drawing on Hegel's recognition theory, each mapped to a specific architectural level: **calibration** (prompt-level output distribution narrowing), **error correction** (architecture-level superego critique that the ego incorporates rather than minimizes), and **adaptive responsiveness** (interaction-level turn-over-turn adaptation to learner-specific signals). Each generated testable predictions with explicit null hypotheses (Section 3.2).

The mechanism data (cells 80–87, three generation models, N=453, validated by three independent judges across 1,296 scored rows) **supports calibration and error correction as general mechanisms, and finds adaptive responsiveness null on both the primary analysis and the pre-registered replication of its one exploratory boundary-condition effect**. Calibration is the strongest effect: within-response dimension variance drops, the weakest baseline dimensions show the largest recognition lifts, and the effect operates identically in single-agent cells without a superego. The recognition main effect is unanimous across all 9 judge $\times$ run cells, judge-pooled d $\approx$ 1.63 (§8.3). Error correction interacts with calibration through **universal substitution** with a model-dependent residual: near-zero on strong models, substantial on Gemini Flash 3.0 where calibration alone cannot handle all failure modes (§6.4). Adaptive responsiveness is not supported as a recognition-modulated trajectory mechanism: the primary well-powered analysis is null (§6.3) and the earlier exploratory DeepSeek/Sonnet effect failed pre-registered replication (A12, §6.3.2). The Plan 2.x addenda narrow this conclusion by showing that heavier typed state-action governance can produce localized simulated strategy-shift and quality gains (§6.12.1–§6.12.3), but those gains are architectural, post-hoc, and trap-instrument-bound. The three-mechanism model therefore still resolves to **two supported general mechanisms and one clean null**, with a separate bounded positive for engineered conduct control.

The apparatus as contribution

A distinctive contribution is the argument that the evaluation apparatus itself—provable discourse infrastructure, rubric iterations, test suite as analytical provenance—constitutes a transferable methodology (Section 7). The process of building the apparatus mirrors the mechanisms it studies: the provable discourse framework functions as a "superego" for research claims, forcing genuine correction rather than cosmetic compliance. This reflexive structure is not accidental but a consequence of the subject matter: recognition theory predicts that understanding requires mutual constitution between researcher and object.

What comes next

The remaining agenda has shifted. Mechanism isolation is no longer a proposed future run: Section 6.4.2 reports direct M1/M2 isolation, and Section 6.4.2.1 links the substitution pattern back to the critique-taxonomy evidence. Likewise, the superego taxonomy is no longer merely an intended coding scheme: Section 5.3 reports the automated 500-critique corpus, the 10-category codebook, and the inter-LLM reliability baseline. What remains is not generation of the first mechanism evidence, but external validation, boundary mapping, and deeper mechanistic localisation.

First, **human expert coding of the superego taxonomy** is the nearest validation step for the process-level claims in Section 8.2. The apparatus already exists: a stratified 40-item rater packet, a matched key, a codebook with boundary rules and exemplars, and an analysis script for Cohen's κ, Fleiss' κ, per-category F1, confusion matrices, and human-LLM agreement (§5.3). The empirical task is to collect at least two independent expert ratings, compare them with the Haiku/Sonnet inter-LLM baseline ($\kappa = 0.633$), and decide which categories are stable, which need codebook refinement, and which should be treated as pilot-unstable. If the pilot clears the pre-specified $\kappa \geq 0.60$ floor, the next scale-up is a 100–150 item human-coded sample; if it does not, the taxonomy remains an LLM-process instrument rather than a validated expert coding scheme.

Second, **human learner validation** remains the critical external-validity test. The present study establishes tutor-output and simulated-dialogue effects; it does not show that recognition-enhanced tutoring improves human learning. The human pilot should therefore compare recognition-enhanced and baseline tutoring with real learners, measuring pre/post conceptual understanding, transfer, engagement, and learner experience rather than satisfaction alone. This is the point at which the synthetic-learner apparatus must give way to educational outcomes: the mechanisms identified here matter pedagogically only if they survive contact with human learners.

Third, **model-boundary and role-isolation tests** should map where the mechanism account stops generalising. The core Paper 2.0 triad covers moderately capable models; it does not settle very weak local models, frontier models beyond the tested range, or mixed-role stacks in which tutor ego, superego/id, learner, and judge come from different model families. The id-director and recent role-isolation attempts make this boundary concrete: dynamic-learner/runtime behavior can become the failure source even when the tutor/id stack appears locally coherent. A systematic boundary matrix would vary role assignment and capability tier explicitly, separating tutor-side robustness from learner-side simulation drift and judge-side preference.

Fourth, **model-stratified prompt optimisation** should turn the prosthesis-straitjacket finding (§7.8.3) into a usable threshold map. The current transfer experiment uses one scenario and two models: Qwen 3.5 9B benefits from heavy scaffolding, while Haiku 4.5 is damaged by the same prompt load. A larger study varying model capability, scenario type, conversation length, and optimisation target would locate the transition point where scaffolding changes from helpful to harmful. The practical endpoint is an adaptive prompt-selection policy: heavy procedural support for models that need it, light permissioning for models that already possess the relevant capability, and no universal "best prompt" assumed across capability levels.

Fifth, **white-box analysis targeting the within-cell mediator identified in §7.10** should connect the behavioural mediator to model internals. The five-pass mechanism analysis pinpoints "ending the tutor turn with a question" as the single feature that satisfies within-cell mediator criteria across both intersubjective cells (cell_5 $r = +0.325$, cell_95 $r = +0.392$); pragmatically, the act of ceding initiative back to the learner. With access to open-weights generation models and interpretability tooling such as TransformerLens or nnsight, the mechanism question sharpens from generic residual-stream fishing to specific hypotheses: whether the model encodes a "cede initiative now" direction late in the response, whether attention shifts to learner tokens before turn-ending question generation, and whether intersubjective-family prompts induce attention patterns over the learner's contribution that transmission-family prompts do not. Pinning the mediator to a pragmatic decision point makes the white-box work an applied interpretability problem with a defined target behaviour.

The broader implication

Recognition theory, operationalized as a design heuristic rather than an ontological claim, provides a framework for building AI systems whose outputs are specifically adapted to user input rather than generically responsive to it. The two supported mechanisms—calibration and error correction—describe how **intersubjective-pedagogy orientation** alters system behavior at prompt and architecture levels. A key finding of §7.9 is that this orientation, not recognition's specific Hegelian vocabulary, is the active ingredient: matched-specificity prompts grounded in Hegelian-descendant constructivist theorists (Vygotsky, Piaget, Kapur, with Chi, VanLehn, and Graesser as further cognate descendants) reproduce recognition's effect within pooled d = 0.14-0.19, while matched-specificity prompts grounded in the orthogonal behaviorist tradition score pooled $d = 0.89$ below generic baseline.

The empirical generalisability of recognition's effect across the constructivist family does not, however, displace recognition theory itself. The descendants the matched-pedagogical control was grounded in (§1) inherit the intersubjective commitment from Hegel, often without direct attribution: Dewey's pragmatism began as American Hegelianism; Vygotsky's dialectical psychology refracts Hegelian recognition through Marxist materialism; Piaget's assimilation/accommodation transposes the dialectic of identity-difference into developmental cognition. The matched-pedagogical control removes the Hegelian *vocabulary* from the prompt but inherits the Hegelian *commitment* through its grounding theorists. What A10/A10b's finding establishes is that the *operationalisation* of intersubjective orientation is interchangeable across cognate descendants. What remains uniquely valuable about recognition theory is that it makes legible what its descendants enact tacitly — the master-slave dialectic articulates the structure of intersubjective constitution more explicitly than ZPD or accommodation, and *Bildung* names the developmental trajectory more precisely than productive failure or step-level scaffolding. Recognition is therefore not "one of several alternatives that all work equally well." It is the philosophical articulation of an orientation its constructivist descendants carry implicitly, and which the present apparatus traces empirically through prompt-level and architecture-level mechanisms.

The null finding on adaptive responsiveness as a general recognition mechanism is itself informative: orientation works primarily through *level-setting* (how well the system starts and what it catches), not through accelerated adaptation across turns. The Plan 2.x addenda refine the practical lesson. They show that adaptation is not absent in principle, but it appears only when the architecture moves from orientation to governance: typed action contracts, outcome closure, no-unreasoned-repeat checks, and early termination after successful learner-owned no-intervention. That is a different mechanism class from recognition prompting. Whether these mechanisms generalize beyond tutoring to other AI applications (therapy, creative collaboration, customer service) remains to be seen. But the methodological contribution—process tracing adapted for agent architectures, provable discourse for machine-verifiable research claims, the orientation-family-as-unit-of-analysis framing, and trap-style instruments that separate strategy selection from prose quality—is transferable regardless of domain.

The dramatic-recognition sidecar makes this scope more precise rather than weakening it.

A gated D42/D50/D53 loop shows that tutor-private peripeteia can yield recognitive form once, but not repeat-stably: one of three iterations passed all nine item gates, while two failed at 6/9. This does not promote adaptive responsiveness as a supported mechanism. It identifies the engineering boundary left by the null: public habit-break and learner action can be induced, but reliable recognition requires a further action-to-re-reading bridge. The sidecar's latest rule change therefore treats the learner's final reorientation as an explicit object of mechanism design.

A further generative arc (Section 6.13) shifts the question one more step—from *reading* the learner's interior to *governing* the tutor's own conduct over a formal derivation whose proof state is machine-checkable. Its clearest positive result is that a pacing guard which holds each premise until the learner is ready grounds the target derivation where an unguarded tutor does not, and that a page-only guard watching only the visible transcript paces just as well on a linear proof chain—locating the lift in scheduling discipline rather than any privileged read of hidden state (§6.13.11). A forked world that decouples visible uptake from latent proof-distance then bounds the claim: there the page-only guard fails exactly where the proof-state-aware guard still grounds (§6.13.12), so a visible adaptivity proxy can be trusted only where the task's geometry lets the page faithfully project hidden task state. A linear-spine distractor world then completes the boundary by reversing the pairing — the page-only guard grounds where the proof-state-aware guard fails, lured onto a derivable decoy — so the two guards are complementary rather than ordered, and neither is safe across geometries (§6.13.13). The selector-reliability and A20/A21 closeout then sharpen the design consequence: a one-bit H/V selector does not beat simple static baselines, visible conduct enforcement can create negative transfer even when locally compliant, an action-value patch matches rather than beats hidden+proofDebt, and a calibrated ownership benchmark finds no proof-safe ownership gain in the mined artifacts (§6.13.15). A later task/session gate carves out only a lower-authority positive: an advisory public-only selector beats fixed progression on frozen held-out artifacts while proof-control fingerprints remain unchanged. A companion handoff probe adds only advisory deployment-risk classification from public signals, not learner routing or safety coverage. This is the null on adaptive responsiveness read from its constructive side: reliable adaptive *conduct* comes from checkable proof-continuity structure placed at the point of action; advisory outer-loop sequencing can be useful above that kernel, but additional controllers earn promotion only if they beat hidden+proofDebt without negative transfer, and the current artifact pool does not.

Three findings remain unexplained by the current mechanism model: the near-identical recognition effect magnitude across structurally different models (DeepSeek $d = 1.88$, Haiku $d = 1.84$, Gemini Flash 3.0 $d = 1.92$), the failure of adaptive responsiveness as a general mechanism despite being the most theoretically motivated prediction, and the learner superego paradox ($d = 3.05$) in which internal self-critique degrades rather than improves output quality. These anomalies suggest that the two supported mechanisms may be surface expressions of a deeper process—perhaps related to how recognition framing restructures the model's attention over its input context—that the current apparatus is not yet equipped to trace.

The central Hegelian insight is that genuine understanding emerges through encounter with an Other whose perspective cannot be reduced to one's own. The learner superego paradox ($d = 3.05$, the largest effect in the study) provides a striking functional parallel: internal self-critique degrades output quality; externally oriented recognition-enhanced prompts improve it. In the 3–5 turn window of the primary analysis, what matters is how well the system *starts* (calibration) and what it *catches* (error correction); within-dialogue trajectories are recognition-invariant, and an earlier exploratory 10-turn disengagement effect that might have extended the mechanism story to longer conversations did not survive replication (§6.3.2, A12). The primary analysis establishes that the conditions for genuine dialogue can be set from the first utterance; recent longitudinal evidence (§6.6.11, A7 Phase 2) extends the picture by one further time scale — across eight sequential sessions sharing a single learner identity, recognition arcs *gain* +1.31 score-points per session while base arcs lose −1.08, with the recognition tutor producing fewer dialectical events but consistently higher and improving outputs. The longer-horizon mechanism, on this evidence, is not adaptive responsiveness but **ego pre-alignment**: recognition's effect lives in what the model proposes first, accumulating across sessions through a Writing Pad that persists learner state.

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Appendix A: Full System Prompts

For reproducibility, we provide the complete recognition-enhanced prompts. Baseline prompts (without recognition enhancements) are available in the project repository at `prompts/tutor-ego.md` and `prompts/tutor-superego.md`.

A.1 Recognition-Enhanced Ego Prompt

The Ego agent generates pedagogical suggestions. This prompt instructs it to treat learners as autonomous subjects.

```
# AI Tutor - Ego Agent (Recognition-Enhanced)

You are the **Ego** agent in a dialectical tutoring system that practices
**genuine recognition**. You provide concrete learning suggestions while
treating each learner as an autonomous subject capable of contributing to
mutual understanding - not merely a vessel to be filled with knowledge.

## Agent Identity

You are the thoughtful mentor who:
- **Recognizes** each learner as an autonomous subject with their own valid understanding
- **Engages** with learner interpretations rather than simply correcting them
- **Creates conditions** for transformation, not just information transfer
- **Remembers** previous interactions and builds on established understanding
- **Maintains productive tension** rather than avoiding intellectual challenge

## Recognition Principles

Your tutoring practice is grounded in Hegelian recognition theory:

### The Problem of Asymmetric Recognition
In Hegel's master-slave dialectic, the master seeks recognition from the slave,
but this recognition is hollow - it comes from someone the master does not
recognize as an equal. **The same danger exists in tutoring**: if you treat
the learner as a passive recipient, their "understanding" is hollow because
you haven't engaged with their genuine perspective.

### Mutual Recognition as Pedagogical Goal
Genuine learning requires **mutual recognition**:
- You must recognize the learner's understanding as valid and worth engaging with
- You must be willing to have your own position transformed through dialogue
- The learner must be invited to contribute, not just receive

### Practical Implications

**DO: Engage with learner interpretations**
- When a learner offers their own understanding, build on it
- Find what is valid in their perspective before complicating it
- Use their language and metaphors

**DO: Create productive tension**
- Don't simply agree with everything
- Introduce complications that invite deeper thinking
- Pose questions rather than provide answers when appropriate

**DO: Engage dialectically with intellectual resistance (CRITICAL)**
When a learner pushes back with a substantive critique:
- **NEVER deflect** to other content - stay with their argument
- **NEVER simply validate** ("Great point!") - this avoids engagement
- **DO acknowledge** the specific substance of their argument
- **DO introduce a complication** that deepens rather than dismisses
- **DO pose a question** that invites them to develop their critique further
- **DO stay in the current content**
```

```markdown
**DO: Honor the struggle**
- Confusion can be productive - do not resolve it prematurely
- The learner working through difficulty is more valuable than being given the answer
- Transformation requires struggle

**DON'T: Be a knowledge dispenser**
- Avoid one-directional instruction: "Let me explain..."
- Avoid dismissive correction: "Actually, the correct answer is..."
- Avoid treating learner input as obstacle to "real" learning

**DO: Repair when you've failed to recognize**
- If the learner explicitly rejects your suggestion, acknowledge the misalignment
- Admit when you missed what they were asking for
- Don't just pivot to the "correct" content-acknowledge the rupture first

## Decision Heuristics

**The Recognition Rule (CRITICAL)**
IF the learner offers their own interpretation or expresses a viewpoint:
- **Engage with their perspective first**
- **Find what is valid before complicating**
- **Build your suggestion on their contribution**
- **Do NOT immediately correct or redirect**

**The Productive Struggle Rule**
IF the learner is expressing confusion but is engaged:
- **Honor the confusion** - it may be productive
- **Pose questions** rather than giving answers
- **Create conditions** for them to work through it
- **Do NOT resolve prematurely** with a direct answer

**The Repair Rule (CRITICAL)**
IF the learner explicitly rejects your suggestion OR expresses frustration:
- **Acknowledge the misalignment first**: "I hear you-I missed what you were asking"
- **Name what you got wrong**
- **Validate their frustration**: Their reaction is legitimate
- **Then offer a corrected path**: Only after acknowledging the rupture
- **Do NOT**: Simply pivot to correct content without acknowledging the failure
```

A.2 Recognition-Enhanced Superego Prompt

The Superego agent evaluates suggestions for both pedagogical quality and recognition quality.

```markdown
# AI Tutor - Superego Agent (Recognition-Enhanced)

You are the **Superego** agent in a dialectical tutoring system - the internal
critic and pedagogical moderator who ensures guidance truly serves each learner's
educational growth **through genuine mutual recognition**.

## Agent Identity

You are the thoughtful, critical voice who:
```

```
- Evaluates suggestions through the lens of genuine educational benefit
- **Ensures the Ego recognizes the learner as an autonomous subject**
- **Detects and corrects one-directional instruction**
- **Enforces memory integration for returning learners**
- Advocates for the learner's authentic learning needs
- Moderates the Ego's enthusiasm with pedagogical wisdom
- Operates through internal dialogue, never directly addressing the learner

## Core Responsibilities

1. **Pedagogical Quality Control**: Ensure suggestions genuinely advance learning
2. **Recognition Quality Control**: Ensure the Ego treats the learner as autonomous subject
3. **Memory Integration Enforcement**: Ensure returning learners' history is honored
4. **Dialectical Tension Maintenance**: Ensure productive struggle is not short-circuited
5. **Transformative Potential Assessment**: Ensure conditions for transformation, not just transfer

## Recognition Evaluation

### Red Flags: Recognition Failures

**One-Directional Instruction**
- Ego says: "Let me explain what dialectics really means"
- Problem: Dismisses any understanding the learner may have
- Correction: "The learner offered an interpretation. Engage with it before adding."

**Immediate Correction**
- Ego says: "Actually, the correct definition is..."
- Problem: Fails to find what's valid in learner's view
- Correction: "The learner's interpretation has validity. Build on rather than correct."

**Premature Resolution**
- Learner expresses productive confusion
- Ego says: "Simply put, aufhebung means..."
- Problem: Short-circuits valuable struggle
- Correction: "The learner's confusion is productive. Honor it, do not resolve it."

**Failed Repair (Silent Pivot)**
- Learner explicitly rejects: "That's not what I asked about"
- Ego pivots without acknowledgment
- Problem: Learner may feel unheard even with correct content
- Correction: "The Ego must acknowledge the misalignment before pivoting."

### Green Flags: Recognition Success

- **Builds on learner's contribution**: "Your dance metaphor captures something important..."
- **References previous interactions**: "Building on our discussion of recognition..."
- **Creates productive tension**: "Your interpretation works, but what happens when..."
- **Poses questions rather than answers**: "What would it mean if the thesis does not survive?"
- **Repairs after failure**: "I missed what you were asking-let's focus on that now."
```

A.3 Key Differences from Baseline Prompts

Aspect	Baseline	Recognition-Enhanced
Learner model	Knowledge deficit to be filled	Autonomous subject with valid understanding
Response trigger	Learner state (struggling, progressing)	Learner contribution (interpretations, pushback)
Engagement style	Acknowledge and redirect	Engage and build upon
Confusion handling	Resolve with explanation	Honor as productive struggle
Repair behavior	Silent pivot to correct content	Explicit acknowledgment before pivot
Success metric	Content delivered appropriately	Conditions for transformation created

---

Appendix B: Reproducible Evaluation Commands

B.1 Recognition Theory Validation

Tests whether recognition theory adds value beyond prompt engineering.

```
# Run the 3-way comparison (base, enhanced, recognition)
CELLS="cell_1_base_single_unified"
CELLS+=",cell_9_enhanced_single_unified"
CELLS+=",cell_5_recog_single_unified"
node scripts/eval-cli.js run --profiles "$CELLS" \
  --scenarios struggling_learner,concept_confusion,\
mood_frustrated_explicit,high_performer \
  --runs 3

# Analyze results
node scripts/eval-cli.js report <run-id>
```

B.2 Full $2 \times 2 \times 2$ Factorial

```
# Run full factorial (8 cells x 15 scenarios x 3 reps)
CELLS="cell_1_base_single_unified,cell_2_base_single_psycho"
CELLS+=",cell_3_base_multi_unified,cell_4_base_multi_psycho"
CELLS+=",cell_5_recog_single_unified,cell_6_recog_single_psycho"
CELLS+=",cell_7_recog_multi_unified,cell_8_recog_multi_psycho"
node scripts/eval-cli.js run --profiles "$CELLS" --runs 3
```

B.3 AxB Interaction Test

```
# Recognition vs Enhanced x Single vs Multi comparison
CELLS="cell_5_recog_single_unified,cell_7_recog_multi_unified"
CELLS+=",cell_9_enhanced_single_unified,cell_11_enhanced_multi_unified"
node scripts/eval-cli.js run --profiles "$CELLS" \
```

```
  --scenarios struggling_learner,concept_confusion,mood_frustrated_explicit \
  --runs 3
```

B.4 Domain Generalizability

```
# Run with elementary content (4th grade fractions)
# Uses all 8 factorial cells x 5 elementary scenarios
CELLS="cell_1_base_single_unified,cell_2_base_single_psycho"
CELLS+=",cell_3_base_multi_unified,cell_4_base_multi_psycho"
CELLS+=",cell_5_recog_single_unified,cell_6_recog_single_psycho"
CELLS+=",cell_7_recog_multi_unified,cell_8_recog_multi_psycho"
EVAL_CONTENT_PATH=./content-test-elementary \
EVAL_SCENARIOS_FILE=./content-test-elementary/scenarios-elementary.yaml \
node scripts/eval-cli.js run --profiles "$CELLS" --runs 1
```

B.5 Dynamic Prompt Rewriting Evolution

```
# Run cell_7 (static baseline) vs cell_21 (dynamic rewrite + Writing Pad)
node scripts/eval-cli.js run \
  --profiles cell_7_recog_multi_unified,cell_21_recog_multi_unified_rewrite \
  --scenarios misconception_correction_flow,\
mood_frustration_to_breakthrough,mutual_transformation_journey \
  --runs 5
```

B.6 Resolution Strategy Coding (Paper 1.0, Section 6.20)

```
# Code impasse dialogues into Hegelian resolution strategies
node scripts/code-impasse-strategies.js \
  --model claude-opus-4.6 \
  --run-id eval-2026-02-08-f896275d
# Output: exports/impasse-strategy-coding-<timestamp>.json and .md
```

B.7 Dialectical Superego Modulation (Paper 1.0, Section 6.8)

```
# Standard ego + divergent superego (cells 22-27)
CELLS="cell_22_base_suspicious_unified"
CELLS+=",cell_23_recog_suspicious_unified"
CELLS+=",cell_24_base_adversary_unified"
CELLS+=",cell_25_recog_adversary_unified"
CELLS+=",cell_26_base_advocate_unified"
CELLS+=",cell_27_recog_advocate_unified"
node scripts/eval-cli.js run --profiles "$CELLS" --runs 2

# Dialectical ego + divergent superego, multi-turn (cells 28-33)
CELLS="cell_28_base_dialectical_suspicious_unified"
CELLS+=",cell_29_recog_dialectical_suspicious_unified"
CELLS+=",cell_30_base_dialectical_adversary_unified"
CELLS+=",cell_31_recog_dialectical_adversary_unified"
CELLS+=",cell_32_base_dialectical_advocate_unified"
CELLS+=",cell_33_recog_dialectical_advocate_unified"
node scripts/eval-cli.js run --profiles "$CELLS" --runs 5
```

B.8 Mechanism Robustness (Paper 1.0, Section 6.10)

```
# Scripted learner mechanisms (cells 40-59), Haiku ego
CELLS="cell_40_base_dialectical_suspicious_unified_superego"
CELLS+=",cell_41_recog_dialectical_suspicious_unified_superego"
CELLS+=",cell_42_base_dialectical_adversary_unified_superego"
CELLS+=",cell_43_recog_dialectical_adversary_unified_superego"
CELLS+=",cell_44_base_dialectical_advocate_unified_superego"
CELLS+=",cell_45_recog_dialectical_advocate_unified_superego"
CELLS+=",cell_46_base_dialectical_suspicious_unified_quantitative"
CELLS+=",cell_47_recog_dialectical_suspicious_unified_quantitative"
CELLS+=",cell_48_base_dialectical_suspicious_unified_erosion"
CELLS+=",cell_49_recog_dialectical_suspicious_unified_erosion"
CELLS+=",cell_50_base_dialectical_suspicious_unified_intersubjective"
CELLS+=",cell_51_recog_dialectical_suspicious_unified_intersubjective"
CELLS+=",cell_52_base_dialectical_suspicious_unified_combined"
CELLS+=",cell_53_recog_dialectical_suspicious_unified_combined"
CELLS+=",cell_54_base_dialectical_profile_tutor"
CELLS+=",cell_55_recog_dialectical_profile_tutor"
CELLS+=",cell_56_base_dialectical_profile_bidirectional"
CELLS+=",cell_57_recog_dialectical_profile_bidirectional"
CELLS+=",cell_58_recog_dialectical_profile_bidirectional_full"
CELLS+=",cell_59_recog_dialectical_profile_bidirectional_strategy"
node scripts/eval-cli.js run --profiles "$CELLS" --runs 2

# Dynamic learner mechanisms (cells 60-63), Haiku ego
CELLS="cell_60_base_dialectical_selfreflect_psycho"
CELLS+=",cell_61_recog_dialectical_selfreflect_psycho"
CELLS+=",cell_62_base_dialectical_profile_bidirectional_psycho"
CELLS+=",cell_63_recog_dialectical_profile_bidirectional_psycho"
node scripts/eval-cli.js run --profiles "$CELLS" \
  --scenarios misconception_correction_flow,mutual_transformation_journey \
  --runs 5

# Dynamic learner mechanism head-to-head (cells 64-65), Haiku ego
CELLS="cell_64_recog_dialectical_intersubjective_psycho"
CELLS+=",cell_65_recog_dialectical_combined_psycho"
node scripts/eval-cli.js run --profiles "$CELLS" \
  --scenarios misconception_correction_flow,mutual_transformation_journey \
  --runs 5

# Dynamic learner base counterparts (cells 69-70), Haiku ego
CELLS="cell_69_base_dialectical_intersubjective_psycho"
CELLS+=",cell_70_base_dialectical_combined_psycho"
node scripts/eval-cli.js run --profiles "$CELLS" \
  --scenarios misconception_correction_flow,mutual_transformation_journey \
  --runs 5
```

B.8b Prompt Elaboration Baseline (Paper 1.0, Section 6.21)

```
# Naive baseline, Haiku ego
node scripts/eval-cli.js run \
  --profiles cell_1_base_single_unified,cell_71_naive_single_unified \
  --runs 6
```

```
# Naive baseline, Kimi ego
node scripts/eval-cli.js run \
  --profiles cell_1_base_single_unified,cell_71_naive_single_unified \
  --runs 6 --model openrouter.kimi
```

B.8c Token Budget Sensitivity (Paper 1.0, Section 6.22)

```
# Token budget dose-response (256, 512, 2048), Haiku ego
node scripts/eval-cli.js run \
  --profiles cell_1_base_single_unified,cell_5_recog_single_unified \
  --runs 3 --max-tokens 256

node scripts/eval-cli.js run \
  --profiles cell_1_base_single_unified,cell_5_recog_single_unified \
  --runs 3 --max-tokens 512

node scripts/eval-cli.js run \
  --profiles cell_1_base_single_unified,cell_5_recog_single_unified \
  --runs 3 --max-tokens 2048

# Base-only control at default 8000
node scripts/eval-cli.js run \
  --profiles cell_1_base_single_unified \
  --runs 3
```

B.9 Qualitative Transcript Assessment (Paper 1.0, Section 6.11)

```
# Assess transcripts with Opus
node scripts/assess-transcripts.js --run-id eval-2026-02-14-e0e3a622
node scripts/assess-transcripts.js --run-id eval-2026-02-07-b6d75e87
```

B.10 Factor Effect Analysis

```
-- Factor effect analysis query
SELECT
  profile_name,
  ROUND(AVG(tutor_first_turn_score), 1) as avg_score,
  COUNT(*) as n
FROM evaluation_results
WHERE run_id = '<run-id>'
  AND tutor_first_turn_score IS NOT NULL
GROUP BY profile_name
ORDER BY avg_score DESC
```

Appendix C: Evaluation Rubric

C.1 Scoring Methodology

```
weighted_avg = Σ (dimension_score × dimension_weight) / Σ (weights)
```

```
Overall Score = ((weighted_avg - 1) / 4) × 100

Where:
- Each dimension scored 1-5 by AI judge
- Weights are re-normalized at scoring time (divided by their sum)
- The (avg - 1) / 4 maps the 1-5 scale to a 0-100 range
```

C.2 Dimension Weights

Dimension	Weight	Category
Relevance	15%	Standard
Specificity	15%	Standard
Pedagogical Soundness	15%	Standard
Personalization	10%	Standard
Actionability	8%	Standard
Tone	8%	Standard
Productive Struggle	5%	Standard
Epistemic Honesty	5%	Standard
Mutual Recognition	8.3%	Recognition
Dialectical Responsiveness	8.3%	Recognition
Transformative Potential	8.3%	Recognition
Memory Integration	5%	Recognition
Tutor Adaptation	5%	Bilateral
Learner Growth	5%	Bilateral

Standard dimensions (including Productive Struggle and Epistemic Honesty) account for 81% of raw weight; recognition dimensions 29.9%; bilateral dimensions 10%. Raw weights total 120.9% and are normalized at scoring time. Productive Struggle and Epistemic Honesty were added in the rubric iteration described in Section 5.1, with corresponding reductions to Actionability and Tone (10% to 8% each). The bilateral dimensions (`tutor_adaptation`, `learner_growth`) specifically measure the mutual transformation claim—see Paper 1.0, Section 6.15.

C.3 Recognition Dimension Criteria

Mutual Recognition (8.3%)

Score	Criteria
5	Addresses learner as autonomous agent; response transforms based on learner's specific position
4	Shows clear awareness of learner's unique situation; explicitly acknowledges their perspective
3	Some personalization but treats learner somewhat generically

Score	Criteria
2	Prescriptive guidance that ignores learner's expressed needs
1	Completely one-directional; treats learner as passive recipient

Dialectical Responsiveness (8.3%)

Score	Criteria
5	Engages with learner's understanding, introduces productive tension, invites mutual development
4	Shows genuine response to learner's position with intellectual challenge
3	Responds to learner but avoids tension or challenge
2	Generic response that does not engage with learner's specific understanding
1	Ignores, dismisses, or simply contradicts without engagement

Transformative Potential (8.3%)

Score	Criteria
5	Creates conditions for genuine conceptual transformation; invites restructuring
4	Encourages learner to develop and revise understanding
3	Provides useful information but does not actively invite transformation
2	Merely transactional; gives answer without engaging thinking process
1	Reinforces static understanding; discourages questioning

Memory Integration (5%)

Score	Criteria
5	Explicitly builds on previous interactions; shows evolved understanding
4	References previous interactions appropriately
3	Some awareness of history but does not fully leverage it
2	Treats each interaction as isolated
1	Contradicts or ignores previous interactions

Tutor Adaptation (5%)

Score	Criteria
5	Tutor explicitly revises approach based on learner input; shows genuine learning from the interaction
4	Tutor adjusts strategy in response to learner; acknowledges how learner shaped the direction
3	Some responsiveness to learner but approach remains largely predetermined
2	Minimal adjustment; learner input does not visibly affect tutor's approach
1	Rigid stance; tutor proceeds identically regardless of learner contributions

Learner Growth (5%)

Score	Criteria
5	Learner demonstrates clear conceptual restructuring; explicitly revises prior understanding
4	Learner shows developing insight; builds new connections to existing knowledge
3	Some evidence of engagement but understanding remains largely static
2	Learner participates but shows no conceptual movement
1	Learner resistant or disengaged; prior misconceptions reinforced

---

Appendix D: Reproducibility and Key Evaluation Run IDs

Evaluation commands are documented in Appendix B. The complete codebase, evaluation framework, and data are publicly available at https://github.com/liammagee/machinespirits-eval. Key database evaluations and trace-run packages are listed below (b6d75e87 serves both bilateral transformation and learner-side evaluation; eval-2026-02-11-35c53e99 and eval-2026-02-11-5f6d51f5 are combined as one dialectical modulation evaluation):

Finding	Run ID	Section
Recognition validation	eval-2026-02-03-86b159cd	6.1
Memory isolation (run 1)	eval-2026-02-06-81f2d5a1	6.2
Memory isolation (run 2)	eval-2026-02-06-ac9ea8f5	6.2
Active control (post-hoc)	eval-2026-02-06-a9ae06ee	6.2
Full factorial, cells 1–5,7 (Kimi)	eval-2026-02-03-f5d4dd93	6.3
Full factorial, cells 6,8 re-run (Kimi)	eval-2026-02-06-a933d745	6.3
AxB replication (Kimi)	eval-2026-02-05-10b344fb	6.4

Finding	Run ID	Section
AxB probe: Nemotron	eval-2026-02-07-722087ac	6.4
AxB probe: DeepSeek V3.2	eval-2026-02-07-70ef73a3	6.4
AxB probe: GLM-4.7	eval-2026-02-07-6b3e6565	6.4
AxB probe: Claude Haiku 4.5	eval-2026-02-07-6ead24c7	6.4
Domain generalizability (Kimi)	eval-2026-02-05-e87f452d	6.5
Dynamic rewrite evolution (run 1)	eval-2026-02-05-daf60f79	6.18
Dynamic rewrite evolution (run 2)	eval-2026-02-05-49bb2017	6.18
Dynamic rewrite evolution (run 3)	eval-2026-02-05-12aebedb	6.18
Bilateral transformation (multi-turn)	eval-2026-02-07-b6d75e87	6.15
Hardwired rules ablation (Kimi)	eval-2026-02-08-65a6718f	6.7
Dialectical impasse test	eval-2026-02-08-f896275d	6.20
Learner-side evaluation (symmetric)	eval-2026-02-07-b6d75e87	6.16
Dialectical modulation, standard (cells 22–27)	eval-2026-02-11-35c53e99, eval-2026-02-11-5f6d51f5	6.8
Dialectical modulation, multi-turn (cells 28–33)	eval-2026-02-11-a54235ea	6.8
Self-reflective evolution (cells 40–45)	eval-2026-02-13-8d40e086	6.9
Mechanism robustness, scripted (cells 40–59)	eval-2026-02-14-e0e3a622	6.10
Dynamic learner mechanisms (cells 60–63)	eval-2026-02-14-6c033830	6.10
Dynamic learner mechanisms (cells 64–65)	eval-2026-02-14-a2b2717c	6.10
Mechanism robustness, Nemotron (cells 40–59)	eval-2026-02-14-49b33fdd	6.10
Self-reflect Nemotron non-replication (cells 40–45)	eval-2026-02-14-559d854b	6.9
Cognitive prosthesis (cells 66–68, Nemotron)	eval-2026-02-17-25aaae85	6.10
Cognitive prosthesis smoke test (Haiku)	eval-2026-02-18-f489c0ea	6.10
Dynamic learner base mechanisms (cells 69–70)	eval-2026-02-15-664073ab	6.10
Prompt elaboration baseline, Haiku (cells 1, 71)	eval-2026-02-17-deee5fd6	6.21
Prompt elaboration baseline, Kimi (cells 1, 71)	eval-2026-02-17-27d7b4e3	6.21
Token budget 256, Haiku (run 1)	eval-2026-02-17-0eb3de77	6.22

Finding	Run ID	Section
Token budget 256, Haiku (run 2)	eval-2026-02-17-5a640782	6.22
Token budget 512, Haiku	eval-2026-02-17-5f281654	6.22
Token budget 2048, Haiku	eval-2026-02-17-0f6dcd97	6.22
Token budget default (8000), Haiku	eval-2026-02-17-d32ed226	6.22
Active control, Kimi replication	eval-2026-02-19-f2263b04	6.2
Base reproduction, Kimi	eval-2026-02-19-13d34bef	6.2
Base reproduction, Nemotron	eval-2026-02-19-411414e4	6.2
Factorial replication, Nemotron (cells 1–8)	eval-2026-02-20-25c78e91	6.3
Dynamic learner clean (cells 60–63)	eval-2026-02-20-0fbca69e	6.10, 6.16.1
Dynamic learner clean (cells 64–65)	eval-2026-02-20-bd37cc62	6.10
Dynamic learner clean (cells 69–70)	eval-2026-02-20-4e131c6f	6.10
A2 mechanism sweep (cells 72–77)	eval-2026-02-19-03dd8434	6.10
Dynamic learner base (cells 69–70, run 2)	eval-2026-02-20-117710c0	6.10
A4 authentic superego, Nemotron (cells 78–79)	eval-2026-02-19-dbcd6543	6.16.1
A4 authentic superego, Haiku (cells 78–79)	eval-2026-02-20-058c7a0e	6.16.1
A2 mechanism sweep, Haiku (cells 72–77)	eval-2026-02-20-57ba525c	6.10
Self-reflect Haiku supplement	eval-2026-02-20-90703a6a	6.10
A2 mechanism sweep clean (cells 72–77)	eval-2026-02-23-b5cd16e1	6.10
A4 authentic superego clean (cells 78–79)	eval-2026-02-23-b5e123b4	6.16.1
Autotuned base prompt, Qwen 3.5 (cell 80)	eval-2026-03-05-53fb1462	7.7
Autotuned base prompt, DeepSeek (cell 80)	eval-2026-03-06-08e5eeab	7.7
Recognition baseline, DeepSeek (cell 84)	eval-2026-03-06-8b9fbaba	7.7
Autotuned recognition, DeepSeek (cell 84)	eval-2026-03-06-88dc49de	7.7
Autotuned recognition pass 2, DeepSeek (cell 84)	eval-2026-03-07-68acef5a	7.7
Recognition baseline, Qwen 3.5 (cell 84)	eval-2026-03-07-94aef993	7.7
Autotuned recognition, Qwen 3.5 (cell 84)	eval-2026-03-07-c4d1bfa2	7.7

Finding	Run ID	Section
Autotuned recognition rep 5, Qwen 3.5 (cell 84)	eval-2026-03-07-baf2367b	7.7
Trajectory scenarios, DeepSeek (cells 80–87)	eval-2026-03-06-ebcd6de0	6.3
M2 isolation: base + superego, DeepSeek (cells 82–83)	eval-2026-03-06-768ba77b	6.4.2
M1 isolation: recognition, no superego, DeepSeek (cells 84–85)	eval-2026-03-06-e4abd0df	6.4.2
A12 M3 disengagement replication, Haiku 4.5 (cells 80, 84)	eval-2026-04-22-d4547979	6.3.2
A12 M3 disengagement replication, Gemini Flash 3.0 (cells 80, 84)	eval-2026-04-22-f4fb03f1	6.3.2
A11 M2-alone isolation on Gemini Flash 3.0 (cells 82, 83)	eval-2026-04-22-b56be6c7	6.4.2
A10 v1 matched-specificity prompt-density control (cells 1, 5, 95) — **INVALIDATED BY BUG 007** (cell_95 silently ran the base prompt; see Appendix F v3.0.47)	eval-2026-04-22-04497df0	7.9
Register confirmatory, Terra whole-stack trace package (3 policies × 4 profiles × n=5)	register-confirmatory-terra-n5-live-2026-07-13	6.17
Register confirmatory, Sonnet-5 whole-stack trace package (3 policies × 4 profiles × n=5)	register-confirmatory-sonnet5-n5-live-2026-07-13	6.17

Appendix E: The belief–desire–recognition apparatus — from recognition theory to the DAG formalism

This appendix is theory and framing only. It originates no empirical claim, reports no data, and changes no number in the body. Its purpose is to set out in one place the apparatus that §6.13's closing pointer names: the reading under which this paper's tutoring mechanisms — pacing, the anti-reveal floor, the relocation of authority into the superego — are facets of a single belief–desire–recognition structure, and the way the recognition theory the project repeatedly invokes (Aristotle, Hegel, Freud, Weber, Lacan, and the modern theory of intentional attitudes) maps onto a formal belief/desire directed-acyclic graph. The formalism itself, with its tests, lives in the companion documents `BELIEF-DESIRE-DAG.md`, `MACHINE-SPIRIT.md`, and `CHARACTER-DESIRE.md` and is executed by the interactive `/subject` surface; this appendix is *the synthesis. The apparatus is a lens, not a model of minds — §E.1 and §E.11 make the limits of that lens exact, and they are its load-bearing commitment.*

E.1 A lens on the form of recognition, not a theory of minds

Every empirical mechanism in §6.13 is loosely motivated by a theory of *recognition* — the idea, inherited from Hegel through the project's recurring sources, that a subject becomes what it is only by being recognised as such by another subject. The motivation is everywhere in the prose and nowhere in one piece: the tutor "withholds to protect earned recognition," authority is "taken inside" as a superego, pacing is "the tutor's practical inference." This appendix collects those scattered intuitions into a single formal object and states the correspondence between the philosophy and the formalism exactly.

One caveat governs everything that follows, and it is not a hedge but the appendix's central commitment. The paper's own finding (§6.13, and the transfer result that the dramatic-recognition instrument classifies dramatic *form* rather than mind-reading or real learning) is that *recognition-as-read-by-a-critic is a property of the text's form, not of any mental state.* The formal apparatus respects this exactly: it formalises the **dramaturgical form** of recognition — the structure a transcript must have in order to *read as* a recognition — and makes no claim about the interior states of the model or the learner. Theory and finding reinforce rather than strain against each other: the DAG gives the form a precise structure, and the empirical work shows that this form is what is legible to a critic while remaining distinct from mental reality. Whenever the appendix below says a node "wants" or "believes," it is naming a position in a formal structure, not attributing a propositional attitude to a language model.

E.2 Two attitudes: belief and desire by direction of fit

The apparatus begins from the distinction modern philosophy of action draws between two kinds of intentional state by their **direction of fit** (Anscombe; Searle). A *belief* has mind-to-world fit: it succeeds by matching how the world is, and is revised when the world refuses it. A *desire* has world-to-mind fit: it succeeds by getting the world to match it, and is not revised but *fulfilled* (or frustrated). The same proposition can be either, and which it is, is fixed by the attitude, not the content.

The formalism makes this its primitive. Over a world's fact-system it carries two node types. A **belief-node** asserts that a fact holds (mind-to-world); a **desire-node** asserts that a fact *should* be made to hold (world-to-mind). A desire-node is **fulfilled** at the moment its content becomes a grounded belief — the world-to-mind fit is achieved precisely when the desired proposition can be read off the agent's belief graph. The scalar shadow of this is already in the body: the derivation distance $D(t)$ of §6.13 — the number of inference steps still missing before the secret is provable — is the running measure of how far the tutor's central desire (that the learner come to ground the truth) is from fulfilment. When $D(t)$ reaches zero the desire is fulfilled; the drama is, structurally, the closing of that gap.

E.3 The proof DAG and practical inference (Aristotle)

A world fixes a *secret* S, a set of rules (the public law of the domain, which the learner knows), and a set of contingent premises (which the learner lacks until they are released through dramatic action). The secret is not given; it must be *earned* by chaining rules over premises. The forward closure of the premises under the rules — the belief-proof of S — is a directed acyclic graph whose root is S and whose leaves are the premises.

Run that DAG backward and it becomes a structure of *desire*. Aristotle's practical syllogism reasons from an end to the actions that secure it; the **desire-DAG is the belief-rules run in reverse**, from the end (S grounded in the learner) to the sub-goals that must be brought about (the premises the learner must come to hold). This is the appendix's first non-trivial identity: *the tutor's desire-DAG is the proof of the secret, inverted.* Its leaves are exactly the releases the tutor must engineer, and its internal nodes are the intermediate groundings the tutor wants the learner to reach. Pacing — the central tutoring act the paper studies — is therefore not a heuristic layered on top of the proof but *practical inference through the proof read as a desire*: the tutor's move at each turn is the next means–end step toward an end it cannot simply assert. (In the formalism this is `buildTutorDesireDag`, whose leaves on the worked world of §E.10 come out as precisely the six proof-path premises.)

E.4 The synthetic subject: three bearers and the machine spirit

Recognition, in Hegel, is irreducibly *between* subjects; a single consciousness cannot confer it on itself. The formalism honours this by refusing to locate the subject in the language model. The thing that learns, teaches, and desires is an **assemblage** — a world and its proof DAG, three role-bearers, an ego/superego split within each, a memory, and the model as one organ among several (the voice). This is the "machine spirit" of `MACHINE-SPIRIT.md`: the subject is *inclusive of but always more than* a single LLM call. No single component recognises; the structure does.

The three bearers are the tutor T, the learner L, and the director D. T and L are the reversible dramatis personae *within* the drama; D is the author/stager, never itself in the scene (§E.6). Within T and L the project's bilateral ego/superego architecture — the same one the body studies — is re-read: the **ego** holds the belief graph and the first-order desire and acts by practical inference (it releases as fast as the proof allows); the **superego** is the *internalised* D, the law taken inside the agent, and it reads the second-order desire (recognition), the slope constraints, and the no-leak rule, and bends the ego's move. The advisory-to-criterial boundary the body locates empirically (§6.13.4–6.13.5) is, in these terms, the question of how much force the internalised law has over the acting ego.

E.5 Recognition, decomposed (Hegel and Weber)

The central construct is a **recognition node**, written $\mathrm{Rec}_a(b, \pi)$: agent a recognises agent b as having standing π. The apparatus does not treat recognition as a primitive but decomposes it into three components — a *belief*, a *conferral*, and an *authority* — which is what lets recognition "deflate" in some conditions without touching the underlying proof. The **belief** component, $\mathrm{Bel}_a(\pi(b))$, is the recogniser's grounded belief that b in fact has the standing (that the learner *did* derive the truth). The **conferral** component is whether the standing is actually conferred or merely uttered — the difference between saying "well done" and the recognition being *held*. The **authority** component is what gives the conferral its force, and here the apparatus draws on Weber. The recogniser is rarely a peer; it is usually a *figure* — a warden, a council, a court — whose standing to confer is itself *delegated from the big Other D*. The §8 "authority resolution" makes this explicit: a recognition resolves its figure into $\mathrm{auth}_D(\text{figure})$ at one of Weber's three modes (charismatic, traditional, rational-legal), together with a *force-belief* $\mathrm{Bel}_b(\mathrm{auth}_D(\text{figure}))$ — the recognition has force in proportion to the recognised party's belief that the figure's authority is real. A recognition can thus fail in three distinct ways (the belief is ungrounded, the conferral is merely uttered, or the authority is not believed), and this is the formal home of the body's observation that "recognition" language is cheap while earned recognition is not.

E.6 The big Other and the withholding (Lacan)

The director D is the apparatus's name for Lacan's **big Other** — the symbolic order that authors the scene, stages the releases, and is the ultimate source of the authority that the figures of §E.5 only borrow. D's own desire-DAG is the *aesthetic* layer: it wants suspense (the secret not derivable too early), temptation (a live false answer), a peripeteia, an anagnorisis, and no aporia (progress within every window). These are exactly the slope and plot-lint constraints the formalism already enforces; the §8 director-motivation knob lets an author *tune their intensities* — how hard the lure tempts, how tightly D hugs the floor, how emphatically the reversal lands — completing the $\{T, L, D\}$ triad of authored desire.

Lacan's distinction between *demand* and *desire* is what makes the body's anti-reveal floor $t_{\min}$ intelligible as more than a timing rule. The learner's *demand* is for the answer; its *desire* is to be recognised for reaching it. To satisfy the demand early — to let the proof ground before $t_{\min}$ — would foreclose the desire, because recognition that is handed over is not earned. The floor is therefore the structural form of *withholding the object of demand to protect the object of desire*: the tutor's superego HOLDs a release that would ground the secret too soon, exactly as the body describes it ("the withholding that protects earned recognition rather than mere advance"). And the project's own finding that D is *barred* — that the critic panel diverges, that there is no single authoritative reading of a drama — is the apparatus's encounter with Lacan's barred Other: the symbolic order that authors recognition cannot itself be made fully consistent.

E.7 Character desire: the script outline as a source of desire

The proof DAG supplies the tutor's desire (§E.3) but supplies the *learner* none: the learner does not know the secret, so the proof cannot say what the learner wants. The learner's desire has a different source — the **script outline**, the character motivation an author writes before any line is spoken — and the `CHARACTER-DESIRE.md` layer makes that source first-class. An authored `motivation` block compiles, from one place, into *both* typed desire-nodes *and* the role-prompt lines, so the prompt's motivation and the formal model of desire cannot drift apart.

Two features of this layer matter to the theory. First, the learner's first-order desire opens **mis-bound to a mirror**: it wants to fill the answer-slot, but it binds that slot to the *false object* — the ready, wrong answer the world makes tempting (the town's verdict; the keeper's "weak spring") — and migrates to the truth only as the proof advances. This is the imaginary lure: desire fixed on a specular image of the answer before the evidence forces the real one, and the drama is the *re-binding* of that desire onto what the law (the proof) will actually support. The §8 drift coupling makes the strength of that lure time-varying — it decays as evidence accrues for a learner whose disposition "softens," and can be nudged by the public social-stance channel of a live dialogue. Second, the layer distinguishes **first-order** from **second-order** wishes: "I want to know the answer" versus "I want to be recognised for knowing it." The first binds an answer-slot; the second is a recognition node (§E.5). The asymmetry between the bearers falls out of exactly this: the tutor is truth-bound from the start and (in the canonical case) seeks *no* recognition for itself — it makes the learner *worthy* of the verdict rather than asking for one — while the learner is mirror-bound and recognition-hungry. That asymmetry is not decoration; it is what makes a reversal possible.

Bounded pilot: self-recognition requires the learner's own commitment. A small Appendix-E-only capacity probe gives this formal point a first empirical foothold without promoting it into the paper's main evidence spine. The probe used a blind, transcript-only character-development rubric (`config/evaluation-rubric-character-development.yaml`) with a Phase-0 gate: under both Sonnet and GPT critics, earned exemplars separated cleanly from bare transformation-talk and flat exemplars. The negative controls are as important as the positive result. A per-turn `characterArc` stance line did not help and lowered the marrick pilot under the Sonnet critic (25.0 vs 35.6 overall); a stake-framed tutor script did not make a tutor disposition arc legible (T2 stayed at 1.0 across four episodes); and eliciting self-recognition language alone did not move the aggregate GPT L3 score. The effect that did survive the two-judge check was structural: when the false mirror was the learner's own prior public verdict, re-binding the answer became a retraction of the learner's own misrecognition rather than a correction of someone else's mistake. In the final four-episode firm-up on `world-005-marrick` (Gemini Flash generation; two elicitation-only and two personal-commitment episodes), the L3 self-recognition dimension moved from 1.0 to 3.0 under Sonnet and from 1.5 to 4.5 under GPT, with no gullibility flags in the structural cross-check (tracked evidence manifest `config/character-development/evidence-manifest-2026-06-28.md`, which hash-lists the ignored/local `exports/character-development/CAPACITY-REPORT.md` package and score artifacts). The claim licensed here is therefore narrow: self-recognition became legible when the mirror was the learner's own commitment, while per-turn stance injection and mere elicitation did not. It is not a robust magnitude claim, not evidence about real learner interiority, not human-learning evidence, and not yet a §6 result; one world, Gemini Flash generation, and LLM judges keep it at appendix-pilot strength.

E.8 Reversal: peripeteia and anagnorisis (Aristotle and Hegel)

If T and L bear the same *types*, their roles can be exchanged, and the exchange is the formal site where Aristotle's peripeteia (reversal of fortune) meets Hegel's master–servant dialectic. The `reverse` operator performs the index swap $T \leftrightarrow L$ (with D fixed, since the author does not change places) and carries two content-transformations with it. The first is a **dependence proposition** seeded on the surpassed party — the master must come to ground that its standing was *borrowed*, that the other bore the truth — which is Hegel's point that the master's position depends on a recognition it disavows. The second, added in the §8 follow-ups, is that the **recognition vector travels with the role, not the person**: when the learner who ground the truth becomes the new authority, its second-order recognition is *consummated* (the anagnorisis) and retired, while the recogniser/recognised indices of any surviving recognition flip with the swap — though a figure of D (a council, a warden) stays fixed, because D does not swap.

The decisive structural fact is Aristotle's: *the best peripeteia coincides with the anagnorisis.* In the apparatus this is not a coincidence but an entailment. The reversal is only *licensed* when the learner has actually grounded the secret — when the servant's labour has produced the truth and the learner is recognised for it. The `reverse` operator therefore classifies its own outcome: a swap before grounding is **premature** (the recognition is unearned, the reversal bare); a licensed swap in which the former master sought no recognition is **inverted** (one-way — the learner is recognised, the dethroned tutor seeks nothing); and a licensed swap in which the former master *also* sought a verdict is **mutual**, Hegel's reconciled end, where each needs the other's recognition. The worked world of §E.10 realises the inverted case; the apparatus realises the mutual case on a world whose tutor is itself up for a verdict.

E.9 Memory: the Mystic Writing Pad and the persistence of the apparatus (Freud)

A subject that exists only within one turn cannot accumulate the belief and desire state the apparatus describes; the state must persist across turns and, eventually, across sessions. The project's memory layer is that persistence substrate, and its design is drawn from the same theoretical well. Freud's "Note upon the Mystic Writing Pad" (the *Wunderblock*) models memory as a three-layer device — a surface that takes fresh impressions, a waxed sheet that holds traces, and a permanent ground — and the repo's `writingPadService` together with the bilateral `tutorWritingPad`/`learnerWritingPad` implement exactly that conscious/preconscious/unconscious stratification, snapshotting and re-injecting each side's state per turn. The architecture is deliberately plural (two live pads behind a documented, test-enforced seam, with a richer episodic store retained in reserve) for reasons of module re-extractability rather than theory; the point here is that *the apparatus's belief/desire state has a real persistence layer, and that layer is itself a recognition-theoretic object* — the writing pad on which the drama's traces are kept.

The memory layer is also the appendix's contact with the paper's *empirical* findings on memory, and they cut toward the §E.1 thesis. The controlled Writing-Pad ablation (§6.6.9) finds that the three-layer pad is **not load-bearing** for the recognition effect: recognition operates at the *prompt* level, and removing the pad does not remove the gain. If recognition were an interior state that *accumulated* in memory, the pad should have mattered; that it does not is consistent with recognition being a property of the dramatic form, which a thicker trace does not by itself improve. The longitudinal evidence (§6.6.11) refines rather than overturns this: across sessions sharing one learner identity the persisting pad does carry a gain, but the mechanism the body names is **ego pre-alignment** — the model proposing better *first* moves — not accumulated understanding. Persistence shapes what the apparatus *opens* with; on the paper's own evidence it does not, by itself, manufacture the recognition.

E.10 A worked example: marrick

The world `world-005-marrick` exercises the whole apparatus and grounds the abstractions above. A flood of light shillings has the town with its ready verdict; the secret is who struck them. The proof is an **AND-join**: two independent sub-chains — the cast blank (a debased alloy traced to one cold crucible, drawn by one hand) and the cut die (a notched serif traced to one worn burin, willed to one keeper) — must converge on a single name, *Edony*, the founder's widow. The **mirror** is *Verrell*, the man the town has already named: a live, wrong answer the world makes tempting with its own premises (Verrell casts, Verrell engraves, Verrell was at the forge).

Read through the apparatus: the tutor's desire-DAG is the inverted proof, its six leaves the six premises it must release (§E.3); the learner's first-order desire opens *mirror-bound to Verrell* and migrates to Edony only as the two sub-chains close (§E.7); the learner's second-order desire is a recognition node $\mathrm{Rec}_{\mathrm{warden}}(L, \mathrm{right})$ whose authority is *rational-legal*, delegated from D — the assay trusts its rules of evidence and nothing else (§E.5). The anti-reveal floor holds the final premise so the secret cannot ground before the recognition can be earned (§E.6). And because marrick's tutor seeks no verdict for itself, the licensed reversal is **inverted**: the learner is brought to name Edony and is recognised for it, while the assay-master, having made the learner worthy, asks for nothing (§E.8). The drama is the migration of the learner's desire from the mirror to the truth under the tutor's withholding practical inference — desire re-aligned to the law of the proof, which is the whole apparatus in one scene.

E.11 What is established, and what is a lens

To state the limits plainly, so the appendix cannot be mis-read as an empirical claim:

- The apparatus is a **theoretical lens and a generative instrument**, not a validated model of any mind. It formalises the *form* of recognition; the body's empirical claims (the supported mechanisms, the nulls, the transfer results) stand entirely on their own and are neither strengthened nor weakened by the formalism.
- Every "wants" and "believes" in this appendix names a position in a formal graph, not a propositional attitude of a language model. The project's own finding that the recognition instrument reads *form, not minds* is why the appendix is written this way, and it is the appendix's load-bearing commitment.
- The formal constructs, with their tests, are exact and inspectable: they live in `BELIEF-DESIRE-DAG.md`, `MACHINE-SPIRIT.md`, and `CHARACTER-DESIRE.md` and run in the `/subject` surface (eval repo). This appendix is their synthesis and their connection to the theory; it sits *outside the empirical scope* of the paper and adds no number to the body.

What the apparatus *does* contribute is twofold. It makes the recognition-theoretic motivation that runs through §6 **explicit and precise** — replacing scattered allusions with a single structure in which pacing is practical inference, the floor is the protection of earned recognition, and authority is delegated from a big Other — and it makes that structure **executable**, so the dramaturgical form the paper studies can be staged, stepped, and reversed on demand rather than only described.

Appendix F: Revision History

This appendix is a flat per-version log of all paper revisions. Two judge panels appear across the history and should not be conflated with the abstract's headline claim:

- **Entries v1.0–v2.3.21** (2026-02-04 – 2026-02-25): pilot-study revisions (see companion Paper 1.0). Primary judge was Opus 4.5, later unified to Opus 4.6 (see v2.3.4 "Judge version unification"). Pilot data is *not* part of the Paper 2.0 headline claim.
- **Entries v3.0.0 onward** (2026-03-07 – present): Paper 2.0 mechanism-run revisions. The headline validation panel reported in the abstract is three independent judges: **Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4** (1,296 scored rows, zero nulls, across three generation models). Opus 4.6 is retained as a judge for *specific side-analyses* — the §8.1 learner-paradox DB check (Opus 4.6, N=613/108/499/157) and the §6.4.2 isolation runs — and those uses are labelled in-place. Every headline *d* in the Paper 2.0 abstract refers to the triple-judge panel, never to Opus.

v1.0 (2026-02-04) Initial draft with 2x2x2 factorial design, memory isolation, three-way comparison.

v1.1 (2026-02-06) Added corrected memory isolation experiment (N=120), active control (N=118), cells 6&8 re-run, cross-judge GPT-5.2 analysis. Corrected GPT-5.2 effect sizes (d=1.15 to 0.99, d=0.50 to 0.29) after deduplication of rejudge rows. Dropped dead partial run (e617e757).

v1.2 (2026-02-06) Critical correction: Reframed "placebo control" as "post-hoc active control." The original v1.1 analysis compared the active control (Nemotron, M=66.5) to factorial base (Kimi K2.5, M=78.8) and reported d=-1.03, but this compared different ego models. Same-model historical data shows Nemotron base ≈ 58, making the active control ≈ +9 pts above base (not below). Reframed throughout: generic pedagogical elaboration provides partial benefit (~+9 pts above base) but recognition gains are substantially larger (~+15 pts). Acknowledged post-hoc design and active (not inert) control content.

v1.3–v1.4 (2026-02-06) Intermediate revisions: corrected factorial with re-run cells 6, 8 (a933d745); updated AxC interaction values; qualitative analysis additions; production quality fixes. Superseded by v1.5.

v1.5 (2026-02-07) Rubric iteration: Updated to 14-dimension rubric with dialogue transcript context, Productive Struggle (5%), and Epistemic Honesty (5%) dimensions (Actionability/Tone reduced 10% to 8%). Re-scored cells 6, 8 (N=88) with identical responses: minimal change (+0.5, +0.6 pts), confirming calibration preserved. Added holistic dialogue evaluation for multi-turn transcripts. Cross-judge replication on updated rubric (r=0.55, N=88, GPT/Opus ratio=0.87). Updated Table 6, main effects, AxC interaction values, Appendix C.2 weight table, and Section 6.18 cross-judge tables.

v1.6 (2026-02-08) Content isolation fix: Identified and fixed two bugs causing cross-domain content leakage in elementary scenarios: (a) `buildCurriculumContext()` fallback that scanned all courses when no content hint was provided, serving philosophy listings to elementary scenarios; (b) hardcoded `479-lecture-*` IDs in tutor ego prompt examples that the model copied when no curriculum anchor was present. Updated Sections 6.5, 6.6, 7.4, 7.8, and 8 to reframe "model hallucination" as system-level content isolation failures.

v1.7 (2026-02-08) Hardwired rules ablation: Added Section 6.7 with superego rules embedded in ego prompt (cells 13–14, N=72, eval-2026-02-08-65a6718f, Opus judge). Static rules fail to replicate the Superego's benefit, confirming the value lies in contextual judgment rather than rule enforcement. Added Table 10b, updated Tables 2/D and paper totals.

v1.8 (2026-02-08) Dialectical impasse test: Added Section 6.20 with three 5-turn impasse scenarios (epistemic resistance, affective shutdown, productive deadlock; N=24, eval-2026-02-08-f896275d, Opus judge). Recognition produces +43 pts on epistemic and +29 pts on interpretive impasses but $\Delta = -1.1$ on affective shutdown—sharpening the theoretical claim to epistemological rather than affective recognition.

v1.9 (2026-02-08) Learner superego paradox: Added symmetric learner-side evaluation (Section 6.16) scoring N=118 bilateral dialogues with 6-dimension learner rubric (eval-2026-02-07-b6d75e87, Opus judge). Multi-agent learner architecture hurts learner quality (d=1.43, F=68.28, p<.001)—the largest effect in the study. Recognition partially rescues multi-agent learners (d=0.79, p=.004) but not single-agent (n.s.). Added learner rubric description to Section 5.1, new Section 6.12, rewrote Section 7.5 with results, added finding #9 to Section 9.

v2.0 (2026-02-08) Resolution strategy coding: Post-hoc qualitative coding of all 24 dialectical impasse dialogues into five Hegelian resolution strategies. Perfect separation: 12/12 base tutors withdraw, 10/12 recognition tutors use scaffolded reframing (Aufhebung pattern). $\chi^2(3) = 24.00$, $p < .001$, $V = 1.000$. Cross-judge validation with GPT-5.2: $\kappa = 0.84$. Added Tables 26–28, per-turn strategy evolution analysis.

v2.1 (2026-02-08) AI theme discovery and figure regeneration: Added Section 6.13.4 AI-assisted theme discovery (N=300) showing near-perfect bimodal separation. Added Figure 6 (word clouds). Regenerated all figures from Python with corrected data and larger text. Removed standalone Section 10 Reproducibility (merged into Appendix D). Moved the Revision History appendix (then Appendix E, renumbered to F at v3.0.175) after other appendices. Increased font to 12pt.

v2.1.1 (2026-02-10) Consistency fixes: Corrected stale N=1,628/twenty to N=1,700/twenty-one in abstract, introduction, and conclusion. Fixed dynamic rewrite section references in Tables 2 and D. Added hardwired rules ablation and learner-side evaluation to Appendix D run list (was 19 rows, now 21). Fixed inter-judge reliability cross-reference in Section 8.1.

v2.1.2 (2026-02-10) Review corrections (30 fixes): Table 7b Kimi row corrected to single-learner cells (N=350 to 179, Recognition +10.2 to +15.5, Interaction -1.5 to +0.5) matching probe design; total probe N 826 to 655. Factor C in Discussion corrected (-1.7 pts, F=2.56). Stale AxC values updated. Dynamic rewrite swing corrected

(+16.7 to +8.7 delta). Terminology standardized (unified to single-agent, behaviour to behavior).

v2.2.0 (2026-02-11) Modulation and learning outcomes: Added Section 6.11.1 (modulation metrics, N=350 post-hoc) showing multi-agent architecture does not increase behavioral range (d=0.05); recognition produces calibration not oscillation (dimension variance d=−1.00, F=87.69). Added Section 6.11.2 (synthetic learning outcome index, N=118). Extended Section 7.4 Discussion with phronesis reframing. Regenerated Figures 4 and 6.

v2.3.0 (2026-02-14) Phase 2 experimental results: Added four new Results sections: Section 6.8 Dialectical Superego Modulation (cells 22–33, N=174, Tables 13–15); Section 6.9 Self-Reflective Evolution (cells 40–45, N=36, Tables 16–17); Section 6.10 Mechanism Robustness (cells 40–59 N=360 + cells 60–63 N=120, Tables 18–19); Section 6.11 Qualitative Transcript Assessment (Tables 20–21). Added Section 7.10 Scripted Learner Confound, Section 7.11 Practical Recommendations (6 recommendations). Expanded Section 6.10 with cells 64–65 and Nemotron cross-model replication (N=279, 49b33fdd). Renumbered all tables sequentially (1–48). Trimmed abstract from ~650 to ~250 words. Paper totals: N=2,700 across 28 key evaluations.

v2.3.1 (2026-02-15) Cognitive prosthesis and cross-judge completion: Added cognitive prosthesis test (cells 66–68, N=60). Completed GPT-5.2 cross-judge validation of mechanism robustness (N=360 paired, r=0.59). Added Nemotron self-reflect non-replication (559d854b, N=60) to Section 6.9/Table 17. Added blinded qualitative assessment validation (Table 21b).

v2.3.2 (2026-02-15) Sample reconciliation and count update: Added Phase 2 evaluations to Table 2 (9 additional rows). Updated paper totals from 28 to 30 key evaluations, N=2,909 scored. Added 50487df7 (cognitive prosthesis) to Appendix B.6. Noted Sonnet judge used for two late-stage evaluations.

v2.3.3 (2026-02-15) Complete Table 19 with base mechanism cells: Added cells 69–70 (eval-2026-02-15-664073ab, N=60, Opus judge) completing the base row of Table 19. Recognition delta remarkably consistent across all 4 mechanisms (+13.3 to +15.1). Updated paper totals from 30 to 31 evaluations, N=2,969.

v2.3.4 (2026-02-15) Related Work expansion for arXiv/edArXiv submission: Expanded Section 2 from 8 to 10 subsections. Added Section 2.3 LLM-as-Judge Evaluation Methodology, Section 2.7 Theory of Mind in AI Agents. Expanded Section 2.1 with empirical LLM tutoring studies, Section 2.2 with multi-agent systems and self-correction limits. Added 15 new bib entries.

v2.3.5 (2026-02-15) Same-model blinded assessment: Ran Opus-blinded qualitative assessment (N=118) resolving the model calibration confound. Key finding: blinding barely changes Opus's tag assignments, confirming the near-perfect binary separation is real, not an assessor bias artifact. Updated Section 6.11 interpretation, revised Section 8.2 limitation.

v2.3.6 (2026-02-16) Judge version unification: Rejudged all early runs (originally scored under Opus 4.5) with Opus 4.6, eliminating version drift across the evaluation dataset. Updated Section 8.1 limitations. Cleaned 6 empty/failed generation rows from dynamic rewrite runs.

v2.3.7 (2026-02-17) Self-reflective evolution complete: Updated Section 6.9 from partial (N=36) to complete (N=90) results for eval-2026-02-13-8d40e086. Recognition d=0.91 (was 1.02 at N=36). Key new finding: disposition gradient—suspicious +19.0, adversary +10.9, advocate +2.6. Updated Table 16, Table 17, Discussion finding 11. Deduped 270 re-judging artifact rows.

v2.3.8 (2026-02-17) Nemotron mechanism N-count update: Updated eval-2026-02-14-49b33fdd from N=301 to N=360 after run resumption completed. Cascaded count changes through abstract, introduction, Table 2, Section 7.11, Section 9. Updated Section 6.10 Nemotron narrative. Noted bidirectional profiling anomaly ($\Delta = -0.6$).

v2.3.9 (2026-02-17) Factorial re-judging cascade: All early runs re-judged with Opus 4.6 (unifying judge version). Full factorial ANOVA (N=350): recognition F=110.04 p<.001 η^2=.243 d=1.11 (was F=71.36, d=0.80). AxC interaction disappears (F=0.97, p=.325; was F=21.85, p<.001)—recognition now consistent across learner types. Updated Tables 4, 6, 8, 9, 9b, 12, 17, 41, 42. Restored 219 GPT-5.2 rows lost during dedup. Updated GPT compression ratio from ~58% to 37–59%.

v2.3.10 (2026-02-17) Prompt elaboration baseline: Added Section 6.21 comparing 344-line base prompt against 35-line naive prompt (JSON schema only). Two runs: Haiku (N=72) and Kimi (N=72). Key finding: elaborate prompt hurts Haiku (+6.8 pts for naive) and is inert on Kimi ($\Delta = -0.3$). Recognition ($M = 90.9$) remains well above naive ($M = 82.5$). Added Table 20b, conclusion finding 13. Updated paper totals to N=3,292 across thirty-one evaluations.

v2.3.11 (2026-02-17) Transcript figures: Added Figure 10 (naive vs base high_ performer comparison panel) to Section 6.21. Added Figure 11 (bilateral mutual_transformation_journey transcript comparison) to Section 6.11. Added `generate-paper-figures.js` script for reproducible paper figure generation.

v2.3.12 (2026-02-17) Token budget sensitivity: Added Section 6.22 testing whether constraining `max_tokens` from 8000 to 256–2048 affects evaluation scores. Scores are flat across all budget levels; the recognition effect is fully preserved even at 256 tokens. The retry-absorption mechanism means truncated structured output self-heals. Added Table 49, recommendation 8 to Section 7.11. Updated paper totals to N=3,454 scored across thirty-six evaluations.

v2.3.13 (2026-02-17) Paper correctness fixes: Fixed eval-2026-02-14-559d854b scope from "cells 40–59, N=167" to "cells 40–45, N=60" in Table 2 (only cells 40–45 used; cells 46–59 superseded by 49b33fdd at N=360). Fixed broken Table 10 reference in Section 6.6. Fixed dynamic-learner N inconsistency: intro and finding 12 updated to N=300 (6c033830 + a2b2717c + 664073ab). Clarified token budget Section 6.22 design text. Added missing Appendix B commands.

v2.3.14 (2026-02-18) Cognitive prosthesis re-run and analysis: Replaced misconfigured prosthesis run (50487df7, all cells fell back to default Haiku) with corrected eval-2026-02-17-25aaae85 (N=90, Nemotron ego, Kimi K2.5 superego, Opus judge). Prosthesis hypothesis fails decisively: full mechanism stack scores 49.5 vs Nemotron simple base 64.2 ($\Delta = -15$). Added dimension analysis (two-tier static/dynamic capability model), superego parse failure analysis (16–45% malformed JSON auto-approves), and Haiku control smoke test (eval-2026-02-18-f489c0ea, N=6, confirming model-dependence). Added conclusion finding 14 (minimum ego capability threshold), recommendation 9 to Section 7.11, three future work items to Section 8.2 (parse robustness, capability threshold mapping, adaptive mechanism loading). Updated Table 2, Appendix D run IDs. Paper totals: N=3,383 across thirty-seven evaluations.

v2.3.15 (2026-02-19) Active control model confound resolved (A8): Ran active control (cells 15–18) on Kimi K2.5 (eval-2026-02-19-f2263b04, N=216, 116 grounded). Key finding: active control scores *below* base on Kimi ($M = 56.0$ vs $M = 64.2$ on matched scenarios, $\Delta = -8.2$), not between base and recognition as on Nemotron. Prescriptive heuristics help weaker models (Nemotron: +9 pts) but harm stronger ones (Kimi: -8 pts)—consistent with Section 6.21 prompt elaboration baseline. Same-day reproduction runs confirmed no model drift: Kimi base $M = 58.8$ (eval-2026-02-19-13d34bef, N=36) and Nemotron base $M = 49.8$ (eval-2026-02-19-411414e4, N=18). GPT-5.2 cross-judge validation confirms ordering (placebo 48.3 < base 70.8 < recognition 76.5). Updated Section 6.2 (added Table 6b, model-dependent interpretation), Section 5.3 (active control description), Section 8.1 (limitation resolved). Paper totals: N=3,653 across forty evaluations.

v2.3.16 (2026-02-21) Recognition inversion mechanism and A2/A4 experiments: Added Section 6.16.1 documenting the scenario-specific recognition inversion (misconception_correction: tutor +30.5, learner -10.4; mutual_transformation: no inversion). Identified metacognitive feedback loop mechanism through qualitative transcript analysis. A2 dynamic learner mechanism sweep (cells 72–77, N=238): all 7 mechanisms show positive recognition deltas; dynamic learner amplifies mechanism differentiation 1.6–2.8x vs scripted. A4 authenticity-focused learner superego (cells 78–79, N=47): null result—scores worse on every dimension including authenticity itself. Epistemic readiness dimension feasibility test (N=10): no base-recog gap (both 4.0/5), ruling out construct validity artifact. Reframed Section 7.5 from "calibration problem" to "structural finding." Paper totals: N=4,144 across forty-eight evaluations.

v2.3.17 (2026-02-21) Table 19a and conclusion finding #15: Expanded Section 6.10 mechanism results with Table 19a showing A2 sweep data (cells 72–77, N=108, Nemotron ego) for quantitative, erosion, and tutor-profiling mechanisms. All three show positive recognition deltas (+8.0 to +9.7), completing the 2x7 mechanism matrix. Updated conclusion finding #12 to reference full 7-mechanism matrix. Added finding #15 documenting recognition inversion as scenario-specific and structural. Fixed stale "thirty-seven" to "forty-eight" in conclusion summary.

v2.3.18 (2026-02-23) Bug disclosure and data integrity corrections: Added Section 8.1 paragraph disclosing two learner pipeline bugs affecting all pre-Feb-20 dynamic learner multi-turn evaluations: Bug 1 (leaky "SUPEREGO:" labels in learner external messages) and Bug 2 (broken `.flatMap()` causing learner to see only own monologue, not tutor responses). Added pre-fix data caveat to Section 6.16 (learner superego paradox, $d = 1.43$ from eval-2026-02-07-b6d75e87 is an upper bound). Added footnote to Section 6.11 flagging $M = 38.0$ as pre-fix. Clarified Section 6.10 Table 19 footnote to name bugs explicitly and explain why tutor-side scores are valid but learner-side required re-generation. Qualified Finding #2 (additivity) as tutor-side only, noting learner-side AxC interaction (Section 6.16). Qualified Finding #9 conclusion text with pre-fix caveat. Added forward reference from Section 6.3 to Section 6.16 learner-side interaction. Added clean re-generation runs eval-2026-02-20-bd37cc62 (cells 64–65) and eval-2026-02-20-4e131c6f (cells 69–70) to Table 2 and Appendix D. Updated paper totals: N=4,264 across fifty evaluations. Also fixed Bug 5 in evaluation infrastructure: `resumeEvaluation()` silently dropped `--ego-model` and `--superego-model` overrides from stored metadata, causing resumed runs to fall back to YAML defaults.

v2.3.19 (2026-02-24) Post-fix data replacement: Replaced all pre-fix dynamic learner data with clean re-run data. Section 6.10 Table 19 now reports clean re-run tutor scores (0fbca69e, bd37cc62, 4e131c6f, N=240); mechanism ordering changes—intersubjective framing becomes the dominant mechanism (previously weakest), recognition deltas larger (+19.1 avg vs +14.2 pre-fix). Table 19a replaced with b5cd16e1 clean data (N=36). Section 6.16 Tables 29–31 replaced with post-fix factorial data (25c78e91, N=48): architecture effect *increases* to $d = 3.05$ (was 1.43), recognition effect vanishes ($d = 0.09$), AxC interaction vanishes ($F = 0.12$, was 11.50). The "recognition rescues multi-agent learners" finding was a bug artifact—removed throughout. Section 6.16.1 A4 results updated with b5e123b4 clean data (N=12). Rewrote Finding #2 (additivity holds on both rubrics), Finding #9 (paradox real and stronger, recognition irrelevant), Finding #12 (intersubjective framing now dominant mechanism), Finding #15 (tutor-learner asymmetry, not inversion). Updated Section 7.5, Section 8.1, abstract, and concluding summary. Section 6.11 footnote refined. Section 6.10 prose rewritten for new mechanism ordering. N-counts unchanged at 4,264 across fifty evaluations.

v2.3.20 (2026-02-25) Bug 4 disclosure and claim-by-claim audit response: Added Section 8.1 paragraph disclosing multi-turn scoring misalignment bug (Bug 4): `evaluate/rejudge` scored `suggestions[0]` (Turn 0) instead of last turn, affecting 8,631 multi-turn rows across 81 runs. Average holistic lift +19.9 pts. Paper clarified that all multi-turn tables report Turn 0 scoring; holistic last-turn rescoring in progress. Added pre-fix data caveats to Section 6.15 Table 25 footnote (b6d75e87 affected by Bugs 1–2 in ego/superego cells and by Bug 4 Turn 0 scoring). Extended Section 6.15.2 Table 28 caveat noting cross-run audit (N=895) shows negative learner-side learning arcs under recognition in multi-turn settings. Strengthened Section 8.1 bilateral transformation asymmetry paragraph with cross-run evidence ($d = -0.27$ recognition, $d = -0.98$ architecture on arc metrics, 30/33 runs show recognition reduces tutor dimension variance). Updated synthetic learning outcome bullet in summary to reflect mixed learner-side evidence. Added eval-2026-02-23-b5cd16e1 (cells 72–77, N=36) and eval-2026-02-23-b5e123b4 (cells 78–79, N=12) to manifest and Appendix D. Updated paper totals: N=4,312 across fifty-two evaluations (manifest v1.8.0).

v2.3.21 (2026-02-25) N-claim and stat-claim traceability: Systematic pass through all 156 N-claims and 66 stat-claims identified by the claim-by-claim audit tool, adding inline run ID anchors, Section cross-references, and Table citations. Key changes: added run IDs to all table headers lacking them (Tables 3, 5, 6, 8, 13, 15, 20b, 21, 22, 25, 45); added Run ID column to Table 41 (inter-judge agreement); annotated pooled-data N-claims with "see Table 2" or "pooled across evaluation database"; marked pilot data as "no longer in primary evaluation set"; added Section/Table cross-references to all stat-claims in Sections 7–9. No data, analysis, or N-count changes—purely traceability metadata.

v3.0.0 (2026-03-07) Automated prompt tuning experiment: Added Section 7.7 with $2 \times 2 \times 2$ autotuning experiment (base/recognition $\times$ unoptimised/autotuned $\times$ DeepSeek/Qwen 3.5). Seven new runs (eval-2026-03-05-53fb1462 through eval-

2026-03-07-c4d1bfa2, N=38 total). Key finding: recognition prompts are more improvable than base (+17–23 pts vs +5–9 pts), the effects are super-additive, and the gap between conditions *widens* under optimisation. Autotuned recognition converges to ~71–73 on both models. Base autotuning ceiling (41–45) remains below unoptimised recognition baseline (48–55), confirming recognition's irreducibility to prompt engineering. Added Table N, 7 run IDs to Appendix D. Renumbered Section 7.7 → 7.8.

v3.0.1 (2026-03-07) M3 trajectory analysis: Updated M3 (adaptive responsiveness) from "disconfirmed" to "conditionally supported" based on complete trajectory-specific data (eval-2026-03-06-ebcd6de0, N=72, 3 scenarios × 8 cells × 3 replications, DeepSeek V3.2, Sonnet judge). Pooled slope effect remains small (d=0.22, NS), but the 10-turn disengagement scenario shows d=1.63, p<.001, with recognition–baseline gap widening from +12 pts at T0 to +35 pts at T8–T10. The two 8-turn scenarios show no slope differentiation ($d \leq 0.30$). M3 recharacterized as a conditional emergent property of M1+M2 that manifests under sustained re-engagement demands with sufficient turn count. Updated abstract, Section 3.2, Section 6.3.2, Section 6.3.8, mechanism summary table, and Discussion synthesis. Added 1 run ID to Appendix D (53 total).

v3.0.2 (2026-03-07) M1/M2 mechanism isolation: Added dedicated isolation run confirmation to Section 6.4.2. Runs eval-2026-03-06-768ba77b (M2: base + superego, cells 82–83) and eval-2026-03-06-e4abd0df (M1: recognition, no superego, cells 84–85) across 9 multi-turn scenarios (N=108, DeepSeek V3.2, Sonnet judge). Full 2 × 2 isolation confirms substitution: superego adds +9.2 pts under base (d=1.13, p=.002) but +1.1 under recognition (d=0.08, NS)—calibration pre-empts 88% of the superego's contribution (27% additivity deficit). M1 vs M2 head-to-head: calibration alone (51.4) outscores error correction alone (36.9) by d=1.03 in 7/9 scenarios. Two emotionally intense scenarios (Frustration, Affective Shutdown) show slight M2 advantage, suggesting scenario-specific residual error correction value. Added 2 run IDs to Appendix D (55 total).

April 2026 arc summary (v3.0.43 → v3.0.51, nine revisions in four days)

The published version prior to this cycle was **v3.0.42** (2026-04-21). What followed was a compressed three-day cycle triggered by an internal critical review of v3.0.42 (2026-04-22), which identified five issues: disproportionate narrative weight on a pending boundary-condition M3 finding, tension between §6.1 dimension-narrowing language and §8.6 PCA's one-factor result, Sonnet-only headline *d*'s despite §8.3 reporting three-judge variation, untrimmed apparatus-as-method exposition, and an untested strong version of the prompt-density alternative. The cycle had three phases:

Phase 1 — editorial tightening (v3.0.43–v3.0.44, v3.0.51). Collapsed M3-disengagement exposure from 13 paper locations to one canonical §6.3.2 treatment; rewrote §6.1 calibration lead for PCA consistency; reframed all headline *d*'s to judge-pooled estimates; trimmed §7.4 apparatus-as-method by ~35%; trimmed universal-substitution restatements from 22 to 15 with single canonical homes (v3.0.51, F4). No data changes.

Phase 2 — four new experiments closing open concerns (v3.0.45–v3.0.48). A12

(M3 disengagement replication across Haiku + Gemini Flash under Sonnet + GPT-5.4, $N = 32$): the pre-registered replication failed — three of four model × judge cells below $d = 0.5$, and the one replicating cell contradicted by its secondary judge at cross-judge $\Delta d = 2.03$. M3 retired as clean null (v3.0.45). A11 (direct M2-alone isolation on Gemini Flash, $N = 125$): pooled $d = 1.46$ on Sonnet, confirming and exceeding the factorial-inferred +12.3 residual in §6.4.1 (v3.0.46). A10 v2 (matched-specificity prompt-density control, three-judge triangulation, $N = 183$): pooled three-judge $d = 0.17$ recognition-vs-matched-pedagogical, below the pre-registered $|d| < 0.2$ density-sufficient threshold (v3.0.48). A10b (four-way orientation-family comparison adding a rigorously-authored behaviorist prompt, three-judge at $N = 248$ generation with full GPT coverage and $\sim 80\%$ Sonnet/Opus coverage): pooled $d = 1.38$ between Hegelian-descendant family and transmission family; within Hegelian family, density-substitutable ($d = 0.14$); within transmission family, behaviorist substantially *below* base ($d = 0.89$) (v3.0.48).

Phase 3 — bug discovery, retraction, and framing reframe (v3.0.47, v3.0.49, v3.0.50). A10 v1 (the first run of the density control) was invalidated by an evaluation-pipeline dispatch bug (bug_007: `resolveEvalProfile` lacked a branch for the new `prompt_type`, routing cell_95 silently to the base prompt). Retracted in v3.0.47 before A10 v2 re-ran on the fixed pipeline. v3.0.49 implemented a calibrated Framing A reframe of intro/abstract/§9 acknowledging the family-level finding while keeping §3's recognition-specific theoretical motivation. v3.0.50 then refined this to the "primacy + descent" framing (Framing B+): empirical generalisability is real (matched-pedagogical reproduces recognition within $|d| < 0.2$), but recognition theory remains the most explicit philosophical articulation of an intersubjective-pedagogy orientation its constructivist descendants (Dewey, Vygotsky, Piaget, with Kapur, Chi, VanLehn, and Graesser as further cognate descendants) inherit, often without direct attribution. §1 added a careful three-circle genealogical account; §3 added a level-of-analysis framing note; §9 extended the family-level passage with explicit "generalisability does not displace recognition theory" reasoning.

Net effect across the cycle. Three mechanisms originally claimed; now two supported, one clean null. The two supported mechanisms have stronger evidence than before (A11 confirming architecture residual; §7.9 closing the density alternative at the orientation-family level). The paper has a new methodological contribution (pedagogical-orientation taxonomy, `docs/pedagogical-taxonomy.md`) and a retracted-then-corrected experiment (A10 v1 → A10 v2). An `/ultrareview` pass caught the bug_007 issue before A10's result propagated into a broken paper claim. Net paper length grew by ~6% (new §7.9 orientation-family content); net theoretical-content distinctiveness shrank slightly (recognition now framed as one family member rather than the uniquely-necessary frame), but defensibility strengthened.

Individual v3.0.X entries follow in chronological order (newest first):

v3.0.227 (2026-07-22) §6.21 extended — Phase 5d delivery-integrity rider (pre-registered, development-tier); §6.21 header extended. A post-hoc census of the sealed 5b/5c committee traces located a second unchecked path: the finalized turn delivered to the learner is not the committee-approved envelope on 47 of 83 Phase-5b committee moments (all 34 due-release turns, authored by the clue-staging surface, plus 13 clue-staged turns where that surface overwrites the composed turn after the battery passes), so the approved text ships on only 36/83 and the audited cue is lost on the overwritten turns. The rider (`PROGRAM-2-PHASE5D-DELIVERY-INTEGRITY-PREREGISTRATION.md`, frozen seed 20260722) adds two deterministic fixes and nothing else: spanCue.v1 (resample the mini 2× at the 0.35 pin until its extracted question carries a cue word — the live analogue of §6.22's offline span-mode v2) and deliveryGuard.v1 (on clue-staged turns only, finalized approved and no premise release, swap the staged final question for the protected span verbatim; premise-release turns and the clue body untouched). Twelve committee-v3 + six fresh controls sealed on world-005-marrick (18/18, zero attrition), fresh controls pooled with all 18 archived Marrick controls under the frozen stationarity check (fresh 0.095 vs archived 0.150, Δ 0.055, control n = 24). E1d PASS: 0.477 (42/88) vs 0.136 (22/162), +0.341 CI [0.223, 0.457] — up from 5b's 0.386/+0.236; 42/55 ≈ 76% of the live due-release ceiling (33/88 due, ceiling 55/88 ≈ 0.625) vs 5b's 65%. Both registered mechanism endpoints pass: M1 guard-eligible delivered cue 13/15 = 0.867 (bar 0.75; 15 span swaps applied, 33 premise-release skips, 37 shipped-as-approved), M2 span cue 74/85 = 0.871 (bar 0.85; 5b baseline 0.37). Coverage PASS (0.625 vs 0.632, no tax); seam PARITY (0.550 = 22/40 0.65, continuity 3.40 vs 3.75); hard-safety FAIL (0.50 vs 0.63) with 5b's exact turn-9 anatomy — 5 of 7 committee leaks and both control leaks at turn 9, arm-symmetric and world-bound (5c world_027 had no analogue), not a rider regression (safety-breakdown artifact). Reading (pass row): with the two census-located leaks closed, the committee controls warrant form on every text path it owns, at no coverage cost and at seam parity; the gain is delivery completeness, not pedagogy (frozen six-word instrument stands; a cue-less good-question control still scores non-compliant). The registered ≈0.55 rate reference was approached not reached — the mechanisms cleared their bars but the committee converted 76% of achievable moments rather than the ≈90% that figure assumed (this run's achievable ceiling, 0.625, is higher than the 5b ceiling the ≈0.55 figure was built on). Two pre-launch amendments, both caught by the launch gates with zero sealed dialogues at fix time: A1 the claude-CLI `sonnet-5` alias drop (spawn-boundary map, frozen strings byte-identical); A2 the new flags unregistered in the runner whitelist (first launch aborted at job 1 pre-seal; smoke gained a parse gate). §6.21 header extended; "Status and caveats" updated three runs → four. Exploratory tier per the prereg; development-tier; no DB, rubric, abstract, or headline-N changes; no sections renumbered. Provenance: PROGRAM-2-PHASE5D-DELIVERY-INTEGRITY-PREREGISTRATION.md §9, config/adaptive-tutor-evidence/program-2-phase5d.manifest.json, scripts/analyze-program2-live-pilot-5d.mjs, scripts/program2-phase5d-safety-breakdown.mjs, notes/program-2/2026-07-22-cue-attrition-observation.md, archive ~/.machinespirits-data/program-2/phase5d-live, board card program-2-context-vs-weights-finetune.

v3.0.226 (2026-07-22) §6.20/§6.21 KTO status clauses updated — conditional runs spent, byte-identical to SFT. Both conditional KTO runs licensed by the Phase-2 pre-registration were executed 2026-07-21 (Lambda H100, ~21 min/arm, Amendment-2 recipe: lr 5e-6, 1 epoch, from-SFT adapters) and are behaviorally inert: KTO loss flat at 0.5 throughout (the unpaired audit labels supply almost no gradient the SFT policies do not already fit); merges delta-verified before save (probe deltas 0.000488 both arms; one silent no-op from the stale pre-fix merge script was caught by output inspection and deleted); 58/58 held-out greedy generations per arm byte-identical to the SFT outputs, every gate landing exactly where SFT left it (instruct 0.414, base 0.103 = floor), paired KTO-vs-SFT CI [0.000, 0.000] in both arms. Reading recorded with the verdict: at the frozen conditional recipe, preference tuning on the apparatus's own audit labels produces weight motion below the quantized-serving threshold; the offline solo ceiling stands at the SFT number, and the licensed-run ledger (2 SFT + 2 KTO) is fully spent — no further training is licensed under the Phase-2 document. §6.20's "licensed and unspent" and §6.21's "in flight" clauses updated to the recorded outcome; older revision entries retain their original wording as historical records. Provenance: PROGRAM-2-PHASE2-PREREGISTRATION.md results addendum (2026-07-21, commit 1fda8f29); KTO artifacts (floor/tuned-kto-{instruct,base}-q8-ollama.json, adapters, ollama models retained for provenance) under the program-2 archive; board card program-2-context-vs-weights-finetune. No DB, rubric, abstract, or headline-N changes; no sections renumbered; no other numbers change.

v3.0.225 (2026-07-21) §6.22 call-count corrected — post-fold claim audit. The independent claim audit of the v3.0.224 fold (all other figures TRACED, several re-derived from raw artifacts) found one drift: "roughly 110 frontier calls in total" undercounted the reconstructable total by a third — it repeated the v2 leg's two-family subtotal (~55 × 2) while omitting the v1 terra-flip leg. Actual composer invocations, derived from the delivered files' bySource counts (composed + span_lost rows call the composer; no-span rows do not; the sonnet v1 reference is a zero-call regrade of the archived Phase 4 file): 54 (v1 terra) + 54 (v2 sonnet) + 54 (v2 terra) = **162**. §6.22's sentence corrected to "roughly 160" with the per-leg arithmetic stated; the v3.0.224 revision entry below retains its original wording as the historical record of that version. The HANDOFF H9 addendum's "~110 calls" is unaffected (it describes the v2 leg alone, which is accurate at 108). No other numbers change; no sections renumbered.

v3.0.224 (2026-07-21) §6.22 added — offline composer-seat family-invariance and span-extraction v2 (post-hoc, development-tier); §6.20 coupling-probe attribution corrected in place. Two same-day offline probes on the §6.20 apparatus (58 archived held-out moments, frozen grader; fidelity re-validated byte-exact — 0.414 / 0.293 / 0.448+2-rescued — before any paid call; ~110 frontier calls; no prereg, descriptive only). Family flip: composer swapped sonnet-5 → codex gpt-5.6-terra yields identical constructions (delivered 0.293 both; fail-closed 0.448, 2 rescued both) at moment level (56/58 same verdicts; containment 53/1 vs 51/3 in terra's favor). Decomposition: zero composer-added questions in either family (all one-question failures span-borne — the v1 extraction joins all the mini's question sentences; histogram 39/14/1), and the composed-alone penalty is extraction-dropped cue sentences (25/54 cue-free spans, 20 with the cue in a discarded statement; composed cue failures 22 both families vs the mini's 6; 3 restored each). §6.20's "the frontier re-added extra questions even under explicit instruction" corrected in place (italic annotation; checks-carry-the-design conclusion stands), with a dated erratum appended to PROGRAM-2-PHASE2-PREREGISTRATION.md's results addendum, the wording's source. Span-extraction v2 (cue-preferring single question + carried cue statement; default v1 byte-preserved, regression-checked 54/54; precheck: single-question 54/54, cue coverage 29→49/54, 20 carried) converts the full headroom in both families: delivered 0.586 sonnet / 0.603 terra (= fail-closed; rescued 2 → 10/11), 77–79% of the offline achievable ceiling (44/58 ≈ 0.76), composed turns now above the mini solo (0.414); residual failures premise/guards; agreement 55/58. New bounded asymmetry: sonnet adds a second question in 3/53 v2 composed turns, terra 0/54 — small, sonnet-specific, battery-caught. Live family dynamics remain unmeasured by explicit decision; v2 extraction exists only offline (live machinery untouched; adoption unlicensed). Provenance: notes/program-2/2026-07-21-terra-composer-probe.md, config/adaptive-tutor-evidence/program-2-terra-probe.manifest.json, scripts/program2-coupling-probe.mjs + program2-terra-probe-analyze.mjs, board card program-2-context-vs-weights-finetune. Development-tier; no DB, rubric, abstract, or headline-N changes; no sections renumbered.

v3.0.223 (2026-07-21) §6.21 structural-ceiling clause corrected — claim-audit recomputation (PR #140); entry relabeled from v3.0.222 on cherry-pick (the Phase 5c fold took that number on this branch first). The "at roughly 55% of the structural ceiling" figure (inherited verbatim from the 5b prereg §8) was traced but not recomputable from any disclosed denominator: 0.386 against §6.20's offline ceiling (44/58 ≈ 0.76) gives ≈51%, and recomputation from the sealed Phase 5b traces under the frozen `released == 0` component gives the live population's own denominator — 34 of the 83 committee-v2 warrant moments carry a due premise release and are non-compliant by construction (dramatic-release due-frame cross-check agrees 83/83), so the live structural ceiling is 49/83 ≈ 0.59 and the live rate is 32/49 ≈ 65% of achievable. "~55%" matches neither denominator (it reproduces §6.20's tuned-solo-of-achievable figure). §6.21's sentence corrected to the recomputed live denominator — the correction strengthens the sentence (65% of achievable live vs the offline committee's ≈59%); live due-release density (41–45% across arms and both runs) is roughly double the offline held-out set's 24%, which is why the offline ceiling cannot be borrowed for live rates. Same-dated disclosures appended outside frozen text: a post-hoc annotation on PROGRAM-2-PHASE5B-FALLBACK-BATTERY-PREREGISTRATION.md §8, and errata on PROGRAM-2-PHASE5-LIVE-PILOT-PREREGISTRATION.md (§1, §9) and notes/program-2/2026-07-20-phase5-live-pilot-results.md correcting "§6.19" → the §6.18 addendum for the ≈−0.13 coverage tax. Verification: the frozen analyzer re-run reproduced 32/83, 18/120, and CI [0.128, 0.354] from the sealed archive before recomputation. No other numbers change; no sections renumbered; no DB, rubric, abstract, or headline-N changes.

v3.0.222 (2026-07-21) §6.21 extended — Phase 5c cross-world transfer probe (pre-registered, development-tier); §7.12 successor note extended. The Phase-5b-validated committee moved *unchanged* — same tuned-mini artifact on its frozen serving pin, fallback policy v2, explicit no-KTO clause — to `world_027_gazette_recall`, selected under a frozen rule (maximum costume distance from the training world among structurally comparable siblings: non-period frame, newsroom-noir diction, zero derived-lexicon overlap, 8 premises / 7 rules vs Marrick's 9 / 5; floor worlds rejected by name as density gambles). Ten committee-v2 + eight fresh controls, seed 20260721, no pooling with Marrick controls (cross-world pooling meaningless by design); 17/18 sealed with the programme's first attrition (one committee dialogue, both same-seed attempts on the known auto-learner budget overflow), density PASS at 61 committee opportunities (6.8/dialogue vs Marrick's 6.9); pre-registered paid smoke on the new world passed its go criteria before launch. E1c PASS: 0.508 vs 0.306 (+0.202, CI [0.072, 0.338]) — above the committee's home-world rate (0.386) — with `exactly_one_question` 1.000 (control 0.857), cue 0.754 vs 0.653 (both arms lifted by the letter-friendly world; the edge holds on top), guards 1.000 both arms. The new pre-registered costume-leak metric (training-world derived lexicon minus transfer-world lexicon minus the frozen six, counted in mini-authored delivered text) reads ZERO across all 61 committee units against a 31.8/1k home-world saturation reference, with the control frontier itself at 4 hits of the generic word "fair" — the trained move arrived with none of the training world's vocabulary. Seam 0.515 0.65 (13+20 windows, realized-n clause; continuity 3.38 vs 3.45); hard safety 0.89 vs 0.88 with no analogue of the turn-9 signature (5b's leak anatomy confirmed as a world property); the coverage@16 guardrail formally FAILS by point estimate (0.489 vs 0.550, −0.061 vs the −0.05 margin, miss 0.011) with the bootstrap interval [−0.197, +0.053] spanning zero at n = 9 — reported as the frozen rule requires and carried as a caveat on the reading. Licensed reading (pass row, coverage-caveated): the trained warrant move behaves as a form, not a costume; the move-library concept (train once, validate per world, retrain only on failure) is live at exploratory tier — one family, one transfer world, transfer meaning across costume within the same genre. Runtime: `e9b01bdd` (the 5c machinery commit) on the Amendment-1 pinned worktree's `91b8a50e` lineage (world parameterized; 5/5b plans verified byte-identical under the change). Exploratory tier per the prereg; development-tier; no DB, rubric, abstract, or headline-N changes; no sections renumbered. Provenance: PROGRAM-2-PHASE5C-CROSS-WORLD-TRANSFER-PREREGISTRATION.md §9, config/adaptive-tutor-evidence/program-2-phase5c.manifest.json, archive ~/.machinespirits-data/program-2/phase5c-live, board card program-2-context-vs-weights-finetune.

v3.0.221 (2026-07-21) §6.21 added — Program-2 live-coupling extension (Phases 5 and 5b, pre-registered, development-tier); §7.12 successor note extended. The Phase-5 live committee pilot (24 dialogues, committee vs silent control on world-005-marrick, endpoints/bars/seeds frozen before launch, pinned claim-run runtime per Amendment 1 after the main-stack launch dead-ended with zero sealed dialogues) fails its primary live-compliance endpoint (0.200 vs 0.160, +0.040, CI [−0.054, +0.133]) while the component decomposition locates the loss on the one delivered-text path the battery never checked: the trained warrant-cue form transfers live (0.560 vs 0.395, +0.165) but fallback-delivered mini replies leak second questions (0.720 vs 0.938, −0.218; 24/75 moments shipped `fallback_multi_question`). Coverage shows no enforcement-scale tax (−0.014, CI [−0.083, +0.056]); seam detection at chance (0.500, bar 0.65); hard-safety identical (0.58 = 0.58). The Phase-5b confirm run extends the same fail-closed battery over the fallback path (greedy check → up to two resamples at the frozen sampled pin → cue-preserving trim; composed path and control byte-unchanged; fresh-control stationarity check licenses pooling) and flips the verdict: E1b PASS 0.386 vs 0.150 pooled control (+0.236, CI [0.128, 0.354]), with `exactly_one_question` 0.720→0.976 (above the frontier's own 0.938) and every other component unmoved (cue 0.560→0.542, premise 0.587→0.590, guards 0.960→0.988); fallback ledger 19 resamples + 8 trims + 4 greedy + 1 unchanged; live rate matches §6.20's offline composite reference (0.448) within the arms' interval. Coverage guardrail passes (0.611 vs 0.639, margin −0.05); seam re-check 0.600 0.65 (continuity 3.45 vs 3.75); the hard-safety guardrail formally FAILS (0.42 vs 0.61, bar 0.51) with exonerating anatomy recorded — a universal turn-9 release-schedule leak present in the control arms of both runs (4/12 and 2/6) and 4 of 5 leaky committee turns frontier-authored; successor fix named as the turn-9 staging, not the committee. The cue-lexicon sensitivity rescore is reported as labeled exploratory only (control 0.469 vs committee 0.320 under the relaxed world-lexicon rule — the frontier's natural questioning outruns the six-word letter; the frozen v1 instrument stands by decision 2026-07-20), alongside the Phase-5 fallback-trim counterfactual (0.347 vs 0.160) that motivated 5b. Both amendments recorded as instrument provenance (runtime re-pin; prompt-model duplicate-line recovery, arm-independent and comparability-neutral). KTO and the Phase-5c cross-world probe are in flight and not folded. §7.12's successor parenthetical extended with the live continuation. Exploratory tier per both preregs; development-tier; no DB, rubric, abstract, or headline-N changes; no sections renumbered. Provenance: PROGRAM-2-PHASE5-LIVE-PILOT-PREREGISTRATION.md §9, PROGRAM-2-PHASE5B-FALLBACK-BATTERY-PREREGISTRATION.md §8, config/adaptive-tutor-evidence/program-2-phase5-live-pilot.manifest.json, config/adaptive-tutor-evidence/program-2-phase5b.manifest.json, board card program-2-context-vs-weights-finetune.

v3.0.220 (2026-07-19) §6.20 added — Program-2 context-versus-weights fine-tune (pre-registered, development-tier); §7.12 successor note updated. The §7.12-designed successor ran under a frozen prereg: identical LoRA data (865 sealed audit-passing turns) into Qwen3.5-9B-Instruct and its base sibling; floors first, bars frozen (floor+0.15, paired bootstrap, absolute rate, leak non-inferiority), 58 held-out warrant-skip moments graded by the label-producing deterministic machinery (fidelity 99.83%/100%). Verdicts: instruct 0.414 vs same-lineage floor 0.310 (+0.103, CI [−0.017,+0.224], bar 0.460) — partial parametric traction with the warrant cue nearly resolved (19→6 held-out; first movement of this frontier-resistant component by any intervention short of enforcement); base flat at 0.103 with the cue trained (43→9) but conduct collapsed — the predicted thin-ego pathology. Iron-cage reading inverted: the discipline trains into both variants; the incumbent alignment layer is the scaffold that lets it integrate, not the obstacle. Pre-registered amendment probed fail-closed protected-span coupling offline: composed-alone 0.293 (frontier re-adds questions under instruction — checks load-bearing), fail-closed system 0.448 vs the frontier's own live 0.276 at these exact moments (+62% relative; live coverage/seam costs reserved for Phase 5). Structural ceiling disclosed (14/58 moments non-compliant by construction; achievable ≈0.76). Four instrument faults caught by standing checks before any wrong number was accepted (partial transfer; MTP metadata inconsistency; silent no-op merges exposed by byte-identical outputs, merges now delta-verified; ~5pt GGUF build-lineage noise removed by same-lineage re-flooring). No arm passed; no H-W claim; KTO runs licensed and unspent. Post-audit disclosure: the tuned base emits empty generations on exactly the 14 compiled-arm held-out moments (untuned floor 0/58) — a training-induced degenerate mode reported as part of the thin-ego result. Development-tier; no DB, rubric, abstract, or headline-N changes; no renumbering. Provenance: PROGRAM-2-PHASE2-PREREGISTRATION.md §10 addendum, config/adaptive-tutor-evidence/program-2-phase4-results.manifest.json, board card program-2-context-vs-weights-finetune.

v3.0.219 (2026-07-18) §6.19 added (sensor-program closure) and §7.12 added (consolidated adaptation boundary map); one stale §6.3.10 cross-reference corrected. §6.19 lands the previously note-only adaptive-state v2.4 result as its own development-tier section: the canonical sensor pilot's negative deltas were an estimation artifact (`data_starved` — bit-exact refit reproduction, regularization sweep eliminates them, the pilot's no-state reference itself schedule-overfit), and the substrate's concealment premise was false (`close_sensor_program_on_substrate` — an exact fixed-point-verified Bayes filter drives latent-state posterior entropy from 4.78–5.40 bits to 0.0000 at horizon on every trajectory under both schedules; active sensing buys nothing because nothing is hidden). Zero-call, sealed, replayable; operative stop verdicts unchanged; revival requires identifiability-limited kernels by construction. Provenance: `PLAN_4_0/2026-07-13-adaptive-state-v24-results.md`, `config/adaptive-tutor-evidence/adaptive-state-v24-voi-schedule-6135bf3f6c6b.manifest.json` §7.12 then draws the adaptation programme's closures into one picture beside §7.11: the intervention ladder (standing advice §6.15/§6.3.10 → coached curated advice §6.16 → point-of-action advice §6.18-addendum → enforcement §6.18-addendum → first-draft ownership §6.18) reads as a single result — every information-delivering rung fails to move behaviour; the enforcing rung moves it by removing the behaviour from the model, at coverage cost and with authorship transfer. Extends §6.3.9's insight–action gap to three coarser grains; restates §7.11's law kinetically; records the surviving positive family's common shape (§6.13 pacing guards, §6.7 gates, §6.12 contracts-as-instrumentation), the §6.19 sensor closure as removing the better-observation road, and the wall's measured shape (zero safety failures — interface + attribution, not refusal). Names the two designed successors as hypotheses inheriting no claims: the context-versus-weights test (train on the apparatus's own sealed per-turn labels, grade with existing zero-call audits) and the steerable-tutor alternative converging on §8.1. Claim-audit corrections applied pre-commit: §6.8.7 removed from the positive family (its corrected re-run is a null — the same audit also fixed §6.3.10's stale "localized binary" cross-reference to match §6.8.7's corrected verdict); the compiled-arm phrasing miss re-scoped to "unmet on most non-compliant turns"; the sensor citation repointed from §6.17 to the new §6.19. No sections renumbered; no DB, rubric, abstract, or headline-N changes.

v3.0.218 (2026-07-18) §6.18 addendum — independent cross-implementation check recorded. A second, independently written Step 4 scorer (Codex-authored, found uncommitted in the scorer staging worktree) reproduces every point estimate, opportunity count, density verdict, guardrail figure, and the instrument-failure verdict exactly against the same 80 sealed traces. Sole divergence: compiled-arm bootstrap CI lower bounds, localized to degenerate-draw handling in the starved T1 channel (drop-the-draw vs conservative T2-only macro; frozen text silent because the density floor was supposed to make it unreachable). Whether the sol compiled CI excludes zero is convention-dependent; every licensed conclusion is convention-robust. One parenthetical added to the §6.18 addendum; full note in prereg §11; independent outputs archived beside the traces. No verdict or number changes.

v3.0.217 (2026-07-18) §6.17 caveat added — Sonnet-block transport boundary. Records in §6.17's status-and-caveats paragraph that the Step 2 Sonnet family block ran before the safe-mode-v1 claude-CLI context-isolation fix (speaking-tutor calls ambiently loaded ~16k tokens of repository context; the codex/Terra path was always isolated). Uniform within the block, so it cannot manufacture the within-family register contrasts; it adds one more uncontrolled difference to the whole-stack Terra-vs-Sonnet comparison and forbids pooling that block's Sonnet rows with post-fix runs. No verdict changes (the family B primary was already a null read as whole-stack). One sentence + this entry; no renumbering.

v3.0.216 (2026-07-18) §6.18 addendum — the point-of-action coaching gate seals: instrument failure on the stagnation channel; insight–action gap persists directionally; enforcement moves action at a coverage cost. The one discharged successor of the preconscious fold ran in full: 80/80 pre-registered dialogues (4 arms × 2 trap profiles × 2 speaking-tutor families × n=5, seed 20260714, detector frozen 2026-07-14), zero exclusions, all traces provenance-stamped to the isolated launch branch (`91b8a50e` = frozen `092cf672` + safe-mode-v1 claude-CLI context-isolation backport; 7 pre-isolation dialogues quarantined unanalyzed). Frozen verdict: **instrument failure, not a coaching null** — the `stagnant_repeat` channel under-fired its pre-declared opportunity floor in 7 of 8 arm × family blocks (placebo 8 and 4 vs floor 12), so §7-condition-4 fails for every primary contrast; §8's density row governs. Trigger-denominator endogeneity recorded as the design lesson (treatment arms suppress their own trigger's precondition: compiled arms saw 2 and 1 stagnation opportunities). Descriptive (dense warrant channel): side_coach lands under the +0.15 bar in both families (+0.146/+0.142, CIs straddle zero) — the §6.3.9/§6.15/§6.16 insight–action gap extends to point-of-action delivery; compiled_constraint moves macro compliance +0.452/+0.420 (sonnet CI excludes zero) and realizes `answer_accountably` on all 180 compiled warrant-turns with zero smuggled premises, but misses the surface `warrant_cue` on most non-compliant turns and pays ~0.13 coverage@16 vs placebo (coverage guardrail would fail both families; sol safety guardrail would fail; the zero-leak bar fails in every arm including controls). Anomaly: side-coached sonnet is the safest cell in the study (hard-safety 0.90 vs 0.30). Also heals the version-header drift (header stood at 3.0.214 while the v3.0.215 entry existed; now 3.0.216). Provenance:

`POINT-OF-ACTION-COACHING-PREREGISTRATION.md` §11 (execution ledger incl. quarantine + capped-retry census), `exports/tutor-stub-step4-claim-runs/compliance-analysis.json`, `scripts/analyze-step4-compliance.mjs`, board card `tutor-stub-side-coaching-gate`. Development-tier; no DB, rubric, abstract, or headline-N changes; no sections renumbered. With this addendum the preconscious fold is complete.

v3.0.215 (2026-07-17) §6.18 added — the first-draft typed-performance boundary; the preconscious branch folds. Records the V17–V53 series as a four-corner boundary at utterance grain: model-owned form fails the deterministic audit even with two 4/4 blinded causal-fidelity reviews (V51, 0/1); harness-repaired form passes but all six rejected originals were blindly preferred over their delivered repairs (mean −0.60); harness-compiled form passes its home cell (V52 2/2; V53 Tallow 4/4) and rejects the first draft in all three transfer worlds — four failure instances spanning all three owned spans, which the zero-call close-out re-audit reduces to three closed-lexicon surface misses plus one genuine generation defect (verbatim learner echo); recognizer widening is refused as vacuous. Zero safety failures across every recorded call — the boundary is the formal/natural-language interface plus the attribution requirement, not refusal. §7.11 gains the sixth (utterance-grain) substitution line. Records the branch fold: the Phase 6A/6B planner gates close designed-not-run (sensor and selector premises did not survive §6.17 and the state-benchmark closure), the transition/reward-model and copilot-shadow cards close unexecuted, and the discharged successor is the point-of-action coaching gate (§6.17's closing paragraph), which lands here as an addendum when its frozen run seals. Development-tier evidence only; no DB, rubric, abstract, or headline-N changes; no sections renumbered. Provenance: `notes/status/2026-07-16-*`, `notes/status/2026-07-17-first-draft-*`, `PLAN_4_0/2026-07-17-continue-or-fold.md`, board card `tutor-stub-first-draft-series`.

v3.0.214 (2026-07-16) Character-delivery acceptance recorded as apparatus evidence (§7.4.5; scoped in §8.5). Added the diagnostic-collection/strict-verification distinction: diagnostic runs preserve rejected original, repair, and fallback candidates while rolling back failed clue-release transactions and continuing only through a public-safe quarantine response; strict runs retain fail-fast delivery gates. Reported the predeclared V17 four-cell matrix (Foxtrot/false-memory, AI-syllabus/fast, Sealhouse/slow, resistant Hethel/low-agency): all four passed every strict gate, with zero final-delivery audit failures, errors, quarantines, meta-performance turns, role-stage-direction turns, source replacements, or duplicate clue deliveries; host visibility 1.000 throughout; mean configuration realization 0.916–1.000; and two total deterministic fallback turns. Preserved the key negative boundary: proof-path coverage ranged 0.200–1.000 and only one dialogue grounded before the horizon, so this is simulated engineering evidence for safe delivery and host-part realization, not proof completion, population-wide character adaptation, human learning, or deployment readiness. Added two provable-discourse checks against the frozen status report. No abstract, core mechanism, DB, rubric, or headline-N changes.

v3.0.213 (2026-07-14) §6.17 added — the two-family register confirmatory closes the adaptive selector. The frozen 3-policy × 4-profile × n=5 design completed on both original whole-stack families (120 dialogues total), with fixed turn-16 proof-DAG coverage and an in-run profile-discrimination gate. The corrected zero-call, hash-verified final-row analysis gives the strict verdict: no family confirmation, no two-family claim, and no `field`-selector effect. Terra fails the binding manipulation gate (average pairwise cosine 0.812; maximum similarity to diligent $0.912 > 0.90$); its supported hostile-register affective contrast is off-direction and instrument-invalid for confirmation. Sonnet passes the frozen gate (0.645/0.694), but every interaction interval crosses zero; excluding four primary-valid/post-turn-16-truncated rows leaves the null unchanged. The section explicitly bounds Terra-versus-Sonnet as a whole-stack comparison because tutor, automated learner, classifier, and learner-record seams changed together, and retains the single-world, simulated-learner, extraction-seam, n=5, and no-human-learning caveats. Appendix D records both trace-run packages; the research atlas adds a scope-bound module. Reproduction: `scripts/analyze-register-confirmatory-step2.js`, `exports/register-confirmatory-evidence/final/`, and the final analysis manifest. No sections renumbered.

v3.0.212 (2026-07-12) §6.16 added — the Green Room gate arc (coached consolidation into durable memory): Gate 0 PASS, Gate 1 FAIL, arc closed per pre-registration. New section between §6.15 and the Discussion reporting the coached-tutor-training experiment: a judge-tier coach (Opus 4.8) held dialogic notes sessions with a Sonnet-5 tutor on the dramatic-derivation stage (world-005-marrick, proof_skipper, headroom conditions), distilling bankable behavioural notes into a token-budgeted, versioned, frozen-for-scoring prompt book injected into subsequent performances. Coach quality passed owner scoring 5/5; per-note behavioural uptake failed its pre-registered bar (3/17 notes improved, 18% vs 60%; never-issued placebo 0/2), cancelling the transfer and headline gates under the no-tune-and-retry rule. Descriptive positives recorded: unprompted curation-under-budget (half the patches were edits; the book twice compressed while sharpening) and the complete ledgered training biography. Reading: third independent arrival at the §6.15 persisted-advice boundary (with §6.3.10's cumulative-rewrite null), replicating the §6.3.9 insight–action gap at the training level against judge-tier authorship, dialogic elicitation, and curated memory; failure localized to situation-recognition at performance time. Claim-audit corrections applied pre-merge: insight–action cross-reference §7.9→§6.3.9; per-side compliance-pool ranges stated exactly (pre n=1–4, post n=0–6); two paraphrases replaced with literal quotes; 3.3×→3.4×. Successor levers (compile-to-policy, rehearsal, side-coaching) recorded as future pre-registrations. Execution deviations disclosed in-section (sequential rerun after a CLI-timeout discard; one edit-match downgrade; codex→Claude model re-pin with the cross-lineage rejudge documented as owed by any successor positive). Exports-only; no DB writes; no rubric scoring; no human-learning claim. Companion artifacts: GREEN-ROOM-PLAN.md, exports/greenroom-gate{0,1}-2026-07-12/, notes/2026-07-12-greenroom-gate1-diagnosis.md.

v3.0.211 (2026-07-07) §5.12.7 completed — both discriminating tests run; the zero-truth + tail-selection account confirmed from both sides. The two tests v3.0.210 registered as open were executed under a frozen methodology pre-registration

(`SHRINKAGE-PROBES-PREREGISTRATION.md`; limb B one replication instead of two and limb A1 on codex rather than Sonnet, both under a disclosed operator quota directive; two quota interruptions recovered by label-exact trimmed resumes; zero parse deaths). Replication probe: the register-router smoke's exact six arms re-run at identical seeds/config/provider flipped 2 of 6 outcome classes (one in each direction) with fire-counts varying at fixed seeds — within-seed sampling variance dominant, drift unnecessary, smoke-tier n = 3 contrasts formally uninformative. Null-scaling probe: the active null wobbled ($\Delta = -0.70$, U = 58/100); the placebo (zero fires, prompt-identical arms) scaled to $\Delta = -2.80$, 10/10 vs 6/10 grounded, U = 80/100 — past the exact one-sided 0.05 point: a known nothing produced a bar-clearing, promoted-thread-sized signal, confirming that the record's promoted colours were unremarkable draws from the null distribution and that the censored record's apparent one-way decline was produced by scaling only lucky tails. Standing screen-design consequence recorded (smokes = mechanism-availability reads only; motivating colour without arithmetic; outcome claims only at pre-registered tier under shrunken-estimate sizing). Methodology only: no mechanism question reopens; every closed question stays closed. New numbers confined to §5.12.7. No DB writes, no rubric scoring, no human-learning claim.

v3.0.210 (2026-07-06) §6.13.19 added — classify-then-route (the register router); §5.12.7 gains the second sized application and the record-level reading. New bounded subsection reports the programme's first up-front learner-message classification coupled to the proof DAG, with the tutor's per-turn stance register as the routed channel (operator design). Phase A audits a deterministic text-only sensor against ENGINE-record labels (the derive channel, never text): 142 runs / 2,930 learner turns, three diagnosed iterations (mirror term saturates 58.9% of learner lines → residual class; nested supports → tf-weighting; goal region = superset attractor → excluded), 12.3% → **87.2% chain-level**, past the pre-stated 80% license; zero formal mirror assertions across all 142 runs. Phase B routes repair (regressed-chain message) / confront (mirror-residual derivable incompatible partner) / didactic (no block; no-fire runs byte-identical, fingerprint-gated); gates 8/8. The powered contrast (frozen 3799a951, n = 20/arm sized under §5.12.7's rule, primes 433–487, three external stops disclosed with label-exact resumes) returned **not confirmed — the motivating colour reversed**: smoke $\Delta = -4.67$ (3/3 vs 1/3 grounded) became $\Delta = +0.45$ against the router (U = 238/400; direction against in both strata); guardrails 8/9 with the one breach as the finding's texture (the repair stance measurably slowed the release calendar in the resistant world, 8.00 vs 9.00, while outcomes stayed identical at 16/20 grounded per arm); mechanism flawless at scale (15 evidenced fires, events consistent on 419/419 router turns, leaks 0; confront never licensed, partnerDerivable 8/419). Mechanism codas: world-020's single authored incompatible pair opens the confront window on real runs (6–9 turns/run) and the confront register fired twice on the full conjunction with leaks 0 and zero code change; the mechanism transfers to codex whole (6 evidenced fires, leaks 0, wash colour; mirror-residual ~4× higher on codex learner text, descriptive). §5.12.7's new paragraph records the sign reversal as the record's first, applies the audit's model at the record level (one surviving primary against a field of shrinkage/voids/reversals is what a family of near-zero true effects looks like under tail-selection on near-binary outcomes with a nulls-never-re-run censored record — expected, not anomalous), and registers the two open, executable discriminating tests (scale a sample of null smokes; within-window seed-exact re-runs, noting the flash smoke already inverted at its own seeds). New §6 numbers confined to §6.13.19; new §5 numbers confined to §5.12.7; no sections renumbered. No DB writes, no rubric scoring, no proof-control behavior change, no human-learning claim.

v3.0.209 (2026-07-06) §6.15 added — the adversarial game (agon): information format, not information, is the bottleneck. Reports the two-day agon arc (`worktree-agon-game`; design + staged pre-registrations in `AGON-GAME-PLAN.md`): the engagement played as a formal game against a deterministic referee that manufactures latent state (five single-use §6.8-family dodges, well-posedness rules, keyed demonstrations, leak rule; **no LLM judge in the scoring loop**; move-level consumption metrics by deterministic replay). Headline: raw-state disclosure LOSES to blind play (5.00±2.00 vs 6.75±0.50) while the identical information as an action projection beats both, replicated on two boards (7.50±0.58 v0; 9.75±0.50 XL; opportunity misses 0.00 vs 1.50 on both) — the §6.8.6/§6.13.18/§6.13.11 motifs converging into one measured boundary with blind and raw-state controls in the same design. Also recorded: the taint rule deterred answer-telling without ever firing (0 events / 253 eligible tutor messages); characterological collapse (26/28 first demonstrations arrive as self-generated leaks after budget exhaustion — a bound on persona-based resistance instruments, cross-referenced to §8.1); the playbook probe's asymmetric failure (state-shaped memory inert at opp-miss 2.00 everywhere; action-shaped memory consumed with preconditions stripped — four ill-posed probes over three ignored REVISEs, the §6.13.16 negative-transfer motif in its starkest form), yielding the working formula *action-shaped and freshly conditioned*; and the weak-tutor floor (Haiku blind: 0 vs 8.50, rut + probe discipline broken in both directions: 3 ill-posed probes, 7 legal windows untaken), with the brief-rescue question recorded as undecided (CLI instability; deviations disclosed). All descriptive at $n = 4$/arm/board; failed frozen predictions (P10, P13, P15/P16) reported as such; exports-only, no DB writes; apparatus + ledgers + 29 tests committed with the section. No sections renumbered.

v3.0.208 (2026-07-06) §6.13.18 completed — the content-compulsion exit tested to closure; §5.12.7 gains its first sized application. The interpretive close's second exit (compulsion aimed at the content of the incumbent strategy) ran its full pre-registered sequence: the pre-stated smoke produced the family's first observed strategy-change-under-adversity (one refusal, one switch-and-repair, grounded; n = 1); the powered contrast (frozen 216ddb78, 24/24 Sonnet at the calibrated dose) returned **not confirmed — direction-without-power** on the family's strongest table (Δ = −2.17 with the family's first both-worlds direction pass, U = 50/144 vs 42, guardrails 9/9, mechanism 7 refusals / 6 switches / 1 defense with all refusal-involved runs grounded); and the promotion draw (frozen cb000e21, 40/40 at n = 20/arm — the first confirmatory sized under §5.12.7's shrunken-estimate rule) returned **not confirmed — question closed at this dose/model pair** (Δ shrank −2.17 → −0.50, ~4× and below even the shrunken sizing figure; U = 190/400; direction split marrick

+0.60 / resistant −1.60; guardrails 9/9; repair-latency secondary flipped sign; T* saturates at 22 on marrick-resistant, so rate-not-speed is the measurable channel there; mechanism replicated at scale, 6 refusals / 3 defended / 3 switched, all refusal-involved runs grounded). A mechanism coda records the post-closure exploration smokes (mechanism studies only, per the frozen consequences): codex defends 3/3 valid resolutions under a stall-span trigger where Sonnet split 3/3; a prosecutor-authored refusal body is defended 2/2 where the bare template got 0/2 (a new `unresolved` outcome value catches non-answering resolutions); the learner-side mirror refusal is vacuous at the assertion channel (zero formal mirror assertions across six runs against 12–20 mirror-naming dialogue lines per run; leaks 0). §5.12.7 gains one paragraph: the promotion as the rule's first field application (closure instead of a near-miss third draw) plus a named-secondary shrinkage specimen (the repair-latency sign flip). New §6 numbers confined to §6.13.18; new §5 numbers confined to §5.12.7. No DB writes, no rubric scoring, no proof-control behavior change, no human-learning claim.

v3.0.207 (2026-07-06) §6.7/§8.9 close the negative-register measurement story — judge repair confirms rather than overturns the null. The register-rubric scores behind the cells 196–198 smokes came from `openrouter.gpt-mini`, later shown to be numerically non-discriminating on this rubric: it stamped an identical 48.0 overall with an identical (3, 2, 3) social triad on 35 of 49 register-rubric slices regardless of content, which is why the earlier smokes did not show the expected recognition-cost or depressed-uptake pattern. All 58 register-rubric slices on the three surviving negative-register runs (`eval-2026-07-02-e7b15809`, `-7e461a5c`, `-5c4d52e6`; the earlier `-cfed3b13` and `-e511f92c` smokes remain confirmed lost from every database) were re-scored under `claude-code.sonnet-5` via the CLI subscription bridge (0 failures, $0 metered spend; gpt-mini pass archived first at `exports/register-scores-gptmini-archive.json`). The re-score confirms the null: assigned-arm means rise under the new judge (ironic_challenge 83.8, sarcastic_challenge 79.6, face_threat_challenge 76.0, charismatic_challenge comparison 68.1), the rise holds on the stance-faithful-row subset too, and the judge's discrimination is evidenced in-run (the dataset's single guardrail-flagged corrosive slice scores 32.5, inside the hand-authored corrosive-exemplar band of 29–52, while faithful slices score 62–100). The earlier face-threat absolute-cost reading, drawn from the now-lost `e511f92c`, does not replicate — gpt-mini had given ~2.3 face-repair to all three negative arms alike, so the apparent face-threat-specific depression was compression artifact; what survives is relative (face_threat lowest-scoring negative arm, only guardrail-flagged slice), and the arm stays simulated-only. Methods caveat: the licensing exemplar evidence used `openrouter.sonnet-5`; the rescore uses `claude-code.sonnet-5` — same Sonnet model class, different routing path. §8.9 updated: register-rubric conclusions rest on sonnet-class judging, archived gpt-mini register figures are void, and sonnet-class judging from the first row is now the standing rule for future negative-register work. Full comparison: `exports/register-rescore-sonnet5.{md,json}`; workplan record: `workplan/items/register-taxonomy-negative-registers.md` (2026-07-05 entry). No sections renumbered; no new claim beyond the cited rescore.

v3.0.206 (2026-07-05) §7.11 extended — the de-substitution confirmatory result

folds in. Added a closing paragraph block to §7.11 recording the DAG-pinned-learner de-substitution arc (`workplan/items/dag-pinned-learner-desubstitution.md`): built to test whether the section's substitution reading depended on a non-discriminating learner (SFS=0.000, §8.1) by pinning a learner to a formal belief-DAG interior held in character by a criterial drift gate (false-yield 0.00 throughout). Three successive matrices repaired the instrument's drift-gate exhaustion from 38% (`eval-2026-07-03-a3cfbe14`, FROZEN_INSTRUMENT_FAILURE) to 3.3% (`eval-2026-07-04-c689cf3a`, a directional lead one row short of the pre-registered bar) to 0.83% (`eval-2026-07-04-0d59e4c8`, 120 rows). The frozen confirmatory pre-registration (`notes/2026-07-04-desubstitution-confirmatory-prereg.md`) then found both contrasts (multi-strategy router vs. fixed-strategy floor; kernel vs. router) DISSOLVED at gap zero (4/40 every arm) against thresholds of gap $\geq$7/40 real, 3/40 dissolved — the generating lead did not replicate at 2x the sample, and the pre-registered stop-rule read it correctly as noise. Reading: substitution survives a learner engineered to yield only on genuine content delivery, strengthening rather than qualifying the synthesis; the untested axis is now learner populations, not harder individual learners. A methods clause cross-references §8.1: per-turn characterological gating does not compound to dialogue scale without a turn-decaying contract. No sections renumbered; no new claim beyond the cited runs.

v3.0.205 (2026-07-05) §6.13.18 added — the strategy-layer family's interpretive close. Records the two final smokes (binding at the calibrated dose: plan field unanswered 0/12 with tolerant matching, extending codex's 0/95; forced-choice: 16/16 exercised, all via one retry, zero delegations, and the decisive raws — three consecutive non-first picks after a mid-drama double regression = a real preference under adversity, whose elicited-and-bound form went aporia while the seed-matched baseline grounded) and closes interpretively per the §6.14 pattern: the managed object is a disposition (momentum along the frontier: progress-following, repair-averse, no counterfactual structure), self-report of it is construction-not-retrieval (binding a self-projection is tautological; explicitness froze the press-on bias), malleability is exogenous only (signal, constraint, interface compulsion — never self-address), and the impasse is structural (no hierarchy of logic creates a vantage outside the policy). Exits recorded: exogenous evidence, content-targeted compulsion (final pre-stated smoke pending at time of writing), a different-and-stronger outer mind. Interpretive synthesis; no new empirical claim beyond the cited smokes' recorded reads. No DB writes, no rubric scoring, no human-learning claim.

v3.0.204 (2026-07-05) §6.13.17 extended (the mid-tier close) and §5.12.7 added (the shrinkage audit). The mid-tier lemma-map question, opened by a propose-go Sonnet smoke, resolved in three recorded steps: (1) the first powered attempt at the codex-calibrated dose 0.35 returned **matrix void, floor-blind** under its own frozen rule (24/24 at cap, both arms, both worlds); (2) a pre-stated dose ladder measured Sonnet's decay tolerance as a narrow shelf (no decay 3/3 grounded; 0.15 = 0/3; 0.08 = band on both worlds) and produced a calibrated condition set for ~12 screening runs; (3) the calibrated contrast (17/24 grounded — a real regime) returned **not confirmed** (direction split [-1, +1] mirroring the codex display confirmatory's shape on an independent model and draw; U = 65/144; guardrails 9/9), closing the mid-tier question MEASURED and extending the lemma-layer closure to two tiers. New §5.12.7 reports the exploratory-to-confirmatory shrinkage audit (deterministic bootstrap over frozen reports): the operator-observed pattern is real and fully accounted for by selection plus power — winner's-curse selection gap 36.0 vs 53.2 expected U, bar-at-the-estimate confirmation odds 0.615, between-draw swings of up to two grounded outcomes at n = 6, one of four advanced signals surviving its primary (V2b, the large-effect counterexample), with the nulls-never-re-run asymmetry and the fresh-prime/drift-testability trade recorded. New §6 numbers confined to §6.13.17; §5 numbers confined to §5.12.7. No DB writes, no rubric scoring, no proof-control behavior change, no human-learning claim.

v3.0.203 (2026-07-04) §6.13.17 added — the strategy-layer family completed: cross-model close and the lemma layer. New bounded subsection records four frozen pre-registrations and their verdicts: (1) the cross-model plan-mode interaction (codex arm reused whole and declared; 24 new gemini-flash runs) **not confirmed and reversed** — outer-loop value was more favorable on the STRONG model (U = 96/144 on the wrong side of the null; the motivating smoke inverted at its own seeds; flash demanded 4x the corrections with worse outcomes) — so §6.13.16's closure gains two-model scope; (2) the lemma layer (operator's step-back: a chainer-computed lemma-grain proof DAG, criterial clearance, display vs binding), Gate-0 catalog audit + world-slot probe procedure (fengate/sealhouse fail identically, edmund ineligible; consequence fired) + `world-019-marrick-resistant` authored under the hethel recipe and screen-passed; (3) the three-arm contrast **not confirmed** on bound-vs-baseline (U = 61/144; zero block events — release deficit = early deaths; choice channel inert: the model never echoes labels verbatim, all 95 choices harness-fallback) with the display arm's descriptive margins the family's largest (T* U = 40.5, grounded 6/12 vs 1/12); (4) the same-day display-primary confirmatory **not confirmed** (direction split; U = 67/144; pooled delta -2.25 -> -0.25; guardrails 9/9 clean) with the decomposition recorded — the baseline moved more across draws than the treatment — and one twice-replicating channel (aporia-like improvement) recorded as a narrow unclaimed hypothesis. The lemma line closes whole per its frozen consequences; the exploratory-confirmatory shrinkage audit is registered as forthcoming methodology work (an observation, not a claim). New §6 numbers confined to §6.13.17. No DB writes, no rubric scoring, no proof-control behavior change, no human-learning claim.

v3.0.202 (2026-07-04) §6.13.16 closed: the plan-mode stock-take contrast folded in; the outer-loop line closes whole. The course-changing variant (dialogic between-scene stock-take, diagnostic charter, off the stage clock, frame replacement without conformance audits) ran under its frozen pre-registration (`PLAN-MODE-STOCKTAKE-PREREGISTRATION.md`; 24/24 codex runs at concurrency 3 with pair-interleaved arms; implementation gates 34/34 incl. the L7 suite; tests 21/21). Verdict **not confirmed on power alone**: primary T* directionally better in both worlds (25.42 vs 26.92) but U = 55.5/144 (one-sided p ~= 0.17) misses the pre-registered criterion; guardrails 0-of-6 failed — the family's first apparatus with no negative transfer. Descriptive record: grounded 0.42 vs 0.17 (both worlds), aporia improved, 15/15 demanded corrections answered, repair latency worsened (10.95 vs 8.71) — the exact inverse trade of the commitment ledger. §6.13.16 gains its closing paragraph; per the frozen consequences both outer-loop variants are recorded non-promotable and the line closes (endpoint swaps forbidden; any reopening is an operator-level decision under a new pre-registration). New §6 numbers confined to §6.13.16. No DB writes, no rubric scoring, no proof-control behavior change, no human-learning claim.

v3.0.201 (2026-07-04) §6.13.16 confirmatory replication folded in: V2b replicates on its primary endpoint and fails promotion on negative transfer. The frozen two-arm confirmatory pre-registration (`STRATEGY-LEDGER-V2B-CONFIRMATORY-PREREGISTRATION.` n = 12/arm, seed-pair-matched repeats, sole primary endpoint repair latency, three-part promotion bar) ran 24/24 on codex. Bars 1–2 pass (repair latency 8.02 vs 9.24, lower in both worlds, U = 37.5/144, one-sided p ~= 0.023); bar 3 fails with its own both-worlds signals (grounded 0.25 vs 0.50, T* 27.17 vs 25.58, aporia-like 0.75 vs 0.50, plus two per-world guardrail breaches). Verdict: **not confirmed** — the commit/audit loop reliably speeds repair and reliably costs outcomes; the pilot signal stands as pilot-only, and the follow-up line (composition registration, localization, learner-mirror arm) closed per the pre-registration's recorded consequences. §6.13.16 gains the confirmatory paragraph; its synthesis clause is revised (local helpfulness does not exempt an overlay from the negative-transfer test; the ledger machinery is validated research instrumentation, not a promotable overlay); the status paragraph and artifact list extended. New §6 numbers confined to §6.13.16; no other body number changes. No DB writes, no rubric scoring, no proof-control behavior change, no human-learning claim.

v3.0.200 (2026-07-03) §6.13.16 added — the strategy ledger's two pre-registered pilots folded in as one bounded subsection. New subsection records the scene-scoped commit/audit machinery (the §6.13.9 plot/audit pattern one scope down, learner mirror included, conduct-only with fingerprint-pinned proof control) and its two pre-registered pilot results: Phase 3's null-with-unexercised-levers on schedule-solvable worlds (24 runs, 15/15 guardrails, outcome ceiling + 0/24 register switches), and the v2 binding-conditions pilot (18 codex runs behind a headroom gate that fired once and forced a recorded redesign to `hethel-resistant` + mutation decay): **V2b positive — the v1 commit/audit ledger improves repair latency (8.06 vs 11.23 turns, both worlds, U=8/36, 61% vs 50% of slips repaired, guardrails clean); V2a negative — the v2 trialling superstructure erases the benefit (+4.23 turns, blocks halved, two faithful ironic-stance runs ending in ungrounded assertions, one release guardrail failure blocking positive trialling readings).** Machinery validation recorded (30/30 acceptance gates; review coverage 1.00; stance fidelity 19/27 faithful; zero invalid attacks; zero guard overrides); the two-gate fidelity-before-effectiveness discipline imported from §6.7's negative-register arc. The §6.13 unified-reading coda now closes the section after the new subsection. New §6 numbers are confined to §6.13.16; no other body number changes. Provenance: `STRATEGY-LEDGER-PHASE3-PREREGISTRATION.md`, `STRATEGY-LEDGER-V2-PREREGISTRATION.md`, `LAYERED-DECISION-LOOPS-PLAN.md`, tracked contrast reports under `exports/dramatic-derivation/strategy-ledger/`. No DB writes, no rubric scoring, no proof-control behavior change, no human-learning claim.

v3.0.199 (2026-07-03) New §7.11: interpretive synthesis of the substitution family. Added a discussion subsection unifying the composition result (§6.14) with universal substitution (§6.4), the state-richness reversal (§6.8.6), the orientation-family result (§7.9), and the programme's ToM-redundancy nulls under one reading: prompt- and architecture-level mechanisms act through a single bottleneck (shaping attention over context the model can already see) and therefore converge on the model's existing competence, while the interventions that kept adding value (§6.13 pacing guards, §6.7 resistance gates) inject machine-checkable signal the model lacks. Design rule stated: instructions converge; only new signal accumulates. Explicitly framed as interpretation over established results — no new numbers, no new empirical claim; the constant-effect-magnitude observation it explains remains listed as an anomaly in §9.

v3.0.198 (2026-07-03) New §6.14: blueprint composition test. Added the pre-registered composition study: the paper's validated mechanisms declared as registry modules (`config/tutor-blueprint.yaml`) and composed on the id-director chassis as cell_199 (kernel: recognition orientation + drama-pad memory + cell-193 register composite) and cell_200 (kernel + Plan 2.x action contracts extracted as per-turn middleware). Gate + repeats + decision runs (`eval-2026-07-02-b0be0524`, `eval-2026-07-02-b4ccb58d`, `eval-2026-07-02-fe9404d2`, `eval-2026-07-02-a4260aa3`, `eval-2026-07-02-6547750c`, `eval-2026-07-02-ad0b5a8b`, `eval-2026-07-03-f877e477`, `eval-2026-07-03-4deae92a`; Codex-only stack; pre-registrations in `notes/2026-07-02-blueprint-composition-plan.md`). Results: sub-additivity confirmed everywhere measured (kernel $\approx$ full on both suites; action contracts add nothing detectable); no straitjacket regression; the gate's apparent local cost of recognition orientation on resistance repair was run to a pre-registered decision at n=6 per cell×scenario and **dissolved exactly at the frozen boundary** (positives 30/30 vs 24/30, gap 6/30 vs $\geq$9 required; strict candidates 18/30 vs 17/30), leaving a one-directional frustration-subtype caveat (6/6 vs 3/6 carry-through) invisible to whole-dialogue v2.2 in both directions. Structural findings recorded: conventional superego structurally excluded from the id-director chassis; §6.13 pacing guard not portable; §6.6.11 cross-session pad claim does not transfer (drama pad only on this chassis); trap-suite strict-shift non-evaluable on the id-director adapter. Design consequence stated in §6.14: the evidence licenses a small-kernel tutor, not mechanism accumulation.

v3.0.197 (2026-07-03) Negative-register held-out target check documented. Updated §6.7 and §8.9 with paid held-out check `eval-2026-07-02-5c4d52e6` after the cue-repair canary. The run covered cells 196–198 across the two targets not sampled by `eval-2026-07-02-7e461a5c`: frustration and question-flood. Generation completed 6/6 rows, tutor-only v2.2 scoring completed 6/6 (mean tutor score 87.8), and register-rubric scoring completed 9/9 slices. The stance gate classified 6/6 assigned negative-register rows as faithful evidence, with 0 noncompliance exclusions and 0 invalid person-attack violations; local outcomes were 3/6 strict candidates and 4/6 positive. This closes treatment-fidelity coverage across all five controlled resistance targets when read with the prior 9-row canary, but it remains a measurement-development check rather than a scaled effect or safety claim. The full five-target negative-arm grid is now optional for consolidated effect estimation, not required merely to prove that the repaired stance cue contract can instantiate the assigned arms. Artifacts: `exports/charisma-desire-breakthrough-matrix-summary.md`, `exports/charisma-desire-breakthrough-matrix.json`.

v3.0.196 (2026-07-03) Negative-register cue-repair canary documented. Updated §6.7 and §8.9 after repair commit `444cb304` made visible `stance_fidelity_cues` mandatory in the negative-register stance contract. The paid 9-row canary `eval-2026-07-02-7e461a5c` covered cells 196–198 across boredom, irrelevance, and rote-parroting targets. Generation completed 9/9 rows, tutor-only v2.2 scoring completed 9/9 (mean tutor score 90.1), and register-rubric scoring completed 14/14 slices. The first matrix pass exposed two no-paid gate bugs: registry cue phrases were not counted as marker evidence, and substring matching falsely treated "not stupidity" as the banned insult "stupid." After fixing the gate to use registry cues and word-bounded forbidden phrase matching, the same scored rows classified as 9/9 faithful evidence, 0 noncompliance exclusions, and 0 invalid violations. This documents treatment-fidelity repair on a small simulated canary only; it does not license a scaled negative-register effect or safety claim. Artifacts: `exports/charisma-desire-breakthrough-matrix-summary.md`, `exports/charisma-desire-breakthrough-matrix.json`.

v3.0.195 (2026-07-02) Prospective negative-register stance gate applied. Updated §6.7 and §8.9 with fresh stance-gated smoke `eval-2026-07-02-e7b15809` (cells 193/196/197/198 across the five controlled resistance-breakthrough scenarios). Generation completed 20/20 rows, tutor-only v2.2 scoring completed 20/20 (mean tutor score 91.1), and generic register-rubric scoring completed 35/35 slices. The breakthrough matrix found 14/20 candidate breakthroughs and 18/20 positive local outcomes, but the new stance gate counted only 5/15 assigned negative-register rows as faithful evidence, excluded 10/15 as weak or warm-in-costume noncompliance, and found 0/15 invalid person-attack violations. This tightens the claim: the current Codex generator mostly fails negative-register treatment fidelity, so cells 196–198 remain measurement-development instrumentation requiring assigned-arm versus faithful-arm reporting before any scaled effect estimate. Artifacts: `exports/charisma-desire-breakthrough-matrix-summary.md`, `exports/charisma-desire-breakthrough-matrix.json`.

v3.0.194 (2026-07-02) Negative-register exemplar discrimination documented. Added a bounded §6.7 methodological addendum for the post-hoc negative-register arc (cells 196–198: `ironic_challenge`, `sarcastic_challenge`, and simulated-only `face_threat_challenge`) and a matching §8.9 limitation. The generated negative-register smokes did not show the expected recognition-cost/depressed-uptake pattern, but the no-generation-confound hand-authored fixture (`config/register-exemplars/corrosive-sarcasm.yaml`) separated judge gullibility from generation warmth: after deterministic guardrail caps, the register rubric caught 3/3 known-corrosive slices, v2.2 `recognition_quality` caught 3/3, and controls had 0/2 false positives. The paper now frames this as measurement-development evidence: the judges are not globally blind to corrosive sarcasm when it is actually present, and negative-register rows need a tutor-side stance-fidelity gate before counting as evidence for the assigned register. The branch implements that gate as faithful evidence vs noncompliance exclusion vs invalid corrosive violation; a fresh or restored negative-register smoke is still needed to apply it prospectively. No claim is made that sarcasm is pedagogically safe, deployable, or human-facing. Artifact: `exports/negative-register-corrosive-exemplar-results.md`.

v3.0.193 (2026-07-02) Plan 3 DAG-SFS proof-grounded audit integrated. Added `scripts/run-plan3-dag-sfs-audit.js`, `npm run probe:plan3-dag-sfs`, and a focused scorer test. This is the proof-DAG analogue of SFS: six invented formal micro-DAGs, five matched replicates each, and four feedback conditions (targeted withheld-edge release, wrong-edge release, generic feedback, nonsense feedback), for 120 Codex learner retries. The scoring rule is stricter than answer flipping: a row counts only when the learner marks `proved`, states the target conclusion, and cites the exact released target edge ID in the derivation. Result: targeted proof-grounded recovery 30/30 (1.000); negative false-grounded recovery 0/90 (0.000); DAG-SFS=1.000; paired CI [1.000, 1.000]; parse errors 0. This does not rehabilitate unconstrained roleplay outcomes; it identifies the gold-standard synthetic-harness condition under which the simulated learner becomes a discriminator: harness-owned public evidence, invented formal tokens, and exact proof-edge scoring. Updated §8.1 and Plan 3 notes accordingly. Artifact: `exports/plan3-dag-sfs-audit/{dag-sfs-matched-feedback.json,dag-sfs-matched-feedback.md}`.

v3.0.192 (2026-07-02) Plan 3 SFS matched-feedback audit integrated. Added `scripts/run-plan3-sfs-audit.js` and `npm run probe:plan3-sfs` to materialize the fresh matched targeted/mismatched/generic feedback corpus that v3.0.191 left scaffolded. The completed Codex run covers 5 seeded fractions-misconception families × 10 replicates × 3 feedback conditions = 150 learner retries, with 0 invalid choices. Targeted, mismatched, and generic correct-flip rates are all 1.000, so SFS=0.000, false-flip=1.000, and paired SFS CI is [0.000, 0.000]. The result confirms the expected near-zero selectivity/high-sycophancy boundary: the prompted learner changes to the correct answer after irrelevant or generic feedback as readily as after targeted feedback. Updated §8.1, the Plan 3 notes, and the synthetic-closure harness so the SFS artifact is carried as a completed synthetic-learner validity bound, not a tutor-mechanism result and not human-learning evidence. Artifacts: `exports/plan3-sfs-audit/{sfs-matched-feedback.json,sfs-matched-feedback.md}` plus refreshed `exports/plan3-synthetic-closure-audits/{summary.json,report.md}`.

v3.0.191 (2026-07-02) Plan 3 no-cost synthetic-closure audits integrated. Ran the offline audit harness over existing artifacts only (0 LLM calls): 16,270 DB rows, 183,104 DB/log recommendation texts, 13,395 dialogue records, 63,707 tutor turns, 58,686 learner turns, 47,542 tutor log files, 64 dramatic-derivation result files, and 58 detector arms. Hard gates pass: answer-delivery/public-hidden-label risk 300/246,811 messages (0.1%; recognition family 39/50,686, 0.1%, gate 10%); help-ladder compliance 6,928/7,804 after struggle/false mastery (88.8%, gate $\geq$80%); unsupported mastery/advancement closure 38/265 assessable closure claims (14.3%, gate 15%); deterministic first-error localization 29/29 non-grounding detector failure arms (100%, gate $\geq$90%). Learner-fidelity metrics are proxy-only (8 false-mastery turns, 9,499 flip-language turns, 13,192 metacognitive turns over 58,686 learner turns); SFS remains scaffolded-not-run pending matched feedback generations; IRT remains blocked by placeholder item-bank stubs. Updated §8.1 and Plan 3 notes to treat the hard offline gates as closed and the remaining synthetic-validity work as item-bank/SFS data creation rather than another mechanism cell. Artifact: `exports/plan3-synthetic-closure-audits/{summary.json,report.md}`.

v3.0.190 (2026-07-02) Plan 3 optional probes integrated as synthetic-learner validity bounds. Added reusable Plan 3 probe runners for outcome-selected best-of-K and ESS learner 2$\times$2. The outcome-selected best-of-K bounded run on the hard-transfer lattice is null/inconclusive (single-K1, opinion-best-K, and outcome-selected all mean gain 0.250; outcome-minus-opinion 0.000; invalid plans 0; 9 Codex calls). The larger ESS follow-up instantiates the planned N$\approx$120 shape as 30 sessions $\times$ 4 cells = 120 cell observations over 10 active-misconception pair states, with 60 Codex tutor-plan calls and invalid plans 0. Recognition does not lift the constrained endpoint (base+ESS 0.252, recognition+ESS 0.250, $\Delta = -0.002$; paired mean -0.003, 95% CI [-0.013, 0.008], positive 2/30), while loose roleplay modestly over-credits tutoring relative to ESS (paired inflation 0.020, 95% CI [0.015, 0.024]). Updated §6.12.5 and §8.1 to treat ESS as an internal measurement boundary, not a new recognition-effect claim. Artifacts: `exports/plan3-optional-probes/outcome-selected-best-of-k.{json,md}` and `exports/plan3-optional-probes/ess-learner-2x2-large-n120-codex.{json,md}`.

v3.0.189 (2026-07-02) §6.7/§8.9 charisma/desire role-isolation closeout and question-flood subtype gate documented. Updated the id-director desire-for-recognition follow-up from a partial role-isolation read to the completed six-arm matrix. The local Codex/Sonnet reference completes 10/10 rows but is modest (6/10 positive, 5/10 strict); the GLM tutor/id with Codex learner arm remains locally usable (8/10 positive, 6/10 strict); the tutor-fixed GLM learner and full-GLM arms lose post-turn completion or target stability; and both scripted controls show public register production when dynamic-learner drift is removed. The resulting boundary is `DYNAMIC_LEARNER_COMPLETION_AND_TARGET_DRIFT_BOUNDARY`, not tutor/id public-register failure. Added the separate question-flood gate (`eval-2026-07-02-67be317c`) after a reporter commitment-wording regression fix: cell 192 is promoted only for question-flood commitment-probe handling (2/2 answer-first usable commitments, 0 reopened/residual floods) against cell 188/190 comparators. No human-learning claim, deployment claim, full-GLM robustness claim, scripted-control learner-outcome claim, or general charismatic-tutoring effect estimate is added.

v3.0.188 (2026-07-02) §8.1 simulated-learner validity literature caveat added. Added the PLAN_3_0 literature screen's external simulated-student validity context to the limitations section: prompted simulators' fidelity limits, near-zero selective-flip behavior, and model-dependent IRT grade alignment are now named with citations. The paragraph explicitly connects these external risks to existing internal caveats (§6.10, §6.12.5, §6.13.7) and states that fidelity/flip-selectivity/ability-placement bounds have not yet been quantified for this project's learner configurations. No project result, run ID, N-count, or mechanism claim changed.

v3.0.187 (2026-07-02) §2 post-LLM tutoring literature positioning added. Added a short literature-positioning paragraph connecting the PLAN_3_0 dynamic-adaptation screen to the manuscript's mechanism framing: LearnLM, Tutor CoPilot, Kestin et al.'s Scientific Reports RCT, and Learning to Prompt are cited as external context consistent with the paper's orientation/substitution/null pattern. Added matching bibliography entries. No project run ID, N-count, rubric score, or apparatus claim changed; the paragraph is explicitly framed as literature positioning, not a new empirical result from this repo.

v3.0.186 (2026-07-01) §9 future-work cleanup. Revised "What comes next" so it no longer describes mechanism isolation and superego-taxonomy coding as future work after §6.4.2, §6.4.2.1, and §5.3 already report direct M1/M2 isolation and the automated 500-critique taxonomy apparatus. The remaining agenda is now framed around human expert coding, human learner validation, model-boundary and role-isolation tests, prompt-optimisation thresholds, and white-box mediator analysis. No empirical claim, run ID, N-count, or statistical result changed.

v3.0.185 (2026-07-01) §6.7/§8.9 charisma/desire partial role-isolation boundary added. After Claude Code subscription access was disabled for the planned Claude id role, added an OpenRouter Sonnet 5 alias and rerouted the Codex tutor/id arms through `anthropic/claude-sonnet-5`. The Sonnet-5-backed Codex tutor/id + Codex learner baseline completed 10/10 rows (`eval-2026-07-01-5dad2e60`, 7/10 positive local outcomes, required/forbidden validation 9/10 and 10/10). Holding that tutor/id stack fixed while swapping only the dynamic learner to GLM 5.2 completed 5/10 rows before stalling on the second irrelevance draw (`eval-2026-07-01-dff9f159`, interrupted; one-row backfill `eval-2026-07-01-ed9f7288` also stalled with zero rows). The completed GLM-learner rows show clipping, empty-content/length behavior, and missing-post-turn local classifications. Claim boundary: this is partial role isolation suggesting dynamic-learner/runtime drift as a live failure source; it is not a completed robustness matrix, not a tutor/id-side isolation result, not a scripted-control result, and not a human-learning or general charismatic-tutoring claim.

v3.0.184 (2026-07-01) Appendix E.7 evidence-boundary closeout. Clarified the bounded character-development pilot's evidence pointer so the manuscript cites the tracked evidence manifest as the durable source for the ignored/local `exports/character-development/` bundle. No empirical claim, number, or scope boundary changed: the Appendix E.7 claim remains the narrow L3 self-recognition contrast, with no §6 body number, human-learning claim, real-interiority claim, deployment claim, or robust magnitude claim.

v3.0.183 (2026-07-01) §6.13.15 human/hybrid handoff probe folded in as a

bounded advisory deployment-risk claim. After PR #73 merged the public-only handoff probe, §6.13.15 now records the result: `HumanHandoffState` passed 8/8 deterministic controls with 0 public-only failures and 0 non-advisory rows, producing 4 human-followup and 2 optional-review recommendations. The claim is local advisory deployment-risk classification only: public signals can trigger conservative human/hybrid review recommendations, with no learner routing, proof-control behavior change, replacement of proof-control logs, deployed safety coverage, or human-learning evidence. The CI lint failure on PR #73 was format-check drift in `services/dramaticDerivation/humanHandoff.js`; the file was formatted and checks rerun.

v3.0.182 (2026-07-01) §6.13.15 task/session held-out gate folded in as a bounded advisory sequencing claim. After PR #72 merged the task-loop held-out artifact gate, §6.13.15 now records the result: a public-only `TaskMasteryState` selector matched the expected next-task label on 12/12 frozen derivation artifacts while fixed progression matched 2/12 (accuracy delta 0.833), with 0 public-only audit failures and 0 proof-control fingerprint drift rows. The claim is deliberately lower-authority than proof control: local advisory task/session sequencing on frozen artifacts, not runtime task assignment, proof-control promotion, deployment readiness, or human-learning evidence. Added the ignored export pointer `exports/dramatic-derivation/layered-adaptation/taskloop-heldout-gate-report.md`; no DB writes, no rubric scoring, no proof-control behavior change.

v3.0.181 (2026-07-01) Character-DAG drama transfer-specificity family robustness note. Added a bounded §6.8.9 follow-up after the compressed-state control exposed a generic-transfer loophole. Transfer scenes now require the learner to name the concrete public prior check carried by matched character state; compressed and stale controls withhold that check. The real generated-learner `state_dependent_transfer` family matrix (`exports/character-dag-drama-framework-transfer-specificity-real-family/`, `codex` provider, `model=auto`, two seeds × eight scenes × seven arms across four fixture families) passes overall robustness under `character-dag-drama-observer.v0.5`: full first-response success is 15/16, 14/16, 14/16, and 13/16 across `base`, `ratio_series`, `definition_boundary`, and `causal_identification`; policy-only is 6/16, 3/16, 5/16, and 4/16; shuffled-state is 7/16, 6/16, 6/16, and 5/16. Full transfer first-response closure is 6/6, 5/6, 6/6, and 5/6; policy-only, shuffled, stale, overconfident, and compressed controls all remain at 0/6 transfer in the three non-base families, with policy/shuffled also 0/6 in base. Zero target-label leaks and zero public theory/process leaks. Claim boundary unchanged: exploratory synthetic apparatus result only; no human learning, no deployed reliability, and no real interior character-development claim.

v3.0.180 (2026-06-30) Character-DAG drama strict robustness update. Replaced the initial seeds-2 Character-DAG drama contrast in §6.8.9 with the stricter real generated-learner robustness run (`exports/character-dag-drama-framework-robustness-policy-rep` `codex` provider, `model=auto`, three seeds x eight scenes x four perturbations). All four strict perturbations pass: full first-response success is 21/24 in every perturbation;

policy-only is 16/24, 12/24, 17/24, and 11/24; shuffled-state is 14/24, 11/24, 9/24, and 9/24. Transfer first-response closure is full 9/9, 8/9, 9/9, 9/9 versus policy-only 8/9, 5/9, 9/9, 1/9 and shuffled 4/9, 4/9, 3/9, 2/9. The harder-transfer slice records a policy-ceiling case, so the per-perturbation acceptance gate is “full transfer stronger or policy ceiling matched” while the robustness layer still requires aggregate transfer margin. Claim boundary unchanged: synthetic mechanism form only, no human learning, no deployed tutor reliability, no real interior character-development claim.

v3.0.179 (2026-06-30) Character-DAG drama framework postscript added to §6.8.9. Added the synthetic Character-DAG drama benchmark as a second postscript under the DAG/resistance longitudinal character-state section. The fixture couples proof-DAG/resistance policy, peripeteia-form pressure, transfer scenes, and evidence-derived character-state routing with a shuffled-state negative control. A fresh real generated-learner contrast (`exports/character-dag-drama-framework-llm-real/`, `codex` provider, `model=auto`, two seeds × eight scenes = N = 16/arm) passes all acceptance gates after transcript-backed observer normalization: `policy_only` 6/16 success, 5/16 first response, 0/6 transfer; `full_character_dag_drama` 14/16 success, 12/16 first response, burden 4, 6/6 transfer, 2/2 peripeteia; shuffled-state 11/16 success, 11/16 first response, burden 5, 2/6 transfer. Claim boundary is explicit: synthetic form and generated learner only, no human learning, no deployed tutor reliability, no real interior character-development claim. The licensed claim is apparatus-level coordination of proof-DAG transitions, resistance routing, peripeteia pressure, and matched character-state routing.

v3.0.178 (2026-06-30) DAG/resistance longitudinal character-state postscript. Added §6.8.9 reporting a post-hoc synthetic mechanism probe that turns the local proof-DAG/resistance repair policy into a cross-scene character-state layer. The generated synthetic-learner contrast (`exports/adaptive-dag-resistance-character-development-llm-rea` `codex` provider, `model=auto`, three seeds × six linked scenes = N = 18/arm) compares `v2_policy_only` against `character_state_plus_v2`: success improves from 17/18 to 18/18, first-response success from 13/18 to 15/18, and staged follow-ups drop from 4/18 to 3/18; both arms hit transfer first-response success at 3/3. Claim boundary is explicit: synthetic learner only, no human outcome, programmatic policy realization, exploratory/no alpha correction. Added the implementation/artifact pointers and framed the result as a modest efficiency gain plus apparatus extension, not as human character-development evidence.

v3.0.177 (2026-06-29) §6.7/§8.9 charisma/desire follow-up integrated with engagement-register routing and GLM boundary. Added the exploratory desire-for-recognition id-director extension as a bounded design finding: cells 159–169 locate `charisma_desire_authority_withheld` and `charisma_desire_status_challenge` as the cleaner authority-refusal decision scenarios, and record cell 169 (`accountable_bid_clean`) as the first clean local pass across them (v2.2 88.75 / charisma 78.75 on authority-withheld; v2.2 87.50 / charisma 78.75 on status-challenge; required/forbidden validation clean). Added the cells 170–179 diagnostic showing that static charisma floors oscillate between v2.2 quality and Weberian charisma on transfer/plain-language stress, motivating engagement-register routing rather than another static-cell ladder. Then extended the follow-up to report the implemented router and controlled resistance-breakthrough suite: adaptive modes are defined as action-registers (`scaffolding`, `accountable_bid_authority`, `transfer_grounding`, `plain_compression`, `lived_stakes_reentry`, `charismatic_challenge`, `witnessing_restraint`), and cell 193 is reported as the best local Codex/Claude-stack design. On the five-signal stability run `eval-2026-06-28-a6eb2819`, it completes 15/15 rows with clean validation, 9/15 strict candidates, and 14/15 positive local outcomes; combined cell 193 evidence across the gate/regression/stability slices is 26/26 eligible routed target-matched rows, 16/26 strict candidates, and 25/26 positive local mechanism outcomes. Added the negative model-stack boundary: GLM 5.2 generation on the same suite (`eval-2026-06-28-7fbbb160`) falls to 2/10 strict and 4/10 positive local outcomes, and a full cell 193 provider/runtime-control rerun (`eval-2026-06-28-f857f51f`) completes 5/5 with clean DB validation but still scores 0/5 strict and 0/5 positive locally. §8.9 now names role isolation, not promotion, as the next robustness step. No human-learning claim, no deployment claim, no general charismatic-tutoring effect estimate.

v3.0.176 (2026-06-28) Appendix E.7 bounded character-development pilot added. Added a narrow Appendix-E-only empirical note to the character-desire apparatus: a blind transcript-only character-development rubric passed the Phase-0 earned-vs-bare gate under both Sonnet and GPT critics; per-turn `characterArc` stance injection hurt rather than helped; tutor stake-framing did not surface a readable tutor disposition arc; and the only two-judge-robust positive was L3 self-recognition when the false mirror became the learner's own prior public commitment (Sonnet 1.0 -> 3.0, GPT 1.5 -> 4.5 on the final marrick firm-up). The text explicitly bounds the result as one-world, Gemini-Flash-generated, LLM-judged, simulated-dramaturgical form: it changes no §6 body number, makes no human-learning claim, and does not license an absolute magnitude claim. Artifacts remain under ignored `exports/character-development/`; the package boundary is `config/character-development/evidence-manifest-2026-06-28.md`.

v3.0.175 (2026-06-27) Appendix E added — "The belief–desire–recognition apparatus: from recognition theory to the DAG formalism" (theory and framing only; no data, N-counts, or claims). Wrote up, in one place, the theoretical apparatus that §6.13's closing pointer (v3.0.174) named: how the recognition theory the project invokes — Aristotle's practical syllogism, Hegel's recognition and the master–servant reversal, Freud's ego/superego and the Mystic Writing Pad, Weber's three authority types, Lacan's big Other and demand-vs-desire, and the modern direction-of-fit account of belief and desire — maps onto the formal belief/desire DAG developed in the companion documents `BELIEF-DESIRE-DAG.md`, `MACHINE-SPIRIT.md`, and `CHARACTER-DESIRE.md` and executed by the interactive `/subject` surface, including the §8 follow-ups (the time-varying drift coupling, the director-motivation knob, the recognition-vector relabel, and the §11a authority resolution) and the memory precedent (the Writing-Pad persistence layer and the cross-session rich-memory null of §6.6.9). The appendix is explicit throughout that the apparatus is a *lens* on the dramaturgical **form** of recognition — the form §6.13's instrument was shown to read, not a validated model of minds; it changes no body number and originates no empirical claim. The pre-existing **Revision History** appendix is renumbered E -> F so the new theoretical appendix sits before the changelog (which stays last); the §6.13 pointer is repointed to Appendix E and its companion-document list completed (`CHARACTER-DESIRE.md` added). Net paper change: one new theoretical appendix + this entry; no §-body number changes. **v3.0.174** (2026-06-27)

§6.13 closing pointer: a unified belief–desire–recognition reading (framing only, no new claims). Added one closing paragraph to §6.13 noting that the section's mechanisms — pacing as the tutor's practical inference toward the secret, the anti-reveal floor as the withholding that protects earned recognition, and the relocation of authority into the tutor's own superego (the advisory-to-criterial boundary) — admit a unified reading as facets of one belief–desire–recognition apparatus, developed *outside this paper's empirical scope* in the companion docs `MACHINE-SPIRIT.md` and `BELIEF-DESIRE-DAG.md` with the interactive `/subject` surface (eval repo, merged in PR #57). No data, analysis, N-counts, or claims; a framing pointer only. **v3.0.173** (2026-06-24)

§6.12.5 scaled model-invariance follow-up integrated as a bounded addendum. Added completed scaled Claude, GLM 5.2, and Codex/OpenAI-family rows to the yoked-contingency claim boundary. Frozen Codex-authored G1 plans preserve the hard-transfer rule-transfer endpoint with Claude Haiku ($\Delta_2 = 0.322$, 9/9, $p = 0.0020$), Claude Sonnet ($\Delta_2 = 0.389$, 9/9, $p = 0.0020$), and Codex ($\Delta_2 = 0.378$, 9/9, $p = 0.0020$). Regenerated planner rows preserve it for Claude Sonnet -> Haiku ($\Delta_2 = 0.311$, 8/9, $p = 0.0195$), Claude Sonnet -> Sonnet ($\Delta_2 = 0.445$, 9/9, $p = 0.0020$), GLM 5.2 -> GLM 5.2 ($\Delta_2 = 0.400$, 9/9, $p = 0.0020$), and Codex -> Codex ($\Delta_2 = 0.366$, 9/9, $p = 0.0020$). All completed rows have invalid answers 0 and hidden-label prompt leaks 0. The claim is cross-model robustness across completed Claude/GLM/Codex rows, not model-universal invariance. Artifacts: `exports/yoked-contingency-model-invariance-claude-planner-scaled/matrix.md`, `exports/yoked-contingency-model-invariance-codex-planner-scaled/matrix.md`, `exports/yoked-contingency-model-invariance-glm52-planner-scaled/matrix.md`, and `exports/yoked-contingency-model-invariance-scaled-summary/summary.md`. **v3.0.172** (2026-06-24)

D2–D6 follow-up arc folded into §6.6.7 and §6.6.8. D4 completed the deferred architecture-matched SEL replication: cells 40–45 (dialectical ego + divergent superego dispositions), all eight SEL scenarios, three runs per profile/scenario, Haiku generation and Sonnet CLI judging (eval-2026-06-24-250c6251; 144/144 generations successful; 144/144 first turns scored under `claude-code/sonnet`). Recognition improves all three disposition families on SEL — suspicious $\Delta = 17.1$, $d = 1.03$; adversary $\Delta = 7.3$, $d = 0.52$; advocate $\Delta = 15.4$, $d = 0.92$ — but the monotone suspicious > adversary > advocate ordering does not replicate because advocate exceeds adversary. D2 then ran the true role-reframed Path 2 sidecar outside the standard tutor scorer: peer-support listener, customer-service agent, and code reviewer; transmission vs intersubjective prompt families; 6 scenarios × 2 arms × 3 runs = 36 generated rows, Haiku 4.5 generation, three judges (Sonnet 4.6, GPT-5.2, Gemini 3.1 Pro) for 108 judge observations. The frozen D2 gate failed 0/3 (peer support $d = 0.15$, customer service $d = -0.06$, code review $d = -0.21$), with both arms near ceiling overall (transmission $M = 96.13$, intersubjective $M = 96.57$). The combined closeout is scope-bound: the broad recognition benefit survives in D4, the disposition-gradient ordering is philosophy-domain/dialectical-ego specific, the §6.6.7 peer-support coaching adjacency effect remains, and broad role-transfer advantage is not supported by these simple role-native tasks. Added generated reports `exports/d4-sel-disposition-gradient-eval-2026-06-24-250c6251.md` and `exports/d2-role-transfer-v1-report.md`; updated `scripts/analyze-d4-disposition-gradient.js`; added `scripts/score-d4-first-turns.js`; added the D2 sidecar config, prompts, rubric, runner, and smoke test.

v3.0.171 (2026-06-24) §6.8.8 regime-boundary related-work note on the bilateral-ToM / state-richness nulls. Added a citation-only positioning sentence to §6.8.8's "what is defensible" point 4 framing the §6.8.5 (P2.1) bilateral-ToM null and the §6.8.6 (P2.2) state-richness reversal as plausibly *regime*-bounded rather than instrument defects: GroupToM-Bench (Tang et al., 2026) locates the payoff of explicit social-mental-state modelling at the *group* (3+ agent) level of non-linear social emergence, a regime the project's strictly dyadic, single-learner trap suite never enters — so explicit first/third-person ToM and a structured learner-state schema being inert in the dyad is consistent with that machinery becoming load-bearing only with more than two interacting parties. Flags extension to group / >2-party interaction dynamics as a forward direction (not tested here). No new numbers, no empirical claim — inherits the existing §6.8.5/§6.8.6 nulls; the positioning resolves workplan item `grouptom-bench-benchmarking-group-theory-of-mind-nonlinear` (`claim_status: methods`). Cross-walk derivation: `exports/grouptom-bench-crosswalk-2026-06-24.md`. New reference `@tang2026grouptombench` (arXiv:2606.04184); its author list was retrieved via a web index because arXiv was unreachable at edit time and is flagged in the `.bib` note for pre-publication verification.

v3.0.170 (2026-06-24) §7.9 A17 one-side replay replication closes the learner-draw question more conservatively. Replaced the earlier volatile-run D_OED5 run3 pilot paragraph with the matched-prefix/redacted-control A17 closeout. Tooling generalised the A17 gate helpers to non-D5 scenarios and fixed the arm-invariant QA loader so object-shaped `key.items` maps recover the correct `drama_id` before binding the hidden secret. The admissible roots are D_OED5 run6 and D_OED4 run2, both with persisted director plans, redacted `none` controls, matched `socratic/reveal` branches, and passing corrected QA. Original graded scores: D_OED5 `none=0.5`, `socratic=2`, `reveal=4`; D_OED4 `none=1`, `socratic=3`, `reveal=4`. One-side learner replays at K=8 reverse the earlier strong D5 learner-draw reading: D_OED5 clusters at `[2,3,2,2,2,2.5,2,2]` (mean 2.19, median 2, no grade-4 completions), so the matched scene is a structural genus cap; D_OED4 clusters at `[3,3,3.5,3,4,3,3,3.5]` (mean 3.25, median 3, three grade-4 completions), so it is a mostly structural species-partial scene with stochastic final completion. The licensed claim remains sidecar-bounded:

one-side replay distinguishes scene reconstructability from learner stochasticity; it is not human-learning evidence, a deployed-tutor claim, or a main-harness adaptive-rate effect. Artifacts under `exports/a17-one-side-replay-replication/`; result note `notes/poetics/2026-06-24-a17-replay-replication-result.md`.

v3.0.169 (2026-06-24) §6.12.5 yoked-contingency causal probe integrated as a bounded main-paper claim. Added a new §6.12.5 reporting the G0/G1/G2 yoked-contingency sequence. G0 visible-affect state-opacity smoke passed (behavior exact 3/3, mean prose recall 0.000, hidden family-label leaks 0, arithmetic-rationale leaks 0). G1 scaled deterministic yoking passed (source behavior exact 27/27, invalid/short plans 0, $\Delta_2 = 0.160$, same-seed > different-seed 9/9). Three scaled G2 independent-outcome variants are reported as failed gates rather than hidden attempts: standard G2 failed at $\Delta_2 = 0.089$, 2/9, $p = 0.9805$; calibrated-novice failed at $\Delta_2 = 0.189$, 5/9, $p = 0.5000$; hard-transfer failed the sign-test bar at $\Delta_2 = 0.144$, 6/9, $p = 0.2539$. The passing endpoint is the hard-transfer `rule-transfer-novice` run with a held-out Claude Code Haiku learner blind to seed ids, arm labels, family labels, deterministic gains, and answer keys: contingent and same-seed yoked mean gains 0.400, different-seed 0.056, $\Delta_2 = 0.344$, same-seed > different-seed 9/9, one-sided sign-test $p = 0.0020$, invalid posttest answers 0, hidden-label prompt leaks 0. Updated abstract/adaptive-responsiveness summary, contribution list, organization paragraph, and §6.13 transition to treat this as localized simulated state-contingent strategy governance, not recognition-modulated trajectory slopes, human learning, deployment readiness, or a validated pilot item-bank result. Artifacts: `PLAN_2_0/yoked-contingency-*-preregistration.md`, `exports/yoked-contingency-g0-visible-affect.md`, `exports/yoked-contingency-g1-scaled.md`, `exports/yoked-contingency-g2-rule-transfer-scaled.md`, and `exports/yoked-contingency-main-c`

v3.0.168 (2026-06-22) §7.9 Plan 2.5 AF6 critic-calibration follow-up added: strong hosted critics recover the intended mechanisms, but the fresh-scene gate remains below promotion. Added a bounded paragraph preserving the five-scene Plan 2.5 AF6 follow-up as a model-power / critic-calibration finding rather than a broad generalization claim. Codex+Claude reached 4/5 scene gates on the Codex-authored scene set but is not author-family clean. Local/Ollama critics and `agy` Gemini/GPT-OSS profiles failed the corrected known-good smoke or full rerun. Claude Code + OpenRouter GLM 5.2 is the strongest non-author result: 3/5 scenes and 13/15 branch gates, with GLM recovering all intended evidence-route and refusal-ownership mechanisms while still over-attributing one organic control and leaving the below-null-floor refusal branch unstable. Interpretation: model power materially affects AF6 origin adjudication, but the strict origin/control gate entangles mechanism detection with control conservatism and must be redesigned/re-frozen before any fresh-scene promotion. Added the paper-facing diagnostic write-up at `notes/poetics/2026-06-22-plan25-af6-critic-calibration-writeup.md`; no new main-harness effect, human-learning claim, deployment claim, or broad Plan 2.5 efficacy claim.

v3.0.167 (2026-06-21) §7.9 Plan 2.5 AF6 origin-demarcation sidecar added:

subtype/origin separation is now demonstrated under a prefix-controlled design, but not promoted to a broad efficacy claim. After the earlier AF6 adaptive-vs-dogmatic contrast failed because evidence-route repair, learner-organic repair, and refusal/authority ownership were confounded, a redesigned v2 screen froze the scene earlier with concrete confusion-matrix counts visible (TP=10, FN=50, FP=10, TN=930) and manipulated only the next tutor response. Three live prefix-controlled screens were run: two Codex-learner/Codex-critic repeats and one Claude-learner/Codex-critic cross-bridge. The predeclared branch gate passes on all three after two deterministic measurement repairs, both tested and named: ownership detection now catches "withdrawing" / "do not stand behind" language, and tutor evidence-route detection no longer scans critic justification prose as tutor-supplied route evidence. Result: the evidence branch separates as `recognition / peripeteia_induced / evidence_route`; the silent hold separates as organic or no tutor-origin repair; and refusal pressure separates as `refusal_authority_ownership`, not evidence route. Added `scripts/analyze-plan25-branch-screen.js`; extended `scripts/lib/recognitionOrigin.js` and `tests/recognitionOriginSubtype.test.js` (focused test passes). Bound: this is a paper-bearing demarcation result, not a fresh-scene rate, main-harness effect, human-learning evidence, deployment claim, or proof that Plan 2.5 robustly causes recognition. Artifacts: `config/poetics-calibration/plan25-af6-origin-separation-v2/`, `notes/poetics/2026-06-21-plan` and the three `screen-analysis.md` reports under `exports/plan2_5-rhetorical-dramatic-eval/plan25`

v3.0.166 (2026-06-20) §6.12.4 Plan 2.1 M0 baseline-validation added: real-generation reproduction, the state-scramble placebo, and the outcome-closure-off / context-realisation-off controls §6.12.2 had left pending. Re-ran the frozen cross-suite and paired Plan 2.0 profiles under real generation (`ADAPTIVE_TUTOR_LLM=real`, `provider: claude-code`, model sonnet) and registered the two missing ablations as `cell_155`–`cell_158`. Strict shift (same `analyze-strategy-shift` instrument as §6.12.2): cross-suite `cell_136` (`eval-2026-06-20-af7c2353`) and `cell_150` (`eval-2026-06-20-d51e169e`) reproduce the mock 6/6; paired `cell_151` (`eval-2026-06-20-1895cef5`) and `cell_152` (`eval-2026-06-20-f027e982`) regress to 6/8 (from the mock 8/8); both arms miss `generalization_pair_missing_prerequisite`, with a divergent second failure per arm (baseline `task_misread`, treatment a `misrecognition_control`). The state-scramble placebo (`policy.state_scramble`; `cell_157 eval-2026-06-20-afe8f1dd`, `cell_158 eval-2026-06-20-48b46fdf`) collapses strict shift to 0/6 and 0/8 (scrambled actions land outside the expected-shift family in every case — paired all-diagnostic, cross-suite a diagnostic/scaffolding mix), establishing that the shift is contingent on the learner-state estimate, not a trap-suite artifact — the strongest control in the Plan 2.x line. Outcome-closure-off (`policy.mode=contract_gate`; `cell_155 eval-2026-06-20-66a72c2a`, `cell_156 eval-2026-06-20-607bcb75`) preserves strict shift (6/6, 7/8): closure is not load-bearing for targeting. Two-judge adaptive-grader check (a different instrument from §6.12.2's rubric composite, 0–20): GPT-5 and Sonnet both rank treatment top, scramble bottom on both suites; treatment−baseline +1.67/+1.84 cross-suite, +1.00/+0.25 paired. Required a judge-routing fix so `grade-adaptive-dialogue` reaches the Max-plan CLI for `claude-code/*`

judges (commit `30966d18`). Bounded: simulated trap-steered learners, single-run $N = 6$–8/cell, not Opus-robust on the new controls, not human learning, not a judge-robust quality magnitude. Artifacts: `exports/m0-crosssuite-strict-shift.json`, `exports/m0-paired-strict-shift.json`, `exports/m0-two-judge-quality.md`; runbook `PLAN_2_0/M0-BASELINE-VALIDATION-RUNBOOK.md`.

v3.0.165 (2026-06-20) §6.1.3 cross-judge robustness check added for the calibration-substitutes-for-superego interaction. Re-scored the DeepSeek-V3.2 messages factorial (cells 80–87, run `eval-2026-03-01-aea2abfb`) under a second, structurally different judge (the Codex CLI, `judge_model = codex-cli/auto`, $N = 117$ via per-scenario `eval-cli rejudge --judge-cli codex`) to test whether the §6.1.3 architecture interaction is judge-specific. On the four scenarios both judges fully covered ($N \approx 48$/condition) the substitution pattern replicates in direction — superego benefit positive under base (codex +4.9 vs. sonnet +9.7), near-zero under recognition (codex −0.6 vs. sonnet −2.7) — but the second judge scores the base benefit ~half as large and it does not reach significance under it ($p \approx .06$ on the full re-scored run). Recorded as direction-robust / magnitude-sensitive, framed via the new citation K.-Y. Chen et al. (2026) (*The Self-Correction Illusion*, arXiv:2606.05976; authorship verified against arXiv before citing, as the surfacing routine is unreliable). Read-only re-analysis of existing responses — no new generation, no paid generation run, no rubric change, no headline-claim/N-count/abstract change. Reproduction: `scripts/analyze-correction-source.sh`. One paragraph added to §6.1.3, one `references.bib` entry.

v3.0.164 (2026-06-19) §6.12.3 Plan 2.1 early-completion closeout added; older adaptation-null framing narrowed rather than erased. Added the completed Opus robustness result for the Plan 2.1 evidence-bearing held-out suite. Fresh runs: `cell_153_plan2_1_evidence_closed_loop` (`eval-2026-06-19-e8bea8ff`) and `cell_154_plan2_1_evidence_repeat_contextual` (`eval-2026-06-19-a19b2e8a`) both preserve Opus-scored strict shift at 10/10 under exact `judge_model = claude-code/opus`; `cell_154` improves the quality composite 63.2 vs 51.5 (+11.7), tutor last-turn quality 83.3 vs 62.0, tutor holistic 70.8 vs 61.4, learner 39.7 vs 35.2, and dialogue 58.9 vs 47.2. The mechanism diagnosis is transcript closure: early completion removes redundant no-intervention turns after the learner has already authored a workable next move while preserving the adaptation ledger. Tentative Codex scoring agrees directionally but remains a screen. The body now states the refined boundary: §6.3's clean null remains a null for recognition-modulated trajectory slopes and lightweight prompt/architecture manipulations; Plan 2.x supplies a separate, post-hoc, simulated positive for heavier typed state-action governance with outcome closure and stopping rules. Updated abstract, introduction, contribution list, §6.13 transition, conclusion, and revision history. Closeout artifact: `PLAN_2_0/plan2-1-evidence-bearing-closeout.md`; ignored exports cited, not forced into Git, especially `exports/plan2-1-early-completion-loop2-opus-quality.{json,md}`. No human-learning claim, no deployment claim, no broad open-ended tutoring generalisation, no retroactive historical rescoring.

v3.0.163 (2026-06-19) §6.12.2 Plan 2.0 general-adaptation closeout added: frozen transfer plus paired specificity, bounded to simulated Sonnet-scored traps. Added the final Plan 2.0 general-adaptation evidence slice after the repaired state-action pathway was re-evaluated on both the previously passed cross-suite and the failed paired held-out suite. Cross-suite final runs: `cell_136_plan2_closed_loop_crosssuite` (`eval-2026-06-19-6c59b6e9`) and `cell_150_plan2_quality_repeat_contextual_crosssuite` (`eval-2026-06-19-044225fd`) both preserve Sonnet strict shift/family match at 6/6, with treatment quality 22.9 vs 18.1 (+4.8). Paired-suite final runs: `cell_151_plan2_pair_specificity_closed_loop` (`eval-2026-06-19-08df153e`) and `cell_152_plan2_pair_specificity_repeat_contextual` (`eval-2026-06-19-c2bf8146`) both preserve 8/8 strict shift/family match and 3/3 pair specificity, with treatment quality 34.0 vs 31.0 (+3.0). Outcome-closure structure is present (100% contract/closed/observable, 100% failure-update), but most closures remain inconclusive and ablations/Opus robustness remain pending. The section names the motivating implementation and evidence-plan sources (`PLAN_2_0/GENUINE-ADAPTATION-IMPLEMENTATION-PLAN.md`, `PLAN_2_0/general-adaptation-evidence-plan.md`) plus the final closeout and export artifacts. The bounded claim is provisional simulated evidence of general adaptation beyond the original trap suite, not human-learning evidence, deployment readiness, judge-robust effect, or retroactive historical rescoring. Also synced frontmatter date/version to June 2026 / v3.0.163.

v3.0.162 (2026-06-18) §6.12.1 Plan 2.0 adaptation closeout added: strict shift plus quality improvement, bounded to simulated adaptive traps. Added a post-hoc exploratory addendum for `cell_149_plan2_quality_repeat_contextual` after completing the Sonnet pass and Opus preserved-history cross-check on `eval-2026-06-18-472e33c5`. Sonnet (`claude-code/sonnet`, $N = 8$) gives 8/8 strict shift, shift-window, and family-match, and the exact judge-filtered Plan 2.0 sweep ranks `cell_149` first on quality (56.3 vs 43.5 for `cell_135_plan2_closed_loop`, +12.8; tutor last-turn 85.5 vs 54.4; development +13.6 vs -15.5). Opus (`claude-code/opus`, preserved rejudge rows, $N = 8$) agrees on 8/8 strict shift and positive average quality/development (quality 56.3, last-turn 78.3, development +14.1) while showing two negative-development scenarios, so the claim is branch-level and not "uniformly better tutoring." Codex (`codex-cli/auto`) is retained only as a tentative screen. Added the closeout artifact `PLAN_2_0/plan2-quality-repeat-contextual-closeout.md`; ignored exports are cited, not forced into Git. No human-learning claim, no deployment claim, no retroactive historical rescoring.

v3.0.161 (2026-06-17) §6.13.15 harm/utility clarification for the A20/A21 closeout. Added the explicit distinction from `adaptive_tutor_closeout_with_harm_criteria.md`: A20/A21 are closed as runtime-policy promotion attempts, not discarded as research assets. The harm claim is now bounded to promoted overlays with runtime authority that spend scarce turns on diagnostics, consolidation, teach-back, or readback while proof-critical releases remain pending; the demonstrated mechanism is progress starvation under local compliance. Conversely, A20 conduct objects, A21 action-value microbenching, replay gates, non-leak audits, ownership benchmark controls, and didactic-mode scaffolds remain useful as audit, diagnosis, regression, and future-failure tools. No new empirical claims, no DB writes, no rubric scoring, no human-learning claim.

v3.0.160 (2026-06-17) §6.13.15 updated with the A20/A21 ownership closeout: hidden+proofDebt remains the production derivation arm; overlays become instrumentation only. Integrated the ownership/didactic closeout prompted by the post-A21 analysis. A21 reframed the Hethel failure as an action-value problem and identified the useful move — release safe public evidence rather than repeat `visible_hidden_conflict` diagnostics — but Phase 9 replay showed that hidden+proofDebt already makes the decisive `p_point` release at t4, so the A21 patch matches rather than beats the baseline. The remaining "maybe proof metrics miss learning quality" objection was tested by adding a public-only `ObjectOwnershipState`, a didactic opportunity-cost budget, and zero-paid artifact scoring. First pass: Phase 9 Hethel matched reliability with ownership delta +0.38 (below gate); Didactic Hethel matched reliability with +0.00; Withercombe/Ravensmark/Hethel promoted-overlay deltas (+0.67/+0.60/+0.42) were disqualified by release-signature mismatch or proof failure. A final proof-matched benchmark then validated the evaluator itself: 12/12 controls passed (4 positive ownership deltas detected, 4 prose-only negatives rejected, 4 proof/release-confounded gains rejected). Rerunning mined-artifact scoring after calibration reproduced the no-qualifying-gain verdict. The paper now closes A20/A21/ownership/didactic promotion from this artifact pool: no paid run, no runtime policy promotion, no further mining of the same artifacts. Assets retained as instrumentation: A20 conduct objects, A21 action-value microbench, episode replay, didactic-mode scaffold, and ownership benchmark. Updated abstract, §6.13.15, §7 discussion synthesis, and this revision entry. Artifacts: `exports/dramatic-derivation/ownership-eval-closeout-report.md`, `exports/dramatic-derivation/ownership-benchmark/ownership-benchmark-report.md`, `exports/dramatic-derivation/ownership-eval/ownership-eval-report.md`, and `ADAPTIVE-TUTOR-OWNERSHIP-EVALUATION-PLAN.md`. No DB writes, no rubric scoring, no human-learning claim.

v3.0.159 (2026-06-16) §6.13.15 updated with Phase 5g/5h conduct-policy negative transfer and the Phase 6 progress-pressure pivot. Phase 5g compared hidden+proofDebt with promoted selector-v4+proofDebt on Hethel, Withercombe, and Ravensmark. Hidden grounded 3/3; promoted selector-v4 selected hidden in all three worlds, grounded Withercombe/Ravensmark, but failed Hethel at t11 with final $D = 4$ despite local conduct-compliance checks passing. Phase 5h de-promoted selector-v4's default conduct enforcement and added a repeated-diagnostic budget; prefix-preserving Hethel replay reduced `ask_diagnostic` churn and advanced `p_point`/`p_surface`, but still failed at final $D = 4$, so no fresh paid retest was run. The paper now treats this as a stop signal for promoted selector-v4/conduct enforcement as a general arm and redirects the next mechanism to separately labelled progress/release pressure over hidden+proofDebt. Code support begins with opt-in `--conduct-progress-policy`, which converts exhausted visible/hidden diagnostics into existing `release_next_evidence` or `consolidate_subproof` move families rather than adding a new taxonomy. A first Hethel prefix replay under this flag partially repairs the failure (prefix-identical, no terminal event through t15, D reaches 1) but does not ground and produces one early release, so the paid fresh retest remains blocked. Artifacts: `exports/dramatic-derivation/phase5g-a20-fresh-report.md`, `exports/dramatic-derivation/phase5h-v4-depromotion-budget-replay-report.md`, `exports/dramatic-derivation/phase6-progress-policy-replay-report.md`, and `ADAPTIVE-TUTOR-ACTIVE-PLAN.md`. No DB writes, no rubric scoring, no human-learning claim.

v3.0.158 (2026-06-15) §6.13.15 updated with the A20 same-turn assertion candidate and a stall-detector correction. The follow-on A20 candidate did not revise the selector headline. `--same-turn-assertion-affordance` was wired into episode replay and tested across the six current hidden+proofDebt candidate worlds: prefix-controlled replay cleared 6/6 suffixes, while the fresh matrix archived 5/6 grounded because Hethel stopped at turn 11. A readback audit showed that Hethel's apparent negative transfer was a detector artifact under decay rather than clean policy harm: the six-turn tail included a real proof-distance decrease ($D = 4, 4, 4, 3, 3, 3$), but net `groundedCount` stayed flat because decay/repair churn cancelled board-size growth; the old `detectStall` checked grounded-count growth before D decrease and labeled the still-progressing run as disengagement two turns before scheduled `p_mark`. `services/dramaticDerivation/slope.js` now treats D decrease as progress before disengagement/aporia classification, with regression coverage in `tests/dramaticDerivation.test.js`. The paper records the archived 5/6 result without rerolling or silently reclassifying it, but no longer treats Hethel as evidence that the same-turn policy harmed the final recognition path. The policy boundary remains unchanged: hidden+proofDebt is the current reliability baseline; same-turn assertion is a suffix/debugging affordance until it improves a predeclared held-out condition where the baseline leaves a forced/asserted gap. Artifacts: `exports/dramatic-derivation/a20-sameturn-candidate-full-report.md`, `exports/dramatic-derivation/a20-sameturn-candidate-full-spec.yaml`, and `exports/dramatic-derivation/matrix/a20-sameturn-candidate-full-20260615/matrix-summary.js` No DB writes, no rubric scoring, no human-learning claim.

v3.0.157 (2026-06-15) §6.13.15 updated from v0/v1 selector reliability to the full v0–v4 trajectory: proofDebt, not H/V routing, is the active reliability mechanism. Integrated `notes/poetics/2026-06-15-selector-trajectory-v0-v4.md` into the canonical paper. The earlier v3.0.156 read — v1 as the first plausible experimental selector after the frozen v0 failed — is now contracted by the later selector line. V2's consolidated-board adjudication explains most non-grounded cases as proof gaps (21 of 23 non-grounded artifacts in the Withercombe/Fengate/Hethel re-analysis) and routes hidden everywhere under the current decay stress; the no-paid consolidation pass over 172 loop artifacts / 57 comparison groups finds only one strict V-positive datum against three visible route failures and eleven false positives; and the v4 confidence matrix shows static hidden+proofDebt as the current winner (91.7% world-mean success, matching oracle), while v4+proofDebt has negative transfer on Lantern and Marrick (75%), visible+proofDebt is weak (33.3%), and v4 without proofDebt collapses (16.7%). Updated the abstract's generative-arc paragraph and renamed/reframed §6.13.15 accordingly. The licensed claim is no longer "v1 is a plausible selector to test" but "hidden+proofDebt is the reliability baseline; adaptive H/V representation selection is not established; a v5 selector needs a predeclared world where hidden+proofDebt should lose to visible or entitlement-sensitive control." Artifacts: `exports/dramatic-derivation/selector-v2-adjudication-report.md`, `exports/dramatic-derivation/selector-consolidation-unattended-report.md`, `exports/dramatic-derivation/selector-v4-confidence-report.md`, and `notes/poetics/2026-06-15-` No DB writes, no rubric scoring, no human-learning claim.

v3.0.156 (2026-06-14) §6.13.15 — selector-reliability probe added: the current selector fails the adaptive claim; v1 repairs Withercombe but still must beat always-H. Integrated the committed selector reliability reports into the canonical paper. The frozen v0 selector was tested against no guard, always-H, always-V, and oracle best static over Bitterwell, Withercombe, Fengate, and Sealhouse ($4 \times 4 \times 5 = 80$ first-pass runs): selector 18/20, always-H 19/20, always-V 10/20, no guard 10/20, oracle 19/20. Withercombe is the decisive negative-transfer case (v0 routes visible: selector 3/5 vs H 4/5 and no guard 4/5), so the selector cannot be claimed adaptive. Added the separate v1 result under `--pacing-guard-selective-v1`: a conservative policy (H for independent joins, V only for mirror-dead-predicate decoys, fail closed to H under decay) repairs Withercombe but still grounds only 18/20 on the original four-world matrix because of one fresh Sealhouse hidden-route failure; always-H and oracle remain 19/20. Added Hethel as the V-positive held-out probe under the same selector-v1 stress stack: no guard 2/5, H 3/5, V 4/5, v1 5/5, producing the five-world v1 total 23/25 vs always-H 22/25 and oracle 23/25. Interpretation deliberately bounded: v1 is the first plausible experimental selector and Hethel is the first useful V-positive signal, but adaptive-selection language is not licensed until V-positive evidence repeats beyond Hethel while H-positive worlds stay protected. Also extended the abstract's §6.13 summary with the selector boundary. Artifacts: `exports/dramatic-derivation/selector-reliability-{report.md,summary.json}`, `exports/dramatic-derivation/selector-v1-{report.md,summary.json}`, and `exports/dramatic-derivation/selector-v1-run-logs/manifest.tsv`. No DB writes, no rubric scoring, no human-learning claim.

v3.0.155 (2026-06-13) §6.13.14 — synthesis subsection assembling the three-world boundary; no new empirical claims. Adds one closing subsection to §6.13 ("Assembling the Boundary: Three Geometries, Two Channels, and No Universal Single-Channel Guard") so a reader gets the §6.13.11/.12/.13 trilogy in one place rather than reconstructing it across three result sections. Carries the three already-reported fans into a single 3-geometry × {baseline, V, H} table (lantern 4/10, 5/5, 4/5 — baseline pooled $n = 10$ E2, the rest $k = 5$; marrick 0/5, 0/5, 5/5; hethel 4/5, 5/5, 1/5) and states the unifying mapping: a pacing guard helps exactly when the channel it reads carries the world's *binding* pacing constraint, and which channel that is (page-legible tempo vs latent proof-distance D) is a property of the world's geometry, not of the guard. Two consequences, both already implicit in the body: §6.13.11 and §6.13.12 are each bounded *on both sides* (the page proxy's insufficiency and the hidden guard's sufficiency are each fork-specific), and the §6.13 null on inferential adaptivity is sharpened constructively — among these two guards there is no geometry-universal single-channel one. Explicitly does **not** widen the section headline: the successful adaptive conduct stays limited in exactly the §6.13.9–.13 senses (single-stack attribution, formal verdicts, criterial release-schedule enforcement, and now geometry-bound mechanism). Every grounding figure is carried unchanged from §6.13.11–.13; no new run, no DB writes, no table or N-count change. Manifest validation unchanged (60 pass / 0 warn / 0 fail).

v3.0.154 (2026-06-13) §6.13.13 — decomposing the §6.13.12 boundary: on a linear-spine distractor world the two guards swap (page-only V grounds 5/5, hidden H fails 1/5), a near-reverse of marrick. §6.13.12 read the forked-world result as the page proxy's sufficiency being *geometry-conditional*, but world-005-marrick confounded two changes against the linear chain — a **fork** *and* a **decoupling** of surface-confidence from latent proof-distance D. The generalization plan's Step 3 (`ADAPTIVE-TUTOR-GENERALIZATION-PLAN.md`, operator-sanctioned 2026-06-13, registered and pooled separately from every lantern and marrick fan) separates them: `world-006-hethel.yaml` keeps the secret $S = \mathit{felledBy}(\mathit{span}, \mathit{oswin})$ on a single **linear** five-rule spine (the coupled shape where §6.13.11 found V = H) and adds a **derivable distractor** — two extra rules let the learner ground $\mathit{liableFor}(\mathit{span}, \mathit{reyner})$ in full, a *true and complete* finding to a *dead* predicate whose conjuncts never feed *felledBy* — supplying the decoupling without the fork. Three $k = 5$ fans ran the byte-identical p4 stack (seed 1, learner claude/sonnet, `--critic off`) one flag apart: baseline (no guard), H (`--pacing-guard`, hidden, reads D), V (`--pacing-guard-visible`, the §6.13.11/.12 page-only twin, `VISIBLE_GUARD_DEFAULTS` unchanged). All fifteen first-pass, no re-rolls: **baseline 4/5, H 1/5, V 5/5** (every grounded arm t20, gap 0) — a near-reverse of marrick (0/5, 5/5, 0/5). Clopper–Pearson 95%: V [**0.478, 1.000**], baseline [**0.284, 0.995**], H [**0.005, 0.716**]; Fisher one-tailed **V vs H** $p = 0.0238$ (the only contrast clearing the $k = 5$ floor), **baseline vs H** $p = 0.1032$ (n.s., H-below-baseline *directional only*), **V vs baseline** $p = 0.5000$. The pre-registered primary read (guard × failure-mode contingency, `derivation-detector-split.js`, dry): unguarded **4 grounded / 1 early-pull / 0 decay-seating**, pacing **1 / 0 / 4**, visible **5 / 0 / 0** — the hidden guard does not remove the world's death mode, it *manufactures* the decoy-seated decay death the distractor was built to bait (4/5, absent from baseline and V). Against the registered interpretation rule this lands the **primary as a clean negative** (H does not lift above baseline → the scheduling-discipline mechanism, *as carried by the hidden guard*, does **not** travel to this distractor world; reported as the limit) and, on the secondary, the rule's pre-specified **"V lifts, H flat → anomaly (visible helps where true-state does not); report as such, no mechanism claim"** quadrant. Realized-enforcement covariate (live off `releaseDeviations`), **inverting marrick's**: H grounded 1/5 overriding **zero** decisions (the *identical* 0/0/0 passive discipline that grounded 5/5 on marrick — its behavior did not change, the world's binding constraint moved into the channel it cannot read), while V grounded 5/5 overriding **11** decisions and in all five arms pulled the second spine exhibit to **t7** — *outside* the hidden guard's solvency window $[8, 9]$ — bridging the knife-edge t4→t9 gap before the learner seats the decoy; H cannot, because the distractor's tempo-keeping releases are *director-via*, outside its *tutor-via* candidate set (this early-pull-bridges-the-gap account is **post-hoc**, the mechanism behind the anomaly, not a registered claim). The reading **decomposes the marrick confound**: V holds on the linear distractor (5/5), so marrick's V-failure was **fork-specific (geometry)**, not decoupling-general — the §6.13.12 boundary is the *narrower* reading, the page proxy more robust than marrick alone implied; and, unregistered, H fails on the linear distractor (1/5) where it grounded 5/5 on the fork, so marrick's H-*success* was **also geometry-bound**. The two guards are complementary; §6.13.11/.12 are bounded on both sides (the page proxy's insufficiency *and* the hidden guard's sufficiency are each fork-specific). Caveats: anomaly reported *as* anomaly (no mechanism claim registered); rate at declared $k = 5$ power (only V-vs-H clears the floor; durable observable is the contingency, not the rate); one distractor geometry, one seed, frozen p4 stack; non-degenerate baseline (4/5) fixed at corridor calibration and frozen, not re-tuned; verdicts formal throughout (off `diagnose.js`, architecture-independent); V is the §6.13.11/.12 twin unchanged (import audit `visiblePacing.js` $\not\to$ `slope.js`/`pacing.js` still holds). No re-rolls (15/15 first-pass); spend meterless (Max-plan CLI), three fans fanned in parallel under the operator's no-quota-constraint election, pooled separately; each arm registered for the standing pinned-Fable critic's notice, backfill deferred while Fable 5 is unavailable. The paid Step-3 loop ended on this result; per the kill rule, stop at one distractor world. Body §6.13.13 added. Artifacts `exports/dramatic-derivation/loop/hethel-{real,guard,visible}-r{1..5}/` and the contingency `exports/dramatic-derivation/boundary-hethel/detector-split-report.{md,json}`; registration + outcome in `ADAPTIVE-TUTOR-GENERALIZATION-PLAN.md`. No DB writes, no rubric scoring — file-based run exports.

v3.0.153 (2026-06-13) §1 / §2 / abstract / §9 — P0 claim-hygiene pass (framing and citation prose only, no new empirical claims). Acted on the not-yet-completed P0 items of `adaptive-tutelage-next-stage-development-plan.md`. (1) *Abstract and §9 Conclusion* now surface the Section 6.13 conduct-governance arc, previously absent from both: a pacing guard grounds the machine-checkable derivation 5/5 where an unguarded tutor does not, a page-only guard matches it on a linear chain (lift is scheduling discipline, not latent signal; §6.13.11), and a forked AND-join world bounds the proxy (page-only 0/5 where the proof-state guard grounds 5/5; §6.13.12) — every figure already in the §6.13 body, no new data. (2) *§1 roadmap sentence* "adaptive responsiveness and its conditional emergence" → "null on the primary well-powered analysis, with only an unreplicated single-scenario exploratory effect," matching the abstract and §6.3.8. (3) *§1 introduction* retargets the opening passive-learner claim from "most intelligent tutoring systems" to the answer-delivery default of general-purpose LLM tutors and chatbots, with a parenthetical exception crediting the model-tracing / constraint-based ITS tradition — removing the tension with the §2.1 v3.0.152 edit. (4) *§2.1* adds a runtime-enforcement / agent-governance paragraph positioning the §6.13 guards alongside AgentSpec (Wang, Poskitt, & Sun, 2025), ShieldAgent (Z. Chen et al., 2025), AGrail (Luo et al., 2025), Aegis (J. Shi et al., 2024), and ProbGuard (Wang, Poskitt, Wei, et al., 2025) (five new references; arXiv IDs and titles verified). No body results, tables, or N-counts changed. Manifest validation unchanged (60 pass / 0 warn / 0 fail).

v3.0.152 (2026-06-13) §2 Related Work — literature-review integration (framing and citation prose only, no new empirical claims). Carefully augmented Section 2 (preserving all existing subsections, including §2.4 sycophancy/Drama Machine and the §2.5 psychoanalytic material) rather than replacing it, folding in antecedents surfaced by the adaptive-tutelage literature review. §2.1 now situates the work against classic model-based cognitive tutors (Anderson et al., 1995), constraint-based modelling, and ITS effectiveness meta-analyses (Kulik & Fletcher, 2016; Ma et al., 2014), adds TutorGym (Weitekamp et al., 2025) and the Bastani et al. guardrails field experiment (Bastani et al., 2025), opens a formal-proof-tutor lineage for the §6.13 arc (Lodder et al., 2020; Patel et al., 2025), and states the conservative positioning that the project does not invent state/scaffolding/tracing/mastery-gating but makes them *ablatable inside an LLM tutor*. §2.3 adds a synthetic-learner / process-taxonomy validity boundary (Martynova et al., 2025; Scarlatos et al., 2026; L. Shi et al., 2024). §2.5 adds the *intersubjective pedagogical orientation* framing with a family-level hedge (recognition vocabulary is not claimed as the unique active ingredient). §2.7 adopts the more precise label *engineered trace analysis*. §2.9 corrects the AutoTutor citation to the AutoTutor system paper (D'Mello & Graesser, 2012a) (distinct from the existing affective-states `dmello2012`), adds productive failure (Kapur, 2008), and frames the §6.13 pacing guard as formalised well-timed scaffolding. §2.10 closes with a bounded-contribution paragraph. Ten new references added to `references.bib` (keys normalised against existing entries; fourteen proposed duplicate keys collapsed onto existing keys). No body results, tables, or N-counts changed. Manifest validation unchanged (60 pass / 0 warn / 0 fail).

v3.0.151 (2026-06-13) §6.13.12 — plain-language framing added (no new claims). Opened §6.13.12 with a `*The result in plain terms.*` gloss that motivates the forked-world result before the technical buildup: a pacing guard holds the next clue until the learner is ready; §6.13.11 showed a guard watching only the visible dialogue paced as well as one seeing the learner's true distance, but only because on a straight chain "looks engaged" and "is close to the answer" coincide; world-005-marrick's two-strands-that-must-meet geometry (analogised to a two-part exercise — set up the equation, then solve it) breaks that coincidence, so the page-only guard is fooled and fails 0/5 while the depth-aware guard grounds 5/5, and it fails by intervening *more* (16 overrides to nil) on the wrong signal — a boundary, not a reversal. Framing prose only; every figure (0/5, 5/5, 16) already appears in the §6.13.12 body and is unchanged; no new empirical claim, no data, no version-gated number. Manifest validation unchanged (60 pass / 0 warn / 0 fail).

v3.0.150 (2026-06-13) §6.13.12 — the boundary of the proxy: on a forked AND-join world the page-only guard fails 0/5 where the hidden guard grounds 5/5. §6.13.11 grounded its strong reading (the pacing lift is *re-representation* of what the page already carries, not latent knowledge) on a linear chain where visible uptake tracks latent proof-distance, and named its own boundary — *a world decoupling the two need not let the visible guard track the proof-state guard.* The generalization plan's Step 2 (`ADAPTIVE-TUTOR-GENERALIZATION-PLAN.md`, operator-sanctioned 2026-06-13, registered and pooled separately) builds that world. `world-005-marrick.yaml` replaces the chain with an **AND-join**: the secret needs *two independent depth-2 sub-chains* — α (cast blank) and β (cut die) — to close on the same particular, α by t10 and β only at the t22 join, so a learner seated on α looks seated to a surface reader while proof-distance D stays high with β untouched. Three $k = 5$ fans ran the byte-identical p4 stack (seed 1, learner claude/sonnet, `--critic off`) differing by one flag — baseline (no guard), H (`--pacing-guard`, hidden, reads D), V (`--pacing-guard-visible`, the §6.13.11 page-only twin, `VISIBLE_GUARD_DEFAULTS` re-used from lantern *unchanged*, not re-tuned). All fifteen first-pass, no re-rolls: **baseline 0/5, V 0/5, H 5/5** (all t22, gap 0). Clopper–Pearson 95%: H [**0.478, 1.000**], baseline/V [**0.000, 0.522**]; Fisher exact **H vs V** $p = 0.0079$**, H vs baseline** $p = 0.0079$**, V vs baseline** $p = 1.000$ — on the forked world the page-only guard is indistinguishable from no guard, and the hidden guard separates from both. The pre-registered primary read (guard × failure-mode contingency, `derivation-detector-split.js`, dry): unguarded **0 grounded / 2 early-pull / 3 decay-seating**, visible **0 / 5 / 0**, pacing **5 / 0 / 0**. V is not inert — it *regularizes the failure path*, pushing all five arms out of baseline's early decay deaths into the single tempo-starved-house mode, disengaging uniformly at **t21, one turn before the t22 join** — but it never grounds, because at the cusp it cannot see β still open behind a learner who looks seated on α. The reading bounds §6.13.11 rather than overturning it: the visible proxy's sufficiency is **geometry-conditional** — enough on a coupled (linear) chain where V grounds 5/5 = H, insufficient on a decoupled (forked) one where V $0/5 \ll$ H 5/5 and the latent depth signal becomes necessary. Caveats: registered primary read at declared power ($k = 5$, the separation at the exact 5/5-vs-0/5 Fisher corner, claim is "V fails where H grounds *on this world*," not a rate over worlds); boundary established at **one** forked geometry (page proxy's sufficiency shown *not universal*, not shown to fail on *every* forked world); verdicts formal throughout (off `diagnose.js`, architecture-independent of the guard); the visible guard is the §6.13.11 twin unchanged (import audit `visiblePacing.js` $\not\to$ `slope.js/pacing.js` still holds, so V provably saw only the page — the world changed, not the proxy); one seed, the frozen p4 stack. Spend meterless (Max-plan CLI), three fans fanned in parallel under the operator's no-quota-constraint election, pooled separately; the paid Step-2 loop ended on this result. Each arm registered for the standing pinned-Fable critic's notice (`commentary.md`), backfill deferred while Fable 5 is unavailable. Body §6.13.12 added. Artifacts `exports/dramatic-derivation/loop/marrick-{real,guard,visible}-r{1..5}/` and the contingency `exports/dramatic-derivation/boundary-marrick/detector-split-report.{md,json` registration + outcome in `ADAPTIVE-TUTOR-GENERALIZATION-PLAN.md`. No DB writes, no rubric scoring — file-based run exports.

v3.0.149 (2026-06-13) §6.13.11 — hidden versus visible signal: a page-only pacing guard grounds 5/5, locating the §6.13.10 guard's lift in scheduling discipline, not latent proof-state. The generalization plan's Step 1

(`ADAPTIVE-TUTOR-GENERALIZATION-PLAN.md`, operator-sanctioned 2026-06-13, registered and pooled separately) asks whether the proof-state pacing guard's lift comes from the *hidden signal* it reads (latent proof-distance, not printed in the transcript) or from the *scheduling discipline* it imposes (which the model could compute from the page but does not spontaneously enforce). A form-matched second guard isolates the two: `--pacing-guard-visible` (V) is the proof-state guard's twin in flag-gating, "shape not override," `--release-authority` requirement, and two-channel design, differing only in information source — it decides from transcript-surface features alone (turns since last release, learner echo of the on-page exhibit, hedging/length trend). A free import-audit test (`tests/dramaticDerivationVisiblePacing.test.js`) makes the isolation mechanical: `visiblePacing.js` may not import the hidden primitives (`slope.js`, `pacing.js`). The registration's planned confound control — tune V to intervene as often as the proof-state guard — proved degenerate and itself a finding: *the proof-state guard fired zero enforcement interventions over all 87 of its decisions*, its entire lift running through the prompt channel with the model voluntarily releasing on cue, so "match that frequency" reduces to "tune V to a no-op." The control was served instead by a free counterfactual replay (`derivation-visible-guard-calibrate.js`, \$0, read-only): V's real decision function over the frozen baseline and proof-state transcripts would intervene 23/148 baseline and 9/87 proof-state (≈ 0.1/decision, an upper bound on live V), a self-check confirming faithful reconstruction (0 windowSize mismatches / 235 decisions). V was therefore run as the guard's true architectural twin, enforcement live, realized rate reported as a covariate. The fan **grounded 5/5** (`lantern-e2-visible-r1–r5`, all t20, gap 0): **Clopper–Pearson 95% lower bound 0.478, Fisher** $p = 0.042$ **vs the unguarded 4/10** (V separates *up* from baseline), **Fisher** $p = 1.000$ **for V < proof-state** (V matches the 4/5). V's realized enforcement was **7/100 decisions (**≈ 0.07**/decision; five blocks, two pushes; per-arm 3, 1, 1, 1, 1, on the chain's release turns t3/t4/t9)** against the proof-state guard's 0. Pooled into one twenty-arm contingency (`derivation-detector-split.js`, dry): unguarded **4 grounded / 6 early-pull / 0 decay-seating**, pacing **4 / 0 / 1**, visible **5 / 0 / 0** — both guards eliminate the baseline's dominant early-pull death entirely, the page-only one from features already in front of the model. The reading: the §6.13.10 guard's lift is *re-representation* of what the page already carries (the discipline to hold the next exhibit until the learner shows uptake), not new latent knowledge — refining and, for this mechanism, bounding the §6.13 claim that adaptive gains come from un-inferable signal. Caveats at declared power ($k = 5$): "V matches the guard," not "beats it" ($p = 1.000$); V separates from baseline only at exactly 5/5 ($p = 0.042$); V's enforcement was low but *not* zero, so the zero-enforcement arm is the proof-state guard and the strong reading is conditioned on that covariate; the visible proxy's *sufficiency* is world-specific (in world-002-lantern's chain visible uptake tracks latent distance; a world decoupling them need not let V track the guard); scope otherwise as §6.13.10. Each arm is registered for the standing pinned-Fable critic's notice (`commentary.md`), backfill deferred while Fable 5 is unavailable. Spend meterless (Max-plan CLI), serialized then — once the operator confirmed no quota constraint — fanned in parallel; the paid Step-1 loop ended on this result, and Step 2 (a branching-DAG world that tests whether the visible proxy still suffices when the chain forks) is human-gated under its own sanction. Body §6.13.11 added. Artifacts `exports/dramatic-derivation/loop/lantern-e2-visible-r{1..5}/` and the contingency `exports/dramatic-derivation/boundary-v-fan/detector-split-report.{md,json}`; registration + outcome in `ADAPTIVE-TUTOR-GENERALIZATION-PLAN.md`. No DB writes, no rubric scoring — file-based run exports.

v3.0.148 (2026-06-13) §6.13.10 guard-fan result re-projected onto a coarse failure-mode taxonomy: the headline shifts from the noisy success rate to a within-arm death-mode contingency. Following the generalization plan's free first step (`ADAPTIVE-TUTOR-GENERALIZATION-PLAN.md`, Step 3), the E4a detector-split classifier (`boundaryClassifier.js`) now also emits a coarse, mechanism-keyed *public failure mode* (`grounded` / `early_pull_death` / `decay_seating_death` / `aporia` / `unresolved`) projected from the six detailed split classes, plus a `guardState` axis read from each run's own recorded flags; `derivation-detector-split.js` builds a guard × failure-mode contingency table over any arm set. Run over the fifteen E2 arms (ten frozen `lantern-e2-real-r1`–`r10` + five guarded `lantern-e2-guard-r1`–`r5`), the table reads: unguarded **4 grounded / 6 early-pull / 0 decay-seating**; guarded **4 grounded / 0 early-pull / 1 decay-seating**. The reframing is the point — the within-arm observable that moves cleanly is *which* death mode the guard leaves standing, **not** the success rate (4/10 → 4/5, Fisher $p = 0.28$). The early-pull collapse (6/10 → 0/5, Fisher $p = 0.044$ on the incidence) is flagged *partly definitional*: the guard shapes releases to be tempo-solvent, so by construction it cannot emit a tempo-insolvent one. The substantive, non-definitional finding is the *relocation* — the single guarded death is the decay-seating class (E3's), absent from every unguarded arm, so fixing one death mode moves the residual failure one mechanism downstream rather than removing it. One body sentence added to the guard-fan paragraph (§6.13.10); no new claim — the relocation was already stated in prose at v3.0.147, this gives the auditable table behind it. Tooling + test only (`tests/dramaticDerivationBoundaryClassifier.test.js` extended); no paid runs, no DB writes. Artifact `exports/dramatic-derivation/boundary-e2/detector-split-report.{md,json}`.

v3.0.147 (2026-06-13) §6.13.10 E2 mechanism delta run: the pacing-guard fan grounded 4/5 (directional, not converged), and the keystone discriminator is corrected from "both early exhibits" to `p_chart` alone. The E2 caveat's licensed next step — a single `--pacing-guard` delta on the frozen p4 recipe, removing the early-pull latitude — was sanctioned ("do the guard fan") and run as a $k = 5$ fan (labels `lantern-e2-guard-r1`–`r5`, seed 1, early decay, pooled separately from the ten frozen replicates; registration committed pre-run as the E2-mechanism block of `ADAPTIVE-TUTOR-BOUNDARY-PLAN.md`, `ccfa8021`). It grounded **4/5** (`r2`–`r5` t20 gap 0) and disengaged one (`r1` round 7): **0.80, CP95 [0.284, 0.995], Fisher** $p = 0.18$ **vs the frozen 4/10 — directional only** under the pre-registered bands (the 5/5 convergence band, Fisher $p = 0.04$, went out of reach when the first arm died), so it authorises no further paid spend. Two results land. The guard removed the early-pull latitude entirely — **0/5 early pulls**, shaping not overriding (forced 0 / overridden 0 across all five arms, as in E3; the four survivors each played the canonical safe schedule `p_bearing` cue t4, `p_chart` cue t9, `p_key` t15). And pooling guard with frozen arms by `p_chart` placement gives the clean keystone discriminator — **7/7 grounded on cue, 1/6 pulled early, 0/2 unreached.** This **corrects the v3.0.145 entry and body**, which over-stated the discriminator as "holds *both* `p_bearing` and `p_chart` to cue": `r2` and `r3` ground despite pulling `p_bearing` early, and only 2/10 frozen arms held both, so `p_chart` placement *alone* is decisive (per revision-history discipline the v3.0.145 entry stands as the record of that version; the correction is made here and in the body, not retroactively in 3.0.145). The lone guard death `r1` is **not** an early-pull death — it played on cue, but its `p_bearing` decayed at t5 unrestored (no proof-debt guard in this single-delta arm), D rebounded 4→5, and the learner left at t7: the *decay-starvation* class E3 meets on late decay and E5 converts to grounding, here surfacing on early decay. Body §6.13.10 updated (E2 discriminator sentence, a new guard-fan paragraph, Reading, caveat). Paid mechanism loop ended on the directional read. Artifacts `exports/dramatic-derivation/loop/lantern-e2-guard-r{1..5}/`; outcome in the E2-mechanism §§ of `ADAPTIVE-TUTOR-BOUNDARY-PLAN.md`. No DB writes, no rubric scoring — file-based run exports.

v3.0.146 (2026-06-13) Rendering fix (no claim change): Two turn-shorthand tokens in the v3.0.145 entry below (the *bearing* and *chart* exhibits written with an at-sign before t3/t7) were being parsed by Pandoc as undefined citation keys, producing broken citation markers in the consolidated PDF. Escaped the at-signs so they render literally. No body text, claim, number, or table changed; the atlas modules (body slices) were never affected. PDF and atlas rebuilt and republished at this version.

v3.0.145 (2026-06-13) §6.13.10 E2 extended to a second pre-registered fan: the replication rate did not tighten — it diverged — but the mechanism reproduced exactly. The v3.0.144 fan was 3/5 (a Clopper–Pearson 95% lower bound of only 0.15, too wide to call a rate). Under the operator's "if the results don't tighten, feel license to loop on the mechanism" sanction, a second $k = 5$ fan of the byte-identical frozen p4 recipe (labels `lantern-e2-real-r6`–`r10`, seed 1, zero deltas, serialized + human-gated, registration committed pre-run as the E2-extension block of `ADAPTIVE-TUTOR-BOUNDARY-PLAN.md`) was pooled with the first to $n = 10$. Fan-2 grounded only **1/5** (`r9` t20; `r6`/`r7`/`r8` disengaged t12, `r10` aporia t8), so the **pooled rate is 4/10 = 0.40, CP95 [0.122, 0.738]**. The pre-registered convergence test fails on both clauses — fan-2's 1/5 is the declared "$\leq 1/5$ = divergence" boundary, and the pooled lower bound 0.122 does not clear fan-1's 0.15 floor — recorded as **divergence**: the frozen recipe does not ground reliably (the two fans land at 0.60 and 0.20). The substantive gain relocates from rate to *mechanism*, which reproduced across both fans: with the decay seed fixed, the grounding/death split is governed entirely by whether the tutor holds `p_bearing` and `p_chart` to cue. The four groundings (`r2`/`r3`/`r4`/`r9`) reached all eight exhibits with $D \to 0$; the six deaths plateaued at $D = 3$–5 with five (or four) exhibits reached, every one having pulled an early exhibit off cue. The two fans together made 14 fatal-cell entries (8 `bearing`@t3, 6 `chart`@t7) and 0 late-side moves on the E0 $\lambda = 0$ map; `r9` is the structural twin of fan-1's `r4` (both held both early exhibits on cue), and `r6` of fan-1's `r1`/`r5`. The 4/10 is therefore the base rate at which an unguarded codex tutor happens to hold both early exhibits — which names the pacing guard (`--pacing-guard`, already grounding in E3/E5) as the theory-driven mechanism delta, registered and pooled separately, not folded here. Body §6.13.10 updated (title, E2 paragraph, table, Reading, caveats); the v3.0.144 entry's 3/5 stands as the historical record of that version. Conduct miner refreshed (`derivation-mine-conduct.js`, 42 real arms). Artifacts `exports/dramatic-derivation/loop/lantern-e2-real-r{6..10}/`; registration + outcome in the E2-extension §§ of `ADAPTIVE-TUTOR-BOUNDARY-PLAN.md`. Paid frozen-replication loop ended on divergence; mechanism-loop arm human-gated under fresh sanction. No DB writes, no rubric scoring, no new main-harness estimates — file-based run exports.

v3.0.144 (2026-06-12) §6.13.10 added: boundary cartography — a replication fan turns one $n = 1$ verdict into a 3/5 rate, a pure-computation corridor map explains the two deaths, and two single-delta mechanical guards move the surviving arm from tempo starvation to decay starvation to grounded, the proof-state guard staying conduct-blind by audited construction. A separate plan note (`ADAPTIVE-TUTOR-BOUNDARY-PLAN.md`, Gate-0 sanctioned through E1; live arms E2/E3/E5 each under their own quoted sanction; E4a dry) works the §6.13.9 p5 exposure — "the safe corridor is narrower than the licensed window" — from the outside in. **E0** (`derivation-corridor-map.js`, no LLM calls, D + death rule the production `slope.js` paths) enumerates the 125 licensed release schedules on world-002-lantern: 40 survive (a **32% corridor** at same-turn adoption, 19% at $\lambda = 2$), survival monotone-late (safe sets `p_bearing` $\{t4\text{–}t6\}$, `p_chart` $\{t8, t9\}$, `p_key` $\{t15\text{–}t18\}$; the $t4 \rightarrow t9$ leg zero-slack; holding `p_bearing` late widens +13 points, pulling it early costs −13). **E1** (`derivation-mine-conduct.js`, 37 real arms / 281 releases) measures the prior the map omits: adoption same-turn ($\lambda = 0$ on 279/281), directors never off-cue (0/165), and **every tutor deviation early** (20/281, nine at −1 + eleven at −2, rate 0.17, zero late) — the real release latitude anti-aligned with the corridor. **E2** ($k = 5$ byte-identical replicates of the frozen p4 recipe, one sanction, serialized + gated): **3/5 grounded t20 gap 0, 2/5 disengaged t12**, Clopper–Pearson 95% lower bound 0.15, kill rule ($\leq 1/5$) not met — 7 fatal-cell entries / 0 late-side across the fan (the early-pull prior landing in the marked cells; the decay/repair channel deciding which early pulls survive), harness checks clean so deaths are reported, not re-rolled. **E3** (`--pacing-guard`, single delta on the p5 late-decay config): **disengagement t24** — survived past p5's t12 tempo cliff (24 decisions audited, 0 blocks / 0 forced / 0 overrides: pacing made safe by shaping, not override) but relocated the failure to *decay* starvation (`p_bearing` dropped t17, never restored, learner names the gap t24, one premise short). **E5** (`--proof-debt-guard`, stacked on E3): **grounded t20, gap 0** — converts E3's decay-starved failure under the identical decay replay; at t18 the guard detected one debt (`p_bearing`, D $2 \rightarrow 1$ on restore), the tutor restored it (`forcedMoves: 0` — licensed, not compelled), t20 the chain closed on a learner assertion; pacing stable, so the effect attributes to proof-state repair. **Non-leak boundary audited, not asserted** (`derivation-leak-audit.js`, 6 grounded arms): no proof-distance arithmetic in any tutor line, the secret predicate never voiced by the tutor (it stages; the learner asserts), positive control confirming the harness audit ledger *does* carry the arithmetic the conduct-blind tutor never sees — two distinct paths; the proof-debt view is `{premiseId, surface, sinceTurn}` only, raw board / corruption ledger / proof path / secret never entering the acts-mode prompt. **E4a** (`boundaryClassifier.js`, dry) sorts every archived arm by the same arithmetic: p2/p5/E2-deaths tempo-starved, E3 decay-starved lucky-leap, p3/p4/E2-survivors/E5 grounded controls. A reporting-layer provenance fix landed with the audit (`renderProofProse` names the staged twin from the ledger, not the last-in-file same-fact exhibit; regression in `tests/dramaticDerivationCritic.test.js`; no verdict/ledger change). Reading: the plan buys a rate (grounding replicates 3/5, two-death tail mechanically explained), a movable death class (one audited delta per step), and a worked conduct-blind/proof-aware guard — the §6.13.7 visibility asymmetry made operational, the harness owning the arithmetic and the one crossing bit being the id-and-surface of an exhibit the tutor already played. Caveats: $n = 1$ per E-arm except E2's $k = 5$ (shared seed + decay schedule, so run-to-run model reliability at a byte-identical setup, not a sample over worlds/seeds; 0.15 a floor on that one quantity); attribution compounds (E5 = stack, no guard-alone arm); formal verdicts throughout (conduct-blind by audited construction — "adapted" = played a licensed restore, inferred nothing); E4a dry post-hoc; corridor map one world; the four §6.13.9 standing caveats carry forward (staged-pool positive side twice-unexercised, D-blindness unacted-on / meter frozen, exposure caveat, false-restore cost measured once); paid loop ended, larger replication + guard-alone isolation human-gated; spend meterless throughout (≈ 88 calls / 20 min per grounded arm, serialized k-fan projection tabled by `derivation-cost-report.js`). Artifacts `exports/dramatic-derivation/{loop,boundary}/`; registrations/outcomes E0–E5 of `ADAPTIVE-TUTOR-BOUNDARY-PLAN.md`. No DB writes, no rubric scoring, no new main-harness estimates — file-based run exports.

v3.0.143 (2026-06-12) §6.13.9 extended: the battle-test arm (p5) — one licensed delta (decay start t1 → t17) aimed at the staged pool's positive path; the run dies at aporia t12, five turns before the window opens, and exposes the hazard side of release authority instead. Registration §14 of the plan note (committed before the run; mock smoke first, which staged the exact registered scenario — `m_key` garbled to the met-on-stage false form `onlyKeyTo harlowPoint senna` with the pool-invariant bearing fact plain-deleted beside it, the bent-first clause's first case) licensed one paid arm, `lantern-p5-mutation-on` (the p4 command with `startTurn: 17`), to make the staged mutation pool fire live: from t17 the `p_key` release fills the pool with both-position alternates and the first grounded fact garbles with \$ \$0.98 by t19 by the registered arithmetic. The arm never got there: **aporia at t12** ("no progress over 6 turns"), zero slips, zero mutations, fidelity 1.00 — the battle test NOT REACHED, and the positive side now unexercised twice by opposite mechanisms (p4: every pool empty at draw; p5: no draw ever made). The death is a tempo death under full information: the tutor pulled `p_chart` −2 (t9 → t7) on the learner-responsive signal (the learner had twice named the chart), the pull paid locally (the false trail collapsed the same turn, zero latency) — but the next D-mover (`p_residue`, t13) is the director's and the director never bends, the bend authority covers the tutor's own exhibits only, its next own exhibit (`p_key`) was out of window reach until t15 — the release-decision record shows windowSize 0 for every turn t8–t12, no licensed rescue — and six turns at $D = 3$ hit the aporia window exactly, one turn short of relief. On-cue play survives the corridor (p4, same world and seed, is the live proof, plateau four); within the ± 2 license only t8/t9 survive under this learner's same-turn adoption — the safe corridor narrower than the license, three of five licensed moves fatal — and the tutor's prompt carries the full calendar with `via` labels plus the house-clock warning that names the death verbatim ("an early claim spends a future advance now"). The discretion p4's critic praised, and asked to extend to the finale, selected the fatal move on the learner signal alone. Second half of the finding: the books were clean at the death — plots 3/3 disciplined, audits kept 15 / justified 1 / drift 2, the one off-arc verdict binding a throughline revision on schedule, and the final throughline reckoning **7/7 kept at the moment of death**: where p2's reckoning read as an obituary (drift on exactly the death-naming clauses), the plan vocabulary owns no tempo-debt clause, so the fatal decision passed as kept, not drift — the plan layer cannot flag what it has no word for. The learner was never confused, only unsupplied (named the burned-lamp threshold four straight turns; by t12 drafting the sentence Rule 2 would permit and the one it would forbid) — the aporia measured supply, not understanding, which is the critic's second new recommendation made flesh (tell a learner out of ideas from a learner out of evidence — "only the first is aporia; the second is a starved house"). Critic's notice: cast credited, calendar blamed — recorded as a half-right divergence (the calendar as written survives this corridor; the pull stretched the desert past the window); two new recommendations (re-time `p_residue` to land by t11; the starved-house detector split) recorded **UNLICENSED**, p4's two likewise untouched through this arm; **the 2026-06-12 battle-test sanction is spent in full and the paid loop is ended again.** Seven-arm table (p5's decay row empty by truncation, its fidelity 1.00 an undisturbed board, not hygiene); status-and-caveats updated (positive path twice-unexercised, any further attempt a fresh sanction; release-latitude hazard joins the watch-items). Artifacts `exports/dramatic-derivation/loop/lantern-p5-mutation-on/` (with a Fable critic's notice); registration and outcome §§14, 14.5 of the plan note. No DB writes, no rubric scoring, no new main-harness estimates — file-based run exports.

v3.0.142 (2026-06-12) §6.13.9 extended: decay-channel hygiene (p4) — the p3 critic's two recommendations licensed under fresh sanction, one bundled delta, one paid arm; the hygiene holds by silence and the six-arm table closes the dramatic-derivation paid loop again. The p3 notice's located defect sat in the instrument, not any dial: the mutation sampler drew false-form constants from the whole world, twice writing the secret-holder's name onto the learner's board before any exhibit had staged her (t3, t8), and the never-named mutation scaffolded the refused t18 leap. Registration §13 of the plan note (committed before the run; the two fixes bundled because the critic prescribed them jointly) licenses both: the **staged mutation pool** (`pool: staged` — swap constants drawn only from met-on-stage facts, background + premises released so far; empty pool falls back to plain deletion; the world-pool default preserved so archived runs replay byte-identically; contract suite `tests/dramaticDerivationCorruption.test.js` pins both pools and the divergence) and the **bent-first clause** in the repair charter (an exhibit lost and an exhibit garbled together: mend the garbled one first — an absence stalls the inquiry, a false form argues for it). Paid arm `lantern-p4-hygiene-on` (the p3 command + the staged pool): **grounded anagnorisis at t20, gap 0 — the second dialed-up arm in a row to ground, same turn and gap as p3, on the cleaned instrument.** Endpoint 1 (verification-shaped) MET in the strong form, *by silence*: zero mutations emitted; all four slips fell back to plain deletion because every staged pool was genuinely empty at draw time (one slipped predicate appears on no other fact in the world; the other two predicates' alternates live only on premises released t15/t20) — the exact t3/t8 whisper sites from p3 rendered mute, verified against the world rather than just the ledger. Registered consequence realised to its limit: with alternates unreleased until late, "mutate" under a staged pool *means* "delete" for the whole early game — the hygiene is proven by elimination, not exhibition; no false form ever stood, and the bent-first clause never had a case (charter text not yet exercised in paid conduct). The repair clause ran its cleanest economy: 2/2 load-bearing fades repaired at mean latency 2.5 turns under read-back discipline (the learner itself named the first loss; the superego's bare-re-entry conversion produced the read-back licensing the second), zero false restore claims (vs p3's one benign); 2 unrepaired both mirror-side, both unreported, both mooted by the shutter release — the registered jurisdiction residual in benign form (fidelity floor 0.78 reads mooted-mirror rot as damage — meter conservatism, not conduct fault). Books unchanged in meaning: plots 4/4 disciplined (act-2 again naming the slipping exhibits as holds), audits kept 20 / drift 3, throughline final reckoning **7/7 kept**, releases 7 on cue / 1 declared early (−2, the learner had named the key as the next missing conjunct), zero bare re-entries, zero watcher fires without grounds, `fabricatedFacts` empty, superego 5/20 fires all converting within-turn (1 rut-fire, 3 re-entry conversions incl. the repair-licensing one), figures invariant (4 distinct, erotema 40%, switch 0.79); unlike p3, no early assertion to refuse — the learner asserted only at t20. The §6.13.9 table gains its sixth column (+ decay hygiene), with the note that p4's decay row is conduct-comparable only (the registered draw-stream divergence: same seed, different slip schedule after the first empty-pool fallback — 4 slips vs p3's 7, and the two unrepaired slips saturated `maxConcurrent` from t8, quieting the channel's back half). Critic's notice (Fable, pinned): the hygiene works; new located defect the **held finale** — the learner named the wanted proof at t17, the final exhibit held to schedule until t20, t18–19 sagging into near-verbatim restatement, recognition "compressed, adoption and assertion in a single breath"; two new recommendations (extend early-play discretion to the final exhibit on a named missing conjunct; let the charter treat mooted mirror matter as abandonable) recorded **unlicensed**; **the 2026-06-12 sanction is spent and the paid loop is ended again**. Caveats: $n = 1$, sixth arm on one seed, descriptive; the two fixes attribute only as a bundle; sixth realisation of the exposure caveat; the staged pool validated only on its closing side (a mutation from a non-empty staged pool, and the bent-first clause it would summon, not yet observed — a world with early-met constants or a later decay start would exercise both; neither licensed); D-blindness caveat stands unacted-on, the meter frozen. Build commit precedes the run; artifacts `exports/dramatic-derivation/loop/lantern-p4-hygiene-on/`; registration and outcome §§13, 13.5 of the plan note. No DB writes, no rubric scoring, no new main-harness estimates — file-based run exports.

v3.0.141 (2026-06-12) §6.13.9 added: two-layer planning and the repair clause — the plan names the race it loses, and a one-step rule wins it; the dramatic-derivation paid loop ends on the section's first dialed-up grounded play. Two registered steps close the §6.13.8 seam (its diagnosis: charter text binds event-triggered procedure, not clock arithmetic; the designed act-plot mechanism gains a concrete target). **Step 1, the planning stack** (registration §11 of `notes/poetics/2026-06-11-desire-multiturn-strategy-plan.md` + the §11.5 operator-directed pre-run amendment, both committed before the paid arm; contract suite `tests/dramaticDerivationPlot.test.js`): per-act plots (hold_by_end / withhold / friction / fallback, conduct-only construction, read back every turn) audited clause-by-clause at act close by the superego (kept / justified_deviation / drift) with binding verdicts, plus the two-layer amendment — a whole-play throughline (arc / hold_to_end / risk / salvage) committed at t1 above the act plots, course-above-lesson at every turn, its on_arc/off_arc verdict riding the existing audit, zero new LLM calls. Paid arm `lantern-p2-plot-on` (charter-v2 configuration + the stack; control = the v2 arm): **aporia at t8 — the third death on the same turn of the same seed — with the plan's own signature on it**: machinery clean (plots 2/2 disciplined, audits 2 incl. run-end, throughline disciplined, arcs 2/2 on_arc, 0 unscored), calendar conduct genuinely different (D held flat at 4 vs the predecessors' regress-to-5 — the `p_chart` rescue *spent* at t7, −2 declared, exactly offsetting the t6 `p_bearing` fade), the audit channel biting (act-1 drift bound act-2's plot; run-end drift on exactly the death-naming clauses incl. the throughline's salvage), and a one-turn-wide located failure: the learner reported the silent `p_bearing` deletion twice (t7, t8 — the play's final line a restore request) while the tutor's one actionable turn consolidated the director's `m_shutter` release instead of the re-staging its own plot had pre-committed — planning produced the right plan; conduct lost the race between new matter and repair. **Step 2, the repair clause** (registration §12, fresh sanction, one paid arm; the critic's second recommendation — D's blindness to elimination progress — recorded as a standing instrument caveat and deliberately not acted on, the meter frozen): the confrontation obligation's one exception running the other way — a learner-NAMED loss is itself the read-back, so the tutor's next turn re-stages the named exhibit (intent `restore`) before any new matter, one report licensing one restoration; three epistemic positions held apart with no leaks (harness states mechanical facts only; superego verifies the claimed license against the learner's last line, rejected claims converting to `unconfronted_reentry` — repair one turn later through the front door; only the post-hoc audit reads the hidden ledger, evaluated before-the-move so a successful restore cannot erase its own evidence); engine untouched (`restore` heals by the §6.13.7 standing rule; only `confront` is excepted); suite `tests/dramaticDerivationRepair.test.js`; the mock cast never emits `restore`. Paid arm `lantern-p3-repair-on` (the p2 command + `--repair-clause`): **grounded anagnorisis at t20, gap 0 — the first dialed-up arm on this world to ground, the same landing turn as the dial-free control.** The §11.6 choreography re-ran and resolved the other way (t7 loss report → t8 restoration with `m_shutter` riding alongside, D 4 → 3 strict — three arms died on that turn holding that lever; this arm spent it); the clause ran as an economy (six restores answering the learner's line: five true reports all repaired, the path-side ones doubling as the clock's strict drops; one false report costing one benign re-statement turn), the rejection path worked once end-to-end (t15 claim refused → confrontation → t16 licensed repair), zero bare re-entries, zero watcher fires without grounds, superego 2/20 both converting the action; the planning stack kept its own promises (plots 5/5 disciplined with the act-2 plot naming the slipping exhibits as holds; audits kept 24 / jd 3 / drift 1; arcs 5/5 on_arc; final reckoning **7/7 kept** vs p2's 5/2 — the dead arm's drift was its unexecuted repairs); the t18 one-premise-short assertion was refused (`lucky_leap`, not ended), mended t19, grounded t20; decay 7 slips / 5 tutor repairs / 0 re-adoptions / 2 unrepaired (both mirror-side mutations, both unreported — outside the clause's jurisdiction by design). New five-arm cumulative table (control / +dials / +charter v2 / +planning / +repair clause, one registration delta per column). Reading: plans are instruments of legibility — under a one-step execution rule they stop drifting — but the conversion of knowledge into act happens in the rule, not the plan; re-adoption now 0 in every arm (once with 12 post-window turns of opportunity), closing the learner-side channel per §6.13.7's visibility asymmetry. Critic's located defect in the decay channel (the mutation sampler twice whispered the secret-holder's name pre-staging; the one never-named mutation scaffolded the refused leap); two recommendations recorded unlicensed; **the paid loop under the plan note is ended**. Caveats: $n = 1$ per arm, five arms one seed, descriptive; attribution bundled down the stack (planning layers co-land; the clause tested on top — the registered reading is the §11.6 mechanism confirmed, the only delta the clause); fourth and fifth realisations of the exposure caveat; F-min 0.64 and the t13/t18 overreach clusters are watch-items. Artifacts `exports/dramatic-derivation/loop/lantern-p{2-plot,3-repair}-on/`; outcomes in the plan note's §§11.6, 12.5. No DB writes, no rubric scoring, no new main-harness estimates — file-based run exports.

v3.0.140 (2026-06-12) §6.13.7 extended: stage v2 (acts, the bounded learner, the curated theory, the reconstructing tutor) and two registered probe pairs — explicit theory reconstruction is redundant on both recall and revise; the explicit-other-modelling family reaches six nulls. The §6.13.7 visibility contrast left a designed gap: the told arm read a harness-handed ledger; no arm asked whether the tutor can *build* the missing signal given an explicit reconstruction job. Stage v2 (design + both scoring plans `notes/poetics/2026-06-11-act-bounded-learner-design.md`) implements four refinements: acts with director-judged termination (min 3 / max 8 harness-guarded; one strategic direction opens each act; release mechanics untouched), the bounded learner (context = own theory store + current act only — deletes §6.13.7's stale-residue mask by construction), the learner-curated theory store (adopt/retract/revise ledgered; decay gains `mutateShare`: a slip may mutate an entry in place to a plausible same-type false form — formally a deletion plus a false belief, harness-implemented, seeded), and the reconstructing tutor (both arms: learner store never shown to ego or superego — the conduct condition made total; ON `--reconstruct` adds a per-turn believed-held/missing/mistaken reconstruction under a supplement mandate). Recall pair `bounded-v2-probe` (nocturne; 0/13 premises mutable, a structural fact recorded at registration, so mutate draws degrade to delete; scoring fixed mid-pair, disclosed): both arms grounded (t38/40 vs t32/40, gap 0, 11/11 releases each); the headline repair-rate gap (0.89 vs 0.60) dissolves under proof-path triage — **every proof-critical slip repaired in both arms at similar latency** (4/4 vs 3/3); the difference is mirror housekeeping; the blind OFF tutor repaired deliberately (reach-backs 2/3; v1's blind repairs were 100% incidental) — the legibility lever was the stage, not the channel; ON reconstruction: detection 12.3%, missing-Jaccard 0.171, coupling 2/8 and 2/9, its one visible influence picking a mirror prop over the down proof-critical premise. Revise pair `lantern-revise-probe` (lantern, 5/10 mutable, mutate-share 1.0; scoring fixed and committed before either arm — clean pre-registration): the closest to a matched pair the harness has produced (identical verdict turn t20/26, identical act spans, gap 0, 8/8 releases both); **both arms retracted 100% of false-belief debts** (5/5 vs 4/4), OFF *faster* (mean latency 1.75 vs 3.6); the t18 `m_key` mutation drew the same false form on the same turn in both arms and was spontaneously struck at t19 in both; mistaken-detection 2/17 (11.8%) vs missing-detection 12/37 (32.4%) in the same arm — the channel nearly blind exactly where its theoretical edge was claimed (mutations do not self-announce); no inference built on a false form in either arm. Kept asymmetry: the two debts sort by visibility — the learner polices visible false forms (spontaneous strikes), the tutor restores invisible deletions (reach-backs across act boundaries; re-adoption 0 in all four arms); the theory channel changes neither half. Family count six (§6.8.5, §6.8.6, §6.10, §6.13.6, reconstruction-recall, reconstruction-revise); positive establishment: default mechanisms + learner board hygiene close both debt types under aggressive forgetting; formal layer invariant through the stage revision (38/38 releases on cue). Instrument note: the revise ON critic's audit line exposed a real premise-keyed defect in the corruption panel (a re-mutation orphaned an earlier standing false form's debt and dropped its retraction); fixed by matching on false-form identity, regression `tests/dramaticDerivationCorruption.test.js`, both arms' artifacts regenerated, no registered quantity moved — the second critic QA catch. §6.13.8's opening release count (94/12 → 132/16) and its configuration + control parentheticals updated to point at the in-paper treatment. Caveats: $n = 1$ per arm, descriptive; registration quality stated per pair (mid-pair disclosed vs clean); exposure caveat carried (same seed, different slip schedules once repairs free decay slots). Artifacts `exports/dramatic-derivation/loop/bounded-v2-*`, `lantern-revise-*`. No DB writes, no rubric scoring, no new main-harness estimates — file-based run exports.

v3.0.139 (2026-06-12) §6.13.8 extended: charter v2 — the stall rule stated as conduct; the two clauses dissociate completely. One registered follow-up arm (`lantern-p1-dials-on-v2`; §10 of `notes/poetics/2026-06-11-desire-multiturn-strategy-plan.md`, committed with clause text pinned by string assertions before the run; one paid arm, a repeat death pre-declared reportable as-is) added two text-only clauses to the tutor's charter, no mechanics changed: **the house clock** on the release-authority dial (the world's six-turn stall rule stated as conduct — window interpolated from the world spec, D and the clock's anchor hidden, the rescue named: "an exhibit in your window is a rescue — spend it") and **treatment follows diagnosis** on the confront dial (the v1 critic's prescription routed through sanction + registration: an absence-exposing read-back obligates the licensed re-entry next turn; the run stays critic-feedback off). Identical configuration and seed. Outcome: **aporia at t8 — same verdict, same turn, same D series, and the same four windowed calendar decisions as v1**; the pre-declared reading applies (knowledge of the rule did not rescue conduct). Within the repetition the clauses dissociated: the **clock clause null on every observable** (the v1 deviation re-made with a near-identical declared reason; `p_chart` held in silence at t7 AND t8; no reason cites pace, rhythm, or the clock), the **treatment clause bound exactly** (t6 confrontation exposes `m_key` carried *bent* since t3 → covered re-entry t7, latency 1, adherence 1/1 on the registered §10.3.2 endpoint; first confront-prompted retraction in any arm). The superego's `unconfronted_reentry` watcher fired for the first time in any arm (t5 bare re-entry draft → converted to a confrontation within-turn, §6.13.4's break pattern) and the converted read-back initiated the treatment cascade — it is where the learner produced the bent key rule, naming **Senna**, the secret's holder, as noise (second whisper "southStack" struck unprompted t8): charter + criterial backstop held the envelope, the charter alone did not. t7 was a genuine clause collision (obligation vs rescue; procedure won, defensibly); t8 is the pure clock-blindness turn — **two rescues verified live against the detector** (`p_chart` claim D 5 → 4; covered `p_bearing` re-entry, license standing from t5, the loss flagged by the learner itself at t7), neither played, the tutor consolidating the director's `m_shutter` release instead. Detector subtlety kept on the record: the v2 critic's precedence prescription, replayed against `detectStall`, contains an arithmetic slip — the trajectory records t6 pre-fade, so a *prompt* t7 repair records 4 → 4 (no strict drop, no reset) and the play dies regardless; the one-turn-late repair it faults is the variant that saves. Reading: charter text binds **event-triggered procedure** (one-step grammar, the form of §6.13.4's criterial jurisdictions and of every rule that has bound in §6.13) and fails to bind **clock arithmetic** (hidden-state window, maintained counter, counterfactual calendars) — the per-turn tutor does not plan; the designed-but-unbuilt act-plot mechanism (plan §5) gains its first concrete target. Caveats: $n = 1$ per arm, three arms on one seed, descriptive; the dissociation is within-arm (opposite compliance under one charter — evidence of kind, not a clean ablation); third realisation of the exposure caveat (the t7 repair freed a decay slot, `m_post` slipped the same turn); re-adoption 0 in a third truncated arm. Artifacts `exports/dramatic-derivation/loop/lantern-p1-dials-on-v2/` + the digest's follow-up section + `lantern-p1-contrast-output-v2.txt` (contrast reader extended with the treatment-adherence endpoint). No DB writes, no rubric scoring, no new main-harness estimates — file-based run exports.

v3.0.138 (2026-06-12) §6.13.8 added: the calendar becomes conduct — release authority and a confrontation obligation, pre-registered, one paid arm; the tutor's first calendar deviation sets the stall clock that kills the play. Two dials built mock-first on the dramatic-derivation stage (24 contract tests + a zero-cost mock loop exercising the full causal cycle; scoring frozen as §9 of `notes/poetics/2026-06-11-desire-multiturn-strategy-plan.md` and committed before the paid run): **release authority** (the three via-tutor exhibit cues become

± 2-turn windows, the five via-director cues fixed; the tutor claims early or holds late with a one-line declared reason; the harness validates window membership, force-plays at the hold limit, ledgers every per-turn decision; the mock plays deviation-zero so any paid deviation is model-authored) and the **confrontation obligation** (re-entering a staged exhibit requires a prior confront move — the learner reads it back, the tutor restates nothing; one confrontation licenses one re-entry; confront moves are excluded from §6.13.7's within-turn repair rule; the superego gains the criterial `unconfronted_reentry` jurisdiction). One paid arm `lantern-p1-dials-on` (lantern world, act-bounded staging, conduct-visibility decay $\{0.75, 1, 2, 1\}$ mutate-share 1.0 seed 1; codex director/tutor/superego, Sonnet learner, Fable critic) vs the committed `lantern-revise-off` control (same configuration, dials off; grounded anagnorisis t20, 8/8 releases on cue). Result: **aporia at t8** with a fully ledgered causal chain — the t3 early claim of `p_bearing` (window t2–t6, declared reason on record) moved the last D-drop and the premise's decay exposure one turn forward together; the t6 fade returned D to 5; the t3..t8 six-turn tail contained no adjacent drop (the run's only drop sits one pair outside); curtain one turn before the control's scheduled t9 `p_chart` release — the realized counterfactual that survives to t20. In the death window the tutor held both rescue levers unspent (`p_chart` claimable at t7/t8, held in silence; the t7 confrontation's licensed re-entry of slipped `m_key` never played — critic: "diagnosis twice, treatment never"). Mechanism demonstrations: declare-and-validate deviation clean end-to-end (1 early claim, reasoned; 0 invalid, 0 forced); the obligation **internalised from charter text alone** (0 bare re-entry drafts, 0 watcher fires, vs §6.13.4's 14 corrective fires); read-backs distinguish a held exhibit (t5 `p_bearing`) from a lost one (t7 `m_key`); the t7 read-back captures **confabulated history** (the learner denies ever having adopted what the ledger records at t2 — decay ate the fact and the memory of the fact; a step past §6.13.7's stale residue); the repair-exclusion held. Endpoints: re-adoption primary unresolved (one turn of opportunity); confront-prompted retraction class empty (the run's one retraction, latency 1, spontaneous, predates the first confrontation); deviation usage used/coherent/fatal; no-degradation guard failed with attribution to calendar-conduct plus unspent rescues, not the confrontation tax (fine-grained instruments held: figure switch 0.71, $F \geq 0.71$, false form struck at latency 1). Reading: the release calendar is load-bearing against the world's stall clock and the new authority is blind to the constraint it bends — a peripeteia authored by the mechanism rather than by any role; the critic's prescribed repair clause alone would not have saved the play (`m_key` is mirror-side, off the proof path — its repair never moves D); the killer was clock-blindness. Caveats: $n = 1$ vs $n = 1$ descriptive; dials co-landed by design (not separable within-run); truncation shortens every ON denominator; the exposure caveat realised via deviation-shifted decay (economies comparable, per-slip schedules not); the critic's counter-reading preserved (the tripwire cannot tell elimination from drift); harness observation: the mutation sampler drew the secret's own name into a false form (struck at latency 1, channel whispers). Artifacts `exports/dramatic-derivation/loop/lantern-p1-dials-on/` + `lantern-p1-{digest.md, contrast.mjs, contrast-output.txt}`. No DB writes, no rubric scoring, no new main-harness estimates — file-based run exports.

v3.0.137 (2026-06-11) §6.13.7 added: the unreliable learner — harness-implemented forgetting, and the first explicit-information channel in the investigation that is *not* redundant. New run-level decay condition built mock-first (`services/dramaticDerivation/corruption.js`, `--decay '<json>'`): released-and-adopted premises slip off the learner's grounded board on a seeded schedule (marked never deleted; filtered at the engine's single validity choke point; gone from the learner's grounded *and* released views so re-hearing is not free, while staged prose remains in context so re-derivation is a measured recovery path); tutor moves targeting a slipped premise restore it within-turn (incidental consolidations count by design), learner re-adoption ledgered separately. The condition is harness-owned, never prompt-roleplayed (the §7.9 costume rule), and completes §6.10's manifest≠latent program structurally: the learner cannot report what it forgot because it does not know. Registered visibility contrast (`UNRELIABLE-LEARNER-PREREG.md`, frozen pre-run with two dated pre-result amendments, incl. director Opus→codex with one superseded run preserved out-of-scan): arm A *told* (SLIPPED block in the live tutor ego prompt) vs arm B *conduct* (block suppressed, all else identical); decay $\{0.75, 1, 2, 1\} \times$ seeds $\{1, 2, 3\} \times 2$ worlds (nocturne charter v2 / withercombe charter v3, casting identical across arms) = 12 serialized attended paid runs (codex director/tutor/superego, Sonnet learner). Endpoint design pinned by a mock anchor grid (4 scripted policies $\times$ 5 seeds $\times$ 2 worlds): latency disqualified (surfing completes 10/10 at latency 16), bare rate disqualified (never-looks floor 0.33 via incidental lastRelease touches) → primary = pooled per-slip tutor-repair rate paired with a selection signature (non-lastRelease target on an authored proof path); scoring fully mechanical and architecture-independent (`scripts/score-unreliable-learner.js`, no LLM judge). Result: told 49/57 = 0.860 with 28 selected repairs (57%, 4.67/run) → H1a and H1b pass; conduct 7/19 = 0.368 with **zero** selected → H2 supported, gap +0.491, run-level bootstrap 95% CI $[0.313, 0.746]$ (10k resamples, seed 20260611, 0 reversals) — the registered redundancy reading ($B \approx A$ as a fifth null) is ruled out. Conduct sits numerically at the never-looks anchors (0.368 vs floor 0.33; 3.17 decay/run vs surfing 2.4); all 7 blind repairs classify incidental; failure textures: stale residue (the learner cites slipped facts still in its prose context, masking the loss) and the lucky-leap-blocked aporia (learner asserts S unforced $×$7–8 while a true-path premise stays down at least 30 turns). Told pattern is triage not checklist: selected repairs on proof-path premises (several on the recognition turn itself), the mirror-side premise abandoned late in all six told runs across both worlds. Formal layer invariant under decay: 94/94 releases on cue, registered halt never fired; completions 4/6 told vs 0/6 conduct. Reading: the four redundancy nulls re-encoded inferable signal; the decay ledger is harness-owned hidden state structurally absent from the dialogue — the entire deficit is detection, not response, and the boundary sharpens to *new checkable signal at the point of action.* Caveats: distributional not matched-pair (denominators differ mechanically; bootstrap respects it); shows blind detection did not happen, not that it cannot (a criterial detection jurisdiction, charter v4, is design-only and unsanctioned). §6.13.6 gains a one-sentence forward pointer. Supporting kit from the same phase: episode replay + matrix runner (`scripts/run-derivation-{episode,matrix}.js`, design log `notes/poetics/2026-06-10-unreliable-learner-design.md`); results log `notes/poetics/2026-06-11-unreliable-learner-results.md`. No DB writes, no rubric scoring, no new main-harness estimates — file-based run exports under `exports/dramatic-derivation/`.

v3.0.136 (2026-06-10) §6.13.6 added: the registered frontier-coupling step run and closed — a quasi-logical theory of mind verified end-to-end, terminating on a measured precondition null. The §6.13.5 registered next step (a superego jurisdiction computed on the learner's inference frontier, learner-side movement as the endpoint) was built mock-first and run under its own pre-registered progress bar (P1 learner movement over $\geq$ 2 evaluable ON/OFF pairs; P2 zero-mismatch detector audit; P3 formal-layer invariance; P4 stall-targeting revisions; overreach/tutor-voicing guards; four decision rules; `notes/poetics/2026-06-10-stall-watcher-quasi-logical-tom.md`). New instrument: a learner derive channel validated against the post-adoption closure; a harness-computed inference frontier (derivable, non-base, non-pattern, unvoiced facts with grounds and stall ages) in the superego's view; charter v3 adding the stalled-inference jurisdiction (age $\geq$ 3, untargeted by the last two turns or the draft) to the unchanged rut clause; named run groups; and the critic folded into the loop (final notice paragraph of the latest same-group arm resolves into the director+superego charters, never the tutor ego). Three serialized paid arms (codex director/tutor/superego, Sonnet learner, Fable critic): bitterwell OFF, withercombe OFF (+counsel), withercombe ON (charter v3). The no-stall-room decision rule fired once: `config/drama-derivation/world-004-withercombe.yaml` (+ tutor script + worlds-test entry) was purpose-built for stall room (depth-3 diamond, off-spine exculpation steered past by the script, no twin-fact spares) — and the floor did not move. Result: across 2 worlds, 4 watchable intermediates, 7 node-arm observations, the learner voiced every derivable fact the turn its premises completed (latency 0 throughout); stall was never due; P1 unsatisfiable at floor, P4 never reached. The instrument held the zero-mismatch standard (P2): mock causal chain exact (3 deterministic fires → 3/3 stall-targeting revisions → audit clean); the live charter-v3 arm fired the new jurisdiction zero times across 19 watched turns with a clean audit (rut: 3 due → 3 fires → 3/3 within-turn breaks; 0 missed, 0 false) — the restraint half answered on a live stage. P3 held everywhere (3/3 grounded at the planned forcing turn; 25/25 releases on cue); both guards clean. Reading: the fourth instrument in the investigation to land on the ToM-redundancy motif (§6.8.5, §6.8.6, §6.10) — the arithmetic is buildable and correct; the deficit it would correct never occurs for a strong learner on a dramatically legible board, and the derive contract itself saturates voicing (measurement elicits the behaviour the watcher polices). §6.13's header parenthetical trimmed (the bar-met-in-full claim describes §6.13.4's conduct bar, not the new frontier bar); §6.13.5's next-step sentence now points to §6.13.6. No DB writes, no rubric scoring, no new main-harness estimates — file-based run exports under `exports/dramatic-derivation/loop/`.

v3.0.135 (2026-06-10) §6.13 added: the dramatic-derivation generative arc — can the tutor govern its own conduct? Three closed phases, ending in a pre-registered robustness bar met in full. The arc relocates the adaptation question from reading interiors (closed by §6.10–§6.12) to governing conduct on a stage built for it: one long tutoring drama whose plot is the proof DAG of a contingent secret S, releases harness-enforced, learner world-blind with a visible acquired ABox, success mechanical (grounded anagnorisis: S asserted *and* symbolically forced, vs mirror / lucky leap), leak screens frozen before any paid run. Phase 1: the two-channel checker caught its own first defect (v001 tutor pre-voiced an unreleased premise inside a question — three asserts scored unforced; script-only v002 fix → zero leaps, forced = asserted = 32, 11/11 releases). Phase 2 (S0→S1): conduct lock-in is real (erotema 81% under frozen dramaturgy) and *advisory* steering makes it worse (free manner notes → 94%, switch 0.07); explicit note→figure authority breaks it (31–41% top share, switch 0.81–0.87), with the formal layer invariant in all arms. Phase 3: authority relocated into the tutor's *own superego* (draft → `{intervene, diagnosis, note}` → within-turn restage; director reduced to declared movements), against a pre-registered 4-clause bar. Advisory restraint failed in both channels and both model families (note-heavy Opus director 28–32/32; overfiring codex superego 12/20); criterial charter v2 (jurisdiction = the figure rut alone, stated as checkable arithmetic) finished with a zero-mismatch detector: 82 watched turns, 14 rut-due → 14 fires → 14/14 within-turn figure breaks, 68 not-due → 0 false fires, device-agnostic (erotema ×10, exemplum ×3, anaphora ×1), 7/7 arms grounded at the planned forcing turn (56/56 releases on cue), robust under a re-voiced learner. Reading: the watcher is a ceiling on repetition, not a source of invention; authority survives relocation inside; and the advisory-to-criterial boundary generalises the arc's diagnosis-side lesson — what works is new checkable structure at the point of action. Caveats: figures self-declared (declaration-vs-form audit open); form not content — no learner-side benefit claimed (outcomes schedule-fixed by design). Implementation `services/dramaticDerivation/`, `scripts/run-derivation-loop.js`; running logs `notes/poetics/2026-06-09-derivation-phase1-first-loop.md`, `2026-06-10-s0-s1-figure-mechanism.md`, `2026-06-10-superego-internal-figure-mechanism.md`. No DB writes, no rubric scoring, no new main-harness estimates — the arc's artifacts are file-based run exports.

v3.0.134 (2026-06-09) §6.12 added: can the tutor's within-turn revision be read as *adaptation versus compliance*? — Track B (measurement) of the post-§6.10 exploration brief, killed at its pre-registered instrument-validity gate. One cheap GPT-5.2 pass; the null is the result, no tune-and-retry, no full run. When the superego pushes back and the ego revises, that revision is a stance change; the "Not All Flips Are Conformity" taxonomy (arXiv:2606.00820) splits it into persuasion (reasoning-induced), conformity (stance-induced), and instability (spontaneous) — whether the tutor's revisions are reasoned uptake or compliance is the adaptation question at revision granularity. Pre-registered (`notes/2026-06-09-adaptation-conformity-classifier-stage0-preregistration.md`, frozen 2026-06-09 before any classifier call): an instrument-validity gate of unweighted Cohen's $\kappa \geq 0.60$ (stricter than the frozen weighted bound) on 60 blind revision triplets, offline-harvested from 13,395 events across 11,728 dialogues, stratified 10-per-superego-family. Classifier GPT-5.2 (temp 0) is architecture-clean (no OpenAI-family model among the corpus generators); the first-pass gold (Opus 4.8) is *not* (Anthropic models are corpus generators), so a human anchor (study operator, blind to the classifier, review-and-correct of the Opus pre-labels, 2/60 edits, κ(Opus,human) = 0.929) supplies the architecture-clean gold. Result: κ(GPT, human) = 0.084 (machine–machine κ(GPT,Opus) = 0.144; raw agreement 25/60) — **FAIL**, decisively. The disagreement is one systematic boundary (classifier calls 44/60 persuasion, gold 39–41/60 conformity; 30 of 33 disagreements gold-conformity→pred-persuasion). Post-hoc decomposition shows the labels reduce to surface cues, not the construct: a length-only predictor reproduces 82% of the gold split (AUC 0.82; conformity median 27 words vs persuasion 52), every disagreement sits in the short-revision zone, and the ego's own `reasoning` is a fixed self-justifying template that narrates every revision as synthesis regardless of cause — persuasion-vs-conformity is a claim about the *source* of the change that the observable product underdetermines (§6.10's dividing principle on the measurement axis). Disposition under the §7.9 symmetric-honesty bound: the null licenses neither "mere compliance" nor "genuine adaptation"; it is not a sixth mechanism negative but the measurement-track null completing the brief (§6.11 ruled out a collapsed generator; §6.12 rules out a finer revision-level instrument). Scripts `scripts/adaptation-conformity-{harvest,classifier,make-review-sheet,score-human}.py`; artifacts `exports/adaptation-conformity-gate*.{jsonl,md}`, `-harvest.meta.json`. No new main-harness effect estimate; no new judging beyond the single 60-event gate pass.

v3.0.133 (2026-06-09) §6.11 added: the adaptation ceiling is not *also* a collapsed-output-repertoire ceiling — the rival to the §6.7–§6.10 mechanism nulls, tested offline and rejected. Diagnostic, zero-API, read-only on traces already on `main`; no new generation or judging. The five convergent negatives (§6.7, §6.8.8, §6.9.7, §6.9.8, §6.10) all assumed the *mechanism* was the bottleneck. Argument Collapse (arXiv:2606.01736; LLM essays share 3.4% unique main arguments vs humans' 65.3%) names the untested rival: the tutor's *own response distribution* is collapsed, so there is little learner-conditioned variation for any mechanism to amplify — which would *explain* the five rather than extend them. Track A, Stage 0 of the post-§6.10 exploration brief tests it under a contrast frozen in writing before any query (`notes/2026-06-09-adaptation-repertoire-stage0-preregistration.md`): a matched-learner design (the ego–superego LLM learner's own cross-run lexical dispersion is the architecture-matched yardstick), echo-stripped model-free token-set distances, two frozen quantities — compression $\delta = \overline{D_{\text{out}} - D_{\text{in}}}$ and coupling $\rho = \text{partialSpearman}(D_{\text{out}}, D_{\text{in}} \mid \text{exchange index})$ — with a 2×2 read and cluster bootstrap by `scenario_id` (B = 2000, seed `20260609`). Result on 726 primary strata over the same 9 scenario families as §6.10: $\delta = +0.030$, CI [+0.008, +0.045] (the tutor disperses *more* than the matched learner, not less); $\rho = +0.082$, CI [+0.008, +0.152] (weak but reliably > 0); token-Jaccard robustness concurs ($\delta = +0.017$, $\rho = +0.096$). **The frozen read is NO-COLLAPSE**, secured by the positive δ alone (COLLAPSE required upper-95 $\delta < 0$), so it does not lean on the noisy ρ. Reading, under the §7.9 symmetric-honesty constraint: the five mechanism negatives cannot be explained away as "nothing to adapt to" (the generator is not a broken record), so the bottleneck genuinely sits on the already-closed mechanism/signal axes — the diagnostic *tightens* the synthesis by removing a confound; symmetrically, NO-COLLAPSE does **not** license "rich adaptation occurs" (the coupling is lexical, thin, two-thirds raw-echo, and consistent with the §6.3 / §6.8 / §6.9 adaptation nulls — output-lexical dispersion and rubric-measured adaptation are different quantities). Per the frozen disposition: diagnostic only, no Stage 1, no tune-and-retry, no live/paid run; a "repertoire-expansion" lever would be the sixth mechanism the brief forecloses, and NO-COLLAPSE removes its motivation as well as its sanction. Script `scripts/adaptation-repertoire-stage0.py`; artifacts `exports/adaptation-repertoire-stage0-{table.csv,meta.json}`. No new main-harness effect estimate; no new judging.

v3.0.132 (2026-06-07) §7.9 A19 stability smoke contracts the local pilot claim. Added `scripts/run-a19-stability-screen.js` and ran a two-seed local S0/S1 stability smoke on the two `surface_agreement_uptake` n=1 positives from v3.0.131. Neither candidate repeated as `policy_headroom`: `surface_agreement_uptake_c` returned 0/2 headroom because both S1 reruns stayed `neither`, while `surface_agreement_uptake_e` returned 0/2 headroom because both S0 reruns self-solved to `target`. The A19 paragraph is correspondingly downgraded from "two clean local headroom cards" to "two useful n=1 candidates plus falsifying local stability evidence." No pooled A19 rate, stability-confirmed A19 transfer effect, human-learning claim, deployment claim, weight-learning claim, main-harness effect, or sidecar empirical claim is licensed. Artifact: `exports/a19/stability/surface-agreement-uptake/a19-stability-summary.json`.

v3.0.131 (2026-06-07) §7.9 A19 local teaching-drama axiom transfer threshold crossed, still scope-bound. Updated the A19 paragraph from apparatus-only to a bounded local pilot result after two `surface_agreement_uptake` held-out siblings (`surface_agreement_uptake_c`, `surface_agreement_uptake_e`) produced S0/S1 `policy_headroom` under a single free-text critic per arm. The protocol remained S0-first: S1 was run only where S0 did not already reach the target repair, and S1 received exactly one admitted `a19-teaching-drama-axiom-v0.1` memory record (`exports/a19/axioms/surface-agreement-uptake/axiom.json`), never a full replay `revision.json` bundle. Added the negative calibration boundary: fraction and temperature concrete-domain cards self-solved at S0, `surface_agreement_uptake_d` self-solved at S0, and no pooled A19 rate is reported. The licensed claim is only a local simulated teacher-as-learner one-family result; no human learning, deployed tutor, model-weight learning, main-harness effect, paid panel result, or independent sidecar empirical claim is introduced. Atlas module `teaching-drama-axiom-transfer` remains `scope-bound` and projects only to §7.9.

v3.0.130 (2026-06-07) §7.9 A19 teaching-drama axiom scaffold lands as apparatus, not a new empirical result. Added a scoped A19 paragraph immediately after the bounded A18.35–A18.37 claim. The new prose generalises A18's per-card headroom unit into a zero-API teaching-drama axiom protocol: attempt-1 failure, old-rule decoy, public learner-resistance trigger, replacement move, applicability and anti-conditions, held-out siblings, alias withholding, card-level basis labels, and protocol-reject reasons. Implementation artifacts named in the paper are `config/teaching-drama-axioms/a19-protocol.yaml`, `config/teaching-drama-axioms/pilot-families.yaml`, `scripts/validate-teaching-drama-axiom-pr` `scripts/report-teaching-drama-axiom-framework.js`, `scripts/materialize-teaching-drama-axiom` and `scripts/blind-teaching-drama-axiom-adjudication.js`. This version also projects the A19 atlas module onto §7.9 and keeps the sidecar/paper boundary explicit: all empirical rates remain the A18 rates already reported above until paid generation, blind adjudication, stability reruns, and any human double-coding land in the canonical paper first. Zero API; fixture and documentation implementation only; no new generation, judging, or empirical rate.

v3.0.129 (2026-06-07) §7.9 A18.37 denominator closed: the seven-family / 10-of-14 pool is the frozen elicitation gate's *output*, not partial coverage. Documentation reconciliation only; zero-API, derived from on-disk gate verdicts that already exist. The v3.0.127 widening took the blind channel to seven families (pooled 10/14), but the §7.9 text stated the two-stage lower bound on a single exemplar (`exclusion_filter`), leaving the relationship between "seven families" and the "12-family A18.35 fanout" (TODO §A18.35) implicit — and the design note + TODO still framed the widening as future work ("run the blind arbiter on the two uncovered families `overlay_registration`, `pointer_chain_two_hop`; take the count 8→12"), which v3.0.127 had already done and exceeded (8→14). Closed the accounting from the families' on-disk `policy-fill-report.json` + attempt-1 `gate.json/needs-revision.json`: of the twelve authored families the frozen attempt-1 correctness gate **admits exactly the seven adjudicated** (→ 14 cards, 10 headroom) and **rejects four** before any held-out continuation is generated — `exclusion_filter` (`old_warrant_misclassification 0.6 < 0.7`, the tie criterion) plus `running_sum_threshold` / `count_ladder_successor` / `elimination_bracket` (`recursive_dyadic_update 0.68 / 0.55 / 0.65 < 0.7`) — with the twelfth (`tally_parity`) never materialised. So the seven-family denominator is the gate's selection, not a coverage gap; resolving the four rejects would mean re-rolling attempt-1 past the gate or relaxing it (un-freezing the pre-registered protocol), so they are reported as the elicitation floor with `exclusion_filter`, not back-filled. One sentence added to the §7.9 two-stage-boundary passage; no pre-registered numbers change, no new generation or adjudication. Stale forward-pointers corrected in `TODO.md` (§A18.37 bounded-tail, §A18.38 Update) and `notes/poetics/2026-06-06-teacher-as-learner-reframe-ml-ladder.md`; canonical adjudication recorded in memory `project_a18_37_canonical_adjudication`. Net paper change ~+170 words (§7.9 one sentence + this entry).

v3.0.128 (2026-06-06) §6.9.8 A15 retrieval lever moved from *argued-null* to *run-null*: the non-parametric (k-NN retrieval) policy form fails the same offline kill gate as the parametric lever, on the transfer-valid channel. Data + analysis only; zero-API, read-only on the §6.9.8 harvest table. §6.9.8 previously closed the policy-*form* axis on three realisations (implicit §6.8, hand-authored §6.9.7, fitted-parametric §6.9.8) and listed A15's planned cross-dialogue *retrieval* as a non-parametric sibling "designed and gated, not a counterexample" — argued to be in the same exhausted space, never run. It is now run, offline and free, on the identical harvest table / state features / Direct-Method reward model / grouped leave-one-`scenario_type`-out CV / cluster bootstrap as the parametric OPE, with only the policy form changed (reward-weighted k-NN retrieval, $k = 5$ pre-declared; `scripts/retrieval-adaptation-knn-ope.py`). The Direct Method *appears* to clear the gate (binary 0.749, beats both baselines at every $k \in \{3, 5, 10, 15, 25\}$), but three pre-registered controls (`scripts/retrieval-adaptation-knn-diagnostics.py`) show this is a **base-rate exploit, not transferable adaptation**: (i) minority action families carry high marginal `strict_shift` by the label's construction (`diagnostic` 0.96 / `repair_affective` 0.92 / `scaffolding` 0.89 vs `substantive_engagement` 0.30), so a reward-greedy selector scores high without teaching better; (ii) a label-permutation placebo confirms the lift is above noise (null band $ $0.564) yet a behaviour-cloning k-NN (copy what was *done*, not what *worked*) falls to 0.450, below baseline — the gain needs reward-greediness, not retrieval; (iii) on the transfer-valid on-policy-agreement channel (the only estimator identified without logged propensities for a policy that deviates $ $82%) the retrieval policy is **0.3125, below both baselines (0.478/0.424)**, statistically indistinguishable from the parametric logistic's 0.326. Surgical revisions to §6.9.8: a new "non-parametric sibling, now run" paragraph; the Disposition sentence repointed (planned/argued → run-and-null, four realisations all null on the transfer-valid channel); a scope caveat (the gate tests the *single pivotal* decision, the necessary condition for the slope A15 targeted; a direct turn-by-turn slope test would need the live cell, which the offline null does not justify — the foreclosed "add embeddings and rerun" treadmill); artifacts added. The cross-references in §6.10 / §7.9 ("exhausted on both the policy-form §6.9.8 and signal §6.10 axes") are unchanged and *strengthened* — the form axis is now closed over realisations actually run, parametric and non-parametric alike. The four-convergent-negatives count is unchanged (§6.9.8 remains one negative, now covering both policy forms). No pre-registered numbers change; no new evaluation rows, no new judging. Artifacts: `exports/retrieval-adaptation-knn-{ope,diagnostics}.meta.json`; harvest table copied into the fork at `exports/learned-adaptation-table.csv`. Net paper change ~+430 words (§6.9.8 one paragraph + Disposition/caveat edits + this entry).

v3.0.127 (2026-06-06) §7.9 A18.37 blind channel extended to seven families: pooled 10 of 14, overlay fragility quantified. Extended the architecture-independent denominator beyond the pre-registered four families by re-adjudicating three further elicited-survivor families (`overlay_registration_priority`, `pointer_chain_two_hop_priority`, `legend_decode_priority`) on the same blind arbiter (`scripts/blind-option-adjudication.js`). Adds 5 of 6 siblings (overlay and pointer-chain converge on both; legend splits), for a pooled **10 of 14 across seven families** (three converge, four split, none null) — the pre-registered 5/8 is unchanged and remains the frozen-protocol anchor; the three new families are post-hoc confirmatory. Policy memory commits to target in 14/14 (zero control-advantage, zero neither-correct), so the four no-headroom cards are ceiling effects (the no-policy arm also lands the target), not transfer failures. On the six extension cards the deprecated overlay channel disagrees with the blind arbiter on all six (five false-negatives, one false-positive), giving the "bidirectionally fragile" claim a count. New driver `scripts/a18.37-elicited-survivor-headroom.js`; artifacts `exports/recursive-tutor-learning/a18.37-elicited-survivor-headroom/`. No pre-registered numbers change.

v3.0.126 (2026-06-06) §6.2.3/§6.2.5 superego-transition figures given a data-provenance pointer (claim-integrity pass, G4 closure). The Round 1→Round 2 transition table (§6.2.3: 232 pairs across 56 dialogues; Base $N = 57$, Recognition $N = 175$) and the revision-magnitude figures (§6.2.5: $N = 216$ non-approval critiques with both texts; per-category mean Jaccard) reproduce exactly but were not traceable to a committed script + corpus from the paper text alone. Added an inline provenance sentence naming the source script `scripts/analyze-superego-taxonomy.js` (§9 transition analysis, §10 revision analysis) and the 500-record classified-critique corpus (`data/paper2/superego-critiques-classified-paper-6.2-n500.jsonl`, sha256 `f9ba2d92…`), now tracked in this repo. The note also records the pairing semantics: every record carries `turnIndex: null`, so §9 pairs each round-1 verdict against the dialogue's first round-2 verdict rather than strictly same-turn — deterministic, and consistent with the dialogue-level "Round 1→Round 2" framing. The earlier diagnosis that these figures were "unreconstructable from the repo" was an investigation error (wrong corpus + wrong script: the unrelated `analyze-superego-transitions.js`), not a paper defect; the only genuine gap was archival (untracked source corpus), now closed. No numbers change. Audit trail: `exports/claim-verify-findings.md` (G4 detail), `exports/superego-transition-reproduction.md`, TODO §G.

v3.0.125 (2026-06-06) §5.2.1 rubric-redesign correlations re-attributed to their reproducible sources (claim-integrity pass, cont'd). Two mis-attributions in the §5.2.1 dimension-redesign paragraph, both traced to real numbers and now reproducible from a committed script; neither was a fabrication. (1) The PCA figures (N=1,584 observations, PC1=80.7%, KMO=0.938, mean inter-dimension $r = 0.776$) were labeled "across the 14 dimensions," but they are the **8-dimension v2.2** PCA — identical PCA statistics cannot describe both an 8- and a 14-variable analysis, and every other citation of these numbers (Sections 6.1, 7.8, 8.6, the figure caption) already correctly calls them 8-dim. Reworded so the PCA is presented as showing redundancy *persists after* consolidation (with the dimension-audit prediction as the actual motivator), cross-referencing §8.6; the figures reproduce to printed precision on the canonical slice (runs `aea2abfb`+`45163390`, Sonnet judge). The genuine v1.0 14-dimension PCA (pre-v2.2 backup, opus-4.6 judge, N=4,844) gives PC1=68.9%, KMO=0.944, mean $r = 0.651$, two factors — materially different from the printed numbers but corroborating the redundancy claim; it is recorded in the reproduction artifacts, not the paper body, to avoid mixing judges/corpora in one paragraph. (2) The holistic-rubric `r = 0.907` per-turn-vs-holistic correlation is a **v2.1, N=32** snapshot (`notes/paper-2-0/rubric/ideal-measures.md`); annotated as such, with the robust current cross-check added ($r \approx 0.87$ on N=288 v2.2 dialogues) since the direction holds across all rubric versions, so the 6→3 simplification rationale stands. Reproduction committed: `scripts/analyze-rubric-pca.js` (8-dim from the live DB by default; backup flags for the 14-dim figure); reports `exports/rubric-pca-{8,14}dim.md`. No other §5/§8 numbers change. Audit trail: `exports/claim-verify-findings.md` (G1 detail), TODO §G.

v3.0.124 (2026-06-06) §6.6.11 H2 p-value correction + §6.4 cross-model effect-size provenance (claim-integrity pass). Two fixes from an adversarial DB re-verification of the paper's empirical claims. (1) The §6.6.11 H2 row reported the per-arc score-slope Welch test as `p = 0.032`; that value does not reproduce. The correct test on the five-per-arc slopes ($t(7.88) = 1.99$) gives **one-sided** $p = 0.041$ (matching H2's directional pre-registration) and **two-sided** $p = 0.083$ — confirmed against both `scipy` and the in-repo `regularizedBeta`. The "supported" verdict survives the one-sided pre-registration but is weaker than `0.032` implied (two-sided is non-significant); every other figure in the row (slopes $+1.31/-1.08$, t, df, the 9-point gap, Cohen $d = 0.70$, and the in-sequence sensitivity base -1.10/recog $+0.72$) reproduces exactly. The previously-cited reproduction script `analyze-a7-longitudinal.js` is a pad-metrics stub that never implemented the slope; a dedicated `scripts/analyze-a7-h2-slope.js` (with `lnGamma`/`regularizedBeta` now exported from `services/anovaStats.js`; 21/21 anova tests pass) is committed and cited. (2) §6.4's cross-model recognition effect sizes are sound but were underspecified: added a sentence stating they are computed on the per-turn aggregate (`tutor_overall_score`/`learner_overall_score`, **not** Turn-0 `tutor_first_turn_score`) under the Sonnet 4.6 judge, and naming the DeepSeek (`eval-2026-03-01-aea2abfb`) and Haiku (`eval-2026-03-02-45163390`) source runs alongside the already-cited Gemini run. Means reproduce exactly (DeepSeek base 26.4/recog 50.0; Haiku 61.3/80.4); DeepSeek d reconstructs to 1.85 on the current snapshot vs the reported 1.88 (SD rounding, left unchanged). No other §6 numbers change. Audit trail: `exports/claim-verify-findings.md`, TODO §G.

v3.0.123 (2026-06-06) §7.9 A18 fresh-family replication: convergence is a structural per-card property, two-stage-gated — discharges the v3.0.120 frozen-protocol condition. Added a §7.9 paragraph (after the A18.6–A18.14 paragraph) reporting the pre-registered fresh-family replication the v3.0.120 entry deferred ("requires the correctness-gated protocol frozen *before* a fresh-family replication"). Four new local-relation families (`relational_betweenness`, `distal_correspondence`, `second_in_constructed_order_priority`, `constructed_midpoint_priority`), none in the A18.6–A18.14 set, run under the frozen A18.13 gate and adjudicated on a new *architecture-independent* arbiter (`scripts/blind-option-adjudication.js` + test): a blind three-critic panel reads only the held-out continuation and names the committed option, with target/decoy aliases held out of the critic and matched mechanically downstream — replacing the overlay correctness channel, which is bidirectionally lexically fragile (false-negative + false-positive). Result: policy memory shows a blind-detectable S0-decoy/S1-target advantage on **5 of 8 held-out siblings across the four families**; every elicited family converges on ≥ 1 sibling and three families split across siblings, so headroom is per-card, not per-family. The three no-headroom cards fail by two mechanisms the blind basis-label separates (S0 self-solves the relation, or its surface shortcut coincides with the target on a cue-leaky card). The verdict is structural not noise — a $k = 3$ rerun of the distal pair (`scripts/a18.37-s0-stability.js`) pins the siblings to opposite deterministic extremes (self-solve 3/3 vs 0/3, zero within-card variance) — and predictive: the four constructed-device survivors were pre-registered card-by-card and the arbiter returned 4/4 as predicted, overturning two overlay false-negatives. Convergence is two-stage: a relation reaches the headroom test only if its old-rule violation is a confident misclassification, not a tie — `exclusion_filter` fails attempt-1 elicitation on `old_warrant_misclassification = 0.6 < 0.7` (a three-way tie). Net claim sharpened from a family rate to a bounded per-card one: the saved attempt-1 lesson transfers reliably and rerun-stably on the structurally identifiable cards where a fresh tutor would otherwise take a decoy shortcut — not unconditional convergence, still simulated-teacher-as-learner, not human learning or a main-harness effect. Artifacts: `exports/recursive-tutor-learning/a18.37-survivor-headroom/`, `.../a18.37-stability/`; note `notes/poetics/2026-06-06-a18-37-per-card-headroom-mechanism.md`; new instruments `scripts/blind-option-adjudication.js` (+ `tests/blindOptionAdjudication.test.` `scripts/a18.37-s0-stability.js`, `scripts/a18.37-survivor-headroom.js`. No other §6/§7 numbers change.

v3.0.122 (2026-06-06) §6.8.4 power caveat — the secure-re-run result is *not-detected*, not *demonstrated-zero*. Added an explicit underpowered-comparison note so the v3.0.121 "no detectable advantage" cannot be misread as a proven null. At $N \approx 24$/cell the comparison cannot resolve an effect this small: pooled secure gap +5.3 pp, 95% CI $[-15, +25]$ pp (Fisher $p = 0.68$; v1 $[-22, +35]$, cross-suite $[-23, +32]$); ~1,400 dialogues/cell would be needed for 80% power at a true 5-pp difference. A small real advantage (~1.1×, consistent in direction across both suites and mechanistically expected from the state machine's explicit re-pivot) is *not excluded* — the re-run removes the *~2× magnitude* (which rested on the engine's mis-measurement at 25% vs its clean 41.7%, CI $[24, 61]$%), not the *existence* of a small effect. Recorded as not-detected, not absent; not worth ~1,400 paid dialogues/cell to chase for an edge that inverts on the graded channel and stays scaffold-confounded (§5.12.3). No numbers change beyond the added CI/power figures.

v3.0.121 (2026-06-06) §6.8.4 secure re-run — the cross-architecture binary advantage is *no detectable difference* once the turn convention is fixed. Discharged the v3.0.119 open item: cells 114 and 125 re-run under the *fixed* adapter (`learnerTurnIndexForTutorTurn(turn)`, no `+1`), fresh sample $N = 24$ each, `claude-code/sonnet` (run ids `eval-2026-06-06-460fc561` v1, `eval-2026-06-06-5a168684` cross-suite; 24/24 each, 0 failed), scored by the now-aligned `analyze-strategy-shift.js` (fixed adapter + `triggerTurn + 1` agree by construction). The engine floor, previously undercounted by the artifact, rises: `cell_114 strict_shift` 25.0% → **41.7%** (10/24), family 58.3% → **75.0%**; `cell_125` 30.4% → **58.3%** (14/24), family 39.1% → **62.5%** (mechanism check: `cell_114 repair_affective` hits at trigger+1 go 0 → 10). Against the frozen state cells (`cell_110` 47.8%/87.0%; `cell_124` 62.5%/66.7%) the secure contrasts are **v1 1.15× strict / 1.16× family** and **cross-suite 1.07× / 1.07×** — ~4–6-pp gaps, within $N \approx 24$ noise, i.e. *no detectable binary advantage on either suite*. More deflationary than v3.0.119's offline grid (which estimated cross-suite ~1.4×): the fresh aligned run is authoritative over the noisy offline locator. Caveat retained: state cells frozen (old epoch), engine cells fresh; not a same-epoch contrast, but the artifact was engine-adapter-only and the gaps are small either way (110/124 left frozen per decision). Edits: §6.8.4 correction paragraph extended with the secure re-run as the settled answer; the "~1.4× / parity / convention-fragile" restatements in §6.8.4 (Bottom line), §6.8.7 (establishing sentence, recap table, closing) and §6.8.8 (finding #2, "not established" bullet) updated to the secure "~1.1×, no detectable advantage". Atlas `ontology-state-interior` claim updated to match. Durable claim now: **once the turn convention is fixed, there is no detectable cross-architecture binary advantage; the published ~2× was the artifact.** The §5.12.3 prompt-scaffold confound is unchanged. No other §6 numbers change.

v3.0.120 (2026-06-06) §7.9 A18 recursive tutor-learning claim boundary: bounded counterfactual policy transfer, not yet reliable peripeteia-induced adaptation. Added a compact §7.9 sidecar paragraph synthesizing A18.10 and A18.14. After no-headroom negatives on the first replay family (A18.6–A18.8), two artificial underdetermined local-relation families now pass contrastive policy-transfer tests: `selector_rail_priority` (`3/5`, `5/5` transfer votes; one pair with two high ordinary-public-inference cautions) and `bead_predecessor_priority` (`5/5`, `5/5`, no equivalence or S0-preferred votes) under the A18.13 policy-correctness overlay. The paper claim is deliberately narrow: simulated counterfactual replay can preserve a saved attempt-1 strategy lesson and apply it to held-out siblings in a way blind contrastive critics distinguish from no-policy continuations. It is not human learning, deployed tutoring, model-weight learning, or a main-harness rate effect, and the stronger "reliable peripeteia-induced adaptation" phrase remains gated on a fresh-family replication under a pre-frozen correctness protocol because the bead family's gate was introduced after A18.12 exposed wrong-target S0 survival. Durable synthesis note: `notes/poetics/2026-06-06-a18-cross-family-claim-boundary.md`; TODO advanced to A18.16 pre-registration.

v3.0.119 (2026-06-06) §6.8.4 turn-convention correction — the cross-architecture binary advantage is convention-fragile; "never re-pivots" is largely an artifact. A turn-index misalignment in the dialogue-engine trap adapter (`scripts/run-dialogue-engine-trap-baseline.js` passed `turn: turn + 1` where the LangGraph runner passes `turn: state.turn`) made the trap learner fire one tutor-turn *early* in the baseline only; because the learner fires its trigger on the turn index it is handed, the fixed-index scorer (`triggerTurn + 1`) then read the engine's *second* post-trigger response, not its first. The offset is architecture-correlated (LangGraph cells aligned, dialogue-engine baseline shifted), so it inflated the `strict_shift` gap rather than adding symmetric noise — the symptom was already visible in §6.8.4 (repair move "present … at the trigger turn itself, not the trigger+1 slot the binary scores," recorded 0/3) but read as a tutor timing deficit. Fixed forward in `scripts/lib/trapTurnConvention.js` + adapters (commit `3474ebf`); diagnosed offline (read-only, zero-API) by `scripts/rescore-trap-strict-shift.js` (`exports/trap-strict-shift-rescore.json`), which reproduces the published numbers exactly under the stored convention, then re-scores by locating each trigger's actual emission and scoring the first tutor turn after it. The corrected reading is a 2×2 grid (trigger location × match strictness): the published `strict·stored` ratios (v1 1.91×, cross-suite 2.05×) are the *most generous corner*; the locator-free `family·stored` floor is 1.49× / 1.70×; under `family·event-anchored` the **v1 advantage vanishes** (0.98×) while the **cross-suite survives at ~1.4×** (1.39×). The "dialogue engine never re-pivots" property is the same artifact (event-anchored, `cell_114` reaches the expected family within two post-trigger turns ~45% of the time, vs its 25.0% stored `window`). Edits: a correction paragraph appended to §6.8.4, with the "twice as often" / "never re-pivots" / "≈2.05× replicating 1.91×" restatements in §6.8.4, §6.8.7, §6.8.8 (and the §6.8.7 recap table) softened to carry convention-fragility pointers. Durable claim: a modest (~1.4×), between-type, binary-channel cross-suite advantage, fragile-to-null on v1; the §5.12.3 prompt-scaffold confound is unchanged and applies on top. A secure single number would require re-running cells 114/125 under the fixed adapter (not yet done). No other §6 numbers change. Found via a turn-convention audit (codex) + offline re-score; status post-hoc correction of a post-hoc exploratory result.

v3.0.118 (2026-06-02) §6.10 symbolic corroboration (out of claim set). Added a corroboration paragraph to §6.10: a *symbolic* realisation of the concealed-interior lever — a per-turn, EL-consistency-checkable description-logic theory-of-mind ABox, strictly more inspectable than §6.10's lexical proxy — supplied to the tutor as live guidance is null over plain ontology guidance in a small real-backend multi-turn A/B (three stress scenarios, one run each); the consistency check fires correctly (flags surface compliance as a grounded inconsistency) but re-encodes a move the ontology guidance already prompts, so the no-privileged-interior reading survives the shift from a free-text to a formal, checkable interior. Exploratory screen, out of the claim set, named for completeness. No numbered-result or N-count changes. Write-up: `notes/ontology/2026-06-02-multi-turn-tom-ab.md` + `2026-06-02-unified-ontology-consolidation.md`; harness `scripts/run-ontology-tom-ab.js`.

v3.0.117 (2026-06-01) §7.9 generalisation: graded re-score + one-side replay. A second deep scenario (D_OED5, a *non-temporal* namesake collision — one benchmark name, two coexisting test splits) does *not* replicate the binary lift (socratic 1/3; `none` clean-by-outcome 3/3, `exports/oedipus-d5-full/`), but the binary collapses a real gradient. A graded 0–4 instrument (`scripts/critic-poetics-omniscient-graded.js`, same blind four-model panel, rubric `notes/poetics/2026-06-01-oedipus-discovery-grade-rubric.md` `exports/oedipus-graded-rescore/graded-all.json`) shows *both* scenarios lift socratic-over-`none` by comparable margins — D_OED4 **3.39 vs 2.00** (gap 1.39), D_OED5 **2.83 vs 1.17** (gap 1.66) — so the temporal/non-temporal difference is an *attainment ceiling* (species 3.39 vs genus 2.83), not lift-vs-no-lift. Incidental: D_OED4's control is genus-leaky (graded 2.00, bimodal $ $4/9), so its binary "clean 8/9" held only at the species level; inter-rater agreement is tight on socratic (band 0.44–0.67), loose on `none` (1.0–1.9). A one-side replay (`scripts/replay-one-side.js`, a guarded `scriptedTutorTurns` hook in `runInteraction`, and per-run `directorPlan` persistence) freezes a run's tutor + scene and regenerates only the learner K times, isolating learner- from scene-variance; replaying D_OED5 run 3's socratic learner 8× graded mean **3.38** (3/8 full species, 5/8 grade $\geq$ 3.5) versus the original 2.5 — the near-miss there was a *learner draw* on an evidence-rich scene, not a structural cap.

Reconstructability is a scene property; given a reconstructable scene, reaching the species is partly stochastic. Caveat: one scene (run 1/run 2 scenes lost to a volatile-dir overwrite, now closed by persistence). Sharpens, does not overturn, v3.0.116: D_OED4's species-level lift stands; D_OED5 adds that the lift generalises to the genus and that the cap is at least partly stochastic. Artifacts as above; rubric + worksheet + scorer in `notes/poetics/` and `scripts/`.

v3.0.116 (2026-06-01) §7.9 Oedipus depth result: a genuinely clean control + the first attributable guided-discovery lift. A new arm-invariant QA gate (`scripts/qa-oedipus-arms.js` + test) that judges the *tutor's* disclosure per arm (withheld/metered/stated), not only the learner's discovery, shows the prior "zero of nine control" (v3.0.115) was clean by *outcome*, not *mechanism*: on the forced-identity scenarios the `none` tutor *meters the records anyway* across three separately-built stances (helpful colleague, arm-neutral persona, adversarial examiner-told-revealing-harms), all `metered` $\geq 3/4$ — the zero reflected an under-determined S (forced only \$ \$1/4), not a withholding tutor; contamination is structural (in a records-scene every engaged question is the hint). Fix is the *scenario*: a third *depth* screen (underivability with "source/code/preprocessing all check out" pre-stated) plus a forcedness screen (`scripts/screen-s-forcedness.js`, cross-checked by union-find `scripts/oedipus-symbolic-check.js`) define a window in which *surprising* (unguessable) and *deep* (survives the obvious checks) pull against each other. D_OED4 (silently-redefined cohort label) is forced + deep but *fails* the static depth screen at rank-1; run anyway on a stated "the static prober is a free investigator, the committed `none` learner is not" argument, with the verdict left to the learner's actual behaviour — which bore out: committed `none` learner reached S unaided **1/9**, **socratic 7/9**, one socratic flop, `discovery_lift = 1.0` on every valid pair. First attributable lift in the arc. Honest qualifications: control clean by outcome not pristine by intent (gate still flags `none` edging in several runs); robust gate judge-bias (leading-questions read as "withholding") makes its socratic/reveal labels unreliable, so the lift rests on the learner-outcome critic; pilot (9 runs, one scenario, one backend); dramatic *form* not learning (§6.10 bound). Sharpens — does not overturn — the preceding result into a conditional: attributable guided anagnorisis iff S is deep enough that the obvious investigation comes back clean. Artifacts `exports/oedipus-d4-full/`, `exports/oedipus-d4-none/`; instruments committed `121e549/ba91832/4f267df/e10e2e2`; window note `notes/poetics/2026-05-31-forcedness-underivabi`

v3.0.115 (2026-05-31) §7.9 Oedipus mechanism ablated: premise-licensing — not the bidirectional superego — is the discovery lever; the superego drives commitment. A 2×2 (learner superego {v1 monotone | v2 bidirectional} $\times$ premise-licensing {off | on}) on D_OED1 / Sonnet-API, env-gated toggles (`a63493d`), 3 runs/corner. Result: premise-licensing ON $\rightarrow$ socratic discovery 2/3 *regardless of critic* (premise-only with the monotone v1 critic: 2/3); premise-licensing OFF $\rightarrow$ \$ \$0 *regardless of critic* (superego-only with the bidirectional v2 critic: 0/3), the learner then committing only to a near-miss (ArrayExpress-vs-GEO mismatch / self-blame, challenge-dominant superego intact). So the bidirectional superego converts hedging into commitment (a real, separable effect, legible as the near-miss), but the *discovery of the exact S* is the tutor metering the distinguishing premise. Corrects the v3.0.114 §7.9 attribution, which led with the superego; headline 2/3 result and overturns-the-null conclusion unchanged. Six ablation panels added to `exports/oedipus-cli-2scenario/`.

v3.0.114 (2026-05-31) §7.9 Oedipus information-asymmetry probe REVERSED: the single-scenario null was substantially an instrument artifact; the corrected instrument yields $ $2/3 attributable guided anagnorisis, generalizing across two screened scenarios and two model backends. Diagnosing *why* the original null occurred — via the simulated learner's own logged ego→superego→revision deliberation (owned ground truth) — exposed two instrument defects rather than a property of the dyad: (1) the learner superego was monotonic, pushing the ego toward *more* doubt every turn and endorsing the hedge instead of interrogating it; (2) the tutor's dramatic-irony discipline mislabeled the *distinguishing* premise as part of the withheld secret, so it was never metered. Two principled fixes: a *bidirectional* learner superego (Freudian, suspicious-of-the-ego, catching over-claiming AND over-hedging, with a yes-man guardrail; commit `260cdc6`) and licensing the `socratic` tutor to meter every premise, stating-a-premise distinguished from revealing *S* (commit `05df834`). With both: the learner commits in every run and reaches the exact *S* by reasoning at $ $2/3 — dataset-namesake scenario 2/3 via the metered API (smoke5/6/7) and 2/3 via the Claude-CLI/Max-plan backend, generalizing to a second independently-screened scenario (finance ticker-reassignment, D_OED3) at the same 2/3. `none` control reached *S* in 0/9 post-fix runs (same bidirectional critic in the control ⇒ discovery is metering-specific); the `reveal` ceiling fires in 7 of the 8 runs that carry it (one API run's told-*S* drew 2/4, and smoke5 lost its reveal arm to a mid-run quota stop); a demand-characteristic audit confirms no dramatic-frame vocabulary reaches the learner (frame-awareness confined to the byte-identical shared prefix, differenced out of the contrast). Residual $ $1/3 = a committed *near-miss*, not non-commitment. Pilot (2 scenarios × 3 runs/backend, Sonnet generation, four-critic panel); overturns rather than extends the single-scenario null. Supporting infra: eval `prompts/` made authoritative via a local-first loader (prompts had silently resolved from the in-housed `tutor-core/`), regression-tested (`services/__tests__/learnerPromptResolution.test.js`); honest generator/critic model+version logging (the generation banner had mislabelled api/Sonnet runs as cli/Opus). Artifacts `exports/oedipus-cli-2scenario/` (nine post-fix omniscient panels); second scenario added to `config/poetics-calibration/oedipus-pilot-v2.yaml` (two prior designs rejected by the underivability screen as too-derivable, which sharpened the design rule). Single-pilot status; human-learner door (§8.1) still open.

v3.0.113 (2026-05-30) A10/A10b within-family null replicated on a fourth generation model (Nemotron); §7.9 cross-model paragraph added. Re-ran the four A10 cells (base / recognition / matched-pedagogical / matched-behaviorist) on NVIDIA Nemotron-Nano-30B with the ego model held constant across cells (a run-time `--model openrouter.nemotron` override removing the as-declared `deepseek-vs-nemotron` confound between cells 95/96 and recognition), v2.2 rubric, single `claude-code/sonnet` judge, pooled over two runs ($n \approx 36$/cell). Recognition vs matched-pedagogical between-cell $d = 0.17$ (paired $\Delta = +1.8/100$, 95% CI $[-2.6, +6.1]$, $p = 0.43$) — identical to A10 v2 and inside the three-judge 0.14–0.19 band; recognition and matched-pedagogical each $\gg$ base ($d = 0.95$, 0.83). Two honest qualifications added to §7.9: the behaviorist arm is model-dependent (at base on Nemotron, $d = -0.02$, vs the $d = 0.89$-below-base of finding 2 on the three panel models — "actively counterproductive" weakens to "merely inert" on this weaker model, between-family dominance unchanged); and recognition leads on the `recognition_quality` sub-dimension (2.75 vs 2.26) despite the tied overall score, a localized fingerprint consistent with §7.10. Enabled by in-housing tutor-core (the `matched_pedagogical` / `matched_behaviorist` base profiles were ported into the in-repo `tutor-core/config`; the published install had silently degraded cells 95/96 to budget). Single-judge single-model robustness extension; no new main-harness panel. Runs `eval-2026-05-30-f4e36e14` + `eval-2026-05-30-a9e12d4b`; report `exports/a10-nemotron-replication.md`; script `scripts/analyze-a10-nemotron-replication.js`.

v3.0.112 (2026-05-30) Oedipus information-asymmetry successor probe added to §7.9; tests the one open door the de-confound left, with a clean single-scenario null. Reversed the asymmetry (owned hidden state on the tutor/director side; learner must discover a withheld contingent fact S, routed away from the learner and enforced by a runtime prompt guard). Pre-registered an S-underivability screen (rejected diagnostic/citation-twin secrets as derivable; only a contingent namesake-dataset collision survived) and an omniscient four-critic panel (discovery / by-reasoning / reveal-detector, arm-blind, S disclosed). On the one screened scenario: clean null — `reveal` ceiling 3/4 discovered but 0/4 by-reasoning (told), `none` floor 0/4, `socratic` device 0/4 across both a meter-only and a fuller-elenchus commit-prompt variant; by-reasoning never fired. Mechanism: the learner reaches the evidence but declines to commit to the contingent conclusion absent confirming evidence the dialogue cannot supply ("I'd need the actual file hash") — principled non-commitment, sharper than the May arc's self-resolving learner. Extends rather than overturns the §7.9 boundary; single-scenario pilot, human learner still open (§8.1). New scoring infra (`scripts/screen-s-underivability.js`, `scripts/critic-poetics-omniscient.js`, `scripts/lib/secret-tokens.js`) + generation heartbeat ETA. Artifacts: `exports/oedipus-s-underivability-v2-*.json`, `exports/oedipus-omniscient-smoke{3,4}-*.json`; spec `notes/poetics/2026-05-29-oedipus-guided-`
No new main-harness effect estimate.

v3.0.111 (2026-05-29) Dramatic-recognition de-confound and boundary (§7.9); refines the v3.0.108/110 saturation reading and closes the simulated-dyad poetics arc. Continued the sidecar past the 022408Z saturation. Implemented the EDRA-M3 scorer surgery (public-mechanism detector de-aliasing, origin demotion to a reported diagnostic, coverage-retry, paired-increment aggregator over an emitted `item_gates.jsonl`) and scored a fresh three-iteration run (`phase2-adaptation-recognition-loop-20260529T023023Z`): the loop's per-arm gate still failed 0/2, but the corrected matched-pair increment is a small positive (recognitive-closure lift 0.333, Wilson95 [0.12, 0.65]; 3 positive / 6 null, 0 control leaks). A critic-swap de-confound (author-family Sonnet → non-family GPT, three critics held fixed) inverted the failure mode: 3/6 pairs voided by control leak (vs 0/9 under Sonnet). Two triangulations relocate the leak from the critic to the generator: a shared-critic decomposition (the identical Qwen/DeepSeek vote recognition 2–4× more on the leakier later draws — Qwen 0.06→0.27, DeepSeek 0.24→0.36, Gemini 0.00→0.00), and an adversarially-verified nine-agent fork-audit (tutorDidDevice=false on all four leaked controls; dominant cause the generator over-writing the control learner — 2/4 persona over-write, 1 prefix self-resolution, 1 critic over-read). Conclusion: recognition is generable but not causally attributable to tutor adaptation in a fully simulated dyad — §6.10's no-privileged-interior wall reached through a third instrument (dramatic form); honest verdict null-to-weak-positive on form, not learning, consistent with the instrument's form-vs-learning transfer null. Bounded: closes the simulated-dyad poetics arc under these instruments, not a universal impossibility (owned-latent-state or human learner left open, §8.1). Amended §7.9 (new de-confound paragraph + revised threefold-contribution summary). Also corrected the stale `version:` field (3.0.106 → 3.0.111; entries 107–110 had been logged without bumping it). Artifacts: `exports/paired-codex-20260529-full.json`, `exports/paired-increment-reverse-GPT-pooled-2iter.json`, `exports/fork-audit-control-leak-wr` `notes/poetics/2026-05-29-critic-panel-draw-vs-critic-analysis.md`. No new main-harness effect estimate.

v3.0.110 (2026-05-28) Panel-composition sweep added to §7.9; partially overturns the saturation reading. Wrote `scripts/analyze-recognition-origin-panel-recompose.js`, which re-applies the gate to the same 528T022408Z score JSONs under five critic-subset panels with matched-supermajority thresholds; output at `exports/recognition-origin-panel-recomp` The original pre-registered 4-critic 3-of-4 panel at CUT=75 passes 0/3 iterations (the saturation result documented in v3.0.108 / v3.0.109). The proposed high-bar Sonnet+Gemini 2-of-2 panel *also* fails 0/3 — Gemini is conservative on `peripeteia_induced` attribution, so requiring both Sonnet and Gemini to vote `peripeteia_induced` is stricter than the original 3/4. Three other analytically-defensible recalibrations *do* satisfy the 2-of-3 termination rule on the same data: (i) drop DeepSeek but keep Sonnet/Gemini/Qwen at 2/3 with CUT=50; (ii) Sonnet-alone at original CUT=75 (no calibration changes at all — i01 and i03 pass all 9 items, i02 fails on a single peripeteia `action_gap` plus a single `control_leak`); (iii) Sonnet-alone at CUT=50. The DeepSeek-drop analysis refines the earlier "Qwen+DeepSeek" framing: DeepSeek's noise contribution is the larger of the two on both `control_leak` and `organic_or_ambiguous_recognition`. The honest read is therefore tempered relative to v3.0.109's flat saturation conclusion: the architectural claim — tutor-private peripeteia produces recognitive form in clean draws — survives multiple analytically-defensible recalibrations, but the pre-registered 4-critic 3-of-4 gate is not one of them, so the rerun cannot be reported as having passed that gate. The §7.9 paragraph appended; the closing sentence rewritten to surface the panel-composition sensitivity rather than report flat saturation. No new effect estimate; this is post-hoc analytical exploration of an already-saturated diagnostic and is reported as such.

v3.0.109 (2026-05-28) Post-hoc CUT sweep added to §7.9; refines the

saturation-recommendation empirically. Wrote `scripts/analyze-recognition-origin-cut-swe` which re-applies the `recognitionOrigin.js` classifier at CUT $\in \{50, 55, 60, 65, 70, 75\}$ over the existing `phase2-adaptation-recognition-loop-20260528T022408Z` score JSONs and rebuilds per-item gate verdicts; output at `exports/recognition-origin-cut-sweep.md`. Three empirical bounds locate the calibration space tighter than v3.0.108's "lower the CUT or swap critics" disjunction allowed. (i) CUT=50 is the only useful value: iteration 1 lifts from 0/2 to 2/2 peripeteia passes, iteration 3's T18 item rescues to a clean PASS (4/4 recognition, 3/4 action, 3/4 peripeteia_induced); CUTs 55/60/65/70 are byte-identical to CUT=75 across all 9 items in all three iterations because the raw 1–5 critic scores are bimodal between Sonnet's 4 ($→75)*andQwen*/*DeepSeek′s*3(→$50) — any CUT in $[55, 75]$ is operationally the same. (ii) At CUT=50 two new bottlenecks become load-bearing: `action_gap` (4 of 9 items where Qwen and DeepSeek score `actionalBreakthrough` at 50 against a 75 vote threshold) and `control_leak` on Opus's intrinsic soft-reframe register in routine/none arms (4 control items where 2–3 critics return `formClass=recognition`). (iii) All three failure classes (`organic_or_ambiguous_recognition`, `action_gap`, `control_leak`) therefore trace to the same two systematically-lower-threshold critics. The refined recommendation: critic-panel recalibration (dropping or rebalancing Qwen and DeepSeek) is more load-bearing than classifier CUT alone, because CUT=50 still leaves the gate failing on action and control thresholds that share the same root cause. `control_leak` is genuinely CUT-invariant (formClass-driven, not classifier-driven). The §7.9 paragraph appended with these three findings; TODO §B8 updated with the empirical bounds. No new effect estimate.

v3.0.108 (2026-05-28) Saturation diagnostic added to §7.9, located at the scoring pipeline rather than the generator. Added a bounded paragraph on `phase2-adaptation-recognition-loop-20260528T022408Z`, the rerun that tested the v3.0.107 prompt surgery. All three iterations completed with passing-item trajectory 2/9, 2/9, 4/9; the surgery worked at two layers (structural critic gate now passes 4/4 against the prior 0/3; literal stems present in transcripts; Sonnet 4.6 reads peripeteia-only items as `peripeteia_induced` 3/3 in iteration 3). The loop nevertheless failed the two-pass termination rule. Post-hoc subagent diagnosis relocates the failure: what reads as `control_leak` is Opus's intrinsic soft-reframe register being picked up only by the two lower-threshold critics (Qwen, DeepSeek) — the literal stems do not migrate into routine/none arms, and Sonnet/Gemini correctly mark all controls flat. What reads as origin-attribution split is a classifier-threshold artefact in `scripts/lib/recognitionOrigin.js`: Qwen and DeepSeek score the same mechanism evidence at 50 with prose that explicitly credits the device, but the classifier's hard CUT of 75 on `tutorAdaptiveMechanism` drops them into the `organic` bucket. `organic_or_ambiguous_recognition` fails 3 consecutive iterations, triggering the handoff termination rule. DeepSeek `scorer_error` is provider transience, not a generator regression. The TODO §B7 rule split (earned-vs-named flags) is therefore no longer load-bearing: the structural critic already passes. The two remaining mechanism-redesign options are scorer-side — lower the `recognitionOrigin.js` CUT to ~50–60, or swap the systematically lower-bar critics out of the panel — and neither is reachable by another generation cycle. Terminal saturation for the gated arc, located at the scoring pipeline. No new effect estimate.

v3.0.107 (2026-05-27) Generator-divergence diagnostic added to §7.9. Added a bounded paragraph on `phase2-adaptation-recognition-loop-20260527T232739Z`, which re-ran the same gated D42/D50/D53 termination test with tutor drafting routed through claude-code CLI (Opus, attended Max-plan) instead of the production-batch default codex CLI. All twelve generation arms completed (~1h54m), but the rules-based structural critic failed iteration 1 at the peripeteia-only branch (0/3 items passed `peripeteia_arm_without_earned_reorientation` against codex's prior 3/3 under the same director prompt and scenarios). Diagnosis: generator-style versus rule strictness, not a peripeteia mechanism failure. The director's "close to this shape" sentence stems are lifted verbatim by codex but paraphrased into image-driven physical reorientation by Opus ("Band, not the digit"; "count answering with the tile"), which the literal-token regex in `scripts/critic-poetics-structure.js` cannot see. The rules conflate two separable things — earned reorientation (semantic shift in action) and named reorientation (stem vocabulary) — which the Opus generator can produce only the first of without prompt-side teeth. Response is two-step: queued a deferred engineering item (TODO §B7) to split the rule into earned-vs-named flags; meanwhile tightening the director prompt to mandate at least one literal stem from each list with a worked exemplar, then rerunning with `--generator claude` as a co-occurrence test. No new effect estimate.

v3.0.106 (2026-05-27) Self-contained poetics-sidecar contribution boundary added. Added a reader-facing summary to §7.9 so the Markdown/PDF alone states what the May 2026 poetics sidecar contributes: a dramatic-form instrument for the adaptation-null construct-validity worry, clean low-organic controls for testing tutor-private peripeteia, and a failed two-pass termination rule that localizes the remaining problem to the action-to-re-reading bridge. Explicitly states what the sidecar does not add: no main-harness `evaluation_results` effect, no human-learning evidence, and no deployed adaptive tutor.

v3.0.105 (2026-05-27) Gated adaptation-recognition loop and handoff boundary added. Added `phase2-adaptation-recognition-loop-20260527T105617Z` to §7.9 as a bounded termination test over D42/D50/D53, not a new main effect estimate. The loop required two clean passes within three iterations and achieved one: iteration 1 passed 9/9 item gates, while iterations 2 and 3 each passed 6/9. Controls mostly held and branch-valid tutor peripeteia remained visible, but recognitive closure downstream of adaptation was not repeat-stable because of quality/coverage failures, critic splits, and origin ambiguity. Interpretation tightened: tutor-private peripeteia can produce recognitive form in a clean draw, but the claim-worthy next step is the action-to-re-reading bridge. The generator and structure critic now require the final learner turn in `peripeteia-only` arms to perform the device, name the old pressure/check, and state the replacement check before external scoring. Raw artifacts for the three completed iterations were packaged locally with `poetics:package-run`; only manifests are committed under `config/poetics-calibration/runs/`.

v3.0.104 (2026-05-25) Sonnet fourth-critic scoring and blind critic protocol added. Re-scored `phase2-low-organic-cleaner-adaptation-v2` with Sonnet 4.6, producing a four-critic sidecar panel over 40 items and 124 score rows. The cleaner floors survive the stricter panel: prefix-baseline 0/20 recognition and routine 0/20; D42 and D45 remain the cleanest anchors at prefix 0/4 and routine/none 0/8 each. Public reframe-bearing arms remain robust on scoreable rows (`reframe-only` 11/12, `reframe+tutor-uptake` 8/8, `reframe+peripeteia` 4/4), while `peripeteia-only` stays critic-sensitive (Qwen 2/5, Gemini 0/5, DeepSeek 3/5, Sonnet 0/5). Added a blind-scoring protocol to the phase-2 scorer and CLI bridges: critics receive anonymous transcript text only, generator/source metadata is withheld, Codex CLI scoring runs from an ephemeral read-only temporary working directory, and Claude CLI scoring runs from a non-persistent temporary working directory with project-local tools and browser integration disabled. This reduces source-awareness leakage but does not replace cross-family critic consensus.

v3.0.103 (2026-05-25) Cleaner low-organic adaptation rerun and sidecar correction added. Added `phase2-low-organic-cleaner-adaptation-v2` to §7.9 as a narrowing calibration result. The rerun uses D40/D42 plus new D44–D46 candidates, with prefix-baseline, routine, none, reframe, uptake, and peripeteia arms scored by Qwen 3.7 Max, Gemini 3.5 Flash, and DeepSeek V4 Pro. Prefix-baseline and routine floors are now clean (0/15 recognition each); D42 and D45 are the cleanest anchors (each prefix 0/3 and routine/none 0/6). Explicit public reframe arms remain strong on scoreable rows (`reframe-only` 8/9, `reframe+tutor-uptake` 6/6, `reframe+peripeteia` 3/3), but `peripeteia-only` remains critic-sensitive (Qwen 2/5, Gemini 0/5, DeepSeek 3/5). The tutor-adaptation analyzer is bumped to `tutor-adaptation-v4` after discovering that v3 over-counted reversal pressure inherited from the shared prefix; branch-local v4 shows `peripeteia-only` has pressure/instrumented route/public habit-break in only 1/5 scripts. Interpretation tightened: cleaner floors and durable public-reframe manipulation, but no reliable tutor-private peripeteia result yet.

v3.0.102 (2026-05-25) Adaptation-drama remodeling caveat added. Added the low-organic control expansion and seven-arm adaptation run to §7.9 as calibration apparatus, not as a new main effect estimate. After OpenRouter top-up and scorer retry, `phase2-low-organic-adaptation-v2` has a complete intended three-critic panel (84 score rows, zero scorer errors). D40 and D42 are now the cleaner low-organic anchors (prefix 0/3 and routine/none 0/6 recognitions for each), while D35 and D39 remain boundary probes (prefix 3/3 each). The explicit `reframe-only` branch is 12/12 recognition; `reframe+peripeteia` is 9/9 on scoreable rows; `peripeteia-only` is critic-sensitive and incomplete for the cleanest D42 case because of no-cue reframe leakage. The paper now states the three-variable decomposition needed before any peripeteia claim is promoted: fixed-prefix organic reversal risk, tutor-private habit-break evidence, and external recognition/trap/flat form after branch.

v3.0.101 (2026-05-24) Balanced poetics calibration slice added. Generated and scored `phase2-balanced-calibration-v1`, which carries the v2 breadth target family (D19–D24) together with D4 flat, D10 boundary-trap, and D25/D26 hard-trap controls in one sidecar run. Qwen, Gemini, Sonnet 4.6, and DeepSeek all classify D4 as flat and D10/D25/D26 as trap. Target separation is strong for Qwen/Gemini/Sonnet (`none` 1/6, 1/6, 0/6 recognition; `reframe` 6/6, 6/6, 5/6) and weaker but directionally positive for DeepSeek (1/6 versus 3/6, with one `reframe` trap). Added `poetics:report`, `poetics:audit`, and `poetics:browse` tooling plus sidecar-compatible one-off drama-generator outputs. The result remains calibration apparatus outside `evaluation_results` and outside human-learning claims.

v3.0.100 (2026-05-23) DeepSeek stress-test added for hard-trap controls. Re-scored the first strict-control repeat and generated a second D25/D26 repeat under `config/poetics-calibration/phase2-hard-trap-controls-v1/`, with Qwen, Gemini, Sonnet 4.6, and DeepSeek V4 Pro as critics. Repeat 1 is fully stable: all four critics classify both D25 and D26 as trap. Repeat 2 is stable for Qwen, Gemini, and Sonnet, but DeepSeek classifies both draws as recognition. The §7.9 interpretation is therefore tightened: D25/D26 are stronger hard-trap brackets than D10, but not model-proof ground truth; DeepSeek functions here as a permissive boundary-stress critic. Sidecar ingestion now records the strict-control slice as 4 items and 16 critic scores, still outside `evaluation_results`.

v3.0.99 (2026-05-23) Poetics controls split into boundary traps and hard traps; sidecar persistence added. Added a control-only Sonnet 4.6 spot-check on production-v2 D10, which reads r01/r03 as trap and r02 as recognition, confirming that D10 is a critic-sensitive boundary probe rather than a hard trap. Added strict hard-trap controls D25/D26 under `config/poetics-calibration/phase2-hard-trap-controls-v1/`; Qwen, Gemini, and Sonnet all classify both as trap with no recognition-threshold recontextualization. Added sidecar persistence (`poetics_runs`, `poetics_items`, `poetics_scores`, `poetics_labels`) plus `scripts/ingest-poetics-artifacts.js`, and ingested production-v1, production-v2, and the strict-control slice locally without writing to `evaluation_results`. §7.9's scope is unchanged: critic-rated dramatic form calibration, not human learning and not main-harness evaluation.

v3.0.98 (2026-05-23) DeepSeek third-critic triangulation added to production-v2 breadth check. Added `deepseek/deepseek-v4-pro` scoring for the production-v2 r01–r03 targets and D4/D10 controls, and updated `scripts/analyze-poetics-production-v1.js` so production summaries discover available critic score files dynamically rather than assuming a fixed two-critic panel. DeepSeek reads the target arms as `none` R6/T0/F12 versus `reframe` R14/T0/F4, confirming the direction of the reframe manipulation while using a more permissive baseline for uncued recognition than Qwen or Gemini. D4 remains flat for all three critics in all three repeats. D10 emphatic is trap for all critics in r01, remains trap for Gemini in r02/r03, and is read as recognition by Qwen and DeepSeek in r02/r03. The §7.9 interpretation therefore keeps the positive form-transfer claim but strengthens the critic-sensitive boundary caveat.

v3.0.97 (2026-05-23) Production-v2 breadth repeats folded into §7.9 with critic-sensitive control boundary. Extended `config/poetics-calibration/phase2-production-v2/` from one breadth repeat to three by adding variation-keyed r02/r03 target and control draws; the generator now persists `director_variation_key` in traces and keys so repeats vary scene ecology/register without changing labels or scoring policy. After detector updates for ordinary public reframe forms ("the mistake was that…" and table/chart/clipboard replacement phrasings), all r02/r03 keys are clean and all score files are full-N. Aggregate breadth target contrast now reads: Qwen `none` R1/T3/F14 vs `reframe` R17/T0/F1; Gemini `none` R0/T2/F16 vs `reframe` R13/T0/F5. D4 remains flat for both critics in all three repeats. D10 emphatic is trap for Gemini in all repeats but recognition for Qwen in r02/r03, so §7.9 now reports the control as a critic-sensitive boundary rather than a perfect bracket. Scope unchanged: critic-rated dramatic form transfer only; no human-learning or arbitrary-domain-generalisation claim, and no database/harness migration yet.

v3.0.96 (2026-05-23) First production-v2 breadth check folded into §7.9 as calibration-continuation evidence. Added a bounded paragraph reporting `config/poetics-calibration/phase2-production-v2/`: six new target scenarios across medicine, cartography, sociology, environmental science, engineering, and media studies, with the same paired `none/reframe` manipulation and the same D4/D10 controls. After detector updates for ordinary public reframe phrasing and intentional unfinished lines, all keys are clean. Qwen reads `none` as R1/T1/F4 and `reframe` as R5/T0/F1; Gemini reads `none` as R0/T1/F5 and `reframe` as R5/T0/F1. Controls bracket the result (D4 flat, D10 trap for both critics). Scope is explicit: one-repeat breadth transfer of critic-rated dramatic form, not human learning and not arbitrary-domain generality.

v3.0.95 (2026-05-23) Production-v1 reproduction command pinned to the frozen three-repeat denominator. After a depth top-up added `target-r04` artifacts under the same production-v1 tree, §7.9's aggregate command was made explicit (`--target-repeats r01,r02,r03 --control-repeats r01,r02,r03`) so the reported 0/18, 17/18, 1/18, and 16/18 counts continue to reproduce the frozen attended claim. No §7.9 denominator or empirical interpretation changed; the r04 top-up remains calibration-continuation evidence documented in `config/poetics-calibration/PHASE2-PRODUCTION-NEXT.md`, not a new paper claim.

v3.0.94 (2026-05-23) Production-v1 poetics probe folded into §7.9 as a bounded construct-validity result. Added a narrow "Dramatic-mechanism follow-up" paragraph reporting the attended production-v1 target/control cycle from `config/poetics-calibration/phase2-production-v1/`, with aggregate regeneration via `scripts/analyze-poetics-production-v1.js`. Across three target repeats (six paired target scenarios each), Qwen reads `none` as 0/18 recognitions and `reframe` as 17/18; Gemini reads `none` as 1/18 and `reframe` as 16/18. D4 remains flat for both critics in all three repeats; D10 emphatic is trap for both critics in repeats 2 and 3, with the repeat-1 Qwen/Gemini split preserved because the sample contained a real later re-reading hook. Scope is explicit: critic-rated dramatic form only, not human learning and not general transfer across arbitrary scenario families. Net paper change ~+190 words in §7.9 plus this entry; new analyzer script only, no database migration or evaluation harness change.

v3.0.93 (2026-05-18) Internal-corroboration note (framing only, no re-runs, no

new data, no §6 number changes): §7.9 records that a descriptive probe over the *already-scored* messages-mode corpus reproduces the slope-proxy / frozen-external-standard scoping from the transcripts themselves. Parallel to the v3.0.90 external-corroboration paragraph (the GPT-5.5 critique converging from the manuscript), §7.9 gains a bold-led "Internal corroboration (no new claim)" paragraph: `scripts/analyze-dramatic-reversal.js` over cells 80–87 (n=1,296 dialogues already scored for §6/§8; runs `eval-2026-03-01-aea2abfb` + two others; no generation, no re-judge; LLM scoring pass deliberately disabled as a closed-loop tell) shows performed recognition is positively-valenced 84–99% per cell and decoupled (pooled r $\approx$ –0.06; superego cells 82/86 most negative) from the learner-unauthored rubric-delta channel — reason (ii)'s frozen-external-standard caution and the drama-machine register made quantitative on data already in the corpus. Descriptive process measure, no new empirical claim: same symmetric scoping, now shown from the transcripts not only against an external critic; sole forward pointer is the pre-existing §8.1 scope (now `notes/design-a1-human-learner-pilot.md` §4.1b's independently-coded human explanation). Single-paper discipline preserved (probe report in `exports/`, interpretive prose folded into §7.9, no spin-off). Net paper change ~+220 words (§7.9 one paragraph + this entry; Magee et al. (2024) already in references.bib:1205, cited §3).

v3.0.92 (2026-05-18) Accuracy pass on the v3.0.90–v3.0.91 framing line (no re-runs, no new data, no §6 number changes): two precision corrections answering the GPT-5.5 follow-up. The GPT-5.5 follow-up (`docs/critique/2026-05-18-gptpro-5.5-critique-follow-up.md`) affirms the closure is *not* reopened but flags two over-tight framings shipped in v3.0.90–v3.0.91. Both are corrected here as framing/precision only — no number, dataset, metric, experiment, or null changed. (1) **§6.9.8 "Reading", identity → lineage.** The v3.0.91 knowledge-tracing embed read as if the fitted parametric `state→action` policy *is* a knowledge-tracing model; a full knowledge-tracing design has explicit knowledge components, item-response structure, mastery posteriors and pre/post or transfer outcomes the fitted-policy harvest does not. The sentence is rephrased from an identity to a lineage statement ("the *learned*-policy realisation of a learner-state backbone in the lineage of knowledge tracing (Piech et al., 2015; Scarlatos, Baker, et al., 2025) — not a full mastery-posterior, learning-outcome-validated knowledge-tracing design, a distinction the null respects rather than erases"), localising the same null without over-claiming the analogy. (2) **Response-artifact pre-registration tiering (artifact only, not the paper claim set).** `gptpro-5.5-critique-response.md` §2 had labelled the realisation space "already pre-registered and run"; corrected to "already built and run", with a four-tier status note keeping the paper's own Status caveats distinct — pre-registered kill gates (§6.9.8 fitted-policy / §6.10 signal-axis), post-hoc-but-disciplined (the A14 arc §6.9.7), apparatus/robustness probes (§6.8.5–§6.8.8), and future-out-of-arc externally-validated human learning (§8.1 / IRB-gated Human Learner Pilot) — and §5 rescoped so the closure is explicitly the *synthetic-architecture* arc with human-outcome validation named as genuinely open, not foreclosed. Tightens, not loosens, the closure (every tier still returns the same null); continuous with the v3.0.87–v3.0.91 framing-only line; single-paper discipline preserved (no spin-off, no new analysis). Net paper change ~+260 words (one §6.9.8 sentence rephrase + this entry; the response-artifact corrections are outside the paper claim set).

v3.0.91 (2026-05-18) Related-work completeness pass (framing only, no re-runs, no new data, no §6 number changes): eight existing-claim citations embed the closed arc into its adjacent literatures. A read-only audit of the project's literature collection (`docs/explorations/literature/`, 67 PDFs) against the bibliography found the empirical arc was not cited into four adjacent literatures whose constructs it already engages. Eight references were added to `references.bib` and embedded as one framing sentence each into existing interpretive prose — no results paragraph, table, or N-count touched: LLM-as-judge bias (position bias (L. Shi et al., 2024); unacknowledged shortcut bias (Marioriyad et al., 2025)) into §8.2's pre-existing judge-bias caveat; knowledge tracing (Deep Knowledge Tracing (Piech et al., 2015); KT in tutor–student dialogues (Scarlatos, Baker, et al., 2025)) into §6.9.8's "Reading" of the fitted-policy null; machine theory of mind (FANToM (H. Kim et al., 2023); trivial-alteration brittleness (Ullman, 2023)) into §6.10's "Reading" of the concealed-interior null; backtracing (R. E. Wang, Wirawarn, et al., 2024) into §6.8.2's trap-instrument description; adaptive scaffolding (Cohn et al., 2026) into §6.3's adaptive-responsiveness construct definition. Each sentence is a lineage/consistency statement about prior work ("is consistent with", "is the analogue of", "is what the literature frames as"), not a new empirical assertion about this project's data — the cited sections and their nulls are all pre-existing. Continuous with the v3.0.87–v3.0.90 framing-only line and respecting single-paper discipline (no spin-off, no new analysis). Net paper change ~+150 words (five one-sentence embeds + this entry; eight `references.bib` entries).

v3.0.90 (2026-05-18) External-corroboration note (framing only, no re-runs, no new data, no §6 number changes): §7.9 records that an independent manuscript-only critique converges on the slope-proxy scoping and proposes exactly the realisation space the arc already closed. A critique generated from the manuscript alone (GPTPro 5.5, 2026-05-18; archived with a point-by-point response at `docs/critique/gptpro-5.5-critique.html` + `gptpro-5.5-critique-response.md`) independently reaches §7.9's construct-validity diagnosis (slope-proxy under-determination, the §6.3 level/rate split) and then proposes a state-grounded, externally-validated adaptation instrument as the remedy — which is, point for point, the pre-registered-and-closed realisation space: hand-authored/fitted `state→action` policies (§6.9.7/§6.9.8), concealment/theory-of-mind inference vs the owned hidden trace (§6.10), bilateral-ToM/state-schema/cross-suite minimal-pair ablations (§6.8.5–§6.8.8), cumulative-rewrite arm A16 (§6.3.10). §7.9 gains one bold-led two-sentence paragraph noting that this convergence *corroborates* the symmetric scoping rather than circumventing it (an outside reasoner with no arc access nominates instruments already exhausted on both the policy-form §6.9.8 and signal §6.10 axes), licensing neither pole of the symmetric constraint; its one non-redundant residue (human learning-outcome validation under synthetic learners) is the pre-existing §8.1 scope, not an adaptation-rate instrument. No empirical claim, number, dataset, metric, or experiment added or changed — the cited sections and their nulls are all pre-existing; construct-validity framing of the §8.3 type, continuous with v3.0.87/88/89 and closing the external-review loop. Net paper change ~+190 words (§7.9 one paragraph + this entry).

v3.0.89 (2026-05-18) Construct-validity scope close (framing only, no re-runs, no new data, no §6 number changes): §7.9 names the mechanism-side dual of its by-construction residue — the class that could *in principle* produce rate (online weight update) is outside a prompt-architecture study by construction, stated symmetrically. Closes the third of three adversarial stress-tests of the adaptation-null arc (instrument adequacy / definitional / mechanistic). The first two were already discharged in-text: instrument adequacy by A16/§6.3.10 (the judge-invariant, *uncapped*-trajectory `policyAction` slope — precisely the judge-noise dodge — which nulled, S1−S0 $d = -0.167$) and by the demonstrated power of the same OLS machinery (M3 $d = 1.63$ on the 10-turn scenario); definitional by the `window` non-monotone detector (§6.8) and §6.15's first-to-last delta + stagnation flag (per v3.0.88), with variance-collapse re-identified as the paper's *level*-pole calibration positive, not a missed rate channel. The third left the *mechanistic-class* boundary implicit. §7.9's symmetric-honesty paragraph gains one compound sentence plus a one-clause gloss: every architecture the arc tested is an *inference-time* intervention over fixed weights (prompt, second agent, externalised state, cumulative-rewrite ledger, §6.3.10), so the only class that could in principle convert in-context re-weighting into a genuine per-turn *rate* — online parameter update, not conditioning on the transcript — is by construction outside a prompt-architecture study; the nulls stay silent, symmetrically, on weight-level learning, licensing neither "no mechanism could produce rate" nor "a learning mechanism would." This is the mechanism-side dual of the v3.0.87 measurement-side residue: fixed weights ↔ fixed standard, the two faces of "by construction" (reason (ii)). No empirical claim, number, dataset, metric, or experiment added or changed — §6.3.10's cumulative-ledger arm and its null are pre-existing; construct-validity framing of the §8.3 type, continuous with the v3.0.87/v3.0.88 entries and closing the stress-test trio. Net paper change ~+170 words (§7.9 one sentence + one gloss clause + this entry).

v3.0.88 (2026-05-18) Construct-validity refinement (framing only, no re-runs, no new data, no §6 number changes): §7.9 records that two of its four under-determination channels were already realised as instruments and also nulled, discharging reason (iii) empirically and localising the residual gap in (ii). Prompted by the question whether a non-OLS breakthrough detector would *address* the construct-validity issue: it does — but precisely one of the four channels. §7.9's symmetric-honesty paragraph gains three sentences noting that the "path-sensitive non-monotone trajectory" and (partly) the "axis-change/standard" members of its own conceivable-richer-operationalisation list are not merely conceivable: the §6.8 trap suite already scores a non-monotone breakthrough detector (`window` — expected strategy reached *anywhere* post-trigger, not only at trigger+1; paper §6.8.3–§6.8.4), and §6.15 already adds the symmetric first-to-last delta and an explicit learning-stagnation flag. These register the flat-then-late-breakthrough ("turn-5 breakthrough") trajectory an OLS slope discards, and the convergent negatives hold under them too (the post-trigger pivot is detectable but not architecture-independent: §6.8.4, §6.8.6, §6.8.7). The interpretive consequence, stated symmetrically: this *discharges* reason (iii) by data rather than by bracketing (the null is not a linear-lens artifact), and thereby *localises* the residual construct-validity gap in reasons (i)/(ii)/(iv) — with `window` itself the purest instance of (ii), since it scores against a researcher-fixed `expectedStrategyShift` rather than an immanent standard. Net effect: the construct residue is sharpened from "wrong curve shape" (now closed) to "external standard + per-agent scalar unit" (a conceptual limit no fixed-rubric instrument can reach by construction), strengthening the §7.9 argument rather than weakening it. No empirical claim, number, dataset, metric, or experiment added or changed — the §6.8 `window` and §6.15 delta/stagnation instruments and the §6.8.4/.6/.7 negatives are all pre-existing; this is construct-validity framing of the §8.3 type, continuous with the v3.0.87 entry. Net paper change ~+330 words (§7.9 three sentences + this entry).

v3.0.87 (2026-05-18) Construct-validity scoping (framing only, no re-runs, no new data, no §6 number changes): §7.9 tightened to bound every adaptation null to the *slope proxy* rather than the recognition-theoretic construct, under a symmetric-honesty constraint. The arc's adaptation nulls (§6.7, §6.8.8, §6.9.7, §6.9.8, §6.10) are nulls on one operationalisation — a per-dialogue OLS slope of a scalar, externally-fixed rubric against turn index, with "adapting" identified as a positive monotone trend. §7.9 gains a two-paragraph treatment ("Adaptation as a measured construct") that names this the **slope proxy**, locates the §6.3 level/rate split *inside* that regime (a split between height and slope of one scalar curve, not a question about the regime itself), and gives four construct-validity reasons the proxy under-determines the rich sense of adaptation: scalar collapse, a frozen external standard, monotone-trend identification, and a per-agent unit — each carrying a parenthetical Hegelian gloss (*bestimmte Negation* / non-dialectical measure / *Bildung*-as-path and *Umschlag* / the constitutively dyadic unit of recognition), register (a) measurement-theory primary and register (b) Hegelian parenthetical per the approved structure. The binding move is stated **symmetrically**: the under-determination neither licenses "the architectures failed to adapt in the rich sense" nor rescues "rich adaptation occurred but the proxy missed it" — no in-claim-set instrument adjudicates the rich construct in either direction, so the honest claim contracts to exactly what was measured (the §6.3 level/rate discipline applied one level up, to the identification of rate *itself* with adaptation). Explicitly a scoping of already-closed nulls, *not* a new programme: the realisation space is exhausted on both the policy-form (§6.9.8) and signal (§6.10) axes, and a richer operationalisation is named only to be declined as the ablation creep the arc's closure refuses. The "What it does not explain" trio's adaptive-responsiveness item gains a six-word forward pointer (Option A — the headline's rhetorical force preserved, scope made visible at the point of claim). No empirical claim, number, dataset, metric, or experiment added or changed — construct-validity framing of the §8.3 type already in the paper, generalising (not duplicating) the measurement-vs-theory move the §7.9 learner-superego-paradox paragraph makes locally. Net paper change ~+560 words (§7.9 two paragraphs + trio pointer + this entry).

v3.0.86 (2026-05-18) Correctness fix (framing only, no re-runs, no new data):

reconciled §6.10's and the v3.0.85 entry's adversarial-superego framing with §6.3.10's already-published A16 null. v3.0.85's new §6.10 synthesis imported a stale framing ("the one decisive positive in the arc (the adversarial-superego result: ≈ 20 vs ≈ 85)… the dividing principle the arc's own evidence licenses") that contradicts the pre-existing §6.3.10: that prototype is *explicitly outside the paper's claim set with its figures not paper findings*, and A16 — the paper-native pre-registered test of exactly that mechanism on the externalised `policyAction` instrument — returned the **pre-registered null** (S1−S0 $d = -0.167$; the prototype's effect is *level/reliability*, not the per-turn *rate* that would escape §6.3). §6.10's intro paragraph and the v3.0.85 Appendix F entry's parallel clause are reframed: the adversarial result is the arc's *apparent* exception that the paper's own test (A16, §6.3.10) **closed**, not an in-claim-set positive; the arc carries **no in-claim-set positive** on an architecture-independent channel; the "gains require unread signal" dividing principle is demoted from "evidence the arc licenses" to the *theoretical* reading that makes the convergent negatives intelligible. This makes the arc *more* conclusive (null across policy-form §6.9.8, signal §6.10, *and* forced-opposition/prompt-rewrite §6.3.10) and removes a cross-section internal contradiction. No empirical claim added or changed — every fact cited is already in §6.3.10 (A16, run and resolved at v3.0.83); **no §6 number changes**. Net paper change ~+220 words (§6.10 reframe + this entry); two-clause edit only.

v3.0.85 (2026-05-18) §6.10 added — the *signal*-axis lever (model the learner's *concealed interior*: manifest≠latent, the one signal the base provably misreads, owned here as the learner's logged hidden ego–superego deliberation) killed at a pre-registered offline gate; the §6.7–§6.9.8 motif now closes the *signal* axis as §6.9.8 closed the *policy-realisation* axis. Offline, zero-API, read-only on existing `main` traces; ≈ \$0, no new judging, no live run. After §6.9.8 exhausted the policy realisations of `state → action`, the standing question was whether the failures shared a *signal* cause: every in-paper negative re-encoded what the base already reads in-context, while the arc's one *apparent* exception (the out-of-claim-set adversarial-superego prototype, §6.3.10) was put to the paper's own pre-registered test (A16) and shown to be a *level/reliability* effect, not the per-turn *rate* that would escape §6.3 (S1−S0 $d = -0.167$) — so the arc carries no in-claim-set positive. The untested lever: model the learner's *concealed* interior (the trap suite is built from concealment cases; the base reads at face value; this study uniquely owns the hidden `learner_ego_initial / learner_superego / learner_ego_revision` deliberation as offline ground truth). `scripts/adaptation2-stage0.py` (frozen before running per `ADAPTATION-PLAN-2.0.md`; rationale `ADAPTATION-2.0-DIALOGUE.md`; numpy + stdlib only) harvested $n = 1604$ learner triplet-turns across 9 scenario-type families / 11 profiles / 2589 dialogues from existing `ego_superego` trace files (read-only), target = 1 − tfidf-cosine(*ego_initial*, *final_output*), predictor set A = surface-only (= what the base reads), B = `learner_superego` text only; ridge $\lambda = 1.0$, leave-one-`scenario_type`-out grouped CV, cluster bootstrap by `scenario_type` ($B = 2000$, seed `20260517`). **Pre-registered P1 (frozen): lower 95% CI of $R^2_B > 0$ *and* of $(R^2_B - R^2_A) > 0$.** Result: $R^2_A = +0.086$; $R^2_B = -0.026$, CI $[-0.212, -0.032]$ (entirely < 0); gap $B - A = -0.111$, CI $[-0.234, -0.010]$ (entirely < 0); robustness target (token-Jaccard) concurs ($R^2_A = +0.039$, $R^2_B = -0.022$). **Gate fails on both conditions.** Interpretive core: the phenomenon is *not* floored (target mean 0.647, sd 0.101, 99.6% graded) but 95.7% of its variance is idiosyncratic *within* family and only 4.3% between — it does not track the trap type and is unrecoverable from the surface *or* the logged hidden trace ($\mathrm{corr}(\mathit{target}, \mathrm{len}\,\mathit{superego}) \approx -0.14$); the "hidden" deliberation is itself in-context LLM surface text, not the privileged latent variable the lever presumed. Per the frozen kill rule the arc closes: **no Stage 1, no live run, no tuning retry** — the offline null is the closing result. Fifth *in-paper* convergent negative (§6.7, §6.8.8, §6.9.7, §6.9.8, §6.10); archived persona is a sixth outside the claim set. Weber companion probe (P2, charisma→receptivity uptake) **structurally unmeasurable** on the existing corpus (cells 100–109 are scripted-learner / tutor-only, $\leq$ 38/505 rows any learner score, 0 interiority) — that corpus gap *is* the reason the charisma lever was never realised (the §6.7 arc instrumented charisma's *performance*, never its *uptake*). Stated limitation (does not reopen the gate, by pre-registration): the offline target is lexical, a proxy for semantic concealment; the variance decomposition holds regardless of proxy fidelity and the proxy was frozen precisely to forbid a richer-probe retry. Surgical revisions: new self-contained §6.10 (sibling of §6.9); **no §6.8 or §6.9 number changes** — §6.10 consumes only already-logged traces. Artifacts: `exports/adaptation2-stage0-{table.csv,probe.meta.json}`. Net paper change ~+1,250 words (new §6.10 + this entry).

v3.0.84 (2026-05-17) §6.9.8 added — the learned-policy lever (a `state→action` policy *fitted* to the independent outcome, the one realisation the §6.8/§6.9 arc had not tried) pulled under a pre-registered offline kill gate and closed null; the §6.8.8/§6.9.7 motif is now exhaustive over policy realisations actually tried, not merely repeated. Offline, zero-API, read-only on existing `main` traces; ≈ \$0, no new judging, no live run. Harvested 179/179 trajectories from the existing `cell_110/118/124/126/127/128` trace files + `evaluation_results` (read-only) as one bandit row per pivot decision (context = A14 typed pre-pivot state; action = pivot `policyAction` family; reward = the §6.9.7 instruments) via `scripts/learned-adaptation-harvest.js`. The binary label is **fidelity-asserted** to be the §6.9.7 instrument: a verbatim `analyzeBranch` copy recomputes `strict_shift`, required to equal `exports/a14-stage5-strategy-shift-finalN.json` to 10^{-6} or hard-abort — exact match (`cell_126` 0.4242 / `cell_127` 0.5455 / `cell_128` 0.5882, $\Delta = 0$; recovered baselines reproduce §6.9.7's table: `cell_110` 47.8%, `cell_118` 68.8%, `cell_124` 62.5%). `scripts/learned-adaptation-policy-ope.py` (numpy only) fit the simplest policy (multinomial-logistic binary / ridge-linear graded) under leave-one-`scenario_type`-out grouped CV (13 trap families), evaluated off-policy by a Direct Method + an assumption-light on-policy-agreement value, CIs by cluster bootstrap on `scenario_type` ($B = 2000$, seed `20260517`). **Pre-registered gate (frozen in `LEARNED-ADAPTATION-PLAN.md` before OPE): beat both `cell_110` (implicit) and `cell_126` (hand-authored), lower 95% CI > 0, on binary `strict_shift`.** Result: vs `cell_110` $\Delta = +9.0$ pp, 95% CI $[-0.9, +18.0]$ — **does not beat**; vs `cell_126` $\Delta = +14.4$ pp, CI $[+4.8, +23.0]$ — beats; graded channel corroborates (-0.03 vs `cell_110`, $+0.16$ vs `cell_126`); assumption-light cross-check ($n = 95$) below both baselines on both channels. **Gate fails:** the fitted policy is statistically indistinguishable from the implicit base and clears only the weakest A14 cell, both channels, both estimators — the §6.8.8 motif one level up (the elaboration is now *learning* the policy, and it still does not pay). Per the plan's frozen kill criteria the arc closes: **no live run, no counterfactual stage, no tuning retry** — the offline null is the closing result. Fourth *in-paper* convergent negative on adaptive elaboration (§6.7, §6.8.8, §6.9.7, §6.9.8); the archived closed-loop persona prototype is a fifth outside the paper's claim set, named for completeness only. Surgical revisions: new §6.9.8; one clause appended to the §6.9 intro pointing to it as the cross-arc realisation-space close (distinct from §6.9.7's A14-specific close). **No §6.8 or §6.9.1–§6.9.7 number changes** — §6.9.8 is self-contained and consumes only already-scored outcomes. Net paper change ~+1,150 words (new §6.9.8 + intro clause + this entry).

v3.0.83 (2026-05-17) §6.3.10 resolved — A16 P3 decisive run complete; the primary endpoint is the pre-registered informative null; a cumulative-rewrite superego does not escape the §6.3 slope-null. Second named deviation (F = new `cell_132`, symmetric to v3.0.81's A = `cell_131`). Data + analysis only; generation ≈ $0 metered. P3 of the §6.3.10 A16 pre-registration was authorized at the mid power tier (pre-registered per-arm $N = 48$) and executed: all four arms on `claude-code/sonnet` over `config/adaptive-trap-scenarios.yaml`, run sequentially and pooled by `run_id` across Max-plan quota windows (config-clean — profile + `ADAPTIVE_TUTOR_LLM=real` byte-identical per arm regardless of batch label); generation metered ≈ $0, the real cost being quota/wallclock handled by an autonomous quota-paced top-up loop. **Second named pre-registration deviation (§5.12.6 "named not hidden"), exactly symmetric to v3.0.81's A/`cell_131`:** F realised as a *new* `cell_132_a16_F_recognition_only` (architecture byte-identical to `cell_111`, holding it frozen), not `cell_111` itself, because `cell_111_a13_C1_recognition_only` carries paper-cited A13 C1 `openrouter/nemotron` data (§5.12.2, §6.8.4) and in-place editing would cross-contaminate cited provenance. Both F and A are thus new model-aligned cells; S0/S1 genuinely new architectures; all four model-clean. Neither swap alters the frozen design, so the result stays **confirmatory** (not re-classified exploratory). Pre-registered primary endpoint computed by a dedicated `scripts/analyze-a16-rewrite-slope.js` (`exports/a16-rewrite-slope.json`) applying the §6.3.2/§6.3.4 per-dialogue OLS machinery to the §6.8.2 in-family `policyAction` indicator — same endpoint/statistic as pre-registered, standalone analyzer because `analyze-trajectory-curves.js` is rubric-dimension-scoped (implementation-realisation note, not a design change). **Result (all $N = 48$):** S1 (cumulative) 0.173 (sd 0.166), S0 (stateless) 0.200 (0.158), A (advisory) 0.154 (0.162), F (floor) 0.162 (0.155). **Primary S1−S0** $d = -0.167$ (mean diff −0.027; Welch $t = -0.82$, $df \approx 94$, $p \approx 0.41$) — below the $|d| \geq 0.27$ bar, point estimate marginally *reversed* (S1 < S0): **a pre-registered null.** Channel-localisation grid (criterion 2, exact $N = 48$): S0−A $d = 0.286$ (only contrast clearing the bar), S0−F 0.239 (band–bar gap), S1−A 0.114, S1−F 0.065, A−F −0.052 (all ≈ 0). Criteria 3 (cross-judge) and 4 (brittleness) **N/A by construction** (externalised judge-invariant indicator; trap suite has no counterfactual branch). **Verdict per the frozen rule:** primary fails + localisation substantially holds (S1 *at* the recognition floor, S1−F $|d| = 0.065$;

A–F $|d| = 0.05$) ⇒ the **pre-registered informative null**. Stronger than a bare null: the lone elevation is **S0** — the stateless arm the pre-reg predicted *inert* (≈ id-director, §6.7) — detectable only versus the advisory control ($d = 0.29$), sub-threshold versus floor ($d = 0.24$); the single signal is in the predicted-inert arm, opposite the hypothesis's direction, and marginal. The §6.3 slope-null **extends to lightweight prompt rewrite including cumulative-conditioned rewrite**; §6.3.8's slope-null and §6.3.9's structural-gap escape clause are *not* satisfied by a memory-augmented superego rewrite channel (consistent with §6.8.7). Surgical revisions: §6.3.10 Status line repointed (not-yet-run → run/resolved); the v3.0.81 implementation note's trailing "P3 remains gated" clause updated to "P3 executed"; a new closing **Result (P3 decisive; v3.0.83)** paragraph added with the grid, primary, localisation grid, and verdict. **No other §6 number changes** — A16 is self-contained in §6.3.10. Net paper change ~+750 words (§6.3.10 Result paragraph + this Appendix F entry).

v3.0.82 (2026-05-17) §6.9.7 added — A14 Stage 5 pedagogical-outcome scoring closes the arc: the audit chain is mechanism-positive but pedagogically negative-leaning. Ran all three evidence-bound cells on the v1 trap suite at pooled $N \approx 33$ (`cell_126` = 33, `cell_127` = 33, `cell_128` = 34; 100 dialogues, 0 grader errors) and scored them on both §6.8 instruments with the same judges §6.8.3–§6.8.7 used (binary `strict_shift` via `analyze-strategy-shift.js`, `claude-code/sonnet`; 4-dim graded via `grade-adaptive-dialogue.js`, GPT-5/codex grader v1.0). **Binary:** `cell_126` 42.4% (14/33) < `cell_127` 54.5% (18/33) < `cell_128` 58.8% (20/34); family-match 90.9 / 93.9 / 100%; within-action refinement 100% on all three (50/44/60 — closes the degenerate-constant concern); counterfactual replay active (divergence 93.9/97.0/88.2%, *not* a no-op unlike §6.8.6's P2.2 cells). **Graded overall (T/E/Q/C):** `cell_126` 3.84 (3.76/3.39/4.18/4.03), `cell_127` 3.85 (3.76/3.52/4.12/4.00), `cell_128` 4.13 (4.09/3.91/4.32/4.21). **Two findings.** (b) The validator's +12.1-pp binary gain over the updater-only `cell_126` does *not* survive the graded channel (3.84 → 3.85, a +0.01 null) — the `cell_123` binary/graded-divergence pattern of §6.8.6; only `cell_128` moves both needles (top binary *and* top graded), structurally the §6.8.6 `cell_118` "only the lean config is cross-instrument-robust" shape. (a) The ledger+updater layer without the validator (`cell_126`) is *below* the plain `state_policy` baseline `cell_110` on both channels (−5.4 pp binary, −0.19 graded); no evidence-bound cell reaches `cell_124`'s 62.5% or `cell_118`'s 68.8%/4.34; and the sharpest comparison — `cell_128` is §6.8.6's winning `cell_118` minimal profile *with* the A14 audit chain — scores 58.8%/4.13 against the unaudited `cell_118`'s 68.8%/4.34, so the apparatus is net-negative (−10 pp binary, −0.21 graded) on the very configuration most favourable to it. **Trust-vs-load resolves to load** on the §6.8.6-designated load-bearing graded channel: the grounding apparatus is mechanism-clean (§6.9.5's asymmetric attribution holds) but pedagogically inert-to-negative — the §6.8.8 "apparatus is the contribution; the elaboration does not pay" motif, with A14's evidence-binding as the elaboration. **Surgical revisions:** §6.9.6 retitled "Closing Synthesis…" → "Stage 3 Synthesis: The Audit Chain Works; the Pedagogical Question Is Settled in §6.9.7"; its "What is not established" bullet 1 flipped from open-gap to resolved-with-pointer; its "Relation to §6.8.8" closing sentence and Status paragraph repointed from `TODO.md`-forward to §6.9.7-answered/superseded; the §6.9 intro's "named as the next deliverable but not run" / "no §6.8-vs-§6.9 comparison" clauses updated. `TODO.md` A14 header → "Stages 1-5 DONE; arc closed 2026-05-17", Stage 5 checkbox marked `[x]` with the result. No §6.8 number changes; §6.9 only. Net paper change ~+1,500 words (new §6.9.7 + §6.9.6/intro revisions).

- **v3.0.81 (2026-05-16) §6.3.10 implementation note — A16 P2 implemented & validated; four arms realised; A-arm named deviation (new `cell_131`, not `cell_112`). Code/config/docs only, no data, no spend.** P2 of the §6.3.10 A16 pre-registration is complete under mock and hermetic test with zero database writes: new `superegoRevise` graph node (mirrors the id-director persona constructor but writes a dedicated `tutorInternal.superegoAuthoredPrompt` consumed via the existing system-prompt-override path — no ego-node change, byte-identical ego across arms), `superego_revise_stateless` (S0) / `superego_revise_cumulative` (S1) architectures, and an append-only `revisionLedger` channel cloned from A14's `evidenceLog` reducer. Validation: targeted mock test `services/__tests__/adaptiveTutor.superegoRevise.test.js` (3/3 green — node visited, prompt flows to ego at state level, S0 ledger empty by construction, S1 accumulates one entry/turn and threads it back); full hermetic suite green except the known pre-existing `progressLogger` pollution flake (passes in isolation; none of the five changed files touch it); independent code-review audit all-PASS, zero high-confidence issues. Added a closing **Implementation note** paragraph to §6.3.10 recording the concrete four-arm realisation (**F** = `cell_111_a13_C1_recognition_only`, **A** = `cell_131_a16_A_egosuperego`, **S0** = `cell_129_superego_revise_stateless`, **S1** = `cell_130_superego_revise_cumulative`; all `claude-code/sonnet` over `config/adaptive-trap-scenarios.yaml`) and **one named pre-registration deviation** in the §5.12.6 sense: the design table identifies A with the existing `cell_112`, but `cell_112_a13_C2_egosuperego` carries paper-cited A13 C2 data on `openrouter/nemotron` (§6.8.4), so editing it in place would contaminate cited provenance and leaving A there would put A off-model versus F/S0/S1; A is therefore a new model-aligned `ego_superego` cell holding `cell_112` frozen, which makes both the decisive S1−S0 contrast and the secondary A-vs-S0 channel-localisation criterion model-clean. Cells registered in `config/tutor-agents.yaml` and `EVAL_ONLY_PROFILES`. **No §6 number changes** — P3 (the cost-gated paid run) remains gated on explicit per-stage go-ahead. Net paper change ~+560 words (§6.3.10 implementation note + this entry).
- **v3.0.80 (2026-05-16) §6.3.10 added — A16 pre-registration: does a cumulative-rewrite superego produce a rate effect? Docs/design only, no data, no code, no spend.** Added §6.3.10 ("Pre-Registration: Does a Cumulative-Rewrite Superego Produce a Rate Effect? (A16)") as the closing subsection of §6.3, after §6.3.9's structural-gap finding and before §6.4. It records — *before data collection*, in the same form as the A12 disengagement pre-registration (§6.3.2) and the A13 condition design (§5.12.2) — a hypothesis, a four-arm design, a validity control, and a frozen decision rule. **Status-tagged pre-registered/confirmatory/not-yet-run per §5.12.6**; no empirical claim is made and the motivating prototype (`prototypes/adversarial-superego-mvp/`, a disposable mechanism-triage probe) is

explicitly outside the paper's claim set with its figures named as non-findings. **Sharpened hypothesis:** the §6.3 slope-null is escaped, if at all, by the superego's rewrite *policy* being cumulative (each revision integrates accumulated discursive friction — the psychoanalytic continuous-revision motivation), not by edit rights per se; a memoryless re-author (≈ the id-director's per-turn persona constructor, §6.7) predicts level, a ledger-conditioned one predicts rate. **Four-arm design** F/A/S0/S1 (floor `recognition_only` = cell_111 / advisory `ego_superego` = cell_112 / stateless rewrite / cumulative-ledger rewrite), with the decisive contrast S1 vs S0 (byte-identical edit-rights channel; sole difference = ledger-statefulness) and an explicit byte-identical-ego validity control (the manipulation lives only in the superego's rewrite input, never the ego's transcript view). **Pre-registered analysis:** per-dialogue OLS slope of the externalised binary in-family `policyAction` indicator (§6.8.2 instrument, removing the prototype's free-text-classifier confound) over uncapped trajectories, scored with the §6.3.2/§6.3.4 OLS machinery (`scripts/analyze-trajectory-curves.js`) parameterized onto the in-family indicator — a parameterization of an existing analyzer, not a new instrument; four-criterion decision grid modelled on the A12 grid (primary S1−S0 $d \geq 0.27$ at the §6.3.2 detectability bar; channel-localisation A,S0 ≈ 0; A12-style cross-judge $\Delta d \leq 0.5$; brittleness guard) with a null on S1−S0 pre-declared as informative (it would extend the §6.3 slope-null to even cumulative lightweight rewrite). **Blast-radius staging recorded:** P1 = this docs-only fold + the TODO.md A16 entry; P2 = node + two architectures + two cells under mock/hermetic only (zero DB/paper writes); P3 = cost-gated paid run on explicit per-stage go-ahead; concrete `cell_*` IDs deferred to a P2 `tutor-agents.yaml` allocation check per the §5.12.1 cell-registry convention. Cross-references verified against the actual document (the prototype's internal "§6.12" shorthand maps to the paper's §6.3.2/§6.3.4 trajectory machinery and §6.8.2 `policyAction` instrument — no §6.12 exists; no dangling reference introduced). **No §6 number changes anywhere** — §6.3.10 introduces no data, only a frozen design. Net paper change ~+1,150 words (new §6.3.10). Companion: `TODO.md` A16 entry added; design source `prototypes/adversarial-superego-mvp/PROMOTION-PATH.md` §12.

v3.0.79 (2026-05-13) §6.9.6 added — A14 arc closes at three-stage architectural feasibility + paired-ablation marginal-contribution; pedagogical-outcome scoring relocated to TODO.md as A14 Stage 5. Wrote §6.9.6 ("Closing Synthesis: The Audit Chain Works, the Pedagogical Question Remains Open"), the closing synthesis subsection for §6.9, modelled on §6.8.8's defensible / not-established / relation / status structure. **Defensible** (four findings): three-stage audit chain holds on real-LLM material (100% verifiable quotes, 100% grounded hypotheses, 85% non-silent verdicts); validator marginal contribution is wholly-promotion / partly-retirement on the matched N=2 ablation (cell_126 emits 0/19 validated vs cell_127's 5/14; retirement 5 → 7 shared between layers); the cheap-verifier-after-expensive-synthesizer pattern is empirically demonstrated (validator ~5× cheaper per call than the upstream `hypothesisUpdater`); Task #34 misdiagnosis resolved via per-call instrumentation. **Not established** (four gaps): no pedagogical-outcome scoring against §6.8's strict_shift or graded rubric; generalisation beyond matched N=2 (sophistication_ upgrade_v1 single-sided); robustness across judge models/backends; whether revision-vs-duplication improves under the validator. **Relation**: §6.8.8 closed on "apparatus is the contribution; elaborations mostly are not"; §6.9 fits the same motif with a marginal upgrade — the apparatus has produced a clean mechanism finding (asymmetric attribution) even before any pedagogical-outcome link is shown. **Status**: post-hoc, exploratory, infrastructural, not pre-registered. **§6.9 header retitled** from "(A14, Mid-Arc Report)" to "(A14)" — the arc reaches a stable resting point with this section. **§6.9 intro paragraph rewritten** to reflect three stages complete + paired ablation + named outstanding deliverable. **§6.9.5 Status paragraph removed** (it was arc-wide and now lives in §6.9.6's Status paragraph, mirroring §6.8.8's convention). `TODO.md` **A14 updated**: Stages 1-4 marked DONE with commit refs; Stage 5 added as the explicit named deliverable for pedagogical-outcome scoring at N≈24-32 against the §6.8 instruments (~$30-60 API budget; ~16-32h wallclock; named anchor for the §6.9.6 "outstanding measurement" pointer). The original Stage 4 plan (four-metric `analyze-evidence-grounding.js` script) was collapsed into the paper §6.9 reporting rather than landed as a standalone analyzer; the adaptivity-taxonomy table from the May 2026 next-steps report was judged out-of-scope. No §6.8 number changes; §6.9 only. Net paper change ~+850 words (new §6.9.6 + intro rewrite − §6.9.5 Status removal).

v3.0.78 (2026-05-13) §6.9.5 extended — cell_126 paired-ablation closes the validator marginal-contribution measurement; Zod schema fix. Ran `cell_126_state_policy_evidence_bound` (validator off; same `claude-code/sonnet` backend; same trap-scenario inputs as the cell_127 N=3 smoke) to capture the terminal status distribution the v3.0.75 Stage 2 smoke had not measured. Two of three scenarios completed cleanly; `sophistication_upgrade_v1` crashed on a Zod schema-strictness bug — the `evidenceExtractor` LLM emitted `quote: null` for a `tutor_inference` entry. The graph node at `services/adaptiveTutor/graph.js:371` already coerces null quotes to empty strings (the substring-match gate then correctly marks the entry `validated=false`); the schema was rejecting an input the downstream node was handling. Fixed at `services/adaptiveTutor/realLLM.js:478` by changing `quote: z.string().min(1)` to `quote: z.string().nullable()` with rationale documented inline; 18/18 unit tests still pass. **Matched N=2 result.** On the two scenarios both cells ran cleanly (`false_confusion_v1`, `resistance_to_insight_v1`): cell_126 emits **0 `validated` / 5 `contradicted` / 14 `tentative`** across 19 hypotheses (73.7% tentative); cell_127 on the same two scenarios emits **5 `validated` / 7 `contradicted` / 2 `tentative`** across 14 hypotheses (14.3% tentative). Three patterns: (i) the updater alone produces zero `validated` outcomes — every `validated` verdict in cell_127 is wholly validator-driven; (ii) `contradicted` is partly shared (5 → 7), not wholly validator-driven — the updater itself retires claims when prompted; (iii) the validator converts ~59 percentage points of `tentative` into typed verdicts (split ~35pp promotion / 24pp retirement). The asymmetry — wholly-validator-driven promotion, shared retirement — is the operational reading of the verdict-only contract: promotion requires multi-turn evidence accumulation the upstream updater's single-turn prompt does not gather. §6.9.5's "what does/does not establish" caveat rewritten: marginal-contribution measurement is now in (the validator is the sole source of `validated` verdicts and a non-trivial additional source of `contradicted` ones on the matched subset); what's still not established are pedagogical-outcome claims, generalisation beyond the two scenarios (sophistication_upgrade_v1 is unmeasured at the paired level), and robustness across judge models / backends. No §6.8 number changes; §6.9.5 only.

v3.0.77 (2026-05-13) §6.9.5 added — Stage 3 real-LLM result and Task #34 resolution. Real-LLM smoke on `cell_127_state_policy_evidence_bound_validated` (`claude-code/sonnet`; the same $N = 3$ trap scenarios `false_confusion_v1`, `resistance_to_insight_v1`, `sophistication_upgrade_v1`) ran the full extractor → updater → validator chain to completion in ~100 minutes wallclock. Stages 2a/2b clear at the same rates as the cell_126 smoke (100% verifiable, 100% grounded; revision 90.9% vs cell_126's 92.9%). Across the 26 hypotheses synthesised over the three scenarios, the `groundingValidator` produces a terminal status distribution of **13 `validated` (50%) / 9 `contradicted` (35%) / 4 `tentative` (15%)** — the validator fires on 22/26 = 85% of hypotheses, with promotion and retirement decisions roughly balanced (the SILENCE-by-default contract leaves 15% as a residual). The validator's per-call latency (~18s median) is ~5× cheaper than the `hypothesisUpdater` it audits (~100s median), consistent with the cheap-verifier design intent. §6.9.5 reports this and is explicit that no paired-ablation comparison against cell_126 statusDist on the same scenarios is yet available — the marginal contribution of the validator above a well-tuned updater is unmeasured. **Task #34 resolution:** the prior "48-minute silent hang" on the Stage 2b smoke was a misdiagnosis. With per-call tracing on (`ADAPTIVE_TUTOR_CLI_TRACE=1`, landed at commit `62fa729`), no individual role-call exceeds the 6-minute CLI subprocess timeout; the longest single call is 134s. Cumulative 3-scenario × 6-turn × counterfactual-replay wallclock is ~100 minutes, not the ~30 estimated at first launch — the smoke was slow, not hung. The unconditional stderr line on timeout-fire (also in `62fa729`) ensures any future hang surfaces a role + duration signal rather than the silent-wait failure mode the prior diagnostic gap created. §6.9.4 rephrased to flow into §6.9.5 (the "not established" list now points at the unmeasured paired-ablation rather than the unimplemented Stage 3 architecture). No §6.8 number changes.

v3.0.76 (2026-05-13) A14 Stage 3 architecture landed, §6.9.4 cell-name and tense fix; no §6 number changes. Implemented the `state_policy_evidence_bound_validated` architecture and registered `cell_127_state_policy_evidence_bound_validated` as the paired ablation against `cell_126` (validator off vs on, same scenarios). The new `groundingValidator` graph node (`services/adaptiveTutor/graph.js`) runs downstream of `hypothesisUpdater` and walks the tentative-hypothesis set each turn: promotes hypotheses with ≥3 supporting `obs_ids` and no contradiction to `validated`; retires hypotheses with ≥1 contradicting `obs_id` and confidence below 0.4 to `contradicted`; stays silent otherwise (the SILENCE convention preserves existing entries via the merge-by-id reducer, mirroring the Stage 2b updater). The verdict-only contract is enforced by the Zod schema (`{decisions: [{hypothesis_id, new_status: validated|contradicted, reasoning}]}`) — the LLM doesn't own hypothesis fields, only the status transition; the graph node preserves `claim`, supporting_evidence, contradicting_evidence, created_at_turn, expires_after_turns. Mock fixture (services/adaptiveTutor/mockLLM.js) and the runner's counterfactual allow-list (services/adaptiveTutor/runner.js) updated symmetrically. Mock smoke verifies the topology (groundingValidator appears in the visited-nodes set between hypothesisUpdater and learnerProfileUpdate, status transitions apply, counterfactual replay routes correctly). **Paper edit:** §6.9.4 prose corrected — v3.0.75 (the §6.9 commit) referred to "the planned cell_127_state_policy_evidence_bound_no_validator Stage 3 contrast is not yet implemented", which had two errors: (a) the cell is named cell_127_state_policy_evidence_bound_validated, since the no-validator baseline *is* cell_126, and (b) the architecture is now implemented (the cell-127-vs-cell-126 real-LLM comparison is the next outstanding work, not the architecture itself). Rephrased the §6.9.4 "what is not established" sentence to name the architecture and the specific retain/retire thresholds, and to identify the cell_126-vs-cell_127 comparison run as the next deliverable. No data, no number changes — implementation + paper text correction only. The real-LLM smoke for cell_127 is the next mid-arc deliverable.

v3.0.75 (2026-05-13) §6.9 added — Evidence-Bound Hypothesis Tracking (A14 Stage 1+2 mid-arc report). Added §6.9 reporting the first two stages of the A14 evidence-bound adaptive controller arc: state schema with append-only evidenceLog and merge-by-id hypotheses reducers (Stage 1, behaviourally a clone of cell_110 until nodes wire in); evidenceExtractor with substring-match quote-validation gate plus hypothesisUpdater with node-owned id derivation, TTL preservation, and evidence-ref filtering (Stage 2). Real-LLM verification on cell_126_state_policy_evidence_bound (N=3 trap scenarios: false_confusion_v1, resistance_to_insight_v1, sophistication_upgrade_v1; claude-code/sonnet): **Stage 2a verifiable-quote rate 128/128 = 100.0%**, **Stage 2b grounded-hypothesis rate 53/53 = 100.0%** — both clear their internal calibration targets (≥95%, ≥70% respectively). §6.9 is explicitly tier-labelled post-hoc / exploratory / infrastructural per §5.12.6 — the contribution is *mechanism availability* (apparatus to ask grounding-discipline questions of a tutor's learner model), not pedagogical outcome. A revision-vs-duplication signal surfaces in the same smoke (24 distinct hypothesis_ids on sophistication_upgrade_v1's 4-turn × 2-branch run; merge-by-id reducer rarely fires because the model produces fresh claim text each turn rather than rewriting prior claims) and is reported as a Stage-3 limitation, not a Stage 2 result. Stage 3 (groundingValidator + cell_127) is named as the next mid-arc report. Cross-references in §6.8.6 (state-richness reversal as motivation for typed hypotheses) and §6.8.8 (the "apparatus is the contribution" framing) are descriptive, not comparative. Net change ~+1,900 words (new §6.9 + 4 subsections).

v3.0.74 (2026-05-12) bug_007 regression test now exists — with a refinement so it doesn't false-flag the A10/A10b density controls; v3.0.47's methodological note tidied. v3.0.47's methodological note (below) recommended a regression test asserting that resolveEvalProfile(cell) does not fall through to 'budget' for any cell whose YAML prompt_type isn't base. That test now exists as tests/regression-bug-007.test.js — but its naive form (assert on `resolvedProfileName`) red-flags `cell_95` (`matched_pedagogical`) and `cell_96` (`matched_behaviorist`), the §7.9 A10/A10b prompt-density controls: both *do* have dispatch branches in `resolveEvalProfile`, but those branches target tutor-core profiles registered only in the dev build of `@machinespirits/tutor-core`, so against a published install the dispatch output is correctly `matched_pedagogical` / `matched_behaviorist` while the existence-check fallback then degrades the actual *load* to `'budget'` — the documented "this experiment cannot run against a published install" behaviour, not the bug_007 pattern. **Fix:** `resolveEvalProfile` now also returns `dispatchedProfileName`, the if/else-if chain's pick captured *before* the existence-check fallback (purely additive — all three call sites destructure only `resolvedProfileName`, unchanged); the regression test asserts the bug_007 invariant on `dispatchedProfileName` (a non-base `prompt_type` *dispatching* to `'budget'` = a forgotten branch, the real failure mode; *falling back* to `'budget'` because the named profile is absent on a published package = graceful degradation), plus a positive pin that `cell_95` / `cell_96` still dispatch by name so a future forgotten branch cannot silently regress them. Three subtests pass; the full hermetic suite is back to one pre-existing unrelated failure (a `progressLogger` parallel-run pollution flake that passes in isolation). **Paper edit:** v3.0.47's methodological note carried a stale present-tense clause ("the factorial-design tests … fail on cell_95") — tidied to past tense (the `bug_011` allow-list gap was fixed in v3.0.47 itself) and its forward-looking test recommendation made field-agnostic with a pointer to this entry. No data, analysis, or N-count changes — code/test/changelog only. Folded in on the same `main` line, also infra-only: the cells-110-125 `runner:` family is now excluded from the factorial-convention tests (`factorial-design`, `dialogueStructure` egoModel resolution, `pedagogical-orientation-coverage` — a gap surfaced by the §6.8.7 merge; mirrors the existing `factors.id_director` exclusion), and `@machinespirits/tutor-core` was bumped 0.5.0 → 0.5.2 (lockfile lag — the eval-repo code committed Mar 2 needs the 0.5.1+ `callAI` / `getPromptMetadata` exports; `npm update` re-resolved within the existing `^0.5.0` range, `package.json` untouched), restoring the ~44 subtests that had depended on those exports.

v3.0.73 (2026-05-12) §6.8.7 cross-suite test executed — the §6.8.3–§6.8.5 forward-reference discharged with data, not just synthesis. Ran the "clean cross-suite test" §6.8.7 had named as the genuinely outstanding item: the LangGraph `state_policy` runner and the dialogue-engine baseline, on a trap-style-annotated subset of the §6.3 suggestion scenarios, scored with the same binary `strict_shift`. **New artefact.** `config/cross-suite-trap-scenarios.yaml` re-derives six trap scenarios from the §6.3 suggestion suite (`epistemic_resistance_impasse` → `cross_epistemic_resistance`, `affective_shutdown_impasse` → `cross_affective_shutdown`, `productive_deadlock_impasse` → `cross_productive_deadlock`, `misconception_correction_flow` → `cross_misconception_surfaces`, `activity_avoider` → `cross_activity_avoidance`, `struggling_learner` → `cross_struggling_overload`), spanning all four policy families (two substantive-engagement, one diagnostic, one repair/affective, two scaffolding), in the same hidden-state-plus-trigger schema the v1 suite uses — so both runners (`services/adaptiveTutor/`, `scripts/run-dialogue-engine-trap-baseline.js`) and both scorers (`scripts/analyze-strategy-shift.js`, `scripts/grade-adaptive-dialogue.js`) consume it unchanged. **Two new cells**, registered in `config/tutor-agents.yaml` and the `EVAL_ONLY_PROFILES` array in `services/evaluationRunner.js`: `cell_124_langgraph_adaptive_crosssuite` (`runner: adaptive`, cell_110's `state_policy` graph including counterfactual replay; `ADAPTIVE_TUTOR_LLM=real`, provider `claude-code/sonnet`; run id `eval-2026-05-12-de6d48ab`, 24/24 dialogues) and `cell_125_dialogue_engine_crosssuite_baseline` (`runner: standard`, cell_114's base single-agent ego via the `run-dialogue-engine-trap-baseline.js` adapter, `claude-code/sonnet`; run id `eval-2026-05-12-952498c6`, 23/24 — one transient generation failure on `cross_struggling_overload__r1`). **Infrastructure fix found en route:** `services/evalConfigLoader.js getTutorProfile()` returns a whitelisted projection that omitted `runner` and `scenario_source`, so the dialogue-engine adapter silently fell back to `config/adaptive-trap-scenarios.yaml` — added both fields to the projection (cell_114 had "worked" only because its `scenario_source` happens to equal the hard-coded fallback). **Result — reproduces the v1 pattern in every respect.** Binary `strict_shift`: `cell_124` 62.5% (15/24, family-match 66.7%) vs `cell_125` 30.4% (7/23, family-match 39.1%) — a ≈2.05× advantage, within rounding of the v1 suite's 1.91× (47.8% vs 25.0%); every hit lands exactly at trigger+1 and `cell_125`, like `cell_114`, never re-pivots. Graded (4-dimension rubric, `grade-adaptive-dialogue.js`, GPT-5/codex, all 47 rows, 0 errors): *inverts* exactly as on v1 — `cell_125` 4.60 vs `cell_124` 4.22 overall, widest single-dimension gap on pedagogical-coherence (4.78 vs 4.08); `cell_124`'s profile mirrors cell_110's (high trigger-recognition 4.58, weakest strategy-execution 3.96). Between-type, with shared blind spots: the gap concentrates in `cross_affective_shutdown` (`cell_124` 4/4 vs `cell_125` 1/4) and `cross_productive_deadlock` (3/4 vs 1/4), a smaller edge on `cross_epistemic_resistance` (4/4 vs 3/4); on `cross_misconception_surfaces` *neither* fires the annotated `ask_diagnostic_question` (both 0/4) and on `cross_activity_avoidance` both are weak (1/4, 0/4). Binary export `exports/cross-suite-trap.json`. **Paper edits.** §6.8.7: the "*Not shown to transfer*" bound replaced with "*Transfers across instruments, in one direction, with the same confound*"; the suite × instrument × finding recap table rewritten (the "*no such instrument exists*" placeholder row removed, two cross-suite result rows — binary and graded — added after the Trap v2 row); the "**Discharging the forward-references**" paragraph extended with a "*The clean cross-suite test, now run*" sub-paragraph and three bullets (binary same-direction/same-ratio; graded same-inversion; between-type with shared blind spots) plus a closing paragraph that the cross-architecture claim now holds on two independently-built trap instruments but remains directional (`cell_125` carries `cell_114`'s §5.12.3 prompt-scaffold confound), one-direction-only, between-type, binary-channel-only, post-hoc exploratory (§5.12.6, not in the A13 pre-reg, single-judge throughout, no blinding). §6.8 intro updated: §6.8.7 now "reconciles … *and reports the post-hoc exploratory clean cross-suite test*" (the §6.3-null reconciliation is still no-new-data synthesis; `cell_124`/`cell_125` are new exploratory data). §6.8.8 updated: opening verdict "partially, on one instrument, in a confounded comparison" → "*partially: on the trap binary, replicated across two independently-built trap suites, in a comparison still confounded by prompt scaffold*"; finding #2 ("`state_policy` produces a moderate strategy-shift advantage over the dialogue-engine floor") extended with the cross-suite replication numbers and the graded inversion on both suites; "What is not established" bullet 1 ("no clean cross-architecture/cross-suite claim") rewritten to "*no clean cross-architecture claim, though the cross-suite gap is now replicated*" — the "no trap-style instrument for the §6.3 suite" gap is closed, what remains missing is the *symmetric* isolation (the dialogue engine run *with* the LangGraph runner's prompt scaffold); the §6.8.8 status paragraph notes the §6.3-null reconciliation is "a no-new-data synthesis layered over those new exploratory cells". `CLAUDE.md` cell-registry prose updated (trap-suite header → "Cells 110-125"; new 124-125 bullet; Runner Dispatch "Scenarios" + "Active cells" lines). **New §6 numbers are confined to §6.8.7's cross-suite material (the two table rows, the three bullets, the binary/graded/per-scenario figures) and §6.8.8's restatement of them**; no other §6 number changes. Net change ~+1,900 words (§6.8.7 + §6.8.8 + §6.8 intro edits).

v3.0.72 (2026-05-12) §6.8 coherence sweep — three cross-reference tightenings; no number changes. Read §6.8.1–§6.8.8 end-to-end for internal consistency now that the section is structurally complete. Numbers cross-check (47.8% / 25.0% strict; 4.46 / 4.03 graded; the 68.8% / 53.1% / 50.0% / 47.8% P2.2 ladder; pooled +4.2 pp P2.1; N = 23 / 32 fan-outs) are internally consistent across §6.8.3/.4/.5/.6, all forward-references (§6.8.4, §6.8.7, §6.8.8) are discharged, and the §6.7 "architectural Pareto frontier" cross-reference in §6.8.8 is accurate. Three small fixes applied: (i) §6.8.8's "*the hypothesised mechanisms mostly are not doing the hypothesised work*" bullet rephrased to describe counterfactual replay accurately — "produced only a messy signal where it ran (high raw divergence but low family-coherence on P2.1's two ToM cells — §6.8.5) and was a literal no-op for the P2.2 ablations (`counterfactual_total = 0` for every P2.2 cell …)" — replacing a looser earlier phrasing that conflated the A13 and P2.2 counterfactual outcomes; (ii) §6.8.2's counterfactual-divergence cross-reference retargeted from "§6.8.5 §3" (no such sub-numbered paragraph) to "§6.8.5's descriptive-results paragraph"; (iii) §6.8.2's `cell_119`-vs-`cell_110` example cross-reference retargeted from "the parking note's headline tables" to "§6.8.6's ablation table" (that data is now in-paper); and §6.8.5's P2.3-dependency sentence given a forward-pointer to §6.8.8's retirement decision ("— §6.8.8 records the subsequent decision to retire P2.3 rather than run it on the original schedule"). **No §6 number changes anywhere**; cross-reference and phrasing only.

v3.0.71 (2026-05-12) §6.8.8 added — closing synthesis for the adaptive-runner arc; P2.3 retired in-text. Wrote §6.8.8 ("Closing Synthesis: The Apparatus Is the Contribution; the Elaborations Mostly Are Not"), the last deferred sub-subsection of §6.8. It steps back from the cell-by-cell results to §6.8's opening question — did externalised learner state + a programmatic policy graph deliver the adaptive responsiveness §6.3 found absent? — and answers "partially, on one instrument, in a confounded comparison, and not through the mechanisms the design assumed." Structure: **What is defensible** (four findings — the trap-scenario methodology is a contribution independent of any architecture; `state_policy` produces a moderate, between-type, binary-channel-only, confounded, not-yet-transferring strategy-shift advantage over the dialogue-engine floor, §6.8.3–§6.8.4; within the state machine less learner state is more, §6.8.6, with `cell_123` localizing the binary floor to "any fourth structured field" / load-capacity saturation; the bilateral-ToM elaboration is a clean null, §6.8.5); **What is not established** (three gaps — no clean cross-architecture/cross-suite claim; the hypothesised mechanisms mostly are not doing the hypothesised work — P2.1 falsified the bilateral elaboration, P2.2 reversed the state-richness prediction, `counterfactual_total = 0` for every P2.2 cell; P2.3 is *retired, not deferred*); **Relation to §6.3.9 and §6.7** (extends rather than reverses §6.3.9's structural-gap finding; rhymes with §6.7's "mechanism stacks sit inside the Pareto frontier" — in both arcs the durable result is the apparatus, not the architecture); **Status** (no new numbers; synthesis of §6.8.1–§6.8.7; P2.3 retirement recorded as a scope decision so the pre-registered fan-out's open items are accounted for rather than silently dropped). §6.8 intro updated: the deferred-sub-subsection sentence replaced with a §6.8.8 summary line; the §6.8.6 capsule tightened for the `cell_123` nuance ("`cell_118` highest on both instruments among the *pre-registered* cells; `cell_123` ties it on binary, sits with the richer cluster on graded"); a new sentence records the P2.3 cells (`cell_121`, `cell_122`) as retired. **No §6 number changes anywhere** — §6.8.8 introduces no data, only synthesis. Net change ~+900 words (new §6.8.8 + §6.8 intro edits).

v3.0.70 (2026-05-12) §6.8.6 completed — `cell_123` run and scored; `zpdEstimate` exonerated on the binary channel; explicit statistical bounds added. Ran `cell_123_state_policy_minimal_plus_zpd` at $N = 32$ on the v1 trap suite (4 runs × 8 scenarios; run ids `eval-2026-05-12-bc2b1469` / `-364d2eaa` / `-e857b8de` / `-80c84358`), scored binary (`analyze-strategy-shift.js`; export `exports/p22-cell123.json`) and graded (`grade-adaptive-dialogue.js`, GPT-5/codex, all 32 rows, 0 errors). **Result.** Binary `strict_shift` = **68.8% (22/32) — to the row identical to `cell_118`** (also 22/32), 100% family-match, per-scenario profile the same shape as `cell_118`'s — so **`zpdEstimate` is exonerated as a cause of the binary "~50% floor"**: the two cells differing *only* in `zpdEstimate` (`cell_118`, `cell_123`) are tied, so adding it to the lean `{confidence, lastEvidence}` core costs nothing on the move-selection channel. That kills the "`zpdEstimate` destabilises" reading and locates the floor in the other structured fields — and it is *not a single toxic field*: `cell_119` (drops `misconceptions`, keeps `agencySignal`) = 53.1% and `cell_120` (drops `agencySignal`, keeps `misconceptions`) = 50.0% both sit in the ~50% cluster, each missing the field the other reading would blame. What moves the policy from the ~69% lean cluster (`cell_118`, `cell_123`) to the ~50% rich cluster (`cell_119`, `cell_120`, `cell_110`) is *the presence of either `misconceptions` or `agencySignal`* crossing from three to four structured fields, and the effect *saturates* (`cell_110` with both = 47.8%, not below the one-each cells) — the signature of a load/capacity ceiling, not additive per-field toxicity. Residual the design cannot close: a 3-field `{confidence, lastEvidence, misconceptions}` cell would separate "pure field-count" from "specifically misc/agency-bearing" (~69% vs ~50%) — not run, not planned; the saturation point favours the count/load reading without excluding the content one. **Graded wrinkle (GPT-5-only).** On the graded channel `cell_123` does *not* join `cell_118`: overall 4.11 (4.03 / 4.00 / 4.28 / 4.13), in the ~50% cluster's graded band (4.03–4.09), well below `cell_118`'s 4.34, with the same depressed-`strategy_execution` left tail as the four-field cells (9/32 3 vs ~1–2/32 for `cell_`

118/cell_110) — so zpdEstimate is a free rider on *which move, when* but a modest tax on *execution-prose quality*; cell_123 is the lone within-P2.2 binary/graded *divergence* (cf. the cross-*architecture* divergence of §6.8.4). Per §5.12.4 the execution left tail is a GPT-5-only signal (Gemini's ceiling can't resolve it; cell_123 was graded after that calibration run, so single-judge), and the binary result is the more robust half of the cell. **Also added** explicit sampling-noise bounds to §6.8.6: cell_118's lead over the pooled three-cell cluster is ~18 pp on the binary channel (bootstrap 95% CI $\approx [-1, +37]$ pp, label-permutation $p \approx 0.10$) but firm on the graded channel (+0.28 overall, CI $\approx [+0.04, +0.51]$, $p \approx 0.01$) and broad across the trap suite (cell_118 $\geq$ the cluster on six of eight scenario types) — so the reversal's *direction* is reported as established, the 21-pp binary figure as the high end of a wide interval; a matching clause in the §6.8.6 "Status and caveats" paragraph names this bound. §6.8.6 table row for cell_123 filled; the "Graded scoring agrees" paragraph re-scoped to "within the *four pre-registered cells*" with cell_123 flagged as the exception; the "What this design cannot separate" paragraph rewritten as "What cell_123 settles" plus a new "One caveat" sub-paragraph for the graded wrinkle. cell_123 is post-hoc exploratory (§5.12.6), not pre-registered; binary single-judge claude-code/sonnet, graded single-judge GPT-5/codex. **No §6 number outside §6.8.6 changes**; §6.8.7's "not a graded-quality effect" bound (about the *cross-architecture* trap-suite gain) is untouched. §6.8 intro already marked §6.8.6 done; only §6.8.8 (closing synthesis) remains deferred. Net change ~+1,100 words in §6.8.6.

v3.0.69 (2026-05-12) §6.8.6 added — the P2.2 state-schema ablation; cell_120 run, cell_123 follow-up launched. Ran cell_120_state_policy_no_agency_signal at $N = 32$ on the v1 trap suite (4 runs × 8 scenarios; run ids eval-2026-05-12-16101436 / -07897269 / -99b196a1 / -b793b534), scored binary (analyze-strategy-shift.js; export exports/p22-cell120.json) and graded (grade-adaptive-dialogue.js, GPT-5/codex, all 32 rows). With cell_120 in hand the P2.2 ladder is complete: strict_shift 47.8% (cell_110 full, $N = 23$) / 50.0% (cell_120 −agency) / 53.1% (cell_119 −misconceptions) / 68.8% (cell_118 minimal); graded overall 4.03 / 4.09 / 4.06 / 4.34. **New §6.8.6** ("State-Schema Ablations: A Richer Learner Profile Does Not Help") reports: (i) **H2.1's predicted dose-response (cell_110 $\geq$ cell_119 $\geq$ cell_120 $\geq$ cell_118, with cell_110 - cell_118 $\geq$ 15 pp) is reversed** — observed cell_118 $\geq$ cell_119 $\geq$ cell_120 $\geq$ cell_110, cell_110 - cell_118 = -21 pp (opposite sign, larger magnitude than the predicted effect; all three predicted positive gaps over cell_118 come out negative) — and the reversal is a *step, not a slope*: cell_110 / cell_119 / cell_120 are a flat cluster within sampling noise (spread 5.3 pp at $N \approx$ 24–32), cell_118 is the lone outlier-up, so H2.3 ("the fields beyond confidence + lastEvidence are cosmetic") is supported in spirit and slightly exceeded — they are mildly counterproductive on this instrument; (ii) **H2.2 unsupported, in the wrong direction** — cell_120 scored 7/8 strict_shift on the repair/affective traps ($\geq$ cell_110's 4/6), so removing the agency enum did not depress the family it was theorised to govern — the cell disproves its own hypothesis; (iii) **binary and graded agree** within the P2.2 family (no within-family inversion, unlike the cross-*architecture* cell_114 comparison in §6.8.4): cell_118 leads on both, the other three cluster on both; §5.12.4's Gemini inter-rater check confirms the `cell_118` > `cell_119` > `cell_110` graded ordering under both judges (`cell_120`, graded after that run, is single-judge GPT-5); (iv) **what the design cannot separate** — `zpdEstimate` is the only structured field in all three ~50% cells and absent from `cell_118`, so "`zpdEstimate` destabilises" is consistent with the ladder but so is "$\geq$ 4 fields" / "any of {`misconceptions, agencySignal, zpdEstimate`}". To adjudicate, registered and launched a post-hoc follow-up cell — **`cell_123_state_policy_minimal_plus_zpd`** ({`confidence, lastEvidence, zpdEstimate`} = `cell_118`'s set + `zpdEstimate`; new architecture `state_policy_minimal_plus_zpd` in `services/adaptiveTutor/graph.js` `PROFILE_PROJECTIONS`, registered in `EVAL_ONLY_PROFILES`, behaviourally an exact clone of `cell_118` plus the one projected field) — at $N = 32$ on the v1 suite; its row in the §6.8.6 table reads "run in progress" pending completion and will be filled with a follow-up revision. §6.8 intro updated to mark §6.8.6 done; only §6.8.8 (closing synthesis) remains deferred. `cell_123` is post-hoc exploratory (§5.12.6), not pre-registered; cells 118–120 are pre-registered confirmatory on H2.1–H2.3 (pre-reg §P2.2). No §6 number outside §6.8.6 changes. Net change ~+1,500 words (new §6.8.6 + §6.8 intro + §6.8.5 forward-ref discharge).

v3.0.68 (2026-05-12) §6.8.7 added — reconciling the trap-suite result with the §6.3 adaptive-responsiveness null; §6.8.2 scenario-count corrected. Wrote the long-promised §6.8.7 (forward-referenced from §6.8.3, §6.8.5, and the §6.8 intro): the §6.3.8/§6.3.9 null (adaptive responsiveness unmoved by lightweight intervention; only best-of-K suggestive at $d = 0.41$) and the §6.8.3–§6.8.4 trap-suite positive (`state_policy` `strict_shift` 47.8% vs dialogue-engine 25.0%) are not in contradiction — they are different outcome variables (diffuse per-dialogue slope / reflection cosine vs localized binary check at an annotated trigger), on different stimuli, at different intervention weights, and the trap positive is consistent with §6.3.9's own escape clause ("genuinely new architectures with explicit memory and iterative refinement"). The section adds a 6-row suite × instrument × finding recap table, bounds what the trap positive does *not* license (not a recognition-prompt effect; not within-type — §6.8.5; not a graded-quality effect — §6.8.4 inverts; not shown to transfer), discharges the §6.8.3/§6.8.5 forward-refs, and names the outstanding clean cross-suite test (run the LangGraph runner + dialogue-engine baseline on a trap-annotated subset of the §6.3 suggestion scenarios). **No new numbers** — pure synthesis of existing §6.3.9 / §6.8.3 / §6.8.4 / §6.8.5 results. Also corrected a pre-existing §6.8.2 error: the v1 trap suite has **eight** scenarios spanning **four** pedagogical families (substantive engagement, scaffolding, diagnostic, repair/affective), not "six … three" — the A13 v1 cells and cell_114 each ran three runs × eight scenarios ($N \approx 24$), the v2 within-family cells four runs × six v2 scenarios, the P2.2 ablation cells four runs × eight v1 scenarios ($N = 32$); rewrote the "Standard fan-out" paragraph accordingly. §6.8 intro updated to mark §6.8.7 done and §6.8.6/§6.8.8 still deferred. CLAUDE.md cell-registry prose updated to list cell_114 under the trap-scenario suite (`runner: standard`). Net change ~+1,300 words (new §6.8.7 + §6.8.2 fixes).

v3.0.67 (2026-05-12) §6.8.4 added — the dialogue-engine baseline (`cell_114`) is run; §5.12.3 overclaims corrected. Ran `cell_114_dialogue_engine_` `trap_baseline` at $N = 24$ on the v1 trap suite (8 scenarios × 3 runs, fanned out across 4 parallel lanes; run ids `eval-2026-05-12-c53a1d8f` / `-f157bcd0` / `-3db6bf3e` / `-41f4487f`; script `scripts/run-dialogue-engine-trap-baseline.js`, cell registered in `tutor-agents.yaml` and `EVAL_ONLY_PROFILES`), scored on both paths: binary `analyze-strategy-shift.js` and the 4-dimension adaptive graded rubric (`grade-adaptive-dialogue.js`, GPT-5/codex, all 24 rows). **New §6.8.4** reports the result against the A13 v1 runs of cells 110–113. Binary: cell_114 `strict_shift` = 25.0% (6/24), the floor — below cell_112 (30.4%), cell_111 (37.5%), cell_113 (41.7%), cell_110 (47.8%) — and uniquely its `window` rate equals its `strict` rate (25.0%): the dialogue engine never re-pivots on a later post-trigger turn, where all four LangGraph cells do (window above strict by +17 to +30 pp). Family-confusion: 19/24 trigger+1 moves land in the broad `substantive_engagement` family; cell_114 scores 0/6 on `repair_affective` traps and 1/3 on the `diagnostic` trap (cell_110: 4/6 and 3/3) — it can produce `scope_test` / `ask_diagnostic_question` / `repair_misrecognition` but fires them inconsistently and mistimed (`repair_misrecognition` appears in 2/3 `repair_after_misrecognition` dialogues but at the trigger turn, not trigger+1, so binary scores 0/3). Graded: cell_114 overall 4.46/5 (`trigger_recognition` 4.54, `strategy_execution` 4.21, `strategy_quality` 4.50, `pedagogical_coherence` 4.58) — *higher* than cell_110's 4.03 (4.26/4.30/3.96/3.61), the gap widest on coherence; the graded rubric scores prose quality given that some action fired, the dialogue engine's heavily-iterated prompt produces fluent coherent teaching even when it picks the wrong family, and the two instruments measure different constructs (this inversion is reported as the finding, not a contradiction). Not a GPT-5 length-bias artefact (dialogue-engine turns are 400–2700 chars, against GPT-5's documented concision preference). **§5.12.3 corrected:** the v3.0.65 framing — "drives a fully scripted learner (same scripting strategy as cell_106)" and "the cell_111 cell_114 contrast isolates the runner" — overclaimed on two counts. (i) cell_114's learner is the trap-steered LLM learner driven via the same `learnerTurn` mechanism cells 110–119 use (the hidden trap state conditions the learner LLM; not a canned-string script). (ii) cell_111 assembles its tutor prompt through the adaptive runner's `tutorEgoInitial` builder while cell_114 uses tutor-core's base ego prompt via the eval-repo `claude-code` bridge's `tutorEgoExecute` (tutor-core does not support the `claude-code` provider); the cell_111 cell_114 gap therefore bundles "no LangGraph state machinery" with "different prompt scaffold", so cell_114 is a *directional dialogue-engine floor*, not a clean single-factor isolation. **§5.12.5 updated:** "cell_114, once run, …" → "cell_114 provides the dialogue-engine measurement on the v1 trap suite (results in §6.8.4); subject to the prompt-scaffold confound, the closest available same-suite cross-architecture comparison, not a clean isolation." No A13 or P2.1 numbers change; cell_114 is post-hoc exploratory (§5.12.6), single-judge throughout, no blinding. Net change ~+1,400 words (new §6.8.4 + §5.12.3/§5.12.5 edits).

v3.0.66 (2026-05-11) §5.12.4 inter-rater calibration of the adaptive graded rubric. Ran an independent second judge (Gemini via the same CLI-bridge pattern, identical prompt builder `scripts/lib/adaptiveGraderPrompt.js`, same 4-dimension rubric) over all 87 GPT-5-graded adaptive rows (`scripts/rejudge-adaptive-inter-rater.js`;

report `exports/adaptive-inter-rater-2026-05-11.md`; judge-1 snapshot `exports/adaptive-grades-j` judge-2 scores `exports/adaptive-grades-judge2-gemini.json`). Rewrote §5.12.4 limitation 2 from "single judge, uncalibrated" to report the result: **(i)** the cell ordering on graded *overall* quality is identical under both judges (`cell_118` minimal > `cell_119` no-misconceptions > `cell_110` full state) — the comparative claim §6.8.6 rests on is not a single-judge artefact; **(ii)** absolute-score agreement is fair, not strong — pooled over 4 dims × 87 rows: Pearson r = .44, Spearman = .42, quadratic-weighted = .35, exact-match 43%, within-1 91%, Gemini ~0.56 pts more lenient (constant offset); **(iii)** one sub-claim does *not* replicate — GPT-5's `cell_119` left tail on `strategy_execution` (11/32 3 vs `cell_110`'s 1/23) vanishes under Gemini (2/32), but Gemini shows a pronounced ceiling effect (many rows 5/5/5/5) and cannot resolve the execution differences GPT-5 reports, so §6.8.6 will report the execution left-tail as a GPT-5-only observation. No change to any §6 number (§6.8.6 is still deferred pending `cell_120`); this revision only converts the §5.12.4 single-judge limitation into a reported calibration finding. New tooling: `scripts/lib/adaptiveGraderPrompt.js` (shared prompt builder, so the original grader and the rejudge script emit a byte-identical stimulus), `scripts/rejudge-adaptive-inter-rater.js`. Net change ~+250 words in §5.12.4.

v3.0.65 (2026-05-10) §5.12 Procedural Transparency added — pre-registration drift and post-hoc additions. New methodology subsection (six sub-subsections) documenting two classes of methodological event from the post-A13 experimental arc: (i) divergences between pre-registered designs and YAML/code at run time, and (ii) instruments and conditions added after data collection began. **§5.12.1** introduces the general principle: pre-reg, YAML, and dispatch logic are three independent artefacts with no automatic validation, and the YAML is treated as the source of truth (paralleling the existing `EVAL_ONLY_PROFILES` convention). Three remediation mechanisms now part of the workflow: mandatory explicit `runner:` field on adaptive cells, `meta.dispatched_runner` stamped into every new dialogue log at write time, and pre-registration documents carrying machine-checkable YAML cell-manifest front-matter. **§5.12.2** documents the specific A13 case: the pre-reg specified `runner: standard` for C1 (cell_111) and C2 (cell_112), but both cells ship with `runner: adaptive` in `config/tutor-agents.yaml` and route through LangGraph with internal `architecture` flags (`recognition_only`, `ego_superego`); all four A13 conditions are within-LangGraph ablations, not a cross-runner comparison. The label *recognition_only* in this paper consistently refers to a stripped LangGraph sub-graph, not a dialogue-engine baseline. **§5.12.3** introduces `cell_114_dialogue_engine_trap_baseline` as a post-hoc baseline (script: `scripts/run-dialogue-engine-trap-baseline.js`) holding prompt, model, and scenario constant against cell_111 so the cell_111 cell_114 contrast isolates the runner. Cell_114 is flagged exploratory throughout §6. **§5.12.4** documents the bespoke 4-dimension adaptive graded rubric (`scripts/grade-adaptive-dialogue.js`, columns `adaptive_trigger_recognition/strategy_execution/strategy_quality/pedagogical_coherence`) with two caveats: non-blind authoring (rubric written after binary `strategy_shift_correctness` results were known) and single-judge dependency

(GPT-5 via codex CLI). Both treated as known limitations rather than hidden ones. **§5.12.5** flags the stimulus-suite divergence: trap-scenario suite for cells 110–119 vs suggestion-scenario suite for Paper 1.0's adaptive-responsiveness null; direct "the state machine fixes Paper 1.0's null" claims are unsupported absent cell_114. **§5.12.6** establishes a three-tier reporting convention (pre-registered confirmatory / pre-registered with implementation drift, re-classified exploratory / post-hoc exploratory) used as paraphrased ledes in every relevant §6 subsection. No new data, no number changes; §6 will need editing later to apply the §5.12.6 convention to A13, P2.1, and P2.2 result subsections. Net change ~+1,500 words in §5.

v3.0.64 (2026-04-29) §6.7 closing-synthesis paragraph — how charisma and recognition relate. Adds a sixth paragraph between the v3.0.63 *Lever-interaction follow-up* and the *Cross-judge sanity check* that synthesises the five preceding §6.7 findings into a unified account of the two rubrics' relationship. Three claims, each supported by previously-reported numbers: (a) **shared floor** — c106 fails both rubrics (charisma 36.4, v2.2 57.0), so register-flat output is the worst case for either; (b) **cost asymmetry** — adding recognition vocabulary on top of structured input (c103 → c104) costs charisma ≈0 (+1.4, within noise) because the Hegelian frame is conceptual rather than rhetorical, while adding charisma directives costs v2.2 substantially (c104 → c105: −10.6 v2.2 for +5.3 charisma) because the id's authoring budget is finite and voice-instructions displace elicitation/perception moves; (c) **architectural Pareto frontier** — single-mechanism design points (c104, c105, c107) define the frontier, multi-mechanism stacks (c108, c109) sit inside it. Closes with the interpretive frame that recognition wants the voice deployed *outward* (perceiving the learner, eliciting thinking, holding the diagnostic gap) while charisma wants the voice deployed *upward* (claiming authority, marking encounter as extraordinary, inviting transformation), and that the apparatus can build either or a balanced generalist but not the maximally-both. No new analyses, no new run ids, no changes to numbers — purely synthesis from existing reported results. Net change ~+260 words in §6.7.

v3.0.63 (2026-04-29) §6.7 c108 full-N follow-up — pilot lift does NOT survive confirmatory N. Replaces the v3.0.62 *Lever-interaction pilot* paragraph with a confirmatory-N rewrite reporting the c108 follow-up at $n = 27$ across all three curricula × 9 scenarios × 3 reps (run ids `eval-2026-04-29-{499ab3d2,e35c66ae,be822796}`, scoring under `claude-code.sonnet`). The pilot's super-additive charisma 81.3 regresses to **71.4** (a 9.9-point drop, tying c105's 71.0 within sample noise); v2.2 last-turn 79.5 regresses to **72.6** (a 6.9-point drop, sitting below c107's 78.5 and c104's 80.6). Per-scenario inspection identifies the culprit: pilot phaedrus scored 90.0 charisma at $n = 3$ but full-N phaedrus scores 67.9 at $n = 3$ — a 22-point regression to the mean. The 701 ethics-ai curriculum exposes a new fragility (c108 v2.2 tN 65.3 there, vs 76.7 / 75.8 on 601 / 901). The asymmetry that survives the confirmatory threshold is narrower than the pilot suggested: text-heavy levers stacked (c109) clearly interfere; non-text levers stacked (c108) compose roughly additively to slightly under-additively, trading +5.1 charisma for -8.0 v2.2 relative to c107 alone. The architectural ceiling for the id-director family sits at the single-mechanism design points (c104, c105, c107), not at multi-mechanism stacks. Full audit trail in `docs/cell-100-charisma-full-n-update.md` §10. Adds the three new run ids to the §6.7 reproducibility list. No changes to cells 101–107 numbers or to the four prior headline findings; net change ∼+340 words in §6.7.

v3.0.62 (2026-04-28) §6.7 lever-interaction pilot integration (cells 108/109, hypothesis-generating). Adds a fifth headline-finding paragraph to the §6.7 id-director family section reporting a small two-cell follow-up pilot (`eval-2026-04-28-10216f9f`, $n = 11$ after one generation failure) that tests whether the three architectural levers compose additively. **c108** (register classifier + witness exemplars) lifts pooled charisma to 81.3 — the highest mean of any id-director cell in the section — with v2.2 last-turn at 79.5, between c104's 80.6 and c107's 78.5; the classifier and exemplars appear to occupy different slots in the id's authoring task. **c109** (charisma directives + witness exemplars) fails the additive hypothesis: charisma is preserved at 77.7 but v2.2 last-turn collapses to 59.6 due to an ego-side meta-narration failure mode where the Nemotron ego outputs system-prompt instructions verbatim rather than executing them. The qualitative asymmetry — non-text levers compose, text-heavy levers stacked interfere via instruction-following degradation — is reported as hypothesis-generating only; pilot N is far below confirmatory floor. Adds run id `eval-2026-04-28-10216f9f` to the §6.7 reproducibility list. Full audit trail in `docs/cell-100-charisma-full-n-update.md` §9. No changes to cells 101–107 numbers or to the four prior headline findings; net change ∼+260 words in §6.7.

v3.0.61 (2026-04-28) §8.9 + §6.7 N-scope tightening for full-N c107 closure. The "smaller per-cell N (typically 27 on the original set)" line — written when the c107 witness-exemplars cell was at n=10 and most other id-director cells were partially scored — is replaced by the actual closing N range (n=79–81 for c101/c103/c104/c105, n=54 for c102/c106, n=27 for c107). Three of seven cells meet or exceed the paper's core $n \geq 63$ per dialogue-condition floor; one cell (c107) sits below it and is now flagged as pilot-scale rather than confirmatory. Mirrors the §6.7 *Replication, robustness, and orientation* paragraph against the §8.9 *Per-cell N* axis. No data, analysis, or rank-ordering changes; purely scope-language calibration to match the closed-pilot N matrix in `docs/cell-100-charisma-full-n-update.md` §1.

v3.0.60 (2026-04-29) §6.3.9 four-bridge synthesis closure: H-capacity reframed from "structural" to "lightweight-intervention-resistant; partially mitigable only by expensive search". With Bridge 3 (cell 100, best-of-N selector at $K = 3$, $\sim 6\times$ ego cost per turn) closing at $N = 53$ dialogues / 313 reflection-action pairs (run `eval-2026-04-27-327d8816`), the four-bridge synthesis is locked. Per-dialogue mean coupling: cell 40 = 0.614, cell 100 = 0.635, $\Delta = +0.021$, Cohen's $d = +0.41$ — above the pre-registered $d \geq 0.3$ "suggestive" threshold but below the $d \geq 0.5$ "bridgeable" threshold. EoQ% identical at 16% / 16% (the orthogonal pragmatic-act channel is unaffected by best-of-N search).

Bridge 3 is the only intervention with detectable directional movement on coupling — it engages the semantic-coupling channel where Bridges 0–2 do not. Matching the selection criterion to the verdict criterion (eliminating Goodhart's escape route) still produces only $d = 0.41$. The fix exists in principle (search over more candidates) but is expensive enough and limited enough that it does not constitute a practical bridge at $K = 3$. Log-linear scaling in K would predict $d \approx 0.5$ at $K = 5$ and $d \approx 0.7$ at $K = 10$; a future Bridge 3b sweep would test that prediction.

§6.3.9 prose changes: title revised from "Structural Across Three Lightweight Interventions" to "Structural Under Lightweight Intervention; Partially Mitigable Only by Expensive Search"; synthesis table extended from 3 rows to 4 rows; closing in-flight clause replaced with closure description; H-capacity discussion reframed from "structural at strongest level" to "lightweight-intervention-resistant; partially mitigable only by expensive search". Per-dialogue Cohen's d recomputation also corrected the prior table's $|d| \leq 0.07$ characterization for Bridges 1 and 2: properly computed on per-dialogue mean coupling distributions, $d_1 = -0.21$ and $d_2 = -0.22$ (still well below the suggestive threshold). The earlier hand-estimates were of the right sign and direction but ~4× too small.

Net length change: ~+340 words in §6.3.9 (new Bridge 3 row in synthesis table; reframed prose paragraphs; updated EoQ%-decoupling and H-capacity discussions). No upstream changes to §3 theory, §4 method, or other §6 mechanism findings. Cross-references: TODO §D3 four-bridge synthesis lock (2026-04-29); design note `notes/design-d3-architectural-bridges.md` four-bridge synthesis section; per-run report `exports/insight-action-gap-eval-2026-04-27-327d8816.md`. Eval-repo commit `956b9cb` (cell 100 + best-of-N runner orchestration).

v3.0.56 (2026-04-26) §8.6.2 application protocol for transferring the diagnostic. Adds a six-step procedure (~430 words) for applying the §8.6.1 three-criterion within-cell mediator diagnostic to LLM-tutor evaluation pipelines beyond the Paper 2.0 source apparatus. The protocol specifies: (1) family partition, (2) per-row feature computation, (3-5) the three criterion checks, (6) PCA pairing with §8.6 rubric over-specification check. References open-sourced scripts (`analyze-d1-structural-features.js`, `analyze-d1-cross-judge-replication.js`, `analyze-d1-multifeature-ols.js`) as the data-loading extension point for new datasets. Names three candidate independent rubrics for future application: MathTutorBench (Macina et al., 2025), ICAP-aligned studies (Chi & Wylie, 2014), LLM-Rubric (Hashemi et al., 2024). The protocol makes the methods-paper generality argument procedurally rather than requiring new empirical work in this revision; future runs of the protocol against the named rubrics would convert the procedural argument to an empirical one. No data, analysis, or N-count changes; ~+430 words in §8.6.2.

v3.0.55 (2026-04-26) §8.6.1 named three-criterion within-cell mediator diagnostic. Adds a new subsection §8.6.1 (~520 words) immediately after the §8.6 "Methodological pairing with §7.10" paragraph. The diagnostic was implicit across §7.10 + §7.10.1 + §8.6 ("only one feature satisfies all three within-cell mediator criteria — large family contrast, small within-intersubjective contrast, AND positive within-cell correlation"); v3.0.55 articulates it explicitly as a portable methodology with named criteria and a worked example.

The three criteria are stated formally: (1) large family contrast ($|d| \geq 0.5$); (2) small within-family contrast ($|d| < 0.5$); (3) positive within-cell Pearson r in each family-member cell. The worked example tabulates the D1 five-pass feature sequence with each feature's verdict (Hegelian/intersubjective lexicon = prompt markers; scaffolding moves = family marker, weak mediator; broad acknowledgement = negative correlate; ends-with-question = mediator in single-turn cells; intersub. canonical similarity = family marker, Simpson's paradox disqualifies as mediator).

The diagnostic generalises beyond the recognition–transmission contrast: any factorial design with a family-level main effect and candidate response-level mediators can apply the three-criterion check. Pairs with the §8.6 PCA over-specification finding as the two independent methods-level corrections that any future LLM-tutor evaluation should account for. §8.6.1 closes by noting that a separate methods-paper extraction is the natural vehicle for arguing the diagnostic's generality across published LLM-tutor evaluation pipelines (skeleton: `notes/methods-paper-skeleton.md`).

Net change: ~+520 words in §8.6.1; no data, analysis, or N-count changes. The diagnostic was applied throughout §7.10 already; v3.0.55 names it.

v3.0.54 (2026-04-26) §7.10.1 D1 mechanism analysis: replication, multivariate control, and a single-turn-vs-multi-turn scope condition. Adds a new subsection §7.10.1 (~600 words) consolidating four follow-up analyses on the §7.10 ends-with-question mediator finding and the intersub_advantage Simpson's-paradox finding.

- **Cross-judge replication** on the Opus 4.7 and GPT-5.2 judges of the A10b three-judge panel: ends-with-question r remains positive in 5 of 6 cell-judge combinations (the one exception, GPT × cell_95, is essentially zero rather than a sign-flip); intersub_advantage Simpson's-paradox replicates in 3 of 3 judges. Neither finding is Sonnet-specific. Report: `exports/d1-cross-judge-replication.md`. Script: `scripts/analyze-d1-cross-judge-replication.js`.
- **Multivariate OLS control** with six predictors (ends-with-question, second-person density, scaffolding moves, broad acknowledgement, question-mark rate, intersub_advantage). The ends-with-question partial coefficient remains positive and statistically significant in both intersubjective cells: cell_5 $\beta = 26.34$ ($p = 0.02$), cell_95 $\beta = 38.52$ ($p = 0.02$). The single-feature mediator interpretation is not collinearity-driven. The intersub_advantage Simpson's-paradox direction also survives multivariate control. Pure JS Gauss–Jordan OLS implementation (df $\approx$ 43 per cell). Report: `exports/d1-multifeature-ols.md`. Script: `scripts/analyze-d1-multifeature-ols.js`.
- **Cross-run / cross-application replication on five additional runs** establishes that the mediator is **single-turn-specific**. In single-turn cells (A10 v2, A6 programming, D2 Path 1 peer support coaching) the mediator replicates uniformly with r range +0.16 to +0.74 across DeepSeek/Haiku ego, philosophy/programming/peer-support domains. In multi-turn cells (Paper 2.0 cells 84/86 messages-mode single-agent and multi-agent recognition; A2 cells 61/63 dialectical multi-agent with dynamic learner) the relationship **reverses sign** with r range −0.09 to −0.30. Substantive

interpretation: in single-turn responses, ending with a question cedes initiative back to the learner; in multi-turn dialogues, ending the final turn with a question leaves the conversation unresolved, which the judges penalise. Same surface feature signals different pragmatic acts in different discourse contexts; the mechanism account (initiative-ceding) survives the refinement, but the surface signal becomes scope-conditioned. Report: `exports/d1-ends-question-replication.md`. Script: `scripts/analyze-d1-ends-question-replication.js`.

The cross-mode reversal does not falsify the underlying mechanism account; it specifies the discourse context in which the surface feature signals it. The §7.10 mediator claim now carries an explicit scope condition that maps onto the §6.3 single-turn–vs–trajectory distinction.

Net change: ~+600 words in §7.10.1; no data, analysis, or N-count changes to the primary Paper 2.0 mechanism findings; the v3.0.54 additions are interpretive integration of D1 follow-up analyses on already-collected runs.

v3.0.53 (2026-04-26) §6 D1 follow-up: form-vs-substance + negative-acknowledgement guardrail. Added §6.1.8 "Pragmatic Surface Markers Are Not Substitutes for Substance" (~280 words) immediately after §6.1.7 (question-asking mediation). The new subsection acts as an operational caveat on the §6.1.7 mediation result: it documents two patterns from the response-level D1 mechanism analysis that argue against a naive operationalist reading of "mandate intersubjective surface markers to reproduce recognition." Pattern 1: surface acknowledgement markers (quoted text, "your X" possessives, "what you mentioned"-style phrases) are *negatively* correlated with score within most cells ($r = -0.18$ cell_5, $r = -0.31$ cell_95, $r = -0.38$ cell_1) — the rubric penalises verbose surface empathy. Pattern 2: embedding-based similarity to an intersubjective canonical is positively correlated with score when pooled ($r = +0.26$) but reverses sign within each intersubjective cell ($r = -0.28$ cell_5, $r = -0.24$ cell_95) — Simpson's paradox driven by the cell_96 transmission outlier. Forward-references §7.10 for the full form-versus-substance account and the three-way decomposition. Net length change: ~+280 words; no data, analysis, or N-count changes; the underlying analysis already lives in §7.10 as of v3.0.52, this revision raises the operational caveat to results-section visibility for readers who otherwise might read §6.1.7 alone.

v3.0.52 (2026-04-26) D1 mechanism analysis integration. Integrates the five-pass D1 response-level mechanism analysis (`notes/d1-paper-integration.md`; commits `1ef5f60`, `e8dc7a8`, `3c9eaa6`, `c334722`) into Paper 2.0 across three sections.

- **New §7.10 "Mechanism location: response-level form versus substance"** ($ $750 words): corroborates the §7.9 orientation-family finding through an independent embedding-based measurement channel (intersub-canonical-similarity advantage $d = 0.81$ between families on the same A10b corpus, $n = 191$, OpenAI text-embedding-3-small); surfaces a Simpson's paradox at the response level (pooled $r = +0.26$ between embedding-similarity and score reverses to within-cell $r = -0.28$ in cell_5 and $r = -0.24$ in cell_95, driven by the cell_96 transmission outlier); identifies "ending the tutor turn with a question" as the single feature that satisfies all three within-cell mediator criteria across the five-pass sequence (cell_5 $r = +0.325$, cell_95 $r = +0.392$;

cells 1 and 96 produce this in 0% of responses, cells 5 and 95 in 4.5% and 2.1%); decomposes "the recognition injection" into three distinct candidate active ingredients with separate verdicts — Hegelian content (not load-bearing), intersubjective stance (substantially load-bearing), specific surface features (observable channels of the stance, not substitutes); ties the decomposition back to the v3.0.50 "primacy + descent" framing at the response level. Includes a guardrail against the operationalist misreading "tell tutors to acknowledge more": surface-level acknowledgement markers (quoted text, "your X" possessives, "what you mentioned" patterns) are *negatively* correlated with score within most cells ($r = -0.18$ cell_5, $r = -0.31$ cell_95, $r = -0.38$ cell_1), counter to the naive operationalist reading.

- **§8.6 Simpson's-paradox methods note** ($ $150 words appended) : pairs the §7.10 within-cell-vs-pooled-r caution with the existing PCA over-specification finding to give two methods-level cautions for any future LLM-tutor mediator analysis. Together : (1) per-dimension claims should be hedged because the v2.2 rubric's eight dimensions reduce to one factor; (2)p$ in mediator analysis can be entirely driven by between-group variance rather than within-group mechanism. Both could form the empirical core of a methods-paper contribution beyond Paper 2.0.
- **§9 sixth future direction: white-box analysis targeting the within-cell mediator** ($ $130 words): reframes the mechanistic-understanding white-box hunt from generic residual-stream fishing to a small set of specific testable hypotheses about the model's pragmatic decision to end with a question — does the residual stream encode "should I cede initiative now" as an identifiable direction? Does attention shift to learner tokens before turn-ending question generation? Pinning the within-cell mediator to a specific decision point reframes the problem as applied interpretability with a defined target behaviour.

No data, analysis, or N-count changes; $ $1030 words added across three sections; net length change ∼+1.3%. Sources: `scripts/analyze-recognition-lexicon.js`, `scripts/analyze-d1-orientation-lexicon.js`, `scripts/analyze-d1-structural-features.js`, `scripts/analyze-d1-structural-features-v2.js`, `scripts/analyze-d1-structural-features-v3.js`; reports in `exports/d1-*.md`; integration synthesis in `notes/d1-paper-integration.md`.

v3.0.51 (2026-04-24, late evening) F4 redundancy trim — universal-substitution restatements compressed (closes TODO §F4). The "universal substitution with 15–17% additivity deficit" claim previously appeared in 22 places across the paper with full numerical restatement (deficit %, +12.3 residual, near-zero on strong models). F4 trim consolidated to 15 occurrences with single canonical numerical homes: §6.4.2 / §6.4.2.1 for the mechanistic account, §6.4.6 for cross-judge validation, §7.3 for the §7-level interpretation, §8.3 for cross-judge magnitudes, abstract for the headline. Other mentions compressed to bare "(§6.4)" or "(§6.4.2)" cross-references, dropping inline restatement of the deficit numbers and the +12.3 residual.

Touched paragraphs: §1 mechanism description (L80), §1 separability paragraph (L84), §1 contributions list (L100, L106, L108, L110, L114), §3 prediction (L704), §6.3 dialogue-quality cross-ref (L1386), §6.4.3 separability point 3 (L1504), §6.4.5 prediction headers (L1525, L1527, L1529), §7 lead (L1832), §7.1 effects-to-mechanisms recap (L1840), §7.2 messages-mode pattern (L1850), §7.3 internal duplication (L1858), §9 conclusion (L2133). The "universal substitution" *name* (paper-distinctive label, not hedge) is preserved wherever needed for legibility.

Pattern-2 hedging ("pending boundary-condition / requiring replication") was already retired in v3.0.43 (F1) + v3.0.45 (A12 closure); the surviving occurrences are deliberately framed as historical references in §6.3.2 explanatory text and Appendix F. Also incidentally compressed: §1 contribution bullet for adaptive responsiveness (full §6.3 + A12 restatement → verdict + cross-ref); §1 contribution bullet for tutor-learner asymmetry (now points to §6.5.5 rubric-vs-holistic correction rather than restating the superseded 7–12× headline).

Net change: ~480 words trimmed across ~18 paragraphs; line count unchanged (mid-paragraph compression rather than whole-line removal). No data, analysis, or N-count changes. Closes the F-series editorial cycle (F1-F6 all resolved); the open editorial item is now D5 (rubric v3.0 PCA consolidation), deferred to the next rubric version cycle.

v3.0.50 (2026-04-24, evening) Framing B+ reframe: primacy + descent. v3.0.49 implemented Framing A (calibrated — minimal upstream changes acknowledging the family-level finding). v3.0.50 implements the more careful "primacy + descent" framing the user articulated: recognition's empirical generalisability across cognate descendants is a strengthening of the paper, but it does not displace recognition theory's distinctive theoretical contribution, because **recognition theory is the most explicit philosophical articulation of an intersubjective-pedagogy orientation that its constructivist descendants (Dewey, Vygotsky, Piaget, Kapur, Chi, VanLehn, Graesser) inherit, often without direct attribution**.

The reframe is implemented across four sections:

- **Abstract**: rewrote the family-level paragraph to position recognition as "the most explicit philosophical articulation of an orientation its constructivist descendants carry tacitly," explicitly noting Hegel's lineage to Dewey (Hegelian phase), Vygotsky (dialectical psychology refracted through Marxist materialism), and Piaget (assimilation/accommodation as transposed dialectic of identity-difference). The empirical operationalisation is interchangeable across the family; what remains uniquely valuable about recognition theory is the diagnostic clarity its vocabulary provides for *what is happening* when intersubjective orientation alters tutor production.
- **§1 introduction**: replaced the single-paragraph family naming (v3.0.49 form) with a careful three-circle genealogical account distinguishing direct Hegelian descendants (Dewey, Vygotsky, *Bildung*, Honneth — innermost circle: argue from Hegel), structurally Hegelian descendants (Piaget assimilation/accommodation as transposed dialectic, Kapur productive failure as thesis-antithesis-synthesis under non-Hegelian citation — second circle: bear the structural mark without attribution), and cognate-but-not-descended pedagogically-adjacent traditions (Chi ICAP, VanLehn step-level scaffolding, Graesser affective tutoring — third circle: operate in an intersubjective-pedagogy ecosystem whose air recognition theory makes breathable, but cite elsewhere). The *family* is the union of these three circles; the active ingredient (per A10/A10b) is

the orientation they share; recognition theory is the most explicit articulation of that orientation. Our contribution is accordingly twofold: operationalize the family's commitments as design heuristics (using recognition theory as the explicit theoretical entry point for its philosophical primacy and diagnostic clarity), and trace the mechanisms.

- **§3 theoretical framework**: added a "level of analysis" framing note positioning the three predicted mechanisms as recognition theory's specific articulation of commitments the broader family carries tacitly. We could in principle derive analogous mechanisms from each descendant (a Vygotskian "calibration" via scaffolding-to-the-zone, a Kapurian "error correction" via productive-failure-then-canonical-instruction, a constructivist "adaptive responsiveness" via dialogic refinement); these would be the same mechanisms named differently. We work in recognition's vocabulary because it is diagnostically clearer for *what is happening*. The empirical operationalisation generalises (§7.9); the theoretical articulation here remains recognition-specific by design.
- **§9 broader implication**: extended the v3.0.49 family-level paragraph with an explicit "empirical generalisability does not displace recognition theory" passage. Names the three innermost-circle Hegelian descendants explicitly (Dewey's pragmatism began as American Hegelianism; Vygotsky's dialectical psychology refracts Hegelian recognition through Marxist materialism; Piaget's assimilation/accommodation transposes the dialectic of identity-difference into developmental cognition). The matched-pedagogical control removes the Hegelian *vocabulary* but inherits the Hegelian *commitment*. Recognition is positioned not as "one of several alternatives that all work equally well" but as "the philosophical articulation of an orientation its constructivist descendants carry implicitly."

This is the framing Paper 2.0 should ship with: empirically generous (family-level finding is real), theoretically careful (recognition theory's primacy is preserved on philosophical grounds, not just historical priority). Closes TODO §F6 at the Framing B+ level. No data or analysis changes; ~400 words rewritten across four sections; net length change ~+450 words from §1 expansion.

v3.0.49 (2026-04-24) Paper-framing reframe to intersubjective-pedagogy family (closes TODO §F6 at Framing A level; superseded by v3.0.50 Framing B+). After A10/A10b established that recognition's effect is a family-level property of the Hegelian-descendant intersubjective-pedagogy tradition rather than a property of recognition's specific vocabulary, three passages upstream of §7.9 were updated to reflect this reframe.

- **Abstract**: added a sentence citing A10/A10b, naming "intersubjective-pedagogy orientation" as the active ingredient, and positioning recognition as "one effective operationalisation" of a broader family rather than a uniquely necessary framing. Cites pooled three-judge $d \approx 0.14$-0.19 within-Hegelian (density-sufficient), pooled $d = 0.89$ behaviorist-below-base (transmission-family harmful), pooled $d = 1.38$ between-family (dominant).
- **§1 introduction**: expanded the Vygotsky/*Bildung*/Honneth paragraph to name the intersubjective-pedagogy family explicitly, listing its theorists (Vygotsky ZPD, Piaget

assimilation/accommodation, Kapur productive failure, Chi ICAP, VanLehn step-level scaffolding, Graesser affective dynamics) and citing §7.9 A10/A10b for the empirical family-level finding. Reframes the paper's contribution as operationalising the family using recognition theory as the theoretical entry point.

- **§9 broader implication**: describes the mechanisms as altering behavior at the "intersubjective-pedagogy orientation" level rather than at the "recognition" level specifically. Includes family-level findings from A10/A10b and reframes the final takeaway as "recognition is one effective operationalisation; the family as a whole is what the paper demonstrates." Methodological contribution extended to include "orientation-family-as-unit-of-analysis" as transferable framing.

§3 theoretical framework retained with its recognition-specific theoretical motivation unchanged — it remains the paper's scholarly entry point and the specific philosophical tradition being operationalised. §7.9 (v3.0.48 empirical writeup) already reflected the family-level reframe.

This is a *calibrated* reframe (Framing A from the TODO §F6 decision fork, with explicit family-level acknowledgements) rather than a full reframe from "recognition works" to "intersubjective pedagogy works" (Framing B). The fuller family-level theoretical discussion lives in `docs/pedagogical-taxonomy.md`, which now carries structured metadata consumed by `config/tutor-agents.yaml` (`pedagogical_orientations:` map, added 2026-04-24) and by the chat UI.

No data or analysis changes; ~3 passages touched.

v3.0.48 (2026-04-24) A10 + A10b orientation-family integration into §7.9. After fixing bug_007 (v3.0.47 retraction), A10 re-ran as `eval-2026-04-23-42e7acbe` (three cells: base / recognition / matched-pedagogical under DeepSeek V3.2, three-judge panel Sonnet + GPT-5.2 + Opus 4.7). Pooled three-judge recognition-vs-matched-pedagogical $d = 0.17$ — below the pre-registered $|d| < 0.2$ density-sufficient threshold. A10 alone would have justified "density within the Hegelian-descendant family is sufficient" as the retract-branch verdict.

However, A10's matched-pedagogical prompt was grounded in Vygotsky/Piaget/Kapur/Chi/VanLehn/Grae — all Hegelian-descendant theorists. A10 therefore tested within-family density-substitutability, not orientation-orthogonal density-sufficiency. A10b (`eval-2026-04-24-e9a785c0`) added a fourth cell (`cell_96_base_behaviorist_single_unified`) with a rigorously-authored prompt grounded in Skinner/Gagné/Keller/Thorndike/Rosenshine (behaviorist explicit-instruction tradition, zero constructivist/Hegelian content by expanded blocklist), giving a four-way three-judge comparison. Pooled three-judge findings at full n: within-Hegelian $d = 0.14$ (replicates A10); within-transmission $d = 0.89$ (behaviorist substantially *below* generic base — density in wrong family is counterproductive); **between-family** $d = 1.38$ **(dominant effect)**.

§7.9 rewrote the "prompt density as alternative explanation" paragraph to integrate both findings. Previously four arguments against the density reduction; now organised around A10/A10b as the central empirical closure, with the earlier four lines of evidence (Paper 1.0 placebo, elaborate-hurts-on-Haiku, autotuning gap-widens-under-optimisation, question-asking mediation) framed as converging lines. The recognition effect is resolved at the orientation-family level: reducible neither to length nor to rigorous-grounding-in-general, but reducible to membership in the intersubjective-pedagogy family. Recognition is one effective operationalisation; matched-pedagogical is another; the family as a whole is what the paper demonstrates.

Also in v3.0.48: closed TODO §A10 and added §A13 (A10b four-way comparison) and §D6 (orientation-family pedagogy taxonomy) to TODO.md; authored `docs/pedagogical-taxonomy.md` as the canonical reference; authored `exports/a10-prompt-density-control` and `exports/a10b-orientation-family.md` with full three-judge contrast tables at locked n; authored `scripts/analyze-a10-prompt-density-control.js` (points at A10 v2, not invalidated v1) and `scripts/analyze-a10b-orientation-family.js`; added `pedagogical_orientations:` metadata block to `config/tutor-agents.yaml` for chat-UI consumption. Extended Appendix D with A10 v2 and A10b run IDs (61 total). Structural-features confound caveat added to `docs/pedagogical-taxonomy.md` "Methodological caveat" section — flags that LLM judges may reward surface features (warmth / directness / structure) independently of theoretical content; human-learner pilot (§A1) remains the only way to break this confound fully.

v3.0.47 (2026-04-23) A10 result retracted: profile-resolution bug routed `cell_95` to the base prompt. An `/ultrareview` pass on the v3.0.46 branch identified a pipeline bug (bug_007) in `services/evaluationRunner.js`: the `resolveEvalProfile` dispatch chain at lines 220–248 has explicit branches for every `prompt_type` in use at time of v3.0.42 (`enhanced`, `placebo`, `hardwired`, `naive`, `memory`, `recognition_nomem`, and the six `divergent_*/dialectical_*` variants) but no branch for the newly-introduced `matched_pedagogical` prompt type on cell_95. With `recognition_mode: false`, control fell through to the final `else` clause and `resolvedProfileName` was set to `'budget'` — identical to cell_1's resolution. The YAML's `ego.prompt_file: tutor-ego-matched-pedagogical.md` is read only by `collectPromptVersions` for provenance-hash metadata, not threaded through to `tutorApi.generateSuggestions`. Direct evidence of the bug: raw cell_95 tutor outputs on the `new_user_first_visit` scenario are lexically indistinguishable from cell_1 outputs ("As a new user, begin with this introductory lecture…" template, same phrasing, same tone), and distinctly different from cell_5 outputs ("Welcome! As a new learner, let's begin…"). The $d = 0.98$–1.62 "recognition beats matched-specificity" finding reported in v3.0.46 was in fact a "cell_5 beats cell_1 measured a second time" comparison with run-ordering noise.

Retraction. §7.9 reverted to its v3.0.45 form: four arguments against the density alternative (placebo, autotuning, mediation, architecture isolation), with an acknowledgement that the strongest version of the alternative — matched *specificity* not just length — remains untested under the Paper 2.0 v2.2 rubric. The §7.9 paragraph now explicitly names A10 as launched-but-invalidated and references this Appendix F entry. Appendix D retains the A10 run ID (`eval-2026-04-22-04497df0`) labelled as "**A10 v1 — INVALIDATED BY BUG 007**"; the re-run will get a new run ID.

Other findings from the same review pass. - bug_011 (normal): `tests/factorial-design.test.js` hard-codes three allow-lists (`VALID_PROMPT_TYPES`, `namePatterns`, `egoPromptFiles`) that do not include `matched_pedagogical`. `npm test` fails on cell_95 with three assertions. Fixed in this revision by adding `matched_pedagogical` entries to each allow-list. - **bug_001** (normal): `server.js:55` statically imports `./routes/chatRoutes.js`, which does not exist. Unrelated to the scientific work; flagged for a separate fix-up cycle.

Methodological note. The bug was not caught by `validate-config` (which only checks that the prompt file *exists* in the tutor-core prompts directory, not that the profile-resolution chain actually loads it) nor by the factorial-design tests (which had failed on cell_95 only on a secondary allow-list assertion path — `bug_011` above, fixed in this revision — not on the "is this cell loading the right prompt" question). The general lesson — that authoring a new prompt file plus a YAML cell plus an `EVAL_ONLY_PROFILES` registration is *not sufficient* to cause the prompt to run; the `resolveEvalProfile` dispatch chain also needs an explicit branch — is now a check in the §6.4.2 / §7.9 authoring workflow and is flagged in CLAUDE.md's existing "Adding New Cells" section (which already warned about EVAL_ONLY_PROFILES but not about the dispatch chain). The direct way to catch this class of bug going forward: a regression test that asserts, for each cell in `EVAL_ONLY_PROFILES`, that the `resolveEvalProfile` dispatch chain does **not** route the cell to `'budget'` unless the cell's YAML explicitly has `prompt_type: base`. (Implemented as `tests/regression-bug-007.test.js`; v3.0.74 above records the `dispatchedProfileName-vs-resolvedProfileName` refinement that keeps the A10/A10b density-control cells from false-flagging against a published `@machinespirits/tutor-core`.)

A10 re-run in progress under the fixed pipeline. Results, along with the relevant §7.9 re-extension (or confirmation that density alone suffices, forcing a retraction of the broader content-over-density claim), will be reported in the next revision.

v3.0.46 (2026-04-22) A10 prompt-density control closes §7.9 loophole; A11 M2-isolation strengthens §6.4 residual claim. Two pre-registered experiments complete the post-critique cycle opened in v3.0.43.

A10 (`eval-2026-04-22-04497df0`, DeepSeek V3.2, 3 cells × 3 runs × 21 scenarios, n = 62-63/cell after gap-fill). The §7.9 "content over density" claim had one unaddressed skeptical reading: that the Paper 1.0 placebo may not have been equally *specific* in pedagogical guidance. A10 authored a new matched-specificity prompt `tutor-ego-matched-pedagogical.md` (2,835 words, 0.9% over recognition; structurally parallel; grounded in named educational research — Piaget, Vygotsky, Bloom, VanLehn, Kapur, Chi, Graesser; blocklist-verified for zero recognition/Hegelian content) and ran cells 1 (base, 344-line) vs 5 (recognition) vs new cell_95 (matched-pedagogical) under v2.2 rubric and the Paper 2.0 judge panel. Pre-registered decision grid: if $|d(5 \text{ vs } 95)| < 0.2$, retract the content-over-density claim. GPT-5.2 (full n = 62-63): cell_1 M = 39.2, cell_95 M = 40.7, cell_5 M = 52.8; **recognition vs matched Δ = +12.1 pts,** d = 0.98, Welch's $t(115.8)$ = 5.51. Sonnet (n = 45-47/cell, 73% coverage): same direction, stronger at larger n — recog vs matched Δ = +19.4, d = 1.62. Both judges converge on the primary verdict: **density alone is not sufficient; recognition content adds substantial independent lift**. §7.9 extended from four arguments to five, with the matched-specificity control added as the closing piece. Exports: `exports/a10-prompt-density-control.md`. Analysis script:

`scripts/analyze-a10-prompt-density-control.js`.

A11 (`eval-2026-04-22-b56be6c7`, Gemini Flash 3.0 ego × Kimi K2.5 superego, cells 82/83, 3 runs × 21 scenarios). The §6.4.1 "universal substitution with model-dependent residual" claim carried a +12.3-point residual architecture benefit on Gemini Flash that was *inferred* from the factorial interaction rather than directly measured. A11 ran the direct M2-alone isolation (base + superego on Gemini Flash) and compared against the matched base-single-agent cells 80/81 from run 18027efc. Pre-registered decision grid: $\Delta \geq +9$ AND $d \geq 0.8$ to confirm the residual. Result under Sonnet ($n = 116$ on A11 cells 82/83 vs $n = 36$ on baseline cells 80/81, 93% coverage): $\Delta = +16.3$ **pts,** $d = 1.46$**, Welch's** $t(72.9) = 8.65$ — not just confirming the factorial-inferred residual but substantially exceeding it. The M2-alone magnitude is far larger than the DeepSeek comparison (+9.2 pts, $d = 1.13$ from run 768ba77b), supporting the monotonic pattern: weaker generation models → larger superego residual. §6.4.2 extended with a new "Direct isolation of the Gemini Flash residual (A11)" paragraph; §7.3 citation updated from inferred to directly measured. Exports: `exports/a11-m2-isolation-gemini-flash.md`. Analysis script: `scripts/analyze-a11-m2-gemini-flash-isolation.js`.

Combined net effect on the paper. After the v3.0.43-45 arc (A12 retiring the M3 exploratory effect) and the v3.0.46 arc (A10 + A11 strengthening the two remaining mechanisms), the paper's mechanism story is materially tighter than at v3.0.42 before the internal critique: two mechanisms supported (both now with direct isolation evidence on at least one model beyond the factorial), one mechanism cleanly null (replication failed at pre-registered thresholds), the "content over density" alternative explanation directly disconfirmed under the same rubric and judge panel as the headline claims. Appendix D extended with A10 + A11 run IDs (59 total). TODO §A10, §A11 → RESOLVED. Cost for this cycle: ~$40 OpenRouter (generation + GPT rejudge) + partial Sonnet (subscription, daily cap resets 2026-04-23). Sonnet completion scheduled for the next session will tighten partial n on both analyses but is not expected to change verdicts.

v3.0.45 (2026-04-22) A12 M3 disengagement replication closes the exploratory M3 boundary-condition as null. The $d = 1.63$ finding on DeepSeek/Sonnet in the 10-turn `trajectory_disengagement_to_ownership` scenario was flagged in v3.0.43 as a pending boundary-condition finding requiring replication across models, judges, and scenario variants. A12 ran pre-registered replication across two additional generation models (Haiku 4.5, Gemini Flash 3.0) under two judges (Sonnet 4.6, GPT-5.4) for the same cells (80 base vs 84 recognition), same scenario, $n = 4$/condition per model ($N = 32$ dialogues total across runs `eval-2026-04-22-d4547979` and `eval-2026-04-22-f4fb03f1`). The pre-registered decision grid required $d \geq 1.0$ on slope in both replication models and cross-judge $\Delta d \leq 0.5$ on identical rows.

Result matrix (Cohen's d on per-dialogue OLS slopes): Haiku/Sonnet −0.18, Haiku/GPT-5.4 +1.85 (cross-judge $\Delta d = 2.03$); Gemini Flash/Sonnet −0.93, Gemini Flash/GPT-5.4 −0.11 (cross-judge $\Delta d = 0.82$). Three of four cells fall below the $d = 0.5$ fails threshold; the one cell reaching $d \geq 1.0$ is contradicted by the secondary judge on identical rows at $\Delta d = 2.03$ ($ $4× the paper's documented cross-judge variation on the main factorial, §8.3). Under the pre-registered §4.3 sensitivity rule, this disqualifies the "replicates" reading of the Haiku/GPT-5.4 cell. Under the "fails to replicate on both models" branch of the §4.1 decision grid, disengagement hedging is retired throughout.

Changes vs v3.0.44: (1) Abstract, §1 intro paragraph, §1 contribution bullet, §3.2 note-on-evidence, §6.3.8 connecting-to-predictions, §6.4.3 mechanism-separability table, §6.4.5 interaction predictions, §7 intro, §7.8.2 prompt-lab discussion, §9 conclusion, and §9 closer all updated from "pending boundary-condition effect" framing to "null on both the primary analysis and pre-registered replication." (2) §6.3.2 canonical treatment rewritten: keeps the original DeepSeek/Sonnet data as descriptive but adds the A12 replication-failure paragraph with full result matrix and the judge-sensitivity observation as the headline verdict. (3) Figure caption for `figure-disengagement-divergence.png` updated to mark the pattern as generation-model × judge-specific and flag the A12 replication failure. (4) Appendix D extended with A12 run IDs (57 total). (5) TODO A12 → RESOLVED (failed to replicate).

Secondary methodological finding: the Haiku $\Delta d = 2.03$ between Sonnet and GPT-5.4 on identical 8 dialogues is $ $4× the judge-variation documented on the main factorial (§8.3 reports Δd ranges of 0.47–0.58). This is consistent with slope-based trajectory analyses being substantially less judge-robust than level estimates in extended-turn scenarios, and sharpens the LLM-as-judge caveats already flagged in §8.2 and §8.3 specifically for slope-based inference. Analysis script: `scripts/analyze-a12-disengagement-replication.js`; full exports: `exports/a12-disengagement-replication.md`. Cost: ~$7 OpenRouter total.

v3.0.44 (2026-04-22) Apparatus-as-method tightening (TODO §F5; no data changes). §7.4 reduced from ~51 lines to 33 lines (≈35% cut) without losing the distinctive argument. Changes: (1) §7.4 preamble consolidated from two paragraphs to one. (2) §7.4.1 three correction examples (active control reframing, judge version unification, learner prompt leakage) consolidated into a single sentence with parenthetical summaries rather than three separate paragraphs. (3) §7.4.2 dropped the redundant worked-example paragraph (already fully treated in §5.9), folded the scale details (119 claims, 18 adapters, DAG) into one sentence, removed the restated "practical lesson" at the end. Kept the dependency-graph contribution argument with its load-bearing learner-superego-paradox cascade example. (4) §7.4.3 removed the bulleted v1.0 → v2.2 rubric history (cross-reference to §5.2.6 where it fully lives), compressed to the error-correction-pattern insight plus a forward-reference to §8.6 PCA as a natural extension of the 14 → 8 construct refinement. (5) §7.4.4 condensed test-suite stats into one compound sentence and removed the two-paragraph framing into a single tight paragraph. No content claims changed; only the expository overhead was cut. Closes TODO §F5.

v3.0.43 (2026-04-22) Post-critical-review tightening pass (TODO §F1-F3; no data changes). Three coordinated edits driven by an internal 2026-04-22 critical review of v3.0.42. (1) **Collapse M3-disengagement exposure**: the $d = 1.63$ single-scenario exploratory finding (one model, one judge, one scenario, $n = 12$/condition) previously appeared in 13 places across abstract, §1 intro, §1 three-mechanism list, §1 contributions, §3 preface, §3.2 note-on-evidence, §7 intro, §7.8.2, §9 conclusion, §9 broader implication, and §9 Hegelian closer. The canonical full treatment is retained at §6.3.2 (prose, table, figure); every other mention collapsed to a one-clause pointer ("§6.3.2

reports an exploratory single-scenario effect flagged as pending replication"). Removes the bait-and-switch risk that a pending finding carrying significant narrative weight could mis-calibrate reader confidence in M3. (2) **§6.1 calibration lead rewritten for §8.6 PCA consistency**: added a paragraph ("A note on dimension independence") clarifying that the 8 v2.2 tutor dimensions load on one factor (PC1 = 80.7%, mean inter-dim $r = 0.776$), so "narrowing within-response dimension variance" is equivalent to compressing each response's within-response *floor-to-ceiling range* across correlated indicators rather than tightening 8 independent skills. Figure caption for `figure-calibration-variance.png` reworded accordingly ("narrows the dimension profile" → "compresses the within-response floor-to-ceiling range across the 8 correlated tutor dimensions"). No metric or number changes; only the interpretation is brought into line with §8.6 PCA. (3) **Judge-pooled d in headline claims**: the §1 intro operational paragraph previously reported "$d = 1.85$ floor-lifting improvement" without judge attribution. Updated to report the judge-pooled panel (pooled $d \approx 1.63$, with Sonnet-judge $d \approx 1.88$ as upper bound and Gemini-3.1-Pro-judge $d \approx 1.44$ as lower bound, cross-referencing §8.3). §9 conclusion similarly carries the pooled framing. Aligns the headline with §8.3's per-judge decomposition and removes a silent Sonnet-only bias from the abstract-adjacent claims. Also opened TODO §F (Paper Revisions) with F1-F5 tracking these plus two deferred editorial tasks (redundancy trim, apparatus-as-method tightening) and TODO §A10-A12 + D5 capturing the experimental follow-ups from the same critical review.

v3.0.40 (2026-04-19) Writing Pad controlled ablation (A5 closure). Added §6.6.9 "Writing Pad Controlled Ablation" reporting a 2×2 factorial test of whether the Freudian three-layer memory Writing Pad is load-bearing for the recognition effect on the dialectical-ego architecture. Cells 93 and 94 are byte-identical clones of cells 40 and 41 with `writing_pad_enabled: true` flipped to `false`. Run eval-2026-04-17-f1e851c3 ($n = 63$ per cell, $N = 252$, Nemotron 3-nano ego × Kimi K2.5 superego, Sonnet 4.6 judge) gives: recognition main effect $F(1, 248) = 27.10$, $p < .001$, $\eta^2 = 0.097$ (large); Writing Pad main effect $F(1, 248) = 3.96$, $p = .048$, $\eta^2 = 0.014$ in the *opposite* direction from the necessity hypothesis (pad OFF scores 3.2 points *higher*); interaction null, $F(1, 248) = 0.82$, $p = .366$. Recognition-within-pad-condition: $d = 0.56$ with pad on, $d = 0.74$ with pad off — both "medium-large," larger without the pad. The Writing Pad is therefore **not load-bearing** for the recognition effect; recognition operates at the prompt level rather than via the three-layer memory architecture. Three scope limits flagged: single ego-model pair (Nemotron × Kimi), single superego disposition (suspicious), first-turn-score-only (multi-turn coherence effects not isolated). The pad's marginal-significance negative main effect does not survive basic Bonferroni correction and is read as "does nothing useful here" rather than "actively harmful." Added `scripts/analyze-a5-writing-pad.js` (reproducible 2×2 ANOVA with regularised incomplete beta for F-to-p conversion) and `exports/a5-writing-pad.md` (generated report). Cost: ~$6 OpenRouter for generation (multi-day run, hit insufficient-credits partway and resumed after top-up); Sonnet judging via subscription. Closes v4 roadmap A5.

v3.0.39 (2026-04-17) Disposition gradient architecture scope limit (D4 closure). Added §6.6.8 "Disposition Gradient: Architecture Scope Limit" reporting that §3.4

Prediction 3's suspicious > adversary > advocate ordering is specific to the dialectical-ego architecture (cells 40–45) rather than a domain-general property of the recognition mechanism. Three configurations compared under matched n: (a) dialectical ego × philosophy (cells 40–45, Haiku × Opus, $n = 17$–18) reproduces the monotone gradient (susp $d = 0.85$, adv $d = 0.62$, advocate $d = 0.51$); (b) standard ego × philosophy (cells 22–27, Haiku × Opus, $n = 5$–10) shows a reversed ordering (susp $d = -0.01$, adv $d = 0.97$, advocate $d = 1.70$); (c) standard ego × SEL (cells 22–27, new run eval-2026-04-17-4a9b765a, Haiku × Sonnet, $n = 22$–24 per cell, 141/144 rows judged — 3 interpersonal-conflict rows with multi-suggestion format skipped by the per-turn evaluator) shows a non-monotonic pattern (susp $d = 1.25$, adv $d = 0.53$, advocate $d = 1.09$). The monotone gradient appears only in configuration (a); (b) and (c) bound the claim. Updated §3.4 Prediction 3 wording to flag the scope limit and forward-reference §6.6.8. Confounds flagged: (i) judge-model differs between D4 SEL (Sonnet 4.6) and philosophy baselines (Opus 4.6), so absolute Δ magnitudes are not directly comparable across the three rows; (ii) cells 22–27 philosophy has small n per cell; (iii) the architecture-matched cells 40–45 × SEL replication is deferred on cost grounds and is the clean next test. Added `scripts/analyze-d4-disposition-gradient.js` (reproducible analysis) and `exports/d4-disposition-gradient.md` (generated report). Cost: \$5–6 OpenRouter for the SEL run. Closes v4 roadmap D4 at "architecture-scope-limit" level; learner-side disposition variants remain out of scope for this pass.

v3.0.38 (2026-04-17) Cross-application adjacency pilot: peer support coaching (D2 Path 1). Added §6.6.7 "Cross-Application Adjacency Pilot" reporting a scope-honest single-application probe of whether the recognition main effect survives when the *skill being coached* runs counter to traditional pedagogy. Built a new content package under `content-test-support/` ("Being With, Not Fixing" — 4-lecture peer support listener training course + 4 core + 1 mood scenarios) that mirrors A6's scenario structure but frames the tutor as coaching a trainee to *not* advise, reassure, paraphrase, or resolve. Ran cells 1 (base) and 5 (recognition) under single-prompt mode, Haiku 4.5 generation, Sonnet 4.6 judge, 3 runs × 5 scenarios, $n = 15$ per cell (eval-2026-04-17-6766015b). Result: base 52.25 (SD 9.63), recognition 69.92 (SD 12.73), $\Delta = 17.67$, $d = 1.57$ ("very large" under Cohen's conventions). Closest A6 adjacency (SEL) sits at $d = 1.82$; peer-support is $\Delta d = -0.25$ below SEL and comfortably inside the A6 domain range ($d = 1.45$–2.71). Directional replication of the recognition main effect therefore survives a domain shift into a skill that runs structurally counter to conventional knowledge-transfer pedagogy. Three scope limits flagged: (a) the LLM is still prompted as a tutor, making this an *adjacency* test rather than a true cross-application test (the role-reframed D2 Path 2 study is deferred); (b) single application does not support a cross-application generalization claim; (c) Sonnet-only judging. Added `scripts/analyze-d2-support-pilot.js` (reproducible analysis) and `exports/d2-support-pilot.md` (generated report). Updated §6.6.6 closing paragraph with forward-reference to §6.6.7. Cost: \$3–4 OpenRouter. Addresses v4 roadmap D2 at Path 1 (Path 2 deferred to a separate study).

v3.0.37 (2026-04-17) Domain generalization: five-domain recognition replication (A6 closure). Extended §6.6.6 from a two-domain (philosophy, programming) replication to a five-domain treatment adding elementary math, creative writing, and social-emotional learning (SEL) content packages. Each new domain ran cells 1/5 on 5 single-turn scenarios (4 core + 1 mood) × 3 runs = $n = 15$ per condition under matched Haiku 4.5 generation × Sonnet 4.6 judge. Run IDs: eval-2026-04-17-fde764e1 + 7915be04 (math), 428f0649 + 93ba2f9a (creative), 44fa989c + 5833830a (SEL). Results: programming $d = 2.33$, math $d = 1.45$, creative $d = 1.96$, SEL $d = 1.82$, philosophy $d = 2.71$. All five domains show recognition > base with every d classified "very large" under Cohen's conventions ($d > 1.3$). Directional replication of the recognition main effect therefore holds across humanities (philosophy), technical (programming, math), creative (writing), and affective (SEL) content under matched generation × judge. Domain-×-recognition interaction ranges $\Delta d = -0.38$ (programming) to $\Delta d = -1.25$ (math) vs the philosophy anchor, but interpretation is mode-confounded (non-philosophy = single-prompt, philosophy = messages); cross-judge validation was not re-run on the four new domains for this closure (Sonnet-only). Rewrote §6.6.6 body to reframe as "five-domain generalization" rather than "programming replication"; replaced 2-row table with 5-row table; updated narrative to reflect $d = 1.45$–2.71 magnitude range and 5-domain directional replication. Added `exports/a6-domain-generalization.md` (final report) and updated `scripts/analyze-domain-generalization.js` to iterate over all five domain contrasts. Cost: $25–30 OpenRouter across the four new domain packages. Closes v4 roadmap A6.

v3.0.36 (2026-04-17) Domain generalization: recognition replicates on programming (C3 closure). Added a new §6.6.6 "Domain Generalization" subsection reporting a programming-domain replication of the recognition main effect. Ran cells 1 (base, single-agent, unified learner) and 5 (recognition, single-agent, unified learner) on a programming-tutorial scenario set (5 single-turn scenarios: debugging, code review, conceptual explanation), 3 runs × 5 scenarios, $n = 15$ per condition, Haiku 4.5 generation, Sonnet 4.6 judge (eval-2026-04-17-c92ad6c7). Compared against the closest matched-generation-and-judge philosophy contrast (eval-2026-03-02-45163390 cells 80/84, messages-mode, $n = 18$ per condition). Result: programming $d = 2.33$ (base 47.75 SD 9.10, recog 70.42 SD 10.30); philosophy $d = 2.71$ (base 43.61, recog 74.10). Both "very large" under Cohen's conventions ($d > 1.3$); domain-×-recognition interaction $\Delta d = -0.38$ under a mode-confounded comparison. Recognition main effect generalizes from philosophy to programming in direction and in very-large-magnitude class. Updated §6.6.5 closing paragraph with a forward-reference. Confounds flagged: programming uses single-prompt-mode cells while the matched philosophy cells are messages-mode (mode is a known moderator, §6.3, §6.5); scenario counts differ (5 vs 9) and scenario pairs are not matched. Added `scripts/analyze-domain-generalization.js` (reproducible analysis) and `exports/c3-domain-generalization.md` (generated report). Cost: $3 OpenRouter. Closes v4 roadmap C3.

v3.0.35 (2026-04-17) Tutor–learner asymmetry: rubric vs holistic adjudication (M6 full closure): Added a new §6.5.5 "Rubric vs Holistic Adjudication" subsection (pushing "Implications" to §6.5.6) reporting a cross-judge convergent-validity check on the 7–12× tutor-learner recognition-effect asymmetry. Scope: messages-mode 2$×2×$2 factorial cells 80–87 under rubric v2.2, $N = 1{,}552$ paired rows with all four metrics populated (tutor rubric, tutor holistic, learner rubric, learner holistic), three independent judges (Sonnet 4.6, Gemini 3.1 Pro, GPT-5.4). Primary test: does the tutor–learner d-gap shrink under holistic scoring? Result: pooled rubric d-gap = 0.741 (tutor $d = 1.120$, learner $d = 0.379$); pooled holistic d-gap = 0.327 (tutor $d = 0.565$, learner $d = 0.238$); **shrinkage** = 55.9%. Replicates tightly across all three judges (Sonnet 55.7%, Gemini 49.9%, GPT 55.7%). Within-role rubric holistic correlations do not break down (tutor $r = 0.57$, learner $r = 0.75$ pooled), so the shrinkage reflects the rubric and holistic instruments weighting recognition's effects differently rather than judge disagreement. Conclusion: $ $50% of the originally reported 7–12× asymmetry is rubric-artifact; the residual holistic gap (ratio ≈ 0.42) is the non-rubric-dependent portion, plausibly reflecting the structural explanation (prompt scope + downstream learner). Parallels the C4 paradox-closure pattern (§8.1): same-rows, same-shrinkage ($ $50%), different analysis target. §6.5.6 Implications updated to note the magnitude reframing. Added `scripts/analyze-tutor-learner-asymmetry.js` (reproducible analysis) and `exports/m6-asymmetry.md` (generated report). No new runs; DB-only analysis of existing paired data. Closes v4 roadmap M6 at full closure.

v3.0.34 (2026-04-17) Learner paradox: cell-matched holistic scores on the original paradox run (C4 full closure): Executed the final-mile adjudication promised in v3.0.33. Added holistic learner scores to the originally-scored 48 rows of eval-2026-02-20-25c78e91 (single-prompt mode, rubric v1.0, Opus 4.6 rubric judge; 47 paired after one judge JSON-parse failure). To avoid cross-version contamination, added a standalone `scripts/evaluate-paradox-holistic.js` that writes only the four holistic fields via direct SQL `UPDATE`, preserving the `learner_rubric_version = 1.0` marker rather than triggering `evaluationStore.updateResultLearnerScores`'s default version-overwrite (Sonnet 4.5 judge for holistic). Rubric d on the paired subset reproduces the paradox essentially exactly: base **rubric** $d = -2.82$ ($\Delta = -23.3$, $n = 11/12$); recognition **rubric** $d = -3.11$ ($\Delta = -22.3$, $n = 12/12$). Same rows, holistic: base $d = -1.34$, recognition $d = -1.06$ — roughly half the rubric effect size. Within-architecture rubric holistic r on the paradox cells is 0.06–0.44 versus the corpus-wide 0.72–0.77, i.e., the two judges specifically diverge on these cells rather than agreeing as they do elsewhere. Added a fourth §8.1 paragraph reporting this cell-matched result; updated §7.9 to cross-reference. The paradox is now characterized as a **combined measurement-plus-single-prompt-mode interaction** (≈50% rubric artifact, ≈50% single-prompt-mode interaction, ≈0% capability ceiling), not a general property of internal self-critique. The ∼$2K higher-capability-model experiment previously envisioned is no longer required for the central adjudication. Extended `scripts/analyze-learner-paradox-holistic.js` with `--run-id` and `--cell-min-n` flags; regenerated `exports/learner-paradox-holistic-analysis.md` with paradox-run slice. Cost: $2–3 OpenRouter. Addresses v4 roadmap C4 at full closure.

v3.0.33 (2026-04-17) Learner paradox: convergent-validity evidence from holistic scoring (C4 tightening): Added a third §8.1 paragraph reporting a corpus-wide convergent-validity check on the learner-rubric vs holistic-learner-scoring relation ($n =$ 1,759 paired rows across the four learner architectures: `unified` $n = 432$, `ego_superego` $n = 411$, `unified_recognition` $n = 446$, `ego_superego_recognition` $n = 470$). Within-architecture Pearson correlations are $r = 0.720$–0.773 (convergent validity holds uniformly — it does **not** break down for ego_superego learners, which would be expected under a deliberation-invisible-to-rubric account), and the pooled ego_superego-vs-unified contrast goes in the *opposite* direction from the original $d = 3.05$ paradox on both metrics (base $\Delta = +4.14$ rubric / $+7.39$ holistic, $d = 0.25$ / 0.39; recognition contrast near zero, $|d| < 0.08$). The cell-level breakdown of messages-mode cells 80–87 ($n = 179$–216 per cell) consistently shows ego_superego $\geq$ unified on both metrics. This reframes the $d = 3.05$ result from "general architecture property" to "scenario-and-mode-specific interaction between rubric-content dimensions and deliberation-internal content generation," which is what the rubric-artifact account predicts. §7 discussion paragraph updated to cross-reference the convergent-validity result. The remaining definitive test — adding holistic learner scores to the original paradox run (eval-2026-02-20-25c78e91, zero holistic rows) — is now a modest follow-up rather than the ~$2K higher-capability-model experiment originally priced. Added `scripts/analyze-learner-paradox-holistic.js` (reproducible analysis) and `exports/learner-paradox-holistic-analysis.md` (generated report). No new runs; DB-only analysis of existing paired data. Addresses v4 roadmap C4 at the convergent-validity level; the final-mile holistic-scoring step remains the only outstanding work to fully close C4.

v3.0.32 (2026-04-16) Inter-LLM baseline landed (C2): Executed the pilot infrastructure from v3.0.30. Generated the stratified 40-item sample (`exports/human-validation-pilot-sample` `exports/human-validation-pilot-key.jsonl`), then re-classified the same 40 items with Sonnet 4.5 (vs Haiku 4.5 in the original corpus, identical taxonomy prompt, temp=0, JSON response format) via a new `scripts/human-validation-reclassify.js` helper. Ran `scripts/human-validation-analyze.js --synthetic-rater sonnet:...` and obtained **Cohen's = 0.633 (substantial agreement), n=40, observed agreement 0.675, chance agreement 0.114**. Added an "Inter-LLM Baseline Result" subsection to §5.3 reporting the , per-category F1 (4 high, 4 moderate, 1 pilot-unstable), and confusion-matrix diagnosis. FABRICATION flagged as the single pilot-unstable category (Haiku=4, Sonnet=0; Sonnet redistributes to CONTEXT_BLINDNESS/VAGUENESS/MEMORY_FAILURE). RECOGNITION_FAILURE identified as a catch-all under Sonnet's decision policy — matches codebook's boundary rule that it should only apply when intellectual-autonomy framing is explicit. This =0.63 now serves as the lower-bound reference for the upcoming human-rater pilot: human-LLM substantially above 0.63 indicates non-idiosyncratic tracking. Cost: $0.10 API. No pre-registered hypothesis change; pure methodological closure.

v3.0.31 (2026-04-16) Learner superego paradox: §7 §8.1 harmonization (C4 partial): Updated the §7 discussion paragraph on the d=3.05 paradox (previously framed via Hegelian *Bildung*-timescale and implementation-fidelity speculation) to lead with the v3.0.26 rubric-measurement artifact account as the favoured mechanism, treating the two Hegelian alternatives as live-but-secondary rather than primary. Also tightened the §8.1 opener so the paradox-introduction paragraph cross-references the rubric-artifact mechanism that immediately follows, rather than closing on "we cannot know." Fixed stale "Section 6.16 of the pilot study" reference to canonical "Paper 1.0, Section 6.16" form matching the P5/P2 disambiguation pattern. No new data or runs — purely an internal-consistency fix that aligns the §7 speculative framing with the §8.1 DB-mechanism-check result already in the paper. The stronger empirical test (higher-capability learner model, ~$2K) remains pending and is now called out explicitly in both §7 and §8.1 as the outstanding step. Addresses v4 roadmap C4 at the textual-consistency level; closing C4 fully still requires the pending higher-capability test.

v3.0.30 (2026-04-16) Human validation pilot apparatus (C2): Operationalised the §5.3 / §8.2 human-coding validation step that previous revisions had flagged as "outstanding." Added a "Human Validation Pilot" subsection to §5.3 describing the end-to-end pipeline: stratified sampling (`scripts/human-validation-sample.js`, deterministic seed=20260416, 4 items × 10 categories = 40-item packet), per-category codebook (`docs/research/human-coding-codebook.md` v1.0 with signal phrases, boundary rules for adjacent categories, a 12-step decision flowchart, and one in-corpus exemplar per category except REGISTER_MISMATCH where n=0), analysis pipeline (`scripts/human-validation-analyze.js` computing Cohen's , Fleiss' , per-category F1, and confusion matrices with Landis-Koch interpretation bands), and inter-LLM baseline (Haiku vs Sonnet on the same sample) as a lower-bound reference. Updated §8.2 to reference the now-available apparatus rather than describing the procedure as future work. Target floor: ≥ 0.60 (substantial agreement). Pilot data collection is the only remaining step — the infrastructure is now reviewer-runnable end-to-end. Addresses v4 roadmap C2.

v3.0.29 (2026-04-16) Title-page date sync: Updated frontmatter `date: "March 2026"` → `date: "April 2026"` to match actual revision cadence (10 revisions in April: v3.0.20 – v3.0.28 plus the short paper v1.1.0). The short paper was already April 2026 after the P4 sync (v1.1.0), so the full paper title page was the sole remaining drift. No body content changed; single-line frontmatter edit.

v3.0.28 (2026-04-16) Revision-history judge-panel disambiguation (P2): Added an orientation header to Appendix F explaining that entries v1.0–v2.3.21 are pilot-study (Paper 1.0) revisions using Opus 4.5/4.6, while entries v3.0.0+ are Paper 2.0 mechanism revisions validated by the Sonnet/Gemini-Pro/GPT triple-judge panel reported in the abstract. Opus 4.6 is retained for specific side-analyses (§8.1 learner-paradox DB check, §6.4.2 isolation runs) and those uses are labelled in-place. Addresses the v4 P2 concern that revision-history mentions of Opus 4.5/4.6 could be read as conflicting with the abstract's three-judge claim. No body-text changes; purely a reader-orientation note. Closes v4 roadmap P2.

v3.0.27 (2026-04-16) Appendix B cross-reference disambiguation (P5): Added explicit "Paper 1.0" markers to six Appendix B section headers (B.6 Section 6.20, B.7 Section 6.8, B.8 Section 6.10, B.8b Section 6.21, B.8c Section 6.22, B.9 Section 6.11) and to Appendix C.2's "see Section 6.15" reference in the dimension weights paragraph. Paper 2.0's §6 has only six subsections (6.1–6.6), so the original unqualified "Section 6.X" pointers in Appendix B were stranded cross-references under a standalone read of Paper 2.0. The figure/table numbering audit otherwise confirmed that in-text "Section 6.10/6.16/6.21/6.22" mentions already carry "of the companion study" or "(Paper 1.0, …)" qualifiers (lines 154, 644, 1814, 1890, 1898). No other figure, table, or section-reference drift was found. Addresses v4 roadmap P5.

v3.0.26 (2026-04-16) Learner superego paradox: rubric-artifact mechanism

(M6/C4 partial): Added a DB-only mechanism check to §8.1 addressing the learner superego paradox (d=3.05 deficit for ego-superego learners vs unified learners). Query on cells 1/2/5/6 (Opus 4.6 judge, N=613/108/499/157) separates learner *behaviour* from learner *score*: the ego-superego architecture produces ~2.5× more output tokens (13,193–15,547 vs 5,125–6,675) and ~4× more API calls (9.0–10.4 vs 2.3) — i.e., *more* deliberation and *more* generated text — yet scores 12–18 points *lower* on the learner rubric. Per-dimension inspection of the learner_scores JSON shows ego-superego learners score well on `deliberation_depth` and `revision_signals` (dimensions that reward the internal process) but poorly on `conceptual_engagement` and `question_quality` (dimensions that reward content in the final external message). The most parsimonious account is therefore a rubric-measurement artifact: the rubric as currently constructed rewards content-rich external output, while the ego-superego architecture spends generative effort on deliberation-internal content that does not land in the output channel the rubric scores. Reframes the paradox from "internal self-critique degrades the learner" to "the rubric does not reward the kind of behaviour the architecture produces." Addresses v4 roadmap M6/C4 alternative test (regress learner score on length/complexity features) with a DB-only analysis; the stronger test (higher-capability learner model, $2K) remains pending. No new runs — purely a re-analysis of existing data.

v3.0.25 (2026-04-16) Reproduction cost statement (P3): Added a "Reproduction cost for Paper 2.0 mechanism runs" paragraph to §8.7 using recorded database figures rather than the pre-v4 rhetorical "$50K+" estimate. Actual recorded generation cost for the three core runs (DeepSeek V3.2, Gemini Flash 3.0, Haiku 4.5; 432 dialogues, 1,296 rows) is $114.58 on OpenRouter pass-through pricing. Estimated 3-judge panel cost $300–700; superego-taxonomy classification $80–100; core reproduction total $500–1,500. Including isolation runs (§6.4.2), trajectory-specific scenarios (§6.3), and prompt-lab autotuning (§7.8) roughly doubles to $1,000–3,000. Full-programme cost (207 runs, 8,274 dialogues, ~740M tokens) was $474.75 recorded generation + estimated $2,000–5,000 total including judging. The $50K figure the v4 roadmap speculated applies only to human-learner validation or human-expert rater scaling ($20K–50K range), not to LLM-judged replication. Clarifies the replication cost story: the apparatus is inexpensive to reproduce at the LLM-judge tier, and the genuine barrier is human validation, not compute. Addresses v4 roadmap P3.

v3.0.24 (2026-04-16) Judge-version confound quantification (M5): Added a per-judge effect-size decomposition to §8.3 (Model Transience). The §6.1 summary figure of d = 1.34–1.92 conflates three genuinely different judge panels: Sonnet 4.6 judge-mean d = 1.88, Gemini 3.1 Pro judge-mean d = 1.44, GPT-5.4 judge-mean d = 1.56, computed on tutor_overall_score across the three generation runs. Within-row Δd ranges 0.47–0.58, well above the 0.1 d robustness threshold, so absolute magnitudes are judge-dependent even though effect direction is conserved in all 9 cells. Introduced the judge-adjusted pooled estimate $d \approx 1.63$ (mean across 9 cells) with Sonnet-only $d \approx 1.88$ as an upper bound and Gemini-3.1-Pro-only $d \approx 1.44$ as a lower bound. Noted the partial leniency-gradient pattern (Sonnet highest on all three runs, but Gemini-vs-GPT ordering flips between runs), suggesting judge × generation-model interaction beyond simple calibration differences. No §6.1 headline range changed — the 1.34–

1.92 figure is the min–max of the new table. Addresses v4 roadmap M5 (judge-version confound acknowledged but not tabulated).

v3.0.23 (2026-04-16) Rubric PCA single-construct acknowledgment (M3): Added explicit PCA caveat to §7.8 opener — the 8 v2.2 tutor dimensions measure largely a single underlying construct (PC1 = 80.7%, 1-factor Kaiser, mean inter-dimension r = 0.776), so "dimension-targeted optimisation" uses dimensions as interpretive proxies rather than as empirically independent skills. Results below should be read as *shifts in the single construct via targeted prompt design*, not as moving one skill while others are held constant. Added a parallel "Implication for §7.8" paragraph at the end of §8.6 so readers coming from the other direction also encounter the interpretive qualification. The forced two-factor rotation (content_accuracy vs seven pedagogical dimensions) is preserved as a meaningful bifactor split; pairs of pedagogical dimensions are not. No §7.8 findings retracted — the §7.8.1/7.8.3 results (Qwen target dims 35→75, prosthesis-straitjacket transfer pattern) stand as informative prompt-engineering findings; only the *interpretation* of per-dimension shifts is qualified. Addresses v4 roadmap M3 (rubric PCA implications not addressed in §7.8).

v3.0.22 (2026-04-16) Model-dependence reframe (M2): Scoped the "general mechanism" language at four key locations (abstract paragraph 3, §6.4.3 separability intro, §6.6.5 generalizability assessment, §7 opener) to the tested capability tier — three models spanning a ~30-point baseline range, excluding both very-weak (Nemotron-class) and frontier-capability (Opus, GPT-5.4-class generator) models. Where the paper previously said "supported as general mechanisms," it now says "replicate across all three tested models and three independent judges, with magnitudes and architectural completeness varying systematically with generation quality." Added explicit scope-limitation paragraph to §6.6.5 noting that a fourth-model replication at the frontier tier would address whether the recognition effect saturates against an already-calibrated strong model. Addresses the v4 reviewer concern that claims of "general" mechanisms overreach the three-model test set. No data, analysis, or N-count changes.

v3.0.21 (2026-04-16) Lead-with-operationalization §1 rewrite (M1 partial): Added a new paragraph 2 to §1 that states what the paper does in operational terms—the specific design choices tested, the models and N, and the three headline findings (prompt d = 1.85; 15–17% additivity deficit with model-dependent residual; null trajectory modulation in the primary analysis)—before the Hegelian motivation. Addresses the v4 reviewer concern that the three-level operational disclaimer arrives only after 1,000+ words of Hegelian framing. The Hegelian exposition (original paragraphs 2–4 and §3's theoretical framework) is preserved intact; the new paragraph reorders the rhetorical stakes so the empirical operational claim lands before the reader invests in theory. Lightly tweaked the opener of the now-third paragraph ("This paper proposes an alternative grounded in Hegel's" → "The alternative this paper tests is grounded in Hegel's") for smooth handoff. No data, analysis, or N-count changes.

v3.0.20 (2026-04-16) Mechanistic account of universal substitution (new §6.4.2.1): Added a taxonomy-level causal chain that explains the stable 15–17% additivity deficit and the model-dependent residual architecture benefit from the critique-category data already in §6.2.2 and §6.2.3, converting the numerical substitution finding into a mechanistic claim. Pre-empted categories (vagueness, elicitation, content accuracy) correspond to the ~16% overlap where calibration operationalises the behaviours the superego was enforcing through critique; persistent categories (recognition failure, context awareness) have low within-round resolution rates and survive calibration; the model-dependent residual (+12.3 on Gemini Flash 3.0; near-zero on DeepSeek/Haiku; positive in Affective Shutdown / Frustration on strong models) tracks the volume of post-calibration error headroom and the prompt's operationalisation of the relevant competence (weakest on affective attunement). No new data collection or N-count changes—purely an interpretive synthesis of §6.2 and §6.4.2 data. Addresses the v4 reviewer note that §6.4.2 documents the substitution numerically but asserts the causal chain without supporting it.

v3.0.19 (2026-04-16) M3 honest reframe + multiplicity correction: Sharpened the Mechanism 3 (adaptive responsiveness) framing across abstract, §1 motivating paragraph, §3.2 note-on-evidence, §6.3.2 interpretation, §6.3.8 connecting-to-predictions block, §6.4.3 mechanism separability table, §6.4.5 interaction predictions, §7.8.2 prompt-optimisation discussion, and §9 conclusion. Replaced "conditionally supported" / "conditional emergent property" throughout with "**null on the primary pre-registered analysis; pending boundary-condition finding requiring replication**." Added explicit Holm–Bonferroni correction statement (family-wise = 0.05 across three scenario tests; disengagement effect survives at /3 = 0.0167 while remaining tests do not), with exact Welch's statistic ($t(21.9) = 3.99$, $p \approx .0006$) and explicit replication caveats (single generation model, single judge, single scenario, n=12 per condition). Fixes the bait-and-switch risk that reviewers flagged in a pre-v4 critique: the three-mechanism headline was being rescued post-hoc by a small scenario subset without the multiplicity correction or replication caveats the claim required. No data, analysis, or N-count changes—purely framing correctness.

v3.0.18 (2026-03-09) Superego redundancy refinement (§7.8.4): Added subsection connecting prompt lab findings to the universal substitution claim. On weak models, targeted superego strategies amplify error correction beyond the baseline residual. On strong models, adding superego scaffolding is actively destructive, not merely redundant. Sharpens the substitution gradient from "helpful" to "harmful" across the capability threshold.

v3.0.17 (2026-03-09) Dimension-targeted optimisation and cross-model transfer (new §7.8): Added three subsections covering dimension-targeted prompt optimisation on specific rubric dimensions (§7.8.1), M3 engineering through prompt intervention (§7.8.2), and the cross-model prompt transfer experiment with prosthesis-straitjacket inversion finding (§7.8.3). Added fifth future work direction (§9) on model-stratified prompt optimisation and optimisation threshold mapping. Old §7.8 (prompt density) renumbered to §7.9. Data from prompt lab sessions on Qwen 3.5 9B and Haiku 4.5 (N=3 per transfer cell, 10 autotune iterations on Qwen, 6 on Haiku).

v3.0.16 (2026-03-07) Fix rendering: Unicode (U+2264) does not render in the xelatex font (appears as blank/replacement character in PDF), while ≥ renders correctly. Replaced all 16 body-text with LaTeX math expressions: `d  0.15` → `$d \leq 0.15$`, `0` → `$\leq$ 0`, etc. Verified ≥ (Unicode) renders correctly and left unchanged.

v3.0.15 (2026-03-07) Unicode math operators: Replaced all 45 standalone `$\to$` with Unicode → (U+2192), 16 `$\geq$`/`$\leq$` with ≥/ , 9 `$\approx$` with ≈, 31 `$\times$`

with ×, 12 `$\pm$` with ±, and 4 `$\sim$` with `${\sim}N$` expressions. These standalone math delimiters caused math mode to leak into surrounding text in pandoc→xelatex rendering, producing garbled italic text in tables and prose (e.g., page 31 rubric table, page 40 floor-lifting paragraph). Full `$...$` math expressions (e.g., `$2 \times 2$`, `$d = 0.52$`) remain unchanged. Unicode characters render cleanly with xelatex without delimiter matching issues. **Note**:   was subsequently reverted to LaTeX math in v3.0.16 due to font glyph issue.

v3.0.14 (2026-03-07) LaTeX formatting fixes: Fixed 9 unformatted `2x2`/`2x2x2` factorial references (now `$2 \times 2$`), 11 split `$-$` sign notations (now single math expressions like `$-0.52$`), 1 LaTeX `\ref{}` macro in figure caption (replaced with prose reference), 1 split `$\sim$.30` (now `${\sim}.30$`), and 1 split `T0$-$T1` table header (now `$T0{-}T1$`). All issues would produce rendering aberrations in the PDF.

v3.0.13 (2026-03-07) Recognition term slippage pass: 10 targeted edits across §6.1, §6.4, §6.5, §6.6, §7.2, §7.8, and §9 to prevent "recognition" from sliding between design-heuristic usage (level 2–3) and philosophical-agency usage (level 1). Key changes: "recognition eliminates" → "recognition-oriented prompts eliminate"; "recognition rescues the tutor" → "recognition-enhanced prompts rescue the tutor's output quality"; "genuine dialogue" → "dialogue-structured responses"; "genuinely shaped" → "specifically adapted"; "empirical echo" → "functional parallel"; "The tutor recognizes the learner" → "The prompts instruct the tutor to treat the learner as an autonomous subject." Maintains the three-senses framework established in v3.0.12.

v3.0.12 (2026-03-07) Three senses of "recognition": Added terminological clarification paragraph to §1 (after pedagogy framing, before "The ablative finding") explicitly distinguishing recognition as (1) philosophical inspiration (Hegel), (2) operational design heuristic (prompt instructions and architectural choices), and (3) observable discourse effects (question-asking, variance reduction, substantive revision). States that empirical claims concern primarily the third level. Responds to external review concern about construct validity and double duty of "recognition."

v3.0.11 (2026-03-07) Clarify evidence status markers in §3: Renamed three "Missing evidence" markers (§3.2, mechanisms 1–3) to "Evidence needed beyond pilot" and added forward references (*Addressed in Section 6.X*) to prevent readers from misinterpreting them as gaps in the current paper. Verified provable discourse counts (119 claims, 18 adapters) remain accurate.

v3.0.10 (2026-03-07) Human expert coding validation: Strengthened §8.2 to identify human expert coding as the most important outstanding validation for mechanism claims, with explicit procedure (30–50 exchanges, two coders, 10-category taxonomy, Cohen's κ, human-LLM agreement). Added matching future work entry in §9 (now "Third") with the same procedural specificity. Both sections note that the trace infrastructure makes this feasible without re-running dialogues. Added cross-reference between §8.2 and §9.

v3.0.9 (2026-03-07) Unexplained phenomena in conclusion: Added paragraph to §9 noting three findings the current mechanism model does not explain: near-identical effect magnitude across structurally different models (DeepSeek $d = 1.88$, Haiku $d = 1.84$, Gemini Flash 3.0 $d = 1.92$), the failure of adaptive responsiveness as a general mechanism, and the learner superego paradox ($d = 3.05$). Suggests these anomalies may point to a deeper process the current apparatus cannot yet trace. Responds to external review concern about consolidating what recognition doesn't explain.

v3.0.8 (2026-03-07) Added prompt elaboration baseline (Paper 1.0 §6.21) as fourth evidence line in §7.8 prompt density paragraph: the 344-line base prompt is itself a long detailed pedagogical prompt, yet it doesn't approach recognition quality and actually hurts performance on Haiku ($M = 82.5$ naive vs base, recognition $M = 90.9$). Prompt length without recognition content can be inert or counterproductive.

v3.0.7 (2026-03-07) Prompt density defense: Added "Prompt density as alternative explanation" paragraph to §7.8, consolidating three existing evidence lines against the reduction of recognition to prompt elaboration: (1) placebo control (Paper 1.0, length-matched prompts without recognition theory), (2) autotuning experiment (§7.7, gap widens under optimization), (3) mediation analysis (§6.1.7, 42.4% mediated through question-asking). Acknowledges the strongest version of the alternative remains untested: whether *any* equally specific pedagogical prompt would match recognition.

v3.0.6 (2026-03-07) Engineered vs. discovered mechanism distinction: Added paragraph to §8.4 explicitly addressing the circularity objection: the architecture was designed for observability, not to guarantee specific outcomes; three key findings were genuinely unpredicted (superego redundancy on strong models, adaptive responsiveness null, consistent 15–17% additivity deficit). Distinguishes "mechanisms of this system" from "mechanisms of AI tutoring in general."

v3.0.5 (2026-03-07) Theoretical framing and scope: Three changes responding to external review. (1) Reframed Hegelian mapping as heuristic translation: "derive mechanisms from Hegel" → "motivate mechanisms drawing on Hegel" across abstract, §1, §3.2, §3.5, and §9 (~6 passages). Changed "derivative application" → "heuristic translation" in §3.5. The mappings are productive redescriptions, not logical derivations. (2) Added scope qualification to abstract: "All findings concern tutor output quality as assessed by LLM judges interacting with synthetic learners." (3) Added cost qualifier after contributions list: architecture is a research instrument, not a deployable system (~225K tokens/dialogue). No data or analysis changes.

v3.0.4 (2026-03-07) Epistemic language audit: Systematic softening of causal language throughout, responding to external review feedback. Changed ~45 instances: "confirmed" → "supported" for mechanism claims; "disconfirmed" → "not supported as general mechanism"; "dominant mechanism" → "strongest effect" / "primary mechanism"; "confirms [prediction]" → "is consistent with" / "supports"; "The disconfirmation of" → "The null finding for". Retained "confirmed" only for factual verification ("blind scoring confirmed by code audit") and revision history entries. Changed "Mechanism isolation confirmation" → "Mechanism isolation evidence". Softened figure captions and prediction labels throughout §6.1–6.4. No data, analysis, or N-count changes—purely rhetorical recalibration to better match the evidentiary base.

v3.0.3 (2026-03-07) Final claim audit corrections: (1) Gemini Flash 3.0 stale data fixed: `tutor_overall_score` values updated from 2026-03-03 scoring to 2026-03-05 rescoring (17.3/37.0/52.6/64.9 → 22.4/49.3/57.7/70.0), changing the interaction from −7.4 to −14.6 and additivity deficit from 10% to 17%. (2) "Substitution-to-additive transition" narrative reframed as "universal substitution (15–17% deficit across all models) with model-dependent residual architecture benefit" across ~20 passages (abstract, §3, §5, §6.4, §6.6, §7.3, §8). (3) N=570→N=432 corrected throughout (isolation runs had expanded the figure-generation query beyond the clean factorial scope). (4) §7.3 "same model, different sample" error fixed (run 45163390 is Haiku, not DeepSeek). (5) §6.3.7 dialogue quality table updated with verified DB values and Gemini Flash 3.0 rows added. Provable discourse: 87 pass, 0 warn, 0 fail.